MEAT key-rate engine (meat-core)

Contract, version 1.32.0

The MEAT core of the key-rate engine (the module meat-core) turns a finite protocol declaration, which the protocol compiler emits, and an acceptance set into a convex entropy program and a fixed-length key length; four of its main results form one chain from the single-round protocol channel the declaration generates to that key length, and the fifth is the security of the protocol that outputs it (Corollary 10.8). Marginal-constrained entropy accumulation bounds the entropy of the accepted raw key from below by n times a single-round rate, less a penalty in the acceptance probability, against every adaptive attack with a finite-dimensional quantum memory (Theorem 2.10). A licensed constraint list, one whose entries have a type the declared source mode admits, raises that rate only under the attack-set licence, the producer's certificate that every physical attack reduces to one that respects the list. The single-round rate is bounded below by the value h_low (written as an underlined h_α) of a finite convex program over sector-block states (Theorem 5.4), which a Choi parametrisation writes in solver-friendly variables without deciding the rank of Alice's marginal. The objective at any feasible point is an upper estimate of h_low, never a lower bound on it (Lemma 7.11), while any lower bound on h_low, however obtained, yields a key length (Corollary 10.18) through the fixed-length post-processing model: the key is secret (Theorem 10.7) and correct, and over an authentication scheme the protocol is secure with the authentication error added and its key consumption subtracted (Corollary 10.8), against an adversary whose memory has any dimension. In trusted-source mode Alice's marginal is fixed, which needs the source licence of the transmitter module, the attestation that her system never reaches the adversary; untrusted-source mode leaves it free and never gives a larger value. The Frank-Wolfe submodule supplies the first-order lower bound on h_low, with both linear minima bounded exactly, evaluates it, where the chain carries one, on the reference companion, the declaration of the reference model compiled beside the snapped one, and certifies it where every step from the declaration's data can be enclosed rigorously. The module admits declared data only when they are exactly admissible, the companion aside, which it admits beside its declaration and searches only in a short polish from the declaration's points; it characterises no device and constructs no honest statistics, and only the submodule's certified evaluation bounds the rounding of a computed number; where the honest statistics are supplied rather than computed from the declared devices, or the caller declares a constraint entry, it bounds the mismatch between the acceptance set and the feasible statistics from below, records the cap this proves on the probability that devices as declared pass the acceptance test, and refuses the run where that cap is at most a threshold, one half by default.

Grey text, here and throughout, is non-normative: it explains, and nothing in it is assumed or proved.

The design notes (non-normative) explain why the module is built as it is and draw its processing flowchart. The dependencies between the sections are drawn in the section dependency matrix, and the graph index opens every section's dependency graph. The statements digest holds every definition, assumption, result and requirement without the proofs. How to read a contract explains this layout and its links.

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Main results

The module's most important results, those its overview names as what it establishes, with every assumption each rests on, directly or through the claims it uses. Each links to its statement and proof; the statements digest holds them in full.

Theorem 2.10 Accumulated entropy bound under adaptive attacks

On acceptance, the raw key's Rényi entropy given the adversary's information is at least n times the single-round rate less $\alpha/(\alpha-1)\cdot\log_2(1/Pr[accept])$, for any adaptive

attack with a finite-dimensional memory. Trusted-source mode reads the product marginal of Lemma 2.9; untrusted mode needs none.

Rests on A-ATK, A-ENS, A-LAB, A-RHOM, A-SEED, A-SEQ, A-SRC

Theorem 5.4 Reduction of the rate to the convex program

For a declaration that adds no constraints on the state (an empty constraint list), in either source mode, the value h_low of the finite convex program bounds the single-round rate from below. Every step of the reduction is an equality or lowers the bound, so the computable program is conservative.

Rests on no assumption

Theorem 10.7 Secrecy of the key from the accumulated entropy

For a Rényi order α in (1, 2), a key of length $\ell \leq n{\cdot}h_low - \lambda_EC - \lceil log_2(1/\varepsilon_PA)\rceil - \alpha/(\alpha-1){\cdot}log_2(1/\varepsilon_PA) + 1$ is ε_PA-secret on the accepted state at every all-at-once finite-register attack the reduction chain carries onto the round model (A-LINK). Corollary 10.8 extends it to every attack.

Rests on A-ALICE-MEAS, A-ATK, A-CONFORM, A-ENS, A-FIXED, A-HASH, A-KEYREG, A-LAB, A-LINK, A-RHOM, A-SEED, A-SEQ, A-SPEC, A-SRC

Corollary 10.8 Security of the protocol and its net key

Secrecy and correctness compose to (ε_PA + ε_EV)-security with honest authentication; over an authentication scheme within ε_auth that consumes k_auth bits of pre-shared key, the protocol is (ε_PA + ε_EV + 2ε_auth)-secure; when both parties output a key, its net key gain is ℓ − k_auth bits.

Rests on A-ALICE-MEAS, A-ATK, A-AUTH, A-CONFORM, A-ENS, A-FIXED, A-HASH, A-KEYREG, A-LAB, A-LINK, A-RHOM, A-SEED, A-SEQ, A-SPEC, A-SRC

Corollary 10.18 Key length from any lower bound on the value

Any $h \leq h_low$ may replace h_low in the key-length bound through $max\{0, h\}$, with the same security statement. This is where every solver obtains its key length; a bound $h \leq 0$ yields no positive key when $\varepsilon_EV < 1$.

Rests on A-ALICE-MEAS, A-ATK, A-AUTH, A-CONFORM, A-ENS, A-FIXED, A-HASH, A-KEYREG, A-LAB, A-LINK, A-RHOM, A-SEED, A-SEQ, A-SPEC, A-SRC

Theorem 19.1 The estimate and the first-order lower bound

In exact real arithmetic the best linearisation over the iterates, even ones feasible only up to windows, bounds h_low from below given positive probabilities, maximal branch support, unperturbed gradients and bounds on both linear minima. An exactly feasible point's objective bounds it from above.

Rests on A-ATK, A-ENS, A-LAB, A-RHOM, A-SEED, A-SEQ, A-SRC

Corollary 19.6 First-order bound for the dropped program

In either source mode the first-order bound holds for the dropped program (Definition 9.2), with maximal support asked only of key-active pairs (acceptance value, joint sector) whose entropy is kept. It bounds h_low from below, also at iterates feasible only up to windows.

Rests on A-ATK, A-ENS, A-LAB, A-RHOM, A-SEED, A-SEQ, A-SRC

Corollary 19.7 The first-order bound on the reference companion

At any points, such as those of a search on the snapped declaration or of one continued on the companion, the bracket formed with the reference companion's

objective, gradient, entries and linear minima bounds the companion's value from below; its key length is the reference model's.

Rests on A-ALICE-MEAS, A-ATK, A-AUTH, A-CONFORM, A-CONST, A-ENS, A-FIXED, A-HASH, A-KEYREG, A-LAB, A-LINK, A-MEAS, A-RHOM, A-ROUND-SQ, A-SEED, A-SEQ, A-SNAP, A-SPEC, A-SRC

Corollary 21.3 Key length from the certified bracket

The largest bracket formed by Lemma 21.1 over finitely many points is a first-order bound of Theorem 19.1, so it is at most the value of the declaration it is formed on and gives that declaration's fixed-length key length exactly as h_lb does.

Rests on A-ALICE-MEAS, A-ATK, A-AUTH, A-CONFORM, A-ENS, A-FIXED, A-HASH, A-KEYREG, A-LAB, A-LINK, A-RHOM, A-SEED, A-SEQ, A-SPEC, A-SRC

Assumptions at a glance

Every assumption this contract states or its claims assume, with its category and the claims of this contract that assume it directly. An assumption of another contract links into that contract's PDF.

Assumption	Category	Title	Stated as	Assumed directly by
A-ENS	source	Emission of the specified ensemble	transmitter/ass: ens	Lem. 2.9, Thm. 2.10, Thm. 10.7
A-SRC	source	Retention of the system A in trusted-source mode	transmitter/ass: src	Lem. 2.8, Lem. 2.9, Thm. 2.10, Ex. 3.3, Rem. 3.6, Rem. 5.3, Rem. 10.20
A-LAB	laboratory	Each party's laboratory is sealed	Assumption 13.8	Lem. 2.9, Thm. 2.10, Lem. 10.6, Thm. 10.7
A-AUTH	channel and adversary	Authentication of the classical channel by a keyed scheme	Assumption 13.6	Cor. 10.8
A-EVE-DIM	channel and adversary	The adversary's registers are finite-dimensional	Assumption 13.9	—
A-ALICE-MEAS	protocol structure	Alice's declared measurement where no source certificate is demanded	protocol-compiler/ass: alice-meas	Lem. 10.6, Thm. 10.7
A-FIXED	protocol structure	Protocol parameters fixed before the run	Assumption 13.4	Thm. 10.7, Cor. 10.8

Assumption	Category	Title	Stated as	Assumed directly by
A-RHOM	protocol structure	Round homogeneity of the declaration	protocol-compiler/ass:rhom	Thm. 2.10
A-SEPARATE	protocol structure	Executions that share no key and no device state	Assumption 13.7	Cor. 10.9
A-SEQ	protocol structure	Sequential round structure	transmitter/ass:seq	Lem. 2.9, Thm. 2.10, Lem. 10.6, Thm. 10.7
A-SPEC	protocol structure	Specification faithful to the apparatus	protocol-compiler/ass:spec	Thm. 10.7
A-TSM	protocol structure	Output-state test statistics	reductions/ass:tsm	Rem. 3.6
A-SEED	local randomness	Private, independent and fresh local randomness	Assumption 13.3	Lem. 2.9, Thm. 2.10, Lem. 10.4, Thm. 10.7
A-CONFORM	post processing	The implementation runs the analysed post-processing	Assumption 13.5	Lem. 10.4, Lem. 10.6, Thm. 10.7, Cor. 10.8
A-HASH	post processing	Ideal two-universal privacy amplification	Assumption 13.2	Lem. 10.4, Thm. 10.7
A-KEYREG	post processing	Dedicated key register and one-way reconciliation	Assumption 13.1	Thm. 10.7
A-ATK	supplied artefact	Attack-set licence	protocol-compiler/ass:atk	Lem. 2.8, Rem. 2.12, Thm. 3.1, Ex. 3.3, Rem. 3.6, Thm. 10.7, Cor. 10.19, Rem. 10.20
A-LINK	supplied artefact	Soundness of every reduction link	protocol-compiler/ass:link	Thm. 10.7

Assumption	Category	Title	Stated as	Assumed directly by
A-LMO	numerical trust	Exact linear minimisation over the feasible set	Assumption 22.1	—

Notation

The symbols of the repository's notation catalogue (spec/notation.md) that this contract uses, in the catalogue's order, with their meaning.

Symbol	Meaning
$\mathcal{H}(A)$	Hilbert space of the system label A
$\mathcal{L}(A)$	Bounded linear operators on $\mathcal{H}(A)$
$\mathcal{L}(A, B)$	Linear maps $\mathcal{H}(A) \to \mathcal{H}(B)$
$\mathrm{Pos}(A)$	Positive semidefinite operators
$S_=(A)$	Normalized states
$S_\leq(A)$	Subnormalized states
$\mathbb{1}$, $\mathbb{1}_C$	Identity operator (contextual / named)
id	Identity channel
$\mathfrak{D}$	Protocol declaration $\big(A, \tau_A;\, Q, \mathbb{S};\, \{M_x\};\, \{F_v\};\, \mathfrak{P};\, a_A, a_B;\, \phi;\, r, \mathsf{h};\, \mathfrak{L}\big)$ — the tuple of Def. 1.1: the party partition $\mathfrak{P}$ is a slot, and the marginal slot τ_A takes a state or the no-marginal value (trusted- vs untrusted-source mode)
$\mathbb{S}$	Sector index set of Q
Q	The declared sector-carrying measured system
$\mathfrak{P}$	The declared party partition: the readout maps π_A, π_B onto the alphabets $\mathcal{O}_\mathsf{A}$, $\mathcal{O}_\mathsf{B}$, with a party label for A and, for each sector s, a declared tensor factorisation of $\mathcal{H}(Q_s)$ with a party label per factor
$\mathcal{X}$	Outcome alphabet of the measurement on A — Alice's under the default partition
$\mathcal{V}$	The measuring parties' declared joint outcome alphabet — Bob's alone under the default partition
$\mathcal{C}_A$, $\mathcal{C}_B$	The alphabets of the first and second message of the transcript, which is the $L = 2$ case
$\mathcal{K}$	Transcript alphabet $\prod_{m=1}^{L} \mathcal{C}_m$, which at $L = 2$ is $\mathcal{C}_A \times \mathcal{C}_B$
$\widehat{C}$, $\widehat{\mathcal{C}}$	The acceptance-statistic register $\widehat{C}$ and its alphabet $\widehat{\mathcal{C}}$
$\widehat{\mathcal{C}}_{\mathrm{key}}$	Key-live acceptance values
$\mathcal{Z}$, $\mathcal{Z}_\perp$	Key alphabet $\{0, \ldots, d_Z - 1\}$ and $\mathcal{Z} \sqcup \{\perp\}$
h	Declared key holder, $\mathsf{h} \in \{\mathsf{A}, \mathsf{B}\}$
$\mathfrak{L}$	Declared constraint list $\{(\hat{O}_k, \bowtie_k, \Theta_k)\}$
$\mathcal{S}_{\mathrm{acc}}$	Acceptance set

Symbol	Meaning
V_{msm}	Measurement isometry ($\sqrt{M_x} \otimes \sqrt{F_v}$ form)
V_{ann}	Announcement isometry
V_I	Eve's copy of the announcements
$V_{\widehat{C}}$	Acceptance-statistic isometry
V_R, V_Z	Raw-key isometry; raw key $\to$ key register
W	Full protocol isometry
$\mathcal{M}$	Single-round protocol channel $AQ \to Z\widehat{C}I$
$\mathbf{\Phi}$	The test-statistics map $\mathcal{L}(AQ) \to \mathbb{R}^{\widehat{\mathcal{C}}}$
$\mathcal{Z}$	The key pinching, on the raw-key register R
$\mathcal{P}_{\mathbb{S}}$	Sector pinching on Q
$\mathcal{G}$	Branch (generation) maps $\mathcal{G}_{\hat{c}}$, sector components $\mathcal{G}_{\hat{c},s}$
$\mathcal{W}$	Feasible set of sector-block-diagonal marginals: $\operatorname{tr}\omega = 1$ always; $\omega_A = \tau_A$ in trusted-source mode only; $\mathfrak{L}$ if declared (Def. 5.1)
$h_\alpha^{\uparrow}$	Single-round MEAT rate (up variant)
$h_\alpha^{\uparrow,\mathrm{QKD}}$	Single-round QKD rate — carries the $\uparrow$ arrow
$\underline{h}_\alpha$	Computable relaxation: lower bound on $h_\alpha^{\uparrow,\mathrm{QKD}}$; written `h_low` in plain text
$\underline{h}_\alpha^{\mathrm{FO}}$	First-order lower bound on $\underline{h}_\alpha$: the maximum over iterates of the linearization of the objective, with lower bounds on its two linear minima
τ_{num}	The instance's one declared numerical tolerance: the ceiling on the pre-snap defect an instance may repair in what it declares — a declared block whose measured defect (Hermiticity, positivity, completeness, column-stochasticity, a probability sum) exceeds it is a modelling error and is refused (a probability vector prepared from numerical inputs is admitted by the input preparation's rule instead, foundations/req:declared-tolerance); one within it is snapped onto an exactly admissible family and the snap is recorded with the defect it corrected
$\varepsilon_{\mathrm{PA}}$, $\varepsilon_{\mathrm{EV}}$	Secrecy / correctness budgets
λ_{EC}	Error-correction transcript length, total bits, fixed-length register (shorter transcripts padded)
Ω_{AT}	Acceptance event $\mathrm{freq}_{\hat{c}_1^n} \in \mathcal{S}_{\mathrm{acc}}$
$\widetilde{E}$, $\widehat{E}$	Purifying register; Eve's n-round register
D	Classical KL divergence, base 2; also the quantum relative entropy on operators
$\widetilde{D}_\alpha$	Sandwiched Rényi divergence
$\widetilde{H}_\alpha^{\uparrow}$	Sandwiched Rényi conditional entropy, up
$\widetilde{H}_\alpha^{\downarrow}$	Sandwiched Rényi conditional entropy, down
$\lambda_{\min}$	Minimum eigenvalue
T	Transpose, X^{T}

Terms

The phrases this contract defines, and the terms of other contracts it uses, each with a one-line gloss and the definition that fixes it. In the text, a linked term opens its definition.

Term	Meaning	Defined in
acceptance support	The outcomes of the acceptance alphabet to which some distribution of the acceptance set gives positive probability.	Definition 18.1
acceptance-side inequality	The inequality asking the acceptance-side linear form at the point used to be at most its minimum over the acceptance set.	Definition 18.1
admissible acceptance set	A closed convex subset of the probability simplex on the acceptance values.	Definition 2.5
admissible attack sequence	an attack sequence whose every round lies in the attack set of a given constraint list	foundations/def: round-model
admissible declaration	A protocol declaration satisfying the eight conditions of this module: finite systems and total maps, a well-formed marginal slot, block-diagonal POVMs, announcement-determined sifting, constraints typed by the source mode, a non-empty feasible set and party computability.	Definition 1.2
announcement map	A map from one party's readout alphabet to the alphabet of what that party announces.	protocol-compiler/def: declaration
attack sequence	one channel per round whose output is the per-round system, with Alice's system where the adversary supplies it, and Eve's new memory	foundations/def: round-model
attack set	the channels whose adjoint maps every listed operator into the stated relation with its constant	foundations/def: round-model
authenticated classical channel	The classical channel that delivers each message either exactly as sent or as an authentication failure, never before it was sent, the adversary reading every message and choosing when it is delivered.	Definition 10.2

Term	Meaning	Defined in
authentication post-processing	The closing exchange of accept or abort decisions after which a party that saw an authentication failure, or whose partner did, holds no key.	Definition 10.2
Band A	The direct route: a finite measurement declared on one sector, an empty source segment and one identity receiver link, given as data, with empty slots, that delivers the declared measurement.	protocol-compiler/def: banda
basic format	The format with one Alice sector, one measured system, Alice's measurement as a POVM, two messages neither reading the other, and no party-local entry.	protocol-compiler/def: declaration
binary64 numbers	the finite values of the IEEE 754 binary64 format, each a dyadic rational	foundations/def: binary64
block list	one square complex matrix per block of a sector map, of that block's dimension, identified with their direct sum	foundations/def: sector-map
branch map	The pass map and the fail map of an acceptance value: the sums of the Kraus terms of its pass classes and of its fail classes.	Definition 1.6
certified	the tier of a lower bound on the minimum of the objective of a declaration, formed with every arithmetic step from that declaration's data exact or rigorously enclosed, so that none of its hypotheses concerns arithmetic	foundations/def: tiers
compilation map	The map from a protocol specification to the protocol declaration it fills, slot by slot.	protocol-compiler/def: compile
computed object	an object a module computes itself in the course of a computation, such as an iterate of an optimiser; it is not declared data	foundations/def: declared-data
constrained single-round rate	The infimum, over acceptance distributions and states meeting the declared marginal and constraint list, of the KL penalty plus the key's conditional entropy.	Definition 2.4

Term	Meaning	Defined in
constraint-active	Said of a sector that is key-inert for every key-live acceptance value: it adds no deficit but remains a variable of the program.	Definition 6.1
consumer	a module other than a supplier that reads declared data	foundations/def: declared-data
data sector	A sector of a reduction instance's target whose blocks are the datum's matrices.	reductions/def: reduction-instance
declared basis of A_0	The images under the declared isometry of a fixed orthonormal basis of the Choi input register.	Definition 7.2
declared data	the data an instance states and the data a producer emits as its own declaration, with the sets such data define	foundations/def: declared-data
declared isometry	The isometry from the Choi input register into Alice's system whose range is the support frame.	Definition 7.2
declared model	The handed model with its datum replaced by the declared datum, as the instance's model map defines it.	reductions/def: emitnorm
declared order	a total order on a finite alphabet, fixed together with the alphabet	foundations/def: declared-order
declared tolerance	a binary64 number, at least zero and below one, that an instance states in its run configuration	foundations/def: tolerance
declared value	a binary64 number stated as a datum, which denotes exactly that dyadic rational and approximates nothing	foundations/def: binary64
diagonal frame	The support frame whose declared isometry is the coordinate injection onto the nonzero diagonal entries of a diagonal marginal: exact and tight.	Definition 7.2
domination link	A receiver-stage typed link of kind family-dominating from a finite target family to a family it dominates at the domination weights, whose channel witness is the noise channel with added or shared flag sectors.	reductions/ domination/def: domination-link

Term	Meaning	Defined in
dropped program	The key-rate program with the deficits of a dropped set replaced by zero, over the unchanged feasible set.	Definition 9.2
dropped set	A set of pairs of a key-live acceptance value and a joint sector whose entropy deficits a run replaces by zero.	Definition 9.2
effective outcome set	The outcome pairs of a pass class whose element of Alice's measurement is non-zero and whose element of the measured family is non-zero on the given sector.	Definition 4.5
end of an execution	The time at which both parties of an execution have sent and received every message the post-processing prescribes to them and fixed their outputs.	Definition 10.3
ensemble form	The trusted-source form of Alice's measurement slot that holds positive blocks summing to the marginal, and denotes a POVM-form declaration.	protocol-compiler/def: declaration
ensemble presentation	The certificate form that declares only the weighted virtual ensemble in sector blocks and defines the certificate from it.	transmitter/def: srcert
entrywise enclosure	a Gaussian-rational Hermitian centre together with symmetric non-negative rational radii that bound, entry by entry, the modulus of the difference between a matrix and that centre	foundations/def: exact-reference
estimate	the tier of a key-rate number that is the value of the objective at the point an optimiser returned, claimed as no bound	foundations/def: tiers
evaluated off its snap	said of a block of a snap in a run: no number from which the run computes its key length reads the snapped block, save the points at which those numbers are evaluated and the values other than the output length that the run fixes before its first round	foundations/def: evaluated-off-snap

Term	Meaning	Defined in
exact island	the computations of the engine, its instances' declarations included, by which a consumer of data or the declarer about to declare them decides whether they satisfy the conditions their declaration states, a declarer computes and decides the reference of a measurement family, a consumer's replacement of an emitted scalar bound is decided to lie on the bound's weakening side, a producer constructs the operator data it emits, the acceptance side of the optimisation is constructed and certified, or a lower bound on a linear minimum over the feasible side of the optimisation is certified	foundations/def: exact-island
exactly admissible	said of a declared operator family whose sector map agrees with the given one, whose blocks are exactly Hermitian and positive semidefinite, and whose sum in the declared order equals the constant times the identity on every block and rounds nowhere; the constant is one where none is named	foundations/def: exact-admissibility
factor enclosure	a matrix box of columns together with an exact positive diagonal core, such that the matrix is the product of a matrix L the box contains, the core and the adjoint of L	foundations/def: factor-enclosure
fail class	An announcement class of an acceptance value that the sifting predicate fails.	Definition 1.6
feasible set	The set of normalised block-list states on A and the measured system that meet the declared marginal and every constraint entry, each read exactly.	protocol-compiler/def: feasible
feasible-side inequality	The inequality asking the feasible-side linear form at a block list to be at most its minimum over the exact Choi feasible set.	Definition 18.1
first-order	the tier of a key-rate number that is a lower bound on the minimum of the objective, established by a claim under hypotheses that claim names	foundations/def: tiers

Term	Meaning	Defined in
fixed-length post-processing model	The classical processing after the rounds: acceptance test, one-way error correction with a padded fixed-length transcript, error verification with an announced flag, and privacy amplification, all messages public.	Definition 10.1
full frame	The support frame whose declared isometry is the identity on Alice's system: always available, exact, and tight exactly when her marginal is nonsingular.	Definition 7.2
fully dropped	Said of a joint sector whose pairs with every key-live acceptance value all lie in the dropped set.	Definition 9.2
general	Said of a constraint entry declaring a Hermitian matrix on Alice's system as its right-hand side.	protocol-compiler/def: homog
handed reference	a reference of a snapped measurement family that its declarer passes, with its centres and radii and unaltered, to every consumer of the snap	foundations/def: evaluated-off-snap
homogeneous	Said of a constraint entry, not declared party-local, whose right-hand side is a real scalar; in trusted-source mode it denotes that scalar times the marginal.	protocol-compiler/def: homog
honest-authentication setting	The setting in which every classical message reaches its recipient exactly as sent, in the order sent and after it was sent, the adversary receiving a copy, and no message fails authentication.	Definition 10.1
index map	The emitted choice of a bijection from each alphabet and from the sector set onto an initial segment of the non-negative integers, through which every emitted list is indexed.	protocol-compiler/def: compile
instance	the party that fixes one protocol's device data, protocol data and run configuration	foundations/def: declared-data
joint key-active index set	The joint sectors that are not key-inert with a given acceptance value.	Definition 4.8

Term	Meaning	Defined in
joint sector	A pair of an Alice sector and a sector of the measured system; the joint sectors are ordered with the Alice sector slowest.	protocol-compiler/def: declaration
key pinching	The pinching of the raw-key register onto its computational basis.	Definition 1.4
key-active	Said of a sector that fails to be key-inert for some key-live acceptance value, so that it carries entropy deficits in the objective.	Definition 6.1
key-active index set	The set of sectors that are not key-inert with a given acceptance value.	Definition 4.7
key-inert	Said of a key-live acceptance value and a sector on which the raw-key map is constant on the effective outcome set of every pass class of that value.	Definition 4.5
key-live acceptance value	An acceptance value that some announcement class passing this module's sifting predicate maps to.	Definition 1.1
licensed attack set	The attack set of a licensed constraint list, at the squashed system, in the declared source mode.	Definition 2.3
licensed for the declared source mode	Said of a constraint list every entry of which is in the form the declared source mode admits, or is party-local.	protocol-compiler/def: homog
lifted operator	The operator on the whole measured system that a scoped entry's operator denotes: the identity on every other factor.	protocol-compiler/def: declaration
marginal slot	The slot of a declaration that holds Alice's marginal state, or the value marking untrusted-source mode.	protocol-compiler/def: declaration
matrix box	a matrix of Gaussian-rational centres together with non-negative rational radii that bound, entry by entry, the modulus of the difference between a rectangular complex matrix and that centre	foundations/def: factor-enclosure
maximal branch support	The span of the images, under the pass-class Kraus operators of an acceptance value and a sector, of the admissible Alice subspace tensored with that sector.	Definition 8.1

Term	Meaning	Defined in
maximal key length	The right-hand side of the key-length bound rounded down to an integer and clamped at zero.	Definition 10.13
measured factor	A factor tagged Q: a system one party measures, with its own sector map, alphabet and family.	protocol-compiler/def: declaration
measured system	The quantum system the measuring parties receive in one round, and no other system; Bob's alone where he alone measures.	reductions/def: protocol
measurement channel	The channel that measures a POVM on a quantum input and writes the outcome into a classical register.	reductions/def: msm
moved	a block of a snap of an operator family read as a measurement that is not kept: the snap is not decided exactly to equal on it the data it is made from	foundations/def: snap
objective	The function of an acceptance distribution and a feasible state that the program minimises: a divergence penalty plus one entropy term per key-live acceptance value.	Definition 5.1
output registers	The tuple of registers on which the full outputs of a protocol presentation live.	reductions/def: opreduction
party partition	The readout maps of the two parties, with a party label for Alice's system and for each tensor factor of every sector.	protocol-compiler/def: declaration
party-local	Said of a scoped constraint entry, tagged so, whose condition holds against the marginal of the state on the systems other than its factor.	protocol-compiler/def: homog
party-local form	The form of a constraint entry scoped to one factor of a product measured system: tagged party-local and read as an operator inequality against the marginal actually present.	reductions/def: licensed
pass class	An announcement class of an acceptance value that the sifting predicate passes.	Definition 1.6
POVM form	The form of Alice's measurement slot that holds the POVM itself.	protocol-compiler/def: declaration

Term	Meaning	Defined in
producer	a module other than a supplier that emits numerical data, constructed from declared data it read, to another module as its own declaration	foundations/def: declared-data
protocol declaration	The finite object the compiler emits and the key-rate engine consumes: systems, sector map, marginal slot, measurement families, public maps, raw-key map and constraint list.	protocol-compiler/def: declaration
readout map	A total map from the outcome pairs to one party's declared finite alphabet, saying what that party reads.	protocol-compiler/def: declaration
receiver segment	The list of receiver-stage typed links of a reduction chain.	reductions/def: chain
reduction chain	A pair of a source segment and a receiver segment of typed links.	reductions/def: chain
reduction instance	A handed model with a finite datum and hypotheses about the rest, a target built from the datum and scalar values, a squashing relation with a transported set, and certified entries, all established for every admissible datum.	reductions/def: reduction-instance
reference	a family of Hermitian positive semidefinite block lists complete at the constant on every block, read together with an entrywise enclosure of every block of every element	foundations/def: exact-reference
relaxed feasible set	The feasible set read at a window: positivity and the sector-block structure kept exact, the marginal or coupling condition, the declared constraint rows and the normalisation each relaxed at the window.	Definition 17.1
removable	Said of a sector for which the hypotheses of the sector-removal theorem hold, as certified by a witness meeting the removal checklist.	Definition 6.1

Term	Meaning	Defined in
rigorously enclosed	said of a computation each of whose operations returns a set of numbers, such as an interval or a ball, containing the exact result of the operation at every value its inputs' sets contain	foundations/def: tiers
round channel	The channel of round j: the attack followed by the single-round protocol map, with the announcement placed in the adversary's register.	Definition 2.3
round model	n rounds of a per-round system S, with the attack sequences and attack sets below	foundations/def: round-model
rounds nowhere	said of an accumulation every one of whose binary64 additions is exact	foundations/def: declared-order
scalar list	A constraint list of untrusted-source mode whose right-hand sides are all real numbers, each entry a comparison of real numbers.	Definition 2.2
scope	The measured factor that an entry of the constraint list is declared on.	protocol-compiler/def: declaration
sector map	a block kind together with a non-empty, ordered list of block identifiers, each with a dimension	foundations/def: sector-map
sector set	The finite ordered list of a reduction instance's target sectors, data and scalar, each with a declared dimension.	reductions/def: reduction-instance
sequential sum	the exact sum of a finite sequence, accumulated left to right from zero	foundations/def: declared-order
single-round protocol map	The map from Alice's and the measured system to the key, acceptance and announcement registers that the declaration generates for one round.	Definition 1.4
snap	at a lattice exponent k, data of the same shape as the given data, over the same alphabet in the same order, with every entry a lattice value at k, that satisfy exactly the conditions the declaration of the given data states	foundations/def: snap

Term	Meaning	Defined in
source model	A submodule's named definition of the signal ensemble as mathematical objects, possibly infinite dimensional, rather than as matrices.	transmitter/def: source-spec
source segment	The ordered, possibly empty list of source links leading from the virtual ensemble to the specified one.	transmitter/def: source-segment
source specification	The declared description of Alice's side of a protocol: alphabet, retained system with sectors, measurement, signal ensemble and source mode.	transmitter/def: source-spec
source-replacement certificate	An operator on Alice's retained system and the virtual transmitted system whose marginal on her side is her declared marginal and which, measured on her side by her POVM, leaves the weighted virtual members, with retention attested.	transmitter/def: srcert
squashing channel	A channel from the physical system to the squashed system whose adjoint realises a squashing relation.	reductions/def: squashing
state-blind	Said of a sector on which every measurement operator of the measured system, and every constraint operator when constraints are declared, is a scalar multiple of the identity.	Definition 6.1
statistics map	The map sending a state on Alice's and the measured system to the probabilities of the acceptance values.	Definition 1.4
sum in the declared order	the sequential sum of a family indexed by an alphabet, taken in the alphabet's declared order	foundations/def: declared-order
supplier	a module whose own claims decide their conditions on the data it reads or emits in floating point, at slacks an instance declares to it, rather than exactly	foundations/def: declared-data
support frame	The range of an isometry from the Choi input register into Alice's system that contains the support of her declared marginal.	Definition 7.2

Term	Meaning	Defined in
system metadata	The party label of Alice's system and, per sector, the declared tensor factorisation with a party label per factor.	Definition 1.1
target family	The target POVM of a reduction instance at a datum, assigned one sector block at a time from the datum and the scalar values.	reductions/def: reduction-instance
τ_A-homogeneous list	A constraint list of trusted-source mode whose entries are all declared homogeneous, each right-hand side constructed as its scalar times Alice's declared marginal.	Definition 2.2
tier	one of three labels, estimate, first-order or certified, that a reported key-rate number carries	foundations/def: tiers
tight	Said of a support frame equal to the support of Alice's marginal, equivalently of dimension equal to its rank.	Definition 7.2
transcript protocol	The announcement slot of the general format: an ordered list of messages, each a function of its sender's readout and of the earlier messages.	protocol-compiler/def: declaration
transmitted system	The system Alice sends into the channel, carrying the specified signal ensemble.	transmitter/def: source-spec
trusted-source mode	The source mode in which Alice's retained system never reaches the adversary, with a declared marginal and, above dimension one, a certificate.	transmitter/def: source-spec
untrusted-source mode	The source mode with no certificate and no marginal, in which the adversary may have prepared Alice's retained system.	transmitter/def: source-spec
virtual ensemble	The finite-dimensional ensemble the source segment starts from and the certificate is written against.	transmitter/def: source-segment

Part I
Mathematics

1 The declaration and the protocol channel

This section fixes the data every later result is computed from and the channel those data generate. The protocol declaration is the compiler's, read here with the conditions and names this module's claims rest on (Definition 1.1); an ensemble-form declaration denotes an ordinary one (Lemma 1.3). From the declared measurements, announcements and key map the module generates a single-round protocol channel whose acceptance marginal is the statistics map (Theorem 1.5), and branch maps that split the announcement classes of each acceptance value into pass and fail classes (Definition 1.6, Lemma 1.7). The branch maps carry the orthogonality the entropy calculation uses, and a component of a register other than the raw-key register that is constant across a branch map's Kraus terms can be removed from it without changing any quantity of this module (Remark 1.8).

Imports foundations/def:declared-order, protocol-compiler/def:declaration

The declaration is the compiler's (Definition "Protocol declaration" (protocol-compiler/def:declaration)), in either format, admissible (Definition 1.2), sifting by the announcements and a raw key its holder can compute. This module adds the key-live acceptance values, its reading of the party partition, the summation order and exact admissibility.

The operator, state and channel conventions of this contract, among them that every Hilbert space is finite-dimensional, are those of Definition "Operator conventions and pinching" (foundations/def:pinching), and its divergences and conditional entropies are those of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl) and Definition "Sandwiched Rényi divergence and conditional entropies" (foundations/def:renyi). In the tuple below, which is the compiler's, A is Alice's system and τ_A the content of its marginal slot; Q is the system the family $\{F_v\}$ measures, with sectors $\mathcal{H}(Q) = \bigoplus_{s\in\mathbb{S}} \mathcal{H}(Q_s)$; $\{M_x\}$ is the measurement on A; $\mathfrak{P}$ is the party partition, with readout maps $\pi_{\mathsf{A}}, \pi_{\mathsf{B}}$ onto the alphabets $\mathcal{O}_{\mathsf{A}}, \mathcal{O}_{\mathsf{B}}$; the announcement maps a_A, a_B, or in the general format the messages of (DX5), give each outcome pair (x, v) its joint announcement $a(x, v)$, an announcement class; ϕ maps announcement classes to acceptance values; r is the raw-key map, h the key holder and $\mathfrak{L}$ the constraint list. The definition adds two conditions on r and names the acceptance values that can carry key, then says what this module reads of $\mathfrak{P}$, reads the summation order off the declaration, states exact admissibility, and gives the general format with its reading of exact admissibility. The compiler's conditions on the declaration it emits (Definition "Target conditions of a

well-formed declaration" (protocol-compiler/def:target)) are the counterpart of Definition 1.2, and the two lists are checked against each other at the interface between the two modules. Requirement 14.1 says what becomes of a declaration that is not exactly admissible, and that restricting the states of the program to lists of blocks along Alice's sectors is sound is the subject of Remark "Restriction of the feasible set to block-diagonal states" (protocol-compiler/rem:block-restriction), which Theorem 5.4 discharges. The two source modes that the two values of the marginal slot name are compared on the program in Remark 5.3.

Definition 1.1 (The protocol declaration as the MEAT core reads it)**.** In this module a protocol declaration is a tuple

$$\mathfrak{D} = \Big(A,\ \tau_A;\ Q,\ \mathbb{S};\ \{M_x\}_{x\in\mathcal{X}};\ \{F_v\}_{v\in\mathcal{V}};\ \mathfrak{P};\ a_A,\ a_B;\ \phi;\ r,\ \mathsf{h};\ \mathfrak{L}\Big)$$

of Definition "Protocol declaration" (protocol-compiler/def:declaration), in its basic format or in its general format, (DX1)–(DX6) there, whose components satisfy the well-formedness conditions of Definition 1.2 and the two conditions on its raw-key map below. The two values of its marginal slot, $\tau_A \in S_=(A)$ and $\tau_A = \mathsf{none}$, are the two source modes, trusted-source mode and untrusted-source mode. Write $\mathcal{K}$ for the announcement-class alphabet of that definition, $\mathcal{C}_A \times \mathcal{C}_B$ in the basic format and the product of the message alphabets of (DX5) in the general one, $a(x,v) \in \mathcal{K}$ for the joint announcement it assigns to an outcome pair, and κ for a class. *Sifting is announcement-determined*: there is $\varsigma : \mathcal{K} \to \{\text{pass}, \text{fail}\}$ with $r(x,v) = \perp \iff \varsigma(a(x,v)) = \text{fail}$. *The raw key is party-computable*: for the declared key holder $\mathsf{h} \in \{\mathsf{A}, \mathsf{B}\}$ there is $r_{\mathsf{h}} : \mathcal{O}_{\mathsf{h}} \times \mathcal{K} \to \mathcal{Z}_\perp$ with

$$r(x,v) = r_{\mathsf{h}}\big(\pi_{\mathsf{h}}(x,v),\ a(x,v)\big) \qquad \text{for all } (x,v) \in \mathcal{X}\times\mathcal{V} \tag{1.1}$$

(Definition 1.2 (viii)). We write

$$\widehat{\mathcal{C}}_{\text{key}} \coloneqq \big\{\hat{c} \in \widehat{\mathcal{C}} :\ \exists\, \kappa \in \phi^{-1}(\hat{c}) \text{ with } \varsigma(\kappa) = \text{pass}\big\} \tag{1.2}$$

for the set of *key-live acceptance values*, ς being the predicate of the sifting condition above. That condition fixes ς on every class some outcome pair realises, and on a class $\kappa \notin a(\mathcal{X}\times\mathcal{V})$, which no outcome pair realises, we take $\varsigma(\kappa) = \text{fail}$; so $\widehat{\mathcal{C}}_{\text{key}}$ is the set of acceptance values to which some realised class passing sifting maps, and it is fixed by the declaration.

What this module reads of $\mathfrak{P}$. The party partition $\mathfrak{P}$ of that definition consists of (a) its two readout maps π_{A} and π_{B}, naming what each party physically observes in a round, and (b) *system metadata*: the party label for A and, for each $s \in \mathbb{S}$, the declared tensor factorization $\mathcal{H}(Q_s) = \bigotimes_i \mathcal{H}(Q_{s,i})$ with a party label per factor (a comma-indexed *tensor* factor $Q_{s,i}$, never the parenthesized superscript $X^{(s)}$, which this module reserves for *sector blocks*

of an operator). The *default partition* is that of Definition "Target conditions of a well-formed declaration" (protocol-compiler/def:target): $\pi_{\mathsf{A}}(x, v) = x$, $\pi_{\mathsf{B}}(x, v) = v$ with A labelled A and every Q_s labelled B. Part (a) is consumed by the derivations, through Definition 1.2 (viii). Part (b) is consumed by no derivation; it carries an obligation of the compiler: the declaration's POVMs must be realizable by each party measuring the factors it is labelled with, so that π_{A} and π_{B} are what the parties can in fact read. Part (b) is read by the premises that place the parties' laboratories: Assumption 13.8 takes all its labels as naming the parties' laboratories, and A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)) reads the label of A; so a run attests that obligation there, and the claims that assume those premises rest on it through them. This module never verifies the obligation.

The declared summation order. The declaration fixes, as part of the declared alphabet $\mathcal{V}$ of outcomes, a declared order $v_1 \prec v_2 \prec \cdots \prec v_{|\mathcal{V}|}$ on it. For each sector s the completeness sum is the sum in the declared order of the blocks $F_v^{(s)}$, the sequential sum

$$\Sigma_0^{(s)} \coloneqq 0, \qquad \Sigma_i^{(s)} \coloneqq \Sigma_{i-1}^{(s)} + F_{v_i}^{(s)} \quad (1 \le i \le |\mathcal{V}|), \qquad \Sigma^{(s)} \coloneqq \Sigma_{|\mathcal{V}|}^{(s)}, \tag{1.3}$$

formed left to right, entrywise, with no re-ordering and no re-association. In exact arithmetic (1.3) is the unordered sum and the order is immaterial; in binary64 it is not (Definition "Sums in a declared order and sums that round nowhere" (foundations/def:declared-order)), so the order is part of the datum and this module reads it off the declared alphabet rather than choosing one.

Exact admissibility. A declaration reaches this module with its operator data exactly admissible: every F_v exactly Hermitian and exactly positive semidefinite per sector block, every $\hat{O}_k$ exactly Hermitian per sector block (positivity is asked of no constraint operator anywhere in this module), every M_x and the declared τ_A exactly Hermitian and exactly positive semidefinite, and the completeness sum (1.3) of each sector exactly the identity with every addition rounding nowhere. These are the conditions the compiler decides on the numbers it emits (Definition "Exact admissibility of declared operator data" (protocol-compiler/def:exact)).

The general format. The general format is that of Definition "Protocol declaration" (protocol-compiler/def:declaration), (DX1)–(DX6) there, and the basic format is its case of one Alice sector, one measured factor, POVM form, two messages neither of which reads the other, and no party-local entry. This contract refers to five of that definition's displays under its own labels, each read as it is defined there: the product sector map of (DX1), over the ordered set $\mathbb{F}$ of measured factors, factor f having the sectors $s_f \in \mathbb{S}^f$

with projectors $\Pi^f_{s_f}$ and block dimensions d_{f,s_f},

$$\begin{gathered} Q=\bigotimes_{f\in\mathbb{F}} Q_f, \qquad \mathbb{S}=\prod_{f\in\mathbb{F}} \mathbb{S}^f, \qquad \mathcal{H}(Q_s)=\bigotimes_{f} \mathcal{H}(Q_{f,s_f}), \\ \Pi_s=\bigotimes_{f} \Pi^f_{s_f}, \qquad d_s=\prod_{f} d_{f,s_f}; \end{gathered} \tag{1.4}$$

the product family of (DX3),

$$F_v \coloneqq \bigotimes_{f\in\mathbb{F}} F^f_{v_f}, \qquad F^{(s)}_v=\bigotimes_{f\in\mathbb{F}} F^{f,(s_f)}_{v_f} \qquad \big(v=(v_f)_f,\ s=(s_f)_f\big); \tag{1.5}$$

the two conditions on the declared $\{N_x\}$ of the ensemble form of (DX4), whose blocks lie along the index set $\mathbb{T}$ of Alice's sectors, $\mathcal{H}(A)=\bigoplus_{t\in\mathbb{T}}\mathcal{H}(A_t)$, and the family of the POVM-form declaration an ensemble-form declaration denotes, with P_t the projector onto the support of $\tau^{(t)}_A$ and the inverse square root taken on that support and zero off it,

$$\sum_{x\in\mathcal{X}} N^{(t)}_x=\tau^{(t)}_A \quad\text{and}\quad \tau^{(t)}_A\neq 0 \qquad \text{for every } t\in\mathbb{T}, \tag{1.6}$$

$$M^{(t)}_x \coloneqq \big(\tau^{(t)}_A\big)^{-1/2} N^{(t)}_x \big(\tau^{(t)}_A\big)^{-1/2} + \frac{\operatorname{tr} N^{(t)}_x}{\operatorname{tr}\tau^{(t)}_A}\big(\mathbb{1}_{A_t}-P_t\big); \tag{1.7}$$

and the joint announcement function a transcript protocol of (DX5) derives by composition from its L messages, message m having the sender π_m, whose readout map is π_{π_m}, the alphabet $\mathcal{C}_m$ and the table a_m,

$$\begin{gathered} \kappa_m(x,v) \coloneqq a_m\big(\pi_{\pi_m}(x,v),\ \kappa_1(x,v),\dots,\kappa_{m-1}(x,v)\big), \\ a(x,v) \coloneqq \big(\kappa_m(x,v)\big)_{m=1}^{L} \in \mathcal{K} \coloneqq \prod_{m=1}^{L} \mathcal{C}_m. \end{gathered} \tag{1.8}$$

Exact admissibility in the general format. The exact-admissibility paragraph above holds with these readings: the conditions on $\{F_v\}$ are decided per measured factor, on the family $\{F^f_{v_f}\}$, its blocks and its declared order, and on no product; the conditions on τ_A, on every M_x and on the right-hand side Θ_k of a general entry are decided block by block along $\mathbb{T}$, the trace of τ_A accumulated over the blocks in their declared order and rounding nowhere; and in ensemble form the conditions on $\{M_x\}$ are replaced by those of (DX4) on $\{N_x\}$, the denoted $\{M_x\}$ being a POVM by Lemma 1.3 and decided by nothing.

Uses foundations/def:declared-order, protocol-compiler/def:declaration · *Used by* Def. 1.2, Lem. 1.3, Lem. 1.7, Def. 2.2, Def. 2.3, Def. 4.8, Lem. 4.9, Def. 5.1, Thm. 5.4, Prop. 5.5, Def. 7.2, Lem. 7.4, Thm. 7.5, Lem. 9.1, Def. 9.2, Lem. 9.6, Def. 10.1, Rem. 10.20, Prop. 11.1, Rem. 11.2, Rem. 11.4, Prop. 12.1, Ass. 13.1, Ass. 13.8, Req. 14.1, Req. 14.3, Thm. 19.1, Lem. 20.1, Lem. 21.1, Lem. 21.2 · newly derived · implemented · *Code* certiqs_key.meat_core.declaration.declaration: ProtocolDeclaration

An admissible declaration is finite with total maps, a well-formed marginal slot, block-diagonal POVMs, announcement-determined sifting, constraints typed by the source mode and a non-empty feasible set. Each party announces from its own readout and the key holder computes the raw key from its own.

Condition (viii) is stated for a declared party partition and not only for the default partition, because a declaration that places both parties' data in v, as the $d_A = 1$ embedding of Remark 5.3 (b) does, satisfies the default-partition reading of (viii) vacuously, with $\mathsf{h} = \mathsf{B}$ and r any function of v at all, while guaranteeing nothing about what the key holder can in fact compute; (viii) is what makes the error-correction and verification steps of Theorem 10.7 physically executable. How these conditions compare with the compiler's target conditions is Remark "Relation to the consuming engine's admission conditions" (protocol-compiler/rem:engine-admission). Conditions (i)–(viii) are decided on the declared data by the compiler's validator (Definition "The emitted validator" (protocol-compiler/def:validator), checks V1–V8), condition (vii) one-sidedly (Theorem "Feasibility by an explicit or a verified candidate" (protocol-compiler/thm:feasibility)). This module's admission decides every condition but (vii) again, exactly (Requirement 14.2), and it reports a key length only on a declaration the compiler emitted (Requirement 16.7), so that every condition is decided wherever a key length is reported.

Definition 1.2 (Admissible declaration)**.** The tuple $\mathfrak{D}$ of Definition 1.1 is an *admissible declaration* when it satisfies: (i) all systems finite-dimensional, all alphabets finite, all maps total; (ii) the marginal slot holds $\tau_A \in S_=(A)$ (trusted-source mode) or the value none (untrusted-source mode); in particular $\operatorname{tr}\tau_A = 1$ whenever a marginal is declared, so $\tau_A = 0$ is inadmissible; (iii) $\{M_x\}$ and $\{F_v\}$ are POVMs and every F_v and every $\hat{O}_k$ is sector-block-diagonal; (iv) r maps into $\mathcal{Z} \sqcup \{\perp\}$; (v) sifting is announcement-determined via ς; (vi) $\bowtie_k \in \{\preceq, =, \succeq\}$ and each entry carries one of the two declared forms of Definition 2.2, so that Θ_k is Hermitian on A, in trusted-source mode, respectively $\bowtie_k \in \{\leq, =, \geq\}$ and $\Theta_k = c_k \in \mathbb{R}$ in untrusted-source mode; (vii) $\mathcal{W} \neq \varnothing$; (viii) *party computability:* the declaration names a party partition $\mathfrak{P}$ (Definition 1.1) and

- each party announces a function of its own readout: a_A is defined on $\mathcal{O}_\mathsf{A}$ and a_B on $\mathcal{O}_\mathsf{B}$, with $a(x,v) = (a_A(\pi_\mathsf{A}(x,v)), a_B(\pi_\mathsf{B}(x,v)))$;
- the declared key holder $\mathsf{h} \in \{\mathsf{A},\mathsf{B}\}$ computes the raw key from *its own readout* and the announcements, $r(x,v) = r_\mathsf{h}(\pi_\mathsf{h}(x,v), a(x,v))$ with r_h a declared total function, Eq. (1.1).

Under the default partition ($\pi_\mathsf{A}(x,v) = x$, $\pi_\mathsf{B}(x,v) = v$) these conditions read: $a_A : \mathcal{X} \to \mathcal{C}_A$, $a_B : \mathcal{V} \to \mathcal{C}_B$, and $r(x,v) = r_\mathsf{A}(x, a(x,v))$ resp. $r_\mathsf{B}(v, a(x,v))$.

In the general format of Definition 1.1 (DX1)–(DX6) the same eight conditions read as follows, and at one Alice sector, one measured factor, POVM form and two messages they are the conditions above. (iii): every M_x, the declared τ_A and the right-hand side of every general entry are block diagonal along Alice's sectors $\mathbb{T}$, *by their type as block lists against a declared factor tagged* A, and $\{M_x\}$ is a POVM; in ensemble form the declared $\{N_x\}$ satisfy (1.6) and $\{M_x\}$ is the family (1.7), a POVM block diagonal along $\mathbb{T}$ by Lemma 1.3 (i); $\{F_v\}$ is the product family (1.5) of per-factor POVMs, each block diagonal for its factor's sector map, and every $\hat{O}_k$ is block diagonal for the sector map of its scope. (vi): an entry may in addition carry the party-local form of Definition 2.2, in either mode, with $\bowtie_k \in \{\preceq, =, \succeq\}$ and $c_k \in \mathbb{R}$. (viii): each message of the transcript is a function of its sender's own readout and of the earlier messages, a is the composition (1.8), and (1.1) holds with that a.

Uses Def. 1.1, Def. 2.2 · *Used by* Lem. 1.3, Thm. 1.5, Lem. 1.7, Lem. 2.7, Thm. 2.10, Ex. 3.3, Rem. 3.4, Lem. 4.3, Lem. 4.9, Prop. 5.2, Rem. 5.3, Thm. 5.4, Prop. 5.5, Cor. 5.6, Thm. 5.10, Prop. 6.3, Prop. 6.5, Lem. 7.1, Lem. 7.3, Lem. 7.4, Lem. 7.11, Lem. 8.2, Lem. 8.8, Thm. 8.9, Lem. 9.1, Prop. 9.3, Cor. 9.4, Prop. 9.5, Lem. 9.6, Thm. 10.7, Prop. 10.12, Rem. 10.20, Prop. 11.1, Prop. 12.1, Req. 14.2, Req. 14.3, Req. 15.2, Req. 16.7, Prop. 17.2, Rem. 17.3, Thm. 19.1, Cor. 19.6, Cor. 19.7, Lem. 21.1, Lem. 21.2, Cor. 21.3, Cor. 21.4 · newly derived · accepted

Positive operators summing to Alice's marginal sector by sector denote a POVM, block diagonal along her sectors, that returns them between square roots of the marginal and vanishes on a sector exactly where they do. Conversely a POVM and a marginal, both block diagonal, give such operators.

A declaration in ensemble form holds, in place of Alice's POVM, positive operators N_x that sum to her marginal sector by sector, (1.6). The lemma shows that the operators (1.7) such a declaration denotes are a POVM, so that every claim of this module applies to it as it stands; by part (iii), every predicate asking which outcomes can occur on a sector is decided on the declared numbers. Part (iv) is the converse; the operators it gives need not be exactly representable.

Lemma 1.3 (The measurement an ensemble-form declaration denotes)**.** Let $\mathcal{H}(A) = \bigoplus_{t\in\mathbb{T}} \mathcal{H}(A_t)$, and for every t let $\tau_A^{(t)} \in \mathrm{Pos}(A_t)$ be non-zero and $N_x^{(t)} \in \mathrm{Pos}(A_t)$, $x \in \mathcal{X}$, with $\sum_x N_x^{(t)} = \tau_A^{(t)}$. Let P_t be the projector onto the support of $\tau_A^{(t)}$, let $(\tau_A^{(t)})^{-1/2}$ be the inverse square root on that support, zero off it, and let $M_x^{(t)}$ be the operator of (1.7), $M_x \coloneqq \bigoplus_t M_x^{(t)}$. Then:

(i) $M_x^{(t)} \succeq 0$ and $\sum_x M_x^{(t)} = \mathbb{1}_{A_t}$ for every t, so $\{M_x\}_{x\in\mathcal{X}}$ is a POVM on A and every element is block diagonal along $\mathbb{T}$;

(ii) $(\tau_A^{(t)})^{1/2} M_x^{(t)} (\tau_A^{(t)})^{1/2} = N_x^{(t)}$, and in particular $\mathrm{tr}[M_x \tau_A] = \sum_t \mathrm{tr}\, N_x^{(t)}$;

(iii) $M_x^{(t)} = 0$ if and only if $N_x^{(t)} = 0$;

(iv) conversely, if $\tau_A \in \mathrm{Pos}(A)$ and $\{M_x\}$ is a POVM on A, both block diagonal along $\mathbb{T}$, then $N_x \coloneqq \tau_A^{1/2} M_x \tau_A^{1/2}$ is positive semidefinite and block diagonal along $\mathbb{T}$ and $\sum_x N_x = \tau_A$.

Uses Def. 1.1, Def. 1.2 · *Used by* Def. 4.8, Lem. 7.4, Lem. 9.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.general_validation:validate_general_declaration

Proof. Fix t and drop it from the notation; write $S \coloneqq \tau_A^{-1/2}$, a positive semidefinite operator with $S\tau_A S = P$ and $\tau_A^{1/2} S = S\tau_A^{1/2} = P$, and $w_x \coloneqq \operatorname{tr} N_x / \operatorname{tr} \tau_A$, which is defined because $\tau_A \neq 0$ is positive semidefinite, hence of positive trace, and satisfies $w_x \geq 0$ and $\sum_x w_x = \operatorname{tr} \tau_A / \operatorname{tr} \tau_A = 1$.

(i) $SN_xS \succeq 0$ as a conjugation of a positive semidefinite operator by a Hermitian one, and $w_x(\mathbb{1} - P) \succeq 0$; so $M_x \succeq 0$. By linearity $\sum_x M_x = S\tau_A S + (\mathbb{1} - P) = P + (\mathbb{1} - P) = \mathbb{1}$. Block diagonality of M_x along $\mathbb{T}$ is its definition as a direct sum.

(ii) From $0 \preceq N_x \preceq \sum_{x'} N_{x'} = \tau_A$ it follows that N_x vanishes on the kernel of τ_A: for ξ in that kernel $0 \leq \langle\xi| N_x |\xi\rangle \leq \langle\xi| \tau_A |\xi\rangle = 0$, and a positive semidefinite operator with $\langle\xi| N_x |\xi\rangle = 0$ has $N_x |\xi\rangle = 0$. Hence $PN_xP = N_x$. Then $\tau_A^{1/2} M_x \tau_A^{1/2} = PN_xP + w_x \tau_A^{1/2}(\mathbb{1} - P)\tau_A^{1/2} = N_x$, because $\tau_A^{1/2}(\mathbb{1} - P) = 0$. Taking the trace, $\operatorname{tr}[M_x\tau_A] = \operatorname{tr} N_x$ on the sector, and the sum over t is the second statement.

(iii) If $N_x = 0$ then $w_x = 0$ and both terms of M_x vanish. If $N_x \neq 0$ then $N_x = PN_xP$ is a non-zero operator on the support of τ_A, on which S is invertible, so $SN_xS \neq 0$; and SN_xS is supported on the range of P while $w_x(\mathbb{1} - P)$ is supported on its complement, so their sum is non-zero.

(iv) N_x is a conjugation of $M_x \succeq 0$ by the Hermitian operator $\tau_A^{1/2}$, which is block diagonal along $\mathbb{T}$ because τ_A is, the square root acting block by block; and $\sum_x N_x = \tau_A^{1/2} \mathbb{1}_A \tau_A^{1/2} = \tau_A$. □

The declaration generates coherent measurement, announcement, statistic and key isometries, the single-round protocol map, the key pinching and the statistics map. They give the entropy proof a precise representation of the protocol's classical outputs.

Definition 1.4 (Generated registers, isometries and maps)**.** The classical registers are X (alphabet $\mathcal{X}$), Y (alphabet $\mathcal{V}$), the announcement register C_AC_B and I (alphabet $\mathcal{K}$ each), $\widehat{C}$ (alphabet $\widehat{\mathcal{C}}$), R and Z (alphabet $\mathcal{Z}_\perp$). The announcement register is one register, named after the two messages of the basic format; in the general format it holds the whole tuple $a(x, v)$ of

messages. Define

$$V_{\mathrm{msm}} \coloneqq \sum_{x\in\mathcal{X}}\sum_{v\in\mathcal{V}} |x,v\rangle_{XY} \otimes \big(\sqrt{M_x}\otimes\sqrt{F_v}\big), \quad \mathcal{H}(AQ)\to\mathcal{H}(XY\,AQ), \tag{1.9}$$

$$V_{\mathrm{ann}} \coloneqq \sum_{x,v} |a(x,v)\rangle_{C_AC_B} \otimes |x,v\rangle\langle x,v|_{XY}\,, \qquad \text{controlled on } XY, \tag{1.10}$$

$$V_I \coloneqq \sum_{\kappa\in\mathcal{K}} |\kappa\rangle_I \otimes |\kappa\rangle\langle\kappa|_{C_AC_B}\,, \qquad \text{controlled on } C_AC_B, \tag{1.11}$$

$$V_{\widehat{C}} \coloneqq \sum_{\kappa\in\mathcal{K}} |\phi(\kappa)\rangle_{\widehat{C}} \otimes |\kappa\rangle\langle\kappa|_{C_AC_B}\,, \qquad \text{controlled on } C_AC_B, \tag{1.12}$$

$$V_R \coloneqq \sum_{x,v} |r(x,v)\rangle_R \otimes |x,v\rangle\langle x,v|_{XY}\,, \qquad \text{controlled on } XY, \tag{1.13}$$

$$V_Z \coloneqq \sum_{j\in\mathcal{Z}_\perp} |j\rangle_Z \otimes Z_j, \qquad Z_j \coloneqq |j\rangle\langle j|_R\,, \qquad \text{controlled on } R, \tag{1.14}$$

each acting as the identity on all registers not named. The full protocol isometry and the *single-round protocol map* are

$$\begin{aligned} W &\coloneqq V_R\,V_{\widehat{C}}\,V_I\,V_{\mathrm{ann}}\,V_{\mathrm{msm}}, \\ \mathcal{M}(\omega) &\coloneqq \mathrm{tr}_{AQ\,XY\,C_AC_B\,R}\Big[V_Z\,W\,\omega\,W^\dagger V_Z^\dagger\Big], \qquad \mathcal{M}: AQ\to Z\widehat{C}I. \end{aligned} \tag{1.15}$$

The *key pinching* is $\mathcal{Z}(\tau) \coloneqq \sum_{j\in\mathcal{Z}_\perp} Z_j\,\tau\,Z_j$ on R, the pinching of Definition "Operator conventions and pinching" (foundations/def:pinching) by the projectors Z_j (extended by the identity to any ambient system), and the *statistics map* is

$$\boldsymbol{\Phi}[\omega] \coloneqq \sum_{\hat{c}\in\widehat{\mathcal{C}}} \left(\sum_{(x,v):\,\phi(a(x,v))=\hat{c}} \mathrm{tr}\big[(M_x\otimes F_v)\,\omega\big] \right) \hat{e}_{\hat{c}}, \qquad \boldsymbol{\Phi}: \mathcal{L}(AQ)\to\mathbb{R}^{\widehat{\mathcal{C}}}, \tag{1.16}$$

with $\hat{e}_{\hat{c}}$ the unit vector at position $\hat{c}$.

Used by Thm. 1.5, Lem. 1.7, Thm. 4.1, Lem. 4.3, Lem. 4.9, Prop. 5.5, Thm. 6.2, Lem. 7.4, Cor. 8.5, Lem. 8.6, Lem. 9.1, Cor. 9.4, Prop. 9.5, Lem. 9.6, Prop. 10.12, Req. 14.3 · newly derived · implemented · *Code* certiqs_key.meat_core.objective.branch: generate_maps · *Oracles* registry.py

For an admissible declaration (Definition 1.2) the single-round protocol map is a channel with classical key, acceptance and announcement registers, given by an explicit measurement formula. Its acceptance marginal is exactly the statistics map.

Theorem 1.5 (The declaration generates a protocol channel)**.** Under Definition 1.2:

(i) Each of (1.9)–(1.14), and hence W, is an isometry, and $\mathcal{M}$ is a channel $AQ\to Z\widehat{C}I$ with $\widehat{C}$, I, Z classical.

(ii) For every system E and every $\omega \in S_{\leq}(AQE)$,

$$(\mathcal{M} \otimes \mathrm{id}_E)(\omega) = \sum_{x,v} |r(x,v)\rangle\langle r(x,v)|_Z \otimes |\phi(a(x,v))\rangle\langle \phi(a(x,v))|_{\widehat{C}} \otimes |a(x,v)\rangle\langle a(x,v)|_I \otimes \mathrm{tr}_{AQ}\big[(M_x \otimes F_v \otimes \mathbb{1}_E)\,\omega\big]. \tag{1.17}$$

(iii) The $\widehat{C}$-marginal of $(\mathcal{M} \otimes \mathrm{id}_E)(\omega)$ is the probability vector $\mathbf{\Phi}[\omega_{AQ}]$; $\mathbf{\Phi}$ is affine, and $\mathbf{\Phi}[\omega]$ is a probability distribution on $\widehat{\mathcal{C}}$ for every $\omega \in S_{=}(AQ)$.

Uses Def. 1.2, Def. 1.4 · *Used by* Lem. 1.7, Def. 2.3, Thm. 2.10, Thm. 4.1, Thm. 5.4, Rem. 5.9, Thm. 5.10 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.branch:generate_maps · *Oracles* registry.py

Proof. (i) With $\{|x,v\rangle\}$ orthonormal, $V_{\mathrm{msm}}^\dagger V_{\mathrm{msm}} = \sum_{x,v} \sqrt{M_x}\sqrt{M_x} \otimes \sqrt{F_v}\sqrt{F_v} = \Big(\sum_x M_x\Big) \otimes \Big(\sum_v F_v\Big) = \mathbb{1}_A \otimes \mathbb{1}_Q$ by POVM completeness. Each of (1.10)–(1.13) has the form $\sum_\mu |g(\mu)\rangle \otimes |\mu\rangle\langle\mu|$ for a function g on the control alphabet, so $V^\dagger V = \sum_\mu |\mu\rangle\langle\mu| = \mathbb{1}$ on the control register; $V_Z^\dagger V_Z = \sum_j Z_j = \mathbb{1}_R$. Compositions of isometries are isometries; conjugation by an isometry followed by a partial trace is CPTP. Z, $\widehat{C}$, I are classical because (1.17) exhibits the output diagonal in their computational bases.

(ii) Write $K_{x,v} \coloneqq \sqrt{M_x} \otimes \sqrt{F_v}$, so that $V_{\mathrm{msm}}\,\omega\,V_{\mathrm{msm}}^\dagger = \sum_{x,v,x',v'} |x,v\rangle\langle x',v'|_{XY} \otimes (K_{x,v} \otimes \mathbb{1}_E)\,\omega\,(K_{x',v'} \otimes \mathbb{1}_E)^\dagger$. Every subsequent isometry in W, and V_Z, is controlled on XY directly or on a register whose content is a function of XY, so conjugation by them appends the labels $a(x,v)$, $a(x,v)$ (in I), $\phi(a(x,v))$, $r(x,v)$, $r(x,v)$ (in Z) to the ket side and the primed labels to the bra side of each term. The partial trace in (1.15) includes tr_{XY}, and $\mathrm{tr}_{XY}\,|x,v\rangle\langle x',v'| = \delta_{xx'}\delta_{vv'}$, so only diagonal terms survive; on those, $\mathrm{tr}_{AQ}[K_{x,v}\,\omega\,K_{x,v}^\dagger] = \mathrm{tr}_{AQ}[(M_x \otimes F_v)\,\omega]$ by cyclicity of the partial trace over AQ. This is (1.17).

(iii) Take the $\widehat{C}$-marginal of (1.17) and group terms by $\hat{c} = \phi(a(x,v))$; tracing E replaces the last factor by $\mathrm{tr}[(M_x \otimes F_v)\,\omega_{AQ}]$. Affinity is manifest; normalization follows from $\sum_{x,v} \mathrm{tr}[(M_x \otimes F_v)\omega] = \mathrm{tr}[\omega] = 1$. □

Each acceptance value splits its announcement classes into pass and fail classes, and the two branch maps sum the Kraus terms of each, sector by sector as well. They keep the coherence the entropy calculation needs while separating key-bearing rounds from discarded ones.

Definition 1.6 (Branch maps)**.** For $\hat{c} \in \widehat{\mathcal{C}}$, a class $\kappa \in \phi^{-1}(\hat{c})$ with $\varsigma(\kappa) = \mathrm{pass}$ is a *pass class* of $\hat{c}$, and one with $\varsigma(\kappa) = \mathrm{fail}$ a *fail class*; let $\mathcal{K}_{\mathrm{pass}}(\hat{c}) \coloneqq \{\kappa \in \phi^{-1}(\hat{c}) : \varsigma(\kappa) = \mathrm{pass}\}$ and $\mathcal{K}_{\mathrm{fail}}(\hat{c}) \coloneqq \{\kappa \in \phi^{-1}(\hat{c}) : \varsigma(\kappa) = \mathrm{fail}\}$. For

each class κ define the operator

$$K_\kappa \coloneqq \sum_{(x,v)\in a^{-1}(\kappa)} |r(x,v)\rangle_R \otimes |x,v\rangle_{XY} \otimes |\kappa\rangle_{C_A C_B} \otimes \left(\sqrt{M_x} \otimes \sqrt{F_v}\right) \\ : \mathcal{H}(AQ) \to \mathcal{H}(R\,XY\,C_A C_B\,AQ), \tag{1.18}$$

and the CP *branch maps*, the pass map $\mathcal{G}_{\hat{c}}$ and the fail map $\Theta_{\hat{c}}$,

$$\mathcal{G}_{\hat{c}}(\omega) \coloneqq \sum_{\kappa\in\mathcal{K}_{\mathrm{pass}}(\hat{c})} K_\kappa\,\omega\,K_\kappa^\dagger, \qquad \Theta_{\hat{c}}(\omega) \coloneqq \sum_{\kappa\in\mathcal{K}_{\mathrm{fail}}(\hat{c})} K_\kappa\,\omega\,K_\kappa^\dagger. \tag{1.19}$$

For $s \in \mathbb{S}$, the sector versions $K_\kappa^{(s)}$, $\mathcal{G}_{\hat{c},s}$, $\Theta_{\hat{c},s}$ are (1.18)–(1.19) with $\sqrt{F_v}$ replaced by $\sqrt{F_v^{(s)}}$, acting $\mathcal{H}(AQ_s) \to \mathcal{H}(R\,XY\,C_A C_B\,AQ_s)$.

Used by Lem. 1.7, Thm. 4.1, Prop. 4.6, Def. 4.8, Lem. 4.9, Prop. 5.5, Prop. 6.3, Lem. 7.4, Def. 8.1, Lem. 8.2, Lem. 8.6, Lem. 9.1, Req. 14.3, Req. 14.5, Lem. 21.2 · newly derived · implemented · *Code* certiqs_key.meat_core.objective.branch: ClassBranch · *Oracles* registry.py

The pass and fail maps are trace-non-increasing, their traces add up to the acceptance value's probability, their outputs have orthogonal supports and the fail branch is fixed by the key pinching. They split by sector on block-diagonal inputs, and a value that is not key-live has a zero pass map.

Lemma 1.7 (Properties of the pass and fail branch maps)**.** Under Definition 1.2, for every $\hat{c} \in \widehat{\mathcal{C}}$:

(i) $\mathcal{G}_{\hat{c}}$ and $\Theta_{\hat{c}}$ are CP and trace-non-increasing, with

$$\operatorname{tr}\mathcal{G}_{\hat{c}}(\omega) + \operatorname{tr}\Theta_{\hat{c}}(\omega) = \mathbf{\Phi}[\omega](\hat{c}) \qquad \text{for all } \omega \in \mathrm{Pos}(AQ). \tag{1.20}$$

(ii) $\mathcal{G}_{\hat{c}}(\omega)$ is supported on $\mathrm{span}\{|j\rangle_R : j \in \mathcal{Z}\}$ and $\Theta_{\hat{c}}(\omega) = |\bot\rangle\langle\bot|_R \otimes \theta_{\hat{c}}(\omega)$ for a CP map $\theta_{\hat{c}}$; in particular their supports are orthogonal and $\mathcal{Z}(\Theta_{\hat{c}}(\omega)^p) = \Theta_{\hat{c}}(\omega)^p$ for every $p > 0$.

(iii) (Sector decomposition.) If $\omega = \bigoplus_{s\in\mathbb{S}} \omega_s$ is block-diagonal with respect to the decomposition $\mathcal{H}(AQ) = \bigoplus_s \mathcal{H}(AQ_s)$, then

$$\mathcal{G}_{\hat{c}}(\omega) = \bigoplus_{s\in\mathbb{S}} \mathcal{G}_{\hat{c},s}(\omega_s), \qquad \Theta_{\hat{c}}(\omega) = \bigoplus_{s\in\mathbb{S}} \Theta_{\hat{c},s}(\omega_s), \qquad \mathbf{\Phi}[\omega] = \sum_{s\in\mathbb{S}} \mathbf{\Phi}_s[\omega_s], \tag{1.21}$$

where the direct sums refer to the (mutually orthogonal) subspaces $\mathcal{H}(R\,XY\,C_A C_B) \otimes \mathcal{H}(AQ_s)$ and $\mathbf{\Phi}_s$ is (1.16) with $F_v^{(s)}$ in place of F_v.

(iv) If $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{key}}$ then $\mathcal{G}_{\hat{c}} = 0$.

The complete positivity of $\mathcal{G}_{\hat{c}}$ and $\Theta_{\hat{c}}$ in (i), and all of (iii), hold as well when $\{F_v\}$ is any family of sector-block-diagonal positive semidefinite operators, complete or not, the other components satisfying Definition 1.2.

Uses Def. 1.1, Def. 1.2, Def. 1.4, Thm. 1.5, Def. 1.6 · *Used by* Thm. 2.10, Thm. 4.1, Cor. 4.2, Lem. 4.3, Cor. 4.4, Prop. 4.6, Lem. 4.9, Prop. 5.5, Lem. 7.4, Thm. 8.4, Prop. 17.2, Lem. 18.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.branch: evaluate_branches · *Oracles* registry.py

Proof. (i) CP is Kraus form. For the trace, note that for a single class κ, the register kets $|r(x,v),x,v,\kappa\rangle$ appearing in K_κ are mutually orthonormal as (x,v) ranges over $a^{-1}(\kappa)$ (they differ in XY), so $\operatorname{tr}\left[K_\kappa \omega K_\kappa^\dagger\right] = \sum_{(x,v)\in a^{-1}(\kappa)} \operatorname{tr}[(M_x \otimes F_v)\,\omega]$ by cyclicity. Summing over $\kappa \in \phi^{-1}(\hat{c})$ (pass and fail together) and comparing with (1.16) gives (1.20); each summand of (1.20) is non-negative, so $\operatorname{tr}\mathcal{G}_{\hat{c}}(\omega)$ and $\operatorname{tr}\Theta_{\hat{c}}(\omega)$ are each at most $\mathbf{\Phi}[\omega](\hat{c})$. The remaining step, $\mathbf{\Phi}[\omega](\hat{c}) \le 1$ for a state ω, is not a property of the branch maps: it is Theorem 1.5 (iii), that $\mathbf{\Phi}[\omega]$ is a probability distribution on $\widehat{\mathcal{C}}$, whose proof consumes the completeness of the declared POVMs, $\sum_v F_v = \mathbb{1}_Q$ and $\sum_x M_x = \mathbb{1}_A$ (Definition 1.2). Hence both maps are trace-non-increasing on states.

(ii) By announcement-determined sifting (Definition 1.2 (v)), $r(x,v) \in \mathcal{Z}$ for every (x,v) in a pass class and $r(x,v) = \perp$ for every (x,v) in a fail class. Hence every Kraus operator of $\mathcal{G}_{\hat{c}}$ carries an R-ket in $\mathcal{Z}$ and every Kraus operator of $\Theta_{\hat{c}}$ carries $|\perp\rangle_R$; the $\perp$ value factors out of the sum defining $\Theta_{\hat{c}}$. The pinching $\mathcal{Z}$ acts as the identity on any operator supported on a single R-eigenspace, and powers preserve supports.

(iii) By blockwise functional calculus, $\sqrt{F_v} = \bigoplus_s \sqrt{F_v^{(s)}}$, so $K_\kappa = \sum_s K_\kappa^{(s)}$ with $K_\kappa^{(s)} = K_\kappa(\mathbb{1}_A \otimes \Pi_s)$ mapping $\mathcal{H}(AQ_s)$ into $\mathcal{H}(R\,XY\,C_AC_B) \otimes \mathcal{H}(AQ_s)$ (the operators $\sqrt{M_x} \otimes \sqrt{F_v^{(s)}}$ do not leave the sector). For block-diagonal ω, $K_\kappa\,\omega\,K_\kappa^\dagger = \sum_{s,s'} K_\kappa^{(s)}\,\omega\,K_\kappa^{(s')\dagger}$ and $K_\kappa^{(s)}\omega\,K_\kappa^{(s')\dagger} = K_\kappa^{(s)}(\mathbb{1} \otimes \Pi_s)\,\omega\,(\mathbb{1} \otimes \Pi_{s'})K_\kappa^{(s')\dagger}$ vanishes for $s \neq s'$ because $(\mathbb{1} \otimes \Pi_s)\,\omega\,(\mathbb{1} \otimes \Pi_{s'}) = 0$. The images for different s lie in orthogonal subspaces, giving the direct sums; the $\mathbf{\Phi}$ statement follows from linearity of the trace and block-diagonality of F_v.

(iv) If $\hat{c} \notin \widehat{\mathcal{C}}_{\text{key}}$ there are no pass classes in $\phi^{-1}(\hat{c})$ by (1.2), so the sum (1.19) defining $\mathcal{G}_{\hat{c}}$ is empty.

Any positive family. The Kraus-form argument for complete positivity in (i) and the argument for (iii) read of $\{F_v\}$ only that each F_v is positive semidefinite, so that $\sqrt{F_v}$ is defined, and sector-block-diagonal; neither reads $\sum_v F_v = \mathbb{1}_Q$, which enters (i) only through Theorem 1.5 (iii) for the bound on the traces. □

An isometry on registers other than the raw-key register leaves the pinched trace power unchanged, so no quantity of this module depends on a component of another register that is constant across a branch map's Kraus terms. Varying components and the raw-key register carry what the proofs use.

Theorem 4.1 writes the entropy on an acceptance value through the pinched trace power $\mathrm{tr}\big[(\mathcal{Z}(\mathcal{G}_{\hat{c}}(\omega)^{1/\alpha}))^{\alpha}\big]$ of the pass map's output, the trace of that output and the probability of the value. The remark shows that such a trace power does not see an isometry on registers other than the raw-key register R.

Remark 1.8 (Inert register components of a branch map). If U is an isometry acting only on registers disjoint from R, then $(U\sigma U^{\dagger})^{p} = U\sigma^{p}U^{\dagger}$ and $\mathcal{Z}(U\sigma^{p}U^{\dagger}) = U\mathcal{Z}(\sigma^{p})U^{\dagger}$, so $\mathrm{tr}\big[(\mathcal{Z}((U\sigma U^{\dagger})^{1/\alpha}))^{\alpha}\big] = \mathrm{tr}\big[(\mathcal{Z}(\sigma^{1/\alpha}))^{\alpha}\big]$. Consequently a register component on a register other than R whose value is *constant across all Kraus terms of a given branch map* — for example the $C_A C_B$ factor when a branch has a single announcement class — can be removed from that branch map without changing any quantity in this module: its constant ket $|c\rangle$ enters every Kraus operator through the isometry $|\psi\rangle \mapsto |c\rangle \otimes |\psi\rangle$, which acts only on registers disjoint from R. Components that vary across Kraus terms (the class label κ across classes, the outcome labels within a class) are not of this kind: they carry the orthogonality that the derivations of this module use (Lemma 1.7). Nor is the raw-key register R itself, even where its value is constant, as it is ($\perp$) across every Kraus term of a fail map (Lemma 1.7 (ii)): the pinching $\mathcal{Z}$ acts on R, and R is what separates the pass branch from the fail branch.

Used by Req. 14.5 · newly derived · accepted

2 Accumulation

The imported sequential accumulation theorem bounds the entropy of an accepted multi-round state from below by the round count times a single-round rate, less a penalty in the acceptance probability (Fact 2.1). Instantiated on the generated round channels, with every announcement handed to the adversary and her quantum memory carried between rounds, it gives the section's main result, Theorem 2.10: on acceptance, the entropy of the raw key given the adversary's information is at least n times the single-round rate less $(\alpha/(\alpha-1))\cdot\log_2(1/\Pr[\text{accept}])$, with the product marginal of Alice's systems in trusted-source mode derived from the source's premises, A-SEED (v) and A-LAB at the party holding A (Lemma 2.9), and with no source assumption in untrusted-source mode. Licensed constraint lists, licensed attack sets and the constrained single-round rate are defined here; a constraint list may raise the rate only through a licensed restriction of the attack model (Remark 2.12).

Imports foundations/def:kl, foundations/def:renyi, foundations/def:round-model, protocol-compiler/def:homog, transmitter/def:source-segment, transmitter/def:source-spec, transmitter/def:srcert, transmitter/thm:certificate-sufficiency

On an accepted event the Rényi entropy of n rounds of sequential channels with quantum memory, under a product marginal constraint, is at least n times a single-round rate less (α/(α−1))·log₂(1/p_Ω). The rate minimises a KL penalty plus conditional entropies over the channels' convex range.

The imported theorem is stated below in this module's notation. The locator notes after the statement tie each display to its place in the source, and note (vii) checks that a one-dimensional marginal register is an admissible instance, on which the untrusted-source mode case of Theorem 2.10 rests. Untrusted-source mode on the program is the subject of Remark 5.3, whose part (b) quotes the sentence following [1, Eq. (1)] that note (vii) cites. The Fact is the source's theorem in its published form, on $\alpha < \infty$ (note (vi)), quoted once for both source modes. The source states no separate result for trivial marginal registers: that its hypotheses admit them is note (vii)'s reading of the source at the locators it quotes, and the claim that instantiates the Fact there, Theorem 2.10, is newly derived.

Fact 2.1 (Marginal-constrained entropy accumulation, sequential form)**.** *([1, Thm. 4.2a, Eqs. (100)–(101)].)* For each $j \in \{1,\dots,n\}$ take a state $\sigma^{(j-1)} \in S_=(A_{j-1})$ and a channel $\mathcal{M}_j \in \mathrm{CPTP}(A_{j-1}E_{j-1},\, S_j\widehat{C}_jE_j)$ with $\widehat{C}_j$ classical. Let

$$\rho_{S_1^n\widehat{C}_1^nE_n} = \big(\mathcal{M}_n \circ \cdots \circ \mathcal{M}_1\big)\big[\omega^0\big], \qquad \omega^0 \in S_=(A_0^{n-1}E_0),$$

subject to the product marginal constraint $\omega^0_{A_0^{n-1}} = \sigma^{(0)} \otimes \cdots \otimes \sigma^{(n-1)}$; E_0 is unconstrained, and ω^0 ranges over *all* normalized states compatible with the marginal constraint (purity is not assumed). Suppose $\rho = p_\Omega\,\rho_{|\Omega} + (1-p_\Omega)\rho_{|\bar\Omega}$ with $p_\Omega \in (0,1]$ and $\rho_{|\Omega}, \rho_{|\bar\Omega}$ normalized, all $\widehat{C}_j$ isomorphic to one register $\widehat{C}$ with alphabet $\widehat{\mathcal{C}}$, and let S_Ω be a convex set of probability distributions on $\widehat{\mathcal{C}}$ such that $\mathrm{freq}_{\hat c_1^n} \in S_\Omega$ for every frequency vector with nonzero probability in $\rho_{|\Omega}$. Then, for every $\alpha \in (1,\infty)$ (locator note (vi): the source also covers the endpoint $\alpha = \infty$, which no claim of this module uses),

$$\widetilde{H}^{\uparrow}_{\alpha}\big(S_1^n\big|\widehat{C}_1^nE_n\big)_{\rho_{|\Omega}} \;\geq\; n\,h^{\uparrow}_{\alpha} - \frac{\alpha}{\alpha-1}\log_2\frac{1}{p_\Omega}, \tag{2.1}$$

where

$$h^{\uparrow}_{\alpha} = \inf_{\mathbf{q}\in S_\Omega}\ \inf_{\nu\in\Sigma}\left(\frac{\alpha}{\alpha-1}\,D\big(\mathbf{q}\,\big\|\,\boldsymbol{\nu}_{\widehat{C}}\big) + \sum_{\hat c\in\mathrm{supp}(\boldsymbol{\nu}_{\widehat{C}})} q(\hat c)\,\widetilde{H}^{\uparrow}_{\alpha}(S|E\widetilde{E})_{\nu_{|\hat c}}\right), \tag{2.2}$$

with D the classical (base-2) KL divergence of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl) and $\Sigma = \Sigma_{S\widehat{C}E\widetilde{E}}$ the *marginal-constrained convex range* ([1, Def. 4.2, Eq. (98)]) of

the channels $\mathcal{M}_j \otimes \mathrm{id}_{\widetilde{E}}$ and states $\sigma^{(j-1)}$:

$$\Sigma \coloneqq \mathrm{conv}\Big(\bigcup_{j=1}^{n} \Sigma_j \Big), \qquad \Sigma_j \coloneqq \big\{ (\mathcal{M}_j \otimes \mathrm{id}_{\widetilde{E}})[\omega] : \omega \in S_=(A_{j-1}E_{j-1}\widetilde{E}),\ \omega_{A_{j-1}} = \sigma^{(j-1)} \big\}, \tag{2.3}$$

where every output is embedded in common registers $S\widehat{C}E\widetilde{E}$ (the results hold for any choice of embeddings; [1, footnote 12]) and $\widetilde{E}$ is a register of large enough dimension to serve as a purifying register for any of the $A_{j-1}E_{j-1}$ registers (enlargeable without loss of generality; [1, footnote 17]).

Locator notes. (i) The statement above is the source's Thm. 4.2a with its registers written in this module's fonts; its Eq. (100) is (2.1), its Eq. (101) is (2.2), and its Def. 4.2, Eq. (98) is (2.3). Thm. 4.2a is the special case of the source's Thm. 4.2b in which the per-round secret test registers (the unbarred C_j of the source) are trivial, which is the case Theorem 1.5 generates. The source states its entropies in base 2 (bits). (ii) The divergence in the penalty term is the *KL* divergence D: the source obtains the rate objective as the convex conjugate of a log-sum-exponential ([1, Lemma 4.11, Eqs. (113)–(114)]) and evaluates it in its Eq. (118), stating there explicitly that the first term "is the KL divergence". (iii) An alternative import is the tensor-product specialization [1, Cor. 4.2, Eq. (137)], in which the round channels carry no E_j output and Eve's entire action is the choice of one global input state. The source presents that corollary as a register substitution of the theorems above, and its Sec. 5 applies the resulting analysis when every public announcement is made only after *all* measurements are complete; for per-round ("on-the-fly") announcements — the model Theorem 1.5 generates, whose round channel announces I_j in every round — the source's own route (its Sec. 5, Protocols 1–2 and the reduction between them) is the sequential form quoted here, in which Eve carries a quantum memory E_{j-1} between rounds and may adapt each round's action to all earlier announcements. The present import therefore covers a strictly larger attack model than that specialization would. (iv) The theorem has no announcement register of its own: announcements enter the sequential model inside the E_j outputs (the source's Sec. 5: Eve "immediately incorporates" each I_j into her side information). (v) Indexing is off by one between this module's rounds and the source's registers ([1, Rem. 5.1]): round j here uses the source's $A_{j-1}E_{j-1}$. (vi) The source states the theorem for $\alpha \in (1, \infty]$, and its Def. 2.6 defines the entropies at the endpoint; Definition "Sandwiched Rényi divergence and conditional entropies" (foundations/def:renyi) defines the sandwiched family on $(1, \infty)$ only, so the Fact is quoted, and Theorems 2.10 and 3.1 are stated, on $(1, \infty)$. No claim uses the endpoint: every key-length statement is confined to $(1, 2)$. (vii) *Trivial marginal registers are admissible* — the import check for untrusted-source mode, read off the source at the locators named. The hypotheses that [1, Thm. 4.2a] places on the marginal registers are, in full, that one "take a state $\sigma^{(j-1)} \in S_=(A_{j-1})$,

and a channel $M_j \in \mathrm{CPTP}(A_{j-1}E_{j-1}, S_j\widehat{C}_jE_j)$, such that $\widehat{C}_j$ are classical", together with the product condition $\omega_{A_0^{n-1}} = \sigma_{A_0}^{(0)} \otimes \cdots \otimes \sigma_{A_{n-1}}^{(n-1)}$ on the input state; and [1, Def. 4.2, Eq. (98)] likewise only "consider[s] channels $E_j \in \mathrm{CPTP}(A_{j-1}\bar{A}_{j-1}, Z_j)$ and states $\sigma^{(j-1)} \in S_=(A_{j-1})$" (in the source's symbols, where E_j is a channel and Z_j its output register, not this module's register E_j and raw key Z_j). *No dimension bound and no rank condition is imposed on A_{j-1} at either locator*, and the source's standing convention is explicit that none is imposed globally either: "we will assume that all systems are finite-dimensional, but we will not impose any bounds on the system dimensions unless otherwise specified" ([1, Sec. 2, the convention stated below its Table 1]). A one-dimensional A_{j-1} is therefore an admissible instantiation, and at it the product marginal condition is satisfied by every normalized ω, since $S_=(A_0^{n-1})$ is then a singleton — the hypothesis discharges vacuously rather than failing. Independently, the source reads the same equivalence into its own definition of channel conditional entropy: see the sentence following [1, Eq. (1)]. Nothing else in Fact 2.1 mentions A_{j-1} beyond the purifier requirement, which asks $\widetilde{E}$ to purify $A_{j-1}E_{j-1}$ and is only weakened by shrinking A_{j-1}.

Uses foundations/def:kl, foundations/def:renyi · *Used by* Def. 2.3, Lem. 2.8, Thm. 2.10, Rem. 2.12, Thm. 3.1, Lem. 8.8 · published in [1, Thm. 4.2a, Eqs. (100)-(101)]; [1, Def. 4.2, Eq. (98)]; [1, footnote 12]; [1, footnote 17]; [1, Lemma 4.11, Eqs. (113)-(114)]; [1, Cor. 4.2, Eq. (137)]; [1, Rem. 5.1]; [1, Thm. 4.2a]; [1, Eq. (1)] · accepted · *Oracles* registry.py

A constraint list is licensed when it is licensed for the declared source mode in the sense of Definition "Licensed constraint list" (protocol-compiler/def:homog), a type check that compares no numbers. Its classes are τ-A-homogeneous lists in trusted mode and scalar lists in untrusted mode; the general format adds party-local entries.

The declared forms of a constraint entry and the class of licensed lists are the compiler's (Definition "Licensed constraint list" (protocol-compiler/def:homog)); this definition names the two classes as this module's claims read them. The security chain of this module, Theorems 3.1 and 10.7, admits only licensed lists; every structural and numerical claim is stated and proved for arbitrary Hermitian Θ_k (trusted mode) or arbitrary real c_k (untrusted mode). Remark 3.4 (a) exhibits, in untrusted-source mode, a licensed attack set for which imposing the operator form $\Theta_k = c_k\mathbb{1}_A$ would empty the feasible set; what licenses a party-local entry is Remark "Soundness of a party-local entry" (protocol-compiler/rem:party-local-soundness), and what the constrained accumulation theorem says of it is Remark 3.2.

Definition 2.2 (Licensed constraint list and its two classes)**.** The declared forms of an entry of $\mathfrak{L}$ — homogeneous, carrying a real scalar c_k, general, carrying a Hermitian Θ_k on A, and, in the general format, party-local — are those of Definition "Licensed constraint list" (protocol-compiler/def:homog), and the list is *licensed* when it is licensed for the declared source mode

there. Its two classes for an untagged entry are the following; in the general format an untagged entry is read for its lifted operator, and the cited class also licenses a party-local entry in either mode (the paragraph on party-local entries).

(a) *trusted-source mode:* every entry of $\mathfrak{L}$ is declared in the homogeneous form, carrying $c_k \in \mathbb{R}$; the right-hand side it denotes is then *constructed* as $\Theta_k := c_k\,\tau_A$ from the declared c_k and the declared τ_A, and such a list is a τ_A-*homogeneous list* by construction;

(b) *untrusted-source mode:* $\mathfrak{L}$ is a *scalar list*, i.e. every right-hand side is a real number $\Theta_k = c_k$ and the entry reads $\mathrm{tr}\Big[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega\Big] \bowtie_k c_k$ — a comparison of real numbers, with $\bowtie_k \in \{\leq, =, \geq\}$.

Licensing is therefore a *type* check in both modes and compares no numbers. A general entry remains admissible input, every structural and numerical claim of this module being proved for it, but it is never licensed, so a list containing one carries no key-length claim, which is the safe direction. The two classes agree at $d_A = 1$ (Proposition 5.2 (iii)), and (b) is what the licensing condition (2.8) transports in untrusted-source mode, through (2.14): *the operator form $\Theta_k = c_k \mathbb{1}_A$ is not licensed.*

Party-local entries. In the general format an entry may be declared in the party-local form (Definition "Licensed constraint list" (protocol-compiler/def:homog)); it is written $(\hat{O}_k, \bowtie_k, c_k; f)$, and its state-level reading, in either source mode, is

$$\mathrm{tr}_{Q_f}\Big[\big(\mathbb{1}\otimes(\hat{O}_k - c_k\,\mathbb{1}_{Q_f})\big)\,\omega\Big] \;\bowtie_k\; 0 \qquad \text{as an operator on the systems of } \omega \text{ other than } Q_f, \tag{2.4}$$

those systems being A and the measured factors $Q_{f'}$, $f' \neq f$.

Uses Def. 1.1, protocol-compiler/def:homog · *Used by* Def. 1.2, Def. 2.3, Def. 2.4, Lem. 2.8, Rem. 2.12, Thm. 3.1, Ex. 3.3, Rem. 3.4, Rem. 3.5, Rem. 3.6, Def. 5.1, Prop. 5.2, Prop. 5.5, Cor. 5.6, Prop. 7.6, Rem. 7.9, Thm. 10.7, Rem. 10.20, Prop. 12.1, Req. 16.10 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength:licensed_class_check

The round model is that of foundations at the squashed system, with round channels that hand every public announcement to the adversary's memory. A licensed attack set bounds each attack's adjoint by the declared constraints; whether a protocol reduces to it is the attack-set licence.

The round model is the one the foundations module, which holds the quantum-information preliminaries of this contract, states for a generic per-round system, taken at the squashed system Q: the system the declared family $\{F_v\}$ measures, onto which the squashing channels of the reductions module map the physical system. The reductions module takes the same

model at the physical system for its transfer theorems and states its attack sets at Q (Definition "Round model and attack sequences, typed by source mode" (reductions/def:roundmodel), Definition "Admissible attack sets in the consumer's format" (reductions/def:attset)), so the attack sets of the two modules are the same sets. The definition below adds what this module's accumulation theorems consume: the round channels, the retention of the announcements and the licensed list.

Definition 2.3 (Round model and licensed attack sets)**.** Fix a declaration $\mathfrak{D}$ with single-round protocol channel $\mathcal{M} \in \mathrm{CPTP}(AQ \to Z\widehat{C}I)$ (Theorem 1.5). The round model of this contract is the round model of Definition "Round model, attack sequences and attack sets" (foundations/def:roundmodel) with per-round system Q, Alice's system A and the declared source mode, the round index of Q and A dropped. An attack sequence is thus a family $(\mathcal{A}_j)_{j=1}^{n}$ of channels whose output carries the systems the adversary supplies in the declared source mode, the display being that definition's with its per-round system S taken to be Q:

$$\mathcal{A}_j \in \begin{cases} \mathrm{CPTP}(E_{j-1},\ Q\,E'_j), & \text{trusted-source mode,} \\ \mathrm{CPTP}(E_{j-1},\ A\,Q\,E'_j), & \text{untrusted-source mode,} \end{cases} \tag{2.5}$$

where E_0 is Eve's initial register and E'_j her outgoing memory. In trusted-source mode A is *not* an output of $\mathcal{A}_j$: it is supplied by Alice, inaccessible to the adversary, and carried into the round channel as a separate input — that, and nothing else, is what A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)) licenses. The generated *round channel*s are

$$\mathcal{M}_j \coloneqq \begin{cases} (\mathcal{M} \otimes \mathrm{id}_{E'_j}) \circ (\mathrm{id}_A \otimes \mathcal{A}_j) \ \in\ \mathrm{CPTP}(A\,E_{j-1},\ Z\,\widehat{C}\,E_j), & \text{trusted mode,} \\ (\mathcal{M} \otimes \mathrm{id}_{E'_j}) \circ \mathcal{A}_j \ \in\ \mathrm{CPTP}(E_{j-1},\ Z\,\widehat{C}\,E_j), & \text{untrusted mode,} \end{cases}$$
$$E_j \coloneqq E'_j\,I_j\,: \tag{2.6}$$

the attack delivers the round input on Q — and, in untrusted-source mode, on A as well — together with Eve's new memory, the protocol channel produces $(Z_j, \widehat{C}_j, I_j)$, and the announcement I_j — classical and public — is placed in Eve's register. Content of E_{j-1} that Eve wishes to retain (earlier announcements included) she carries inside E'_j; every physical attack is represented by a sequence that retains *all* announcements, and Theorem 2.10 uses that representation, so that $E_n = I_1^n \widehat{E}$ with $\widehat{E}$ Eve's retained quantum memory. The untrusted-mode round channel has no A input at all, which is what makes it an instance of Fact 2.1 with the marginal register A_{j-1} trivial (Fact 2.1, locator note (vii); Theorem 2.10).

For a licensed list $\mathfrak{L}$ (Definition 2.2) and finite-dimensional registers E', E'', the *licensed attack set* is the attack set of $\mathfrak{L}$ of Definition "Round

model, attack sequences and attack sets" (foundations/def:round-model) at $S = Q$; its displays at that system are, in trusted-source mode,

$$\mathfrak{A}_{\mathfrak{L}}(E',\, QE'') \coloneqq \Big\{\mathcal{A} \in \mathrm{CPTP}(E', QE'') :\ \mathcal{A}^{\dagger}\big(\hat{O}_k \otimes \mathbb{1}_{E''}\big) \bowtie_k c_k\, \mathbb{1}_{E'}\ \ \forall k\Big\}, \tag{2.7}$$

and, with the constraint operator lifted to the larger supplied system,

$$\mathfrak{A}_{\mathfrak{L}}(E',\, AQE'') \coloneqq \Big\{\mathcal{A} \in \mathrm{CPTP}(E', AQE'') :\ \mathcal{A}^{\dagger}\big(\mathbb{1}_A \otimes \hat{O}_k \otimes \mathbb{1}_{E''}\big) \bowtie_k c_k\, \mathbb{1}_{E'}\ \ \forall k\Big\} \tag{2.8}$$

in untrusted-source mode; the two are distinguished by the output system named in the argument. An attack sequence is $\mathfrak{L}$-admissible (an admissible attack sequence for $\mathfrak{L}$) if $\mathcal{A}_j$ lies in the set of (2.7), resp. (2.8), with $E' = E_{j-1}$ and $E'' = E'_j$, for every j. Where $\mathfrak{L}$ carries entries declared party-local (Definition 2.2), each such entry $(\hat{O}_k, \bowtie_k, c_k; f)$ contributes to the licensed attack set, in place of the scalar condition of (2.7) or (2.8), the condition of an entry tagged with the factor f in Definition "Round model, attack sequences and attack sets" (foundations/def:round-model) (iii) at $S = Q$. With Q_{f^c} the product of the measured factors other than f, and R equal to A in untrusted-source mode and trivial in trusted-source mode, its display at that system is

$$\big(\mathrm{id}_{R'} \otimes \mathcal{A}\big)^{\dagger}\Big(X \otimes \big(\hat{O}_k - c_k\, \mathbb{1}_{Q_f}\big) \otimes \mathbb{1}_{E''}\Big)\ \bowtie_k\ 0 \qquad \text{for every finite-dimensional } R' \text{ and every } X \in \mathrm{Pos}(R'R\,Q_{f^c}), \tag{2.9}$$

as an operator on $R'E'$, the tensor factors of the argument written in the order $R'R\,Q_{f^c}$, Q_f, E''; untagged entries, those not declared party-local, keep their conditions, read for their lifted operators (Definition "Protocol declaration" (protocol-compiler/def:declaration) (DX6)). The reference system R' is what carries Alice's retained system in trusted-source mode, where A is no output of $\mathcal{A}$. With R' trivial and $X = \mathbb{1}$ the condition is the scalar one for the lifted operator, so the set with tagged entries is contained in the set with the same entries untagged. For $\mathfrak{L} = \varnothing$ the condition is empty: $\mathfrak{A}_{\varnothing} = \mathrm{CPTP}$, every attack sequence is admissible, and the model is unrestricted. The adjoint condition in (2.7) is exactly the published squashed-attack-set format of [5, Eq. (151)], and restricting attention to a declared subset of channels is the standard modelling step of [5, Def. 8.3, Rem. 8.4]. Whether a given physical protocol reduces to this model with a given $\mathfrak{A}_{\mathfrak{L}}$ is *not* decided here: it is what A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk)) certifies.

Uses Def. 1.1, Thm. 1.5, Fact 2.1, Def. 2.2, foundations/def:round-model · *Used by* Lem. 2.8, Lem. 2.9, Thm. 2.10, Thm. 3.1, Rem. 3.2, Ex. 3.3, Rem. 3.4, Rem. 3.6, Rem. 5.9, Def. 10.1, Thm. 10.7, Cor. 10.9, Ass. 13.8 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.validation: REQUIRED_ASSUMPTIONS · *Oracles* registry.py

The constrained single-round rate minimises a KL penalty plus the key's conditional entropy over acceptance distributions and states meeting the declared marginal and constraint list; at the empty list it is the unconstrained rate. Constraints only raise it, so using it needs a licence.

Definition 2.4 (Constrained target set and single-round rate)**.** For a register $\widetilde{E}'$ and a (possibly empty) declared list $\mathfrak{L}$, let

$$\Sigma^{\mathfrak{L}}(\widetilde{E}') \coloneqq \big\{(\mathcal{M}\otimes \mathrm{id}_{\widetilde{E}'})(\sigma) :\ \sigma \in S_{=}(AQ\widetilde{E}'),\ \sigma \models \mathfrak{D},\ \sigma_{AQ} \models \mathfrak{L}\big\}, \tag{2.10}$$

where $\sigma \models \mathfrak{D}$ abbreviates the declaration's marginal condition,

$$\sigma \models \mathfrak{D} \quad :\Longleftrightarrow \quad \begin{cases} \sigma_A = \tau_A, & \text{trusted-source mode,} \\ \text{no condition beyond } \operatorname{tr}\sigma = 1, & \text{untrusted-source mode,} \end{cases} \tag{2.11}$$

and $\sigma_{AQ} \models \mathfrak{L}$ abbreviates "σ_{AQ} satisfies every entry of $\mathfrak{L}$" in the form Definition 2.2 gives it for the declared mode, i.e. $\operatorname{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\,\sigma_{AQ}] \bowtie_k \Theta_k$ in trusted-source mode and $\operatorname{tr}\big[(\mathbb{1}_A \otimes \hat{O}_k)\,\sigma_{AQ}\big] \bowtie_k c_k$ in untrusted-source mode, for all k; an entry declared party-local is read, in either mode, in the form (2.4) of Definition 2.2, an affine semidefinite condition on σ_{AQ}. $\Sigma^{\mathfrak{L}}(\widetilde{E}')$ is convex in either mode: the state set is the intersection of the convex $S_{=}(AQ\widetilde{E}')$ with affine equalities and operator- or scalar-inequality conditions, each convex, and $\mathcal{M}\otimes\mathrm{id}_{\widetilde{E}'}$ is affine. Define the *constrained single-round rate*

$$h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L}) \coloneqq \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \inf_{\substack{\sigma\in S_{=}(AQ\widetilde{E}) \\ \sigma\models\mathfrak{D},\ \sigma_{AQ}\models\mathfrak{L}}} \left(\frac{\alpha}{\alpha-1}\, D(\mathbf{q}\,\|\,\mathbf{\Phi}[\sigma_{AQ}]) + \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} q(\hat{c})\, \widetilde{H}_\alpha^{\uparrow}(Z|I\widetilde{E})_{\nu^\sigma_{|\hat{c}}} \right), \tag{2.12}$$

with $\nu^\sigma \coloneqq (\mathcal{M}\otimes\mathrm{id}_{\widetilde{E}})(\sigma)$ and $\dim\mathcal{H}(\widetilde{E}) \geq \dim\mathcal{H}(AQ)$; by Lemma 2.7 the value does not depend on that choice. For $\mathfrak{L} = \varnothing$ we write $h_\alpha^{\uparrow,\mathrm{QKD}} \coloneqq h_\alpha^{\uparrow,\mathrm{QKD}}(\varnothing)$, the unconstrained single-round rate, which Theorem 2.10 writes out as (2.17). Since a nonempty $\mathfrak{L}$ shrinks the inner feasible set, $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L}) \geq h_\alpha^{\uparrow,\mathrm{QKD}}(\varnothing)$ always.

Uses Def. 2.2 · *Used by* Lem. 2.7, Lem. 2.8, Thm. 2.10, Thm. 3.1, Rem. 3.2, Rem. 3.6, Def. 5.1, Thm. 5.4, Cor. 5.6 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.objective: objective_value · *Oracles* registry.py

An acceptance set is admissible when it is a closed convex subset of the probability simplex on the acceptance values. Accumulation, the attainment and limit results and the first-order bound all read that geometry.

The acceptance set is the set of acceptance-value distributions on which the acceptance test accepts. Which sets the engine admits, and how it decides that a declared one is admissible, is Requirement 14.4.

Definition 2.5 (Admissible acceptance set)**.** An *admissible acceptance set* is a closed convex subset $\mathcal{S}_{\mathrm{acc}}$ of the probability simplex on $\widehat{\mathcal{C}}$.

Used by Thm. 2.10, Rem. 3.4, Thm. 5.10, Thm. 8.9, Cor. 9.4, Prop. 9.5, Prop. 10.12, Req. 14.4, Lem. 18.2, Thm. 19.1, Cor. 19.6, Cor. 19.7, Lem. 21.1, Rem. 22.2 · newly derived · accepted

The sandwiched conditional Rényi entropy cannot increase when the conditioning system grows, by data processing under the partial trace. The conservative steps that hand the adversary extra information rest on it.

Lemma 2.6 (Conditioning on more side information). For $\alpha \in (1,\infty)$ and any $\rho_{ABC} \in S_{=}(ABC)$: $\widetilde{H}^{\uparrow}_{\alpha}(A|BC)_{\rho} \le \widetilde{H}^{\uparrow}_{\alpha}(A|B)_{\rho}$.

Used by Lem. 2.7, Thm. 5.4, Prop. 5.8 · newly derived · accepted

Proof. For any $\sigma_{BC} \in S_{=}(BC)$, data processing for the sandwiched divergence under the channel tr_C ([4, Thm. 4.1, Eq. (4.82)]; $\alpha > 1 \ge \frac{1}{2}$) gives $\widetilde{D}_{\alpha}(\rho_{ABC}\|\mathbb{1}_A \otimes \sigma_{BC}) \ge \widetilde{D}_{\alpha}(\rho_{AB}\|\mathbb{1}_A \otimes \sigma_B)$. Taking $-\inf$ over σ_{BC} on the left and noting that every σ_B arises as a marginal on the right, $\widetilde{H}^{\uparrow}_{\alpha}(A|BC)_{\rho} = -\inf_{\sigma_{BC}} \widetilde{D}_{\alpha}(\rho_{ABC}\|\mathbb{1}_A\otimes\sigma_{BC}) \le -\inf_{\sigma_B} \widetilde{D}_{\alpha}(\rho_{AB}\|\mathbb{1}_A\otimes\sigma_B) = \widetilde{H}^{\uparrow}_{\alpha}(A|B)_{\rho}$. □

Once the adversary's register is at least as large as Alice's and the measured system together, enlarging it leaves the constrained single-round rate unchanged. The inner infimum may be restricted to pure states, which the branch formula uses.

Lemma 2.7 (Independence of the purifier size)**.** Fix $\mathfrak{L}$ (possibly empty) and write $h(\widetilde{E}')$ for the value of (2.12) computed with a register $\widetilde{E}'$ of $\dim \mathcal{H}(\widetilde{E}') \ge \dim \mathcal{H}(AQ)$ in place of $\widetilde{E}$. Then $h(\widetilde{E}')$ is the same for every such $\widetilde{E}'$, and the inner infimum is unchanged if restricted to pure σ.

Uses Def. 1.2, Def. 2.4, Lem. 2.6, foundations/def:renyi · *Used by* Thm. 2.10, Thm. 3.1, Rem. 3.6 · newly derived · accepted

Proof. Let $\dim \mathcal{H}(\widetilde{E}_2) \ge \dim \mathcal{H}(\widetilde{E}_1) \ge \dim \mathcal{H}(AQ)$.

($h(\widetilde{E}_2) \le h(\widetilde{E}_1)$.) An isometry $V : \mathcal{H}(\widetilde{E}_1) \to \mathcal{H}(\widetilde{E}_2)$ maps every feasible σ for $\widetilde{E}_1$ to the feasible $(\mathbb{1}_{AQ} \otimes V)\sigma(\mathbb{1}_{AQ} \otimes V)^{\dagger}$ for $\widetilde{E}_2$ with the same AQ-marginal, so the divergence term and the constraints are unchanged; and $\widetilde{H}^{\uparrow}_{\alpha}(Z|I\widetilde{E})$ is invariant under an isometry on the conditioning register: data processing ([4, Thm. 4.1, Eq. (4.82)]) under the isometric channel $V(\cdot)V^{\dagger}$ on $\widetilde{E}_1$ and under a reversal channel $\mathcal{R}$ with $\mathcal{R}(V(\cdot)V^{\dagger}) = \mathrm{id}$ (e.g. $\mathcal{R}(X) =$

$V^\dagger X V + \operatorname{tr}\big[(\mathbb{1} - VV^\dagger)X\big]\,\xi$ for a fixed state ξ), applied to the optimisation defining $\widetilde{H}^{\uparrow}_{\alpha}$ in Definition "Sandwiched Rényi divergence and conditional entropies" (foundations/def:renyi), gives both inequality directions. Hence every $\widetilde{E}_1$-bracket value is attained by an $\widetilde{E}_2$-feasible state.

($h(\widetilde{E}_2) \geq h(\widetilde{E}_1)$; purifications suffice.) Let σ be feasible for $\widetilde{E}_2$ and let $\hat{\sigma} \in S_=(AQ\widetilde{E}_1)$ be a purification of σ_{AQ} (it exists: $\dim \mathcal{H}(\widetilde{E}_1) \geq \dim \mathcal{H}(AQ)$). Every extension of σ_{AQ} arises from any fixed purification by a channel on the purifying register: purify σ itself by a further register, relate the two purifications of σ_{AQ} by an isometry on the purifier side, and trace — so $\sigma = (\mathrm{id}_{AQ} \otimes N)(\hat{\sigma})$ for some channel $N : \widetilde{E}_1 \to \widetilde{E}_2$. Then $\nu^\sigma = (\mathrm{id}_{ZI} \otimes N)(\nu^{\hat{\sigma}})$, and data processing on the conditioning register ([4, Thm. 4.1, Eq. (4.82)]; the general-channel form of the argument in Lemma 2.6) gives $\widetilde{H}^{\uparrow}_{\alpha}(Z|I\widetilde{E}_2)_{\nu^\sigma_{|\hat{c}}} \geq \widetilde{H}^{\uparrow}_{\alpha}(Z|I\widetilde{E}_1)_{\nu^{\hat{\sigma}}_{|\hat{c}}}$ for every $\hat{c}$ (conditioning is on a classical value of $\widehat{C}$, which commutes with the channel on $\widetilde{E}$). The divergence term and the constraints agree (same AQ-marginal). Hence the $\widetilde{E}_2$-bracket at σ dominates the $\widetilde{E}_1$-bracket at the pure $\hat{\sigma}$, so $h(\widetilde{E}_2) \geq h(\widetilde{E}_1)$, and the infimum is attained along purifications. Combining the two directions, all values coincide. □

An admissible attack carries the licensed bounds on its adjoint over to the constraint conditions on the round's state, so every round output lies in the constrained target set. A larger output set can only lower the rate, which is the conservative comparison accumulation needs.

Lemma 2.8 (Constraint transport and range inclusion)**.** Let $\mathfrak{L}$ be licensed (Definition 2.2) or empty, and let $\mathcal{A}_j$ be $\mathfrak{L}$-admissible for the declared source mode (Definition 2.3).

(i) *(Transport.) Trusted-source mode:* for every $\omega \in S_=(A\,E_{j-1}\,\widetilde{E})$ with $\omega_A = \tau_A$, the state $\varsigma := (\mathrm{id}_A \otimes \mathcal{A}_j \otimes \mathrm{id}_{\widetilde{E}})(\omega) \in S_=(AQE'_j\widetilde{E})$ satisfies $\varsigma_A = \tau_A$ and $\varsigma_{AQ} \models \mathfrak{L}$. *Untrusted-source mode:* for every $\omega \in S_=(E_{j-1}\,\widetilde{E})$, the state $\varsigma := (\mathcal{A}_j \otimes \mathrm{id}_{\widetilde{E}})(\omega) \in S_=(AQE'_j\widetilde{E})$ satisfies $\varsigma \models \mathfrak{D}$ — which in that mode is only $\operatorname{tr}\varsigma = 1$ — and $\varsigma_{AQ} \models \mathfrak{L}$ in the scalar form of Definition 2.2 (b). *In either mode*, every entry of $\mathfrak{L}$ declared party-local holds of ς_{AQ} in the form (2.4), the attack being admissible for it in the sense of (2.9).

(ii) *(Range inclusion.)* With $\mathcal{M}_j$ the round channel (2.6) and Σ_j its output set (2.3), $\Sigma_j \subseteq \Sigma^{\mathfrak{L}}(E'_j\widetilde{E})$ up to the regrouping of the registers of the output, $(Z\widehat{C})(E'_jI_j)\widetilde{E} \mapsto (Z\widehat{C}I)(E'_j\widetilde{E})$; hence, embedding the finitely many $E'_j\widetilde{E}$ into a common register $\widetilde{E}^*$ with $\dim \mathcal{H}(\widetilde{E}^*) = \max_j \dim \mathcal{H}(E'_j\widetilde{E})$,

$$\Sigma = \operatorname{conv}\Big(\bigcup_j \Sigma_j\Big) \;\subseteq\; \Sigma^{\mathfrak{L}}(\widetilde{E}^*).$$

(iii) *(Antitonicity.)* If $\Sigma \subseteq \Sigma'$ then the value of (2.2) computed over Σ is at least its value computed over Σ'.

Uses Fact 2.1, Def. 2.2, Def. 2.3, Def. 2.4 · *Assumes* A-ATK, A-SRC · *Used by* Thm. 2.10, Thm. 3.1, Rem. 3.2, Rem. 3.4, Rem. 3.5, Rem. 5.9 · newly derived · accepted

Proof. (i) $\varsigma_A = \omega_A = \tau_A$: $\mathrm{id}_A \otimes \mathcal{A}_j \otimes \mathrm{id}_{\widetilde{E}}$ is trace-preserving and acts trivially on A. For the constraints, test against an arbitrary $X \in \mathcal{L}(A)$:

$$\mathrm{tr}\left[X\ \mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\,\varsigma_{AQ}]\right] = \mathrm{tr}\left[(X \otimes \hat{O}_k \otimes \mathbb{1}_{E'_j})\,(\mathrm{id}_A \otimes \mathcal{A}_j)(\omega_{AE_{j-1}})\right] = \mathrm{tr}\left[\left(X \otimes \mathcal{A}_j^\dagger(\hat{O}_k \otimes \mathbb{1}_{E'_j})\right)\omega_{AE_{j-1}}\right]$$

by definition of the adjoint. Hence, as operators on A,

$$\mathrm{tr}_Q\left[(\mathbb{1}_A \otimes \hat{O}_k)\,\varsigma_{AQ}\right] = \mathrm{tr}_{E_{j-1}}\left[\left(\mathbb{1}_A \otimes \mathcal{A}_j^\dagger(\hat{O}_k \otimes \mathbb{1}_{E'_j})\right)\omega_{AE_{j-1}}\right]. \tag{2.13}$$

Write $X_k \coloneqq \mathcal{A}_j^\dagger(\hat{O}_k \otimes \mathbb{1}_{E'_j}) - c_k \mathbb{1}_{E_{j-1}}$. For $\bowtie_k = \succeq$, admissibility (2.7) says $X_k \succeq 0$, and $\mathrm{tr}_{E_{j-1}}[(\mathbb{1}_A \otimes X_k)\,\omega] = \mathrm{tr}_{E_{j-1}}[(\mathbb{1}_A \otimes X_k^{1/2})\,\omega\,(\mathbb{1}_A \otimes X_k^{1/2})] \succeq 0$ — cyclicity under the partial trace for operators on the traced factor, and the middle expression is the partial trace of a PSD operator — so (2.13) equals $c_k\,\omega_A + \mathrm{tr}_{E_{j-1}}[(\mathbb{1}_A \otimes X_k)\,\omega] \succeq c_k\,\tau_A$. For $\bowtie_k = \preceq$, apply the same argument to $-X_k \succeq 0$; for $\bowtie_k = =$, admissibility forces $X_k = 0$, and (2.13) equals $c_k\,\tau_A$ exactly. (For $\mathfrak{L} = \varnothing$ the claim is only $\varsigma_A = \tau_A$.)

(i), untrusted-source mode. Here A is an *output* of $\mathcal{A}_j$, so the computation above has no operator-valued analogue: there is no A factor on the input side to carry an operator inequality through, and the identity (2.13) — whose left-hand side is an operator on A and whose right-hand side is built from $\omega_{AE_{j-1}}$ — has no counterpart. What does survive is its trace. Normalization is immediate: $\mathcal{A}_j \otimes \mathrm{id}_{\widetilde{E}}$ is trace-preserving, so $\mathrm{tr}\,\varsigma = \mathrm{tr}\,\omega = 1$. For the constraints, by definition of the adjoint,

$$\mathrm{tr}\left[(\mathbb{1}_A \otimes \hat{O}_k)\,\varsigma_{AQ}\right] = \mathrm{tr}\left[(\mathbb{1}_A \otimes \hat{O}_k \otimes \mathbb{1}_{E'_j})\,\mathcal{A}_j(\omega_{E_{j-1}})\right] = \mathrm{tr}\left[\mathcal{A}_j^\dagger(\mathbb{1}_A \otimes \hat{O}_k \otimes \mathbb{1}_{E'_j})\,\omega_{E_{j-1}}\right]. \tag{2.14}$$

Write $X_k \coloneqq \mathcal{A}_j^\dagger(\mathbb{1}_A \otimes \hat{O}_k \otimes \mathbb{1}_{E'_j}) - c_k \mathbb{1}_{E_{j-1}}$. For $\bowtie_k = \succeq$ admissibility (2.8) says $X_k \succeq 0$, so $\mathrm{tr}\left[X_k\,\omega_{E_{j-1}}\right] \geq 0$ (the trace of a product of two PSD operators is non-negative), and (2.14) equals $c_k\,\mathrm{tr}\left[\omega_{E_{j-1}}\right] + \mathrm{tr}\left[X_k\,\omega_{E_{j-1}}\right] \geq c_k$, using $\mathrm{tr}\,\omega_{E_{j-1}} = 1$. For $\bowtie_k = \preceq$ apply the same argument to $-X_k \succeq 0$; for $\bowtie_k = =$ admissibility forces $X_k = 0$ and (2.14) equals c_k exactly. This is precisely the scalar entry format of Definition 2.2 (b), and it is the *only* conclusion the licence supports in this mode; the operator form $\Theta_k = c_k \mathbb{1}_A$ is not available (Remark 3.4 (a)). (For $\mathfrak{L} = \varnothing$ the claim is only $\mathrm{tr}\,\varsigma = 1$.)

(i), party-local entries. Let $(\hat{O}_k, \bowtie_k, c_k; f)$ be declared party-local, write $\hat{D} \coloneqq \hat{O}_k - c_k \mathbb{1}_{Q_f}$ and $Y \coloneqq \mathrm{tr}_{Q_f}[(\mathbb{1} \otimes \hat{D})\,\varsigma_{AQ}]$, a Hermitian operator on AQ_{f^c}, and let $X \in \mathrm{Pos}(AQ_{f^c})$. In trusted-source mode, by the definition of the adjoint,

$$\mathrm{tr}[XY] = \mathrm{tr}\left[(X \otimes \hat{D} \otimes \mathbb{1}_{E'_j})\,(\mathrm{id}_A \otimes \mathcal{A}_j)(\omega_{AE_{j-1}})\right] = \mathrm{tr}\left[(\mathrm{id}_A \otimes \mathcal{A}_j)^\dagger\left(X \otimes \hat{D} \otimes \mathbb{1}_{E'_j}\right)\omega_{AE_{j-1}}\right],$$

and (2.9) with $R' = A$ and R trivial says that the operator in the last trace is $\bowtie_k 0$ on AE_{j-1}; since $\omega_{AE_{j-1}} \succeq 0$ the trace is ≥ 0 for $\succeq$, ≤ 0 for $\preceq$ and 0 for $=$. In untrusted-source mode the same computation with $\varsigma_{AQE'_j} = \mathcal{A}_j(\omega_{E_{j-1}})$ gives $\operatorname{tr}[XY] = \operatorname{tr}\left[\mathcal{A}_j^\dagger(X \otimes \hat{D} \otimes \mathbb{1}_{E'_j})\,\omega_{E_{j-1}}\right]$, and (2.9) with R' trivial and $R = A$ gives the same signs. A Hermitian Y with $\operatorname{tr}[XY] \geq 0$ for every $X \succeq 0$ is positive semidefinite, and one with $\operatorname{tr}[XY] = 0$ for every $X \succeq 0$ vanishes, positive semidefinite operators spanning all operators; so $Y \bowtie_k 0$, which is (2.4). The tensor factors are written in the order of (2.9); reordering them is a fixed unitary relabelling on both sides.

(ii) *Trusted-source mode.* Fix $\omega \in S_=(A\,E_{j-1}\,\widetilde{E})$ with $\omega_A = \tau_A$ and set $\varsigma = (\mathrm{id}_A \otimes \mathcal{A}_j \otimes \mathrm{id}_{\widetilde{E}})(\omega)$. By (2.6), $(\mathcal{M}_j \otimes \mathrm{id}_{\widetilde{E}})(\omega) = (\mathcal{M} \otimes \mathrm{id}_{E'_j\widetilde{E}})(\varsigma)$ — the same operator, with the registers of the round-j output regrouped: the announcement I_j, emitted by $\mathcal{M}$, is assigned to $E_j = E'_j I_j$ on the left and to the I-output on the right. By (i), ς is feasible for (2.10) with purifier $E'_j\widetilde{E}$, so the output lies in $\Sigma^{\mathfrak{L}}(E'_j\widetilde{E})$. Isometric embedding of each $E'_j\widetilde{E}$ into the common $\widetilde{E}^*$ preserves feasibility and the AQ-marginal, so every embedded Σ_j lies in the convex set $\Sigma^{\mathfrak{L}}(\widetilde{E}^*)$ (Definition 2.4), and so does the convex hull. *Untrusted-source mode.* The same argument with $\omega \in S_=(E_{j-1}\widetilde{E})$ and $\varsigma = (\mathcal{A}_j \otimes \mathrm{id}_{\widetilde{E}})(\omega)$: the second line of (2.6) gives the same regrouping identity, (i) gives feasibility for (2.10) in the untrusted reading of (2.11), and the embedding step is unchanged. Note that for $\mathfrak{L} = \varnothing$ the inclusion is then immediate for a different reason as well: $\Sigma^{\varnothing}(\widetilde{E}^*)$ is the set of *all* $(\mathcal{M} \otimes \mathrm{id})(\sigma)$ with σ normalized, and every round-channel output is of that form.

(iii) An infimum over a superset is no larger; the bracket of (2.2) is evaluated pointwise. □

In trusted-source mode, under retention, emission of the specified ensemble, sequential rounds, private source randomness and a sealed laboratory, the n rounds start with Alice's systems in the declared marginal in product, and each round finds her system in that marginal.

In trusted-source mode the round model takes Alice's retained systems as inputs of the round channels, and the accumulation theorem (Theorem 2.10) asks that they start in the declared marginal, in product over the rounds. The lemma derives that from the transmitter's premises. The source-replacement certificate describes one round's preparation as Alice's measurement of her half of the certificate (Theorem "Checkability and sufficiency of the source-replacement certificate" (transmitter/thm:certificate-sufficiency) (iii)); A-ENS (Assumption "Emission of the specified ensemble" (transmitter/ass:ens)) makes the n preparations a product, and A-SEED (Assumption 13.3 (v)) makes them independent of what the adversary holds at the start; and retention, A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)), the sealed laboratory of the party that holds

A, A-LAB (Assumption 13.8 (i), (ii)), and the sequential round structure, A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)), let each measurement be moved into its round. Retention alone does not give the product: it fixes what may act on each round's system, and the independence of the source's rounds gives the product. A specification declared by a source model carries, in the place of A-ENS, the premise its submodule declares, which refines A-ENS: for weak coherent pulses it is A-WCP (Assumption "Emission of the declared weak-coherent-pulse states" (transmitter/decoy/ass:wcp)), whose last three clauses are A-ENS's for the model's states. At $d_A = 1$ the conclusion holds of every state (Proposition 5.2 (ii)).

Lemma 2.9 (Product marginal of Alice's retained systems)**.** Let the declaration be in trusted-source mode with $d_A \geq 2$, and let its A, τ_A and $\{M_x\}$ be those of a source specification (Definition "Source specification" (transmitter/def:source-spec)) that carries a source-replacement certificate Ψ for $(A, \mathbb{T}, \tau_A, \{M_x\})$ against its virtual ensemble $\{p_x, \rho_x^{\mathrm{v}}\}$ on A'', in any of the forms of Definition "Source-replacement certificate" (transmitter/def:srcert), an ensemble presentation being read as the block-form certificate it defines; let its source segment satisfy the source identity $\Xi(\rho_x^{\mathrm{v}}) = \rho_x$ for every $x \in \mathcal{X}$, with Ξ the composite channel of the segment and $\Xi = \mathrm{id}$ when the segment is empty (Definition "Source segment and its interface conditions" (transmitter/def:source-segment) (S3)). Let A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)), A-ENS (Assumption "Emission of the specified ensemble" (transmitter/ass:ens)), A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)), A-SEED (Assumption 13.3) and A-LAB (Assumption 13.8) at the party to which the party partition assigns A hold of the physical protocol, let the adversary's initial register E be finite-dimensional, and let σ_E be the state in which it starts. Put

$$\omega^0 := \Big(\bigotimes_{j=1}^{n} \Psi_{A_j A_j''}\Big) \otimes \sigma_E \;\in\; S_=(A_1^n E_0), \qquad E_0 := (A'')_1^n\, E. \tag{2.15}$$

Then:

(i) *(source replacement over the n rounds)* against every attack, the physical protocol and the protocol that starts from ω^0, applies Ξ to A_j'' in round j in the place of the source's emission, and applies $\mathcal{N}_{\mathsf{A}}$, Alice's measurement as the channel of Theorem "Checkability and sufficiency of the source-replacement certificate" (transmitter/thm:certificate-sufficiency) (iii), to A_j at the point of round j at which the declared protocol's measurements are made, produce the same state after every round, on every register that both then hold; in the second protocol A_j is an input of that measurement and of no other operation, which is how

the round model of Definition 2.3 takes Alice's system in trusted-source mode;

(ii) $\omega^0_{A_1^n} = \tau_A^{\otimes n}$;

(iii) for every attack sequence of Definition 2.3 and every j, the state on which the round channel $\mathcal{M}_j$ acts, in a run of the round model from ω^0, has the marginal τ_A on A_j.

At $d_A = 1$, where $\tau_A = [1]$ and no certificate is declared, (ii) and (iii) hold of every state on $A_1^n E_0$.

Uses Def. 2.3, transmitter/def:source-segment, transmitter/def:source-spec, transmitter/def:srcert, transmitter/thm:certificate-sufficiency · *Assumes* A-SRC, A-ENS, A-SEQ, A-SEED, A-LAB · *Used by* Thm. 2.10 · newly derived · accepted

Proof. Idea. The physical preparation is a product over the rounds of one round's classical-quantum source state, which the certificate writes as Alice's measurement of her half of Ψ followed by Ξ; nothing touches her half or her letter before the round's measurements, so the interchange law moves her measurement there; and the marginal of a product of copies of Ψ on Alice's side is the product of their marginals.

At $d_A = 1$. $S_=(A) = \{[1]\}$, so $S_=(A_1^n)$ is a singleton and every state on $A_1^n E_0$, and every state the round channels act on, has the marginal $[1]^{\otimes n} = \tau_A^{\otimes n}$, respectively $[1] = \tau_A$.

Step 1 (the physical source state). Let X_j hold the letter of round j, and write $\rho_{XA'} := \sum_{x\in\mathcal{X}} p_x \, |x\rangle\langle x|_X \otimes \rho_x$ for one round's classical-quantum source state on the transmitted system. By A-ENS (i) and (ii) the letters of the n rounds are independent with the weights p_x and, given them, the emitted systems are in $\rho_{x_1}\otimes\cdots\otimes\rho_{x_n}$, so the letters and the emitted systems are jointly in $\sum_{x_1^n}\prod_j p_{x_j}\,|x_1^n\rangle\langle x_1^n|\otimes\rho_{x_1}\otimes\cdots\otimes\rho_{x_n} = \rho_{XA'}^{\otimes n}$. By A-ENS (iii) neither the letter of round j nor the state emitted in it depends on anything that happens in the other rounds, so preparing all n letters and systems at the start in that state, and releasing the round-j system only in round j, changes nothing the protocol produces. By A-SEED (v) the letters and the prepared systems are jointly independent of the adversary's initial register, and by its last clause no purification of a prepared system is made available to the adversary during or after the protocol, so that no register beyond the letters and the emitted systems carries the preparation; the protocol starts from $\rho_{XA'}^{\otimes n}\otimes\sigma_E$.

Step 2 (the certificate). By the source identity, $\rho_{XA'} = (\mathrm{id}_X\otimes\Xi)(\rho_{XA''})$ with $\rho_{XA''} := \sum_x p_x\,|x\rangle\langle x|_X\otimes\rho_x^{\mathrm{v}}$. By (SR2), in the form of Theorem "Checkability and sufficiency of the source-replacement certificate" (transmitter/thm:certificate-sufficiency) (iii)(b) with the identity channel in the place of $\mathcal{E}$, $(\mathcal{N}_{\mathsf{A}}\otimes\mathrm{id}_{A''})(\Psi) = \rho_{XA''}$. Hence $\rho_{XA'} = (\mathcal{N}_{\mathsf{A}}\otimes\Xi)(\Psi)$, and the initial state of Step 1 is $\big(\bigotimes_j(\mathcal{N}_{\mathsf{A}}\otimes\Xi)_{A_jA_j''}\big)(\omega^0)$: the physical protocol is the protocol

that starts from ω^0 and applies, at the start, $\mathcal{N}_{\mathsf{A}}$ to every A_j and Ξ to every A''_j, the round-j output of Ξ being released in round j. Applying Ξ to A''_j at the start or in round j produces the same state, since nothing else acts on A''_j before round j.

Step 3 (moving the measurement into the round). Fix j. Let $\mathcal{E}_j$ be the composite of every operation of the protocol, of the parties and of the adversary, from the start to the point of round j at which the declared protocol's measurements are made. It acts on neither A_j nor X_j; the measurements of the other rounds, wherever they stand, act on A_k and X_k with $k \neq j$. On A_j: by A-SRC (ii) no operation of the protocol or of the adversary acts on A except the declared measurement on it. On X_j, for the adversary's operations: the letters are records of the party to which the party partition assigns A, and by A-LAB (i) at that party no information about them leaves its laboratory except through the systems handed out of it, which in trusted-source mode are the systems its source emits, and the public messages of the protocol, and by A-LAB (ii) the adversary acts on its laboratory only through the systems delivered into it, as its declared measurements receive them, and the messages the protocol delivers to it, so that nothing she does reads or writes the letter record; no operation of the adversary acts on X_j itself: what the emitted systems carry about the letter is already in the state of Step 2, and the messages are treated next. On X_j, for the protocol's own operations: the letter of round j is a record of that same party, which the declared protocol reads only in the round-j protocol map, at or after the round's measurements, and in the post-processing; by A-SEQ (i) the messages of round j are sent after every measurement of the round, by A-SEQ (iv) no operation on a quantum system of round j depends on them, and by A-SEQ (ii) the post-processing follows all n rounds. So $\mathcal{N}_{\mathsf{A}}$, a map out of A_j alone into X_j, and $\mathcal{E}_j$ act on complementary registers, and by the interchange law of Theorem "Checkability and sufficiency of the source-replacement certificate" (transmitter/thm:certificate-sufficiency) (iii)(a) $\mathcal{E}_j \circ (\mathcal{N}_{\mathsf{A}})_{A_j} = (\mathcal{N}_{\mathsf{A}})_{A_j} \circ \mathcal{E}_j$ as maps on the whole input. Applying this for $j = 1, \ldots, n$ moves each measurement from the start to its round without changing any state the protocol produces after the round, which is (i); in the protocol so obtained A_j enters no operation but the round-j measurement.

Step 4 (the marginals). (ii): σ_E is a state, so $\omega^0_{A_1^n} = \bigotimes_j \operatorname{tr}_{A''_j} \Psi_{A_j A''_j} = \tau_A^{\otimes n}$ by (SR1). (iii): in a run of the round model from ω^0, the state on which $\mathcal{M}_j$ acts is produced from ω^0 by $\mathcal{M}_1, \ldots, \mathcal{M}_{j-1}$ and the attack channel $\mathcal{A}_j$ (Definition 2.3, (2.6)), and in trusted-source mode each $\mathcal{M}_k$ acts on $A_k E_{k-1}$ and $\mathcal{A}_j$ on E_{j-1}, none of them on A_j. A channel on a register complementary to A_j leaves the A_j-marginal unchanged, the partial trace over that register being invariant under it, so the marginal is $\omega^0_{A_j}$, which is τ_A by (ii). □

On acceptance, the raw key's Rényi entropy given the adversary's information is at least n times the single-round rate less $\alpha/(\alpha-1)\cdot\log_2(1/\Pr[\text{accept}])$, for any adaptive attack with a finite-dimensional memory. Trusted-source mode reads the product marginal of Lemma 2.9; untrusted mode needs none.

The imported accumulation theorem, applied to the round channels the declaration generates, bounds the entropy of the accepted raw key from below by n times the single-round rate (2.17), the rate of Definition 2.4 at the empty constraint list, less a penalty in the acceptance probability. The theorem is a statement about attack sequences of the round model, whose registers are finite-dimensional. That every physical attack with finite-dimensional registers whose first channel receives every transmitted signal is such a sequence, with E_n the adversary's whole information about the rounds, is the mapping of the physical protocol onto the model; it rests on the reduction chain from which the declaration was compiled, hypothesis (H-CHAIN) of Theorem 10.7, and, for the round structure, on A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)) (i), (ii) and (iv), Assumption 13.8 at each of the two parties and Assumption 13.5 (ii), and is made where the key length is stated (Theorem 10.7). The security of the key extends from these attacks to every physical attack, whatever the dimension of the adversary's memory, by Corollary 10.8 through Lemma 10.5. Remark 2.11 says which attacks the bound covers and what untrusted-source mode costs.

Theorem 2.10 (Accumulated entropy bound under adaptive attacks). Let $\mathfrak{D}$ satisfy Definition 1.2, let $\mathcal{S}_{\mathrm{acc}}$ satisfy Definition 2.5, and let Assumption A-RHOM (Assumption "Round homogeneity of the declaration" (protocol-compiler/ass:rhom)) hold; in trusted-source mode let A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)), A-ENS (Assumption "Emission of the specified ensemble" (transmitter/ass:ens)), A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)), 13.3 (A-SEED) and 13.8 (A-LAB) at the party to which the party partition assigns A hold as well, and, at $d_A \geq 2$, let the declaration's A, τ_A and $\{M_x\}$ be those of a source specification carrying a source-replacement certificate against its virtual ensemble, with a source segment satisfying the source identity, as Lemma 2.9 states them; together with these, the adversary's initial register being finite-dimensional as every register of the round model is, the five assumptions are the hypotheses of Lemma 2.9, so that by that lemma the protocol's n-round input in that mode has the marginal below (in untrusted-source mode the lemma is not invoked and none of the hypotheses named for that mode is used here). Let $(\mathcal{A}_j)_{j=1}^{n}$ be an *arbitrary* attack sequence (Definition 2.3; no admissibility restriction; its registers E_0 and E'_j finite-dimensional, of any dimensions, as every register of that model is), let

$$\rho_{Z_1^n \widehat{C}_1^n I_1^n \widehat{E}} = \big(\mathcal{M}_n \circ \cdots \circ \mathcal{M}_1\big)\big[\omega^0\big], \qquad \omega^0 \in S_=(A_1^n E_0) \text{ with } \omega^0_{A_1^n} = \tau_A^{\otimes n} \text{ (trusted mode)},$$

respectively $\omega^0 \in S_=(E_0)$ arbitrary (untrusted mode, where the round channels take no A input), with $\mathcal{M}_j$ the round channels (2.6) and Eve's final register written $E_n = I_1^n \widehat{E}$ (Definition 2.3), and let Ω_{AT}, the acceptance event, be the event that the frequency vector $\mathrm{freq}_{\hat{c}_1^n}$ of the acceptance values of the n rounds lies in $\mathcal{S}_{\mathrm{acc}}$, with $\Pr[\Omega_{\mathrm{AT}}] > 0$. Then for every $\alpha \in (1, \infty)$,

$$\widetilde{H}_\alpha^\uparrow\big(Z_1^n \big| \widehat{C}_1^n I_1^n \widehat{E}\big)_{\rho_{|\Omega_{\mathrm{AT}}}} \;\geq\; n\, h_\alpha^{\uparrow,\mathrm{QKD}} - \frac{\alpha}{\alpha-1} \log_2 \frac{1}{\Pr[\Omega_{\mathrm{AT}}]}, \tag{2.16}$$

with $h_\alpha^{\uparrow,\mathrm{QKD}} = h_\alpha^{\uparrow,\mathrm{QKD}}(\varnothing)$ the single-round rate of Definition 2.4 at the empty list, which reads

$$h_\alpha^{\uparrow,\mathrm{QKD}} = \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \inf_{\substack{\omega\in S_=(AQ\widetilde{E}) \\ \omega\models\mathfrak{D}}} \left(\frac{\alpha}{\alpha-1}\, D(\mathbf{q} \,\|\, \mathbf{\Phi}[\omega_{AQ}]) + \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} q(\hat{c})\, \widetilde{H}_\alpha^\uparrow(Z|I\widetilde{E})_{\nu^\omega_{|\hat{c}}} \right),$$
$$\nu^\omega := (\mathcal{M} \otimes \mathrm{id}_{\widetilde{E}})(\omega), \tag{2.17}$$

with $\omega \models \mathfrak{D}$ as in (2.11) — so the inner infimum is over states with $\omega_A = \tau_A$ in trusted-source mode and over *all* normalized states on $AQ\widetilde{E}$ in untrusted-source mode. It ranges over all (not necessarily pure) such states, with $\dim \mathcal{H}(\widetilde{E}) \geq \dim \mathcal{H}(A) \cdot \dim \mathcal{H}(Q)$ (Lemma 2.7: the value is independent of that choice).

Uses Def. 1.2, Thm. 1.5, Lem. 1.7, Fact 2.1, Def. 2.3, Def. 2.4, Def. 2.5, Lem. 2.7, Lem. 2.8, Lem. 2.9, foundations/def:kl, foundations/def:renyi · *Assumes* A-SRC, A-ENS, A-SEQ, A-SEED, A-LAB, A-RHOM · *Used by* Rem. 2.11, Rem. 2.12, Thm. 3.1, Rem. 5.7, Rem. 5.9, Thm. 10.7, Cor. 10.9, Lem. 18.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength:key_length · *Oracles* registry.py

Proof. Idea. The bound is Fact 2.1 applied to the round channels (2.6). Step 1 instantiates the source's marginal register by the declared source mode, Step 2 checks the hypotheses of the import, and Steps 3 and 4 bound the imported rate (2.2) from below by $h_\alpha^{\uparrow,\mathrm{QKD}}$: Step 3 through the range inclusion of Lemma 2.8, Step 4 by reducing the sum over acceptance values to the key-live ones.

Step 1 (the marginal register). Apply Fact 2.1 to the round channels (2.6), with the identifications $S_j \to Z_j$, $S_\Omega \to \mathcal{S}_{\mathrm{acc}}$, and the source's marginal register A_{j-1} instantiated by the declared source mode:

> *Trusted-source mode:* A_{j-1} is the round-j copy of A and $\sigma^{(j-1)} \to \tau_A$ (off-by-one indexing; Fact 2.1, locator note (v)); $\mathcal{M}_j \in \mathrm{CPTP}(A\, E_{j-1},\, Z\, \widehat{C}\, E_j)$.
>
> *Untrusted-source mode:* A_{j-1} is *trivial* ($\dim = 1$) and $\sigma^{(j-1)} \to [1]$, the unique state on it; $\mathcal{M}_j \in \mathrm{CPTP}(E_{j-1},\, Z\, \widehat{C}\, E_j)$, which is the same map read with a trivial tensor factor adjoined to its input. That instantiation is admissible: neither [1, Thm. 4.2a] nor [1, Def. 4.2, Eq. (98)] places

any dimension or rank hypothesis on A_{j-1}, and the source's standing convention imposes no bound on system dimensions (Fact 2.1, locator note (vii), where all three are quoted).

Step 2 (the hypotheses of the import).

- Each $\mathcal{M}_j$ is CPTP with $\widehat{C}_j$ classical: composition and tensoring of the CPTP maps $\mathcal{M}$ (Theorem 1.5 (i), which also makes $\widehat{C}$ classical) and $\mathcal{A}_j$, followed by the output regrouping $E_j = E'_j I_j$. The import takes every system finite-dimensional and places no bound and no other structural requirement on the E_{j-1} registers (Fact 2.1, locator note (vii)); the round model of Definition 2.3 supplies exactly that for the adversary's registers, with no bound on their dimensions, and the bound uses none, which is what lets Corollary 10.8 reach a memory of any dimension through Lemma 10.5; no generality is lost by Q carrying a sector decomposition.

- Product marginal constraint on A_1^n: in trusted-source mode this is the condition on ω^0 in the statement, which Lemma 2.9 (ii) establishes for the protocol's n-round input under its hypotheses, which are those the statement names for that mode with the finite dimension of the adversary's initial register, and at $d_A = 1$ for every state. In untrusted-source mode A_1^n is one-dimensional, $S_=(A_1^n)$ is a singleton, and the condition $\omega^0_{A_1^n} = \sigma^{(0)} \otimes \cdots \otimes \sigma^{(n-1)}$ holds for every normalized ω^0 — it discharges *vacuously*, which is a satisfied hypothesis and not a waived one. Correspondingly Lemma 2.9 is not invoked in that mode and nothing in this proof appeals to it.

- Conditioning registers: the imported bound (2.1) conditions on $\widehat{C}_1^n E_n$, and the physical attack is represented so that $E_n = I_1^n \widehat{E}$ (Definition 2.3: the announcements are classical and public, so the sequence that retains all of them is available for every physical attack and produces the same protocol run). This yields exactly the left-hand side of (2.16), the form Theorem 10.7 uses, where Eve is granted the announcement transcript and the $\widehat{C}$ copies.

- Round structure: all rounds use the same declaration, hence the same $\mathcal{M}$, the same source mode, and (in trusted-source mode) the same τ_A (A-RHOM (Assumption "Round homogeneity of the declaration" (protocol-compiler/ass:rhom))); Eve's channels $\mathcal{A}_j$ are *not* assumed identical across rounds. No $\Sigma = \Sigma_1$ collapse is available or used — the round channels $\mathcal{M}_j$ are genuinely non-isomorphic (the registers E_{j-1} differ across j) — and the homogeneity of the declaration enters only through the common target set of Lemma 2.8 (ii). It is $\mathcal{M}$, not the trustedness of τ_A, that supplies the sharing: in untrusted-source mode the target sets $\Sigma_j \subseteq \Sigma^{\mathfrak{L}}(\widetilde{E}^*)$ are contained in a common set for the same reason, with the marginal clause of (2.11) empty.

- Purifying register: the import requires $\widetilde{E}$ large enough to purify $A_{j-1}\,E_{j-1}$ — a proof-internal requirement, discharged by enlargement ([1, footnote 17]) and only weakened when A_{j-1} is trivial; the defining optimization (2.17) needs only $\dim \mathcal{H}(\widetilde{E}) \geq \dim \mathcal{H}(AQ)$ (the dimension condition in the statement, with Lemma 2.7).

- Acceptance: $\mathcal{S}_{\mathrm{acc}}$ is convex (Definition 2.5) and by definition of Ω_{AT} every frequency with nonzero probability in $\rho_{|\Omega_{\mathrm{AT}}}$ lies in $\mathcal{S}_{\mathrm{acc}}$; $p_\Omega = \Pr[\Omega_{\mathrm{AT}}] > 0$.

- α domain: the import supplies $(1,\infty]$; this module invokes it on $(1,\infty)$, where Definition "Sandwiched Rényi divergence and conditional entropies" (foundations/def:renyi) defines the entropies (Fact 2.1, locator note (vi)); no other import is used in this claim.

Step 3 (the imported rate is at least $h_\alpha^{\uparrow,\mathrm{QKD}}$). It remains to bound the rate (2.2) from below by $h_\alpha^{\uparrow,\mathrm{QKD}}$. By Lemma 2.8 (ii) with $\mathfrak{L} = \varnothing$ — in either source mode — $\Sigma \subseteq \Sigma^{\varnothing}(\widetilde{E}^*)$ for a common register $\widetilde{E}^*$, so by Lemma 2.8 (iii) the rate is bounded below by the value of (2.2) over $\Sigma^{\varnothing}(\widetilde{E}^*)$, whose elements are $\nu^\sigma = (\mathcal{M} \otimes \mathrm{id}_{\widetilde{E}^*})(\sigma)$ with $\sigma \models \mathfrak{D}$; for these, $\boldsymbol{\nu}_{\widehat{C}} = \boldsymbol{\Phi}[\sigma_{AQ}]$ and the conditioning register regroups as $E\widetilde{E} = I\,(E'\widetilde{E})$ (Lemma 2.8 (ii)), so the bracket is bounded from below by that of (2.12) at $\mathfrak{L} = \varnothing$, the $\hat{c}$-sum reduced to $\widehat{\mathcal{C}}_{\mathrm{key}}$ (Step 4), and Lemma 2.7 converts the purifier $\widetilde{E}^*$ to the $\widetilde{E}$ of (2.17) — choosing $\widetilde{E}$ in the application of Fact 2.1 large enough that $\dim \mathcal{H}(E'_j\widetilde{E}) \geq \dim \mathcal{H}(AQ)$ for every j, so that the lemma's hypothesis holds for $\widetilde{E}^*$; [1, footnote 17] permits the enlargement, which only grows Σ and weakens the bound in the safe direction.

Step 4 (reduction of the sum to the key-live values). Fix $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{key}}$ with $q(\hat{c}) > 0$ and $\boldsymbol{\nu}_{\widehat{C}}(\hat{c}) > 0$. By Lemma 1.7 (iv) and (1.17), every outcome pair contributing to $\hat{c}$ has $r(x,v) = \perp$, so $Z = \perp$ deterministically in $\nu^\sigma_{|\hat{c}}$: the conditional state is $|\perp\rangle\langle\perp|_Z \otimes \nu'$, with ν' normalized. For a deterministic classical Z, $\widetilde{H}_\alpha^{\uparrow}(Z|\cdot) \geq 0$: the divergence in the definition of $\widetilde{H}_\alpha^{\uparrow}$ (Definition "Sandwiched Rényi divergence and conditional entropies" (foundations/def:renyi)) evaluates on $|\perp\rangle\langle\perp| \otimes \nu'$ against $\mathbb{1}_Z \otimes \sigma'$, and $\widetilde{D}_\alpha(|\perp\rangle\langle\perp| \otimes \nu' \| \mathbb{1}_Z \otimes \sigma') = \widetilde{D}_\alpha(\nu'\|\sigma')$, since the sandwich is $|\perp\rangle\langle\perp| \otimes \sigma'^{\gamma}\nu'\sigma'^{\gamma}$ with $\gamma = \frac{1-\alpha}{2\alpha}$; the state $\sigma' = \nu'$ is admissible in the infimum and gives $\widetilde{D}_\alpha(\nu'\|\nu') = \frac{1}{\alpha-1}\log_2 \operatorname{tr}\nu' = 0$, so the infimum is at most 0. Hence those terms are non-negative, and dropping them lowers the sum: the sum over $\mathrm{supp}(\boldsymbol{\nu}_{\widehat{C}})$ is at least the sum over $\widehat{\mathcal{C}}_{\mathrm{key}}$, in the extended sense of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl): a term with $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}} \setminus \mathrm{supp}(\boldsymbol{\nu}_{\widehat{C}})$ and $q(\hat{c}) = 0$ is 0 by the $0\log_2\frac{0}{v} = 0$ convention (the undefined conditional state is multiplied by zero), while $q(\hat{c}) > 0$ for such a $\hat{c}$ means $\mathbf{q} \not\ll \boldsymbol{\nu}_{\widehat{C}}$, so the divergence term is $+\infty$ and the bracket equals $+\infty$ on both sides regardless of the convention adopted for the undefined entropy term.

Since Step 3 bounds the imported rate from below, a sum bounded from below by the one over $\widehat{\mathcal{C}}_{\text{key}}$ is all it needs. □

The bound of Theorem 2.10 covers adaptive attacks with a finite-dimensional memory of any size, the tensor-product state model being a special case. In untrusted-source mode it holds with a smaller single-round rate, the price of not assuming a trusted source.

Remark 2.11 (Scope of the accumulated entropy bound). The bound (2.16) of Theorem 2.10 covers every attack in which Eve adapts each round to all earlier announcements and keeps a quantum memory between rounds, finite-dimensional and of any size as every register of the round model is; the security of the key, which this bound feeds, extends to a memory of any dimension (Corollary 10.8, through Lemma 10.5). The tensor-product state model ($\rho = \mathcal{M}^{\otimes n}[\omega]$, $\omega_{A_1^n} = \tau_A^{\otimes n}$) is the special case in which each $\mathcal{A}_j$ routes a pre-prepared register to Q. In untrusted-source mode the left-hand side and the constant $h_\alpha^{\uparrow,\mathrm{QKD}}$ are the same expressions read with ω unconstrained beyond normalization; the bound is then weaker (a larger inner feasible set gives a smaller $h_\alpha^{\uparrow,\mathrm{QKD}}$), which is the direction of conservatism and the price of not assuming a trusted source.

Uses Thm. 2.10 · newly derived · accepted

The imported theorem fixes the inner infimum over the outputs of all attacks, so a constraint list may raise the rate only through a licensed restriction of the attack model (Theorem 3.1). Device knowledge without such a licence enters through the declared measurement.

Remark 2.12 (Where declared constraints may enter the rate). The inner infimum of (2.2) is fixed by the imported theorem: for the unrestricted attack model of Theorem 2.10 it ranges over the outputs of *all* attacks, and restricting it by a nonempty $\mathfrak{L}$ is not licensed by anything in this section. What *is* licensed is a restriction of the attack model itself: when $\mathfrak{L}$ is licensed for the declared source mode (Definition 2.2) and the reduction chain that produced the declaration certifies that every physical attack with finite registers whose first channel receives every transmitted system is represented by a $\mathfrak{L}$-admissible sequence (A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk))), the single-round rate tightens to the constrained $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L})$ — that is Theorem 3.1, and it is the only constrained use this module licenses. Device and squashing knowledge that is not backed by such a certificate enters soundly through the declared POVM $\{F_v\}$ itself, which needs no licence.

Uses Fact 2.1, Def. 2.2, Thm. 2.10 · *Assumes* A-ATK · newly derived · accepted

3 Constrained accumulation

When the reduction chain behind the declaration restricts every attack to a licensed attack set, the accumulated entropy bound holds with the constrained single-round rate of the declared list, with the same penalty (Theorem 3.1). The licensed class is the one the licence transports and not an arbitrary choice: multiples of τ_A in trusted-source mode and scalars in

untrusted-source mode (Remark 3.4), a trusted-mode entry declaring the scalar so that no numerical test is needed (Remark 3.5). For a party-local entry of the general format the theorem stays true, though that reading is not shown canonical (Remark 3.2). The published flag-state squasher is an instance (Example 3.3), and Remark 3.6 records what the licence does not cover.

Imports protocol-compiler/def:homog

Under the attack-set licence A-ATK, the accumulated entropy bound holds with the constrained single-round rate of a licensed list in place of the unconstrained rate, with the same acceptance-probability penalty. The constrained rate is never below the unconstrained one.

Whether an accumulation theorem admits per-round receiver-side constraints is settled by re-importing the accumulation at the sequential level (Fact 2.1): the imported theorem is stated *per channel sequence*; which attack channels Eve is granted is a modelling input outside the theorem ([5, Def. 8.3, Rem. 8.4]); and a declared constraint list that is licensed for the declared source mode is precisely the state-level shadow of an attack-set restriction in the published format [5, Eq. (151)] (Definition 2.3, Lemma 2.8 (i)) — the τ_A-homogeneous entries of Definition 2.2 (a) transported by (2.13), the scalar entries of Definition 2.2 (b) by (2.14). What remains an assumption, and is never asserted here, is that the reduction chain behind the declaration actually delivers such a restriction: A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk)), under which a $\mathfrak{L}$-admissible attack sequence covers every attack available against the physical protocol with finite registers whose first channel receives every transmitted system; the others are reached by a limit (Corollary 10.8).

Theorem 3.1 (Constrained accumulation under the attack-set licence)**.** Let the hypotheses of Theorem 2.10 hold in either source mode, let $\mathfrak{L}$ be licensed for that mode (Definition 2.2), let A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk)) hold, and let the attack sequence be $\mathfrak{L}$-admissible (Definition 2.3). Then for every $\alpha \in (1, \infty)$,

$$\widetilde{H}_\alpha^{\uparrow}\big(Z_1^n \big| \widehat{C}_1^n I_1^n \widehat{E}\big)_{\rho_{|\Omega_{\mathrm{AT}}}} \;\geq\; n\, h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L}) - \frac{\alpha}{\alpha-1}\log_2 \frac{1}{\Pr[\Omega_{\mathrm{AT}}]}, \tag{3.1}$$

with $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L})$ the constrained single-round rate of Definition 2.4, and $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L}) \geq h_\alpha^{\uparrow,\mathrm{QKD}}(\varnothing) = h_\alpha^{\uparrow,\mathrm{QKD}}$.

Uses Fact 2.1, Def. 2.2, Def. 2.3, Def. 2.4, Lem. 2.7, Lem. 2.8, Thm. 2.10 · *Assumes* A-ATK · *Used by* Rem. 3.2, Rem. 3.4, Rem. 3.6, Rem. 5.9, Thm. 10.7, Cor. 10.9, Cor. 10.19 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength: check_key_length_permitted · *Oracles* registry.py

Proof. The proof of Theorem 2.10 applies unchanged, with one refinement: Lemma 2.8 (ii), invoked for the $\mathfrak{L}$-admissible sequence, gives $\Sigma \subseteq \Sigma^{\mathfrak{L}}(\widetilde{E}^*)$, so Lemma 2.8 (iii), the $\widehat{\mathcal{C}}_{\mathrm{key}}$ reduction of the sum (verbatim from the proof of Theorem 2.10), and Lemma 2.7 bound the rate of (2.2) below by $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L})$. Every other hypothesis check is identical. No inequality of the unconstrained proof is reversed anywhere: the improvement over Theorem 2.10 is a restriction of the admissible attack model, purchased entirely by Assumption A-ATK. The final inequality $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L}) \geq h_\alpha^{\uparrow,\mathrm{QKD}}(\varnothing)$ is Definition 2.4 (the constrained inner set is smaller). □

No statement of the constrained accumulation theorem becomes false for a list with a party-local entry: its hypotheses are satisfiable, its conclusion holds and its monotonicity survives. The tagged reading is not shown canonical, and the paragraph before the theorem describes only untagged forms.

Remark 3.2 (Party-local entries and constrained accumulation). Theorem 3.1, which turns a licensed list into a key-length licence, is stated for a list licensed for the declared mode and for an attack sequence admissible for that list. Both readings carry the party-local form of Definition "Licensed constraint list" (protocol-compiler/def:homog), the licensed class being that definition's and the per-round admissibility format of Definition 2.3 carrying the party-local admissibility set beside the two of the untagged forms, with Lemma 2.8 (i) transporting it to the state level in either mode. *So no statement of that theorem becomes false of a list carrying a tagged entry*: its hypotheses are satisfiable, its conclusion holds by that transport, and its monotonicity $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L}) \geq h_\alpha^{\uparrow,\mathrm{QKD}}(\varnothing)$, where $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L})$ is that theorem's single-round QKD rate at order α under the constraints of $\mathfrak{L}$, holds because a tagged entry shrinks the feasible set further than its untagged reading does.

What it does not say is that the tagged reading is canonical: the argument that the licensed class is not an arbitrary choice is stated for the two untagged forms (Remark 3.4) and is not extended to the third here, and the third form's soundness rests on the certificate on the scoped factor's own channel (Remark "Soundness of a party-local entry" (protocol-compiler/rem:party-local-soundness)) and on nothing this contract proves.

The paragraph motivating the theorem describes the untagged forms. That paragraph, before the theorem's statement, rests it on three points: the imported accumulation theorem is stated per sequence of attack channels; which attack channels the adversary is granted is a modelling input outside that theorem; and a constraint list licensed for the declared mode is "precisely the state-level shadow of an attack-set restriction in the published format", the homogeneous entries of form (a) and the scalar entries of form (b) of Definition "Licensed constraint list" (protocol-compiler/def:homog) each being carried to the state level by a transport of its own. *The third point is stated of the untagged forms in its main clause, not only in the two transports it lists*: the published format it cites is the single comparison of a channel adjoint against a multiple of the identity, whereas the admissibility set of a tagged entry is a *family* of operator inequalities, one per reference system and per positive semidefinite operator on it (Definition 2.3). *What covers the third form is the second point*, which makes restricting attention to a declared subset of attack channels a modelling step and is stated for any declared subset, not for a shape of right-hand side. So the theorem's hypotheses, conclusion and proof are unaffected,

and only the description in its motivating paragraph does not reach the third form.

Uses Def. 2.3, Def. 2.4, Lem. 2.8, Thm. 3.1, protocol-compiler/def:homog · newly derived · accepted

The restricted attack set of the published flag-state squasher is exactly the attack set of one licensed entry, a designated outcome's element minus λ_min times the flag projector bounded below by 0, in either source mode. At least one published reduction thus delivers a licensed list.

Example 3.3 (The flag-state squasher as a licensed constraint list). [5, Lemma 9.4, Eqs. (151)–(152)] constructs, for a receiver whose physical POVM is block-diagonal with respect to a photon-number cutoff, a squashed model whose receiver space carries a preserved subspace and a flag sector — matching this module's $\mathbb{S}$ with $\mathcal{H}(Q) = \mathcal{H}(Q_{\text{pres}}) \oplus \mathcal{H}(Q_{\text{flag}})$ — together with the restricted attack set

$$\big\{\mathcal{A} :\ \mathcal{A}^\dagger(F_o \otimes \mathbb{1}_{E''}) \ \succeq\ \lambda_{\min}\, \mathcal{A}^\dagger(\Pi_{\text{flag}} \otimes \mathbb{1}_{E''})\big\},$$

its Eq. (151). The source writes the flag projector Π_{Sq}, defined there as "the projector onto the flag subspace"; here it is Π_{flag}, the projector onto $\mathcal{H}(Q_{\text{flag}})$, a renaming into this module's sector vocabulary. F_o is the squashed POVM element of a designated outcome o and $\lambda_{\min} > 0$ a receiver constant. By linearity of the adjoint this is exactly $\mathfrak{A}_{\mathfrak{L}}$ of (2.7) for the single τ_A-homogeneous entry

$$\hat{O} = F_o - \lambda_{\min}\, \Pi_{\text{flag}}, \qquad \bowtie\ =\ \succeq, \qquad c = 0,$$

whose operators are sector-block-diagonal as Definition 1.2 (iii) requires (F_o by declaration, Π_{flag} a sector projector); the dimension-independent marginal form of the same set is the source's Eq. (152).

The instance carries over verbatim to untrusted-source mode: the source's condition is on operators of Q only, so lifting it as (2.8) prescribes gives $\mathcal{A}^\dagger(\mathbb{1}_A \otimes F_o \otimes \mathbb{1}_{E''}) \succeq \lambda_{\min}\mathcal{A}^\dagger(\mathbb{1}_A \otimes \Pi_{\text{flag}} \otimes \mathbb{1}_{E''})$, i.e. the same single entry $\hat{O} = F_o - \lambda_{\min}\Pi_{\text{flag}}$, $\bowtie = \succeq$, $c = 0$, now read with the scalar right-hand side of Definition 2.2 (b); and $c = 0$ is a scalar in either reading, so no wording of the source is stretched.

The instance shows that A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk)) is not vacuous: at least one published reduction delivers a licensed pair $(\mathfrak{L}, \mathfrak{A}_{\mathfrak{L}})$. Determining $\lambda_{\min}$ for a given receiver, and certifying the reduction itself, are obligations of the receiver and reductions modules; nothing here depends on how they are met.

Uses Def. 1.2, Def. 2.2, Def. 2.3 · *Assumes* A-ATK, A-SRC · newly derived · accepted

The licence transports exactly τ_A-multiples in trusted mode and scalars in untrusted mode, where the operator form c_k·1_A can empty the feasible set. On directions the licence can probe, sound trusted-mode right-hand sides are dominated by homogeneous ones; data stay out of the declaration.

Remark 3.4 (Canonicity of the licensed class). Three observations make Definition 2.2 the natural licensed class rather than an arbitrary restriction.

(a) The licence transports exactly the licensed right-hand sides, and nothing wider. In trusted-source mode the admissibility format (2.7) compares $\mathcal{A}^\dagger(\hat{O}_k \otimes \mathbb{1})$ against multiples of the identity, and Lemma 2.8 (i) turns that into the state-level right-hand side $c_k\, \tau_A$; the trusted marginal is the only operator on A the per-round model supplies.

In untrusted-source mode there is no such operator, and the difference is structural rather than a matter of taste: A is an *output* of $\mathcal{A}_j$ (Definition 2.3), so the input side of the adjoint identity carries no A factor at all and the operator-valued identity (2.13) has no analogue. Only its trace survives, giving (2.14) and the scalar class of Definition 2.2 (b). *The operator form $\Theta_k = c_k \mathbb{1}_A$ is therefore not licensed*, and it is not merely unlicensed but wrong: take $d_A = 2$, $\hat{O} = \mathbb{1}_Q$, $\bowtie = \succeq$ and $c = 1$. Every CPTP $\mathcal{A}$ is admissible, since $\mathcal{A}^\dagger(\mathbb{1}_A \otimes \mathbb{1}_Q \otimes \mathbb{1}_{E''}) = \mathbb{1}_{E'} \succeq 1 \cdot \mathbb{1}_{E'}$ by trace preservation, so $\mathfrak{A}_{\mathfrak{L}}$ is the whole of CPTP and the declaration restricts nothing. The transported scalar constraint is $\mathrm{tr}[(\mathbb{1}_A \otimes \mathbb{1}_Q)\,\omega] = \mathrm{tr}\,\omega = 1 \geq c = 1$, which every feasible ω satisfies — correct and vacuous, as it should be. The operator form would instead demand $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \mathbb{1}_Q)\,\omega] = \omega_A \succeq c\,\mathbb{1}_A = \mathbb{1}_A$, which *no* state satisfies at $d_A \geq 2$ ($\mathrm{tr}\,\omega_A = 1 < 2 = \mathrm{tr}\,\mathbb{1}_A$). It would empty $\mathcal{W}$, violating Definition 1.2 (vii), and would do so for a declaration whose licensed attack set is everything. Writing $\Theta_k = c_k \mathbb{1}_A$ in untrusted-source mode is thus a defect, not a conservative simplification.

The sentence "only its trace survives" is scoped to the per-round model's own operator on A, and the general format shows why the scope matters. What has no analogue in untrusted-source mode is the *operator-valued identity on A against τ_A*, because there is no declared τ_A to compare against; what does survive, beside the trace, is an operator-valued statement on A *tensored with the other measured factors*, and it survives not by weakening the obstruction but by a strictly stronger channel-level condition on the declaring side: the party-local admissibility format of Definition 2.3, which asks $\mathcal{A}_j^\dagger(X \otimes \hat{D} \otimes \mathbb{1}) \bowtie_k 0$ for *every* positive semidefinite X on the retained and unscoped systems, and which Lemma 2.8 (i) transports to $\mathrm{tr}_{Q_f}[(\mathbb{1} \otimes \hat{D})\,\varsigma] \bowtie_k 0$ as an operator, in either mode. *So "only its trace survives" is true of what the* scalar *admissibility format* (2.7) *supplies and is not true of the licensed class as a whole once the party-local format is declared*; the difference is that the scalar format tests one operator inequality on Eve's register and the party-local format tests a family of them, one per X, which is more to license and not less. *And the pathology of the operator-form counterexample does not recur there*: the untrusted-mode counterexample $\hat{O} = \mathbb{1}_Q$, $\bowtie = \succeq$, $c = 1$ read party-locally is not admitted with an empty feasible set, because the same channel-level condition it would need is $\mathcal{A}_j^\dagger(X \otimes 0 \otimes \mathbb{1}) \succeq 0$ for every X with $\hat{D} = 0$, which is satisfied, while the state-level reading is $\mathrm{tr}_Q[(\mathbb{1} \otimes 0)\,\varsigma] \succeq 0$, which every state satisfies — correct and vacuous, as the scalar reading is. *The canonicity argument of (a) is not extended to the third form*: the argument that the licensed class is not an arbitrary choice is made here for the two untagged forms, and the third form's soundness rests on the certificate on the scoped factor's own channel (Remark "Soundness of a party-local entry" (protocol-compiler/rem:party-local-soundness)) and on nothing this remark proves.

(b) In trusted-source mode, sound right-hand sides are dominated by homogeneous ones on the certifiable directions. Restrict attention to the discard-and-prepare channels the licensed set *actually contains*: for every $\mathcal{A} = \varrho\,\mathrm{tr}_{E'}[\,\cdot\,] \in \mathfrak{A}_{\mathfrak{L}}(E', QE'')$ (a state ϱ on QE'' prepared regardless of the input), the reachable marginals include $\tau_A \otimes \varrho_Q$, whose constraint value $\mathrm{tr}\left[\hat{O}_k\,\varrho_Q\right]\tau_A$ is a scalar multiple of τ_A. A sound right-hand side Θ_k for $\bowtie_k = \succeq$ must therefore satisfy $\Theta_k \preceq \left(\inf\{\mathrm{tr}\left[\hat{O}_k\,\varrho_Q\right] : \varrho\,\mathrm{tr}_{E'}[\,\cdot\,] \in \mathfrak{A}_{\mathfrak{L}}\}\right)\tau_A$ whenever that set is nonempty: on the directions the licensed model can probe, Θ_k is dominated by a homogeneous

right-hand side. *Which* discard-and-prepare channels are licensed is decided by the reduction chain behind the declaration, not by this module: Eve may always discard her memory and inject a fresh *physical* state, but what reaches Q after the reduction is its squashed image, which in general ranges over a proper subset of the states on Q. In Example 3.3 the licensed set deliberately excludes some preparations — a fresh flag state $|o'\rangle\langle o'|$ with $o' \neq o$ violates (2.7) — and exactly that exclusion is what makes the attack set restrictive and Theorem 3.1 non-vacuous. The canonical status of the licensed class therefore rests on (a) — the licence transports only τ_A-multiples in trusted-source mode, and only scalars in untrusted-source mode, in the first place — with (b) a domination statement over the licensed subset in trusted-source mode, not a universal necessity claim.

(c) Data stays out of the declaration. $\mathfrak{L}$ is part of $\mathfrak{D}$, fixed before any run; an observed number on a right-hand side would make the declaration data-dependent. The only sanctioned carrier of observed data is the acceptance set $\mathcal{S}_{\mathrm{acc}}$ (Definition 2.5); a constraint whose Θ_k encodes statistics of the current run is unsound here regardless of its provenance.

Uses Def. 1.2, Def. 2.2, Def. 2.3, Def. 2.5, Lem. 2.8, Thm. 3.1 · *Used by* Rem. 3.6, Rem. 10.20 · newly derived · accepted

A trusted-mode entry declares the scalar c_k, not an operator Θ_k said to equal c_k·τ_A, so its right-hand side and the transported bound are one expression. An operator only near c_k·τ_A would need a numerical test, and no relaxation of the program covers admitting it.

Remark 3.5 (Scalar declaration of a homogeneous right-hand side). Licensing is a type check in both modes and compares no numbers (Definition 2.2). This is deliberate, and it is the reason the trusted-mode entry declares "a declared c_k" and not "a declared Θ_k satisfying $\Theta_k = c_k\tau_A$". Under the latter form, a declaration carrying an operator Θ_k merely *near* $c_k\tau_A$ would have to be admitted or refused by a numerical test, and admitting one would be unsafe in a way no relaxation of this module covers: the transport Lemma 2.8 (i) and (2.13) deliver exactly $c_k\tau_A$ on the right-hand side, while (5.2) would impose the declared Θ_k, and the two feasible sets are then neither nested nor related by the relaxation of Definition 17.1, whose window is a spectral floor, or an entrywise ball, on the *relation*, and not an entrywise-relative window on the right-hand side. With the scalar declaration, Θ_k and the transported bound are the same expression in the same c_k and the same τ_A, and the two agree by construction on the declared numbers.

Uses Def. 2.2, Lem. 2.8 · newly derived · accepted

The constrained accumulation theorem licenses only restricting the single-round infimum to marginals meeting a licensed list. Limits on the purifying register, data-dependent or multi-round constraints, a trusted marginal, other proof techniques, the reduction and the test statistics lie beyond it.

Remark 3.6 (Limits of the attack-set licence). Theorem 3.1 licenses exactly one thing: restricting the single-round infimum to marginals satisfying a declared list that is licensed for the declared source mode (Definition 2.2; in the basic format, τ_A-homogeneous in trusted-source mode and scalar in untrusted-source mode), under A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk)). In the general format of Definition 1.1 the licensed class carries a third form, the entries

declared party-local, in either mode, whose right-hand side is a real c_k and whose state-level reading is the operator relation of Definition 2.2. What that adds to the following list is *nothing*: every item (a) to (g) is a statement about what is *not* licensed, each holds of the third form as it holds of the other two, and item (b)'s right-hand sides read $c_k\tau_A$, c_k, or c_k times the marginal that is actually there, all three with *declared* constants. The theorem does *not* license:

(a) *Constraints on the purifying register.* $\mathfrak{L}$'s syntax touches σ_{AQ} only; the inner infimum of (2.12) still ranges over *all* extensions of a feasible AQ-marginal (Lemma 2.7), and no constraint on $\widetilde{E}$ — equivalently, on Eve's side information — is expressible or admitted.

(b) *Data-dependent constraints.* $\Theta_k = c_k\tau_A$ (respectively $\Theta_k = c_k$) with *declared* constants c_k; observed statistics enter only through $\mathcal{S}_{\mathrm{acc}}$ (Remark 3.4 (c)).

(c) *A trusted marginal.* Nothing in Theorem 3.1 licenses declaring τ_A. That is a separate input with a separate attestation, A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)), and in particular a licensed $\mathfrak{L}$ never substitutes for it: a constraint list restricts what *Eve's channel* may do, whereas τ_A asserts that a register never reached her at all.

(d) *Joint multi-round constraints.* Admissibility is per round (Definition 2.3); no constraint on the joint n-round state is licensed, and none is expressible in $\mathfrak{L}$.

(e) *Reuse under other proof techniques.* The restriction is compatible with the accumulation route imported here; it is *not* usable with postselection-based reductions ([5, Rem. 9.4]). A module importing $\mathfrak{L}$ into a different proof technique must re-establish the licence there.

(f) *The reduction itself.* Nothing here establishes a squashing reduction, the existence of a licensed attack set for a given receiver, or the value of a receiver constant such as $\lambda_{\min}$ (Example 3.3); these are obligations of the reductions and receiver modules, assumed through Assumption A-ATK and never verified here.

(g) *Test-statistics changes.* Nothing here licenses a fixed-fraction test-statistics model (Requirement "No fixed fraction outside the measurement operators" (protocol-compiler/req:no-fixed-fraction)) or any modification of $\mathcal{S}_{\mathrm{acc}}$ (A-TSM (Assumption "Output-state test statistics" (reductions/ass:tsm)) stands unchanged).

Uses Def. 2.2, Def. 2.3, Def. 2.4, Lem. 2.7, Thm. 3.1, Rem. 3.4 · *Assumes* A-ATK, A-SRC, A-TSM · newly derived · accepted

4 Branch entropy and sectors

A pure input's conditional entropy on one acceptance value is a function of the protocol marginal alone, through the ratio of a pinched trace-power quantity to the outcome probability, and that quantity splits into sector deficits on block-diagonal inputs (Theorem 4.1). The pinched trace power is concave on the whole positive cone (Lemma 4.3), and every sector deficit is non-positive (Corollary 4.4) and vanishes on a key-inert pair, whose variables

still enter the statistics (Proposition 4.6). With Alice sectors the same holds per joint sector (Lemma 4.9).

Imports foundations/fact:concavity, foundations/fact:pinch-id, foundations/lem:pinchmono

A pure input's conditional entropy on one acceptance value is $1/(1-\alpha)$ times $\log_2$ of K/P, the ratio of a pinched trace-power quantity K to the outcome probability P, and K splits into sector deficits on block-diagonal inputs. The bounds $0 \le K \le P$ hold on the whole positive cone.

The right-hand side of the pinching identity of Fact "Pinched trace-power form of the conditional Rényi entropy" (foundations/fact:pinch-id) depends on the state only through ρ_Q; this is what makes the objective of this module a function of the AQ-marginal alone, and it is why purity of the global state must be arranged before the identity is invoked (Theorem 5.4).

Theorem 4.1 (Branch entropy from the protocol marginal)**.** For $\alpha \in (1,\infty)$, $\omega \in \mathrm{Pos}(AQ)$ and $\hat{c} \in \widehat{\mathcal{C}}$ define

$$P_{\hat{c}}(\omega) \coloneqq \mathbf{\Phi}[\omega](\hat{c}), \qquad T_{\hat{c}}(\omega) \coloneqq \mathrm{tr}\Big[\Big(\mathcal{Z}\big(\mathcal{G}_{\hat{c}}(\omega)^{1/\alpha}\big)\Big)^{\alpha}\Big], \qquad K_{\hat{c}}(\omega) \coloneqq T_{\hat{c}}(\omega)+P_{\hat{c}}(\omega)-\mathrm{tr}\,\mathcal{G}_{\hat{c}}(\omega), \tag{4.1}$$

and, for block-diagonal $\omega = \bigoplus_s \omega_s$, the sector deficits

$$f_{\hat{c},s}(\omega_s) \coloneqq \mathrm{tr}\Big[\Big(\mathcal{Z}\big(\mathcal{G}_{\hat{c},s}(\omega_s)^{1/\alpha}\big)\Big)^{\alpha}\Big] - \mathrm{tr}\,\mathcal{G}_{\hat{c},s}(\omega_s). \tag{4.2}$$

Let $\alpha \in (1,\infty)$, let $\omega \in S_=(AQ\widetilde{E})$ be *pure*, let $\nu^{\omega} = (\mathcal{M}\otimes \mathrm{id}_{\widetilde{E}})(\omega)$, and let $\hat{c} \in \widehat{\mathcal{C}}$ with $P_{\hat{c}} \coloneqq P_{\hat{c}}(\omega_{AQ}) > 0$. Then:

(i) *(the identity)*

$$\widetilde{H}^{\downarrow}_{\alpha}(Z|I\widetilde{E})_{\nu^{\omega}_{|\hat{c}}} = \frac{1}{1-\alpha}\log_2 \frac{K_{\hat{c}}(\omega_{AQ})}{P_{\hat{c}}}; \tag{4.3}$$

(ii) *(the bounds)* $0 \le K_{\hat{c}}(\omega_{AQ}) \le P_{\hat{c}}$, so the right-hand side of (4.3) is non-negative;

(iii) *(the sector form, on the whole cone)* for every $\hat{c} \in \widehat{\mathcal{C}}$ and *every* block-diagonal $\varrho = \bigoplus_s \varrho_s \in \mathrm{Pos}(AQ)$, with none of the hypotheses above on ω and $P_{\hat{c}}$ (no purity, no normalization and no hypothesis $P_{\hat{c}}(\varrho) > 0$),

$$K_{\hat{c}}(\varrho) = P_{\hat{c}}(\varrho) + \sum_{s\in\mathbb{S}} f_{\hat{c},s}(\varrho_s). \tag{4.4}$$

The two bounds of (ii) hold on the whole cone, as a statement separate from the identity (i): for every $\alpha \in (1,\infty)$, every $\hat{c} \in \widehat{\mathcal{C}}$ and *every* $\varrho \in \mathrm{Pos}(AQ)$, with no purity, no normalization, no block-diagonality and no hypothesis $P_{\hat{c}}(\varrho) > 0$, one has $0 \le K_{\hat{c}}(\varrho) \le P_{\hat{c}}(\varrho)$. We call this the theorem's *cone clause*.

Uses Def. 1.4, Thm. 1.5, Def. 1.6, Lem. 1.7, foundations/fact:pinch-id, foundations/lem:pinchmono · *Used by* Cor. 4.2, Lem. 4.3, Cor. 4.4, Prop. 4.6, Def. 4.8, Lem. 4.9, Thm. 5.4, Prop. 5.5, Rem. 5.7, Prop. 5.8, Thm. 5.10, Lem. 7.4, Def. 7.10, Lem. 8.6, Thm. 8.9, Lem. 9.1, Def. 9.2, Cor. 9.4, Lem. 9.6, Prop. 17.2, Lem. 18.2, Thm. 19.1, Rem. 19.3, Lem. 21.1, Rem. 23.9 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.branch: evaluate_branch · *Oracles* registry.py

Proof. *The identity (i).* Write $Q_{\mathrm{tot}} \coloneqq R\,XY\,C_AC_B\,AQ$ and $|\psi\rangle \coloneqq W\,|\omega\rangle_{AQ\widetilde{E}}$, a pure state on $Q_{\mathrm{tot}}\,\widehat{C}\,I\,\widetilde{E}$ (the isometry W of (1.15) acts trivially on $\widetilde{E}$; $\widehat{C}$ and I are classical in ψ in the sense that ψ is invariant under their computational dephasings, both being controlled copies). Let $|\psi_{\hat{c}}\rangle \coloneqq P_{\hat{c}}^{-1/2}\,\langle\hat{c}|_{\widehat{C}}\,|\psi\rangle$, a normalized pure state on $Q_{\mathrm{tot}}\,I\,\widetilde{E}$; normalization holds because $\|\,\langle\hat{c}|\,\psi\|^2 = \Pr\big[\widehat{C}=\hat{c}\big] = P_{\hat{c}}$ by Theorem 1.5 (iii). By construction $\nu^{\omega}_{|\hat{c}}$ on $Z\,I\,\widetilde{E}$ equals $\mathrm{tr}_{Q_{\mathrm{tot}}}[V_Z\;|\psi_{\hat{c}}\rangle\langle\psi_{\hat{c}}|\;V_Z^{\dagger}]$ up to the retained $\widehat{C}$ value, which is the fixed classical symbol $\hat{c}$ and may be dropped from the conditioning without change.

Apply Fact "Pinched trace-power form of the conditional Rényi entropy" (foundations/fact:pinch-id) with $Q \to Q_{\mathrm{tot}}$, $E \to I\widetilde{E}$, projectors $Z_j = |j\rangle\langle j|_R \otimes \mathbb{1}$ ($j \in \mathcal{Z}_{\perp}$; orthogonal, complete), and the pure state $|\psi_{\hat{c}}\rangle\langle\psi_{\hat{c}}|$: hypotheses hold verbatim, and

$$\widetilde{H}^{\downarrow}_{\alpha}(Z|I\widetilde{E})_{\nu^{\omega}_{|\hat{c}}} = \frac{1}{1-\alpha}\log_2 \mathrm{tr}\Big[\Big(\mathcal{Z}\big(\varrho^{1/\alpha}\big)\Big)^{\alpha}\Big], \qquad \varrho \coloneqq \mathrm{tr}_{I\widetilde{E}}\,|\psi_{\hat{c}}\rangle\langle\psi_{\hat{c}}|\,. \tag{4.5}$$

Compute ϱ. Expanding W as in the proof of Theorem 1.5 (ii) and projecting $\widehat{C}$ on $\hat{c}$ leaves the sum over pairs (x,v), (x',v') with $\phi(a(x,v)) = \phi(a(x',v')) = \hat{c}$. The register I holds $|a(x,v)\rangle\,\langle a(x',v')|$; tracing it out annihilates every term with $a(x,v) \neq a(x',v')$, that is, all cross-*class* terms. Tracing $\widetilde{E}$ replaces the rank-one $|\omega\rangle\langle\omega|$ by ω_{AQ} inside each remaining term, since the operators K_κ of (1.18) act only on AQ: $\mathrm{tr}_{\widetilde{E}}\big[(K_\kappa\otimes\mathbb{1}_{\widetilde{E}})\,|\omega\rangle\langle\omega|\,(K_{\kappa'}\otimes\mathbb{1}_{\widetilde{E}})^{\dagger}\big] = K_\kappa\,\omega_{AQ}\,K_{\kappa'}^{\dagger}$, and only $\kappa=\kappa'$ survives. Hence

$$\varrho = \frac{1}{P_{\hat{c}}}\sum_{\kappa\in\phi^{-1}(\hat{c})} K_\kappa\,\omega_{AQ}\,K_\kappa^{\dagger} = \frac{1}{P_{\hat{c}}}\big(\mathcal{G}_{\hat{c}}(\omega_{AQ}) + \Theta_{\hat{c}}(\omega_{AQ})\big). \tag{4.6}$$

By Lemma 1.7 (ii) the two summands have orthogonal supports ($R \in \mathcal{Z}$ versus $R = \perp$), so $\varrho^{1/\alpha} = P_{\hat{c}}^{-1/\alpha}\big(\mathcal{G}_{\hat{c}}(\omega)^{1/\alpha} + \Theta_{\hat{c}}(\omega)^{1/\alpha}\big)$, and $\mathcal{Z}$ preserves both parts, acting as the identity on the second. Therefore

$$\Big(\mathcal{Z}\big(\varrho^{1/\alpha}\big)\Big)^{\alpha} = \frac{1}{P_{\hat{c}}}\Big[\Big(\mathcal{Z}\big(\mathcal{G}_{\hat{c}}(\omega)^{1/\alpha}\big)\Big)^{\alpha} + \Theta_{\hat{c}}(\omega)\Big],$$

again by orthogonality of supports (the two α-th powers act blockwise) and $(\Theta^{1/\alpha})^{\alpha} = \Theta$. Taking the trace and using $\mathrm{tr}\,\Theta_{\hat{c}}(\omega) = P_{\hat{c}} - \mathrm{tr}\,\mathcal{G}_{\hat{c}}(\omega)$ (Lemma 1.7 (i)) gives $\mathrm{tr}\big[(\mathcal{Z}(\varrho^{1/\alpha}))^{\alpha}\big] = K_{\hat{c}}(\omega)/P_{\hat{c}}$, which with (4.5) is (4.3).

The bounds (ii) and the cone clause. $T_{\hat{c}} \geq 0$ because it is the trace of a positive semidefinite operator, and $P_{\hat{c}} - \operatorname{tr} \mathcal{G}_{\hat{c}} = \operatorname{tr} \Theta_{\hat{c}} \geq 0$ by (1.20), which Lemma 1.7 (i) states for *all* $\omega \in \operatorname{Pos}(AQ)$; together these give $K_{\hat{c}} \geq 0$. And $T_{\hat{c}} \leq \operatorname{tr} \mathcal{G}_{\hat{c}}$ is Lemma "Pinching lowers the trace power" (foundations/lem:pinchmono), applied at $\tau = \mathcal{G}_{\hat{c}}(\varrho)$ (that lemma is stated on all of Pos() and consumes no trace hypothesis), giving $K_{\hat{c}} \leq P_{\hat{c}}$. Neither step reads purity, normalization, $P_{\hat{c}} > 0$ or block-diagonality; both read $\varrho \succeq 0$ and nothing else, which is the cone clause.

The sector form (iii). By Lemma 1.7 (iii), $\mathcal{G}_{\hat{c}}(\varrho) = \bigoplus_s \mathcal{G}_{\hat{c},s}(\varrho_s)$; powers and $\mathcal{Z}$ act blockwise on this decomposition ($\mathcal{Z}$ pinches R, which is a tensor factor of every sector subspace), so $T_{\hat{c}}(\varrho) = \sum_s T_{\hat{c},s}(\varrho_s)$, with $T_{\hat{c},s}(\varrho_s) := \operatorname{tr}\big[(\mathcal{Z}(\mathcal{G}_{\hat{c},s}(\varrho_s)^{1/\alpha}))^\alpha\big]$, and $\operatorname{tr} \mathcal{G}_{\hat{c}}(\varrho) = \sum_s \operatorname{tr} \mathcal{G}_{\hat{c},s}(\varrho_s)$; insert into (4.1) and (4.2). This reads $\varrho \succeq 0$ and its block-diagonality and nothing else: neither purity nor $P_{\hat{c}} > 0$. □

For $\alpha > 1$ the pinched trace power is concave on subnormalised operators, so every sector deficit is concave on subnormalised sector states and each K is concave on the feasible set. It is the imported concavity result read at the parameter $1/\alpha$.

Corollary 4.2 (Concavity of the branch quantities on subnormalised states)**.** For $\alpha \in (1, \infty)$, the map $X \mapsto \operatorname{tr}\big[(\mathcal{Z}(X^{1/\alpha}))^\alpha\big] = Q_{1/\alpha}(X)$ is concave on $S_{\leq}()$ of its space; consequently, for every $\hat{c}$, s, the deficit $f_{\hat{c},s}$ of (4.2) is concave on $\{\sigma \in \operatorname{Pos}(AQ_s) : \operatorname{tr} \sigma \leq 1\}$, and $K_{\hat{c}}$ is concave on $\mathcal{W}$.

Uses Lem. 1.7, Thm. 4.1, foundations/fact:concavity · *Used by* Lem. 4.3, Thm. 5.10, Thm. 8.4 · newly derived · accepted · *Oracles* registry.py

Proof. $1/\alpha \in (0, 1)$, so Fact "Concavity of the pinched trace-power functional" (foundations/fact:concavity) applies at parameter $\alpha' = 1/\alpha$ and gives concavity of $Q_{1/\alpha}$ (this is exactly the substitution the source itself performs in its Cor. 27, which states concavity of $Q_{1/\alpha} \circ \mathcal{G}_\delta$ for $\alpha \in [1, \infty)$). The map $\mathcal{G}_{\hat{c},s}$ is linear and maps the stated domain into $S_{\leq}()$ (Lemma 1.7 (i)), and precomposition of a concave function with a linear map preserves concavity; subtracting the linear term $\operatorname{tr} \mathcal{G}_{\hat{c},s}(\cdot)$ preserves concavity. For $K_{\hat{c}} = P_{\hat{c}} + \sum_s f_{\hat{c},s}$: $P_{\hat{c}}$ is affine and a sum of concave functions is concave. The hypothesis check for Fact "Concavity of the pinched trace-power functional" (foundations/fact:concavity): our $\mathcal{Z}$ is a pinching from the complete orthogonal family $\{Z_j\}_{j \in \mathcal{Z}_\perp}$ extended by identity factors, which is again such a family; the arguments $\mathcal{G}_{\hat{c},s}(\sigma)$ are subnormalized PSD operators, inside the source's stated domain $S_{\leq}(Q)$. □

The pinched trace-power functional T is positively homogeneous and concave on the whole positive cone. Hence the sector deficits are concave on the cone and the branch quantities K on its sector-block-diagonal part, for every block-diagonal positive measurement family, complete or not.

Corollary 4.2 gives concavity only on subnormalised operators $S_{\leq}()$, a domain restriction it inherits from Lemma 1.7 (i). Positive homogeneity makes the restriction dispensable: two positive operators, divided by the larger of their traces, lie in $S_{\leq}()$, and T scales back linearly. Dispensing with it is what part (c) needs, and what every use of T off the trace-one slice needs, the relaxed iterate set of the Frank–Wolfe submodule (Proposition 17.2 (c)) among them. Part (c) reads no completeness relation of $\{F_v\}$, which is what makes it available off that slice as well. This lemma is the only place where the positive homogeneity of T is established, and every later use of it, the objective of Theorem 19.1 included, reads it from here.

Lemma 4.3 (Concavity on the positive cone)**.** Let $\alpha \in (1,\infty)$, put $p \coloneqq 1/\alpha \in (0,1)$, and let $\mathcal{Z}$ be a pinching built from a complete family of mutually orthogonal projectors. On the cone $\mathrm{Pos}()$ of the space $\mathcal{Z}$ acts on, consider the functional

$$T(X) \coloneqq \operatorname{tr}\big[\big(\mathcal{Z}(X^p)\big)^{\alpha}\big]. \tag{4.7}$$

(a) T is positively 1-homogeneous: $T(tX) = t\,T(X)$ for every $t \geq 0$.

(b) T is concave on all of $\mathrm{Pos}()$.

(c) Consider a declaration whose components other than $\{F_v\}$ satisfy Definition 1.2 and whose $\{F_v\}$ is any family of sector-block-diagonal positive semidefinite operators, complete or not. Then each deficit $f_{\hat{c},s}$ of (4.2) is concave on all of $\mathrm{Pos}(AQ_s)$, and $K_{\hat{c}}$ of (4.1) is concave on the sector-block-diagonal part of $\mathrm{Pos}(AQ)$, in particular on $\mathcal{W}$.

Uses Def. 1.2, Def. 1.4, Lem. 1.7, Thm. 4.1, Cor. 4.2, foundations/fact:concavity · *Used by* Lem. 4.9, Prop. 5.5, Lem. 7.4, Thm. 8.4, Lem. 8.6, Lem. 9.6, Prop. 17.2 · newly derived · accepted · *Oracles* registry.py

Proof. (a) Let $t > 0$. First, $(tX)^p = t^p X^p$: the map $X \mapsto X^p$ is defined by the functional calculus of $x \mapsto x^p$, and $(t\lambda)^p = t^p\lambda^p$ on each eigenvalue, the eigenprojectors being unchanged by the scaling. Second, $\mathcal{Z}$ is a pinching, hence linear, so $\mathcal{Z}\big((tX)^p\big) = \mathcal{Z}(t^pX^p) = t^p\mathcal{Z}(X^p)$, and $\operatorname{tr}[(t^pY)^\alpha] = t^{p\alpha}\operatorname{tr}[Y^\alpha]$ by the same functional calculus applied to $x \mapsto x^\alpha$. Together, $T(tX) = t^{p\alpha}T(X) = t\,T(X)$ because $p\alpha = 1$. At $t = 0$ both sides vanish.

(b) Let $X, Y \in \mathrm{Pos}()$ and $\lambda \in (0,1)$. If $\operatorname{tr} X = \operatorname{tr} Y = 0$ then $X = Y = 0$, a positive semidefinite operator of zero trace being zero, and the inequality reads $0 \geq 0$. Otherwise put $c \coloneqq \max\{\operatorname{tr} X, \operatorname{tr} Y\} > 0$, so that X/c and Y/c lie in $S_{\leq}()$ and so does their convex combination. Fact "Concavity

of the pinched trace-power functional" (foundations/fact:concavity) applies at $\alpha' = p \in (0,1]$, the substitution Corollary 4.2 makes, and on the source's own stated domain $S_{\leq}()$, so no hypothesis of the import is stretched; it gives

$$T\Big(\lambda\frac{X}{c} + (1-\lambda)\frac{Y}{c}\Big) \;\geq\; \lambda\, T\Big(\frac{X}{c}\Big) + (1-\lambda)\, T\Big(\frac{Y}{c}\Big).$$

Multiply by $c > 0$ and apply (a) once to each of the three arguments: $T(\lambda X + (1-\lambda)Y) \geq \lambda T(X) + (1-\lambda)T(Y)$. No trace hypothesis on X or Y is used: the scaling is by their own traces.

(c) $\mathcal{G}_{\hat{c},s}$ is completely positive by Lemma 1.7 (i) in its Kraus-form clause, which needs $F_v^{(s)} \succeq 0$ and no completeness relation; so it is linear and maps $\mathrm{Pos}(AQ_s)$ into $\mathrm{Pos}()$. Precomposition of the concave T of (b) with a linear map preserves concavity wherever the image lies in $\mathrm{Pos}()$, which is on all of $\mathrm{Pos}(AQ_s)$, and subtracting the linear $\sigma \mapsto \operatorname{tr} \mathcal{G}_{\hat{c},s}(\sigma)$ preserves it; by (4.2) that is concavity of $f_{\hat{c},s}$. For $K_{\hat{c}}$: on sector-block-diagonal ω, Lemma 1.7 (iii) gives $\mathcal{G}_{\hat{c}}(\omega) = \bigoplus_s \mathcal{G}_{\hat{c},s}(\omega_s)$ with mutually orthogonal summands, and powers and $\mathcal{Z}$ act blockwise on that decomposition, so $T_{\hat{c}}(\omega) = \sum_s T_{\hat{c},s}(\omega_s)$ and $\operatorname{tr} \mathcal{G}_{\hat{c}}(\omega) = \sum_s \operatorname{tr} \mathcal{G}_{\hat{c},s}(\omega_s)$. Inserting both into (4.1) gives $K_{\hat{c}}(\omega) = P_{\hat{c}}(\omega) + \sum_s f_{\hat{c},s}(\omega_s)$, which is (4.4) obtained from the definitions and Lemma 1.7 (iii) alone, without Theorem 4.1's entropy statement. $P_{\hat{c}}$ is affine by (1.16), and a sum of concave functions is concave. □

Every sector deficit is non-positive on the whole positive cone, with no trace hypothesis, and vanishes exactly when its passing-branch output is block-diagonal for the key pinching. Proposition 4.6 reads the equality case.

The corollary is stated on the whole cone. It imposes no trace bound $\operatorname{tr}\sigma \leq 1$: such a bound would be inherited from the domain of Corollary 4.2 and is used nowhere, and Proposition 4.6 invokes the equality case for arbitrary $\sigma \in \mathrm{Pos}(AQ_s)$.

Corollary 4.4 (Non-positivity of the sector deficits)**.** For every $\hat{c}$, s and *every* $\sigma \in \mathrm{Pos}(AQ_s)$, with no trace hypothesis (in particular not $\operatorname{tr}\sigma \leq 1$), $f_{\hat{c},s}(\sigma) \leq 0$, and $f_{\hat{c},s}(\sigma) = 0$ if and only if $\mathcal{G}_{\hat{c},s}(\sigma)$ is block-diagonal for the key pinching $\mathcal{Z}$.

Uses Lem. 1.7, Thm. 4.1, foundations/lem:pinchmono · *Used by* Prop. 4.6, Lem. 4.9, Thm. 6.2, Lem. 9.6 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.branch: evaluate_branch · *Oracles* registry.py

Proof. Apply Lemma "Pinching lowers the trace power" (foundations/lem:pinchmono) with $\tau = \mathcal{G}_{\hat{c},s}(\sigma)$ and recall (4.2). That lemma is stated on all of $\mathrm{Pos}()$ and consumes no trace hypothesis, and $\mathcal{G}_{\hat{c},s}$ maps $\mathrm{Pos}(AQ_s)$ into $\mathrm{Pos}()$ by Lemma 1.7 (i), so the statement is on the whole cone. □

A key-live acceptance value and a sector form a key-inert pair when the raw key is constant on the effective outcomes of each passing announcement class. The predicate is finite and read off the declaration, and it marks the pairs whose deficit cancels exactly (Proposition 4.6).

Definition 4.5 (Key-inert pair)**.** For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and $s \in \mathbb{S}$ define the *effective outcome set* of a pass class $\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})$ on sector s as

$$E_\kappa^{(s)} := \big\{(x, v) \in a^{-1}(\kappa) : \ M_x \neq 0, \ F_v^{(s)} \neq 0\big\}.$$

The pair $(\hat{c}, s)$ is *key-inert* if for every $\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})$ the raw-key map r is constant on $E_\kappa^{(s)}$. This is a finite, mechanically checkable predicate on the declaration.

Used by Def. 4.8, Lem. 4.9, Def. 6.1, Prop. 6.5, Lem. 21.1, Lem. 21.2 · newly derived · implemented · *Code* certiqs_key.meat_core.objective.branch: is_key_inert · *Oracles* registry.py

On a key-inert pair the sector deficit vanishes for every positive input, so the sector contributes nothing to the entropy term. Its variables still enter the outcome probabilities, the statistics and the marginal constraint.

On a key-inert pair every pass class writes a single raw-key value, so the branch output never mixes key values and the key pinching leaves it unchanged; the equality case of Corollary 4.4 does the rest.

Proposition 4.6 (Exact cancellation of key-inert sectors)**.** If $(\hat{c}, s)$ is key-inert, then for *every* $\sigma \in \mathrm{Pos}(AQ_s)$ the operator $\mathcal{G}_{\hat{c},s}(\sigma)$ is block-diagonal for the key pinching $\mathcal{Z}$, and consequently $f_{\hat{c},s}(\sigma) = 0$. In particular such sectors contribute nothing to $K_{\hat{c}}$ in (4.4), for every input; they remain optimization variables, entering through $\mathbf{\Phi}$, $P_{\hat{c}}$, and the marginal constraint.

Uses Def. 1.6, Lem. 1.7, Thm. 4.1, Cor. 4.4 · *Used by* Def. 4.7, Def. 6.1, Thm. 6.2, Thm. 7.5, Thm. 8.9 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.sectors: classify_sector · *Oracles* registry.py

Proof. Fix a pass class κ and let r_κ be the constant value of r on $E_\kappa^{(s)}$. Every term of $K_\kappa^{(s)}$ (Definition 1.6) with $(x, v) \notin E_\kappa^{(s)}$ vanishes, because $\sqrt{M_x} \otimes \sqrt{F_v^{(s)}} = 0$ there. Hence $K_\kappa^{(s)} = |r_\kappa\rangle_R \otimes \widetilde{K}_\kappa$ for an operator $\widetilde{K}_\kappa$ carrying no R factor, and $K_\kappa^{(s)} \sigma K_\kappa^{(s)\dagger} = |r_\kappa\rangle\langle r_\kappa|_R \otimes \widetilde{K}_\kappa \sigma \widetilde{K}_\kappa^\dagger$ is supported in a single R-eigenspace. A sum of such operators over κ is R-block-diagonal (each term lies in one block, possibly different blocks for different κ), hence a fixed point of $\mathcal{Z}$. By Corollary 4.4 (equality case), $f_{\hat{c},s}(\sigma) = 0$. The last sentence of the statement restates Lemma 1.7 (i),(iii): the sector still carries statistics weight, so removing it from the *optimization* (rather than from the objective sum) is a different operation, governed by the sector classes of Definition 6.1 and Theorem 6.2. □

The key-active index set of an acceptance value holds the sectors that are not key-inert with it. Restricting the deficit sum to it leaves the objective exactly equal, not bounded, and removes no state variable.

Definition 4.7 (Key-active index set)**.** The *key-active index set* of $\hat{c}$ is $\mathbb{S}_K(\hat{c}) \coloneqq \{s \in \mathbb{S} : (\hat{c}, s) \text{ is not key-inert}\}$. By Proposition 4.6, $K_{\hat{c}}(\omega) = P_{\hat{c}}(\omega) + \sum_{s \in \mathbb{S}_K(\hat{c})} f_{\hat{c},s}(\omega_s)$ for block-diagonal ω; the objective computed with the sum restricted to $\mathbb{S}_K(\hat{c})$ is *exactly equal* to the full sum, not a bound.

Uses Prop. 4.6 · *Used by* Def. 4.8, Thm. 7.5, Req. 14.9, Lem. 21.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective. branch:key_live_values · *Oracles* registry.py

With Alice sectors, the branch maps, statistics and deficit restrict to each joint sector, and a joint pair is key-inert when the raw key is constant on the outcomes that can occur on it, decided on declared data. With one Alice sector these are the sector notions.

Every sector notion of this section is stated on the cone of one measured sector tensored with all of A. Along Alice's declared sectors each of them restricts to a subcone, and the joint objects below are those restrictions; nothing about the objective is defined anew.

Definition 4.8 (Joint-sector deficits and the joint key-active index set)**.** Let $\mathfrak{D}$ be a declaration in the general format of Definition 1.1, with Alice sectors $t \in \mathbb{T}$ and joint sectors $(t, s) \in \mathbb{T} \times \mathbb{S}$. Regard $\mathrm{Pos}(A_t Q_s)$ as the subcone of $\mathrm{Pos}(A Q_s)$ of operators supported on $\mathcal{H}(A_t) \otimes \mathcal{H}(Q_s)$. For $\hat{c} \in \widehat{\mathcal{C}}$ the *joint-sector* maps, statistics and deficit are the restrictions of the sector objects of Definition 1.6 and of (4.2) to that subcone,

$$\mathcal{G}_{\hat{c},(t,s)} \coloneqq \mathcal{G}_{\hat{c},s}\big|_{\mathrm{Pos}(A_t Q_s)}, \qquad P_{\hat{c},(t,s)}(\sigma) \coloneqq P_{\hat{c}}(\sigma), \qquad f_{\hat{c},(t,s)}(\sigma) \coloneqq f_{\hat{c},s}(\sigma) \qquad (\sigma \in \mathrm{Pos}(A_t Q_s)), \tag{4.8}$$

so that on such a σ the Kraus operators (1.18) act through $\sqrt{M_x^{(t)}} \otimes \sqrt{F_v^{(s)}}$. For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and a pass class $\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})$ the *effective outcome set on the joint sector* is

$$E_\kappa^{(t,s)} \coloneqq \big\{(x, v) \in a^{-1}(\kappa) : \; M_x^{(t)} \neq 0, \; F_v^{(s)} \neq 0\big\} \subseteq E_\kappa^{(s)}, \tag{4.9}$$

and the pair $(\hat{c}, (t, s))$ is *key-inert* if r is constant on $E_\kappa^{(t,s)}$ for every $\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})$. The *joint key-active index set* is $\mathbb{J}_K(\hat{c}) \coloneqq \{(t, s) : (\hat{c}, (t, s)) \text{ is not key-inert}\}$.

Key-inertness of a joint pair is decided on declared data. In POVM form $M_x^{(t)} \neq 0$ is a comparison of the declared block with the zero block. In ensemble form it is $N_x^{(t)} \neq 0$, by Lemma 1.3 (iii), again a comparison of a declared block with zero. For a factorised family $F_v^{(s)} = \bigotimes_f F_{v_f}^{f,(s_f)}$ is non-zero exactly when every factor block $F_{v_f}^{f,(s_f)}$ is, a tensor product of matrices vanishing exactly when one of them does; so the predicate is decided per factor and no product is formed. With $|\mathbb{T}| = 1$ one has $E_\kappa^{(t_0,s)} = E_\kappa^{(s)}$,

and the pair $(\hat{c}, (t_0, s))$ is key-inert exactly when $(\hat{c}, s)$ is in the sense of Definition 4.5, with $\mathbb{J}_K(\hat{c}) = \{t_0\} \times \mathbb{S}_K(\hat{c})$.

Uses Def. 1.1, Lem. 1.3, Def. 1.6, Thm. 4.1, Def. 4.5, Def. 4.7 · *Used by* Lem. 4.9, Lem. 7.4, Thm. 7.5, Lem. 8.6, Lem. 9.1, Def. 9.2, Prop. 9.3, Cor. 9.4, Prop. 9.5, Lem. 9.6, Prop. 12.1, Req. 14.3, Cor. 19.6, Cor. 19.7, Lem. 20.1, Lem. 21.1, Lem. 21.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.joint:is_key_inert

On states block-diagonal along Alice's and the measured sectors, the branch maps, the statistics and K split per joint sector; each deficit is non-positive, concave and positively homogeneous. Key-inert joint pairs add zero, so the sum may be restricted to the joint key-active index set.

With Alice sectors, the sector form of Theorem 4.1 (iii) refines once more: a state block diagonal along the joint sectors splits the objective into one deficit per joint sector, and key-inertness is read per joint pair (Definition 4.8).

Lemma 4.9 (The objective split over joint sectors)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 in the general format of Definition 1.1, let $\alpha \in (1, \infty)$ and $\hat{c} \in \widehat{\mathcal{C}}$.

(i) *(decomposition)* For $s \in \mathbb{S}$ and $\sigma = \bigoplus_t \sigma_t \in \mathrm{Pos}(AQ_s)$ with $\sigma_t \in \mathrm{Pos}(A_tQ_s)$,

$$\mathcal{G}_{\hat{c},s}(\sigma) = \bigoplus_{t\in\mathbb{T}} \mathcal{G}_{\hat{c},(t,s)}(\sigma_t), \qquad \Theta_{\hat{c},s}(\sigma) = \bigoplus_{t\in\mathbb{T}} \Theta_{\hat{c},s}(\sigma_t), \qquad \mathbf{\Phi}_s[\sigma] = \sum_t \mathbf{\Phi}_s[\sigma_t], \tag{4.10}$$

the direct sums along the mutually orthogonal subspaces $\mathcal{H}(R\,XY\,C_AC_B)\otimes \mathcal{H}(A_tQ_s)$.

(ii) *(additivity)* For such σ, $f_{\hat{c},s}(\sigma) = \sum_t f_{\hat{c},(t,s)}(\sigma_t)$. Consequently, for every $\omega = \bigoplus_{(t,s)} \omega^{(t,s)}$ block diagonal along the joint sectors,

$$P_{\hat{c}}(\omega) = \sum_{(t,s)} P_{\hat{c},(t,s)}\big(\omega^{(t,s)}\big), \qquad K_{\hat{c}}(\omega) = P_{\hat{c}}(\omega) + \sum_{(t,s)} f_{\hat{c},(t,s)}\big(\omega^{(t,s)}\big), \tag{4.11}$$

and for every $\sigma \in \mathrm{Pos}(A_tQ_s)$: $f_{\hat{c},(t,s)}(\sigma) \le 0$, $0 \le P_{\hat{c},(t,s)}(\sigma) + f_{\hat{c},(t,s)}(\sigma)$, and $f_{\hat{c},(t,s)}$ is concave and positively 1-homogeneous on $\mathrm{Pos}(A_tQ_s)$.

(iii) *(key-inert joint pairs)* If $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and $(\hat{c}, (t, s))$ is key-inert (Definition 4.8), then $f_{\hat{c},(t,s)}(\sigma) = 0$ for *every* $\sigma \in \mathrm{Pos}(A_tQ_s)$. Hence the second sum of (4.11) may be restricted to $(t, s) \in \mathbb{J}_K(\hat{c})$, as an identity and not as a bound.

(iv) *(relation to the sector notions)* If $(\hat{c}, s)$ is key-inert in the sense of Definition 4.5 then $(\hat{c}, (t, s))$ is key-inert for every t; the converse fails in general, and holds when $|\mathbb{T}| = 1$.

Uses Def. 1.1, Def. 1.2, Def. 1.4, Def. 1.6, Lem. 1.7, Thm. 4.1, Lem. 4.3, Cor. 4.4, Def. 4.5, Def. 4.8 · *Used by* Thm. 7.5, Prop. 9.3, Cor. 9.4, Lem. 9.6, Prop. 17.2, Thm. 19.1, Cor. 19.6, Lem. 21.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.joint:evaluate_joint_branch

Proof. (i) By Definition 1.2 (iii) in the general format every M_x is block diagonal along $\mathbb{T}$, and by blockwise functional calculus so is $\sqrt{M_x} = \bigoplus_t \sqrt{M_x^{(t)}}$. Write $\widetilde{\Pi}_t$ for Π_t^A acting on the A factor of $\mathcal{H}(R\,XY\,C_AC_B\,AQ_s)$. Each term of $K_\kappa^{(s)}$ in (1.18) carries the operator $\sqrt{M_x}\otimes\sqrt{F_v^{(s)}}$ on AQ_s, which commutes with $\Pi_t^A\otimes\mathbb{1}$, so $K_\kappa^{(s)}(\Pi_t^A\otimes\mathbb{1}) = \widetilde{\Pi}_t K_\kappa^{(s)}$. Since $\sigma = \sum_t(\Pi_t^A\otimes\mathbb{1})\sigma(\Pi_t^A\otimes\mathbb{1})$ and $\sigma_t = (\Pi_t^A\otimes\mathbb{1})\sigma(\Pi_t^A\otimes\mathbb{1})$,

$$K_\kappa^{(s)}\,\sigma\,K_\kappa^{(s)\dagger} = \sum_t \widetilde{\Pi}_t\,K_\kappa^{(s)}\,\sigma_t\,K_\kappa^{(s)\dagger}\,\widetilde{\Pi}_t,$$

and the t-th term is $K_\kappa^{(s)}\sigma_t K_\kappa^{(s)\dagger}$, supported on the range of $\widetilde{\Pi}_t$. Summing over the pass, resp. fail, classes of $\phi^{-1}(\hat{c})$ gives the two direct sums; the statistics statement is linearity of (1.16) in its argument.

(ii) The subspaces of (i) are invariant under $\mathcal{Z}$, which acts on R alone, and under powers; so for an operator $X = \bigoplus_t X_t$ along them, $X^{1/\alpha} = \bigoplus_t X_t^{1/\alpha}$, $\mathcal{Z}(X^{1/\alpha}) = \bigoplus_t \mathcal{Z}(X_t^{1/\alpha})$, the α-th power is again blockwise, and the trace adds. With $X = \mathcal{G}_{\hat{c},s}(\sigma)$ and (i), (4.2) gives $f_{\hat{c},s}(\sigma) = \sum_t f_{\hat{c},s}(\sigma_t)$, which is the claim by (4.8). For ω block diagonal along the joint sectors, put $\omega_s \coloneqq \bigoplus_t \omega^{(t,s)}$; then $\omega = \bigoplus_s \omega_s$ is block diagonal along $\mathbb{S}$, Lemma 1.7 (iii) gives $\mathcal{G}_{\hat{c}}(\omega) = \bigoplus_s \mathcal{G}_{\hat{c},s}(\omega_s)$ and $\mathbf{\Phi}[\omega] = \sum_s \mathbf{\Phi}_s[\omega_s]$, and by the same blockwise argument along $\mathbb{S}$ together with (4.1), $K_{\hat{c}}(\omega) = P_{\hat{c}}(\omega) + \sum_s f_{\hat{c},s}(\omega_s)$; insert the additivity just proved, and the additivity of $\mathbf{\Phi}_s$ from (i), to obtain (4.11). For the bounds let $\sigma \in \mathrm{Pos}(A_tQ_s)$. $f_{\hat{c},(t,s)}(\sigma) = f_{\hat{c},s}(\sigma) \le 0$ by Corollary 4.4, stated on all of $\mathrm{Pos}(AQ_s)$. Regard σ as an element ϱ of $\mathrm{Pos}(AQ)$ supported on one joint sector: then $\mathcal{G}_{\hat{c}}(\varrho) = \mathcal{G}_{\hat{c},s}(\sigma)$ by Lemma 1.7 (iii), so $K_{\hat{c}}(\varrho) = P_{\hat{c}}(\sigma) + f_{\hat{c},s}(\sigma)$ by (4.1) and (4.2), and the cone clause of Theorem 4.1 gives $K_{\hat{c}}(\varrho) \ge 0$. Concavity on $\mathrm{Pos}(A_tQ_s)$ is the restriction to a convex subcone of the concavity of $f_{\hat{c},s}$ on $\mathrm{Pos}(AQ_s)$, Lemma 4.3 (c); positive 1-homogeneity is part (a) of that lemma applied to $T\circ\mathcal{G}_{\hat{c},s}$, the map $\mathcal{G}_{\hat{c},s}$ and the trace being linear.

(iii) Fix a pass class κ and let r_κ be the constant value of r on $E_\kappa^{(t,s)}$. For $\sigma \in \mathrm{Pos}(A_tQ_s)$ one has $\sigma = (\Pi_t^A\otimes\mathbb{1})\sigma(\Pi_t^A\otimes\mathbb{1})$, so only $K_\kappa^{(s)}(\Pi_t^A\otimes\mathbb{1})$ enters $K_\kappa^{(s)}\sigma K_\kappa^{(s)\dagger}$, and its term for (x,v) carries $\sqrt{M_x^{(t)}}\otimes\sqrt{F_v^{(s)}}$, which vanishes unless $(x,v)\in E_\kappa^{(t,s)}$. Hence $K_\kappa^{(s)}(\Pi_t^A\otimes\mathbb{1}) = |r_\kappa\rangle_R\otimes\widetilde{K}_\kappa$ with $\widetilde{K}_\kappa$ carrying no R factor, and $K_\kappa^{(s)}\sigma K_\kappa^{(s)\dagger} = |r_\kappa\rangle\langle r_\kappa|_R\otimes\widetilde{K}_\kappa\sigma\widetilde{K}_\kappa^\dagger$ lies in one eigenspace of the key pinching. A sum of such operators over κ is a fixed point of $\mathcal{Z}$, so $\mathcal{G}_{\hat{c},(t,s)}(\sigma)$ is block diagonal for $\mathcal{Z}$ and $f_{\hat{c},(t,s)}(\sigma) = 0$ by the equality case of Corollary 4.4. The restriction of the sum follows.

(iv) $E_\kappa^{(t,s)} \subseteq E_\kappa^{(s)}$, since $M_x^{(t)} \neq 0$ implies $M_x \neq 0$; a map constant on a set is constant on a subset. For the failure of the converse take two Alice sectors, $\mathcal{X} = \{0,1\}$ with $M_0 = \Pi_1^A$, $M_1 = \Pi_2^A$, one announcement class, and $r(x,v) = x$: on each joint sector one x occurs and r is constant, on the Q sector both occur and it is not. With $|\mathbb{T}| = 1$ the two sets coincide. □

5 The program and its reduction

The feasible set, the objective and its infimum h_low over the acceptance set form the finite convex program of this module; at d_A = 1 the two source modes give the same program (Proposition 5.2), and untrusted-source mode never gives a larger value. The section's main result, Theorem 5.4, shows that h_low bounds the single-round rate from below for an unconstrained declaration in either source mode, every step being an equality or conservative, and Corollary 5.6 extends it to arbitrary constraint lists. Restricting the variable along Alice's sectors changes no value (Proposition 5.5). The value lies in $[0, +\infty]$, each entropy term being at most $\log_2(\mathrm{d_Z} + 1)$ while the divergence penalty is unbounded (Proposition 5.8), and an infinite value means that no attack the declared list admits is accepted (Remark 5.9). The objective is jointly convex and lower semicontinuous, so a finite infimum is attained (Theorem 5.10).

Imports foundations/def:kl, foundations/def:pinching, foundations/def:renyi, foundations/lem:kernel, protocol-compiler/def:feasible

The program minimises an entropy objective over the compiler's feasible set (Definition "Feasible set" (protocol-compiler/def:feasible)), read as normalised block-diagonal states meeting the declared marginal and constraints. Its infimum over the acceptance set, h_low, is the value a key length is computed from.

The set over which this module's program minimises is the compiler's feasible set (Definition "Feasible set" (protocol-compiler/def:feasible)); the objective and $\underline{h}_\alpha$ are this module's alone.

Definition 5.1 (Feasible set and computable objective)**.** For a declaration $\mathfrak{D}$, $\mathcal{W}$ is the feasible set of Definition "Feasible set" (protocol-compiler/def:feasible). Read as a set of operators on AQ, a list of sector blocks being the block-diagonal operator it denotes, it is

$$\mathcal{W} := \Big\{\omega \in \mathrm{Pos}(AQ) : (\mathbb{1}_A \otimes \Pi_s)\,\omega\,(\mathbb{1}_A \otimes \Pi_{s'}) = 0\ \forall s \neq s', \quad \mathrm{tr}\,\omega = 1, \quad \omega \models \mathfrak{D}, \quad \omega \models \mathfrak{L}\Big\}, \tag{5.1}$$

writing ω_s for the sector blocks, with $\omega \models \mathfrak{D}$ the marginal condition of (2.11) and $\omega \models \mathfrak{L}$ the constraint conditions in the form Definition 2.2 gives them for the declared mode. Written out, $\mathcal{W}$ is

$$\begin{aligned} &\text{trusted:} \quad \omega_A = \tau_A, \quad \mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega] \bowtie_k \Theta_k \ \ \forall k; \\ &\text{untrusted:} \quad \text{(no marginal condition)}, \quad \mathrm{tr}\Big[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega\Big] \bowtie_k c_k \ \ \forall k. \end{aligned} \tag{5.2}$$

The normalization $\operatorname{tr}\omega = 1$ is written *explicitly* in (5.1), and is not derived from anything: in trusted-source mode it also follows from $\omega_A = \tau_A$ (whence $\operatorname{tr}\omega = \operatorname{tr}\tau_A = 1$), but that derivation is unavailable in untrusted-source mode, where normalization is the *only* unconditional constraint the feasible set carries. Leaving it implied would empty the untrusted-mode program of its one defining condition. $\mathcal{W}$ is convex and compact in either mode and in either format (Definition "Feasible set" (protocol-compiler/def:feasible)). For $\alpha \in (1,\infty)$, $\mathbf{q}$ a distribution on $\widehat{\mathcal{C}}$ and $\omega \in \mathcal{W}$, define the *objective*

$$O_\alpha(\mathbf{q},\omega) \coloneqq \frac{\alpha}{\alpha-1}\sum_{\hat{c}\notin\widehat{\mathcal{C}}_{\mathrm{key}}} q(\hat{c})\log_2\frac{q(\hat{c})}{P_{\hat{c}}(\omega)} + \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}}\left[q(\hat{c})\log_2\frac{q(\hat{c})}{P_{\hat{c}}(\omega)} + \frac{q(\hat{c})}{\alpha-1}\log_2\frac{q(\hat{c})}{K_{\hat{c}}(\omega)}\right], \tag{5.3}$$

with the conventions of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl) at vanishing arguments ($0\log_2\frac{0}{v} = 0$; $u\log_2\frac{u}{0} = +\infty$ for $u>0$), and

$$\underline{h}_\alpha \coloneqq \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}}\ \inf_{\omega\in\mathcal{W}}\ O_\alpha(\mathbf{q},\omega). \tag{5.4}$$

The general format. For a declaration in the general format of Definition 1.1, $\mathcal{W}$ is the set of lists over the joint sectors of Definition "Feasible set" (protocol-compiler/def:feasible), $\omega = \bigoplus_{(t,s)}\omega^{(t,s)}$, with the s-block $\hat{O}_k^{\uparrow(s)}$ of the lifted operator of an entry scoped to f_k and, for a party-local entry scoped to f, the split $s = (s_{f^c}, s_f)$ and the product $Q_{f^c,s_{f^c}}$ of the other factors' blocks written there, with every block positive semidefinite; its conditions read, block by block,

$$\begin{aligned}
&\textstyle\sum_{(t,s)}\operatorname{tr}\omega^{(t,s)} = 1 && \text{(normalization)};\\
&\textstyle\sum_{s}\operatorname{tr}_{Q_s}\omega^{(t,s)} = \tau_A^{(t)} \quad \forall t && \text{(marginal, trusted-source mode)};\\
&\textstyle\sum_{s}\operatorname{tr}_{Q_s}\left[(\mathbb{1}_{A_t}\otimes\hat{O}_k^{\uparrow(s)})\,\omega^{(t,s)}\right] \bowtie_k \Theta_k^{(t)} \quad \forall t && \text{(trusted; } \Theta_k^{(t)} = c_k\tau_A^{(t)} \text{ if homogeneo}\\
&\textstyle\sum_{(t,s)}\operatorname{tr}\left[(\mathbb{1}_{A_t}\otimes\hat{O}_k^{\uparrow(s)})\,\omega^{(t,s)}\right] \bowtie_k c_k && \text{(scalar entry, untrusted-source mod}\\
&\textstyle\sum_{s_f\in\mathbb{S}^f}\operatorname{tr}_{Q_{f,s_f}}\left[\left(\mathbb{1}\otimes(\hat{O}_k^{(s_f)} - c_k\mathbb{1})\right)\omega^{(t,(s_{f^c},s_f))}\right] \bowtie_k 0 \quad \forall(t,s_{f^c}) && \text{(party-local, either mode)},
\end{aligned} \tag{5.5}$$

the last as an operator on $\mathcal{H}(A_t)\otimes\mathcal{H}(Q_{f^c,s_{f^c}})$. The objective (5.3) and $\underline{h}_\alpha$ of (5.4) are defined on this $\mathcal{W}$ by the same formulas.

Uses Def. 1.1, Def. 2.2, Def. 2.4, foundations/def:kl, protocol-compiler/def:feasible · *Used by* Prop. 5.2, Prop. 5.5, Cor. 5.6, Prop. 5.8, Thm. 5.10, Def. 6.1, Thm. 6.2, Prop. 6.5, Rem. 6.10, Lem. 7.1, Lem. 7.4, Thm. 7.5, Prop. 7.6, Rem. 7.9, Def. 7.10, Lem. 7.11, Lem. 8.8, Thm. 8.9, Lem. 9.1, Def. 9.2, Prop. 9.3, Cor. 9.4, Prop. 9.5, Lem. 9.6, Thm. 10.7, Prop. 10.12, Rem. 10.20, Prop. 11.1, Rem. 11.3, Prop. 12.1, Req. 16.3, Def. 17.1, Prop. 17.2, Thm. 19.1, Cor. 19.6, Lem. 20.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.objective: check_feasible · *Oracles* registry.py

At d_A = 1 the two source modes have the same feasible set, the n-round product-marginal condition holds for every normalised state, the two licensed constraint classes coincide and the source licence deletes no attack. The two modes are then one declaration.

When Alice's system is one-dimensional, $d_A = \dim \mathcal{H}(A) = 1$, it admits a single state, and the two values of the marginal slot differ only in name. The proposition says so for four objects the source mode touches: the feasible set, the product-marginal condition of Lemma 2.9, the licensed constraint classes and Assumption A-SRC; for the Choi parametrisation it is Definition 7.10 (d). A condition that every normalized state satisfies is satisfied vacuously, which is not the same as violated; Remark 5.3 (b) reads the proposition as an embedding of protocols with no trusted source.

Proposition 5.2 (The two source modes coincide at a trivial Alice system)**.** Let $d_A = 1$. Then trusted-source mode admits exactly one marginal, $\tau_A = [1]$, and:

(i) the two feasible sets of (5.1) coincide;

(ii) the n-round condition $\omega_{A_1^n} = \tau_A^{\otimes n}$ of Lemma 2.9 (ii) holds for every normalized state; it is satisfied vacuously;

(iii) the licensed constraint classes of Definition 2.2 coincide: trusted-mode τ_A-homogeneity is exactly the scalar typing of untrusted-source mode;

(iv) A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)) is vacuous: what it licenses is the condition $\operatorname{tr}\omega = 1$, satisfied by every normalized state, so it deletes no attack.

Uses Def. 1.2, Def. 2.2, Def. 5.1 · *Used by* Prop. 6.3, Def. 7.10, Rem. 10.20, Prop. 11.1, Req. 16.12 · newly derived · accepted

Proof. Since $d_A = 1$, $S_=(A) = \{[1]\}$, so trusted-source mode admits exactly one marginal, $\tau_A = [1]$. For $\omega \in \mathrm{Pos}(AQ)$ the A-marginal is the 1×1 matrix $\operatorname{tr}\omega$, so $\omega_A = \tau_A$ and $\operatorname{tr}\omega = 1$ are the same condition.

(i) The marginal lines of (5.2) agree: that is the equality of conditions just shown. The constraint lines agree as well: at $d_A = 1$ the partial trace tr_Q of an operator on AQ is its trace, a 1×1 Hermitian Θ_k is $c_k[1] = c_k\tau_A$ with $c_k \in \mathbb{R}$ its one entry, and $\preceq$, $=$, $\succeq$ between 1×1 matrices are $\leq$, $=$, $\geq$, so the trusted line $\operatorname{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega] \bowtie_k \Theta_k$ is the untrusted line $\operatorname{tr}\left[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega\right] \bowtie_k c_k$. In the general format the lines of (5.5) agree in the same way, the party-local line being the same in either mode.

(ii) A_1^n is one-dimensional, so $S_=(A_1^n)$ is a singleton and the A_1^n-marginal of every normalized state is its one element $\tau_A^{\otimes n}$.

(iii) In trusted-source mode an untagged entry of a licensed list is declared homogeneous, carrying $c_k \in \mathbb{R}$, and its right-hand side is constructed

as $\Theta_k = c_k\tau_A = c_k[1]$ (Definition 2.2 (a)); in untrusted-source mode an untagged entry of a licensed list carries a real c_k (Definition 2.2 (b)). So the two classes consist of the same lists of entries carrying a real scalar c_k, and by (i) the line of such an entry is in both modes the one comparison of real numbers $\operatorname{tr}\left[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega\right] \bowtie_k c_k$. An entry declared general lies in neither class: its 1×1 Hermitian Θ_k equals $c_k\tau_A$ with $c_k = \Theta_k \in \mathbb{R}$, so it could have been declared homogeneous with the same line, but licensing reads the declared form (Definition 2.2).

(iv) What Assumption A-SRC licenses is the marginal condition $\omega_A = \tau_A$, which by the equality of conditions above is $\operatorname{tr}\omega = 1$; every normalized state satisfies it. □

A declared marginal deletes every feasible state with another marginal, so untrusted-source mode is always available and never gives a larger value. At d_A = 1 the two modes coincide, which lets a protocol with no trusted source be declared there; τ_A = 0 is inadmissible.

Remark 5.3 (The two source modes on the program). *(a) What each mode asserts* is the transmitter's statement (Remark "The two source modes" (transmitter/rem: source-mode) (a): in trusted-source mode the declaration asserts, through A-SRC (Assumption "Retention of the system A in trusted-source mode" (transmitter/ass: src)), that Alice's register A never passes through the adversary's hands, so that $\omega_A = \tau_A$ in every round; in untrusted-source mode it asserts nothing about A). What that does to this module's program is stated here. In untrusted-source mode ($\tau_A = \mathsf{none}$) the adversary prepares AQ jointly (Definition 2.3), the feasible set carries only $\operatorname{tr}\omega = 1$ (Eq. (5.1)), and Lemma 2.9 is not invoked at all. A declared τ_A *deletes* from the feasible set every state whose A-marginal differs, hence every attack that would have produced one. That is a different *kind* of input from an entry of $\mathfrak{L}$, and a more demanding one to justify: an entry of $\mathfrak{L}$ restricts what Eve's channel may do to a register she holds, whereas τ_A asserts that a register never reached her at all. Declaring one without the attestation of Assumption A-SRC overstates the key in the unsafe direction. Untrusted-source mode is always available and always more conservative. Precisely: fix the declaration data $(\hat{O}_k, \bowtie_k, c_k)$ and compare the feasible set (5.1) with τ_A declared against the one with $\tau_A = \mathsf{none}$. Every ω feasible in trusted-source mode is feasible in untrusted-source mode — from $\omega_A = \tau_A$ we get $\operatorname{tr}\omega = \operatorname{tr}\tau_A = 1$, and taking the trace of the operator entry $\operatorname{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega] \bowtie_k c_k\tau_A$ gives the scalar entry $\operatorname{tr}\left[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega\right] \bowtie_k c_k$ (using $\operatorname{tr}\tau_A = 1$, and that $X \succeq Y$ implies $\operatorname{tr}X \geq \operatorname{tr}Y$). Hence the untrusted feasible set is the larger, its infimum the smaller, and $\underline{h}_\alpha$ computed in untrusted-source mode never exceeds the trusted-mode value. Dropping the marginal therefore loses rate; it never gains any, and it can never turn an unsound declaration into a sound-looking one.

(b) At $d_A = 1$ the two modes are the same declaration (Proposition 5.2). This is the *embedding*: a protocol with no trusted source may be declared at $d_A = 1$ by folding the system the sender measures into the sectors of Q and her outcome components into v, with the party partition naming which components she holds. The empty-marginal case is therefore the *safe* direction — it removes an assumption rather than adding one.

The equivalence is also the accumulation paper's own reading of its channel conditional entropy: the sentence immediately following [1, Eq. (1)] records that the definitions of earlier works are recovered "by setting the C register to be trivial and omitting the marginal constraint (equivalently, setting A to be trivial)". That sentence is about the source's Eq. (1), not about its Theorem 4.2a; the derivation of Proposition 5.2 is this module's own and does not rest on it. The corresponding check on the import side is Fact 2.1, locator note (vii).

(c) $\tau_A = 0$ *is inadmissible*, for the two reasons Remark "The two source modes" (transmitter/rem:source-mode) (c) gives; on this module's side Definition 1.2 (ii) refuses it outright, and were the normalization dropped, $\omega_A = 0$ would give $\operatorname{tr}\omega = 0$ and hence $\omega = 0$, so $\mathcal{W} = \varnothing$ in violation of Definition 1.2 (vii) and (5.4) would be an infimum over the empty set. The absent marginal is $\tau_A = \mathsf{none}$; the trivial marginal is $d_A = 1$ with $\tau_A = [1]$; $\tau_A = 0$ is neither.

Uses Def. 1.2 · *Assumes* A-SRC · newly derived · accepted

For a declaration that adds no constraints on the state (an empty constraint list), in either source mode, the value h_low of the finite convex program bounds the single-round rate from below. Every step of the reduction is an equality or lowers the bound, so the computable program is conservative.

Step 1 of the proof is where the block restriction is discharged. The compiler states, as a consumer obligation it does not discharge, that this module's minimisation over the sector-block-diagonal feasible set is sound (Remark "Restriction of the feasible set to block-diagonal states" (protocol-compiler/rem:block-restriction)); Step 1, the dephasing of an arbitrary feasible state by the *joint* pinching along the measured sectors and along Alice's declared sectors, is that statement's proof. It is proved under exactly the typing of Definition 1.1 (DX2), which types Alice's measurement slot, τ_A and every Θ_k against a declared factor. The restriction along $\mathbb{T}$ is licensed by this step, and Proposition 5.5 adds that it costs nothing. Corollary 5.6 extends the theorem to non-empty constraint lists.

Theorem 5.4 (Reduction of the rate to the convex program). Let $\mathfrak{D}$ satisfy Definition 1.2 with $\mathfrak{L} = \varnothing$, in either source mode, and let $\alpha \in (1, \infty)$. Then, with $h_\alpha^{\uparrow,\mathrm{QKD}} = h_\alpha^{\uparrow,\mathrm{QKD}}(\varnothing)$ the single-round rate of Definition 2.4 at the empty list,

$$h_\alpha^{\uparrow,\mathrm{QKD}} \;\geq\; \underline{h}_\alpha. \tag{5.6}$$

Uses Def. 1.1, Def. 1.2, Thm. 1.5, Def. 2.4, Lem. 2.6, Thm. 4.1, foundations/def:renyi · *Used by* Cor. 5.6, Rem. 5.7, Prop. 5.8, Rem. 5.9, Thm. 8.9, Thm. 10.7, Prop. 17.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.objective: objective_value · *Oracles* registry.py

Proof. Idea. Five steps take the bracket of (2.12) at $\mathfrak{L} = \varnothing$, the rate that Theorem 2.10 writes out as (2.17), at a state of its inner infimum to the objective at a point of $\mathcal{W}$: a joint dephasing that changes no output (Step 1), a purification (Step 2), the replacement of the up-arrow entropy by the down-arrow one (Step 3), the branch formula of Theorem 4.1 (Step 4) and

a rearrangement of the logarithms (Step 5). Every step of the proof is an equality or an inequality whose direction lowers the bound (conservative); no step uses $\mathfrak{L}$.

Fix $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and a feasible $\omega \in S_{=}(AQ\widetilde{E})$ with $\omega \models \mathfrak{D}$ ((2.11)); abbreviate $\Phi := \mathbf{\Phi}[\omega_{AQ}]$. We lower-bound the bracket of (2.12) at $\mathfrak{L} = \varnothing$ by $O_\alpha(\mathbf{q}, \tilde{\omega})$ for some $\tilde{\omega} \in \mathcal{W}$; taking infima then yields (5.6). The divergence term of (2.12) is already the KL divergence $\frac{\alpha}{\alpha-1} D(\mathbf{q}\|\Phi)$ of the import, so no divergence relaxation occurs anywhere in this chain.

Step 1 (joint sector dephasing; exact). Let $\mathcal{P}_{\mathbb{T}}(X) := \sum_{t\in\mathbb{T}} \Pi_t^A X \Pi_t^A$ be the pinching along Alice's declared sectors (Definition 1.1 (DX1); the identity map when $|\mathbb{T}| = 1$), $\mathcal{P}_{\mathbb{S}}$ the pinching along $\mathbb{S}$, for a product sector map the tensor product of the measured factors' pinchings, and $\omega' := (\mathcal{P}_{\mathbb{T}} \otimes \mathcal{P}_{\mathbb{S}} \otimes \mathrm{id}_{\widetilde{E}})(\omega)$. Then ω' is feasible: both pinchings are CPTP, so ω' is a state, and in trusted-source mode $\omega'_A = \mathcal{P}_{\mathbb{T}}(\omega_A) = \mathcal{P}_{\mathbb{T}}(\tau_A) = \tau_A$ because τ_A is block diagonal along $\mathbb{T}$ (Definition 1.2 (iii)). Its AQ-marginal is block diagonal along the joint sectors, and $\nu^{\omega'} = \nu^{\omega}$: by (1.17) the output depends on ω only through the operators $\mathrm{tr}_{AQ}[(M_x \otimes F_v \otimes \mathbb{1}_{\widetilde{E}})\,\omega]$, and $\mathrm{tr}_{AQ}[(M_x \otimes F_v \otimes \mathbb{1}_{\widetilde{E}})\,\omega'] = \mathrm{tr}_{AQ}[(\mathcal{P}_{\mathbb{T}}(M_x) \otimes \mathcal{P}_{\mathbb{S}}(F_v) \otimes \mathbb{1}_{\widetilde{E}})\,\omega] = \mathrm{tr}_{AQ}[(M_x \otimes F_v \otimes \mathbb{1}_{\widetilde{E}})\,\omega]$, because both pinchings are self-adjoint for the trace pairing and act on the traced systems, every F_v is block diagonal along $\mathbb{S}$ and every M_x along $\mathbb{T}$ (Definition 1.2 (iii)). *The step is an equality*: ω' is another feasible point with literally the same output state, Eve's register included; no dilation of a pinching is handed to her here, and the purification is Step 2. Hence every entropy term and Φ are unchanged. (If $\mathfrak{L} \neq \varnothing$, every entry still holds at ω'; this is Corollary 5.6.)

Step 2 (purification; conservative). Let $|\hat{\omega}\rangle \in \mathcal{H}(AQ\widetilde{E}\widetilde{E}_2)$ purify ω', with $\widetilde{E}_2$ of dimension $\dim \mathcal{H}(AQ\widetilde{E})$. Enlarging the purifying register is within the scope of the import ([1, footnote 17]). For each $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ with $q(\hat{c}) > 0$: if $P_{\hat{c}}(\omega') = 0$ then $\mathbf{q} \not\ll \Phi$ and the divergence term of the bracket is already $+\infty$, so the claimed lower bound holds trivially; assume $P_{\hat{c}} > 0$. Since $\mathrm{tr}_{\widetilde{E}_2}$ maps $\nu^{\hat{\omega}}_{|\hat{c}}$ to $\nu^{\omega'}_{|\hat{c}}$ (conditioning on the classical $\widehat{C}$ commutes with the partial trace), Lemma 2.6 gives

$$\widetilde{H}^{\uparrow}_{\alpha}(Z|I\widetilde{E})_{\nu^{\omega'}_{|\hat{c}}} \;\geq\; \widetilde{H}^{\uparrow}_{\alpha}(Z|I\widetilde{E}\widetilde{E}_2)_{\nu^{\hat{\omega}}_{|\hat{c}}}.$$

Step 3 (arrow replacement; conservative). By the arrow order of Definition "Sandwiched Rényi divergence and conditional entropies" (foundations/def:renyi), $\widetilde{H}^{\uparrow}_{\alpha}(Z|I\widetilde{E}\widetilde{E}_2)_{\nu^{\hat{\omega}}_{|\hat{c}}} \geq \widetilde{H}^{\downarrow}_{\alpha}(Z|I\widetilde{E}\widetilde{E}_2)_{\nu^{\hat{\omega}}_{|\hat{c}}}$.

Step 4 (branch formula; exact). $\hat{\omega}$ is pure with $\hat{\omega}_{AQ} = \omega'_{AQ} =: \tilde{\omega}$ block diagonal along the joint sectors, and $\tilde{\omega} \in \mathcal{W}$: normalization is preserved by every step ($\mathrm{tr}\,\tilde{\omega} = \mathrm{tr}\,\omega = 1$), and in trusted-source mode additionally $\tilde{\omega}_A = \omega_A = \tau_A$, none of Steps 1–3 having touched the AQ-marginal except through the joint pinching of Step 1, which fixes τ_A. Theorem 4.1, applied

with purifying register $\widetilde{E}\widetilde{E}_2$, gives

$$\widetilde{H}^{\downarrow}_{\alpha}(Z|I\widetilde{E}\widetilde{E}_2)_{\nu^{\tilde{\omega}}_{|\hat{c}}} = \frac{1}{1-\alpha}\log_2\frac{K_{\hat{c}}(\tilde{\omega})}{P_{\hat{c}}(\tilde{\omega})} = \frac{1}{1-\alpha}\log_2\frac{K_{\hat{c}}(\tilde{\omega})}{P_{\hat{c}}},$$

using $P_{\hat{c}}(\tilde{\omega}) = P_{\hat{c}}(\omega') = P_{\hat{c}}$.

Step 5 (rearrangement; exact). Collecting Steps 1–4, the bracket of (2.12) is bounded from below by

$$\frac{\alpha}{\alpha-1}D(\mathbf{q}\|\Phi) + \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}}\frac{q(\hat{c})}{1-\alpha}\log_2\frac{K_{\hat{c}}(\tilde{\omega})}{P_{\hat{c}}}.$$

For each $\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}$ with $q(\hat{c})>0$ and $P_{\hat{c}}, K_{\hat{c}}>0$,

$$\begin{aligned}\frac{\alpha}{\alpha-1}\,q(\hat{c})\log_2\frac{q(\hat{c})}{P_{\hat{c}}} + \frac{q(\hat{c})}{1-\alpha}\log_2\frac{K_{\hat{c}}}{P_{\hat{c}}} & \\ = q(\hat{c})\log_2\frac{q(\hat{c})}{P_{\hat{c}}} + \frac{q(\hat{c})}{\alpha-1}\left[\log_2\frac{q(\hat{c})}{P_{\hat{c}}} - \log_2\frac{K_{\hat{c}}}{P_{\hat{c}}}\right] & \quad (5.7)\\ = q(\hat{c})\log_2\frac{q(\hat{c})}{P_{\hat{c}}} + \frac{q(\hat{c})}{\alpha-1}\log_2\frac{q(\hat{c})}{K_{\hat{c}}}, &\end{aligned}$$

an identity of real numbers. At the degenerate values the identity persists in the extended sense: $q(\hat{c})=0$ makes all terms 0; $P_{\hat{c}}=0$ with $q(\hat{c})>0$ makes both sides $+\infty$; $K_{\hat{c}}=0$ with $q(\hat{c})>0$ makes both sides $+\infty$ (using $K_{\hat{c}}\le P_{\hat{c}}$ from Theorem 4.1's *cone clause*, which is stated without the hypothesis $P_{\hat{c}}>0$ that the degenerate cases here violate, so the $+\infty$ appears on both sides through the K term on the right and through the combination on the left, where $\frac{1}{1-\alpha}\log_2\frac{K}{P}\to+\infty$). Summing (5.7) over $\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}$ and adding the $\hat{c}\notin\widehat{\mathcal{C}}_{\mathrm{key}}$ part of D yields exactly $O_\alpha(\mathbf{q},\tilde{\omega})\ge\underline{h}_\alpha$ after taking the infimum over $\mathcal{W}$ and then over $\mathcal{S}_{\mathrm{acc}}$. □

Pinching a feasible state along Alice's declared sectors keeps it feasible with the same statistics and an objective no larger. So restricting the variable to states block diagonal there changes no value, and the pinching carries minimisers to minimisers.

Step 1 of the proof of Theorem 5.4 shows that restricting the variable to states block diagonal along Alice's declared sectors is sound; the proposition below shows, independently, that the restriction also loses no value.

Proposition 5.5 (Value neutrality along Alice's sectors)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 in the general format of Definition 1.1, in either source mode and with an arbitrary list $\mathfrak{L}$, general and party-local entries included. Let $\mathcal{W}^\circ$ be the set defined as $\mathcal{W}$ is in Definition 5.1 but with block diagonality required along $\mathbb{S}$ only — the feasible set of the same declaration with the factor tagged A forgotten — and let $\mathcal{P}_{\mathbb{T}}(X) \coloneqq \sum_t \Pi_t^A X \Pi_t^A$. Then for every

$\alpha \in (1,\infty)$, every distribution $\mathbf{q}$ on $\widehat{\mathcal{C}}$ and every $\omega \in \mathcal{W}^\circ$, the operator $\omega' \coloneqq (\mathcal{P}_{\mathbb{T}} \otimes \mathrm{id}_Q)(\omega)$ satisfies

$$\omega' \in \mathcal{W}, \qquad P_{\hat{c}}(\omega') = P_{\hat{c}}(\omega), \qquad K_{\hat{c}}(\omega') \geq K_{\hat{c}}(\omega) \quad \forall \hat{c}, \qquad O_\alpha(\mathbf{q},\omega') \leq O_\alpha(\mathbf{q},\omega), \tag{5.8}$$

the last in the extended-real sense. Consequently $\inf_{\mathcal{S}_{\mathrm{acc}}\times\mathcal{W}} O_\alpha = \inf_{\mathcal{S}_{\mathrm{acc}}\times\mathcal{W}^\circ} O_\alpha$, and $\omega \mapsto \omega'$ maps minimisers over $\mathcal{W}^\circ$ to minimisers over $\mathcal{W}$.

Uses Def. 1.1, Def. 1.2, Def. 1.4, Def. 1.6, Lem. 1.7, Def. 2.2, Thm. 4.1, Lem. 4.3, Def. 5.1, foundations/def:pinching, foundations/lem:kernel · *Used by* Rem. 19.8 · newly derived · accepted

Proof. $\mathcal{P}_{\mathbb{T}}$ is a pinching (Definition "Operator conventions and pinching" (foundations/def:pinching)): completely positive, trace preserving, unital and self-adjoint for the trace pairing.

Membership. $\omega' \succeq 0$ and $\mathrm{tr}\,\omega' = \mathrm{tr}\,\omega = 1$. $\mathcal{P}_{\mathbb{T}} \otimes \mathrm{id}_Q$ commutes with $X \mapsto (\mathbb{1}_A \otimes \Pi_s) X (\mathbb{1}_A \otimes \Pi_{s'})$, so ω' is block diagonal along $\mathbb{S}$ as ω is, and along $\mathbb{T}$ by construction; hence along the joint sectors. In trusted-source mode $\omega'_A = \mathcal{P}_{\mathbb{T}}(\omega_A) = \mathcal{P}_{\mathbb{T}}(\tau_A) = \tau_A$, because τ_A is block diagonal along $\mathbb{T}$ (Definition 1.2 (iii)). For an entry read through its lifted operator, the partial trace over Q commutes with a map acting on A alone, so $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\omega'] = \mathcal{P}_{\mathbb{T}}\big(\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\omega]\big)$. In trusted-source mode $\mathcal{P}_{\mathbb{T}}$ is positive and linear, so $X \bowtie_k \Theta_k$ implies $\mathcal{P}_{\mathbb{T}}(X) \bowtie_k \mathcal{P}_{\mathbb{T}}(\Theta_k) = \Theta_k$, every Θ_k being block diagonal along $\mathbb{T}$: for a homogeneous entry $\Theta_k = c_k \tau_A$, for a general one by its type (Definition 1.1 (DX2)). In untrusted-source mode the entry value is the trace of the same operator, which $\mathcal{P}_{\mathbb{T}}$ preserves. For a party-local entry scoped to f, with $\hat{D} \coloneqq \hat{O}_k - c_k \mathbb{1}_{Q_f}$, the partial trace over Q_f commutes with a map acting on A alone, so $\mathrm{tr}_{Q_f}[(\mathbb{1} \otimes \hat{D})\omega'] = (\mathcal{P}_{\mathbb{T}} \otimes \mathrm{id})\big(\mathrm{tr}_{Q_f}[(\mathbb{1} \otimes \hat{D})\omega]\big)$ with the identity map on the other measured factors; a pinching tensored with an identity map is a pinching, hence positive, and $Y \bowtie_k 0$ implies $(\mathcal{P}_{\mathbb{T}} \otimes \mathrm{id})(Y) \bowtie_k 0$. So $\omega' \in \mathcal{W}$.

Linear quantities. For all (x,v), by self-adjointness of the pinching and block diagonality of M_x along $\mathbb{T}$ (Definition 1.2 (iii)), $\mathrm{tr}[(M_x \otimes F_v)\omega'] = \mathrm{tr}[(\mathcal{P}_{\mathbb{T}}(M_x) \otimes F_v)\omega] = \mathrm{tr}[(M_x \otimes F_v)\omega]$. Hence $\mathbf{\Phi}[\omega'] = \mathbf{\Phi}[\omega]$ by (1.16), so $P_{\hat{c}}$ is unchanged, and $\mathrm{tr}\,\mathcal{G}_{\hat{c}}(\omega') = \mathrm{tr}\,\mathcal{G}_{\hat{c}}(\omega)$, that trace being $\sum_{\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})} \sum_{(x,v)\in a^{-1}(\kappa)} \mathrm{tr}[(M_x \otimes F_v)\,\cdot\,]$ by the computation in the proof of Lemma 1.7 (i).

The non-linear term. With $m \coloneqq |\mathbb{T}|$, an enumeration $t \mapsto j_t \in \{1,\dots,m\}$ and $\zeta \coloneqq e^{2\pi i/m}$, the operators $U_k \coloneqq \sum_t \zeta^{j_t k} \Pi^A_t$, $k = 0,\dots,m-1$, are unitaries on A with $\mathcal{P}_{\mathbb{T}}(X) = \frac{1}{m}\sum_k U_k X U_k^\dagger$, by the unitary-mixture form of a pinching in Definition "Operator conventions and pinching" (foundations/def:pinching), with the projectors Π^A_t enumerated by j_t. Each U_k is a function of the projectors Π^A_t and therefore commutes with every operator block diagonal along $\mathbb{T}$, in particular with every $\sqrt{M_x}$. Let $\widetilde{U}_k$ be U_k on the A factor of the output space $\mathcal{H}(R\,XY\,C_A C_B\,AQ)$ of (1.18) and the identity

on the other factors. Every term of K_κ carries $\sqrt{M_x} \otimes \sqrt{F_v}$ on AQ, so $K_\kappa(U_k \otimes \mathbb{1}_Q) = \widetilde{U}_k K_\kappa$ and, $\mathcal{G}_{\hat{c}}$ being linear,

$$\mathcal{G}_{\hat{c}}(\omega') = \frac{1}{m} \sum_{k=0}^{m-1} \widetilde{U}_k \, \mathcal{G}_{\hat{c}}(\omega) \, \widetilde{U}_k^\dagger .$$

For a unitary U that is the identity on R and any $X \succeq 0$: $(UXU^\dagger)^{1/\alpha} = UX^{1/\alpha}U^\dagger$ by functional calculus; $\mathcal{Z}(UYU^\dagger) = U\mathcal{Z}(Y)U^\dagger$ because U commutes with every $Z_j = |j\rangle\langle j|_R$; and $\operatorname{tr}\big[(U\mathcal{Z}(Y)U^\dagger)^\alpha\big] = \operatorname{tr}[\mathcal{Z}(Y)^\alpha]$. So the functional T of (4.7) satisfies $T(UXU^\dagger) = T(X)$, and by its concavity on the whole cone (Lemma 4.3 (b))

$$T_{\hat{c}}(\omega') = T\big(\mathcal{G}_{\hat{c}}(\omega')\big) \geq \frac{1}{m} \sum_k T\big(\widetilde{U}_k \mathcal{G}_{\hat{c}}(\omega) \widetilde{U}_k^\dagger\big) = T_{\hat{c}}(\omega).$$

With (4.1), $K_{\hat{c}} = T_{\hat{c}} + P_{\hat{c}} - \operatorname{tr} \mathcal{G}_{\hat{c}}$, and the two linear terms unchanged, $K_{\hat{c}}(\omega') \geq K_{\hat{c}}(\omega)$.

The objective and the infima. In (5.3) the state enters through $P_{\hat{c}}$, unchanged, and through $\frac{1}{\alpha-1} d(q(\hat{c}), K_{\hat{c}})$ with d the kernel of Lemma "Extended divergence kernel" (foundations/lem:kernel), non-increasing in its second argument; the conventions at vanishing arguments are those of that kernel, so the inequality holds in the extended-real sense. Finally $\mathcal{W} \subseteq \mathcal{W}^\circ$, so the infimum over $\mathcal{W}$ is no smaller, and by (5.8) it is no larger; if $(\mathbf{q}, \omega)$ attains the infimum over $\mathcal{W}^\circ$ then $(\mathbf{q}, \omega')$ attains the same value over $\mathcal{W}$. □

For any declared constraint list, licensed or not, the constrained value bounds the constrained single-round rate from below, with the same list on both sides. A key length from the constrained value still needs the attack-set licence of Theorem 3.1.

No step of the reduction reads the constraint list beyond checking that the joint dephasing keeps every entry, so the comparison holds for any list. Whether the constrained value may enter a key length is a separate question, answered by the attack-set licence (Theorem 3.1).

Corollary 5.6 (The reduction for arbitrary constraint lists)**.** Let $\mathfrak{L}$ be arbitrary: Θ_k an arbitrary Hermitian operator on A in trusted-source mode, $\Theta_k = c_k$ an arbitrary real number in untrusted-source mode, and entries declared party-local in either mode (Definition 1.2 (vi)); no licensing restriction (Definition 2.2) is needed in either mode. Then

$$h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L}) \;\geq\; \underline{h}_\alpha(\mathfrak{L}), \tag{5.9}$$

where $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L})$ is the constrained single-round rate of Definition 2.4 and $\underline{h}_\alpha(\mathfrak{L})$ denotes (5.4) over the feasible set $\mathcal{W}$ of (5.1), which carries the same list ($\underline{h}_\alpha(\mathfrak{L})$ and the unadorned $\underline{h}_\alpha$ are the same quantity; the argument is written when the dependence on the declared list matters, as in $\underline{h}_\alpha(\mathfrak{L}) \geq \underline{h}_\alpha(\varnothing)$).

Uses Def. 1.2, Def. 2.2, Def. 2.4, Def. 5.1, Thm. 5.4 · *Used by* Rem. 5.9, Prop. 9.3, Thm. 10.7, Cor. 10.19, Prop. 17.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.objective: constraint_value · *Oracles* registry.py

Proof. Steps 1–5 of the proof of Theorem 5.4 apply verbatim to a feasible point of (2.12): Step 1 preserves every entry. For an entry read through its lifted operator, block diagonality of $\hat{O}_k$ along $\mathbb{S}$ and self-adjointness of $\mathcal{P}_{\mathbb{S}}$ give $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega'_{AQ}] = \mathcal{P}_{\mathbb{T}}\big(\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega_{AQ}]\big)$; in trusted-source mode $\mathcal{P}_{\mathbb{T}}$ is positive and linear and fixes Θ_k, which is block diagonal along $\mathbb{T}$ ($c_k\tau_A$ for a homogeneous entry, by type for a general one, Definition 1.2 (iii)), so $X \bowtie_k \Theta_k$ implies $\mathcal{P}_{\mathbb{T}}(X) \bowtie_k \Theta_k$; in untrusted-source mode the entry value is the trace of the same operator, which the pinching preserves. For a party-local entry scoped to f, with $\hat{D} = \hat{O}_k - c_k\mathbb{1}_{Q_f}$ block diagonal for the pinching $\mathcal{P}_f$ of that factor and $\mathcal{P}_{f^c}$ the product of the other factors' pinchings, $\mathrm{tr}_{Q_f}[(\mathbb{1}\otimes\hat{D})\,\omega'_{AQ}] = (\mathcal{P}_{\mathbb{T}}\otimes\mathcal{P}_{f^c})\big(\mathrm{tr}_{Q_f}[(\mathbb{1}\otimes\hat{D})\,\omega_{AQ}]\big)$, and a positive map preserves $\bowtie_k 0$; Steps 2–3 do not change the AQ-marginal, and the state $\tilde{\omega} = \omega'_{AQ}$ of Step 4 therefore satisfies every entry of $\mathfrak{L}$ in addition to $\mathrm{tr}\,\tilde{\omega} = 1$, block diagonality along the joint sectors and (in trusted-source mode) $\tilde{\omega}_A = \tau_A$, that is, $\tilde{\omega} \in \mathcal{W}$. Taking infima over the constrained sets on both sides yields (5.9). □

The divergence term passes through the reduction as an identity, purification loses nothing when the optimum is attained on pure states, and the arrow replacement's gap closes as $\alpha \to 1$. No step reverses direction.

Remark 5.7 (Where each step of the reduction loses). The divergence term is passed through as an identity: the imported rate (2.17) already carries the KL divergence D, so nothing is lost there. Step 2 of the proof of Theorem 5.4 loses nothing when the feasible optimum is attained on pure states; Step 3 is the standard down-arrow relaxation (Definition "Sandwiched Rényi divergence and conditional entropies" (foundations/def:renyi)), whose gap closes at $\alpha \to 1^+$ (both variants converge to the von Neumann conditional entropy). No step reverses direction. The bounds $0 \le K_{\hat{c}} \le P_{\hat{c}}$ that Step 4 also supplies are read at points that are not pure, which is why Theorem 4.1 states them as its *cone clause*.

Uses Thm. 2.10, Thm. 4.1, Thm. 5.4, foundations/def:renyi · newly derived · accepted

The objective is non-negative on the feasible set, so h_low lies in [0, +∞]; each entropy term is at most log₂(d_Z + 1), while the divergence penalty has no finite bound. If h_low = +∞, no accepted distribution is absolutely continuous with respect to the statistics of a feasible state.

The objective is a divergence penalty plus one entropy term per key-live acceptance value (Definition 5.1). The two parts behave differently: every entropy term is at most $\log_2(d_Z + 1)$, while the penalty is not bounded at all.

Proposition 5.8 (Range of the value and its finite entropy part)**.** Let $\alpha \in (1, \infty)$.

(i) *(lower bound)* $O_\alpha(\mathbf{q},\omega) \geq 0$ for every distribution $\mathbf{q}$ on $\widehat{\mathcal{C}}$ and every $\omega \in \mathcal{W}$; hence $\underline{h}_\alpha \in [0,+\infty]$.

(ii) *(entropy-part ceiling)* For every $\omega \in \mathcal{W}$ and every $\hat{c} \in \widehat{\mathcal{C}}_{\text{key}}$ with $P_{\hat{c}}(\omega) > 0$, the entropy term $e_{\hat{c}}(\omega) \coloneqq \frac{1}{1-\alpha}\log_2 \frac{K_{\hat{c}}(\omega)}{P_{\hat{c}}(\omega)}$ satisfies $e_{\hat{c}}(\omega) \leq \log_2 |\mathcal{Z}_\perp| = \log_2(d_Z+1)$.

(iii) *(no overall ceiling)* No finite a-priori upper bound on $\underline{h}_\alpha$ exists: the divergence penalty is unbounded already at fixed α.

(iv) *(infinite value)* $\underline{h}_\alpha = +\infty$ whenever no $(\mathbf{q},\omega)$ with $\mathbf{q} \ll \mathbf{\Phi}[\omega]$ is feasible. If $\underline{h}_\alpha = +\infty$, then no $\mathbf{q} \in \mathcal{S}_{\text{acc}}$ is absolutely continuous with respect to $\mathbf{\Phi}[\omega]$ for any $\omega \in \mathcal{W}$.

Uses Lem. 2.6, Thm. 4.1, Def. 5.1, Thm. 5.4, foundations/def:kl, foundations/def:renyi · *Used by* Rem. 5.9, Rem. 6.10, Prop. 9.5, Cor. 10.18, Prop. 11.1, Rem. 11.2, Rem. 11.3, Rem. 11.4, Req. 16.1, Req. 16.2, Prop. 17.2, Lem. 18.2, Thm. 19.1, Rem. 23.7, Rem. 23.9, Req. 24.4 · newly derived · accepted

Proof. *(i)* By Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl) and (5.3)–(5.4), $O_\alpha(\mathbf{q},\omega) = \frac{\alpha}{\alpha-1}D(\mathbf{q}\|\mathbf{\Phi}[\omega]) + \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\text{key}}} q(\hat{c})\, e_{\hat{c}}(\omega)$ with $e_{\hat{c}} \coloneqq \frac{1}{1-\alpha}\log_2 \frac{K_{\hat{c}}}{P_{\hat{c}}}$; the KL term is non-negative by Gibbs' inequality of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl), $\mathbf{q}$ and $\mathbf{\Phi}[\omega]$ being probability vectors, and $e_{\hat{c}} \geq 0$ since $K_{\hat{c}} \leq P_{\hat{c}}$ (Theorem 4.1, *cone clause*: a point of $\mathcal{W}$ is not pure, so the identity (4.3) is not what is read here) and $\frac{1}{1-\alpha} < 0$. Both parts are non-negative at every $(\mathbf{q},\omega)$, so $O_\alpha(\mathbf{q},\omega) \geq 0$ there and $\underline{h}_\alpha \geq 0$.

(ii) (If $P_{\hat{c}} = 0$ and $q(\hat{c}) > 0$, the KL term is already $+\infty$.) Let $P_{\hat{c}}(\omega) > 0$. Purifying ω as in Theorem 5.4, by Theorem 4.1 $e_{\hat{c}}$ equals $\widetilde{H}^{\downarrow}_\alpha(Z|I\widetilde{E}\widetilde{E}_2)$ of a normalized branch state, and by the arrow order $\widetilde{H}^{\downarrow}_\alpha \leq \widetilde{H}^{\uparrow}_\alpha$ of Definition "Sandwiched Rényi divergence and conditional entropies" (foundations/def:renyi) and Lemma 2.6 (dropping all conditioning) it is at most the Rényi entropy $H_\alpha(Z) = \frac{1}{1-\alpha}\log_2 \sum_z p(z)^\alpha \leq \log_2 |\mathcal{Z}_\perp| = \log_2(d_Z+1)$, using that $\sum_z p(z)^\alpha \geq |S|^{1-\alpha}$ for $\alpha > 1$ and $S = \operatorname{supp} p$ (convexity of $t \mapsto t^\alpha$: $\frac{1}{|S|}\sum_{z\in S} p(z)^\alpha \geq \left(\frac{1}{|S|}\right)^\alpha$), so that $\frac{1}{1-\alpha}\log_2 \sum_z p(z)^\alpha \leq \log_2 |S|$.

(iii) Take $d_A = d_Q = 1$ (single Alice outcome, trivial announcement), $\mathcal{V} = \{0,1\}$ with $F_0 = \epsilon$, $F_1 = 1-\epsilon$ for $\epsilon \in (0,\frac{1}{2})$ and $a_B = \mathrm{id}$, $\phi(\kappa) = c_B$, all-fail sifting ($\varsigma \equiv$ fail, $r \equiv \perp$, so $\widehat{\mathcal{C}}_{\text{key}} = \varnothing$), and a singleton acceptance set $\mathcal{S}_{\text{acc}} = \{(q_0, 1-q_0)\}$ with $q_0 \in (0,1]$ fixed. Then $\mathbf{\Phi} \equiv (\epsilon, 1-\epsilon)$ and the objective is the constant $\frac{\alpha}{\alpha-1}D\big((q_0,1-q_0)\|(\epsilon,1-\epsilon)\big) \geq \frac{\alpha}{\alpha-1}\big(q_0\log_2\frac{q_0}{\epsilon} + (1-q_0)\log_2(1-q_0)\big) \to +\infty$ as $\epsilon \to 0^+$, at every fixed $\alpha \in (1,\infty)$.

(iv) If no $(\mathbf{q},\omega) \in \mathcal{S}_{\text{acc}} \times \mathcal{W}$ has $\mathbf{q} \ll \mathbf{\Phi}[\omega]$, the KL term is $+\infty$ at every such pair (the convention $u\log_2\frac{u}{0} = +\infty$ for $u > 0$ of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl)) and the entropy terms are non-negative by (i), so $\underline{h}_\alpha = +\infty$. Conversely, if some

$(\mathbf{q},\omega) \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}$ has $\mathbf{q} \ll \mathbf{\Phi}[\omega]$, its KL term is finite, and every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ with $q(\hat{c}) > 0$ has $P_{\hat{c}}(\omega) > 0$ and an entropy term in $[0, \log_2 |\mathcal{Z}_\perp|]$ by (i) and (ii), so $O_\alpha(\mathbf{q},\omega) < +\infty$ and $\underline{h}_\alpha < +\infty$. Hence $\underline{h}_\alpha = +\infty$ means that no $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ is absolutely continuous with respect to $\mathbf{\Phi}[\omega]$ for any $\omega \in \mathcal{W}$. □

An infinite value of the program is the infeasibility of the declared acceptance set against the declaration: no attack the declared list admits is then accepted and no key is produced. The entropy part of the objective is capped, the divergence penalty is not.

Remark 5.9 (Reading an infinite value). An infinite value is not a large rate. By Proposition 5.8 (iv), $\underline{h}_\alpha = +\infty$ is the infeasibility of the declared acceptance set against the declaration: no $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ is absolutely continuous with respect to $\mathbf{\Phi}[\omega]$ for any $\omega \in \mathcal{W}$. For the attacks the declared list admits this means $\Pr[\Omega_{\mathrm{AT}}] = 0$: for every attack sequence of Definition 2.3 when $\mathfrak{L} = \varnothing$, and for every $\mathfrak{L}$-admissible one when $\mathfrak{L}$ is licensed for the declared source mode, the attacks Theorem 3.1 ranges over, the initial state being in either case that of Theorem 2.10. Indeed, let $\sigma_j \in S_=(AQ)$ be the unconditioned state the protocol channel acts on in round j, the AQ-marginal of the output of the round-j attack; in trusted-source mode the attack acts on the $A_j E_{j-1}$-marginal of the state before round j, whose A_j-marginal is τ_A, the earlier rounds acting on other copies of A. By Lemma 2.8 (i), σ_j satisfies the declaration's marginal condition and every entry of $\mathfrak{L}$, and by Theorem 1.5 (iii) the acceptance value of round j is distributed as $\mathbf{\Phi}[\sigma_j]$. The average $\bar{\sigma} := \frac{1}{n}\sum_j \sigma_j$ satisfies the same conditions, each of them being convex, and $\mathbf{\Phi}[\bar{\sigma}] = \frac{1}{n}\sum_j \mathbf{\Phi}[\sigma_j]$. Pinching $\bar{\sigma}$ along the joint sectors, as in Step 1 of the proof of Theorem 5.4, keeps its statistics and its marginal and, by the proof of Corollary 5.6, every entry of $\mathfrak{L}$, and gives a point $\bar{\omega} \in \mathcal{W}$ with $\mathbf{\Phi}[\bar{\omega}] = \mathbf{\Phi}[\bar{\sigma}]$. If $\Pr[\Omega_{\mathrm{AT}}] > 0$, some sequence $\hat{c}_1^n$ of positive probability has its frequency vector $\mathbf{f}$ in $\mathcal{S}_{\mathrm{acc}}$; every $\hat{c}$ with $f(\hat{c}) > 0$ is the value of some round j of it, so $\mathbf{\Phi}[\sigma_j](\hat{c}) > 0$ and $\mathbf{\Phi}[\bar{\omega}](\hat{c}) \geq \frac{1}{n}\mathbf{\Phi}[\sigma_j](\hat{c}) > 0$. Then $\mathbf{f} \ll \mathbf{\Phi}[\bar{\omega}]$ with $\mathbf{f} \in \mathcal{S}_{\mathrm{acc}}$ and $\bar{\omega} \in \mathcal{W}$, which Proposition 5.8 (iv) rules out. So the hypothesis $\Pr[\Omega_{\mathrm{AT}}] > 0$ of Theorem 2.10, respectively Theorem 3.1, fails, and the accepted state is null: there is no run on which a key would be produced.

For a nonempty list the conclusion reaches no further than the $\mathfrak{L}$-admissible attacks. Take $d_A = 1$, $Q = \mathbb{C}^2$ with one sector, $F_v = |v\rangle\langle v|$ for $v \in \{0,1\}$, the acceptance value the outcome v, $\mathcal{S}_{\mathrm{acc}} = \{(\frac{1}{2},\frac{1}{2})\}$ and for $\mathfrak{L}$ the one homogeneous entry $\mathrm{tr}[(\mathbb{1}_A \otimes |1\rangle\langle 1|)\,\omega] = 0$. Then $\mathcal{W} = \{|0\rangle\langle 0|\}$ and $\underline{h}_\alpha = +\infty$, while the attack that prepares $|+\rangle$ in every round, which is not $\mathfrak{L}$-admissible, is accepted with positive probability at every even n.

The two parts of the objective differ here: the entropy part is capped, each term at $\log_2(d_Z + 1)$ (Proposition 5.8 (ii)), while the divergence penalty is not (Proposition 5.8 (iii)). What the engine does with an infinite value is Requirement 16.1.

Uses Thm. 1.5, Def. 2.3, Lem. 2.8, Thm. 2.10, Thm. 3.1, Thm. 5.4, Cor. 5.6, Prop. 5.8 · newly derived · accepted

The objective is jointly convex and lower semicontinuous in the acceptance distribution and the feasible state, so h_low is a convex minimisation over a compact convex set. A finite infimum is attained.

The objective is a positive combination of the divergence kernel of Lemma "Extended divergence kernel" (foundations/lem:kernel), evaluated at the affine probabilities $P_{\hat{c}}$ and at the concave quantities $K_{\hat{c}}$ (Corollary 4.2); that is what makes the program convex.

Theorem 5.10 (Joint convexity of the program)**.** For every $\alpha \in (1, \infty)$ and every declaration satisfying Definition 1.2 (any $\mathfrak{L}$), the extended-real objective O_α of (5.3) is jointly convex and lower semicontinuous on $\Delta(\widehat{\mathcal{C}}) \times \mathcal{W}$, where $\Delta(\widehat{\mathcal{C}})$ is the probability simplex. Consequently (5.4) is a convex minimization over the compact convex set $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}$, and the infimum is attained whenever it is finite.

Uses Def. 1.2, Thm. 1.5, Def. 2.5, Thm. 4.1, Cor. 4.2, Def. 5.1, foundations/lem:kernel · *Used by* Thm. 7.5, Def. 7.10, Rem. 10.20, Prop. 17.2, Thm. 19.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.objective:objective_from_branches · *Oracles* registry.py

Proof. Write O_α through the kernel d of Lemma "Extended divergence kernel" (foundations/lem:kernel):

$$O_\alpha(\mathbf{q}, \omega) = \frac{\alpha}{\alpha - 1} \sum_{\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{key}}} d\big(q(\hat{c}), P_{\hat{c}}(\omega)\big) + \sum_{\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}} \Big[d\big(q(\hat{c}), P_{\hat{c}}(\omega)\big) + \frac{1}{\alpha - 1}\, d\big(q(\hat{c}), K_{\hat{c}}(\omega)\big) \Big].$$

All prefactors are positive for $\alpha > 1$. Each $P_{\hat{c}}$ is affine in ω (Theorem 1.5 (iii)), so $(\mathbf{q}, \omega) \mapsto d(q(\hat{c}), P_{\hat{c}}(\omega))$ is jointly convex and lsc as the composition of the jointly convex lsc d with an affine map. For the K-terms: $K_{\hat{c}}$ is concave, non-negative (Theorem 4.1, *cone clause* — a point of $\mathcal{W}$ is not pure, and $\mathcal{W} \subseteq \mathrm{Pos}(AQ)$ is all that clause asks), and continuous on $\mathcal{W}$ (Corollary 4.2; continuity holds because $K_{\hat{c}}$ is a composition of continuous maps on a finite-dimensional compact domain). For feasible pairs $(\mathbf{q}_i, \omega_i)$ and $\lambda \in (0, 1)$, concavity of $K_{\hat{c}}$ and monotonicity of d in its second argument give

$$d\big(q_\lambda(\hat{c}), K_{\hat{c}}(\omega_\lambda)\big) \le d\big(q_\lambda(\hat{c}), \lambda K_{\hat{c}}(\omega_1) + (1-\lambda) K_{\hat{c}}(\omega_2)\big) \le \lambda\, d\big(q_1(\hat{c}), K_{\hat{c}}(\omega_1)\big) + (1-\lambda)\, d\big(q_2(\hat{c}), K_{\hat{c}}(\omega_2)\big),$$

the second step by joint convexity of d. Lower semicontinuity of the K-terms follows from lsc of d, continuity of $K_{\hat{c}}$, and monotonicity of d in v. A finite non-negative combination of jointly convex lsc functions is jointly convex and lsc. $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}$ is compact ($\mathcal{S}_{\mathrm{acc}}$ closed in the simplex by Definition 2.5; $\mathcal{W}$ closed and bounded); an lsc function on a compact set attains its infimum when finite. □

6 Sector classes and sector removal

Sectors are classified by exact predicates on the declared data as key-active, constraint-active, state-blind and removable, removability being certified, for a list without party-local entries, by an exhibited witness that meets a six-condition checklist (Definition 6.1). A state-blind sector may be deleted when a mixture of states on distinct other key-inert sectors reproduces its responses and constraint values (Theorem 6.2), and a set of such sectors when every witness targets only sectors that stay (Proposition 6.5). Dropping a constraint-active sector (Proposition 6.3) or averaging sectors (Proposition 6.4) can overstate the key. The remarks show why the index conditions of the checklist are needed, why the witness must be exact, and that a tolerance may only refuse a removal.

Exact predicates on the declared data make a sector key-active, constraint-active, state-blind or removable; for a list without party-local entries, removability is certified by a witness meeting a six-condition checklist. Deleting several removable sectors needs a further condition on the set.

Two operations on the declared sectors cost nothing: evaluating the objective over the sectors that carry a deficit, and deleting a sector whose weight other key-inert sectors can take over. The classes below name the sectors each operation applies to, and the checklist says when a sector may be deleted. Neither half of (W2) is redundant (Remark 6.6); why (W3) and (W4) must hold exactly, and why (W4) names one form, is Remark 6.7; and Remark 6.9 exhibits two removable sectors whose joint deletion overstates the value.

Definition 6.1 (Sector classes and the removal checklist)**.** Let $\mathfrak{D}$ satisfy Definition 1.2. For $s \in \mathbb{S}$:

(a) s is *key-active* if $(\hat{c}, s)$ fails to be key-inert (Definition 4.5) for at least one $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$; these sectors carry the deficits $f_{\hat{c},s}$ in the objective.

(b) s is *constraint-active* if it is key-inert for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$; by Proposition 4.6 it contributes nothing to any $K_{\hat{c}}$ but remains an optimization variable (it enters $\mathbf{\Phi}$, every $P_{\hat{c}}$, the marginal constraint, and $\mathfrak{L}$).

(c) s is *state-blind* if $F_v^{(s)} = c_{v,s}\,\mathbb{1}_{Q_s}$ for scalars $c_{v,s} \geq 0$, for every $v \in \mathcal{V}$ (then $\sum_v c_{v,s} = 1$ by POVM completeness restricted to the sector); if $\mathfrak{L} \neq \varnothing$ we additionally require each $\hat{O}_k^{(s)} = o_{k,s}\,\mathbb{1}_{Q_s}$ for scalars $o_{k,s}$.

(d) s is *removable* if the hypotheses of Theorem 6.2 hold for it.

All four predicates are decided by finitely many linear-algebra checks on the declaration. Items (a) and (b) reduce to key-inertness (Definition 4.5), a finite combinatorial predicate on the declared function tables and on which of the M_x, $F_v^{(s)}$ are the zero operator, decided exactly, at no window. Items

(c) and (d) are decided exactly as well: (c) is a condition on the declared values themselves (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64)), so that a block that is only *nearly* scalar fails it and s is then not state-blind, and (d) is decided by the checklist below.

The removal checklist. A *candidate witness* for a sector s^* is a target list $s_1, \dots, s_L$ as index data, states χ_i as binary64 sector-block data, weights μ_i as binary64 data, and the evaluation order of (W6). It is a *witness* for s^* in the sense of (d) when $\mathfrak{L}$ carries no entry declared party-local and all six of the following hold:

(W1) s^* *is state-blind* in the sense of (c), the $\hat{O}_k^{(s^*)}$ half included whenever $\mathfrak{L} \neq \varnothing$;

(W2) *the witness does not name s^*, and names no sector twice*: $L \geq 1$, $s_i \in \mathbb{S} \setminus \{s^*\}$ for every i, and the s_i pairwise distinct, all compared as index data at no window. A candidate naming s^* fails (W2), and so does one that repeats a target;

(W3) *weights*: $\mu_i \geq 0$ for every i — a comparison of an exhibited binary64 datum against zero — and $\sum_i \mu_i = 1$, the sum formed in the evaluation order the witness exhibits and verified to round nowhere, exactly as (W6) requires of its residuals, so that the computed sum *is* the real sum;

(W4) *states*: $\chi_i \succeq 0$ and $\operatorname{tr} \chi_i = 1$ for every i, decided on the *exhibited form* and never on a computed spectrum: the witness presents each χ_i diagonal in the declared basis of Q_{s_i} — every off-diagonal entry the datum 0 in the sense of (c), every diagonal entry a datum ≥ 0, and their sum formed in the exhibited order, rounding nowhere, and equal to the datum 1 — so that both conditions hold of the declared numbers by construction. Any other presentation fails (W4), which can only *refuse* a removal;

(W5) *key-inert targets*, which is (RM2): every pair $(\hat{c}, s_i)$ with $\hat{c} \in \widehat{\mathcal{C}}_{\text{key}}$ and $i \in \{1, \dots, L\}$ is key-inert (Definition 4.5);

(W6) *both equality families close at residual exactly zero*: (RM1) for every $v \in \mathcal{V}$, and (RM3) for every k (vacuous when $\mathfrak{L} = \varnothing$).

The condition on $\mathfrak{L}$ and (W1)–(W6) are the hypotheses of Theorem 6.2 transcribed, all of them and in the theorem's own words — the condition on $\mathfrak{L}$ from its first hypothesis, (W1) and (W2) from its opening sentence, (W3) and (W4) from its quantifiers over μ_i and χ_i, (W5) from (RM2), (W6) from (RM1) and (RM3) — which is what (d) means by "the hypotheses of Theorem 6.2 hold". Condition (W6) is read on the residual itself: each residual

is formed in an evaluation order that the witness exhibits alongside its data, every elementary operation of which rounds nothing, so that the computed residual *is* the real residual and its comparison against zero decides the real identity.

Deleting a set of removable sectors needs one condition more, and it is not part of this definition. Removability is a property of one sector, and Theorem 6.2 licenses the deletion of one sector; that two sectors are each removable does not license deleting both. Proposition 6.5 states the further hypothesis on the set $\mathbb{S}_{\mathrm{del}}$ a run deletes. Nothing in (W1)–(W6) implies that hypothesis.

In the general format of Definition 1.1 the four classes remain predicates of the sectors $s \in \mathbb{S}$ of Q, decided as above, with $\hat{O}_k^{(s)}$ the block of the lifted operator of an entry. The finer notion along Alice's sectors — which joint pairs $(\hat{c}, (t, s))$ carry a deficit — is Definition 4.8, and a sector that is key-active here may have key-inert joint pairs. Theorem 6.2 is stated for a list carrying no entry declared party-local, so for a list carrying one, clause (d) is unsatisfiable and no sector carries a witness in its sense.

Uses Def. 4.5, Prop. 4.6, Def. 5.1 · *Used by* Thm. 6.2, Prop. 6.5, Rem. 6.6, Rem. 6.7, Rem. 6.8, Rem. 6.9, Rem. 6.10, Req. 14.6, Req. 14.7, Req. 14.8, Req. 14.10 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.sectors:classify_sector · *Oracles* registry.py

A state-blind sector may be deleted when a mixture of states on distinct other sectors, all key-inert, reproduces its responses and its constraint values. Deleting it changes neither the value of the program nor its attainment.

A state-blind sector answers every state alike, so its weight can be moved onto other sectors that reproduce its answers. When those sectors carry no key entropy, the move removes one non-positive deficit and changes nothing else.

Theorem 6.2 (Sound removal of a state-blind sector)**.** Let $\mathfrak{L}$ carry no entry declared party-local, let $s^* \in \mathbb{S}$ be state-blind, and suppose there exist *pairwise distinct* sectors $s_1, \dots, s_L \in \mathbb{S} \setminus \{s^*\}$, states $\chi_i \in S_=(Q_{s_i})$, and weights $\mu_i \geq 0$ with $\sum_{i=1}^{L} \mu_i = 1$, such that

(RM1) *response match:* $c_{v,s^*} = \sum_i \mu_i \operatorname{tr}\left[F_v^{(s_i)} \chi_i\right]$ for every $v \in \mathcal{V}$;

(RM2) *key-inert targets:* every pair $(\hat{c}, s_i)$ with $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ is key-inert;

(RM3) *constraint match* (vacuous for $\mathfrak{L} = \varnothing$): $o_{k,s^*} = \sum_i \mu_i \operatorname{tr}\left[\hat{O}_k^{(s_i)} \chi_i\right]$ for every k.

Then restricting the infimum (5.4) to $\{\omega \in \mathcal{W} : \omega_{s^*} = 0\}$ changes neither its value nor its attainment: the sector s^* may be deleted from the variable list, and its rows deleted from the generated maps, without loss.

Uses Def. 1.4, Cor. 4.4, Prop. 4.6, Def. 5.1, Def. 6.1 · *Used by* Prop. 6.4, Prop. 6.5, Rem. 6.6, Rem. 6.7, Rem. 6.8, Rem. 6.9, Rem. 6.10, Rem. 10.20, Req. 14.8 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.sectors:verify_removal_witness · *Oracles* registry.py

Proof. Given $\omega \in \mathcal{W}$ and $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$, let $m \coloneqq \mathrm{tr}_{Q_{s^*}} \omega_{s^*} \in \mathrm{Pos}(A)$ and define $\hat{\omega}$ by

$$\hat{\omega}_{s^*} \coloneqq 0, \qquad \hat{\omega}_{s_i} \coloneqq \omega_{s_i} + \mu_i\, m \otimes \chi_i, \qquad \hat{\omega}_s \coloneqq \omega_s \quad (s \notin \{s^*, s_1, \dots, s_L\}).$$

The three assignments are mutually consistent, which the two index conditions of the hypothesis ensure: no s_i equals s^*, so the first two do not collide, and the s_i are pairwise distinct, so the second assigns one value to each block it touches. *Feasibility.* Each block is PSD; sector-block-diagonality is manifest. With Alice sectors, $\omega_s = \bigoplus_t \omega^{(t,s)}$ is block diagonal along $\mathbb{T}$, hence so is $m = \bigoplus_t m_t$ with $m_t = \mathrm{tr}_{Q_{s^*}} \omega^{(t,s^*)}$, and so is every $m \otimes \chi_i$: the modified point is block diagonal along the joint sectors, and the move takes place inside each Alice sector separately. *Normalization* — the condition that holds in both source modes, and the only one in untrusted-source mode: $\mathrm{tr}\,\hat{\omega} = \mathrm{tr}\,\omega - \mathrm{tr}\,m + \sum_i \mu_i\, \mathrm{tr}\,m\, \mathrm{tr}\,\chi_i = \mathrm{tr}\,\omega = 1$, using $\mathrm{tr}\,\chi_i = 1$, $\sum_i \mu_i = 1$. *Alice's marginal*, in trusted-source mode only: the same computation at the level of the partial trace, $\mathrm{tr}_Q\,\hat{\omega} = \mathrm{tr}_Q\,\omega - m + \sum_i \mu_i\, m\, \mathrm{tr}\,\chi_i = \mathrm{tr}_Q\,\omega = \tau_A$; the normalization statement is its trace, so replacing the marginal by normalization is exactly the step that carries this theorem into untrusted-source mode. *Constraints.* The s^* contribution to $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\omega]$ is $\mathrm{tr}_{Q_{s^*}}[(\mathbb{1}_A \otimes o_{k,s^*}\mathbb{1})\,\omega_{s^*}] = o_{k,s^*} m$, and the added contribution is $\sum_i \mu_i\, \mathrm{tr}\left[\hat{O}_k^{(s_i)} \chi_i\right] m = o_{k,s^*} m$ by (RM3); all other sectors unchanged. In untrusted-source mode the constraint value is the trace of that same operator, so the identity gives $o_{k,s^*}\,\mathrm{tr}\,m$ on both sides. *Statistics.* For every (x, v), the s^* term of (1.16) is $\mathrm{tr}[(M_x \otimes c_{v,s^*}\mathbb{1})\,\omega_{s^*}] = c_{v,s^*}\,\mathrm{tr}[M_x m]$, and the added terms give $\sum_i \mu_i\, \mathrm{tr}[M_x m]\, \mathrm{tr}\left[F_v^{(s_i)} \chi_i\right] = c_{v,s^*}\,\mathrm{tr}[M_x m]$ by (RM1). Hence $\mathbf{\Phi}[\hat{\omega}] = \mathbf{\Phi}[\omega]$, resolved per outcome pair, so every $P_{\hat{c}}$ is unchanged. *Objective.* The divergence terms depend on $(\mathbf{q}, \mathbf{\Phi}[\cdot])$ only, hence are unchanged. For $K_{\hat{c}}$: by (RM2) and Proposition 4.6, $f_{\hat{c},s_i} \equiv 0$ on *every* input, before and after the modification; $f_{\hat{c},s}$ for other retained s is unchanged; and the term $f_{\hat{c},s^*}(\omega_{s^*}) \le 0$ (Corollary 4.4) is deleted. Hence $K_{\hat{c}}(\hat{\omega}) \ge K_{\hat{c}}(\omega)$, and by monotonicity of the kernel d in its second argument, $O_\alpha(\mathbf{q}, \hat{\omega}) \le O_\alpha(\mathbf{q}, \omega)$. Therefore the restricted infimum is no larger than the unrestricted one; being a restriction it is also no smaller; and the map $\omega \mapsto \hat{\omega}$ carries minimizers to restricted minimizers. □

A constraint-active sector adds no entropy deficit yet shapes the statistics: in a two-sector example, forcing it to zero raises the value from 0 to 1 bit per round. Dropping such a sector from the variable list can therefore overstate the key.

A constraint-active sector contributes nothing to any $K_{\hat{c}}$, which might suggest deleting it along with its variable. Its weight, however, still enters the statistics, and an adversary may place it there.

Proposition 6.3 (Unsoundness of dropping a constraint-active sector)**.** There exists a declaration satisfying Definition 1.2 and a constraint-active sector s_0 such that forcing $\omega_{s_0} = 0$ strictly *increases* $\underline{h}_\alpha$. Dropping a constraint-active sector from the variable list is therefore unsound: it can overstate the extractable key.

Uses Def. 1.2, Def. 1.6, Prop. 5.2 · *Used by* Prop. 6.4, Rem. 6.6, Rem. 6.9, Rem. 6.10 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.sectors:build_sector_report · *Oracles* registry.py

Proof. Take $d_A = 1$ ($\tau_A = [1]$, equivalently untrusted-source mode by Proposition 5.2; single Alice outcome with $M = 1$, announcements trivial, one announcement class, $\varsigma \equiv$ pass, $\phi \equiv \hat{c}_0$, $\widehat{\mathcal{C}} = \widehat{\mathcal{C}}_{\mathrm{key}} = \{\hat{c}_0\}$, $\mathcal{S}_{\mathrm{acc}} = \{\delta_{\hat{c}_0}\}$), $\mathcal{Z} = \{0,1\}$, and $Q = Q_1 \oplus Q_2$ with both sectors one-dimensional. Declare $\mathcal{V} = \{0,1\}$, $r(v) = v$, and

$$F_0^{(1)} = F_1^{(1)} = \tfrac{1}{2}, \qquad F_0^{(2)} = 1, \quad F_1^{(2)} = 0.$$

Sector 2 is key-inert (its effective outcome set is $\{v = 0\}$, a singleton), hence constraint-active; sector 1 is key-active. A feasible state is $\omega = (w_1, w_2)$, $w_i \geq 0$, $w_1 + w_2 = 1$, and $P_{\hat{c}_0} = 1$. A direct computation of (1.19) for sector 1 gives, for weight w_1, the rank-one operator $\mathcal{G}_{\hat{c}_0,1}(w_1) = w_1\,|\eta\rangle\langle\eta|$ with $|\eta\rangle = \frac{1}{\sqrt{2}}\sum_v |v\rangle_R \otimes |x_0, v\rangle_{XY} \otimes |\kappa_0\rangle$, so $\mathcal{G}^{1/\alpha} = w_1^{1/\alpha}\,|\eta\rangle\langle\eta|$, $\mathcal{Z}(|\eta\rangle\langle\eta|) = \frac{1}{2}\sum_v |\xi_v\rangle\langle\xi_v|$ for orthonormal $|\xi_v\rangle$, and $T = w_1 \operatorname{tr}\big[(\frac{1}{2}\sum_v |\xi_v\rangle\langle\xi_v|)^\alpha\big] = w_1\, 2^{1-\alpha}$; hence $f_{\hat{c}_0,1}(w_1) = w_1(2^{1-\alpha} - 1)$ and $f_{\hat{c}_0,2} \equiv 0$, so $K = 1 + w_1(2^{1-\alpha} - 1)$ and $O_\alpha = \frac{1}{\alpha - 1}\log_2 \frac{1}{K}$. The infimum over $w_1 \in [0,1]$ is attained at $w_1 = 0$ with value 0; forcing $\omega_{s_0} = \omega_2 = 0$ (i.e. $w_1 = 1$) gives the strictly larger value $\frac{1}{\alpha-1}\log_2 2^{\alpha-1} = 1$. The value 0 is also the correct one: an adversary that always populates sector 2 makes the raw key deterministic. Hence the drop overstates the key by one bit per round. □

Replacing the sectors of a declaration by one sector with averaged measurement operators can raise the value even where the averaged declaration reproduces the statistics, in the example from 0 to 1 bit per round. Merging sectors is therefore no sound simplification.

Theorem 6.2 deletes a sector by moving its weight onto other sectors that reproduce its responses, and the sector index survives the move. Averaging

sectors into one erases that index, and the following proposition shows that this is not sound.

Proposition 6.4 (Unsoundness of averaging sectors)**.** Coarse-graining the sector decomposition of a declaration — replacing $Q = \bigoplus_s Q_s$ and $\{F_v^{(s)}\}$ by a single sector carrying averaged POVM elements — is unsound: there exists a declaration $\mathfrak{D}$ and an averaged declaration $\bar{\mathfrak{D}}$ reproducing the same outcome statistics on the averaged state space for which $\underline{h}_\alpha(\bar{\mathfrak{D}}) > \underline{h}_\alpha(\mathfrak{D})$.

Uses Thm. 6.2, Prop. 6.3 · *Used by* Rem. 6.10, Req. 14.9 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.sectors:delete_sectors · *Oracles* registry.py

Proof. Take the declaration of Proposition 6.3 with the two sectors replaced by projective responses: sector s responds deterministically with outcome $v = s - 1$, i.e.

$$F_0^{(1)} = 1, \quad F_1^{(1)} = 0, \qquad F_0^{(2)} = 0, \quad F_1^{(2)} = 1$$

(so $\sum_v F_v^{(s)} = 1 = \mathbb{1}_{Q_s}$ in each sector; all other data unchanged, and the raw key $r(v) = v$ takes the value $s-1$ on sector s). Every pair $(\hat{c}_0, s)$ is key-inert (each sector's effective outcome set is the singleton $\{v = s-1\}$), so $f \equiv 0$, $K = P = 1$, and $\underline{h}_\alpha(\mathfrak{D}) = 0$ — correctly: the sector index determines the raw key, and the purifying register knows the sector. Now form $\bar{\mathfrak{D}}$ on the one-dimensional $\bar{Q}$ with $\bar{F}_v := \frac{1}{2}(F_v^{(1)} + F_v^{(2)}) = \frac{1}{2}$ for both v (a valid POVM, and the statistics of $\bar{\mathfrak{D}}$ on its feasible state $\bar{\omega} = 1$ equal those of $\mathfrak{D}$ at the uniform sector weights). The computation of Proposition 6.3 with $w_1 = 1$ applies verbatim to $\bar{\mathfrak{D}}$ and gives $\underline{h}_\alpha(\bar{\mathfrak{D}}) = 1$. An engine that averaged the sectors would thus certify one secret bit per round for a protocol whose key the adversary can know exactly. The obstruction is structural: averaging erases the sector index from the state space while the adversary retains it, so the averaged feasible set no longer contains (an image of) every reachable attack. □

Several removable sectors may be deleted together when every deleted sector's witness targets only sectors that stay. The whole set may then be deleted without changing the value of the program or its attainment.

Theorem 6.2 deletes one sector and moves its weight onto the targets of its witness. Several deletions compose when no deletion removes a sector that another deleted sector's weight is moved onto.

Proposition 6.5 (Deleting a set of removable sectors)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 and let $\mathbb{S}_{\mathrm{del}} \subseteq \mathbb{S}$ be the set of sectors a run deletes. Suppose *every* $s \in \mathbb{S}_{\mathrm{del}}$ *carries a witness in the sense of Definition 6.1 (d) all of whose targets lie outside* $\mathbb{S}_{\mathrm{del}}$. Then restricting the infimum (5.4)

to $\{\omega \in \mathcal{W} : \omega_s = 0 \text{ for all } s \in \mathbb{S}_{\mathrm{del}}\}$ changes neither its value nor its attainment: the whole set $\mathbb{S}_{\mathrm{del}}$ may be deleted from the variable list, and its rows from the generated maps, without loss.

Uses Def. 1.2, Def. 4.5, Def. 5.1, Def. 6.1, Thm. 6.2 · *Used by* Rem. 6.8, Rem. 6.9, Rem. 6.10, Req. 14.6, Req. 14.9, Req. 14.10 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.sectors:delete_sectors

Proof. By induction on $|\mathbb{S}_{\mathrm{del}}|$. The case 0 is vacuous and the case 1 is Theorem 6.2. For the step, delete the members one at a time, in any order, and check at each step that the theorem applies to the *reduced* declaration $\mathfrak{D}'$ obtained by deleting the sectors removed so far. *The sector being deleted is still state-blind, and every key-inertness predicate is unchanged*: both are per-sector properties of that sector's own declared blocks $F_v^{(s)}$, $\hat{O}_k^{(s)}$ and of M_x, a, r, φ, ς (Definitions 4.5 and 6.1 (c)), none of which deleting *another* sector touches. *Its targets are still present*, never having been in $\mathbb{S}_{\mathrm{del}}$, so (W2) and (W5) read the same rows and (RM1), (RM3) the same numbers. *And $\mathfrak{D}'$ satisfies Definition 1.2*: (iii) because $\{F_v\}$ is sector-block-diagonal with $\sum_v F_v^{(s)} = \mathbb{1}_{Q_s}$ per surviving sector, so its restriction is a POVM on $\bigoplus_{s \notin \mathrm{deleted}} \mathcal{H}(Q_s)$, and likewise for $\{M_x\}$, which is untouched; (vi) because Θ_k and c_k are untouched and each $\hat{O}_k$ restricts to a sector-block-diagonal operator; (vii) because Theorem 6.2's own map $\omega \mapsto \hat{\omega}$ carries a point of $\mathcal{W}$ — nonempty by (vii) for the declaration before the step — to a feasible point with $\omega_{s^*} = 0$, whose restriction is a point of the reduced feasible set; and (i), (ii), (iv), (v), (viii) read no sector at all. So Theorem 6.2 applies at each step and leaves the infimum and its attainment unchanged. Finally the theorem's modified point adds mass only to the targets, hence only outside $\mathbb{S}_{\mathrm{del}}$, so no already deleted sector is repopulated by a later step, and the objective on the reduced declaration equals the restricted objective, the deleted sector contributing $f_{\hat{c},s^*}(0) = 0$ and $\mathbf{\Phi}_{s^*}[0] = 0$. □

Without (W2) every state-blind constraint-active sector witnesses its own removal, and deleting it can overstate the value by a bit per round. Distinct targets cost no removal in exact arithmetic: repeated targets merge into one witness certifying the same identities.

Condition (W2) of Definition 6.1 asks two things of a witness for s^*: that it does not name s^*, and that it names no sector twice. Both come from the construction in the proof of Theorem 6.2.

Remark 6.6 (The two index conditions of the removal checklist). *Without (W2), every state-blind constraint-active sector witnesses its own removal.* Let s^* be state-blind and key-inert for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ — that is, state-blind and constraint-active in the sense of Definition 6.1 (b) — and take the *self-witness* $L = 1$, $s_1 := s^*$, $\mu_1 := 1$, and χ_1 any state on Q_{s^*} in the form (W4) requires — a diagonal projector $|i\rangle\langle i|$ will do. Then (W1) holds by hypothesis, (W3) and (W4) hold by construction, (W5) holds because s^* is itself key-inert for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, and *both equality families close*

exactly: state-blindness gives $\operatorname{tr}\left[F_v^{(s^*)}\chi_1\right] = c_{v,s^*}\operatorname{tr}\chi_1 = c_{v,s^*}$ and $\operatorname{tr}\left[\hat{O}_k^{(s^*)}\chi_1\right] = o_{k,s^*}\operatorname{tr}\chi_1 = o_{k,s^*}$, and the evaluation is the product $c_{v,s^*}\cdot 1$, which rounds nothing, so each residual is the datum 0 and (W6) passes as well. Every condition but (W2) is met, and a checklist without (W2) would flag the sector removable and delete a constraint-active sector — the one operation Proposition 6.3 proves unsound. Its counterexample is of exactly this shape: its sector 2 has $F_0^{(2)} = 1$, $F_1^{(2)} = 0$ on a one-dimensional block, hence is state-blind, and is key-inert, hence constraint-active; deleting it *overstates* $\underline{h}_\alpha$ *by one bit per round.* What forbids the self-witness is the construction in the proof of Theorem 6.2: it sets $\hat{\omega}_{s^*} \coloneqq 0$ and $\hat{\omega}_{s_i} \coloneqq \omega_{s_i} + \mu_i\, m \otimes \chi_i$ *simultaneously*, and if some $s_i = s^*$ those two assignments contradict each other, so $\hat{\omega}$ is not defined and no step of the proof is available. The theorem quantifies over $s_1, \ldots, s_L \in \mathbb{S} \setminus \{s^*\}$ for that reason, and the first half of (W2) is that quantifier and nothing else.

Distinct targets, at no cost. With $s_i = s_j$ for $i \neq j$ the proof's second assignment $\hat{\omega}_{s_i} \coloneqq \omega_{s_i} + \mu_i\, m \otimes \chi_i$ would assign two values to one block, the same collision the self-witness produces at $s_i = s^*$, and $\hat{\omega}$ would again not be defined. The theorem therefore quantifies over pairwise distinct targets and (W2) is that quantifier as well. *No removal is lost by it.* A candidate witness with repeated targets *merges*: drop every term with $\mu_i = 0$, and replace the terms at a repeated target s by the single term $(\bar{\mu}_s, \bar{\chi}_s)$ with $\bar{\mu}_s \coloneqq \sum_{i:s_i=s}\mu_i$ and $\bar{\chi}_s \coloneqq \bar{\mu}_s^{-1}\sum_{i:s_i=s}\mu_i\chi_i$, defined because $\bar{\mu}_s > 0$ after the drop. Then $\bar{\mu}_s \geq 0$ and $\sum_s \bar{\mu}_s = 1$, so (W3) holds; $\bar{\chi}_s \succeq 0$ with $\operatorname{tr}\bar{\chi}_s = 1$, a convex combination of states, so (W4) holds; and $\bar{\mu}_s \operatorname{tr}\left[F_v^{(s)}\bar{\chi}_s\right] = \sum_{i:s_i=s}\mu_i \operatorname{tr}\left[F_v^{(s)}\chi_i\right]$ by linearity of $X \mapsto \operatorname{tr}\left[F_v^{(s)}X\right]$, and likewise for $\hat{O}_k^{(s)}$, so (RM1) and (RM3) take the same values and (W6)'s identities are the same identities. *In exact arithmetic the merged witness certifies exactly what the original would have.* The merged witness and its evaluation order must still be *exhibited*, because (W3), (W4) and (W6) are conditions on the exhibited numbers: a merge whose arithmetic rounds sets no flag, which is the safe direction (Remark 6.10).

Uses Def. 6.1, Thm. 6.2, Prop. 6.3 · newly derived · accepted

A witness is computed, not declared, so its weights and states can be made exact by construction; a naively summed weight vector can read 1 while its real sum exceeds 1. The diagonal form of (W4) is one exact presentation of a state, not the only one binary64 admits.

Conditions (W3) and (W4) of Definition 6.1 read exhibited binary64 numbers, and the normalization step of Theorem 6.2 uses $\sum_i \mu_i = 1$ and $\operatorname{tr}\chi_i = 1$ as identities of reals.

Remark 6.7 (Exactness of the exhibited weights and states). *Exact by construction.* (W3) and (W4) are decided at zero defect, but not in the way the declared marginal τ_A is checked at admission. That check decides declared data, which an instance snaps under a producer's obligation and a consumer must take as it finds them. A witness is instead a computed object that the implementation exhibits, and the implementation can make both conditions exact by construction — which (W3) and (W4) as stated require it to. The difference is on the false-positive side, which is the unsafe one. A naively summed $\mu = (1, 2^{-60})$ returns the datum 1.0 while $\sum_i \mu_i > 1$ as reals, and then the normalization step of Theorem 6.2 reads $\operatorname{tr}\hat{\omega} =$

$\operatorname{tr}\omega + (\sum_i \mu_i - 1)\operatorname{tr} m \neq 1$: the modified point is not feasible and the removal would have been licensed on an identity that does not hold. Requiring the sum in the exhibited order and verified to round nowhere closes that window exactly as (W6) closes it for the residuals, and requiring the exhibited form of (W4) closes the corresponding window on a computed $\lambda_{\min}(\chi_i)$ or a computed $\operatorname{tr}\chi_i$. *Neither leaves a residual exposure*: on the forms those two conditions admit, both hold of the real numbers the data denote.

The diagonal form is one exact presentation among others. Any presentation deciding $\chi_i \succeq 0$ and $\operatorname{tr}\chi_i = 1$ exactly by construction would serve, and the diagonal form is *one* such presentation, not the only one binary64 admits. An exhibited unit vector's outer product $|\psi_i\rangle\langle\psi_i|$ with $\langle\psi_i|\psi_i\rangle$ verified to round nowhere is a genuine second: on a four-dimensional Q_{s_i} the vector $\psi_i = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ has exactly representable entries, its norm accumulates $0.25 \to 0.5 \to 0.75 \to 1.0$ rounding nowhere, and its outer product — every entry $\frac{1}{4}$ — is a rank-one state that is *not* diagonal. (W4) does not admit it, and excluding an admissible presentation can only *refuse* a removal. No impossibility is claimed here.

The evaluation order of (W6). The order in which a witness's residuals are formed is exhibited with the witness. This is the one difference from the declared order of (1.3), which is part of the declaration because the datum it governs is declared.

Uses Def. 6.1, Thm. 6.2 · newly derived · accepted

A party-local entry makes clause (d) unsatisfiable, since the removal theorem excludes it; several measured factors leave the checklist no declared value to read, though the theorem can hold. Either way no sector is flagged, so no set of sectors is deleted.

Requirement 14.8 sets the removable flag for no sector of a declaration whose list carries an entry declared party-local, nor of one with more than one measured factor. The two grounds differ in kind.

Remark 6.8 (Removal in the general format). *A party-local entry is excluded by the theorem.* Theorem 6.2 is stated for a list carrying no entry declared party-local, so for a list carrying one, clause (d) of Definition 6.1 is unsatisfiable and no sector carries a witness in its sense.

Several measured factors are excluded by the checklist, not by the theorem. The theorem's proof reads the sector blocks as operators and needs no factorisation of them, so its hypotheses can hold of a declaration with more than one measured factor. What fails there is the decision procedure: the blocks $F_v^{(s)}$ are products (1.5) that are never formed as data, so the bit-level test of (W1) and the residuals of (W6) have no declared value to read. This ground is a refusal of the flag and not a soundness ground.

Both close the deletion of a set. Both refusals retain sectors, which is unconditionally sound, and both are stated of the flag. The flag is what closes the composite deletion: the hypothesis of Proposition 6.5 quantifies over sectors carrying a witness in the sense of (d), and a run deletes a set of sectors only on the strength of the flag, which is set for none of them in either case. So no composite deletion is available for a declaration carrying a party-local entry or measuring more than one factor, and the proposition is not reached there.

Uses Def. 6.1, Thm. 6.2, Prop. 6.5 · newly derived · accepted

Per-sector removal flags do not compose: two sectors can witness each other's removal, and deleting both raises the value by a bit per round. The set hypothesis of Proposition 6.5 is sufficient, not necessary, and is read on the recorded witnesses.

Remark 6.9 (The set hypothesis of the composition rule). *The hypothesis of Proposition 6.5 is on the* set*, not on any one sector, and without it the per-sector flags do not compose.* The flag *removable* is a per-sector flag and Theorem 6.2 licenses the deletion of *one* sector; neither says that deleting every flagged sector at once is sound, and *in general it is not.* Two state-blind constraint-active sectors with identical declared responses each witness the other: take the declaration of Proposition 6.3 and split its sector 2 into two one-dimensional sectors $2a$ and $2b$ carrying the same $F_0 = 1$, $F_1 = 0$ (all other data unchanged, so $f_{\hat{c}_0,2a} = f_{\hat{c}_0,2b} \equiv 0$, $K = 1 + w_1(2^{1-\alpha} - 1)$ and the infimum is still 0, at $w_1 = 0$). Then $\{2b\}$ with $\mu_1 = 1$ and χ_1 the state on Q_{2b} is a witness for $2a$ meeting (W1)–(W6) exactly, and symmetrically for $2b$, so both flags are set with (W2) satisfied at every step — and deleting both forces $w_1 = 1$ and raises the reported value from 0 to 1, the same one bit per round by the same mechanism as Proposition 6.3. The hypothesis of Proposition 6.5 is what fails there, and it is the only thing that fails.

Three readings of that hypothesis. First, it is sufficient and not necessary. With recorded single-target witnesses forming a chain $s_a \to s_b \to s_c \to s_d$ and $\mathbb{S}_{\mathrm{del}} = \{s_a, s_b, s_c\}$ (s_d retained), the hypothesis fails and the deletion is refused — although deleting all three is in fact sound: s_c may be deleted first, and s_b is then witnessed by s_c's own witness, since s_c is state-blind by (W1) and therefore $\mathrm{tr}\left[F_v^{(s_c)}\chi\right] = c_{v,s_c}$ for every state χ, so $c_{v,s_b} = c_{v,s_c} = \sum_j \nu_j \,\mathrm{tr}\left[F_v^{(s_d)}\xi_j\right]$ and (RM1) closes with (ν_j, ξ_j) in place of the original targets, and likewise (RM3). Refusal is the safe side and this module claims sufficiency only. *Second, the hypothesis is decided on the* recorded *witnesses*, so it constrains the record and not the search: a run may re-derive each surviving sector's witness in the reduced declaration after every deletion and record *those. Nothing here forbids sequential re-witnessing*; what is forbidden is deleting a set whose recorded witnesses point into it. *But the strategy is order-sensitive, and the order is part of it*: re-derivation runs on the *surviving* sectors, so it can drop a recorded reference to a sector already deleted and never to one deleted *later*. What a run must therefore achieve is that every recorded witness names only sectors it *retains*, and the way to achieve it is to delete leaf-first — a sector only when its current witness names retained sectors alone. On the chain $s_a \to s_b \to s_c \to s_d$ above with $\mathbb{S}_{\mathrm{del}} = \{s_a, s_b, s_c\}$ the order s_c, s_b, s_a does it: after s_c goes, s_b re-derives onto s_d by the argument just given, then s_a likewise, and the final record is $s_a \to s_d$, $s_b \to s_d$, $s_c \to s_d$, which meets the hypothesis and deletes all three. *The order s_a, s_b, s_c does not*: re-derivation never touches s_a's or s_b's already-recorded witnesses, the record ends as $s_a \to s_b$, $s_b \to s_c$ with both targets inside $\mathbb{S}_{\mathrm{del}}$, and the deletion of the set is refused as a whole. On that record, with $s_c \to s_d$ beside them, a subset of $\mathbb{S}_{\mathrm{del}}$ meets the hypothesis exactly when it does not contain s_b together with s_a or s_c; the largest is $\{s_a, s_c\}$, whose recorded targets s_b and s_d it retains, so the run keeps at most the deletion of s_a and s_c, taken as a set of its own, and retains s_b. *So re-witnessing loses nothing* when such an order exists, and where none does — the two mutually witnessing sectors of the counterexample above, whose witness relation is cyclic — no order helps and the refusal is the correct outcome, that being precisely the case the hypothesis of Proposition 6.5 exists to

refuse. *Third, the comparison is of index data and is decided at no window*, exactly like (W2), and a singleton $\mathbb{S}_{\mathrm{del}}$ satisfies the hypothesis automatically, by (W2).

Uses Def. 6.1, Thm. 6.2, Prop. 6.3, Prop. 6.5 · newly derived · accepted

Retaining every sector computes the value as defined; deleting one on an approximate match computes a restriction whose excess no function of a tolerance bounds, so a tolerance may only refuse a removal. Dropping a constraint-active sector and averaging sectors can both overstate the key.

Remark 6.10 (Retention and deletion of sectors). *The direction of every removal decision.* Removability of Definition 6.1 (d) is a licence to delete, and every outcome other than all six of (W1)–(W6) — a nonzero residual, a rounding operation, an absent witness, a witness that names s^*, a target that is not key-inert, a block that is not exactly scalar — retains the sector. Retention is unconditionally sound and costs nothing: $\underline{h}_\alpha$ is *defined* by (5.4) as the infimum over the whole of $\mathcal{W}$, and Theorem 6.2 says only that one particular restriction of that infimum does not change it. A run that retains every sector computes the definition. A run that deletes one on an *approximate* match computes instead the infimum over $\{\omega \in \mathcal{W} : \omega_{s^*} = 0\}$, a restriction, whose value is $\geq \underline{h}_\alpha$ and which would be reported under the name $\underline{h}_\alpha$ — an overstatement, and one no function of a declared tolerance bounds: the mechanism is a $P_{\hat{c}}$ or $K_{\hat{c}}$ driven towards zero inside a logarithm, which Proposition 5.8 (iii) exhibits driving the objective up without limit and which Proposition 11.1 (i) turns, in the neighbouring setting of a perturbed declared POVM, into two declarations at distance ϵ whose values differ by tens of bits per round — by $\frac{\alpha}{\alpha-1}N$ with N up to 52, that ceiling being the number model's and not the mechanism's. So a tolerance may only ever *refuse* a removal, never license one.

Two unsound operations. Dropping a constraint-active sector (Proposition 6.3) and averaging sectors (Proposition 6.4) are both unsound, and both failure modes overstate the key — the dangerous direction. The first is reachable in two ways from a checklist that looks complete: by dropping (W2), which lets a sector witness its own removal (Remark 6.6), and by deleting every removable sector at once, which lets two sectors witness each other's (Remark 6.9). (W2) refuses the first; the hypothesis of Proposition 6.5 refuses the second, and is the reason removability being a property of one sector does not by itself license the deletion of a set.

Uses Def. 5.1, Prop. 5.8, Def. 6.1, Thm. 6.2, Prop. 6.3, Prop. 6.4, Prop. 6.5 · newly derived · accepted

7 The Choi program

Every feasible state lives on the admissible Alice subspace, the support of the declared marginal (all of Alice's system in untrusted-source mode), tensored with the measured system (Lemma 7.1). In trusted-source mode every state with Alice's marginal is the image of a positive Choi operator on a support frame that need only contain the marginal's support, so no rank of the marginal is decided (Lemma 7.3). Theorem 7.5 rewrites the program exactly in sector Choi blocks, with one coupling constraint per Alice sector, and on a tight frame a homogeneous constraint row may drop its conjugation (Proposition 7.6); in untrusted-source mode the parametrisation is the

identity (Definition 7.10). The section's main result, Lemma 7.11, states that the objective at any exactly feasible point is at least h_low: an upper estimate of the value, never a bound on it.

Imports foundations/def:sector-map

Every feasible state is supported in the admissible Alice subspace, the support of the declared marginal or all of Alice's system, tensored with the measured system; with no constraints the bound is tight. The subspace depends on the marginal slot alone.

This is where the declared marginal enters the support and face rules of the gradients: the maximal branch support of Definition 8.1 is formed from $\mathcal{S}_A$, and for a nonempty $\mathfrak{L}$ the inclusion (i) is all that the bound (8.3) of Lemma 8.2 uses. The marginal also enters the gradient formulas, through the W_τ conjugation in (8.12) of Corollary 8.5, which (8.13) drops.

Lemma 7.1 (The admissible Alice subspace)**.** Let $\mathfrak{D}$ be a declaration, and write

$$\mathcal{S}_A := \begin{cases} \operatorname{supp} \tau_A, & \text{trusted-source mode,} \\ \mathcal{H}(A), & \text{untrusted-source mode,} \end{cases} \tag{7.1}$$

the *admissible Alice subspace*; it is determined by the marginal slot alone and never by $\mathfrak{L}$ or by a particular feasible point. Then:

(i) every $\omega \in \mathcal{W}$ satisfies $\operatorname{supp} \omega \subseteq \mathcal{S}_A \otimes \mathcal{H}(Q)$;

(ii) the bound (i) is tight for $\mathfrak{L} = \varnothing$, and for a nonempty $\mathfrak{L}$ it remains a valid upper bound.

Uses Def. 1.2, Def. 5.1 · *Used by* Thm. 7.5, Def. 8.1, Lem. 8.2 · newly derived · accepted

Proof. (i) In untrusted-source mode this is vacuous. In trusted-source mode, with $\Pi^\perp := \mathbb{1}_A - \Pi_{\operatorname{supp} \tau_A}$, $\operatorname{tr}\big[(\Pi^\perp \otimes \mathbb{1}_Q)\,\omega\big] = \operatorname{tr}\big[\Pi^\perp \tau_A\big] = 0$ forces $(\Pi^\perp \otimes \mathbb{1})\,\omega = 0$ by positivity.

(ii) For $\mathfrak{L} = \varnothing$ take $\omega = \tau_A \otimes \chi$ with χ any full-rank sector-block-diagonal state on Q, for instance $\mathbb{1}_Q/d_Q$, respectively any full-rank sector-block-diagonal state ω on AQ, for instance $\mathbb{1}_{AQ}/(d_A d_Q)$. Such an ω lies in $\mathcal{W}$, and $\operatorname{supp} \omega = \operatorname{supp} \tau_A \otimes \mathcal{H}(Q)$, respectively $\operatorname{supp} \omega = \mathcal{H}(AQ)$, which is $\mathcal{S}_A \otimes \mathcal{H}(Q)$. For a nonempty $\mathfrak{L}$, (i) holds as proved. □

In trusted-source mode a support frame is the range of an isometry ι from a Choi input register A_0 into A that contains the support of τ_A, not necessarily equal to it. Two exact constructions are named: the full frame, always available, and the diagonal frame, tight by construction.

The Choi program parametrises the states with Alice's declared marginal τ_A by positive operators on a register A_0 mapped into A. The frame is asked only to contain the support of τ_A, not to equal it, so that no run has

to decide $\operatorname{rank}\tau_A$ from a computed spectrum at a window. In untrusted-source mode there is no marginal, and no A_0, ι or W_τ is defined or needed (Definition 7.10); at $d_A = 1$ the two modes give the same program (Definition 7.10 (d)).

Definition 7.2 (Support frame and declared isometry)**.** Let the declaration be in trusted-source mode, with declared marginal τ_A. Let A_0 be a register of dimension d_{A_0} and let $\iota \in \mathcal{L}(A_0, A)$ satisfy

$$\iota^\dagger \iota = \mathbb{1}_{A_0}, \qquad \iota\, \iota^\dagger\, \tau_A = \tau_A. \tag{7.2}$$

Write $\Pi_\iota := \iota\iota^\dagger$, which is the orthogonal projector onto $\operatorname{ran}\iota$ ($\Pi_\iota^\dagger = \Pi_\iota$ and $\Pi_\iota^2 = \iota(\iota^\dagger\iota)\iota^\dagger = \Pi_\iota$ by the first condition). The first condition says ι is an isometry; the second says its range *contains* the support of τ_A, since $\Pi_\iota\tau_A = \tau_A$ holds exactly when every column of τ_A lies in $\operatorname{ran}\iota$, i.e. when $\operatorname{ran}\tau_A \subseteq \operatorname{ran}\iota$. We call $\operatorname{ran}\iota$ the *support frame* of the run and ι the *declared isometry*; a support frame is given by any ι satisfying (7.2). The frame is *tight* when $\operatorname{ran}\iota = \operatorname{supp}\tau_A$, equivalently when $d_{A_0} = \operatorname{rank}\tau_A$; in general $\operatorname{rank}\tau_A \le d_{A_0} \le d_A$. The images under ι of a fixed orthonormal basis of A_0 are the *declared basis of* A_0.

Two constructions satisfy (7.2) exactly:

(a) *The* full frame. $A_0 = A$, $d_{A_0} = d_A$, $\iota = \mathbb{1}_A$. Then $\iota^\dagger\iota = \mathbb{1}_A$ and $\Pi_\iota\tau_A = \mathbb{1}_A\,\tau_A = \tau_A$: both conditions of (7.2) hold for *any* admissible τ_A, and both hold *exactly* as binary64 data, a product with the identity rounding nothing. This construction is always available and uses no property of τ_A beyond admission. It is tight exactly when τ_A is nonsingular.

(b) *The* diagonal frame. If the declared τ_A is diagonal in the declared basis of A, $\tau_A = \sum_i t_i\,|i\rangle\langle i|$, then $\operatorname{supp}\tau_A = \operatorname{span}\{|i\rangle : t_i \neq 0\}$, and one may take $d_{A_0} = \#\{i : t_i \neq 0\}$ and $\iota = \sum_j |i_j\rangle\,\langle j|$, the coordinate injection onto those indices in increasing order. Both conditions of (7.2) are then arithmetic in $\{0, 1\}$ and hold exactly, and the frame is tight. Whether $t_i \neq 0$ is a bit-level comparison of a declared binary64 datum against zero and not a threshold: τ_A is admitted only with $\tau_A \succeq 0$ and $\operatorname{tr}\tau_A = 1$ decided exactly (Definition 1.1, the exact-admissibility paragraph), so its diagonal entries *are* its eigenvalues and *are* the declared numbers. The construction is available only for a τ_A that is diagonal in the declared basis, every off-diagonal entry being the datum 0. Construction (a) is always available and exhibits no tightness at all.

Set

$$W_\tau := \tau_A^{1/2}\,\iota \;\in\; \mathcal{L}(A_0, A). \tag{7.3}$$

Uses Def. 1.1 · *Used by* Lem. 7.3, Lem. 7.4, Thm. 7.5, Prop. 7.6, Rem. 7.7, Rem. 7.9, Def. 8.1, Lem. 8.6, Req. 15.1, Req. 15.2, Req. 15.3, Req. 15.5, Prop. 17.2, Req. 27.1, Req. 28.1 · newly derived · implemented · *Code* certiqs_key.meat_core.objective.joint:marginal_frame

On every support frame, W_τ times its adjoint is τ_A, and the Choi map carries the positive operators with identity input marginal onto the states with A-marginal τ_A. It is injective on a tight frame, and only there unless the second system is trivial; being onto, it needs no rank decided.

Whatever support frame a run uses, every state with Alice's marginal is the image of a positive Choi operator on it; a tight frame makes the correspondence one to one. The identities (i)–(iii) are what the parametrisation (iv) and the Choi program of Theorem 7.5 use of W_τ.

Lemma 7.3 (Fixed-marginal parametrisation on a support frame)**.** Let the declaration be in trusted-source mode, let ι be the declared isometry of a support frame, satisfying (7.2), and let W_τ be (7.3) (Definition 7.2). Then:

(i) $\Pi_\iota \tau_A^{1/2} = \tau_A^{1/2}$.

(ii) For every support frame, tight or not,

$$W_\tau W_\tau^\dagger = \tau_A^{1/2}\,\Pi_\iota\,\tau_A^{1/2} = \tau_A^{1/2}\,\tau_A^{1/2} = \tau_A. \tag{7.4}$$

(iii) $\operatorname{ran} W_\tau = \operatorname{supp}\tau_A$, and the Moore–Penrose inverse W_τ^+ satisfies

$$W_\tau W_\tau^+ = \Pi_{\operatorname{supp}\tau_A}, \qquad W_\tau^+ W_\tau = P_\iota := \Pi_{(\ker W_\tau)^\perp}, \tag{7.5}$$

where $\ker W_\tau = \{\xi \in \mathcal{H}(A_0) : \iota\xi \in (\operatorname{supp}\tau_A)^\perp\}$ has dimension $d_{A_0} - \operatorname{rank}\tau_A$. The frame is tight if and only if $\ker W_\tau = 0$, i.e. $P_\iota = \mathbb{1}_{A_0}$, and then W_τ has full column rank and $W_\tau^+ = (W_\tau^\dagger W_\tau)^{-1} W_\tau^\dagger$.

(iv) For any finite-dimensional system B', the map $\chi_{B'}(J) := (W_\tau \otimes \mathbb{1}_{B'})\,J\,(W_\tau^\dagger \otimes \mathbb{1}_{B'})$ maps $\{J \in \operatorname{Pos}(A_0B') : \operatorname{tr}_{B'} J = \mathbb{1}_{A_0}\}$ *onto* $\{\omega \in \operatorname{Pos}(AB') : \operatorname{tr}_{B'}\omega = \tau_A\}$. It is *injective*, hence a bijection with inverse $\omega \mapsto (W_\tau^+ \otimes \mathbb{1}_{B'})\,\omega\,(W_\tau^{+\dagger} \otimes \mathbb{1}_{B'})$, when the support frame is tight, and, if $\dim\mathcal{H}(B') \geq 2$, only then; if $\dim\mathcal{H}(B') = 1$ its domain is the single point $\mathbb{1}_{A_0}$. In general $\chi_{B'}(J) = \chi_{B'}(J')$ if and only if $(P_\iota \otimes \mathbb{1}_{B'})J(P_\iota \otimes \mathbb{1}_{B'}) = (P_\iota \otimes \mathbb{1}_{B'})J'(P_\iota \otimes \mathbb{1}_{B'})$.

Uses Def. 1.2, Def. 7.2 · *Used by* Lem. 7.4, Thm. 7.5, Prop. 7.6, Rem. 7.7, Rem. 7.8, Rem. 7.9, Lem. 8.2, Req. 15.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.joint:marginal_frame · *Oracles* registry.py

Proof. Idea. (i)–(iii) are linear algebra of the two conditions of (7.2). For (iv), the image of an operator with input marginal $\mathbb{1}_{A_0}$ has A-marginal $W_\tau W_\tau^\dagger = \tau_A$; conversely W_τ^+ pulls a state with marginal τ_A back to an operator with input marginal P_ι, which a block annihilated by W_τ completes to $\mathbb{1}_{A_0}$.

Step 1 (identity (i)). For a positive semidefinite X one has $\operatorname{ran} X^{1/2} = (\ker X)^\perp = \operatorname{ran} X$, so $\operatorname{ran}\tau_A^{1/2} = \operatorname{ran}\tau_A \subseteq \operatorname{ran}\Pi_\iota$ by (7.2), and a projector acts as the identity on a subspace of its range.

Step 2 (identity (ii)). The first equality of (7.4) is $W_\tau = \tau_A^{1/2}\iota$ with $\Pi_\iota = \iota\iota^\dagger$, and the second is (i).

Step 3 (identity (iii)). By (ii), $\operatorname{ran} W_\tau = \operatorname{ran}(W_\tau W_\tau^\dagger) = \operatorname{supp}\tau_A$, so the Moore–Penrose inverse W_τ^+ satisfies (7.5). The kernel $\ker W_\tau$ has dimension $d_{A_0} - \operatorname{rank}\tau_A$ because ι is injective and $\operatorname{ran}\iota = \operatorname{supp}\tau_A \oplus \big(\operatorname{ran}\iota \cap (\operatorname{supp}\tau_A)^\perp\big)$. So $\ker W_\tau = 0$ exactly when $d_{A_0} = \operatorname{rank}\tau_A$, which is tightness (Definition 7.2), and then W_τ has full column rank and $W_\tau^+ = (W_\tau^\dagger W_\tau)^{-1} W_\tau^\dagger$.

Step 4 (the map (iv): forward). $\chi_{B'}(J) \succeq 0$, and $\operatorname{tr}_{B'}\chi_{B'}(J) = W_\tau(\operatorname{tr}_{B'} J)W_\tau^\dagger = W_\tau W_\tau^\dagger = \tau_A$ by (7.4).

Step 5 (surjectivity). Let $\omega \succeq 0$ with $\operatorname{tr}_{B'}\omega = \tau_A$ and let $\Pi^\perp := \mathbb{1}_A - \Pi_{\operatorname{supp}\tau_A}$. Then $\operatorname{tr}\big[(\Pi^\perp \otimes \mathbb{1}_{B'})\,\omega\big] = \operatorname{tr}\big[\Pi^\perp \tau_A\big] = 0$, and positivity forces $(\Pi^\perp \otimes \mathbb{1})\,\omega = 0$, i.e. ω is supported on $\operatorname{supp}\tau_A \otimes \mathcal{H}(B')$. Put $J_0 := (W_\tau^+ \otimes \mathbb{1})\,\omega\,(W_\tau^{+\dagger} \otimes \mathbb{1}) \succeq 0$; by (7.4) and (7.5),

$$\operatorname{tr}_{B'} J_0 = W_\tau^+ \tau_A W_\tau^{+\dagger} = W_\tau^+ W_\tau\, W_\tau^\dagger W_\tau^{+\dagger} = P_\iota P_\iota^\dagger = P_\iota,$$

and $\chi_{B'}(J_0) = (W_\tau W_\tau^+ \otimes \mathbb{1})\,\omega\,(W_\tau W_\tau^+ \otimes \mathbb{1})^\dagger = (\Pi_{\operatorname{supp}\tau_A} \otimes \mathbb{1})\,\omega\,(\Pi_{\operatorname{supp}\tau_A} \otimes \mathbb{1}) = \omega$. If the frame is tight, $P_\iota = \mathbb{1}_{A_0}$ and J_0 is already in the domain. In general fix any state φ on $\mathcal{H}(B')$ and put $J := J_0 + (\mathbb{1}_{A_0} - P_\iota) \otimes \varphi \succeq 0$. Then $\operatorname{tr}_{B'} J = P_\iota + (\mathbb{1}_{A_0} - P_\iota) = \mathbb{1}_{A_0}$, and the added term is annihilated by $\chi_{B'}$, because $W_\tau(\mathbb{1}_{A_0} - P_\iota) = W_\tau - W_\tau W_\tau^+ W_\tau = 0$ by the Moore–Penrose identity; so $\chi_{B'}(J) = \chi_{B'}(J_0) = \omega$.

Step 6 (the fibres). $W_\tau = W_\tau P_\iota$ by the same identity, so for any J in the domain $\chi_{B'}(J) = (W_\tau \otimes \mathbb{1})\big[(P_\iota \otimes \mathbb{1})J(P_\iota \otimes \mathbb{1})\big](W_\tau^\dagger \otimes \mathbb{1})$, and W_τ restricted to $\operatorname{ran} P_\iota = (\ker W_\tau)^\perp$ is injective; hence $\chi_{B'}(J)$ determines, and is determined by, the compression $(P_\iota \otimes \mathbb{1})J(P_\iota \otimes \mathbb{1})$. That is the stated fibre description, and it gives injectivity when $P_\iota = \mathbb{1}_{A_0}$, i.e. when the frame is tight; the displayed inverse is then J_0. Conversely, let the frame not be tight, so that $\mathbb{1}_{A_0} - P_\iota \neq 0$, and let $\dim \mathcal{H}(B') \geq 2$, so that B' carries two states $\varphi \neq \varphi'$. With any state φ_0 on B', the operators $J := P_\iota \otimes \varphi_0 + (\mathbb{1}_{A_0} - P_\iota) \otimes \varphi$ and $J' := P_\iota \otimes \varphi_0 + (\mathbb{1}_{A_0} - P_\iota) \otimes \varphi'$ are positive semidefinite with $\operatorname{tr}_{B'} J = \operatorname{tr}_{B'} J' = \mathbb{1}_{A_0}$, have the same compression $P_\iota \otimes \varphi_0$, hence the same image, and differ by $(\mathbb{1}_{A_0} - P_\iota) \otimes (\varphi - \varphi') \neq 0$; so $\chi_{B'}$ is not injective. If $\dim \mathcal{H}(B') = 1$, then $\operatorname{tr}_{B'} J = J$ and the domain is the single point $\mathbb{1}_{A_0}$. □

Any factor W of Alice's marginal parametrises the fixed-marginal states per Alice sector, and the objective reads her side only through her measurement sandwiched by W. On the full frame these are the declared operators of an ensemble-form declaration, so no rank is computed.

Of W_τ, the parametrisation of Lemma 7.3 reads only that it is a factor of τ_A. Stated for any factor, and per Alice sector, it shows that the objective reads Alice's side only through the operators $W^\dagger M_x W$, which on the full frame are the operators an ensemble-form declaration declares.

Lemma 7.4 (Fixed-marginal parametrisation with any factor)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 in trusted-source mode, in the general format of Defini-

tion 1.1.

(i) *(any factor serves)* Let $\tau \in \mathrm{Pos}(A')$ on a finite-dimensional system A', of any trace, and let $W \in \mathcal{L}(A_0, A')$ satisfy $WW^\dagger = \tau$. Let P_W be the projector onto $(\ker W)^\perp$. For every finite-dimensional B' the map $\chi^W_{B'}(J) \coloneqq (W \otimes \mathbb{1}_{B'}) J (W^\dagger \otimes \mathbb{1}_{B'})$ maps $\{J \in \mathrm{Pos}(A_0 B') : \mathrm{tr}_{B'} J = \mathbb{1}_{A_0}\}$ onto $\{\omega \in \mathrm{Pos}(A'B') : \mathrm{tr}_{B'}\, \omega = \tau\}$; it is injective when $\ker W = 0$, and, if $\dim \mathcal{H}(B') \geq 2$, only then; and $\chi^W_{B'}(J) = \chi^W_{B'}(J')$ exactly when $(P_W \otimes \mathbb{1}) J (P_W \otimes \mathbb{1}) = (P_W \otimes \mathbb{1}) J' (P_W \otimes \mathbb{1})$. The operator W_τ of (7.3) is such a factor of τ_A for every admissible frame.

(ii) *(per Alice sector)* For every $t \in \mathbb{T}$ let $W_t \in \mathcal{L}(A_{0,t}, A_t)$ satisfy $W_t W_t^\dagger = \tau_A^{(t)}$, and put $A_0 \coloneqq \bigoplus_t A_{0,t}$, $W \coloneqq \bigoplus_t W_t$. Then $WW^\dagger = \tau_A$, and the assignment $(J_{t,s}) \mapsto \big(\omega^{(t,s)}\big)$, $\omega^{(t,s)} \coloneqq (W_t \otimes \mathbb{1}_{Q_s}) J_{t,s} (W_t^\dagger \otimes \mathbb{1}_{Q_s})$, maps

$$\Big\{ (J_{t,s})_{(t,s)} : \; J_{t,s} \in \mathrm{Pos}(A_{0,t} Q_s), \quad \sum_{s \in \mathbb{S}} \mathrm{tr}_{Q_s}\, J_{t,s} = \mathbb{1}_{A_{0,t}} \;\; \forall t \in \mathbb{T} \Big\} \tag{7.6}$$

onto the set of lists over the joint sectors with positive semidefinite blocks and $\sum_s \mathrm{tr}_{Q_s}\, \omega^{(t,s)} = \tau_A^{(t)}$ for every t, bijectively when $\ker W_t = 0$ for every t, and, if $\dim \mathcal{H}(Q) \geq 2$, only then. The set (7.6) is a Cartesian product over t. If every ι_t satisfies (7.2) for $\tau_A^{(t)}$ on A_t, then $\iota \coloneqq \bigoplus_t \iota_t$ satisfies (7.2) for τ_A and $W_\tau = \bigoplus_t (\tau_A^{(t)})^{1/2} \iota_t$.

(iii) *(the objective reads Alice's side through N_x)* For a factor $W = \bigoplus_t W_t$ as in (ii) put

$$N_x^W \coloneqq W^\dagger M_x W = \bigoplus_t W_t^\dagger M_x^{(t)} W_t \in \mathrm{Pos}(A_0), \tag{7.7}$$

let K_κ^N be (1.18) with $\sqrt{N_x^W}$ in place of $\sqrt{M_x}$, an operator $\mathcal{H}(A_0 Q) \to \mathcal{H}(R\,XY\,C_A C_B\, A_0 Q)$, and let $\mathcal{G}_{\hat{c}}^N$ be (1.19) with K_κ^N. Then for every $J \in \mathrm{Pos}(A_0 Q)$, every (x, v) and every $\hat{c}$, with $\omega \coloneqq \chi_Q^W(J)$,

$$\mathrm{tr}[(M_x \otimes F_v)\, \omega] = \mathrm{tr}\big[(N_x^W \otimes F_v)\, J\big], \qquad \mathrm{tr}\, \mathcal{G}_{\hat{c}}(\omega) = \mathrm{tr}\, \mathcal{G}_{\hat{c}}^N(J), \qquad T\big(\mathcal{G}_{\hat{c}}(\omega)\big) = T\big(\mathcal{G}_{\hat{c}}^N(J)\big), \tag{7.8}$$

T the functional (4.7); hence $P_{\hat{c}}$, $K_{\hat{c}}$ and O_α at ω are functions of J, $\{N_x^W\}$, $\{F_v\}$ and the classical maps alone, and the same holds sector by sector for the joint-sector quantities of Definition 4.8.

(iv) *(ensemble form)* On the full frame, $A_{0,t} = A_t$ and $W_t = (\tau_A^{(t)})^{1/2}$, one has $N_x^W = \tau_A^{1/2} M_x \tau_A^{1/2} = N_x$, the operators of Definition 1.1 (DX4): for an ensemble-form declaration they are the *declared* numbers. So the program of an ensemble-form declaration is the program (7.6), (iii)

in the declared $\{N_x\}$ and $\{F_v\}$, with every constraint row in the conjugated form of (7.9); it is defined, and equals $\underline{h}_\alpha$ of the denoted declaration, at every rank of every $\tau_A^{(t)}$, and no rank is computed.

Uses Def. 1.1, Def. 1.2, Lem. 1.3, Def. 1.4, Def. 1.6, Lem. 1.7, Thm. 4.1, Lem. 4.3, Def. 4.8, Def. 5.1, Def. 7.2, Lem. 7.3 · *Used by* Thm. 7.5, Prop. 17.2, Cor. 19.6 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.joint:joint_choi_parametrization

Proof. (i) Three facts about W. (a) $WW^\dagger = \tau$ is the hypothesis. (b) $\operatorname{ran} W = \operatorname{ran}(WW^\dagger) = \operatorname{supp}\tau$: for any operator, $\operatorname{ran} W \supseteq \operatorname{ran} WW^\dagger$ and $\ker W^\dagger = \ker WW^\dagger$ (if $WW^\dagger\xi = 0$ then $\|W^\dagger\xi\|^2 = \langle\xi|\, WW^\dagger\,|\xi\rangle = 0$), so the two ranges, being the orthogonal complements of equal kernels, agree. (c) Therefore the Moore–Penrose inverse satisfies $WW^+ = \Pi_{\operatorname{supp}\tau}$ and $W^+W = P_W$, and $W = WW^+W = WP_W$. The proof of Lemma 7.3 uses of W_τ exactly (7.4), which is (a), and (7.5) with the identity $W_\tau(\mathbb{1}_{A_0} - P_\iota) = 0$, which are (c); it uses neither $\operatorname{tr}\tau_A = 1$ nor the isometry ι otherwise. We repeat it for W. *Forward:* $\chi^W(J) \succeq 0$ and $\operatorname{tr}_{B'}\chi^W(J) = W(\operatorname{tr}_{B'} J)W^\dagger = WW^\dagger = \tau$. *Onto:* let $\omega \succeq 0$ with $\operatorname{tr}_{B'}\omega = \tau$ and $\Pi^\perp := \mathbb{1} - \Pi_{\operatorname{supp}\tau}$; then $\operatorname{tr}\big[(\Pi^\perp\otimes\mathbb{1})\omega\big] = \operatorname{tr}\big[\Pi^\perp\tau\big] = 0$, and positivity gives $(\Pi^\perp\otimes\mathbb{1})\omega = 0$. Put $J_0 := (W^+\otimes\mathbb{1})\omega(W^{+\dagger}\otimes\mathbb{1}) \succeq 0$. Then $\operatorname{tr}_{B'} J_0 = W^+WW^\dagger W^{+\dagger} = P_W P_W^\dagger = P_W$ and $\chi^W(J_0) = (WW^+\otimes\mathbb{1})\omega(WW^+\otimes\mathbb{1})^\dagger = \omega$. With any state φ on B', $J := J_0 + (\mathbb{1}_{A_0} - P_W)\otimes\varphi \succeq 0$ has $\operatorname{tr}_{B'} J = \mathbb{1}_{A_0}$ and $\chi^W(J) = \omega$, because $W(\mathbb{1} - P_W) = 0$. *Fibres:* $\chi^W(J) = (W\otimes\mathbb{1})[(P_W\otimes\mathbb{1})J(P_W\otimes\mathbb{1})](W^\dagger\otimes\mathbb{1})$ and W is injective on $\operatorname{ran} P_W$, so $\chi^W(J)$ determines and is determined by the compression; injectivity holds when $P_W = \mathbb{1}_{A_0}$. If $P_W \neq \mathbb{1}_{A_0}$ and $\dim\mathcal{H}(B') \geq 2$, two states $\varphi \neq \varphi'$ on B' and any state φ_0 give the two operators $P_W\otimes\varphi_0 + (\mathbb{1}_{A_0} - P_W)\otimes\varphi$ and $P_W\otimes\varphi_0 + (\mathbb{1}_{A_0} - P_W)\otimes\varphi'$ of the domain, which differ and have the same compression, hence the same image, as in the proof of Lemma 7.3. Finally $W_\tau W_\tau^\dagger = \tau_A$ is (7.4).

(ii) $WW^\dagger = \bigoplus_t W_t W_t^\dagger = \tau_A$. Fix t and apply (i) with $A' = A_t$, $\tau = \tau_A^{(t)}$, $B' = Q$. The map $\chi_Q^{W_t}$ and the section $\omega \mapsto J_0$ act on the first tensor factor alone, so they commute with $X \mapsto (\mathbb{1}\otimes\Pi_s)X(\mathbb{1}\otimes\Pi_{s'})$ and preserve block diagonality along $\mathbb{S}$; choosing the state φ of (i) supported on one $\mathcal{H}(Q_{s_0})$ keeps the completed J block diagonal as well. For operators block diagonal along $\mathbb{S}$, $\operatorname{tr}_Q J_t = \sum_s \operatorname{tr}_{Q_s} J_{t,s}$, and likewise for ω. So for each t the map is onto, and injective when $\ker W_t = 0$; if $\ker W_t \neq 0$ and $\dim\mathcal{H}(Q) \geq 2$, the two operators of (i) for W_t, built with a sector-block-diagonal φ_0 and two different sector-block-diagonal states $\varphi \neq \varphi'$ on Q (states on two sectors, or two states on one sector of dimension at least two), are two different block-diagonal preimages of one point. The domain and the codomain are products over t of the per-sector sets, and the map is the product of the per-sector maps, which gives the statement. For the frames: $\iota^\dagger\iota = \bigoplus_t \iota_t^\dagger\iota_t = \mathbb{1}_{A_0}$ and $\iota\iota^\dagger\tau_A = \bigoplus_t \iota_t\iota_t^\dagger\tau_A^{(t)} = \tau_A$, and $\tau_A^{1/2} = \bigoplus_t(\tau_A^{(t)})^{1/2}$ by blockwise functional calculus.

(iii) The first identity is cyclicity of the trace: $\operatorname{tr}\big[(M_x \otimes F_v)(W \otimes \mathbb{1})J(W^\dagger \otimes \mathbb{1})\big] = \operatorname{tr}\big[(W^\dagger M_x W \otimes F_v)J\big]$. The second follows from it and the trace formula in the proof of Lemma 1.7 (i), which reads of the operators in the slot of $\sqrt{M_x}$ only that their squares' traces against the input are $\operatorname{tr}[(M_x \otimes F_v)\,\cdot\,]$, resp. $\operatorname{tr}\big[(N_x^W \otimes F_v)\,\cdot\,\big]$. For the third, fix x. The operator $Y_x \coloneqq \sqrt{M_x}\,W \in \mathcal{L}(A_0, A)$ satisfies $Y_x^\dagger Y_x = W^\dagger M_x W = N_x^W$, so by the polar decomposition $Y_x = V_x\sqrt{N_x^W}$ with $V_x \in \mathcal{L}(A_0, A)$ a partial isometry whose initial space is $\mathcal{I}_x \coloneqq \overline{\operatorname{ran}}\, N_x^W$: $V_x^\dagger V_x$ is the projector onto $\mathcal{I}_x$. Put $\widetilde{V} \coloneqq \sum_{x\in\mathcal{X}} |x\rangle\!\langle x|_X \otimes V_x$, acting as the identity on R, Y, $C_A C_B$ and Q; it is a partial isometry with initial projector $\widetilde{E} \coloneqq \sum_x |x\rangle\!\langle x|_X \otimes V_x^\dagger V_x$. Every term of $K_\kappa(W \otimes \mathbb{1}_Q)$ is $|r(x,v)\rangle\,|x,v\rangle\,|\kappa\rangle \otimes (V_x\sqrt{N_x^W} \otimes \sqrt{F_v})$, so

$$K_\kappa(W \otimes \mathbb{1}_Q) = \widetilde{V}\,K_\kappa^N, \qquad \widetilde{E}\,K_\kappa^N = K_\kappa^N,$$

the second because the term of K_κ^N at x has its A_0 range in $\operatorname{ran}\sqrt{N_x^W} \subseteq \mathcal{I}_x$. Hence $\mathcal{G}_{\hat{c}}(\omega) = \widetilde{V}\,X\,\widetilde{V}^\dagger$ with $X \coloneqq \mathcal{G}_{\hat{c}}^N(J) \succeq 0$ and $\widetilde{E}X\widetilde{E} = X$. For such a pair, $\widetilde{V}$ is isometric on the support of X, so $(\widetilde{V}X\widetilde{V}^\dagger)^{1/\alpha} = \widetilde{V}X^{1/\alpha}\widetilde{V}^\dagger$; $\widetilde{V}$ and $\widetilde{E}$ commute with every $Z_j = |j\rangle\!\langle j|_R$, so $\mathcal{Z}(\widetilde{V}X^{1/\alpha}\widetilde{V}^\dagger) = \widetilde{V}\mathcal{Z}(X^{1/\alpha})\widetilde{V}^\dagger$ and $\widetilde{E}\mathcal{Z}(X^{1/\alpha})\widetilde{E} = \mathcal{Z}(X^{1/\alpha})$; therefore $(\widetilde{V}\mathcal{Z}(X^{1/\alpha})\widetilde{V}^\dagger)^\alpha = \widetilde{V}\mathcal{Z}(X^{1/\alpha})^\alpha\widetilde{V}^\dagger$ and its trace is $\operatorname{tr}\big[\mathcal{Z}(X^{1/\alpha})^\alpha\widetilde{E}\big] = \operatorname{tr}\big[\mathcal{Z}(X^{1/\alpha})^\alpha\big]$. That is $T(\mathcal{G}_{\hat{c}}(\omega)) = T(X)$. The consequences: $P_{\hat{c}}$ is a sum of the first quantities, $K_{\hat{c}} = T + P - \operatorname{tr}\mathcal{G}$ by (4.1), and O_α is a function of these by (5.3). The argument used no property of J beyond $J \succeq 0$; applied to a J supported on one joint sector, where W, M_x and N_x^W act through their t-blocks and F_v through its s-block, it gives the sector-by-sector statement.

(iv) $W_t = (\tau_A^{(t)})^{1/2}$ is Hermitian with $W_t W_t^\dagger = \tau_A^{(t)}$, a factor in the sense of (ii) and the operator (7.3) of the full frame. $N_x^W = \tau_A^{1/2} M_x \tau_A^{1/2}$ is the definition (7.7); for the declaration an ensemble form denotes it is the declared N_x by Lemma 1.3 (ii). By (ii) the assignment is onto the marginal-constrained lists at every rank, a constraint row evaluated at $\omega = \chi^W(J)$ is the conjugated row by substitution, exactly as in the proof of Theorem 7.5, and by (iii) the objective is evaluated from the declared operators; so the infimum over $\mathcal{S}_{\mathrm{acc}}$ and the feasible lists of that program is the infimum of O_α over $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}$ for the denoted declaration, which is $\underline{h}_\alpha$. Nothing in (i)–(iii) reads a rank, an inverse or the operators (1.7). □

In trusted-source mode the program is exactly a jointly convex minimisation over sector Choi blocks, with one coupling constraint per Alice sector and affine or conic rows. Every support frame gives the value h_low, since only surjectivity of the parametrisation enters it.

Applied block by block along the sectors, the parametrisation of Lemma 7.3 rewrites the program of Definition 5.1 in Choi blocks, the variables a solver

works in, without changing its value on any support frame.

Theorem 7.5 (The Choi program)**.** Let the declaration be in trusted-source mode, fix a support frame (Definition 7.2), apply Lemma 7.3 with $B' = Q$, and restrict to sector-block-diagonal operators. Then the assignment $J = \bigoplus_s J_s \mapsto \omega = \chi_Q(J)$, with $\omega_s = \chi_{Q_s}(J_s) \coloneqq (W_\tau \otimes \mathbb{1}_{Q_s}) J_s (W_\tau^\dagger \otimes \mathbb{1}_{Q_s})$, maps

$$\mathcal{W}^J \coloneqq \Big\{ (J_s)_{s\in\mathbb{S}} :\ J_s \in \mathrm{Pos}(A_0 Q_s),\ \sum_{s\in\mathbb{S}} \mathrm{tr}_{Q_s} J_s = \mathbb{1}_{A_0},\ W_\tau \Big(\textstyle\sum_s \mathrm{tr}_{Q_s}[(\mathbb{1}_{A_0} \otimes \hat{O}_k^{(s)}) J_s] \Big) W_\tau^\dagger \bowtie_k \Theta_k \ \forall k \Big\} \tag{7.9}$$

onto $\mathcal{W}$ of (5.1), bijectively when the support frame is tight (Lemma 7.3), and, if $\mathcal{W} \neq \varnothing$ and $\dim \mathcal{H}(Q) \geq 2$, only then. Writing $\mathbf{\Phi}_J[J] \coloneqq \mathbf{\Phi}[\chi_Q(J)]$, $\mathcal{G}^J_{\hat{c},s} \coloneqq \mathcal{G}_{\hat{c},s} \circ \chi_{Q_s}$, $f^J_{\hat{c},s} \coloneqq f_{\hat{c},s} \circ \chi_{Q_s}$, $P^J_{\hat{c}}(J) \coloneqq \mathbf{\Phi}_J[J](\hat{c})$ and $K^J_{\hat{c}}(J) \coloneqq P^J_{\hat{c}}(J) + \sum_{s\in\mathbb{S}_K(\hat{c})} f^J_{\hat{c},s}(J_s)$, the value (5.4) equals

$$\begin{aligned} \underline{h}_\alpha = \inf_{\mathbf{q},\ (J_s)_s} \quad & \frac{\alpha}{\alpha-1} \sum_{\hat{c}\notin\widehat{\mathcal{C}}_{\mathrm{key}}} q(\hat{c}) \log_2 \frac{q(\hat{c})}{P^J_{\hat{c}}(J)} + \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} \left[q(\hat{c}) \log_2 \frac{q(\hat{c})}{P^J_{\hat{c}}(J)} + \frac{q(\hat{c})}{\alpha-1} \log_2 \frac{q(\hat{c})}{K^J_{\hat{c}}(J)} \right] \\ \text{s.t.} \quad & \mathbf{q} \in \mathcal{S}_{\mathrm{acc}}, \qquad (J_s)_s \in \mathcal{W}^J, \end{aligned} \tag{7.10}$$

an *exact* reformulation (not a relaxation), jointly convex in $(\mathbf{q}, (J_s)_s)$, with all constraints affine or conic.

In the general format of Definition 1.1, let the frame be a direct sum $\iota = \bigoplus_{t\in\mathbb{T}} \iota_t$ with every $\iota_t \in \mathcal{L}(A_{0,t}, A_t)$ satisfying (7.2) for $\tau_A^{(t)}$ — either construction of Definition 7.2 applied sector by sector — and put $W_t \coloneqq (\tau_A^{(t)})^{1/2} \iota_t$. Let $\mathcal{W}^J$ be the set of lists $(J_{t,s})$ over the joint sectors with $J_{t,s} \in \mathrm{Pos}(A_{0,t} Q_s)$, the coupling constraints $\sum_s \mathrm{tr}_{Q_s} J_{t,s} = \mathbb{1}_{A_{0,t}}$ for every t, and for every entry the row of (5.5) evaluated at $\omega^{(t,s)} = (W_t \otimes \mathbb{1}) J_{t,s} (W_t^\dagger \otimes \mathbb{1})$: for an entry read through its lifted operator, $W_t \big(\sum_s \mathrm{tr}_{Q_s}[(\mathbb{1}_{A_{0,t}} \otimes \hat{O}_k^{\uparrow(s)}) J_{t,s}] \big) W_t^\dagger \bowtie_k \Theta_k^{(t)}$ for every t, and for a party-local entry the last line of (5.5) at J, conjugated by $W_t \otimes \mathbb{1}$. Then $(J_{t,s}) \mapsto (\omega^{(t,s)})$ maps $\mathcal{W}^J$ onto $\mathcal{W}$ of (5.5), bijectively when every sector's frame is tight, and, if $\mathcal{W} \neq \varnothing$ and $\dim \mathcal{H}(Q) \geq 2$, only then, and (7.10) holds with $K^J_{\hat{c}}(J) = P^J_{\hat{c}}(J) + \sum_{(t,s)\in\mathbb{J}_K(\hat{c})} f_{\hat{c},(t,s)}(\omega^{(t,s)})$. Every constraint of $\mathcal{W}^J$ involves the blocks of one Alice sector only, so $\mathcal{W}^J = \prod_t \mathcal{W}^J_t$, and a linear function of $(J_{t,s})$ is minimised over $\mathcal{W}^J$ by minimising, independently for each t, its part on the blocks of t over $\mathcal{W}^J_t$.

Uses Def. 1.1, Prop. 4.6, Def. 4.7, Def. 4.8, Lem. 4.9, Def. 5.1, Thm. 5.10, Lem. 7.1, Def. 7.2, Lem. 7.3, Lem. 7.4, foundations/def:sector-map · *Used by* Prop. 7.6, Rem. 7.7, Rem. 7.8, Rem. 7.9, Def. 7.10, Lem. 7.11, Cor. 8.5, Lem. 8.6, Rem. 8.10, Lem. 9.6, Rem. 10.20, Req. 15.4, Def. 17.1, Prop. 17.2, Def. 18.1, Lem. 18.2, Lem. 18.3, Thm. 19.1, Cor. 19.6, Rem. 19.8, Lem. 20.1, Lem. 21.1, Lem. 21.2, Rem. 23.2, Req. 28.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.choi: choi_parametrization · *Oracles* registry.py

Proof. Idea. The map χ_Q acts on the A factor alone, so it respects the sector

blocks and carries every constraint row over by substitution. Surjectivity is that of Lemma 7.3, completed on one sector by a block that W_τ annihilates; the value follows by substituting into the objective, and injectivity is not needed.

The map. Lemma 7.3 restricts to sector-block-diagonal operators because χ_Q acts only on the A factor and therefore commutes with the sector projectors $\mathbb{1}\otimes\Pi_s$; the global trace-preservation condition $\mathrm{tr}_Q\, J = \mathbb{1}_{A_0}$ is the single coupling constraint $\sum_s \mathrm{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$, which by the lemma *implies* $\omega_A = \tau_A$ and is implied by it up to the fibre the lemma describes. The $\mathfrak{L}$ rows transfer by substitution, using $\mathrm{tr}_Q[(\mathbb{1}_A\otimes\hat{O}_k)\chi_Q(J)] = W_\tau(\mathrm{tr}_Q[(\mathbb{1}_{A_0}\otimes\hat{O}_k)J])W_\tau^\dagger$ — which holds because $(\mathbb{1}_A\otimes\hat{O}_k)(W_\tau\otimes\mathbb{1}) = (W_\tau\otimes\mathbb{1})(\mathbb{1}_{A_0}\otimes\hat{O}_k)$ — and block-diagonality of $\hat{O}_k$: the k-th row of (7.9) *is* the k-th row of (5.2) evaluated at $\omega = \chi_Q(J)$, so it holds for J exactly when it holds for its image. Normalization $\mathrm{tr}\,\omega = 1$ follows from the coupling constraint and (7.2), since $\mathrm{tr}\,\omega = \mathrm{tr}\Big[W_\tau^\dagger W_\tau\cdot\mathrm{tr}_Q\, J\Big] = \mathrm{tr}\Big[W_\tau^\dagger W_\tau\Big] = \mathrm{tr}\big[\iota^\dagger\tau_A\iota\big] = \mathrm{tr}\big[\tau_A\,\iota\iota^\dagger\big] = \mathrm{tr}[\Pi_\iota\tau_A] = \mathrm{tr}\,\tau_A = 1$. Together with positivity and block-diagonality, which χ_{Q_s} preserves, this gives $\chi_Q(\mathcal{W}^J)\subseteq\mathcal{W}$.

Onto. Fix $\omega\in\mathcal{W}$. Put $J_{0,s} \coloneqq (W_\tau^+\otimes\mathbb{1})\,\omega_s\,(W_\tau^{+\dagger}\otimes\mathbb{1}) \succeq 0$ for every s; then $\chi_{Q_s}(J_{0,s}) = \omega_s$ exactly as in Lemma 7.3 (each ω_s is supported on $\mathrm{supp}\,\tau_A\otimes\mathcal{H}(Q_s)$, which is Lemma 7.1 with $\mathcal{S}_A$ of (7.1)), and $\sum_s \mathrm{tr}_{Q_s} J_{0,s} = W_\tau^+\big(\sum_s \mathrm{tr}_{Q_s}\omega_s\big)W_\tau^{+\dagger} = W_\tau^+\tau_A W_\tau^{+\dagger} = P_\iota$. Choose one $s_0\in\mathbb{S}$ ($\mathbb{S}$ is non-empty, being the index set of a sector map, Definition "Sector maps and block-list operators" (foundations/def:sector-map)) and any state φ on $\mathcal{H}(Q_{s_0})$, and set $J_s \coloneqq J_{0,s} + \delta_{s,s_0}\,(\mathbb{1}_{A_0}-P_\iota)\otimes\varphi$. Each $J_s\succeq 0$; $\sum_s \mathrm{tr}_{Q_s} J_s = P_\iota + (\mathbb{1}_{A_0} - P_\iota) = \mathbb{1}_{A_0}$; and $W_\tau(\mathbb{1}_{A_0}-P_\iota) = 0$ kills the added block under both $\chi_{Q_{s_0}}$ and the conjugated constraint rows, so $\chi_Q(J) = \omega$ and every row of (7.9) reads the corresponding row of (5.2) at ω, which holds. Hence $(J_s)_s\in\mathcal{W}^J$ and $\chi_Q(\mathcal{W}^J) = \mathcal{W}$. Injectivity, when the frame is tight, is the lemma's. If the frame is not tight, $\mathbb{1}_{A_0} - P_\iota \neq 0$; if moreover $\mathcal{W}\neq\varnothing$ and $\dim\mathcal{H}(Q)\geq 2$, there are two different choices of (s_0,φ) — two sectors, or two states on one sector of dimension at least two — and for a fixed $\omega\in\mathcal{W}$ they give two different lists $(J_s)_s$ in $\mathcal{W}^J$, differing in the added block, with the same image ω.

The value. $O_\alpha^J = O_\alpha\circ(\mathrm{id}\times\chi_Q)$ by construction — the objective transfers by substitution of $\omega_s = \chi_{Q_s}(J_s)$ into (5.3), using Definition 4.7 for the restriction of the K-sum to key-active sectors (an identity, by Proposition 4.6) — so $\inf_{\mathcal{S}_{\mathrm{acc}}\times\mathcal{W}^J} O_\alpha^J = \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}}\inf_{\omega\in\chi_Q(\mathcal{W}^J)} O_\alpha(\mathbf{q},\omega) = \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}}\inf_{\omega\in\mathcal{W}} O_\alpha(\mathbf{q},\omega) = \underline{h}_\alpha$, which is (7.10). Only surjectivity enters this chain and injectivity does not, so a non-tight support frame changes no value.

Convexity. Each χ_{Q_s} is linear, so Theorem 5.10 applies composed with a linear map; the constraint set (7.9) is an intersection of the PSD cones with affine conditions (for $\bowtie_k\in\{\preceq,\succeq\}$, a conjugated affine-semidefinite condition), hence convex and compact — compact because $J_s\succeq 0$ with

$\sum_s \operatorname{tr} J_s = \operatorname{tr} \mathbb{1}_{A_0} = d_{A_0}$.

The general format. By Lemma 7.4 (ii), applied with the factors W_t (factors of $\tau_A^{(t)}$ by (7.4) for each sector), the assignment maps the lists satisfying the coupling constraints onto the lists over the joint sectors with positive semidefinite blocks and marginal τ_A, injectively when every $\ker W_t = 0$, which is tightness of every sector's frame. Where some $\ker W_t \neq 0$ and $\dim \mathcal{H}(Q) \geq 2$, the onto step of that lemma's proof, run for W_t with two different sector-block-diagonal states $\varphi \neq \varphi'$ (states on two sectors, or two states on one sector of dimension at least two), gives two different preimages of any point, differing by $(\mathbb{1}_{A_{0,t}} - P_{W_t}) \otimes (\varphi - \varphi')$; having the same image, they satisfy the same rows, which $\mathcal{W}^J$ reads at the image point, so at a point of a non-empty $\mathcal{W}$ both lie in $\mathcal{W}^J$. Normalization follows from the marginal condition and $\operatorname{tr} \tau_A = 1$. The rows of $\mathcal{W}^J$ are by definition the rows of (5.5) at the image point, so a list lies in $\mathcal{W}^J$ exactly when it satisfies the coupling constraints and its image lies in $\mathcal{W}$; for the lifted-operator rows the displayed conjugated form is the substitution made in the paragraph *The map*, sector by sector. Hence the image of $\mathcal{W}^J$ is $\mathcal{W}$. The value: substitute into (5.3) and use Lemma 4.9 (ii), (iii) for $K_{\hat{c}}$ and the restriction of its sum to $\mathbb{J}_K(\hat{c})$; surjectivity gives the equality of the infima as before. Convexity and compactness are as before, each coupling constraint bounding $\sum_s \operatorname{tr} J_{t,s} = d_{A_{0,t}}$. The product structure: the coupling constraint of t, the lifted-operator rows of t and the party-local rows at (t, s_{fc}) each involve only blocks $J_{t,s}$ with that t, and a linear function on a Cartesian product is a sum of linear functions of the factors, minimised factor by factor. □

On a tight support frame a homogeneous row of the Choi program may drop its conjugation by W_τ without changing the feasible set; on any frame the frame-free row only restricts the set, so the infimum over it is at least h_low. The same holds per Alice sector and for party-local rows.

In the Choi program of Theorem 7.5 every constraint row is imposed conjugated by W_τ. For a homogeneous entry the right-hand side $c_k \tau_A$ is itself $c_k W_\tau W_\tau^\dagger$ (7.4), so the conjugation can be taken off both sides. The proposition says when that changes the set $\mathcal{W}^J$ and in which direction.

Proposition 7.6 (Frame-free constraint rows on a tight frame)**.** In (i)–(iii) let the declaration be in trusted-source mode, let W_τ be (7.3) for a support frame (Definition 7.2), and for an entry k of $\mathfrak{L}$ and $(J_s)_{s \in \mathbb{S}}$ with $J_s \in \operatorname{Pos}(A_0 Q_s)$ write $Y_k := \sum_s \operatorname{tr}_{Q_s}[(\mathbb{1}_{A_0} \otimes \hat{O}_k^{(s)}) J_s]$. For a homogeneous entry, whose right-hand side is $\Theta_k = c_k \tau_A$ (Definition 2.2 (a)), call the k-th row of (7.9), $W_\tau Y_k W_\tau^\dagger \bowtie_k c_k \tau_A$, its *conjugated form*, and

$$Y_k \;\bowtie_k\; c_k \, \mathbb{1}_{A_0} \tag{7.11}$$

its *frame-free form*.

(i) If $\ker W_\tau = 0$, that is, if the frame is tight, the frame-free form of a homogeneous row holds for $(J_s)_s$ if and only if its conjugated form does;

so imposing some homogeneous rows in the frame-free form leaves $\mathcal{W}^J$ unchanged.

(ii) On every support frame the frame-free form of a homogeneous row implies its conjugated form. So the set obtained from $\mathcal{W}^J$ by imposing some homogeneous rows in the frame-free form is contained in $\mathcal{W}^J$, and the infimum in (7.10) taken over that set in place of $\mathcal{W}^J$ is at least $\underline{h}_\alpha$.

(iii) In the general format, with $\iota = \bigoplus_t \iota_t$ and W_t as in Theorem 7.5:

(a) for a homogeneous entry read through its lifted operator and every $t \in \mathbb{T}$, (i) and (ii) hold for the row of sector t, with W_t and $A_{0,t}$ in place of W_τ and A_0 and with the frame-free form $\sum_s \mathrm{tr}_{Q_s}[(\mathbb{1}_{A_{0,t}} \otimes \hat{O}_k^{\uparrow(s)})J_{t,s}] \bowtie_k c_k\, \mathbb{1}_{A_{0,t}}$;

(b) for a party-local entry, with Y_k the expression of the last line of (5.5) at J in place of ω, the unconjugated row $Y_k \bowtie_k 0$ is equivalent to the conjugated row $(W_t \otimes \mathbb{1})\, Y_k\, (W_t^\dagger \otimes \mathbb{1}) \bowtie_k 0$ if $\ker W_t = 0$, and implies it on every frame.

(iv) In untrusted-source mode there is no conjugation to remove: every row is a condition on the state variable as written.

Uses Def. 2.2, Def. 5.1, Def. 7.2, Lem. 7.3, Thm. 7.5, Def. 7.10 · *Used by* Rem. 7.7, Rem. 7.9, Req. 15.5, Req. 15.6, Lem. 20.1 · newly derived · accepted

Proof. Idea. By (7.4) the right-hand side $c_k\tau_A$ is $c_k W_\tau W_\tau^\dagger$, so the conjugated form is a congruence of the frame-free one. A congruence preserves each of $\preceq$, $=$ and $\succeq$, and on a tight frame W_τ has a left inverse that undoes it.

Step 1 (the conjugated form as a congruence). By (7.4) (Lemma 7.3 (ii)), $\Theta_k = c_k\tau_A = c_k W_\tau W_\tau^\dagger$, so the conjugated form of the k-th row is $W_\tau X W_\tau^\dagger \bowtie_k 0$ with $X \coloneqq Y_k - c_k \mathbb{1}_{A_0}$, and the frame-free form is $X \bowtie_k 0$.

Step 2 (a congruence keeps the relation; part (ii)). For any linear map V, $X \succeq 0$ implies $VXV^\dagger \succeq 0$, since $\langle v, VXV^\dagger v\rangle = \langle V^\dagger v, XV^\dagger v\rangle \geq 0$ for every v; $X \preceq 0$ is the same statement for $-X$, and $X = 0$ gives $VXV^\dagger = 0$. With $V = W_\tau$ the frame-free form implies the conjugated form, on every support frame. A list satisfying the frame-free forms of some homogeneous rows therefore satisfies their conjugated forms, and so lies in $\mathcal{W}^J$. An infimum over a subset is at least the infimum over the set, and the latter is $\underline{h}_\alpha$ by (7.10) (Theorem 7.5).

Step 3 (part (i)). If $\ker W_\tau = 0$, then W_τ is injective and has a left inverse L, $LW_\tau = \mathbb{1}_{A_0}$; for instance $L = W_\tau^+$, by (7.5) with $P_\iota = \mathbb{1}_{A_0}$ (Lemma 7.3 (iii)). Then $X = L\,(W_\tau X W_\tau^\dagger)\, L^\dagger$, and Step 2 with $V = L$ shows that the conjugated form implies the frame-free one. With Step 2 the two forms are equivalent row by row, so imposing either gives the same set.

Step 4 (part (iii)). (a) In the general format the row of sector t of a lifted-operator entry is, by Theorem 7.5, $W_t\big(\sum_s \mathrm{tr}_{Q_s}[(\mathbb{1}_{A_{0,t}} \otimes \hat{O}_k^{\uparrow(s)})J_{t,s}]\big)W_t^\dagger \bowtie_k$

$\Theta_k^{(t)}$, with $\Theta_k^{(t)} = c_k \tau_A^{(t)}$ for a homogeneous entry (5.5), and $W_t W_t^\dagger = \tau_A^{(t)}$ by (7.4) for each sector. Steps 1 to 3 apply with W_t and $A_{0,t}$ in place of W_τ and A_0, and the value of the general program is $\underline{h}_\alpha$ by the same theorem. (b) For a party-local row, Step 2 with $V = W_t \otimes \mathbb{1}$ shows that $Y_k \bowtie_k 0$ implies the conjugated row on every frame. If $\ker W_t = 0$, then W_t has a left inverse L_t, and $L_t \otimes \mathbb{1}$ is a left inverse of $W_t \otimes \mathbb{1}$; Step 3 with it gives the converse.

Step 5 (part (iv)). In untrusted-source mode the parametrisation is the identity and no W_τ is introduced (Definition 7.10 (a), (b)): the rows of (7.13), and in the general format those of (5.5), are conditions on the blocks of ω as written. □

Deciding tightness of a given τ_A is a rank decision on a computed spectrum; the diagonal frame exhibits it by construction, the full frame exhibits nothing. Read at a positive window, the diagonality test would admit the frame-free rows on a frame that is not tight.

Remark 7.7 (Tightness read off the construction of the frame). Tightness of a support frame is $\ker W_\tau = 0$, equivalently $d_{A_0} = \operatorname{rank} \tau_A$ (Lemma 7.3 (iii)), so establishing it *of a given* τ_A is a rank decision on a computed spectrum: the decision that (7.2) removes, since Lemma 7.3 and Theorem 7.5 hold on every support frame. What can be known without it is whether the construction that produced the frame exhibits $\ker W_\tau = 0$ without computing a rank.

Of the two constructions of Definition 7.2, the diagonal frame (b) does: there $\operatorname{ran} \iota = \operatorname{supp} \tau_A$ holds by construction, the index set $\{i : t_i \neq 0\}$ being fixed by a bit-level comparison of declared binary64 data, which is exact because (b) is available only for a τ_A diagonal in the declared basis, where its diagonal entries *are* its eigenvalues. This covers the case where every $t_i \neq 0$ and construction (b) returns $\iota = \mathbb{1}_A$. The full frame taken as construction (a) is built from $\mathbb{1}_A$ and from no property of τ_A at all, and therefore exhibits nothing about $\operatorname{rank} \tau_A$, however nonsingular τ_A looks.

The diagonality on which (b) rests is itself decided by comparing declared entries with zero. Read at a positive window instead, a singular τ_A with small off-diagonal entries and every $t_i \neq 0$ would pass as diagonal, and (b) would return the full frame, which is not tight. The frame-free form of Proposition 7.6 read off that construction would then be imposed on a non-tight frame, where it imposes the row on the directions of $\ker W_\tau$ as well, cutting $\mathcal{W}^J$ down to a subset, in general proper, whose image under χ_Q need not be all of $\mathcal{W}$; by Proposition 7.6 (ii) an infimum over that subset can only come out higher, the unsafe direction for a key length. For a party-local row of the general format, the unconjugated form $Y_k \bowtie_k 0$ of Proposition 7.6 (iii)(b) is equivalent to the conjugated row when $\ker W_t = 0$, and otherwise implies it and in general restricts further.

Declining the frame-free form costs nothing: the conjugated form is exact on every support frame, tight or not (Theorem 7.5), so imposing it changes no value and forgoes only the redundant directions of $\ker W_\tau$. That is why the condition may be resolved conservatively rather than decided numerically.

Uses Def. 7.2, Lem. 7.3, Thm. 7.5, Prop. 7.6 · newly derived · accepted

In trusted-source mode each Choi block J_s is the Choi operator of a completely positive branch from A_0 to the sector, input factor first, and only the branches' sum preserves trace. The reading is fixed only up to a unitary relabelling of the input, set by the declared isometry.

Remark 7.8 (Ordering convention and channel reading). In trusted-source mode each J_s is the (unnormalized) Choi operator of a CP branch $\mathcal{E}_s : A_0 \to Q_s$ in the convention with the *input* factor A_0 first and trace over the *output* equal to the identity on the input, $\sum_s \mathrm{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$; the induced action is $\mathcal{E}_s(\varrho) = \mathrm{tr}_{A_0}[(\varrho^{\mathsf{T}_{A_0}} \otimes \mathbb{1}_{Q_s})J_s]$ with the transpose taken in the declared basis of A_0, i.e. in the columns of the declared isometry ι of (7.2). This matches the convention of [2, Eqs. (82)–(83) and (85)–(86)]. The reading is interpretive only: the parametrisation of Lemma 7.3 and the program of Theorem 7.5 do not use it. It is relative to the declared ι: replacing ι by ιU for a unitary U on A_0 replaces the block representing a given ω_s by $J'_s = (U^\dagger \otimes \mathbb{1})J_s(U \otimes \mathbb{1})$ (Remark 7.9 (b)), and since $\mathrm{tr}_{A_0}[(\varrho^{\mathsf{T}} \otimes \mathbb{1})J'_s] = \mathrm{tr}_{A_0}[((U\varrho^{\mathsf{T}}U^\dagger) \otimes \mathbb{1})J_s]$ by cyclicity in the traced factor, while $(U\varrho^{\mathsf{T}}U^\dagger)^{\mathsf{T}} = \overline{U}\,\varrho\,\overline{U}^\dagger$, the branch reads $\mathcal{E}'_s(\varrho) = \mathcal{E}_s(\overline{U}\,\varrho\,\overline{U}^\dagger)$ with $\overline{U}$ the entrywise complex conjugate of U in the declared basis. The channel is therefore pinned down only up to that unitary relabelling of its input, while every quantity this module outputs is invariant (Remark 7.9 (d)); the declared ι is recorded with any reported J_s (Requirement 15.3), and Requirement 15.4 fixes the factor order and the transpose basis. The branches are generally trace-non-increasing; only their sum is trace-preserving, which is what couples the sector blocks through the single constraint in (7.9). All declared sectors, including one-dimensional classical (flag-type) sectors, are treated uniformly as output sectors of this parametrized channel.

Uses Lem. 7.3, Thm. 7.5 · *Used by* Rem. 7.9, Req. 15.3, Req. 15.4 · newly derived · accepted

Within one support frame the declared isometry is fixed up to a unitary on A_0, which carries the Choi program onto itself, so the reported estimate, h_lb and key length are invariant. Across frames only h_low and the sector report are frame-free, and h_lb ≤ h_low holds in each frame.

Remark 7.9 (Invariance under the declared basis of the Choi input). In trusted-source mode the isometry ι of (7.2) is not canonical — neither its range, the support frame, nor, within a fixed range, its columns — and this module declares no canonical choice. It needs none, and the two freedoms are settled separately. *Within one range* ι is canonical up to a unitary on A_0, and every quantity this module computes is invariant under that freedom; that is (a)–(d). *Across ranges* the programs genuinely differ — they have different numbers of variables — but their value does not: by Theorem 7.5 each maps *onto* the one set $\mathcal{W}$, whose infimum is $\underline{h}_\alpha$ by (5.4), a definition that never mentions A_0. What is frame-free across ranges is therefore the *mathematical* $\underline{h}_\alpha$ of (5.4), and not the reported numbers: the reported estimate is the objective at the returned *point*, and the reported key length is computed from $\max\{0, \underline{h}^{\mathrm{FO}}_\alpha\}$, so both are functions of the iterates and neither *is* $\underline{h}_\alpha$. Across ranges the programs carry different numbers of variables, a run in one frame is not the image of a run in the other, and those three reported numbers — the estimate, $\underline{h}^{\mathrm{FO}}_\alpha$ and the key length — may all differ. What (a)–(d) establish is the *within-range* statement, and there they are invariant, the two runs corresponding under Ad_U. What holds in every frame regardless is the inequality:

$\underline{h}_\alpha^{\mathrm{FO}} \le \underline{h}_\alpha$ by Theorem 19.1 (ii), respectively (iiib), applied in that frame, against the one frame-free $\underline{h}_\alpha$.

(a) Within one range the freedom is exactly a unitary. Let ι, ι' both satisfy (7.2) *with the same range*, write $\Pi := \iota\iota^\dagger = \iota'\iota'^\dagger$ and put $U := \iota^\dagger\iota'$. Since $\operatorname{ran}\iota' = \operatorname{ran}\Pi$ we have $\Pi\iota' = \iota'$, so $U^\dagger U = \iota'^\dagger\iota\,\iota^\dagger\iota' = \iota'^\dagger\Pi\iota' = \iota'^\dagger\iota' = \mathbb{1}_{A_0}$, and likewise $UU^\dagger = \iota^\dagger\Pi\iota = \mathbb{1}_{A_0}$: U is unitary on A_0, and $\iota U = \iota\,\iota^\dagger\iota' = \Pi\iota' = \iota'$. Conversely ιU satisfies (7.2), with the same range, for every unitary U on A_0. The two admissible Choi factors are therefore related by $W'_\tau := \tau_A^{1/2}\iota' = W_\tau U$, and every identity of Lemma 7.3 (i)–(iii) holds verbatim for either: $W'_\tau W'^\dagger_\tau = W_\tau U U^\dagger W_\tau^\dagger = \tau_A$, and $W'^+_\tau = (W_\tau U)^+ = U^\dagger W_\tau^+$ (the Moore–Penrose inverse of a product with a unitary right factor), so $W'_\tau W'^+_\tau = W_\tau W_\tau^+ = \Pi_{\operatorname{supp}\tau_A}$ and $W'^+_\tau W'_\tau = U^\dagger P_\iota U$, which is $\mathbb{1}_{A_0}$ exactly when the frame is tight. In particular $\ker W'_\tau = U^\dagger(\ker W_\tau)$ has the same dimension, so tightness is a property of the range and not of the columns.

(b) The two parametrizations differ by a relabelling. For a unitary U on A_0 write $\mathrm{Ad}_U(J) := (U\otimes\mathbb{1})\,J\,(U^\dagger\otimes\mathbb{1})$ on $\mathcal{L}(A_0B')$. From $W'_\tau = W_\tau U$,

$$\chi'_{B'} = \chi_{B'} \circ \mathrm{Ad}_U. \tag{7.12}$$

Moreover Ad_U carries $\{J \in \mathrm{Pos}(A_0B') : \mathrm{tr}_{B'}\,J = \mathbb{1}_{A_0}\}$ onto itself, with inverse $\mathrm{Ad}_{U^\dagger}$: it preserves positivity, and $\mathrm{tr}_{B'}\,\mathrm{Ad}_U(J) = U(\mathrm{tr}_{B'}\,J)U^\dagger = U\mathbb{1}_{A_0}U^\dagger = \mathbb{1}_{A_0}$. Hence Lemma 7.3 holds for either choice *with the same image* $\{\omega \in \mathrm{Pos}(AB') : \mathrm{tr}_{B'}\,\omega = \tau_A\}$, and the two parametrizations of one and the same ω are related by $J' = \mathrm{Ad}_{U^\dagger}(J)$.

(c) The Choi program is carried onto itself. Apply Ad_U blockwise to $(J_s)_s$. Positivity and the coupling constraint $\sum_s \mathrm{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$ are preserved by (b). For a constraint entry, $(\mathbb{1}_{A_0}\otimes\hat{O}_k^{(s)})$ commutes with $U\otimes\mathbb{1}$, so

$$\sum_s \mathrm{tr}_{Q_s}\left[(\mathbb{1}_{A_0}\otimes\hat{O}_k^{(s)})\,\mathrm{Ad}_U(J_s)\right] = U\Big(\sum_s \mathrm{tr}_{Q_s}\left[(\mathbb{1}_{A_0}\otimes\hat{O}_k^{(s)})\,J_s\right]\Big)U^\dagger,$$

and conjugating by W_τ gives $W_\tau U(\cdot)U^\dagger W_\tau^\dagger = W'_\tau(\cdot)W'^\dagger_\tau$: the constraint of (7.9) built with ι' and imposed on $(J_s)_s$ is the constraint built with ι and imposed on $(\mathrm{Ad}_U(J_s))_s$. So Ad_U is a bijection from $\mathcal{W}^J$ built with ι' onto $\mathcal{W}^J$ built with ι. It preserves the objective as well, because every ingredient of (7.10) is a pull-back along χ: by (7.12), $P'^J_{\hat{c}}(J) = P^J_{\hat{c}}(\mathrm{Ad}_U(J))$, $f'^J_{\hat{c},s}(J_s) = f^J_{\hat{c},s}(\mathrm{Ad}_U(J_s))$ and $K'^J_{\hat{c}}(J) = K^J_{\hat{c}}(\mathrm{Ad}_U(J))$. The two programs therefore have the same value, and $\underline{h}_\alpha$ is unchanged. (Independently: $\underline{h}_\alpha$ is *defined* by (5.4) as an infimum over $\mathcal{W}$ in the state variable ω, which never mentions A_0; Theorem 7.5 asserts that the Choi program equals that number for any admissible ι, so the invariance is already forced by the theorem.)

(d) Everything downstream is invariant or covariant. The frame-free form (7.11) of a homogeneous row (Proposition 7.6) is preserved, since $c_k\mathbb{1}_{A_0} = U(c_k\mathbb{1}_{A_0})U^\dagger$ and conjugation by a unitary preserves each of $\preceq$, $=$, $\succeq$. The branch operators $X_{\hat{c},s} = \mathcal{G}_{\hat{c},s}(\omega_s)$, the maximal branch supports $W_{\hat{c},s}$ of (8.1) and every value the objective takes are *literally* unchanged, being functions of $\omega_s = \chi_{Q_s}(J_s)$ and of $\operatorname{supp}\tau_A$ alone. The gradients of Corollary 8.5 are covariant: $W'^\dagger_\tau M_x W'_\tau = U^\dagger(W_\tau^\dagger M_x W_\tau)U$ and $\chi'^\dagger_{Q_s} = \mathrm{Ad}_{U^\dagger}\circ\chi^\dagger_{Q_s}$, so both $\Gamma^{(s')}_{\hat{c}}$ and $\Delta^{(s')}_{\hat{c}}$ transform as $G \mapsto (U^\dagger\otimes\mathbb{1})G(U\otimes\mathbb{1})$, and a Frank–Wolfe run in one declared basis is the $\mathrm{Ad}_{U^\dagger}$-image of the same run in the other. Consequently, *within one range*, the reported estimate, $\underline{h}_\alpha^{\mathrm{FO}}$ and the reported key length are independent of which columns were chosen: the two runs

correspond under Ad_U and the objective, the gradients and the pairings are carried along with them. Only the reported J_s and the channel reading of Remark 7.8 are basis-relative, and they are what the frame is recorded for (Requirement 15.3). The sector report is frame-free in the stronger sense, across ranges as well: its predicates read the declared blocks $F_v^{(s)}$, $\hat{O}_k^{(s)}$ and $\{M_x\}$ and the declared tables, and no A_0 enters any of them. *Across ranges the reported numbers are not claimed invariant*; what survives there is the mathematical $\underline{h}_\alpha$, and $\underline{h}_\alpha^{\mathrm{FO}} \le \underline{h}_\alpha$ in each frame, as the opening paragraph says.

Uses Def. 2.2, Def. 5.1, Def. 7.2, Lem. 7.3, Thm. 7.5, Prop. 7.6, Rem. 7.8 · newly derived · accepted

In untrusted-source mode the Choi step is the identity: the variables are the sector blocks of the state, coupled by the single scalar condition that their traces sum to one. At d_A = 1 this is the trusted-mode Choi program.

In untrusted-source mode there is no marginal to fix, so there is nothing to reparametrize. The program is not obtained by substituting $W_\tau = \mathbb{1}_A$ into (7.9): that substitution would give $\sum_s \mathrm{tr}_{Q_s} \omega_s = \mathbb{1}_A$, i.e. ω_A fixed to the *unnormalized* identity — an infeasible condition, and the operator form the licence does not supply (Remark 3.4 (a)). The degeneration runs the other way: W_τ is simply never introduced. Since part (a) below introduces no A_0 and no ι, the declared-basis question of Remark 7.9 does not arise at all, and no support frame is recorded in this mode (Requirement 15.3).

Definition 7.10 (The program in untrusted-source mode)**.** In untrusted-source mode the Choi step is the identity. This is the degenerate case of Theorem 7.5, not a second program:

(a) *The parametrization.* $\chi_{B'} = \mathrm{id}$, $J = \omega$, $J_s = \omega_s$; no A_0, no ι, no W_τ.

(b) *The variables and constraints.* $\mathcal{W}$ of (5.1) is already the feasible set, and by (5.2) it reads

$$\mathcal{W} = \Big\{ (\omega_s)_{s\in\mathbb{S}} : \ \omega_s \in \mathrm{Pos}(AQ_s),\ \sum_{s\in\mathbb{S}} \mathrm{tr}\,\omega_s = 1,\ \sum_{s\in\mathbb{S}} \mathrm{tr}\left[(\mathbb{1}_A \otimes \hat{O}_k^{(s)})\,\omega_s\right] \bowtie_k c_k \ \forall k \Big\}, \tag{7.13}$$

so the coupling constraint that binds the sector blocks is the single *scalar* equation $\sum_s \mathrm{tr}\,\omega_s = 1$, in place of the operator equation $\sum_s \mathrm{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$ of (7.9).

(c) *The objective.* (7.10) holds verbatim with J_s replaced by ω_s and $P_{\hat{c}}^J$, $f_{\hat{c},s}^J$, $K_{\hat{c}}^J$ by their state-form originals $P_{\hat{c}}$, $f_{\hat{c},s}$, $K_{\hat{c}}$ of (4.1): the substitution $\omega_s = \chi_{Q_s}(J_s)$ that Theorem 7.5 performs is the identity here. Hence the value is $\underline{h}_\alpha$ of (5.4) by definition rather than by a theorem, the program is jointly convex by Theorem 5.10 directly, and the feasible set is convex and compact by Definition 5.1.

(d) *Consistency at $d_A = 1$.* There trusted-source mode admits only $\tau_A = [1]$, the one state on a one-dimensional A, and its Choi parametrization degenerates: $d_{A_0} = \operatorname{rank} \tau_A = 1$, $\iota = [1]$, $W_\tau = [1]$, $\chi_{B'} = \mathrm{id}$ and $J_s = \omega_s$. So $\mathbb{1}_{A_0} = [1]$ and $\sum_s \operatorname{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$ *is* $\sum_s \operatorname{tr} \omega_s = 1$, the constraint rows agree because every 1×1 Hermitian Θ_k is $c_k \tau_A$ with $c_k = \Theta_k$ (Proposition 5.2 (iii)), and the two feasible sets of (5.1) coincide (Proposition 5.2 (i)): (7.9) and (7.13) are the same set, and the two modes give literally the same program.

Uses Thm. 4.1, Def. 5.1, Prop. 5.2, Thm. 5.10, Thm. 7.5 · *Used by* Prop. 7.6, Lem. 8.2, Cor. 8.5, Lem. 8.6, Lem. 9.6, Def. 18.1, Cor. 19.6, Lem. 20.1 · newly derived · accepted

The objective at any exactly feasible point, a state or the Choi operators of Theorem 7.5 that parametrise one, is at least the value h_low of the program. It warrants the estimate every run reports, a solver diagnostic and never the lower bound a key length needs.

The number a run reports as its estimate is the objective at the point its optimizer returned. What that number is worth is one line, and it is a property of the program, not of any method that searches it. The membership $x \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ is exactly what must be checked for this warrant to hold, and which check establishes it is the solver submodule's claim, not the core's. Definition 2.5 is not needed: nothing in the lemma reads convexity or closedness, and at an empty $\mathcal{S}_{\mathrm{acc}}$ the statement is vacuous. In either form the number is an *upper* estimate of $\underline{h}_\alpha$ and never a bound on it: nothing here says how far above $\underline{h}_\alpha$ it lies.

Lemma 7.11 (Upper estimate of the value at any feasible point)**.** Let $\mathfrak{D}$ satisfy Definition 1.2, let $\mathcal{S}_{\mathrm{acc}}$ be any set of probability vectors on the acceptance alphabet, and let $\alpha \in (1, \infty)$.

(i) *(State form.)* For every $(\mathbf{q}, \omega) \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}$,

$$O_\alpha(\mathbf{q}, \omega) \;\geq\; \underline{h}_\alpha. \tag{7.14}$$

(ii) *(Choi form.)* Write $O_\alpha^J(\mathbf{q}, J) := O_\alpha(\mathbf{q}, \chi_Q(J))$ for the objective of the Choi program (7.10) of Theorem 7.5, with χ_Q the map of that theorem. For every $x = (\mathbf{q}, J) \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$,

$$O_\alpha^J(x) \;\geq\; \underline{h}_\alpha. \tag{7.15}$$

Uses Def. 1.2, Def. 5.1, Thm. 7.5 · *Used by* Thm. 19.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.objective:objective_value

Proof. (i) $\underline{h}_\alpha$ is by (5.4) the infimum of O_α over $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}$, and $(\mathbf{q}, \omega)$ is a point of that set; an infimum is a lower bound on every value it is taken over.

(ii) By Theorem 7.5, χ_Q maps $\mathcal{W}^J$ onto $\mathcal{W}$ and the value of the Choi program is $\underline{h}_\alpha$; so for $x = (\mathbf{q}, J) \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ the pair $(\mathbf{q}, \chi_Q(J))$ lies in $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}$ and $O_\alpha^J(x) = O_\alpha(\mathbf{q}, \chi_Q(J)) \geq \underline{h}_\alpha$ by (i). □

8 Gradients and the limit of order one

The maximal branch support is the span of the pass-class Kraus images of the admissible Alice subspace tensored with the sector; every branch operator the program forms lies in it, with equality at a positive definite Choi block (Lemma 8.2). On the face of a singular operator the pinched trace power has a gradient in split form, with divided differences over the positive spectrum and pinched operators on the ambient space (Theorem 8.4). The full objective is differentiable along every feasible direction at a point of maximal support with positive probabilities (Corollary 8.5); its branch values and pulled-back gradient depend on the Kraus factors only through their Gram matrices, so exact factors may replace the square roots (Lemma 8.6). As the Rényi order tends to one the value converges to a blockwise relative-entropy program when every acceptable distribution gives each key-live value probability at least some $p_0 > 0$ and some feasible state reproduces an acceptable distribution (Theorem 8.9), a validation reference and never a bound.

Imports foundations/def:kl, foundations/def:pinching, foundations/fact:concavity, foundations/lem:kernel, foundations/lem:matrix-function-perturbation

The maximal branch support of a key-live acceptance value and a sector is the span of the pass-class Kraus images of the admissible Alice subspace tensored with the sector. The side-by-side matrix of those Kraus operators depends on the declaration and the support frame alone.

A branch operator of a key-live acceptance value and a sector cannot reach beyond a subspace that the declaration fixes. That subspace is the face on which Theorem 8.4 differentiates the pinched trace power, and Lemma 8.2 records how the branch operators the program forms sit inside it.

Definition 8.1 (Maximal branch support)**.** For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and $s \in \mathbb{S}$ the *maximal branch support* is the subspace

$$W_{\hat{c},s} \coloneqq \sum_{\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})} K_\kappa^{(s)}\big(\mathcal{S}_A \otimes \mathcal{H}(Q_s)\big) \;\subseteq\; \mathcal{H}(R\,XY\,C_A C_B\,AQ_s), \tag{8.1}$$

the span of the images, under the Kraus operators of $\mathcal{G}_{\hat{c},s}$ (Definition 1.6), of the admissible Alice subspace $\mathcal{S}_A$ of (7.1) tensored with the sector, that is, of the subspace the declaration admits: $\operatorname{supp}\tau_A$ in trusted-source mode, all of $\mathcal{H}(A)$ in untrusted-source mode.

Listing the operators of (8.1) side by side, with W_τ of (7.3) on the support frame of Definition 7.2, put

$$\mathbf{K}_{\hat{c},s} \coloneqq \Big[\, K_\kappa^{(s)}\,(W_\tau \otimes \mathbb{1}_{Q_s}) \,\Big]_{\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})} \;:\; \bigoplus_{\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})} \mathcal{H}(A_0 Q_s) \;\longrightarrow\; \mathcal{H}(R\,XY\,C_A C_B\,AQ_s), \tag{8.2}$$

with $W_\tau \otimes \mathbb{1}_{Q_s}$ replaced by $\mathbb{1}_{AQ_s}$, and the domain by $\bigoplus_\kappa \mathcal{H}(AQ_s)$, in untrusted-source mode. The matrix $\mathbf{K}_{\hat{c},s}$ depends on the declaration and on W_τ alone and on no iterate.

Uses Def. 1.6, Lem. 7.1, Def. 7.2 · *Used by* Lem. 8.2, Thm. 8.4, Cor. 8.5, Req. 15.7, Lem. 18.3, Thm. 19.1, Cor. 19.6, Rem. 23.4 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.branch:branch_support_of

Every branch operator the program forms is supported in the maximal branch support, with equality at a positive definite Choi block; the support is the range of the stacked Kraus matrix. It is always smaller than the ambient space and in general not invariant under the key pinching.

Four facts about the maximal branch support are used by the gradient of Theorem 8.4 and by the first-order argument of the Frank–Wolfe submodule: it bounds the support of every branch operator the program forms, it is the range of the stacked Kraus matrix, it is always strictly smaller than the ambient space, and it is in general not invariant under the key pinching. The bound in (ii) is not a number a run computes: $\operatorname{rank}\tau_A$ enters it only as a mathematical quantity. Hypothesis (H-SUPP) of Theorem 19.1 compares a rank with $\dim W_{\hat{c},s}$, and how that comparison is decided is Requirement 25.2.

Lemma 8.2 (Properties of the maximal branch support)**.** Let $\hat{c} \in \widehat{\mathcal{C}}_{\text{key}}$ and $s \in \mathbb{S}$, and let $W_{\hat{c},s}$ and $\mathbf{K}_{\hat{c},s}$ be as in Definition 8.1.

(i) In trusted-source mode every $\omega_s = \chi_{Q_s}(\Delta)$ with $\Delta \in \operatorname{Pos}(A_0 Q_s)$, in particular every feasible ω_s and the image of the positive and the negative part of every Hermitian direction, satisfies $\operatorname{supp}\omega_s = (W_\tau \otimes \mathbb{1})(\operatorname{supp}\Delta) \subseteq \operatorname{supp}\tau_A \otimes \mathcal{H}(Q_s) = \mathcal{S}_A \otimes \mathcal{H}(Q_s)$; in untrusted-source mode $\mathcal{S}_A = \mathcal{H}(A)$ and the inclusion holds for every $\omega_s \in \operatorname{Pos}(AQ_s)$. In either mode, therefore,

$$\operatorname{supp}\mathcal{G}_{\hat{c},s}(\omega_s) \subseteq W_{\hat{c},s}, \tag{8.3}$$

with equality whenever ω_s attains that maximal support, $\operatorname{supp}\omega_s = \mathcal{S}_A \otimes \mathcal{H}(Q_s)$. In trusted-source mode $J_s \succ 0$ for the Choi block suffices for it, and on a tight support frame is equivalent to it; in untrusted-source mode it is $\omega_s \succ 0$ on $\mathcal{H}(AQ_s)$, which is $J_s \succ 0$, as $J_s = \omega_s$ there (Definition 7.10 (a)).

(ii) $W_{\hat{c},s}$ is the range of one matrix, and its dimension is bounded:

$$W_{\hat{c},s} = \operatorname{ran}\mathbf{K}_{\hat{c},s}, \qquad \dim W_{\hat{c},s} = \operatorname{rank}\mathbf{K}_{\hat{c},s} \;\le\; \left|\mathcal{K}_{\text{pass}}(\hat{c})\right| \cdot \dim \mathcal{S}_A \cdot d_s, \tag{8.4}$$

where $\dim\mathcal{S}_A = \operatorname{rank}\tau_A$ in trusted-source mode and d_A in untrusted-source mode.

(iii) $\dim W_{\hat{c},s}$ is *strictly* smaller than $\dim\mathcal{H}(R\,XY\,C_A C_B\,AQ_s)$, so 0 always lies in the ambient spectrum of a branch operator $X_{\hat{c},s} = \mathcal{G}_{\hat{c},s}(\omega_s)$ of (i)

and "maximal support" is never the statement that $X_{\hat{c},s} \succ 0$ on the ambient space.

(iv) $W_{\hat{c},s}$ is *not* in general $\mathcal{Z}$-invariant.

Uses Def. 1.2, Def. 1.6, Lem. 7.1, Lem. 7.3, Def. 7.10, Def. 8.1 · *Used by* Thm. 8.4, Cor. 8.5, Req. 15.8, Lem. 18.3, Thm. 19.1, Cor. 19.6, Cor. 19.7, Rem. 19.8, Lem. 21.1, Rem. 22.5, Rem. 23.4, Rem. 23.5, Req. 25.2, Req. 28.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.gradient:is_support_maximal

Proof. Support of a sum. For $\varrho \succeq 0$ the support of $\sum_\kappa K_\kappa \varrho K_\kappa^\dagger$ is $\sum_\kappa K_\kappa(\operatorname{supp}\varrho)$: a vector $|w\rangle$ in the kernel of the sum has $\langle w| K_\kappa \varrho K_\kappa^\dagger |w\rangle = 0$, hence $\varrho^{1/2} K_\kappa^\dagger |w\rangle = 0$, for every κ, and conversely each summand has range in $K_\kappa(\operatorname{supp}\varrho)$.

(i) In trusted-source mode, $\operatorname{supp}\omega_s = (W_\tau \otimes \mathbb{1})(\operatorname{supp}\Delta)$ by the support of a sum with the single operator $W_\tau \otimes \mathbb{1}$; it lies in $\operatorname{ran} W_\tau \otimes \mathcal{H}(Q_s)$, and $\operatorname{ran} W_\tau = \operatorname{supp}\tau_A$ (Lemma 7.3 (iii)), which is $\mathcal{S}_A$ by (7.1). In untrusted-source mode $\mathcal{S}_A = \mathcal{H}(A)$ and the inclusion is trivial. The support of a sum then gives $\operatorname{supp}\mathcal{G}_{\hat{c},s}(\omega_s) = \sum_\kappa K_\kappa^{(s)}(\operatorname{supp}\omega_s) \subseteq \sum_\kappa K_\kappa^{(s)}(\mathcal{S}_A \otimes \mathcal{H}(Q_s)) = W_{\hat{c},s}$, with equality where $\operatorname{supp}\omega_s = \mathcal{S}_A \otimes \mathcal{H}(Q_s)$. That is the case when $J_s \succ 0$, since then $\operatorname{supp}\omega_s = \operatorname{ran} W_\tau \otimes \mathcal{H}(Q_s)$, and when $\omega_s \succ 0$ on $\mathcal{H}(AQ_s)$ in untrusted-source mode. On a tight frame the converse holds as well: W_τ is then injective (Lemma 7.3 (iii)), so $W_\tau \otimes \mathbb{1}$ maps $\operatorname{supp} J_s$ injectively onto $\operatorname{supp}\omega_s$, and $\operatorname{supp}\omega_s = \operatorname{ran} W_\tau \otimes \mathcal{H}(Q_s)$, of dimension $d_{A_0} d_s$, forces $\operatorname{supp} J_s = \mathcal{H}(A_0 Q_s)$, that is $J_s \succ 0$.

(ii) The range of a side-by-side listing is the sum of the ranges of its blocks, and $\operatorname{ran}(W_\tau \otimes \mathbb{1}_{Q_s}) = \operatorname{ran} W_\tau \otimes \mathcal{H}(Q_s) = \mathcal{S}_A \otimes \mathcal{H}(Q_s)$ by $\operatorname{ran} W_\tau = \operatorname{supp}\tau_A$ (Lemma 7.3 (iii)), so $\operatorname{ran}\mathbf{K}_{\hat{c},s} = \sum_\kappa K_\kappa^{(s)}(\mathcal{S}_A \otimes \mathcal{H}(Q_s))$, which is (8.1). The rank of a side-by-side listing is at most the sum of the ranks of its blocks, and each block has rank at most $\operatorname{rank}(W_\tau \otimes \mathbb{1}_{Q_s}) = \dim \mathcal{S}_A \cdot d_s$, which gives the inequality. In untrusted-source mode the same two steps apply with $\mathbb{1}_{AQ_s}$ in place of $W_\tau \otimes \mathbb{1}_{Q_s}$.

(iii) For a pass class κ of $\hat{c}$ every $(x, v) \in a^{-1}(\kappa)$ has $r(x, v) \in \mathcal{Z}$, by announcement-determined sifting and Definition 1.2 (iv), (v); so every Kraus operator $K_\kappa^{(s)}$ of (8.1) carries an R-ket $|j\rangle_R$ with $j \in \mathcal{Z}$, and $W_{\hat{c},s} \subseteq \operatorname{span}\{|j\rangle_R : j \in \mathcal{Z}\} \otimes \mathcal{H}(XY\, C_A C_B\, AQ_s)$. The nonzero vectors $|\perp\rangle_R \otimes \xi$ are orthogonal to that subspace, so $\dim W_{\hat{c},s}$ is smaller than the ambient dimension. By (i), $\operatorname{supp} X_{\hat{c},s} \subseteq W_{\hat{c},s}$, so those vectors lie in the ambient kernel of $X_{\hat{c},s}$, and 0 is an eigenvalue of $X_{\hat{c},s}$ on the ambient space.

(iv) The declaration is free to let the raw-key value $r(x, v)$ vary with x inside one announcement class, since Definition 1.2 (viii) constrains only that the key holder can compute it, and then a single Kraus operator (1.18) has range vectors spread over several R-eigenvalues. Z_j applied to such a vector $K_\kappa^{(s)}|\psi\rangle$ retains only the terms with $r(x, v) = j$, which is in general not $K_\kappa^{(s)}|\psi'\rangle$ for any $|\psi'\rangle$, nor in the range of any other Kraus operator (the classes κ occupy orthogonal $C_A C_B$ sectors). An instance: $d_A = 1$,

one Alice outcome x, $Q = \mathbb{C}$ with one sector s, two outcomes v_0, v_1 with $F_{v_0} = F_{v_1} = \frac{1}{2}$, both outcome pairs in one announcement class κ, which passes sifting, and the key held by Bob with $r(x, v_0) = 0$ and $r(x, v_1) = 1$, a function of his readout; let $\hat{c} \coloneqq \phi(\kappa)$ have no other pass class. Then $W_{\hat{c},s}$ is the line spanned by $\big(\,|0\rangle_R\,|x, v_0\rangle + |1\rangle_R\,|x, v_1\rangle\,\big)\,|\kappa\rangle$, and Z_0 maps that vector to $|0\rangle_R\,|x, v_0\rangle\,|\kappa\rangle$, which is not on the line. □

At a Hermitian operator, a matrix function continuously differentiable around its spectrum has as derivative its divided-difference kernel between spectral projectors, self-adjoint under the trace pairing. At trace level only the diagonal kernel survives.

Theorem 8.4 differentiates $X \mapsto X^p$ inside the support of a branch operator with the kernel below, and Lemma 8.8 uses its trace form.

Lemma 8.3 (Divided-difference derivative of a matrix function). Write $p \coloneqq 1/\alpha \in (0, 1)$ and T for the functional (4.7) of Lemma 4.3 on the positive semidefinite cone of a branch space; $\log_2 \mathrm{e} = 1/\ln 2$.

Let X be Hermitian with spectral decomposition $X = \sum_i \lambda_i P_i$, and let f be continuously differentiable on an open interval $I \supseteq \operatorname{spec} X$. Then $X' \mapsto f(X')$, defined on the Hermitian operators with spectrum in I, is Fréchet differentiable at X with

$$Df(X)[H] = \sum_{i,j} f^{[1]}(\lambda_i, \lambda_j)\, P_i H P_j, \qquad f^{[1]}(\lambda, \mu) \coloneqq \begin{cases} \dfrac{f(\lambda) - f(\mu)}{\lambda - \mu}, & \lambda \neq \mu, \\ f'(\lambda), & \lambda = \mu, \end{cases} \tag{8.5}$$

and for Hermitian A, H: $\operatorname{tr}\big[A\, Df(X)[H]\big] = \operatorname{tr}\big[Df(X)[A]\, H\big]$. In particular this applies to $f(x) = x^p$ at every $X \succ 0$, with $I = (0, \infty)$; and, at the trace level, for every Hermitian Y and every g continuously differentiable on an open interval containing $\operatorname{spec} Y$, the map $Y' \mapsto \operatorname{tr} g(Y')$ is Fréchet differentiable at Y with

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \operatorname{tr}\, g(Y + tK) = \operatorname{tr}\big[g'(Y)K\big] \qquad \text{for every Hermitian } K. \tag{8.6}$$

For $\alpha > 1$ the function $g(y) \coloneqq \max\{y, 0\}^\alpha$ is continuously differentiable on $\mathbb{R}$, with $g'(y) = \alpha \max\{y, 0\}^{\alpha-1}$, and $\operatorname{tr} g(Y) = \operatorname{tr}[Y^\alpha]$ for every $Y \succeq 0$.

Used by Thm. 8.4, Lem. 8.8 · newly derived · implemented · *Code* certiqs_key. meat_core.objective.linalg:divided_difference · *Oracles* registry.py

Proof. Polynomials. For a monomial $f(x) = x^m$, $\frac{\mathrm{d}}{\mathrm{d}t}(X + tH)^m\big|_0 = \sum_{a+b=m-1} X^a H X^b$, and $\langle i| \sum_{a+b=m-1} X^a H X^b\, |j\rangle = H_{ij} \sum_{a+b=m-1} \lambda_i^a \lambda_j^b = H_{ij}\, f^{[1]}(\lambda_i, \lambda_j)$, which is (8.5); linearity extends it to polynomials. A polynomial map $X' \mapsto q(X')$ is a finite sum of products of X' with itself, hence Fréchet differentiable at every X, and its Fréchet derivative is the directional derivative just computed.

A Hilbert–Schmidt bound. Let h be continuously differentiable on a compact interval $[c, C]$, and let $A = \sum_i a_i P_i$ and $B = \sum_j b_j P'_j$ be Hermitian

with spectra in $[c,C]$. Then $P_i\big(h(A)-h(B)\big)P'_j=(h(a_i)-h(b_j))P_iP'_j$ and $P_i(A-B)P'_j=(a_i-b_j)P_iP'_j$, so $h(A)-h(B)=\sum_{i,j}h^{[1]}(a_i,b_j)\,P_i(A-B)P'_j$, the terms with $a_i=b_j$ being 0 on both sides. The maps $Z\mapsto P_iZP'_j$ are orthogonal projections of the Hilbert–Schmidt space onto mutually orthogonal subspaces that sum to it, and $|h^{[1]}|\le\|h'\|_\infty$ on $[c,C]^2$ by the mean value theorem, so $\|h(A)-h(B)\|_2\le\|h'\|_\infty\|A-B\|_2$; in the same way $\|\sum_{i,j}h^{[1]}(\lambda_i,\lambda_j)P_iHP_j\|_2\le\|h'\|_\infty\|H\|_2$.

General f. Choose a compact $[c,C]\subset I$ and $\delta>0$ with $\lambda_i\pm\delta\in[c,C]$ for every i. By Weyl's inequality every eigenvalue of $X+H$ lies within $\|H\|$ of one of X, so $\operatorname{spec}(X+H)\subseteq[c,C]$ whenever $\|H\|_2\le\delta$. Pick polynomials $q_n\to f'$ uniformly on $[c,C]$ (Weierstrass) and let Q_n be primitives with $Q_n(c)=f(c)$, so that $\varepsilon_n:=\sup_{[c,C]}|f'-Q'_n|\to0$. Write $L(H):=\sum_{i,j}f^{[1]}(\lambda_i,\lambda_j)P_iHP_j$ and L_n for the same expression with Q_n, which is $DQ_n(X)$ by the polynomial step. For $\|H\|_2\le\delta$ the Hilbert–Schmidt bound, applied to $h=f-Q_n$, whose divided differences are those of f less those of Q_n, gives

$$\big\|\big(f(X+H)-f(X)-L(H)\big)-\big(Q_n(X+H)-Q_n(X)-L_n(H)\big)\big\|_2\;\le\;2\varepsilon_n\|H\|_2.$$

Given $\eta>0$, fix n with $2\varepsilon_n\le\eta/2$ and then $\delta'\in(0,\delta]$ with $\|Q_n(X+H)-Q_n(X)-L_n(H)\|_2\le\frac{\eta}{2}\|H\|_2$ for $\|H\|_2\le\delta'$; then $\|f(X+H)-f(X)-L(H)\|_2\le\eta\|H\|_2$ for $\|H\|_2\le\delta'$. The estimate holds uniformly in H on that ball, so f is Fréchet differentiable at X with $Df(X)=L$, which is (8.5).

Self-adjointness. With $k_{ij}:=f^{[1]}(\lambda_i,\lambda_j)=k_{ji}$ real, $\operatorname{tr}\Big[A\sum_{ij}k_{ij}P_iHP_j\Big]=\sum_{ij}k_{ij}\operatorname{tr}[P_jAP_iH]=\operatorname{tr}\big[\big(\sum_{ij}k_{ji}P_iAP_j\big)H\big]=\operatorname{tr}[Df(X)[A]\,H]$.

Trace level. The general step applies to g at $Y=\sum_i\mu_iP_i$, so $Y'\mapsto g(Y')$ is Fréchet differentiable at Y with derivative $\sum_{i,j}g^{[1]}(\mu_i,\mu_j)P_iKP_j$, and so is its composition with the linear trace. Only the diagonal kernel survives the trace: $\operatorname{tr}[P_iKP_j]=\delta_{ij}\operatorname{tr}[P_iK]$, so $\frac{\mathrm d}{\mathrm dt}\operatorname{tr}g(Y+tK)\big|_0=\sum_i g'(\mu_i)\operatorname{tr}[P_iK]=\operatorname{tr}[g'(Y)K]$. For $\alpha>1$ the function $g(y)=\max\{y,0\}^\alpha$ has derivative $\alpha y^{\alpha-1}$ at $y>0$, 0 at $y<0$, and 0 at $y=0$, since $|g(y)|/|y|\le|y|^{\alpha-1}\to0$; that derivative is continuous because $\alpha-1>0$; and g is y^α on $[0,\infty)$, so $\operatorname{tr}g(Y)=\operatorname{tr}[Y^\alpha]$ for $Y\succeq0$.

□

The pinched trace power is differentiable at a positive definite operator, and on the face of a singular one in split form: divided differences over the positive spectrum, pinched operators on the ambient space. A one-sided derivative is $+\infty$ only along a direction that enlarges the support.

The key-active branch operators are in general singular, and their maximal branch support is in general not invariant under the key pinching (Lemma 8.2 (iv)); part (i) is therefore stated so that no such invariance is needed, and Requirement 15.7 is the rule it gives an implementation.

Part (ii) is why gradients are taken at points of maximal support (Requirement 15.8). At maximal support the split form gives the true one-sided derivative in every feasible direction, because there no feasible direction enlarges the support; that is Lemma 18.3 of the Frank–Wolfe submodule, where Theorem 19.1 consumes it, and Remark 19.4 explains why the explicit-perturbation route has no accounted instance.

Theorem 8.4 (Gradient of the pinched trace power on a face)**.** Let $\alpha \in (1,\infty)$, $p = 1/\alpha$, let T be the pinched trace power (4.7) on the positive semidefinite cone of a branch space (Lemma 8.3), and let $X \succ 0$. Then T is Fréchet differentiable at X with gradient

$$\nabla T(X) = \alpha\, DX^p\Big[\mathcal{Z}\big(Y^{\alpha-1}\big)\Big], \qquad Y \coloneqq \mathcal{Z}(X^p), \tag{8.7}$$

i.e. $\frac{\mathrm{d}}{\mathrm{d}t}T(X+tH)\big|_0 = \operatorname{tr}[\nabla T(X)H]$, where $DX^p[\cdot]$ is the self-adjoint map (8.5) with $f(x) = x^p$ and $Y^{\alpha-1}$ is defined by functional calculus ($0^{\alpha-1} \coloneqq 0$; Y may be singular). Explicitly, in the eigenbasis $X = \sum_i \lambda_i P_i$:

$$\nabla T(X) = \alpha \sum_{i,j} \frac{\lambda_i^p - \lambda_j^p}{\lambda_i - \lambda_j}\, P_i\, \mathcal{Z}\big(Y^{\alpha-1}\big)\, P_j \qquad \text{(diagonal kernel } p\lambda_i^{p-1}\text{)}. \tag{8.8}$$

Moreover:

(i) *(faces)* Let $X \succeq 0$, write $S \coloneqq \operatorname{supp} X$, and let H be Hermitian with $\operatorname{supp} H \subseteq S$. Then $t \mapsto T(X+tH)$ is differentiable at $t = 0$, and (8.7)–(8.8) hold in the following *split* form:

  (a) the divided-difference map $DX^p[\cdot]$ is compressed to S, equivalently, the sum in (8.8) runs over the strictly positive eigenvalues of X only, where X is positive definite and the kernel $f^{[1]}$ is finite;

  (b) $Y = \mathcal{Z}(X^p)$, $Y^{\alpha-1}$ and $\mathcal{Z}(Y^{\alpha-1})$ are formed on the *ambient* branch space and are *not* compressed to S.

  The resulting $\nabla T(X)$ determines the derivative in every direction supported in S; its components outside S are neither determined by this nor needed, support-enlarging directions being the subject of (ii). Compressing the *whole* computation to a subspace S' is licensed when $\sum_{j\in\mathcal{Z}_\perp} Z_j S \subseteq S'$, in particular when $S \subseteq S'$ and S' is $\mathcal{Z}$-invariant ($Z_j S' \subseteq S'$ for every $j \in \mathcal{Z}_\perp$): then the ambient complement contributes 0 to T, and the compressed and ambient computations agree. Compressing to S itself is licensed *only* when S is $\mathcal{Z}$-invariant; otherwise $\mathcal{Z}$ transports weight out of S and the compressed value of T is strictly smaller (the proof gives the argument and a rank-one witness). For the branch operators $X = \mathcal{G}_{\hat{c},s}(\omega_s)$, whose faces are in general not $\mathcal{Z}$-invariant, the relevant face is $S = W_{\hat{c},s}$ of (8.1) at maximal support, and (a)–(b) is what makes the gradient available there.

(ii) *(one-sided derivatives)* For $X \succeq 0$ and any Hermitian H with $X+tH \succeq 0$ for small $t > 0$, the one-sided directional derivative $\lim_{t\downarrow 0} \frac{T(X+tH)-T(X)}{t}$ exists in $(-\infty, +\infty]$, and can equal $+\infty$ only when H strictly enlarges the support of X.

Uses Lem. 1.7, Cor. 4.2, Lem. 4.3, Def. 8.1, Lem. 8.2, Lem. 8.3, foundations/fact:concavity · *Used by* Cor. 8.5, Lem. 8.6, Rem. 10.20, Req. 15.7, Req. 15.8, Lem. 18.3, Thm. 19.1, Rem. 19.3, Rem. 19.4, Cor. 19.6, Lem. 21.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.gradient:state_gradient · *Oracles* registry.py

Proof. Idea. At $X \succ 0$ the gradient is the chain rule through $X \mapsto X^p$ (Lemma 8.3), the linear pinching and $Y \mapsto \operatorname{tr}[Y^\alpha]$. On a face the inner map is differentiated inside the support, where X is positive definite, while the outer map is differentiable at every $Y \succeq 0$ and so stays on the ambient space. Concavity of T locates the only infinite one-sided derivatives.

Step 1 (differentiability at $X \succ 0$). $X \mapsto X^p$ is Fréchet differentiable at $X \succ 0$ (Lemma 8.3 with $f = x^p$ on $I = (0, \infty)$); $\mathcal{Z}$ is linear; and with $g(y) = \max\{y, 0\}^\alpha$ of Lemma 8.3, continuously differentiable on $\mathbb{R}$ because $\alpha > 1$, the map $Y \mapsto \operatorname{tr} g(Y)$, which is $\operatorname{tr}[Y^\alpha]$ on the positive semidefinite operators, is Fréchet differentiable at every $Y \succeq 0$ with derivative $K \mapsto \alpha \operatorname{tr}\big[Y^{\alpha-1}K\big]$ by (8.6). For $|t|$ small $X + tH \succ 0$, so $T(X + tH) = \operatorname{tr} g\big(\mathcal{Z}((X + tH)^p)\big)$. The chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t}T(X+tH)\Big|_0 = \alpha \operatorname{tr}\left[Y^{\alpha-1}\,\mathcal{Z}\big(DX^p[H]\big)\right] = \alpha \operatorname{tr}\left[\mathcal{Z}\big(Y^{\alpha-1}\big)\,DX^p[H]\right] = \operatorname{tr}\left[\alpha\, DX^p\big[\mathcal{Z}(Y^{\alpha-1})\big]\,H\right],$$

using self-adjointness of the pinching and then of DX^p (Lemma 8.3). (8.8) is (8.5) written out.

Step 2 (faces, part (i)). Write $S = \operatorname{supp} X$ and let Π_S be its projector. For Hermitian H with $\operatorname{supp} H \subseteq S$ we have $X + tH = \Pi_S(X + tH)\Pi_S$, and X is positive definite on S, so for $|t|$ small the compression $(X + tH)|_S$ is positive definite and $(X+tH)^p = \big((X+tH)|_S\big)^p \oplus 0$ by the convention $0^p = 0$ of Definition "Operator conventions and pinching" (foundations/def:pinching). Lemma 8.3, applied inside $\mathcal{L}(S)$ with $f(x) = x^p$ on $I = (0, \infty)$, which contains $\operatorname{spec}(X|_S)$, gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(X + tH)^p\Big|_0 = \sum_{i,j:\ \lambda_i,\lambda_j>0} \frac{\lambda_i^p - \lambda_j^p}{\lambda_i - \lambda_j}\, P_i H P_j =: D_S X^p[H], \tag{8.9}$$

a self-adjoint map on the Hermitian operators supported in S. No term with $\lambda_i = 0$ or $\lambda_j = 0$ occurs, which removes the divergence of the kernel at the origin; this is (a). The outer factor is untouched by that compression: with $g(y) = \max\{y, 0\}^\alpha$ as in Step 1, (8.6) makes $Y \mapsto \operatorname{tr} g(Y)$, which is $\operatorname{tr}[Y^\alpha]$ on the positive semidefinite operators, differentiable at every $Y \succeq 0$,

in particular at the *ambient* $Y = \mathcal{Z}(X^p)$, which is not supported in S unless S is $\mathcal{Z}$-invariant. This is (b). Composing the two,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}T(X+tH)\Big|_0 &= \alpha\,\mathrm{tr}\left[Y^{\alpha-1}\,\mathcal{Z}\big(D_S X^p[H]\big)\right] = \alpha\,\mathrm{tr}\left[\mathcal{Z}\big(Y^{\alpha-1}\big)\,D_S X^p[H]\right]\\ &= \mathrm{tr}\left[\alpha\,D_S X^p\big[\mathcal{Z}(Y^{\alpha-1})\big]\,H\right],\end{aligned}$$

by self-adjointness of the pinching and then of $D_S X^p$ (Lemma 8.3 inside $\mathcal{L}(S)$; only $\Pi_S\mathcal{Z}(Y^{\alpha-1})\Pi_S$ enters, because H is supported in S).

Step 3 (compression). Put $S_{\mathcal{Z}} \coloneqq \sum_{j\in\mathcal{Z}_\perp} Z_j S$. It contains S, because $\sum_j Z_j = \mathbb{1}$, and it is $\mathcal{Z}$-invariant, because $Z_i Z_j = \delta_{ij} Z_i$; and for every W supported in $S_{\mathcal{Z}}$, $\mathcal{Z}(W) = \sum_j Z_j W Z_j$ is supported in $\sum_j Z_j S_{\mathcal{Z}} = S_{\mathcal{Z}}$. With X^p supported in S, therefore, $Y = \mathcal{Z}(X^p)$, $Y^{\alpha-1}$ and $\mathcal{Z}(Y^{\alpha-1})$ are supported in $S_{\mathcal{Z}}$. Let $S_{\mathcal{Z}} \subseteq S'$, which holds in particular when $S \subseteq S'$ and $Z_j S' \subseteq S'$ for all j. Compressing the pinching to S', $W \mapsto \Pi_{S'}\mathcal{Z}(W)\Pi_{S'}$, then changes none of these operators, so every trace above may be taken inside S' and the ambient complement contributes 0: the compressed and ambient computations agree.

If S is not $\mathcal{Z}$-invariant, compressing to S itself changes T already. Then $S_{\mathcal{Z}} \neq S$, and $\operatorname{supp} Y = S_{\mathcal{Z}}$: a vector $|w\rangle$ with $\langle w| Z_j X^p Z_j |w\rangle = 0$ for every j has $X^{p/2} Z_j |w\rangle = 0$, hence $Z_j|w\rangle \perp S$, for every j, and conversely each $Z_j X^p Z_j$ has range in $Z_j S$. So Y is not supported in S, and the compressed value $\mathrm{tr}[(\Pi_S Y \Pi_S)^\alpha]$ is strictly smaller than $T(X) = \mathrm{tr}[Y^\alpha]$: for a unit eigenvector $u \in S$ of $\Pi_S Y \Pi_S$, with eigenvalue $\mu = \langle u| Y |u\rangle$, Jensen's inequality for the convex $x \mapsto x^\alpha$ and the spectral distribution of Y in u gives $\mu^\alpha \le \langle u| Y^\alpha |u\rangle$; summing over an orthonormal eigenbasis of S gives $\mathrm{tr}[(\Pi_S Y \Pi_S)^\alpha] \le \mathrm{tr}[\Pi_S Y^\alpha]$, the operator $\Pi_S Y \Pi_S$ vanishing on $S^\perp$; and $\mathrm{tr}[\Pi_S Y^\alpha] < \mathrm{tr}[Y^\alpha]$, because $Y^\alpha \succeq 0$ has the support of Y, which is not contained in S. Witness: let $|\eta\rangle = a\,|j\rangle_R |u\rangle + b\,|j'\rangle_R |u'\rangle$ with $j \neq j'$, $|a|^2 + |b|^2 = 1$, and $X = |\eta\rangle\langle\eta|$, so that $X^p = X$ and $\mathcal{Z}(X^p) = |a|^2\,|j,u\rangle\langle j,u| + |b|^2\,|j',u'\rangle\langle j',u'|$ has rank two while X has rank one. The ambient value is $T(X) = |a|^{2\alpha} + |b|^{2\alpha}$, whereas compressing to $S = \mathbb{C}\,|\eta\rangle$ would return $\big(\langle\eta|\,\mathcal{Z}(X^p)\,|\eta\rangle\big)^\alpha = (|a|^4 + |b|^4)^\alpha$; at $|a|^2 = |b|^2 = \frac{1}{2}$ these are $2^{1-\alpha}$ and $2^{-\alpha}$.

Step 4 (branch operators). For $X = \mathcal{G}_{\hat{c},s}(\omega_s)$, (8.3) gives $S \subseteq W_{\hat{c},s}$, with equality at maximal support, and $W_{\hat{c},s}$ need not be $\mathcal{Z}$-invariant (Lemma 8.2 (iv)). The split form (a)–(b) is therefore the operative one: it needs no $\mathcal{Z}$-invariance, only $\operatorname{supp} H \subseteq S$. The whole computation may be compressed to a $\mathcal{Z}$-invariant $S' \supseteq W_{\hat{c},s}$, for instance $\sum_{j\in\mathcal{Z}_\perp} Z_j W_{\hat{c},s}$, which is $\mathcal{Z}$-invariant because $Z_i Z_j = \delta_{ij} Z_i$.

Step 5 (one-sided derivatives, part (ii)). T is concave on the positive semidefinite cone by Lemma 4.3 (b), which carries the concavity of Fact "Concavity of the pinched trace-power functional" (foundations/fact:concavity), in the form of Corollary 4.2 on $S_{\le}()$, to the whole cone by the

positive 1-homogeneity $T(cX) = cT(X)$ for $c \geq 0$. For a concave function, $t \mapsto \frac{T(X+tH)-T(X)}{t}$ is non-increasing on the feasible ray, so the limit exists in $(-\infty, +\infty]$. If H preserves the support, (i) gives a finite value. The scalar example $x \mapsto x^p$ at $x = 0$ shows the support-enlarging derivative can be $+\infty$. □

At a point with positive probabilities and maximal branch supports, the objective of the Choi program is differentiable along every feasible direction, with an explicit gradient. The point need only be positive semidefinite block by block, so the corollary applies at relaxed iterates.

The gradient of the pinched trace power on a face, Theorem 8.4 (i), combines with the affine maps of the Choi program into the gradient of the whole objective. Of the J factor of the point the hypothesis asks only per-block positive semidefiniteness, and no normalization, coupling condition or declared row, which is what lets Theorem 19.1 (iiib) apply this corollary at an iterate of the relaxed feasible set $\mathcal{W}^J(\varepsilon)$, the only set a run establishes membership in (Remark 23.2). Lemma 18.2 (iii) weakens the positivity hypothesis to the acceptance support $\widehat{\mathcal{C}}_{\mathrm{acc}}$ of (18.1), and to $\widehat{\mathcal{C}}_{\mathrm{key}} \cap \widehat{\mathcal{C}}_{\mathrm{acc}}$ for the K-part, which is the form Theorem 19.1 uses and the only one satisfiable when $\widehat{\mathcal{C}}_{\mathrm{acc}} \subsetneq \widehat{\mathcal{C}}$. Remark 8.7 says why nothing beyond directional differentiability is claimed.

Corollary 8.5 (Gradient of the objective along feasible directions)**.** Let $\alpha \in (1, \infty)$ and let $x = (\mathbf{q}, (J_s)_s)$ be a point with $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and $J_s \in \mathrm{Pos}(A_0 Q_s)$ for every $s \in \mathbb{S}$ — per-block positive semidefiniteness of the J factor and nothing else: no normalization, no coupling condition $\sum_s \mathrm{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$ and no declared row $\bowtie_k$ is assumed of it, so that every point of $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ qualifies, at every $\varepsilon \geq 0$ and the exact $\mathcal{W}^J$ included (Definition 17.1: per-block positive semidefiniteness is the one condition that family does not relax) — and such that $q(\hat{c}) > 0$, $P^J_{\hat{c}} > 0$ and $K^J_{\hat{c}} > 0$ for all $\hat{c}$, and such that each branch operator $X_{\hat{c},s} \coloneqq \mathcal{G}^J_{\hat{c},s}(J_s)$ $(\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}},\ s \in \mathbb{S}_K(\hat{c}))$ has *maximal support*, $\mathrm{supp}\, X_{\hat{c},s} = W_{\hat{c},s}$ of (8.1), for which $J_s \succ 0$ suffices, so that Theorem 8.4 (i) applies on that face. The displayed formulas are written for trusted-source mode; the untrusted-source-mode statement is at the end of this corollary.

Then the objective O^J_α of (7.10) is differentiable at that point *along every feasible direction*, and that is the whole of the conclusion: for every $y \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ the one-variable map $t' \mapsto O^J_\alpha\big(x + t'(y - x)\big)$ is differentiable

at $t' = 0$, with derivative the pairing of $y - x$ against the gradient

$$\frac{\partial O_\alpha^J}{\partial q(\hat{c})} = \begin{cases} \dfrac{\alpha}{\alpha-1}\Big(\log_2 \dfrac{q(\hat{c})}{P_{\hat{c}}^J} + \log_2 \mathrm{e}\Big), & \hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{key}}, \\ \log_2 \dfrac{q(\hat{c})}{P_{\hat{c}}^J} + \log_2 \mathrm{e} + \dfrac{1}{\alpha-1}\Big(\log_2 \dfrac{q(\hat{c})}{K_{\hat{c}}^J} + \log_2 \mathrm{e}\Big), & \hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}, \end{cases} \tag{8.10}$$

$$\nabla_{J_{s'}} O_\alpha^J = -\log_2 \mathrm{e} \sum_{\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{key}}} \frac{\alpha}{\alpha-1} \frac{q(\hat{c})}{P_{\hat{c}}^J} \Gamma_{\hat{c}}^{(s')} \; - \; \log_2 \mathrm{e} \sum_{\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}} \left[\frac{q(\hat{c})}{P_{\hat{c}}^J} \Gamma_{\hat{c}}^{(s')} + \frac{1}{\alpha-1} \frac{q(\hat{c})}{K_{\hat{c}}^J} \Big(\Gamma_{\hat{c}}^{(s')} + \Delta_{\hat{c}}^{(s')} \Big) \right], \tag{8.11}$$

where

$$\Gamma_{\hat{c}}^{(s')} := \sum_{(x,v):\, \phi(a(x,v))=\hat{c}} (W_\tau^\dagger M_x W_\tau) \otimes F_v^{(s')}, \qquad \Delta_{\hat{c}}^{(s')} := \mathbb{1}[s' \in \mathbb{S}_K(\hat{c})]\; \chi_{Q_{s'}}^\dagger \Big(\mathcal{G}_{\hat{c},s'}^\dagger \big[\nabla T(X_{\hat{c},s'}) - \mathbb{1} \big] \Big), \tag{8.12}$$

with $\mathcal{G}_{\hat{c},s'}^\dagger(Y) = \sum_\kappa K_\kappa^{(s')\dagger} Y K_\kappa^{(s')}$ the Kraus adjoint and $\chi_{Q_{s'}}^\dagger(A) = (W_\tau^\dagger \otimes \mathbb{1}) A (W_\tau \otimes \mathbb{1})$. Each $\nabla T(X_{\hat{c},s'})$ is read in the split form of Theorem 8.4 (i): divided differences over the strictly positive spectrum of $X_{\hat{c},s'}$, the pinched operators $\mathcal{Z}(X_{\hat{c},s'}^p)$ and $\mathcal{Z}(Y^{\alpha-1})$ on the ambient branch space.

Untrusted-source mode. There the *state form* is primary, the Choi form being its pullback rather than the other way round: the variables are the sector blocks ω_s themselves (Definition 7.10), $\chi_{Q_{s'}} = \mathrm{id}$, and (8.10)–(8.11) hold verbatim with $J_{s'} \to \omega_{s'}$, $P_{\hat{c}}^J \to P_{\hat{c}}$, $K_{\hat{c}}^J \to K_{\hat{c}}$, $X_{\hat{c},s} = \mathcal{G}_{\hat{c},s}(\omega_s)$ and the pieces (8.12) losing their W_τ conjugation:

$$\Gamma_{\hat{c}}^{(s')} = \sum_{(x,v):\, \phi(a(x,v))=\hat{c}} M_x \otimes F_v^{(s')}, \qquad \Delta_{\hat{c}}^{(s')} = \mathbb{1}[s' \in \mathbb{S}_K(\hat{c})]\; \mathcal{G}_{\hat{c},s'}^\dagger \big[\nabla T(X_{\hat{c},s'}) - \mathbb{1} \big], \tag{8.13}$$

Hermitian operators on $AQ_{s'}$. The derivation is the proof below with $\chi_{Q_{s'}}$ deleted; the positivity hypothesis on the point reads $\omega_s \in \mathrm{Pos}(AQ_s)$ per block, there being no A_0 in that mode; and the maximal-support hypothesis reads $\operatorname{supp} X_{\hat{c},s'} = W_{\hat{c},s'}$ with $\mathcal{S}_A$ now all of $\mathcal{H}(A)$, for which $\omega_{s'} \succ 0$ suffices. In trusted-source mode the same formulas conjugated by χ give the state-form gradients, and at $d_A = 1$ the two agree because $W_\tau = [1]$.

Uses Def. 1.4, Thm. 7.5, Def. 7.10, Def. 8.1, Lem. 8.2, Thm. 8.4 · *Used by* Lem. 8.6, Rem. 8.7, Lem. 18.2, Lem. 18.3, Thm. 19.1, Cor. 19.6, Lem. 21.1, Lem. 21.2, Rem. 23.7, Req. 28.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.gradient:choi_gradient · *Oracles* registry.py

Proof. Idea. The objective is differentiated term by term, through affine maps and, for the deficits, through the chain rule of Theorem 8.4. Of the point x the argument consumes three things. (a) The strict positivity clause, which keeps every logarithm and every denominator of (8.10)–(8.11) away

from its singularity. (b) Maximal support of each key-active branch operator $X_{\hat{c},s}$, which makes Theorem 8.4 (i) applicable on that face; the ambient factors $\mathcal{Z}(X^{p}_{\hat{c},s})$ and $\mathcal{Z}(Y^{\alpha-1})$ it forms need only $X_{\hat{c},s} \succeq 0$, which $J_s \succeq 0$ and complete positivity of $\mathcal{G}^{J}_{\hat{c},s}$ supply. (c) That the J-component of the direction $y - x$ is Hermitian, which per-block positive semidefiniteness of x and of y supplies, a difference of two positive semidefinite operators being Hermitian; this is the hypothesis Lemma 18.3 is stated on, and it names no feasible set, exact or relaxed. Normalization, the coupling condition and the declared rows $\bowtie_k$ are consumed nowhere: not in (8.10), not in (8.11), not in (8.12), and at no step below, which is why the hypothesis asks for per-block positive semidefiniteness and not for feasibility.

Step 1 (the $\mathbf{q}$*-gradient).* Differentiate (7.10) term by term. In $\mathbf{q}$: $\frac{\mathrm{d}}{\mathrm{d}u}\, u \log_2 \frac{u}{v} = \log_2 \frac{u}{v} + \log_2 \mathrm{e}$.

Step 2 (the divergence terms in J*).* $\frac{\partial}{\partial v}\, u \log_2 \frac{u}{v} = -\frac{u}{v} \log_2 \mathrm{e}$ composed with the affine maps $J \mapsto P^{J}_{\hat{c}}$, whose gradients are the constant Hermitian operators $\Gamma^{(s')}_{\hat{c}}$ (expand (1.16) through χ as in the proof of Theorem 7.5 and read off the linear coefficient of $J_{s'}$).

Step 3 (the K*-terms).* $\nabla_{J_{s'}} K^{J}_{\hat{c}} = \Gamma^{(s')}_{\hat{c}} + \Delta^{(s')}_{\hat{c}}$: the P part is as before, and the deficit part is the chain rule through the linear maps $\chi_{Q_{s'}}$ and $\mathcal{G}_{\hat{c},s'}$ and Theorem 8.4, together with $\nabla_X \operatorname{tr}[X] = \mathbb{1}$; the adjoints implement $\operatorname{tr}[Y\, \mathcal{L}(H)] = \operatorname{tr}\big[\mathcal{L}^{\dagger}(Y) H\big]$ for each linear factor.

Step 4 (the chain rule at a singular branch operator). The maximal-support hypothesis is what licenses that chain rule at a singular $X_{\hat{c},s'}$: an admissible direction $\Delta J_{s'} = J'_{s'} - J_{s'}$ is Hermitian, both endpoints being per-block positive semidefinite, and it produces $H = \mathcal{G}^{J}_{\hat{c},s'}(\Delta J_{s'})$ with $\operatorname{supp} H \subseteq W_{\hat{c},s'} = \operatorname{supp} X_{\hat{c},s'}$ by (8.3) applied to the positive and negative parts of $\Delta J_{s'}$ separately (the fact the submodule states as Lemma 18.3, derived here and reading no feasible set), so Theorem 8.4 (i) applies; and the components of $\nabla T(X_{\hat{c},s'})$ outside that face never enter (8.12), because $\chi^{\dagger}_{Q_{s'}} \circ \mathcal{G}^{\dagger}_{\hat{c},s'}$ sandwiches its argument between the operators $K^{(s')}_{\kappa}(W_\tau \otimes \mathbb{1})$ and their adjoints, whose ranges lie in $W_{\hat{c},s'}$ by (8.1). All compositions are differentiable at the stated point, so the sum rule applies. □

Kraus factors with the same Gram matrices give branch operators related by a partial isometry commuting with the key pinching, so the trace power, deficit and pulled-back gradient agree. A subspace containing the support is the support once the compression to it is positive definite.

The Kraus operators of a branch map carry the square roots $\sqrt{M_x} \otimes \sqrt{F_v}$ (Definition 1.6), and in the Choi form the frame W_τ as well. A square root of an exact positive semidefinite matrix is irrational in general, and is computed only approximately. The lemma shows that nothing the objective or its gradient reads depends on that choice: any factor with the same Gram matrix gives the same branch values and the same pulled-back gradient,

through a partial isometry that the key pinching cannot see. The lemma is stated for one pair of factor families at a time, so it holds for every element an enclosure leaves possible, each with its own factors. A certified evaluation may therefore build the branch maps from factors whose rank is fixed: exact factors, or, for an element $LWL^\dagger$ known through an enclosed factor L and an exact positive diagonal core W, the factor $W^{1/2}L^\dagger$, whose Gram matrix is that element for every L the enclosure allows, though L itself is not known (Requirement 28.1). Part (iv) then compresses a branch operator, or a pinched block of one, to its support, where it is positive definite and the divided-difference map of Lemma "Perturbation bounds for matrix functions" (foundations/lem:matrix-function-perturbation) applies: a subspace known to contain the support, such as the range of the factors, is the support as soon as the compression to it is shown positive definite.

Lemma 8.6 (Kraus freedom in the branch maps)**.** Let $\alpha \in (1,\infty)$, $p \coloneqq 1/\alpha$ and $\hat{c} \in \widehat{\mathcal{C}}_{\text{key}}$, and let $\mathcal{H}_{\text{in}}$ be a finite-dimensional Hilbert space. For every outcome pair (x,v) of a pass class of $\hat{c}$, that is $a(x,v) \in \mathcal{K}_{\text{pass}}(\hat{c})$, let S_{xv} and S'_{xv} be linear maps from $\mathcal{H}_{\text{in}}$ into finite-dimensional Hilbert spaces $\mathcal{K}$ and $\mathcal{K}'$ with

$$S'^{\dagger}_{xv} S'_{xv} \;=\; S^{\dagger}_{xv} S_{xv}. \tag{8.14}$$

For $\kappa \in \mathcal{K}_{\text{pass}}(\hat{c})$ put

$$K_\kappa \coloneqq \sum_{(x,v)\in a^{-1}(\kappa)} |r(x,v)\rangle_R \, |x,v\rangle_{XY} \, |\kappa\rangle_{C_A C_B} \otimes S_{xv}, \tag{8.15}$$

a map into $\mathcal{B} \coloneqq \mathcal{H}(R\,XY\,C_A C_B) \otimes \mathcal{K}$, and K'_κ likewise with S'_{xv}, into $\mathcal{B}' \coloneqq \mathcal{H}(R\,XY\,C_A C_B) \otimes \mathcal{K}'$. For every positive semidefinite operator ω on $\mathcal{H}_{\text{in}}$ put $X(\omega) \coloneqq \sum_\kappa K_\kappa \omega K_\kappa^\dagger$ and $X'(\omega) \coloneqq \sum_\kappa K'_\kappa \omega K'^{\dagger}_\kappa$, the sums over $\mathcal{K}_{\text{pass}}(\hat{c})$. Let $\mathcal{Z}$ and $\mathcal{Z}'$ be the key pinchings on $\mathcal{B}$ and $\mathcal{B}'$, by the projectors $Z_j = |j\rangle\langle j|_R \otimes \mathbb{1}$, $j \in \mathcal{Z}_\perp$, and T and T' the functionals (4.7) with these pinchings. For $X \succeq 0$ write $\nabla T(X)$ for the operator of (8.8) with the sum over the strictly positive eigenvalues of X only, $Y = \mathcal{Z}(X^p)$ and $0^{\alpha-1} \coloneqq 0$, which is the split form of Theorem 8.4 (i), and $\nabla T'$ likewise.

(i) *(A partial isometry.)* Let $\mathcal{W} \subseteq \mathcal{B}$ be the span of the subspaces $|r(x,v)\rangle_R \, |x,v\rangle_{XY} \, |a(x,v)\rangle_{C_A C_B} \otimes \operatorname{ran} S_{xv}$. There is a linear map $\mathbf{U} : \mathcal{B} \to \mathcal{B}'$ that is isometric on $\mathcal{W}$ and vanishes on $\mathcal{W}^\perp$, with $\mathbf{U}K_\kappa = K'_\kappa$ for every κ and $(Z_j \otimes \mathbb{1}_{\mathcal{K}'})\,\mathbf{U} = \mathbf{U}\,(Z_j \otimes \mathbb{1}_{\mathcal{K}})$ for every j; and $\mathcal{W}$ is invariant under every $Z_j \otimes \mathbb{1}_{\mathcal{K}}$. Hence $X'(\omega) = \mathbf{U}X(\omega)\mathbf{U}^\dagger$, with $\operatorname{supp} X(\omega) \subseteq \mathcal{W}$, for every $\omega \succeq 0$.

(ii) *(Values.)* For every $\omega \succeq 0$: $T'(X'(\omega)) = T(X(\omega))$, $\operatorname{tr} X'(\omega) = \operatorname{tr} X(\omega)$, $\operatorname{supp} X'(\omega) = \mathbf{U} \operatorname{supp} X(\omega)$ and $\operatorname{rank} X'(\omega) = \operatorname{rank} X(\omega)$.

(iii) *(Gradients.)* For every $\omega \succeq 0$, with $X \coloneqq X(\omega)$ and $X' \coloneqq X'(\omega)$,

$$\nabla T'(X') = \mathbf{U}\,\nabla T(X)\,\mathbf{U}^{\dagger}, \qquad \sum_{\kappa} K'^{\dagger}_{\kappa}\big(\nabla T'(X') - \mathbb{1}\big)K'_{\kappa} = \sum_{\kappa} K^{\dagger}_{\kappa}\big(\nabla T(X) - \mathbb{1}\big)K_{\kappa}. \tag{8.16}$$

(iv) *(Compression to the support.)* Let $A \succeq 0$ be an operator on $\mathcal{B}$.

(a) *(Containment.)* For every $\omega \succeq 0$, $\operatorname{supp} X(\omega) \subseteq \sum_{\kappa} \operatorname{ran} K_{\kappa}$; and $\operatorname{supp}\big((Z_j \otimes \mathbb{1}_{\mathcal{K}})A(Z_j \otimes \mathbb{1}_{\mathcal{K}})\big) \subseteq (Z_j \otimes \mathbb{1}_{\mathcal{K}})\operatorname{supp} A$ for every j; and $\operatorname{supp} A^{q} = \operatorname{supp} A$ for every $q > 0$.

(b) *(A floor proves equality.)* If $Q : \mathbb{C}^{m} \to \mathcal{B}$ is an isometry with $\operatorname{ran} Q \supseteq \operatorname{supp} A$ and $Q^{\dagger}AQ$ is positive definite, then $\operatorname{ran} Q = \operatorname{supp} A$.

(c) *(The compression.)* If $Q : \mathbb{C}^{m} \to \mathcal{B}$ is an isometry with $\operatorname{ran} Q = \operatorname{supp} A$ and $m \geq 1$, then $N \coloneqq Q^{\dagger}AQ$ is positive definite, $A = QNQ^{\dagger}$, $A^{q} = QN^{q}Q^{\dagger}$ for every $q > 0$, and for every operator H on $\mathcal{B}$ the divided-difference sum of (8.9) at A, over the strictly positive eigenvalues of A, equals $Q\,\mathrm{D}_{N}[Q^{\dagger}HQ]\,Q^{\dagger}$, with D_{N} the map of Lemma "Perturbation bounds for matrix functions" (foundations/lem:matrix-function-perturbation) (iii) for $f(x) = x^{p}$.

(v) *(The branch maps.)* With $\mathcal{H}_{\mathrm{in}} = \mathcal{H}(AQ_s)$ and $S_{xv} = \sqrt{M_x} \otimes \sqrt{F_v^{(s)}}$, the K_{κ} are the $K_{\kappa}^{(s)}$ of Definition 1.6 and $X(\omega) = \mathcal{G}_{\hat{c},s}(\omega)$. In trusted-source mode, with $\mathcal{H}_{\mathrm{in}} = \mathcal{H}(A_0Q_s)$, a support frame and $S_{xv} = \sqrt{M_x}\,W_{\tau} \otimes \sqrt{F_v^{(s)}}$, the K_{κ} are the $K_{\kappa}^{(s)}(W_{\tau} \otimes \mathbb{1}_{Q_s})$, $X(J_s) = \mathcal{G}^{J}_{\hat{c},s}(J_s)$, and $\sum_{\kappa} K_{\kappa}^{\dagger}(\nabla T(X) - \mathbb{1})K_{\kappa} = \chi_{Q_s}^{\dagger}\big(\mathcal{G}^{\dagger}_{\hat{c},s}[\nabla T(X) - \mathbb{1}]\big)$, the operator of (8.12); the same holds on a joint sector (t, s) with $M_x^{(t)}$ and the frame of the Alice sector t (Definition 4.8). In each case $S_{xv}^{\dagger}S_{xv}$ is $M_x \otimes F_v^{(s)}$, respectively $(W_{\tau}^{\dagger}M_xW_{\tau}) \otimes F_v^{(s)}$. So for any S'_{xv} with that Gram matrix, the deficit $f_{\hat{c},s}$, the branch operator's rank and the operator $\Delta_{\hat{c}}^{(s)}$ of (8.12) are given, by (ii) and (iii), by X' in place of X.

Uses Def. 1.4, Def. 1.6, Thm. 4.1, Lem. 4.3, Def. 4.8, Def. 7.2, Thm. 7.5, Def. 7.10, Thm. 8.4, Cor. 8.5, foundations/def:pinching, foundations/lem:matrix-function-perturbation · *Used by* Req. 28.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.certified:_deficit

Proof. Isometric conjugation. Let $\mathbf{V}$ be isometric on a subspace $\mathcal{S}$ and vanish on $\mathcal{S}^{\perp}$, or be an isometry and $\mathcal{S}$ its whole domain. Then $\mathbf{V}^{\dagger}\mathbf{V}$ is the orthogonal projector onto $\mathcal{S}$: for ϕ, ψ with components $\phi_{\mathcal{S}}$, $\psi_{\mathcal{S}}$ in $\mathcal{S}$, $\langle \phi, \mathbf{V}^{\dagger}\mathbf{V}\psi\rangle = \langle \mathbf{V}\phi_{\mathcal{S}}, \mathbf{V}\psi_{\mathcal{S}}\rangle = \langle \phi_{\mathcal{S}}, \psi_{\mathcal{S}}\rangle$, an isometry preserving inner products by polarisation. If $A \succeq 0$ has $\operatorname{supp} A \subseteq \mathcal{S}$, write $A = \sum_{k} a_k u_k u_k^{\dagger}$ with

$a_k > 0$ and orthonormal $u_k \in \mathcal{S}$; the $\mathbf{V}u_k$ are orthonormal, so $\mathbf{V}A\mathbf{V}^\dagger = \sum_k a_k (\mathbf{V}u_k)(\mathbf{V}u_k)^\dagger$ is a spectral decomposition with the same positive eigenvalues, its spectral projector for an eigenvalue $\lambda > 0$ is $\mathbf{V}P\mathbf{V}^\dagger$ with P that of A, its support is $\mathbf{V}\operatorname{supp} A$, and $(\mathbf{V}A\mathbf{V}^\dagger)^q = \mathbf{V}A^q\mathbf{V}^\dagger$ for every $q > 0$, with $0^q \coloneqq 0$ as in Definition "Operator conventions and pinching" (foundations/def:pinching); moreover $\operatorname{tr}\big[\mathbf{V}A\mathbf{V}^\dagger\big] = \operatorname{tr}\big[A\,\mathbf{V}^\dagger\mathbf{V}\big] = \operatorname{tr} A$, since $\mathbf{V}^\dagger\mathbf{V}$ is the identity on $\mathcal{S} \supseteq \operatorname{supp} A$.

(i). For each pair (x, v) define V_{xv} on $\operatorname{ran} S_{xv}$ by $V_{xv}(S_{xv}\phi) \coloneqq S'_{xv}\phi$. By (8.14), $\|S_{xv}\phi\|^2 = \langle\phi, S_{xv}^\dagger S_{xv}\phi\rangle = \|S'_{xv}\phi\|^2$, so $S_{xv}\phi = S_{xv}\phi'$ forces $S'_{xv}(\phi - \phi') = 0$: V_{xv} is well defined, linear and isometric. Extend it by 0 on $(\operatorname{ran} S_{xv})^\perp$. Write $e_{xv} \coloneqq |r(x,v)\rangle_R\,|x,v\rangle_{XY}\,|a(x,v)\rangle_{C_AC_B}$; these unit vectors are mutually orthogonal for distinct (x, v), their XY components being. Put $\mathbf{U} \coloneqq \sum_{(x,v)} e_{xv}e_{xv}^\dagger \otimes V_{xv}$. On $\mathcal{W} = \bigoplus_{(x,v)} e_{xv} \otimes \operatorname{ran} S_{xv}$ it acts as V_{xv} on each orthogonal summand, with orthogonal images, so it is isometric there, and it vanishes on $\mathcal{W}^\perp$. Then $\mathbf{U}K_\kappa = \sum_{(x,v)\in a^{-1}(\kappa)} e_{xv} \otimes V_{xv}S_{xv} = K'_\kappa$. Since $Z_j e_{xv}e_{xv}^\dagger = e_{xv}e_{xv}^\dagger Z_j$, both being $e_{xv}e_{xv}^\dagger$ if $r(x, v) = j$ and 0 otherwise, $\mathbf{U}$ commutes with the pinching projectors as stated, and each Z_j maps every summand of $\mathcal{W}$ to itself or to 0. Finally $X'(\omega) = \sum_\kappa \mathbf{U}K_\kappa\omega K_\kappa^\dagger\mathbf{U}^\dagger = \mathbf{U}X(\omega)\mathbf{U}^\dagger$, and $\operatorname{supp} X(\omega) \subseteq \sum_\kappa \operatorname{ran} K_\kappa \subseteq \mathcal{W}$.

(ii). Put $X \coloneqq X(\omega)$. By isometric conjugation, $X'^p = \mathbf{U}X^p\mathbf{U}^\dagger$, $\operatorname{tr} X' = \operatorname{tr} X$, and the support and rank statements hold. By (i), $\mathcal{Z}'(\mathbf{U}A\mathbf{U}^\dagger) = \sum_j (Z_j \otimes \mathbb{1})\mathbf{U}A\mathbf{U}^\dagger(Z_j \otimes \mathbb{1}) = \mathbf{U}\mathcal{Z}(A)\mathbf{U}^\dagger$ for every A, the adjoint of the commutation relation giving $\mathbf{U}^\dagger(Z_j \otimes \mathbb{1}) = (Z_j \otimes \mathbb{1})\mathbf{U}^\dagger$. So $Y' \coloneqq \mathcal{Z}'(X'^p) = \mathbf{U}Y\mathbf{U}^\dagger$ with $Y \coloneqq \mathcal{Z}(X^p) \succeq 0$, and $\operatorname{supp} Y \subseteq \mathcal{W}$, $\operatorname{supp} X^p$ lying in $\mathcal{W}$ and $\mathcal{W}$ being invariant under every Z_j. Isometric conjugation again gives $T'(X') = \operatorname{tr}\big[(\mathbf{U}Y\mathbf{U}^\dagger)^\alpha\big] = \operatorname{tr}\big[\mathbf{U}Y^\alpha\mathbf{U}^\dagger\big] = \operatorname{tr}[Y^\alpha] = T(X)$.

(iii). With Y as in (ii), $Y'^{\alpha-1} = \mathbf{U}Y^{\alpha-1}\mathbf{U}^\dagger$ and so $\mathcal{Z}'(Y'^{\alpha-1}) = \mathbf{U}\mathcal{Z}(Y^{\alpha-1})\mathbf{U}^\dagger$. Let $X = \sum_i \lambda_i P_i$ over the distinct strictly positive eigenvalues of X; the strictly positive eigenvalues of X' are the same, with spectral projectors $\mathbf{U}P_i\mathbf{U}^\dagger$. As $\operatorname{ran} P_i \subseteq \mathcal{W}$, on which $\mathbf{U}^\dagger\mathbf{U}$ acts as the identity, $\mathbf{U}P_i\mathbf{U}^\dagger\,\mathbf{U}\mathcal{Z}(Y^{\alpha-1})\mathbf{U}^\dagger\,\mathbf{U}P_j\mathbf{U}^\dagger = \mathbf{U}P_i\mathcal{Z}(Y^{\alpha-1})P_j\mathbf{U}^\dagger$, and summing with the coefficients $\alpha\,(\lambda_i^p - \lambda_j^p)/(\lambda_i - \lambda_j)$, respectively $\alpha p\lambda_i^{p-1}$, gives the first identity of (8.16). For the second, $K_\kappa'^\dagger A K'_\kappa = K_\kappa^\dagger\mathbf{U}^\dagger A\mathbf{U}K_\kappa$; with $A = \mathbf{U}\nabla T(X)\mathbf{U}^\dagger$ and with $A = \mathbb{1}$, and $\mathbf{U}^\dagger\mathbf{U}$ acting as the identity on $\mathcal{W} \supseteq \operatorname{ran} K_\kappa \cup \operatorname{supp}\nabla T(X)$ (the latter inside $\operatorname{supp} X$), both terms reduce to those with K_κ and $\nabla T(X)$.

(iv)(a). The support of a positive semidefinite operator is its range. $\operatorname{ran} X(\omega) \subseteq \sum_\kappa \operatorname{ran}(K_\kappa\omega K_\kappa^\dagger) \subseteq \sum_\kappa \operatorname{ran} K_\kappa$. With $Z \coloneqq Z_j \otimes \mathbb{1}_\mathcal{K}$, $\operatorname{ran}(ZAZ) \subseteq Z\operatorname{ran}(A)$, since $ZAZw = Z(AZw)$. With $A = \sum_k a_k u_k u_k^\dagger$, $a_k > 0$ and orthonormal u_k, both A and $A^q = \sum_k a_k^q u_k u_k^\dagger$ have range the span of the u_k.

(iv)(b). For $y \neq 0$, $(Qy)^\dagger A(Qy) = y^\dagger(Q^\dagger AQ)y > 0$, so $Qy \notin \ker A$. Hence $\operatorname{ran} Q \cap \ker A = \{0\}$ and $m = \dim\operatorname{ran} Q \leq \dim\mathcal{B} - \dim\ker A = \operatorname{rank} A$.

The inclusion $\operatorname{supp} A \subseteq \operatorname{ran} Q$ gives $\operatorname{rank} A \le m$, so the two subspaces have the same dimension and are equal.

(iv)(c). $QQ^\dagger$ is the orthogonal projector onto $\operatorname{supp} A$, so $A = QQ^\dagger AQQ^\dagger = QNQ^\dagger$. For $y \neq 0$, $Qy \neq 0$ lies in $\operatorname{supp} A$, where A is positive definite, so $y^\dagger Ny = (Qy)^\dagger A(Qy) > 0$. Isometric conjugation with $\mathbf{V} = Q$ and N in the role of A there gives $A^q = QN^qQ^\dagger$ and shows that the strictly positive eigenvalues of A are the eigenvalues μ_a of N, with spectral projectors $Q\Pi_aQ^\dagger$; so the sum of (8.9) is $\sum_{a,b} f^{[1]}(\mu_a,\mu_b)\, Q\Pi_aQ^\dagger HQ\Pi_bQ^\dagger = Q\,\mathrm{D}_N[Q^\dagger HQ]\,Q^\dagger$ by the definition of D_N in Lemma "Perturbation bounds for matrix functions" (foundations/lem:matrix-function-perturbation) (iii).

(v). Substituting S_{xv} in the definition of K_κ gives (1.18) of Definition 1.6, and $X(\omega)$ is then (1.19) for the sector. In the Choi form, $(\sqrt{M_x}\otimes\sqrt{F_v^{(s)}})(W_\tau\otimes\mathbb{1}) = \sqrt{M_x}W_\tau\otimes\sqrt{F_v^{(s)}}$, so the K_κ are the $K_\kappa^{(s)}(W_\tau\otimes\mathbb{1})$, and $X(J_s) = \mathcal{G}_{\hat c,s}(\chi_{Q_s}(J_s)) = \mathcal{G}^J_{\hat c,s}(J_s)$ by the definition of χ_{Q_s} in Theorem 7.5. The Kraus adjoint $\mathcal{G}^\dagger_{\hat c,s}(A) = \sum_\kappa K_\kappa^{(s)\dagger}AK_\kappa^{(s)}$ and $\chi^\dagger_{Q_s}(B) = (W_\tau^\dagger\otimes\mathbb{1})B(W_\tau\otimes\mathbb{1})$ (Corollary 8.5) compose to $\sum_\kappa K_\kappa^\dagger AK_\kappa$. On a joint sector the Kraus operators act through $\sqrt{M_x^{(t)}}\otimes\sqrt{F_v^{(s)}}$ ((4.8)), and the same substitution applies. The Gram matrices are $(\sqrt{M_x}\otimes\sqrt{F_v^{(s)}})^2 = M_x\otimes F_v^{(s)}$ and $(\sqrt{M_x}W_\tau\otimes\sqrt{F_v^{(s)}})^\dagger(\sqrt{M_x}W_\tau\otimes\sqrt{F_v^{(s)}}) = W_\tau^\dagger M_xW_\tau\otimes F_v^{(s)}$. The deficit $f_{\hat c,s} = T\circ\mathcal{G}_{\hat c,s} - \operatorname{tr}\circ\mathcal{G}_{\hat c,s}$ ((4.2)) is unchanged by (ii), and $\Delta^{(s)}_{\hat c}$, the pulled-back operator of (8.12) with ∇T read in the split form, is unchanged by (iii). □

The objective of the Choi program is undefined off the positive cone and the sets Corollary 8.5 quantifies over have empty interior, so only directional derivatives are well posed. Even within the affine hull, a direction off the acceptance support can have one-sided derivative $-\infty$.

Remark 8.7 (Scope of the directional differentiability of the objective). Corollary 8.5 claims no ambient Fréchet differentiability, and the set on which it differentiates is part of its statement. There are two grounds, and the first is decisive on its own.

First, O^J_α is not defined on an ambient neighbourhood of the corollary's point x at all, for either of two independent causes: the objective factors through $X \mapsto X^p$, which is undefined off the positive semidefinite cone, so an ambient neighbourhood of a singular J_s leaves the domain; and the sets the corollary quantifies over have empty interior in the ambient real vector space: the affine equality $\sum_{\hat c} q(\hat c) = 1$ that $\mathcal{S}_{\mathrm{acc}}$ carries forces that on its own, and on the exact $\mathcal{W}^J$ the coupling equality $\sum_s \operatorname{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$ forces it again. So the ambient notion is not well posed here, whereas the directional one is, and it is the one (19.4) consumes.

Second, the restriction cannot be lifted even to directions inside that affine hull. Where the hypothesis is taken in the weakened form of Lemma 18.2 (iii), a coordinate $q(\hat c)$ with $\hat c \notin \widehat{\mathcal{C}}_{\mathrm{acc}}$ sits at 0 throughout $\mathcal{S}_{\mathrm{acc}}$, and $\partial_u\big(u\log_2\frac{u}{v}\big) = \log_2\frac{u}{v} + \log_2 \mathrm{e} \to -\infty$ as $u \downarrow 0$ for $v > 0$, so a direction that raises such a coordinate off 0, one that leaves $\mathcal{S}_{\mathrm{acc}}$ while preserving $\sum_{\hat c} q(\hat c) = 1$, has one-sided derivative $-\infty$;

inside $\mathcal{S}_{\mathrm{acc}}$ the coordinate stays 0 identically and the term is deleted rather than bounded (Lemma 18.2 (ii)). Feasible directions are all that Step 1 of Theorem 19.1 consumes, and no statement of this module reads more.

Uses Cor. 8.5 · newly derived · accepted

At Rényi order one the derivative of the pinched trace power of a subnormalised operator is minus ln 2 times its relative entropy to its pinching. A jointly continuous majorant bounds the derivative just above order one, which carries the limit of order one through the infimum.

The limit of order one, Theorem 8.9, needs the derivative of the pinched trace power at $\alpha = 1$ and a bound on it just above 1 that is continuous in the operator. In the proof, $\Xi(\alpha)$ is the derivative of $\sigma^{1/\alpha}$; the letter Σ is taken by the marginal-constrained range of Fact 2.1.

Lemma 8.8 (Derivative of the pinched trace power at order one). We write $\underline{h}_\alpha(\alpha)$ for (5.4) with its α-dependence explicit, and, for $\sigma \in \mathrm{Pos}()$ with $\operatorname{tr}\sigma \le 1$, $B_\sigma(\alpha) \coloneqq \operatorname{tr}\big[(\mathcal{Z}(\sigma^{1/\alpha}))^\alpha\big]$. Let $\delta > 0$, one number for all σ, and let $\sigma \succeq 0$ with $\operatorname{tr}\sigma \le 1$ (so $\|\sigma\| \le 1$). Then $B_\sigma \in C^1([1, 1+\delta])$, with $B_\sigma(1) = \operatorname{tr}\sigma$ and

$$B'_\sigma(1) = \operatorname{tr}[\mathcal{Z}(\sigma)\ln\mathcal{Z}(\sigma)] - \operatorname{tr}[\sigma\ln\sigma] = -\ln 2\, D\big(\sigma \,\big\|\, \mathcal{Z}(\sigma)\big) \;\le\; 0, \tag{8.17}$$

with the convention $0\ln 0 = 0$. Moreover, for all $\alpha \in (1, 1+\delta]$,

$$B'_\sigma(\alpha) \;\le\; c_\sigma(\alpha) \coloneqq \operatorname{tr}\big[A(\alpha)^\alpha \ln A(\alpha)\big] - \frac{1}{\alpha}\sum_i \lambda_i^{1/\alpha}\ln\lambda_i, \qquad A(\alpha) \coloneqq \mathcal{Z}\big(\sigma^{1/\alpha}\big), \tag{8.18}$$

where λ_i are the eigenvalues of σ; and $(\alpha,\sigma) \mapsto c_\sigma(\alpha)$ is jointly continuous on $[1, 1+\delta] \times \{\sigma \succeq 0 : \operatorname{tr}\sigma \le 1\}$ with $c_\sigma(1) = B'_\sigma(1)$.

Uses Def. 1.2, Fact 2.1, Def. 5.1, Lem. 8.3, foundations/def:pinching · *Used by* Thm. 8.9, Cor. 9.4 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.limit:relative_entropy_to_pinching · *Oracles* registry.py

Proof. Finiteness. $D(\sigma\|\mathcal{Z}(\sigma)) < \infty$ always: a vector in the kernel of $\mathcal{Z}(\sigma)$ satisfies $\sigma^{1/2}Z_j\,|w\rangle = 0$ for all j, hence lies in the kernel of σ, so $\operatorname{supp}\sigma \subseteq \operatorname{supp}\mathcal{Z}(\sigma)$.

Fixed supports. $\operatorname{supp}\sigma^{1/\alpha} = \operatorname{supp}\sigma$ for every α, and the kernel argument above shows $\operatorname{supp}\mathcal{Z}(\tau)$ depends on τ only through $\operatorname{supp}\tau$; hence $S \coloneqq \operatorname{supp}A(\alpha)$ is the same subspace for all α, and S is $\mathcal{Z}$-compatible (its projector is a fixed point of $\mathcal{Z}$, being the support projector of a $\mathcal{Z}$-output, which is block-diagonal). On S, $A(\alpha)$ is positive definite with eigenvalues bounded away from 0 uniformly on compact α-intervals (eigenvalues are continuous in α and positive).

Smoothness. $\alpha \mapsto \sigma^{1/\alpha} = \sum_i \lambda_i^{1/\alpha}P_i$ is C^1 with $\frac{\mathrm{d}}{\mathrm{d}\alpha}\sigma^{1/\alpha} = -\frac{1}{\alpha^2}\sum_i \lambda_i^{1/\alpha}\ln\lambda_i\, P_i =: \Xi(\alpha) \succeq 0$ (each $-\lambda^{1/\alpha}\ln\lambda \ge 0$ for $\lambda \in [0,1]$; the $\lambda = 0$ terms vanish). Hence

$A'(\alpha) = \mathcal{Z}(\Xi(\alpha)) \succeq 0$ is continuous. Restricting to S, where A is smooth and positive definite, $B_\sigma(\alpha) = \mathrm{tr}[A(\alpha)^\alpha]$ is C^1 with

$$B'_\sigma(\alpha) = \mathrm{tr}\left[A^\alpha \ln A\right] + \alpha\ \mathrm{tr}\left[A^{\alpha-1}A'\right], \tag{8.19}$$

by (8.6) (for the second term) and the scalar chain rule in the exponent (for the first; both computed in the joint eigenbasis data of $A(\alpha)$, using that $\frac{\partial}{\partial\alpha}a^\alpha = a^\alpha \ln a$ and the C^1 dependence of eigenvalues and eigenprojectors along the smooth positive-definite path on S; concretely, write $B_\sigma(\alpha) = \mathrm{tr}\exp(\alpha \ln A(\alpha))$ on S and differentiate the two slots).

At $\alpha = 1$. $A(1) = \mathcal{Z}(\sigma)$, $A'(1) = \mathcal{Z}(-\sigma\ln\sigma)$, and $A^{\alpha-1} \to \Pi_S$ as $\alpha \downarrow 1$ (eigenvalues of A on S bounded below). Hence

$$B'_\sigma(1) = \mathrm{tr}[\mathcal{Z}(\sigma)\ln\mathcal{Z}(\sigma)] + \mathrm{tr}[\Pi_S\,\mathcal{Z}(-\sigma\ln\sigma)] = \mathrm{tr}[\mathcal{Z}(\sigma)\ln\mathcal{Z}(\sigma)] - \mathrm{tr}[\sigma\ln\sigma],$$

using self-adjointness of $\mathcal{Z}$, $\mathcal{Z}(\Pi_S) = \Pi_S$, and $\mathrm{supp}(\sigma\ln\sigma) \subseteq \mathrm{supp}\,\sigma \subseteq S$. The identification with the relative entropy uses $\mathrm{tr}[\sigma\ln\mathcal{Z}(\sigma)] = \mathrm{tr}[\mathcal{Z}(\sigma)\ln\mathcal{Z}(\sigma)]$ (again self-adjointness and block-diagonality of $\ln\mathcal{Z}(\sigma)$ on S), and non-positivity is monotonicity of $D \geq 0$.

Majorant. In (8.19), $\|A(\alpha)\| \leq \|\sigma^{1/\alpha}\| \leq 1$ (a pinching is an average of unitary conjugates, Definition "Operator conventions and pinching" (foundations/def:pinching), so it cannot increase the operator norm), hence $A^{\alpha-1} \preceq \Pi_S \preceq \mathbb{1}$, and $A' \succeq 0$ gives $\mathrm{tr}\left[A^{\alpha-1}A'\right] \leq \mathrm{tr}[A'] = \mathrm{tr}[\Xi(\alpha)] = -\frac{1}{\alpha^2}\sum_i \lambda_i^{1/\alpha}\ln\lambda_i$; multiplying by α yields (8.18). Joint continuity: the map $(\alpha,\sigma) \mapsto A(\alpha)$ is jointly continuous; the scalar functions $(a,\alpha) \mapsto a^\alpha \ln a$ (value 0 at $a = 0$) and $(\lambda,\alpha) \mapsto \lambda^{1/\alpha}\ln\lambda$ are jointly continuous on $[0,1]\times[1,1+\delta]$; and a trace of a jointly continuous spectral function of a jointly continuous operator family is jointly continuous. No step restricts δ or makes it depend on σ: the eigenvalue bound on S holds on every compact α-interval, and the norm bound and the two scalar functions hold for every $\alpha \geq 1$. At $\alpha = 1$ the two expressions (8.18) and (8.19) coincide, both equal to (8.17). □

If every acceptable distribution gives each key-live value probability at least some $p_0 > 0$ and some feasible state reproduces an acceptable distribution, the value tends, as $\alpha \to 1$, to a blockwise relative-entropy program. The limit is a validation reference and not a bound.

As the Rényi order tends to one, the weight $\alpha/(\alpha-1)$ of the divergence term of the objective diverges, and the entropy part of each branch tends to a relative entropy to its pinching. Remark 8.10 says what the limit validates and on which declarations it is available.

Theorem 8.9 (Limit of the value as the Rényi order tends to one)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 and let $\mathcal{S}_{\mathrm{acc}}$ satisfy Definition 2.5. Assume that there is $p_0 > 0$ with $q(\hat{c}) \geq p_0$ for every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, and that the

asymptotic feasible set $\mathcal{W}_1 \coloneqq \{\omega \in \mathcal{W} : \mathbf{\Phi}[\omega] \in \mathcal{S}_{\mathrm{acc}}\}$ is nonempty. Then

$$\lim_{\alpha\downarrow 1} \underline{h}_\alpha(\alpha) \;=\; h_1 \coloneqq \inf_{\omega\in\mathcal{W}_1} \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} \sum_{s\in\mathbb{S}_K(\hat{c})} D\Big(\mathcal{G}_{\hat{c},s}(\omega_s) \,\Big\|\, \mathcal{Z}\big(\mathcal{G}_{\hat{c},s}(\omega_s)\big)\Big). \tag{8.20}$$

Uses Def. 1.2, Def. 2.5, Thm. 4.1, Prop. 4.6, Def. 5.1, Thm. 5.4, Lem. 8.8, foundations/def:kl, foundations/lem:kernel · *Used by* Rem. 8.10, Cor. 9.4, Req. 15.9 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.limit:check_limit_hypotheses · *Oracles* registry.py

Proof. Idea. The upper bound evaluates the objective at a state that reproduces an acceptable distribution exactly, where only the entropy part survives, and passes to the limit with the derivative of Lemma 8.8 at $\alpha = 1$. The lower bound takes near-optimal points along $\alpha_n \downarrow 1$, uses the divergence part to force a limit point that reproduces its distribution, and bounds the entropy part from below by the continuous majorant of that lemma.

Notation. Write $\sigma_{\hat{c},s}(\omega) \coloneqq \mathcal{G}_{\hat{c},s}(\omega_s)$ and, for $P_{\hat{c}}(\omega) > 0$, $e_{\hat{c}}(\alpha,\omega) \coloneqq \frac{1}{1-\alpha}\log_2 \frac{K_{\hat{c}}(\omega)}{P_{\hat{c}}(\omega)} \geq 0$.

Upper bound. Fix $\omega \in \mathcal{W}_1$ and take $\mathbf{q} \coloneqq \mathbf{\Phi}[\omega] \in \mathcal{S}_{\mathrm{acc}}$. All divergence terms of O_α vanish, and $q(\hat{c}) = P_{\hat{c}}(\omega) \geq p_0$ for $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$. Then

$$O_\alpha(\mathbf{q},\omega) = \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} P_{\hat{c}}\, e_{\hat{c}}(\alpha,\omega) = \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} \frac{P_{\hat{c}}}{(1-\alpha)\ln 2} \ln \frac{K_{\hat{c}}(\alpha)}{P_{\hat{c}}},$$

and $K_{\hat{c}}(\alpha) - P_{\hat{c}} = \sum_s \big(B_{\sigma_{\hat{c},s}}(\alpha) - B_{\sigma_{\hat{c},s}}(1)\big)$ by (4.4) and $B_\sigma(1) = \operatorname{tr}\sigma$ together with $P_{\hat{c}} - \sum_s \operatorname{tr}\sigma_{\hat{c},s} = \operatorname{tr}\Theta_{\hat{c}}$ cancelling between K and P (both formulas from (4.1)). By Lemma 8.8 and l'Hôpital (or the C^1 mean value theorem), as $\alpha \downarrow 1$,

$$O_\alpha(\mathbf{q},\omega) \longrightarrow -\frac{1}{\ln 2} \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} \sum_s B'_{\sigma_{\hat{c},s}}(1) = \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} \sum_s D\big(\sigma_{\hat{c},s} \| \mathcal{Z}(\sigma_{\hat{c},s})\big),$$

where the sum may be restricted to $s \in \mathbb{S}_K(\hat{c})$ because key-inert sectors have σ already $\mathcal{Z}$-block-diagonal (Proposition 4.6) and hence zero relative entropy. Therefore $\limsup_{\alpha\downarrow 1} \underline{h}_\alpha(\alpha) \leq h_1$-objective of ω, for every $\omega \in \mathcal{W}_1$, so $\limsup \underline{h}_\alpha(\alpha) \leq h_1$.

Lower bound. Let $\alpha_n \downarrow 1$ and pick feasible $(\mathbf{q}_n, \omega_n)$ with $O_{\alpha_n}(\mathbf{q}_n,\omega_n) \leq \underline{h}_\alpha(\alpha_n) + \frac{1}{n}$; by the upper bound this sequence of values is bounded above by some $C < \infty$. Decompose $O_\alpha = \frac{\alpha}{\alpha-1} D(\mathbf{q}\|\mathbf{\Phi}) + \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} q(\hat{c}) e_{\hat{c}}$ (undo (5.7)); both parts are non-negative (the divergence by Gibbs' inequality of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl), $\mathbf{q}_n$ and $\mathbf{\Phi}[\omega_n]$ being probability vectors, and $e_{\hat{c}} \geq 0$ since $K_{\hat{c}} \leq P_{\hat{c}}$ by Theorem 4.1's *cone clause*, the points ω_n being feasible but not pure). Hence $D(\mathbf{q}_n\|\mathbf{\Phi}[\omega_n]) \leq C\frac{\alpha_n - 1}{\alpha_n} \to 0$. By compactness pass to a subsequence

with $(\mathbf{q}_n, \omega_n) \to (\mathbf{q}^*, \omega^*) \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}$: $\mathcal{S}_{\mathrm{acc}}$ is a closed subset of the simplex (Definition 2.5) and $\mathcal{W}$ is compact (Definition 5.1). $D(\mathbf{q}\|\mathbf{p})$ is the sum over $\hat{c}$ of the kernel $d(q(\hat{c}), p(\hat{c}))$ of Lemma "Extended divergence kernel" (foundations/lem:kernel), which is lower semicontinuous, and $\mathbf{\Phi}$ is continuous, so $D(\mathbf{q}^*\|\mathbf{\Phi}[\omega^*]) \le \liminf_n D(\mathbf{q}_n\|\mathbf{\Phi}[\omega_n]) = 0$; by the equality case of Gibbs' inequality (Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl)) $\mathbf{q}^* = \mathbf{\Phi}[\omega^*]$ and $\omega^* \in \mathcal{W}_1$; in particular $P_{\hat{c}}(\omega^*) = q^*(\hat{c}) \ge p_0$ for $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, so $P_{\hat{c}}(\omega_n) \ge p_0/2$ eventually. For each $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, using $\ln(1+u) \le u$ with $u_n := \frac{K_{\hat{c}}(\omega_n) - P_{\hat{c}}(\omega_n)}{P_{\hat{c}}(\omega_n)} \in [-1, 0]$ and $\frac{1}{1-\alpha_n} < 0$:

$$e_{\hat{c}}(\alpha_n, \omega_n) \;\ge\; \frac{u_n}{(1-\alpha_n)\ln 2} \;=\; \frac{1}{\ln 2\, P_{\hat{c}}(\omega_n)} \sum_s \Big[-\frac{B_{\sigma_{\hat{c},s}(\omega_n)}(\alpha_n) - B_{\sigma_{\hat{c},s}(\omega_n)}(1)}{\alpha_n - 1} \Big].$$

By the mean value theorem there are $\xi_n \in (1, \alpha_n)$ with the bracket equal to $-B'_{\sigma_{\hat{c},s}(\omega_n)}(\xi_n) \ge -c_{\sigma_{\hat{c},s}(\omega_n)}(\xi_n)$ by (8.18). Joint continuity of $(\alpha, \sigma) \mapsto c_\sigma(\alpha)$ (Lemma 8.8) and $\sigma_{\hat{c},s}(\omega_n) \to \sigma_{\hat{c},s}(\omega^*)$ (the maps $\mathcal{G}_{\hat{c},s}$ are continuous) give $c_{\sigma(\omega_n)}(\xi_n) \to c_{\sigma(\omega^*)}(1) = B'_{\sigma(\omega^*)}(1)$. Hence

$$\liminf_n \; q_n(\hat{c})\, e_{\hat{c}}(\alpha_n, \omega_n) \;\ge\; \frac{q^*(\hat{c})}{\ln 2\, P_{\hat{c}}(\omega^*)} \sum_s \big(-B'_{\sigma_{\hat{c},s}(\omega^*)}(1) \big) \;=\; \sum_s D\big(\sigma_{\hat{c},s}(\omega^*) \,\big\|\, \mathcal{Z}(\sigma_{\hat{c},s}(\omega^*))\big),$$

using $q^*(\hat{c}) = P_{\hat{c}}(\omega^*)$. Dropping the non-negative divergence part and summing over $\hat{c}$ (a finite sum, so $\liminf$ passes through), $\liminf_n \underline{h}_\alpha(\alpha_n) \ge h_1$-objective of $\omega^* \ge h_1$. Combining with the upper bound proves (8.20). □

The limit of Theorem 8.9 is a blockwise relative-entropy program that checks an implementation as $\alpha \to 1$; it validates no finite-α value and bounds nothing. It is available only where every acceptable distribution is positive on the key-live values.

Remark 8.10 (Validation against the order-one limit). The right-hand side of (8.20) is a blockwise relative-entropy objective, of the structure of the conic von Neumann key-distance program of [7, Eqs. (1)–(6)] with acceptance-set membership in place of the equality constraints used there. Theorem 8.9 is stated over the declared $\mathcal{W}$, so it covers arbitrary $\mathfrak{L}$ unchanged, and affine constraints enter both programs natively. The limit validates the consistency of the $\alpha \to 1^+$ limit of an implementation of (7.10); it does not validate finite-α objective values or the finite-α $K_{\hat{c}}$, and it is a bound of no kind, in particular no lower bound on $\underline{h}_\alpha$ (Requirement 15.9).

On which declarations it is available. Theorem 8.9 assumes a $p_0 > 0$ with $q(\hat{c}) \ge p_0$ for *every* $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, which asks that every element of $\mathcal{S}_{\mathrm{acc}}$ be strictly positive on $\widehat{\mathcal{C}}_{\mathrm{key}}$; on a compact $\mathcal{S}_{\mathrm{acc}}$ that is exactly the existence of p_0, the minimum being attained. A singleton $\mathcal{S}_{\mathrm{acc}} = \{\mathbf{q}^*\}$ with $\widehat{\mathcal{C}}_{\mathrm{key}} \subseteq \operatorname{supp} \mathbf{q}^*$ satisfies it; a polytope with a vertex vanishing somewhere on $\widehat{\mathcal{C}}_{\mathrm{key}}$ does not, and there the limit is unavailable. Theorem 8.9 is a validation statement: no key length passes through it, and a declaration outside its hypothesis loses this check and no claim. The condition is narrower than the acceptance support $\widehat{\mathcal{C}}_{\mathrm{acc}}$ of (18.1), which

asks positivity of *some* element and against which (H-POS) of Theorem 19.1 is stated for exactly that reason.

Uses Thm. 7.5, Thm. 8.9 · *Used by* Req. 15.9 · newly derived · accepted

9 The dropped program

A run may give up the entropy of chosen pairs of a key-live acceptance value and a joint sector, setting their deficits to zero while keeping their variables, statistics and constraints (Definition 9.2). The resulting value bounds h_low and the single-round rate from below (Proposition 9.3); it is attained when finite, and a fully dropped joint sector enters only through its statistics, with no matrix function (Corollary 9.4). With every pair dropped the value is $\alpha/(\alpha-1)$ times the minimal divergence between the accepted distributions and the statistics of the feasible set, a floor under h_low that lies within the entropy ceiling of it, and a positive lower bound on it proves that no feasible state has an accepted distribution as its statistics (Proposition 9.5). On an Alice sector whose marginal has rank one, with default readouts and Alice holding the key, the deficit is linear with constants from declared numbers (Lemma 9.1). The objective and the dropped objective, read on the whole cone, are positively homogeneous of degree one (Lemma 9.6).

Imports foundations/def:kl, foundations/lem:kernel

On an Alice sector of rank-one marginal every feasible block is the fixed pure state times an operator on Q, and with default readouts and Alice holding the key its deficit is linear, with constants from declared numbers. A uniform key bit gives one bit per unit of pass weight.

When Alice's marginal on a sector is pure, her part of every feasible block is fixed. With the default readouts and Alice holding the key, the deficit of the block then reduces to a Rényi quantity of her key value given the announcement, weighted linearly by the block's statistics.

Lemma 9.1 (Linear deficit on a rank-one Alice sector)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 in trusted-source mode and the general format, and let $t \in \mathbb{T}$ with $\tau_A^{(t)} = p_t\,|\hat{v}\rangle\langle\hat{v}|$, $p_t > 0$, $|\hat{v}\rangle \in \mathcal{H}(A_t)$ a unit vector.

(i) *(product blocks)* Every list $(\omega^{(t,s)})_{s\in\mathbb{S}}$ of positive semidefinite blocks with $\sum_s \operatorname{tr}_{Q_s} \omega^{(t,s)} = \tau_A^{(t)}$, in particular the sector-t part of every $\omega \in \mathcal{W}$, has the form $\omega^{(t,s)} = |\hat{v}\rangle\langle\hat{v}| \otimes \sigma_s$ with $\sigma_s \in \operatorname{Pos}(Q_s)$ and $\sum_s \operatorname{tr} \sigma_s = p_t$.

(ii) *(pass classes are product sets)* If the readout maps are the default ones, $\pi_{\mathsf{A}}(x,v) = x$ and $\pi_{\mathsf{B}}(x,v) = v$, then for every transcript protocol and every $\kappa \in \mathcal{K}$ there are sets $\mathcal{X}_\kappa \subseteq \mathcal{X}$, $\mathcal{V}_\kappa \subseteq \mathcal{V}$ with $a^{-1}(\kappa) = \mathcal{X}_\kappa \times \mathcal{V}_\kappa$; if moreover $\mathsf{h} = \mathsf{A}$, then $r(x,v) = r_{\mathsf{A}}(x,\kappa)$ on $a^{-1}(\kappa)$.

(iii) *(linear deficit)* Under the hypotheses of (ii), let $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, $s \in \mathbb{S}$ and $\alpha \in (1,\infty)$, put $\mu_x := \langle \hat{v}|\, M_x^{(t)}\, |\hat{v}\rangle$ and, for $\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})$ and $\mathsf{z} \in \mathcal{Z}$,

$$m_\kappa := \sum_{x \in \mathcal{X}_\kappa} \mu_x, \qquad m_{\kappa,\mathsf{z}} := \sum_{x \in \mathcal{X}_\kappa :\, r_{\mathsf{A}}(x,\kappa)=\mathsf{z}} \mu_x, \qquad \eta_\kappa := \begin{cases} 1 - \sum_{\mathsf{z}} \left(m_{\kappa,\mathsf{z}}/m_\kappa\right)^\alpha, & m_\kappa > 0, \\ 0, & m_\kappa = 0. \end{cases} \tag{9.1}$$

Then for every $\sigma_s \in \mathrm{Pos}(Q_s)$

$$f_{\hat{c},(t,s)}\big(\,|\hat{v}\rangle\langle\hat{v}| \otimes \sigma_s\big) = -\sum_{\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})} \eta_\kappa\, m_\kappa \sum_{v \in \mathcal{V}_\kappa} \mathrm{tr}\left[F_v^{(s)} \sigma_s\right], \tag{9.2}$$

a *linear* function of σ_s, with $0 \le \eta_\kappa < 1$. In ensemble form $\mu_x = \mathrm{tr}\, N_x^{(t)}/p_t$ and $p_t = \mathrm{tr}\, \tau_A^{(t)}$, declared numbers.

(iv) *(two values)* If within every pass class of $\hat{c}$ the key takes d values of equal weight, $m_{\kappa,\mathsf{z}} = m_\kappa/d$, then $\eta_\kappa = 1 - d^{1-\alpha}$ and the sector contributes to $\frac{1}{1-\alpha}\log_2$ of the ratio K/P *as if the conditional entropy were* $\log_2 d$ *on the pass weight*: one bit per unit of pass weight for a uniform key bit. The qualifier matters: what the expression assigns to the pair is its fail weight plus $d^{1-\alpha}$ times its pass weight, so with a non-zero fail weight in the same joint pair the contribution is strictly less than that of a sector of conditional entropy $\log_2 d$, and equals it exactly when the pass weight is the whole of the pair's weight. If the key is constant on every $\mathcal{X}_\kappa \cap \{x : \mu_x > 0\}$ then $\eta_\kappa = 0$ and $f_{\hat{c},(t,s)}$ vanishes on these blocks.

Uses Def. 1.1, Def. 1.2, Lem. 1.3, Def. 1.4, Def. 1.6, Thm. 4.1, Def. 4.8, Def. 5.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.joint:rank_one_deficit

Proof. (i) Let $\omega_t := \bigoplus_s \omega^{(t,s)} \in \mathrm{Pos}(A_t Q)$, so $\mathrm{tr}_Q\, \omega_t = p_t\, |\hat{v}\rangle\langle\hat{v}|$, and let $\Pi^\perp := \mathbb{1}_{A_t} - |\hat{v}\rangle\langle\hat{v}|$. Then $\mathrm{tr}\big[(\Pi^\perp \otimes \mathbb{1})\omega_t\big] = p_t\, \mathrm{tr}\big[\Pi^\perp\, |\hat{v}\rangle\langle\hat{v}|\big] = 0$, and for a positive semidefinite ω_t this forces $(\Pi^\perp \otimes \mathbb{1})\omega_t = 0$. Hence $\omega_t = (|\hat{v}\rangle\langle\hat{v}| \otimes \mathbb{1})\omega_t(|\hat{v}\rangle\langle\hat{v}| \otimes \mathbb{1}) = |\hat{v}\rangle\langle\hat{v}| \otimes \sigma$ with $\sigma := (\langle\hat{v}| \otimes \mathbb{1})\omega_t(|\hat{v}\rangle \otimes \mathbb{1}) \succeq 0$, block diagonal along $\mathbb{S}$ because ω_t is, with blocks σ_s; and $\mathrm{tr}\,\sigma = \mathrm{tr}\,\omega_t = p_t$. For $\omega \in \mathcal{W}$ the hypothesis is the marginal line of (5.5).

(ii) By (1.8), $a(x,v) = \kappa$ holds exactly when $a_m(\pi_{\pi_m}(x,v), \kappa_1, \ldots, \kappa_{m-1}) = \kappa_m$ for every m: by induction on m, if the first $m-1$ components of $a(x,v)$ are $\kappa_1, \ldots, \kappa_{m-1}$ then its m-th component is the left-hand side. With the default readouts the m-th condition is a condition on x alone when $\pi_m = \mathsf{A}$ and on v alone when $\pi_m = \mathsf{B}$. So $a^{-1}(\kappa) = \mathcal{X}_\kappa \times \mathcal{V}_\kappa$ with $\mathcal{X}_\kappa$ the set of x meeting all conditions with $\pi_m = \mathsf{A}$ and $\mathcal{V}_\kappa$ the set of v meeting all those with $\pi_m = \mathsf{B}$. The last statement is (1.1) with $\pi_{\mathsf{A}}(x,v) = x$.

(iii) Fix $\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})$ and write $|m_x\rangle := \sqrt{M_x^{(t)}}\,|\hat{v}\rangle$, so $\langle m_x|m_x\rangle = \mu_x$. By (ii) the operator $K_\kappa^{(s)}$ of (1.18) satisfies, on vectors $|\hat{v}\rangle \otimes |\xi\rangle$,

$$K_\kappa^{(s)}\big(\,|\hat{v}\rangle\otimes|\xi\rangle\,\big) = |a_\kappa\rangle\otimes|\kappa\rangle\otimes B_\kappa\,|\xi\rangle\,, \qquad |a_\kappa\rangle := \sum_{x\in\mathcal{X}_\kappa} |r_{\mathsf{A}}(x,\kappa)\rangle_R\,|x\rangle_X\,|m_x\rangle_A\,, \qquad B_\kappa := \sum_{v\in\mathcal{V}_\kappa} |v\rangle_Y\otimes\sqrt{F}$$

up to the fixed ordering of tensor factors. Hence $G_\kappa := K_\kappa^{(s)}(|\hat{v}\rangle\langle\hat{v}| \otimes \sigma_s)K_\kappa^{(s)\dagger} = |a_\kappa\rangle\langle a_\kappa| \otimes |\kappa\rangle\langle\kappa| \otimes \beta_\kappa$ with $\beta_\kappa := B_\kappa\sigma_s B_\kappa^\dagger \succeq 0$, and $\langle a_\kappa|a_\kappa\rangle = m_\kappa$, the kets $|x\rangle_X$ being orthonormal. The operators G_κ for different κ are supported on the mutually orthogonal subspaces fixed by $|\kappa\rangle$, which are invariant under $\mathcal{Z}$ and under powers, so as in the proof of Lemma 4.9 (ii) the functional T and the trace add over κ, and $f_{\hat{c},(t,s)} = \sum_\kappa \big(T(G_\kappa) - \operatorname{tr} G_\kappa\big)$. If $m_\kappa = 0$ then $G_\kappa = 0$. Otherwise $|a_\kappa\rangle\langle a_\kappa| = m_\kappa\Pi_a$ with Π_a a rank-one projector, so $G_\kappa^{1/\alpha} = m_\kappa^{1/\alpha-1}\,|a_\kappa\rangle\langle a_\kappa|\otimes|\kappa\rangle\langle\kappa|\otimes\beta_\kappa^{1/\alpha}$. The vectors $|a_{\kappa,\mathsf{z}}\rangle := Z_{\mathsf{z}}\,|a_\kappa\rangle$ are mutually orthogonal with squared norms $m_{\kappa,\mathsf{z}}$, and $Z_\perp\,|a_\kappa\rangle = 0$ because κ is a pass class, so $\mathcal{Z}(G_\kappa^{1/\alpha}) = m_\kappa^{1/\alpha-1}\sum_{\mathsf{z}} |a_{\kappa,\mathsf{z}}\rangle\langle a_{\kappa,\mathsf{z}}| \otimes |\kappa\rangle\langle\kappa| \otimes \beta_\kappa^{1/\alpha}$. The summands have mutually orthogonal supports and $(|a_{\kappa,\mathsf{z}}\rangle\langle a_{\kappa,\mathsf{z}}|)^\alpha = m_{\kappa,\mathsf{z}}^{\alpha-1}\,|a_{\kappa,\mathsf{z}}\rangle\langle a_{\kappa,\mathsf{z}}|$, hence

$$T(G_\kappa) = m_\kappa^{1-\alpha}\sum_{\mathsf{z}} m_{\kappa,\mathsf{z}}^\alpha\,\operatorname{tr}\beta_\kappa = \Big(\sum_{\mathsf{z}}\big(m_{\kappa,\mathsf{z}}/m_\kappa\big)^\alpha\Big)\operatorname{tr} G_\kappa, \qquad \operatorname{tr} G_\kappa = m_\kappa\operatorname{tr}\beta_\kappa = m_\kappa\sum_{v\in\mathcal{V}_\kappa}\operatorname{tr}\big[F_v^{(s)}\sigma_s\big].$$

Subtracting gives (9.2). The numbers $m_{\kappa,\mathsf{z}}/m_\kappa$ form a probability vector, so $0 < \sum_{\mathsf{z}}(\cdot)^\alpha \le 1$ and $0 \le \eta_\kappa < 1$. In ensemble form $\mu_x = \operatorname{tr}\big[M_x^{(t)}\tau_A^{(t)}\big]/p_t = \operatorname{tr} N_x^{(t)}/p_t$ by Lemma 1.3 (ii).

(iv) With d equal weights $\sum_{\mathsf{z}}(1/d)^\alpha = d^{1-\alpha}$. If every pass class of $\hat{c}$ has this form then, by (iii), on these blocks $P_{\hat{c},(t,s)} + f_{\hat{c},(t,s)}$ equals the fail weight plus $d^{1-\alpha} = 2^{(1-\alpha)\log_2 d}$ times the pass weight, which is the value the branch formula (4.3) assigns to conditional entropy $\log_2 d$ on the pass weight. If the key is constant where $\mu_x > 0$, one $m_{\kappa,\mathsf{z}}$ equals m_κ and the others vanish. □

A run may give up the entropy of chosen pairs of a key-live acceptance value and a joint sector by setting their deficits to zero, keeping every variable, statistic and constraint. The result is a program of its own; once any pair is dropped, its value is never reported as the undropped one.

A run may prefer a weaker bound that is cheaper to compute: a joint sector whose deficits it gives up for every key-live acceptance value needs no matrix function (Corollary 9.4 (iii)). What the run gives up is only the entropy those deficits carry; the sector stays in the program.

Definition 9.2 (Dropped set and dropped program)**.** Let $\mathfrak{D}$ be a declaration in the general format of Definition 1.1. A *dropped set* is a set $\mathcal{D} \subseteq \widehat{\mathcal{C}}_{\mathrm{key}} \times (\mathbb{T}\times\mathbb{S})$ of pairs of a key-live acceptance value and a joint sector.

For $\omega = \bigoplus_{(t,s)} \omega^{(t,s)}$ block diagonal along the joint sectors put

$$K_{\hat{c}}^{\mathcal{D}}(\omega) \coloneqq P_{\hat{c}}(\omega) + \sum_{(t,s)\colon (\hat{c},(t,s))\notin\mathcal{D}} f_{\hat{c},(t,s)}\big(\omega^{(t,s)}\big) \qquad (\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}), \tag{9.3}$$

with $f_{\hat{c},(t,s)}$ of (4.8); let $O_\alpha^{\mathcal{D}}$ be (5.3) with $K_{\hat{c}}^{\mathcal{D}}$ in place of $K_{\hat{c}}$, with the same conventions at vanishing arguments; and let

$$\underline{h}_\alpha^{\mathcal{D}} \coloneqq \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \inf_{\omega\in\mathcal{W}} O_\alpha^{\mathcal{D}}(\mathbf{q},\omega). \tag{9.4}$$

The minimisation in (9.4) is the *dropped program*. A joint sector (t,s) is *fully dropped* when $\widehat{\mathcal{C}}_{\mathrm{key}}\times\{(t,s)\}\subseteq\mathcal{D}$. For $\mathcal{D}=\varnothing$ these are $K_{\hat{c}}$ ((4.11)), O_α and $\underline{h}_\alpha$.

What is not changed. The feasible set is $\mathcal{W}$, unchanged: a dropped sector's block remains a variable, enters $\mathbf{\Phi}$ and every $P_{\hat{c}}$ with its full weight — the term $P_{\hat{c},(t,s)}(\omega^{(t,s)})$ stays inside $P_{\hat{c}}$ in (9.3) — and enters the marginal condition and every constraint row. Setting a dropped block to zero, omitting its statistics, or removing its weight from $P_{\hat{c}}$ in (9.3) are different operations, and each can overstate the value (Proposition 6.3 for the first). *The dropped set is a formulation choice of a run and not declared data.*

Uses Def. 1.1, Thm. 4.1, Def. 4.8, Def. 5.1 · *Used by* Prop. 9.3, Cor. 9.4, Prop. 9.5, Lem. 9.6, Prop. 10.12, Req. 15.10, Req. 16.2, Cor. 19.6, Lem. 21.1, Lem. 21.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.joint:joint_objective_value

Setting the deficits of a dropped set to zero can only lower the objective, so the dropped value lies between 0 and h_low, hence below the single-round rate, and never grows as the set grows. Dropping only key-inert pairs changes nothing.

Deleting summands that are all non-positive can only raise each K, and the objective falls as K rises. The loss is in the rate and needs no hypothesis on the key map or the key holder, and it is a bound only because the dropped sectors stay in the program.

Proposition 9.3 (The dropped program bounds the value from below)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 in the general format, in either source mode and with an arbitrary list $\mathfrak{L}$; let $\alpha\in(1,\infty)$ and let $\mathcal{D}\subseteq\mathcal{D}'$ be dropped sets (Definition 9.2). For every ω block diagonal along the joint sectors, every $\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}$ and every distribution $\mathbf{q}$ on $\widehat{\mathcal{C}}$:

(i) $0\le K_{\hat{c}}(\omega)\le K_{\hat{c}}^{\mathcal{D}}(\omega)\le K_{\hat{c}}^{\mathcal{D}'}(\omega)\le P_{\hat{c}}(\omega)$;

(ii) $O_\alpha^{\mathcal{D}'}(\mathbf{q},\omega)\le O_\alpha^{\mathcal{D}}(\mathbf{q},\omega)\le O_\alpha(\mathbf{q},\omega)$ in the extended-real sense, and $O_\alpha^{\mathcal{D}'}(\mathbf{q},\omega)\ge 0$ whenever $\operatorname{tr}\omega=1$;

(iii) consequently $0\le\underline{h}_\alpha^{\mathcal{D}'}\le\underline{h}_\alpha^{\mathcal{D}}\le\underline{h}_\alpha\le h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L})$;

(iv) if every pair of $\mathcal{D}$ is key-inert (Definition 4.8) then $K_{\hat{c}}^{\mathcal{D}} = K_{\hat{c}}$ and $O_\alpha^{\mathcal{D}} = O_\alpha$ identically.

Uses Def. 1.2, Def. 4.8, Lem. 4.9, Def. 5.1, Cor. 5.6, Def. 9.2, foundations/def:kl, foundations/lem:kernel · *Used by* Cor. 9.4, Prop. 9.5, Req. 15.10, Cor. 19.6, Lem. 21.1, Cor. 21.3 · newly derived · accepted

Proof. (i) By (4.11) and (9.3), the three quantities $K_{\hat{c}}$, $K_{\hat{c}}^{\mathcal{D}}$, $K_{\hat{c}}^{\mathcal{D}'}$ are $P_{\hat{c}}(\omega)$ plus the sum of $f_{\hat{c},(t,s)}(\omega^{(t,s)})$ over all joint sectors, over those with $(\hat{c},(t,s)) \notin \mathcal{D}$, and over those with $(\hat{c},(t,s)) \notin \mathcal{D}'$ respectively: three nested index sets, the last the smallest. Every summand is ≤ 0 (Lemma 4.9 (ii)), so deleting summands does not decrease the sum, which gives the two middle inequalities, and the last holds because the remaining sum is ≤ 0. $K_{\hat{c}}(\omega) \geq 0$: by Lemma 4.9 (ii), $P_{\hat{c},(t,s)} + f_{\hat{c},(t,s)} \geq 0$ on every joint sector, and $K_{\hat{c}}(\omega)$ is the sum of these over the joint sectors.

(ii) In (5.3) the quantity K enters only through the terms $\frac{1}{\alpha-1} d\big(q(\hat{c}), K\big)$, $\hat{c} \in \widehat{\mathcal{C}}_{\text{key}}$, with d the kernel of Lemma "Extended divergence kernel" (foundations/lem:kernel) — whose conventions at vanishing arguments are those of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl) — and the prefactor is positive. d is non-increasing in its second argument, so (i) gives the two inequalities termwise, in the extended-real sense. For the lower bound let $\operatorname{tr}\omega = 1$, so that $\mathbf{P} \coloneqq \mathbf{\Phi}[\omega]$ is a probability distribution. If some $q(\hat{c}) > 0$ has $P_{\hat{c}} = 0$, or $K_{\hat{c}}^{\mathcal{D}'} = 0$ with $\hat{c} \in \widehat{\mathcal{C}}_{\text{key}}$, the objective is $+\infty$. Otherwise, by $K_{\hat{c}}^{\mathcal{D}'} \leq P_{\hat{c}}$ and monotonicity of d, $d(q(\hat{c}), K_{\hat{c}}^{\mathcal{D}'}) \geq d(q(\hat{c}), P_{\hat{c}})$, so

$$O_\alpha^{\mathcal{D}'}(\mathbf{q},\omega) \;\geq\; \frac{\alpha}{\alpha-1} \sum_{\hat{c}\in\widehat{\mathcal{C}}} q(\hat{c}) \log_2 \frac{q(\hat{c})}{P_{\hat{c}}} = \frac{\alpha}{\alpha-1}\, D(\mathbf{q}\,\|\,\mathbf{P}) \;\geq\; 0,$$

the terms $\hat{c} \notin \widehat{\mathcal{C}}_{\text{key}}$ carrying the prefactor $\frac{\alpha}{\alpha-1}$ already and the terms $\hat{c} \in \widehat{\mathcal{C}}_{\text{key}}$ the sum $1 + \frac{1}{\alpha-1}$ of their two prefactors; the last step is non-negativity of the divergence between probability distributions, Gibbs' inequality of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl).

(iii) Take infima over $\mathcal{S}_{\text{acc}} \times \mathcal{W}$ in (ii); every $\omega \in \mathcal{W}$ has $\operatorname{tr}\omega = 1$. The last inequality is Corollary 5.6.

(iv) For a key-inert pair $f_{\hat{c},(t,s)} \equiv 0$ (Lemma 4.9 (iii)), so the summands deleted in (9.3) are identically zero. □

The dropped objective is jointly convex and lower semicontinuous, its infimum attained when finite, and on a fully dropped joint sector it reads the block only through its statistics. As the Rényi order tends to one the dropped value tends to the relative-entropy program over the kept pairs.

The dropped program keeps the undropped program's feasible set and only removes deficits from each K, so the properties the program's numerics rest on carry over, and a fully dropped joint sector costs no matrix function.

Part (iii) is written in state coordinates; in ensemble form the statistics operators are read in the Choi coordinates of Lemma 7.4, Alice's blocks not being declared there.

Corollary 9.4 (Convexity, gradient and limit of the dropped program)**.** Under the hypotheses of Proposition 9.3, with $\mathcal{S}_{\mathrm{acc}}$ convex and compact (Definition 2.5):

(i) $K_{\hat{c}}^{\mathcal{D}}$ is concave and continuous on the cone of operators block diagonal along the joint sectors;

(ii) $O_{\alpha}^{\mathcal{D}}$ is jointly convex and lower semicontinuous on $\Delta(\widehat{\mathcal{C}})\times\mathcal{W}$ with values in $[0,+\infty]$, and the infimum (9.4) is attained whenever it is finite;

(iii) if (t,s) is fully dropped, then $O_{\alpha}^{\mathcal{D}}$ depends on the block $\omega^{(t,s)}$ only through the real numbers $\operatorname{tr}\Big[(M_x^{(t)}\otimes F_v^{(s)})\,\omega^{(t,s)}\Big]$, $(x,v)\in\mathcal{X}\times\mathcal{V}$; at a point where every $P_{\hat{c}}$ with $q(\hat{c})>0$ and every $K_{\hat{c}}^{\mathcal{D}}$ with $q(\hat{c})>0$, $\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}$, is positive, its derivative in the direction of a Hermitian Δ supported on that joint sector is

$$-\log_2\mathrm{e}\sum_{\hat{c}\in\widehat{\mathcal{C}}}\beta_{\hat{c}}\,\operatorname{tr}\big[\Gamma_{\hat{c}}^{(t,s)}\Delta\big],\qquad \Gamma_{\hat{c}}^{(t,s)}:=\sum_{(x,v):\,\phi(a(x,v))=\hat{c}}M_x^{(t)}\otimes F_v^{(s)},\qquad \beta_{\hat{c}}:=\begin{cases}\frac{\alpha}{\alpha-1}\frac{q(\hat{c})}{P_{\hat{c}}}, & \hat{c}\notin\\ \frac{q(\hat{c})}{P_{\hat{c}}}+\frac{1}{\alpha-1}\frac{q(\hat{c})}{K_{\hat{c}}^{\mathcal{D}}}, & \hat{c}\in\end{cases}\tag{9.5}$$

with $\beta_{\hat{c}}:=0$ where $q(\hat{c})=0$;

(iv) under the additional hypotheses of Theorem 8.9,

$$\lim_{\alpha\downarrow1}h_{\alpha}^{\mathcal{D}}=\inf_{\omega\in\mathcal{W}_1}\sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}}\sum_{\substack{(t,s)\in\mathbb{J}_K(\hat{c})\\(\hat{c},(t,s))\notin\mathcal{D}}}D\Big(\mathcal{G}_{\hat{c},(t,s)}\big(\omega^{(t,s)}\big)\,\Big\|\,\mathcal{Z}\big(\mathcal{G}_{\hat{c},(t,s)}\big(\omega^{(t,s)}\big)\big)\Big).\tag{9.6}$$

Uses Def. 1.2, Def. 1.4, Def. 2.5, Thm. 4.1, Def. 4.8, Lem. 4.9, Def. 5.1, Lem. 8.8, Thm. 8.9, Def. 9.2, Prop. 9.3, foundations/def:kl, foundations/lem:kernel · *Used by* Prop. 9.5, Cor. 19.6 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.joint_gradient:joint_gradient

Proof. Idea. Each $K_{\hat{c}}^{\mathcal{D}}$ is $P_{\hat{c}}$ plus a sum of concave deficits, so (i) holds summand by summand, and (ii) follows from the monotonicity and joint convexity of the kernel as for any concave, continuous, non-negative argument, with attainment from compactness. On a fully dropped joint sector only the linear $P_{\hat{c}}$ read the block, which gives (iii). For (iv) the proof of Theorem 8.9 goes through with the kept summands in place of all of them; we list the changes.

(i) $P_{\hat{c}}$ is linear. Each $f_{\hat{c},(t,s)}$ is concave on its cone (Lemma 4.9 (ii)), and $\omega\mapsto\omega^{(t,s)}$ is linear, so (9.3) is a sum of concave functions. Continuity: $X\mapsto X^{1/\alpha}$ and $Y\mapsto Y^{\alpha}$ are continuous on the positive semidefinite cone by

the continuous functional calculus, $\mathcal{Z}$, $\mathcal{G}_{\hat{c},s}$ and the trace are linear, so each $f_{\hat{c},(t,s)}$ of (4.2) is continuous.

(ii) Write $O_\alpha^{\mathcal{D}}$ as the sum over $\hat{c}$ of $\frac{\alpha}{\alpha-1}d(q(\hat{c}),P_{\hat{c}}(\omega))$ for $\hat{c}\notin\widehat{\mathcal{C}}_{\text{key}}$ and of $d(q(\hat{c}),P_{\hat{c}}(\omega))+\frac{1}{\alpha-1}d(q(\hat{c}),K_{\hat{c}}^{\mathcal{D}}(\omega))$ for $\hat{c}\in\widehat{\mathcal{C}}_{\text{key}}$, with d the kernel of Lemma "Extended divergence kernel" (foundations/lem:kernel). Let g be concave, continuous and non-negative on a convex set, as $P_{\hat{c}}$ and, by (i) and Proposition 9.3 (i), $K_{\hat{c}}^{\mathcal{D}}$ are on the convex set $\mathcal{W}$. For a convex combination with weights λ, $1-\lambda$: $g(\omega_\lambda)\geq\lambda g(\omega_1)+(1-\lambda)g(\omega_2)$, so, d being non-increasing in its second argument and jointly convex, $d(q_\lambda,g(\omega_\lambda))\leq d\big(q_\lambda,\lambda g(\omega_1)+(1-\lambda)g(\omega_2)\big)\leq\lambda d(q_1,g(\omega_1))+(1-\lambda)d(q_2,g(\omega_2))$. For lower semicontinuity let $(q_n,\omega_n)\to(q,\omega)$; then $g(\omega_n)\to g(\omega)$ and lower semicontinuity of d gives $\liminf_n d(q_n,g(\omega_n))\geq d(q,g(\omega))$. Sums with positive coefficients preserve both properties. Non-negativity is Proposition 9.3 (ii). $\mathcal{S}_{\text{acc}}\times\mathcal{W}$ is compact (Definition 2.5, Definition 5.1), and a lower semicontinuous function with values in $(-\infty,+\infty]$ attains its infimum on a non-empty compact set.

(iii) If (t,s) is fully dropped, no $f_{\hat{c},(t,s)}$ occurs in any $K_{\hat{c}}^{\mathcal{D}}$, so the block enters $O_\alpha^{\mathcal{D}}$ through the values $P_{\hat{c}}(\omega)$ alone, and its contribution to $P_{\hat{c}}$ is $P_{\hat{c},(t,s)}(\omega^{(t,s)})=\operatorname{tr}\Big[\Gamma_{\hat{c}}^{(t,s)}\omega^{(t,s)}\Big]$ by (1.16), a sum of the stated numbers. With every relevant argument positive, $\frac{\partial}{\partial v}\,u\log_2\frac{u}{v}=-\log_2\mathrm{e}\,\frac{u}{v}$, and the directional derivative of $P_{\hat{c}}$ and, the deficits of the block being absent, of $K_{\hat{c}}^{\mathcal{D}}$ along Δ is $\operatorname{tr}\Big[\Gamma_{\hat{c}}^{(t,s)}\Delta\Big]$ in both cases. Collecting the prefactors of (ii) gives (9.5); terms with $q(\hat{c})=0$ are identically zero.

(iv) We list the changes to the proof of Theorem 8.9, which is written for $\mathcal{D}=\varnothing$ and the index s. Write $\sigma_{\hat{c},(t,s)}(\omega)\coloneqq\mathcal{G}_{\hat{c},(t,s)}(\omega^{(t,s)})$, an operator of trace at most 1 for $\omega\in\mathcal{W}$, and $e_{\hat{c}}^{\mathcal{D}}\coloneqq\frac{1}{1-\alpha}\log_2\frac{K_{\hat{c}}^{\mathcal{D}}}{P_{\hat{c}}}$, which is ≥ 0 by Proposition 9.3 (i). By (9.3), (4.2) and $B_\sigma(1)=\operatorname{tr}\sigma$,

$$K_{\hat{c}}^{\mathcal{D}}(\omega)-P_{\hat{c}}(\omega)=\sum_{(t,s)\colon(\hat{c},(t,s))\notin\mathcal{D}}\Big(B_{\sigma_{\hat{c},(t,s)}(\omega)}(\alpha)-B_{\sigma_{\hat{c},(t,s)}(\omega)}(1)\Big),$$

which replaces the identity for $K_{\hat{c}}-P_{\hat{c}}$ used there. *Upper bound:* with $\mathbf{q}=\mathbf{\Phi}[\omega]$ the divergence terms vanish and $O_\alpha^{\mathcal{D}}(\mathbf{q},\omega)=\sum_{\hat{c}\in\widehat{\mathcal{C}}_{\text{key}}}P_{\hat{c}}\,e_{\hat{c}}^{\mathcal{D}}$; the limit is taken summand by summand with Lemma 8.8, exactly as there, and gives the sum of relative entropies over the kept pairs; pairs that are key-inert contribute zero because their σ is block diagonal for $\mathcal{Z}$ (Lemma 4.9 (iii) and its proof), which restricts the sum to $\mathbb{J}_K(\hat{c})$. *Lower bound:* the decomposition of the objective into the divergence part and $\sum_{\hat{c}}q(\hat{c})e_{\hat{c}}^{\mathcal{D}}$ is the identity (5.7) of real numbers, valid for any $K\in(0,P]$; both parts are non-negative; compactness of $\mathcal{S}_{\text{acc}}\times\mathcal{W}$, lower semicontinuity of the divergence and the conclusion $\omega^*\in\mathcal{W}_1$ do not involve K; $u_n\coloneqq(K_{\hat{c}}^{\mathcal{D}}-P_{\hat{c}})/P_{\hat{c}}\in[-1,0]$ by Proposition 9.3 (i); and the mean-value step with the majorant (8.18) and

the joint continuity of $c_\sigma(\alpha)$ is applied to each kept summand, the maps $\omega \mapsto \sigma_{\hat{c},(t,s)}(\omega)$ being continuous. The resulting lim inf is the right-hand side of (9.6) at ω^*. □

Dropping every pair leaves $\alpha/(\alpha-1)$ times the minimal divergence between accepted distributions and feasible statistics, and h_low lies between that and it plus $\log_2(d_Z+1)$ times the key-live mass. A positive lower bound proves that no feasible state has accepted statistics.

The objective is a divergence penalty plus one entropy term per key-live acceptance value, and only the entropy terms are bounded (Proposition 5.8); an infinite value is the extreme case (Remark 5.9). Dropping every pair removes the entropy terms altogether, so the dropped program then measures the penalty alone: how far the accepted distributions lie from the statistics of the feasible set.

Proposition 9.5 (The minimal divergence as the value with every pair dropped)**.** Let $\mathfrak{D}$ satisfy Definition 1.2 in the general format, in either source mode and with an arbitrary list $\mathfrak{L}$, let $\mathcal{S}_{\mathrm{acc}}$ satisfy Definition 2.5, and let $\alpha \in (1,\infty)$. Let $\mathcal{D}_{\mathrm{all}} \coloneqq \widehat{\mathcal{C}}_{\mathrm{key}} \times (\mathbb{T} \times \mathbb{S})$ be the dropped set of every pair (Definition 9.2), and put

$$D_{\min} \coloneqq \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \inf_{\omega\in\mathcal{W}} D\big(\mathbf{q} \,\big\|\, \mathbf{\Phi}[\omega]\big) \in [0,+\infty], \qquad m_{\mathrm{key}} \coloneqq \sup_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}} q(\hat{c}) \in [0,1], \tag{9.7}$$

with $\inf \varnothing = +\infty$ and $\sup \varnothing = 0$.

(i) *(The value with every pair dropped.)* For every distribution $\mathbf{q}$ on $\widehat{\mathcal{C}}$ and every $\omega \in \mathcal{W}$, $O_\alpha^{\mathcal{D}_{\mathrm{all}}}(\mathbf{q},\omega) = \frac{\alpha}{\alpha-1}\, D(\mathbf{q}\|\mathbf{\Phi}[\omega])$; hence $\underline{h}_\alpha^{\mathcal{D}_{\mathrm{all}}} = \frac{\alpha}{\alpha-1}\, D_{\min}$.

(ii) *(The value between its floor and its ceiling.)*

$$\frac{\alpha}{\alpha-1}\, D_{\min} \;\le\; \underline{h}_\alpha \;\le\; \frac{\alpha}{\alpha-1}\, D_{\min} + \log_2(d_Z+1)\, m_{\mathrm{key}}. \tag{9.8}$$

(iii) *(When the floor is positive.)* $D_{\min} = 0$ if and only if $\mathbf{q} = \mathbf{\Phi}[\omega]$ for some $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and some $\omega \in \mathcal{W}$. Equivalently, $D_{\min} > 0$ if and only if no distribution of $\mathcal{S}_{\mathrm{acc}}$ is the statistics $\mathbf{\Phi}[\omega]$ of a state $\omega \in \mathcal{W}$.

(iv) *(Bounds at one order serve every order.)* Let b be a real number with $b \le \underline{h}_\alpha^{\mathcal{D}_{\mathrm{all}}}$, and let $(\mathbf{q},\omega) \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}$. Then

$$\frac{\alpha-1}{\alpha}\, b \;\le\; D_{\min} \;\le\; \frac{\alpha-1}{\alpha}\, O_\alpha^{\mathcal{D}_{\mathrm{all}}}(\mathbf{q},\omega), \tag{9.9}$$

and for every $\alpha' \in (1,\infty)$ the value $\underline{h}_{\alpha'}$ of (5.4) at the order α' satisfies $\underline{h}_{\alpha'} \ge \frac{\alpha'}{\alpha'-1} \cdot \frac{\alpha-1}{\alpha}\, b$. If $b > 0$, then $D_{\min} > 0$, and no distribution of $\mathcal{S}_{\mathrm{acc}}$ is the statistics of a state of $\mathcal{W}$. For every $\alpha' \ge \alpha$,

$$\frac{\alpha'}{\alpha'-1}\, D_{\min} \;\le\; \frac{\alpha'}{\alpha'-1} \cdot \frac{\alpha-1}{\alpha}\, O_\alpha^{\mathcal{D}_{\mathrm{all}}}(\mathbf{q},\omega) \;\le\; O_\alpha^{\mathcal{D}_{\mathrm{all}}}(\mathbf{q},\omega), \tag{9.10}$$

so that $\underline{h}_{\alpha'} \le \frac{\alpha'}{\alpha'-1} \cdot \frac{\alpha-1}{\alpha}\, O_\alpha^{\mathcal{D}_{\mathrm{all}}}(\mathbf{q},\omega) + \log_2(d_Z+1)\, m_{\mathrm{key}}$.

Reading: The key-rate program charges a penalty whenever the accepted statistics differ from the per-round statistics of every state the declared devices admit. This result isolates that penalty. If some such state has statistics the acceptance test accepts, the program's value is at most the entropy ceiling, log₂(d_Z+1) times the largest key-live probability an accepted distribution gives. If none has, the value has a floor that grows without limit as the Rényi order approaches one, so a key length can grow with the mismatch rather than with the secrecy of the key. A positive lower bound on the penalty-only program, which the engine computes with its usual solver, proves that no such state has accepted per-round statistics. It does not prove that a finite run of those devices fails the test: its frequencies can still fall in the acceptance set.

Uses Def. 1.2, Def. 1.4, Def. 2.5, Def. 4.8, Def. 5.1, Prop. 5.8, Def. 9.2, Prop. 9.3, Cor. 9.4, foundations/def:kl · *Used by* Prop. 10.12, Req. 16.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.consistency:proven_floor

Proof. Idea. With every pair dropped, K equals P and the two prefactors of a key-live term add up to that of the other terms, so the objective is the scaled divergence. The entropy terms are non-negative and each at most $\log_2(d_Z+1)$, which places $\underline{h}_\alpha$ between the scaled minimal divergence and that plus the entropy ceiling. A divergence vanishes only between equal distributions, and the minimum is attained.

Throughout, $\omega \in \mathcal{W}$ has $\operatorname{tr}\omega = 1$ (Definition 5.1), so $\mathbf{\Phi}[\omega]$ of (1.16) is a probability distribution on $\widehat{\mathcal{C}}$ whose entry at $\hat{c}$ is $P_{\hat{c}}(\omega)$, as in the proof of Proposition 9.3 (ii).

(i). Every pair of $\widehat{\mathcal{C}}_{\mathrm{key}} \times (\mathbb{T}\times\mathbb{S})$ lies in $\mathcal{D}_{\mathrm{all}}$, so the sum in (9.3) is empty and $K_{\hat{c}}^{\mathcal{D}_{\mathrm{all}}}(\omega) = P_{\hat{c}}(\omega)$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$. Fix $\hat{c}$ and write $q = q(\hat{c})$, $P = P_{\hat{c}}(\omega)$. For $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{key}}$ the term of (5.3) is $\frac{\alpha}{\alpha-1} q \log_2 \frac{q}{P}$. For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, with K replaced by P, it is $q\log_2\frac{q}{P} + \frac{q}{\alpha-1}\log_2\frac{q}{P}$: if $q=0$ both summands are 0 by the convention $0\log_2\frac{0}{v}=0$ of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl), as is $\frac{\alpha}{\alpha-1} q\log_2\frac{q}{P}$; if $q>0$ and $P=0$ both are $+\infty$, as is $\frac{\alpha}{\alpha-1} q\log_2\frac{q}{P}$; if $q,P>0$ they add up to $\big(1+\frac{1}{\alpha-1}\big)q\log_2\frac{q}{P} = \frac{\alpha}{\alpha-1}q\log_2\frac{q}{P}$. Summing over $\hat{c}\in\widehat{\mathcal{C}}$ gives $O_\alpha^{\mathcal{D}_{\mathrm{all}}}(\mathbf{q},\omega) = \frac{\alpha}{\alpha-1}\sum_{\hat{c}} q(\hat{c})\log_2\frac{q(\hat{c})}{P_{\hat{c}}(\omega)} = \frac{\alpha}{\alpha-1} D(\mathbf{q}\|\mathbf{\Phi}[\omega])$ in the extended-real sense. Taking the infimum over $\mathcal{S}_{\mathrm{acc}}\times\mathcal{W}$ in (9.4), with the positive factor $\frac{\alpha}{\alpha-1}$ taken out (and $\frac{\alpha}{\alpha-1}\cdot(+\infty)=+\infty$), gives $\underline{h}_\alpha^{\mathcal{D}_{\mathrm{all}}} = \frac{\alpha}{\alpha-1} D_{\min}$.

(ii), the lower inequality. By Proposition 9.3 (iii), $\underline{h}_\alpha^{\mathcal{D}_{\mathrm{all}}} \le \underline{h}_\alpha$, and by (i) the left-hand side is $\frac{\alpha}{\alpha-1} D_{\min}$.

(ii), the upper inequality. If $D_{\min} = +\infty$ the right-hand side is $+\infty$. Otherwise let $(\mathbf{q},\omega)\in\mathcal{S}_{\mathrm{acc}}\times\mathcal{W}$ with $D(\mathbf{q}\|\mathbf{\Phi}[\omega]) < +\infty$; then $q(\hat{c})>0$ implies $P_{\hat{c}}(\omega)>0$. For $\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{key}}$ with $q(\hat{c})>0$ let $e_{\hat{c}}(\omega) := \frac{1}{1-\alpha}\log_2\frac{K_{\hat{c}}(\omega)}{P_{\hat{c}}(\omega)}$; by Proposition 5.8 (ii), $P_{\hat{c}}(\omega)>0$ gives $e_{\hat{c}}(\omega)\le\log_2(d_Z+1)$, so $e_{\hat{c}}(\omega)$ is a real number and $K_{\hat{c}}(\omega)>0$. For such $\hat{c}$, with $q=q(\hat{c})$, $P=P_{\hat{c}}(\omega)$, $K=K_{\hat{c}}(\omega)$, the term of (5.3) is $q\log_2\frac{q}{P} + \frac{q}{\alpha-1}\big(\log_2\frac{q}{P} + \log_2\frac{P}{K}\big) = \frac{\alpha}{\alpha-1}q\log_2\frac{q}{P} + q\,e_{\hat{c}}(\omega)$;

for $\hat{c} \in \widehat{\mathcal{C}}_{\text{key}}$ with $q(\hat{c}) = 0$ the term is $0 = \frac{\alpha}{\alpha-1} q(\hat{c}) \log_2 \frac{q(\hat{c})}{P_{\hat{c}}(\omega)}$. Summing,

$$O_\alpha(\mathbf{q}, \omega) = \frac{\alpha}{\alpha - 1} D(\mathbf{q} \| \mathbf{\Phi}[\omega]) + \sum_{\hat{c} \in \widehat{\mathcal{C}}_{\text{key}} \colon q(\hat{c}) > 0} q(\hat{c})\, e_{\hat{c}}(\omega) \le \frac{\alpha}{\alpha - 1} D(\mathbf{q} \| \mathbf{\Phi}[\omega]) + \log_2(d_Z + 1)\, m_{\text{key}},$$

using $\sum_{\hat{c} \in \widehat{\mathcal{C}}_{\text{key}}} q(\hat{c}) \le m_{\text{key}}$. For every $\varepsilon > 0$ there is such a pair with $D(\mathbf{q} \| \mathbf{\Phi}[\omega]) \le D_{\min} + \varepsilon$, so $\underline{h}_\alpha \le O_\alpha(\mathbf{q}, \omega) \le \frac{\alpha}{\alpha-1}(D_{\min} + \varepsilon) + \log_2(d_Z + 1)\, m_{\text{key}}$; letting $\varepsilon \to 0$ gives (9.8).

(iii). If $\mathbf{q} = \mathbf{\Phi}[\omega]$ with $\mathbf{q} \in \mathcal{S}_{\text{acc}}$ and $\omega \in \mathcal{W}$, then $D(\mathbf{q} \| \mathbf{\Phi}[\omega]) = 0$, and since $D_{\min} \ge 0$ ($\underline{h}_\alpha^{\mathcal{D}_{\text{all}}} \ge 0$ by Proposition 9.3 (iii), and (i)), $D_{\min} = 0$. Conversely let $D_{\min} = 0$. Then $\underline{h}_\alpha^{\mathcal{D}_{\text{all}}} = 0$ by (i) is finite, so by Corollary 9.4 (ii) it is attained at some $(\mathbf{q}, \omega) \in \mathcal{S}_{\text{acc}} \times \mathcal{W}$, where by (i) $D(\mathbf{q} \| \mathbf{p}) = 0$ with $\mathbf{p} \coloneqq \mathbf{\Phi}[\omega]$. Let $S \coloneqq \{\hat{c} : q(\hat{c}) > 0\}$. The divergence being finite, $p(\hat{c}) > 0$ on S, and $0 = D(\mathbf{q} \| \mathbf{p}) = -\sum_{\hat{c} \in S} q(\hat{c}) \log_2 \frac{p(\hat{c})}{q(\hat{c})} \ge -\log_2 \sum_{\hat{c} \in S} p(\hat{c}) \ge 0$, the first inequality by concavity of $\log_2$ (Jensen's inequality for the weights $q(\hat{c})$, $\hat{c} \in S$, which sum to 1), the second because $\sum_{\hat{c} \in S} p(\hat{c}) \le 1$. Both are equalities. The second gives $\sum_{\hat{c} \in S} p(\hat{c}) = 1$, so $p(\hat{c}) = 0$ off S. In the first, $\log_2$ is strictly concave, so equality forces $\frac{p(\hat{c})}{q(\hat{c})}$ to be one constant λ on S; summing $p(\hat{c}) = \lambda q(\hat{c})$ over S gives $\lambda = 1$. Hence $\mathbf{q} = \mathbf{p} = \mathbf{\Phi}[\omega]$. The second sentence is the contrapositive of the first, $D_{\min}$ lying in $[0, +\infty]$.

(iv). By (i), $b \le \frac{\alpha}{\alpha-1} D_{\min}$, which is the lower inequality of (9.9); by (i) again, $\frac{\alpha-1}{\alpha} O_\alpha^{\mathcal{D}_{\text{all}}}(\mathbf{q}, \omega) = D(\mathbf{q} \| \mathbf{\Phi}[\omega]) \ge D_{\min}$, the upper one. The lower inequality of (9.8) at the order α', whose $D_{\min}$ and $\mathcal{W}$ do not depend on the order, gives $\underline{h}_{\alpha'} \ge \frac{\alpha'}{\alpha'-1} D_{\min} \ge \frac{\alpha'}{\alpha'-1} \cdot \frac{\alpha-1}{\alpha} b$. If $b > 0$, the lower inequality of (9.9) gives $D_{\min} > 0$, and (iii) the assertion that follows it. For (9.10), multiply the upper inequality of (9.9) by $\frac{\alpha'}{\alpha'-1} > 0$; the second inequality holds because $t \mapsto \frac{t}{t-1} = 1 + \frac{1}{t-1}$ is decreasing on $(1, \infty)$, so $\frac{\alpha'}{\alpha'-1} \le \frac{\alpha}{\alpha-1}$ for $\alpha' \ge \alpha$, and $O_\alpha^{\mathcal{D}_{\text{all}}}(\mathbf{q}, \omega) \ge 0$ (Proposition 9.3 (ii)). The last bound on $\underline{h}_{\alpha'}$ is the upper inequality of (9.8) at the order α' with the first inequality of (9.10). □

Read on the whole cone, the objective, its Choi form and the dropped objective are positively homogeneous of degree one: scaling the acceptance vector and the state by $c > 0$ scales the value by c. This is what lets a first-order bracket be formed without evaluating the objective.

The objective is defined on the acceptance set and the feasible set, but its formula makes sense on the whole cone, and there it scales linearly: the divergence terms are jointly homogeneous of degree one in their two arguments, the statistics are linear, and the pinched trace power is homogeneous of degree one because $p\alpha = 1$. Differentiated along the point itself, this says that the gradient paired with the point is the value at the point, which is how the Frank–Wolfe submodule forms a bracket from its two linear minima

alone (Lemma 21.1).

Lemma 9.6 (Positive homogeneity of the objective). Let $\alpha \in (1,\infty)$ and let $\mathfrak{D}$ be a tuple of the form of Definition 1.1, in the basic or the general format, that satisfies conditions (i)–(vi) and (viii) of Definition 1.2; its maps are read below. For $\mathbf{q} \in [0,\infty)^{\widehat{\mathcal{C}}}$ and $\omega \in \mathrm{Pos}(AQ)$ block diagonal along the sectors, along the joint sectors in the general format of Definition 1.1, let $O_\alpha(\mathbf{q},\omega)$ be the right-hand side of (5.3), with the conventions of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl) at vanishing arguments, and for a dropped set $\mathcal{D}$ let $K_{\hat{c}}^{\mathcal{D}}$ and $O_\alpha^{\mathcal{D}}$ be those of Definition 9.2 at the same arguments. These are defined on all such $(\mathbf{q},\omega)$, with values in $(-\infty,+\infty]$: there $P_{\hat{c}}(\omega) \geq 0$ and $0 \leq K_{\hat{c}}(\omega) \leq K_{\hat{c}}^{\mathcal{D}}(\omega)$. Let $c > 0$, and read $c \cdot (+\infty) := +\infty$.

(i) $P_{\hat{c}}(c\omega) = c\,P_{\hat{c}}(\omega)$ for every $\hat{c} \in \widehat{\mathcal{C}}$; every sector deficit and every joint-sector deficit satisfies $f(c\sigma) = c\,f(\sigma)$ on its cone; and $K_{\hat{c}}(c\omega) = c\,K_{\hat{c}}(\omega)$ and $K_{\hat{c}}^{\mathcal{D}}(c\omega) = c\,K_{\hat{c}}^{\mathcal{D}}(\omega)$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$.

(ii) $O_\alpha(c\mathbf{q},c\omega) = c\,O_\alpha(\mathbf{q},\omega)$ and $O_\alpha^{\mathcal{D}}(c\mathbf{q},c\omega) = c\,O_\alpha^{\mathcal{D}}(\mathbf{q},\omega)$.

(iii) In trusted-source mode, with χ_Q the assignment of Theorem 7.5 for a support frame, the Choi forms $O_\alpha^J(\mathbf{q},J) := O_\alpha(\mathbf{q},\chi_Q(J))$ and $O_\alpha^{J,\mathcal{D}}(\mathbf{q},J) := O_\alpha^{\mathcal{D}}(\mathbf{q},\chi_Q(J))$ satisfy $O_\alpha^J(c\mathbf{q},cJ) = c\,O_\alpha^J(\mathbf{q},J)$ and the same for $O_\alpha^{J,\mathcal{D}}$, at every $\mathbf{q} \in [0,\infty)^{\widehat{\mathcal{C}}}$ and every J whose blocks are positive semidefinite; in untrusted-source mode the Choi form is the state form (Definition 7.10) and (ii) is the statement.

Uses Def. 1.1, Def. 1.2, Def. 1.4, Thm. 4.1, Lem. 4.3, Cor. 4.4, Def. 4.8, Lem. 4.9, Def. 5.1, Thm. 7.5, Def. 7.10, Def. 9.2, foundations/def:kl · *Used by* Lem. 21.1 · newly derived · accepted

Proof. The hypothesis. The facts cited below, the cone clause of Theorem 4.1, Lemma 4.9 (ii), Corollary 4.4 and Lemma 4.3 (a), are stated for admissible declarations or for the cone of a branch space, and their proofs read the declaration's maps, positivity and block structure only, never condition (vii) of Definition 1.2, the non-empty feasible set; so they hold of $\mathfrak{D}$.

The domain. $P_{\hat{c}}(\omega) = \mathbf{\Phi}[\omega](\hat{c})$ is linear in ω ((1.16)) and non-negative on $\mathrm{Pos}(AQ)$, being a sum of traces of products of positive semidefinite operators. The cone clause of Theorem 4.1 gives $0 \leq K_{\hat{c}}(\omega)$ on the whole cone, and in the general format so does Lemma 4.9 (ii) through (4.11), each $P_{\hat{c},(t,s)} + f_{\hat{c},(t,s)}$ being non-negative. $K_{\hat{c}}^{\mathcal{D}}$ is $K_{\hat{c}}$ with the deficits of the pairs of $\mathcal{D}$ removed ((9.3), (4.11)), and every deficit is non-positive (Corollary 4.4, Lemma 4.9 (ii)), so $K_{\hat{c}}^{\mathcal{D}} \geq K_{\hat{c}} \geq 0$. So every term $u \log_2 \frac{u}{v}$ of (5.3) has $u \geq 0$ and $v \geq 0$, and by Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl) it is 0 when $u = 0$, $+\infty$ when $u > 0 = v$, and

real otherwise; a sum of finitely many such terms with positive coefficients lies in $(-\infty, +\infty]$.

(i). Linearity of $\mathbf{\Phi}$ gives the first claim. A sector deficit is $f_{\hat{c},s}(\sigma) = T(\mathcal{G}_{\hat{c},s}(\sigma)) - \operatorname{tr}\mathcal{G}_{\hat{c},s}(\sigma)$ ((4.2)) with T the pinched trace power of (4.7); $\mathcal{G}_{\hat{c},s}$ and the trace are linear, and $T(cX) = cT(X)$ by Lemma 4.3 (a), so $f_{\hat{c},s}(c\sigma) = c\, f_{\hat{c},s}(\sigma)$. A joint-sector deficit is a restriction of a sector deficit ((4.8)), and Lemma 4.9 (ii) states its homogeneity as well. By (4.1), $K_{\hat{c}}(\omega) = T(\mathcal{G}_{\hat{c}}(\omega)) + P_{\hat{c}}(\omega) - \operatorname{tr}\mathcal{G}_{\hat{c}}(\omega)$, each term homogeneous by the same two facts; in the general format $K_{\hat{c}}$ and $K^{\mathcal{D}}_{\hat{c}}$ are $P_{\hat{c}}$ plus a sum of joint-sector deficits ((4.11), (9.3)), each term homogeneous.

(ii). For $u, v \geq 0$ and $c > 0$, $(cu)\log_2 \frac{cu}{cv} = c \cdot u \log_2 \frac{u}{v}$ under the conventions of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl): both sides are 0 when $u = 0$, both are $+\infty$ when $u > 0 = v$, and otherwise the logarithm is unchanged by the common factor. Each term of (5.3) is such an expression, multiplied by a positive constant ($\frac{\alpha}{\alpha-1}$, 1 or $\frac{1}{\alpha-1}$), with $u = q(\hat{c})$ and $v = P_{\hat{c}}(\omega)$ or $v = K_{\hat{c}}(\omega)$, and by (i) the pair (u, v) at $(c\mathbf{q}, c\omega)$ is c times the pair at $(\mathbf{q}, \omega)$; the same holds with $K^{\mathcal{D}}_{\hat{c}}$. For $c > 0$ and $a, b \in (-\infty, +\infty]$, $c(a + b) = ca + cb$, so the finite sum scales by c.

(iii). χ_Q is linear and maps a list of positive semidefinite blocks to a block-diagonal positive semidefinite operator (Theorem 7.5), so $O^J_\alpha(c\mathbf{q}, cJ) = O_\alpha(c\mathbf{q}, c\,\chi_Q(J)) = c\,O_\alpha(\mathbf{q}, \chi_Q(J))$ by (ii), and likewise for $O^{J,\mathcal{D}}_\alpha$. In untrusted-source mode χ_Q is the identity (Definition 7.10). □

10 Key length

The accumulated entropy feeds a fixed-length post-processing model with a fixed-length error-correction transcript, error verification and privacy amplification by ideal two-universal hashing, every message public (Definition 10.1). The section's main results are Theorem 10.7, an ε_PA-secret key of the stated length on the accepted state for a Rényi order in (1, 2), and Corollary 10.18, which lets any lower bound h ≤ h_low take the place of h_low through max{0, h} and is where every solver obtains its key length. Error verification makes the key ε_EV-correct (Lemma 10.4), the two compose to (ε_PA + ε_EV)-security with honest authentication, and over a real authentication scheme the protocol is (ε_PA + ε_EV + 2ε_auth)-secure with net key gain ℓ − k_auth (Corollary 10.8), all of it against an adversary whose memory has any dimension, by a limit over finite-dimensional ones (Lemma 10.5) and every adversary of the honest-authentication setting read as such an attack (Lemma 10.6); the parameter bounds one execution, separate executions that share no key add their parameters (Corollary 10.9), and a reused authentication key enters through ε_auth, which the earlier execution's parameter does not bound by itself (Remark 10.10). A run whose

rounds follow statistics of feasible states passes the acceptance test with probability at most 2^(−n D_min), D_min the minimal divergence (Proposition 10.12). The reported maximal key length is that bound floored and clamped at 0, and more leakage or a tighter error budget never lengthens it (Proposition 10.17). The constrained value enters under the attack-set licence (Corollary 10.19), behind a two-part gate whose licensed-class half is vacuous in untrusted-source mode and trivially avoidable at d_A = 1 (Remark 10.20).

Imports foundations/def:kl, foundations/def:renyi, foundations/lem:kernel, foundations/lem:register-compression, reductions/def:protocol

After the n rounds the key holder's raw string passes an acceptance test, one-way error correction with a padded fixed-length transcript, error verification with an announced flag and privacy amplification, each message public and delivered unaltered. Secrecy and correctness are stated in it.

The model is the fixed-length specialisation of the generic post-processing of [5, Sec. 4.1, Protocol 1, steps 2–5]: the key length is either ℓ or zero, error correction is one-way with a transcript of fixed length, and the verification outcome is announced. Every message is public, and the model is stated in the setting in which authentication behaves honestly [5, Sec. 3.2]; how a run over a real authentication scheme reduces to it is a separate step.

Definition 10.1 (Fixed-length post-processing model)**.** Fix a declaration $\mathfrak{D}$ with key holder h (Definition 1.1), a round count n, an acceptance set $\mathcal{S}_{\mathrm{acc}}$, an output length ℓ, a transcript length λ_{EC} and error budgets $\varepsilon_{\mathrm{PA}}, \varepsilon_{\mathrm{EV}} \in (0, 1]$. A hash family is a set of functions with a probability law on it [5, Def. 3.5]. The *fixed-length post-processing model* is the following classical processing, applied after the n round channels of Definition 2.3. The raw string Z_1^n is the content of the classical registers $Z_1, \ldots, Z_n$ of the round channels, which the single-round protocol channel $\mathcal{M}$ retains (Theorem 1.5); the key holder computes it from its own outcome record and the announcements I_1^n through the function r_{h} of (1.1), and we call the other party the partner. The acceptance values $\widehat{C}_1^n$ are announced. The processing appends, in order:

(i) the *acceptance test*: the event $\Omega_{\mathrm{AT}} \coloneqq \{\mathrm{freq}_{\hat{c}_1^n} \in \mathcal{S}_{\mathrm{acc}}\}$, a function of $\widehat{C}_1^n$;

(ii) *error correction*, one-way from the holder: a public seed S_{pub}, drawn with the law the error-correction scheme prescribes, and the public transcript $L_{\mathrm{EC}} = g_{\mathrm{EC}}(Z_1^n, S_{\mathrm{pub}})$, which occupies a fixed-length register of λ_{EC} bits: a shorter transcript is padded to length λ_{EC}, so that $\dim(L_{\mathrm{EC}}) = 2^{\lambda_{\mathrm{EC}}}$ exactly; from L_{EC} and its own data the partner forms a guess $\hat{Z}_1^n$ of Z_1^n;

(iii) *error verification*: a public seed S_{EV} selects a function h_{EV}, with the law of the error-verification family, from a hash family of functions with $\lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \rceil$-bit values; the holder announces $T_{\mathrm{EV}} \coloneqq h_{\mathrm{EV}}(Z_1^n)$, the partner compares it with $h_{\mathrm{EV}}(\hat{Z}_1^n)$ and announces the one-bit outcome $F_{\mathrm{acc}} \in \{\mathrm{acc}, \mathrm{rej}\}$, which is acc exactly when the two values agree;

(iv) *privacy amplification*: a function G is drawn, with the law of the privacy-amplification family, from a hash family of functions with ℓ-bit values, and announced; the holder outputs $K \coloneqq G(Z_1^n)$ and the partner $\hat{K} \coloneqq G(\hat{Z}_1^n)$.

We write K_A and K_B for Alice's and Bob's outputs, which are K and $\hat{K}$ in the order the key holder fixes, and

$$S \coloneqq (S_{\mathrm{pub}}, S_{\mathrm{EV}}), \qquad L \coloneqq (L_{\mathrm{EC}}, T_{\mathrm{EV}}, F_{\mathrm{acc}}), \qquad E_{\mathrm{tot}} \coloneqq \widehat{C}_1^n I_1^n \widehat{E}\, S\, L\, G, \tag{10.1}$$

where L has at most $\lambda \coloneqq \lambda_{\mathrm{EC}} + \lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \rceil + 1$ bits and $\widehat{E}$ is the adversary's retained quantum memory of Definition 2.3. The protocol *accepts* on $\Omega_{\mathrm{acc}} \coloneqq \Omega_{\mathrm{AT}} \wedge \{F_{\mathrm{acc}} = \mathrm{acc}\}$; on its complement both parties abort and output the symbol $\perp$ in place of a key. The model is stated in the *honest-authentication setting*: every classical message of the protocol, the round announcements included, reaches its recipient exactly as it was sent, in the order sent and at some time after it was sent, the adversary receives a copy of it, and no message fails authentication.

Uses Def. 1.1, Def. 2.3 · *Used by* Def. 10.2, Def. 10.3, Lem. 10.4, Lem. 10.5, Lem. 10.6, Thm. 10.7, Cor. 10.8, Rem. 10.11, Prop. 10.12, Ass. 13.1, Ass. 13.2, Ass. 13.3, Ass. 13.4, Ass. 13.5, Ass. 13.8 · newly derived · accepted

The authenticated classical channel delivers each message unaltered or as an authentication failure, never before it was sent, and a scheme realises it within ε from a pre-shared key. The authentication post-processing makes both parties abort when either saw a failure.

A real authentication scheme cannot deliver every message: an adversary who tampers with a message makes its recipient reject it. The channel below is the functionality such a scheme can approach [5, Sec. 5, Eqs. (37)–(38)], the approximation is measured by the distinguishing advantage of the composable framework [3, Sec. II.E, Def. 1; Sec. VII.A, Eq. (36)], and the authentication post-processing is step 6 of [5, Sec. 4.1, Protocol 1] together with the rule of that section's opening paragraphs that a party receiving an authentication failure aborts.

Definition 10.2 (Authenticated classical channel and authentication post-processing)**.** Label each classical message of a party by its position i in that party's sequence of sent messages, and the messages a party receives by their position in its sequence of received messages; times are global times, which the parties need not know.

(a) The *authenticated classical channel* between the two parties gives the adversary a copy of every message sent over it and lets her choose when each message is delivered, subject to: (A1) if the i-th message a party receives arrives before the other party sent its i-th message, the received symbol is auth-abort; (A2) otherwise the received symbol is either the i-th sent message, unaltered, or auth-abort.

(b) An *authentication scheme* is a pair of procedures, one per party, that use a pre-shared key and an unauthenticated classical channel, on which the adversary may read, alter, inject, delay and suppress messages. It *ε-realises* the authenticated classical channel if there is a simulator at the adversary's interface such that the distinguishing advantage between the scheme run on the key and the unauthenticated channel, and the authenticated classical channel with that simulator, is at most ε [3, Sec. II.E, Def. 1]. Its *key consumption* is the number of bits of the pre-shared key it uses in one run of the protocol.

(c) The *authentication post-processing* appended to the fixed-length post-processing model (Definition 10.1) is: a party that receives auth-abort for any message aborts, replaces its key by $\perp$ and sends abort in place of each later message; after privacy amplification the partner sends accept if it holds a key and abort otherwise; the key holder sends accept if it received accept and holds a key, and otherwise sends abort and replaces its key by $\perp$; the partner keeps its key if it receives accept and otherwise replaces it by $\perp$.

Uses Def. 10.1 · *Used by* Def. 10.3, Cor. 10.8, Rem. 10.11, Ass. 13.4, Ass. 13.5, Ass. 13.6, Ass. 13.8 · newly derived · accepted

A party has ended an execution once it has sent and received every message the post-processing prescribes to it and fixed its output, a key or an abort; the execution has ended once both parties have. Several executions are compared at that time.

Under the authentication post-processing the two parties do not finish at the same time: the key holder finishes when it sends its last decision, the partner when that decision is delivered, at a time the adversary chooses.

Definition 10.3 (The end of an execution). An *execution* of the protocol is one run of its n rounds, the fixed-length post-processing model (Definition 10.1) and the authentication post-processing (Definition 10.2 (c)). A party has ended an execution once it has sent and received every message that these prescribe to it and has fixed its output, its key or $\perp$: the key holder once it has sent its final accept or abort, the partner once that message, or auth-abort in its place, has been delivered to it. The *end of an execution* is the time at which both of its parties have ended it.

Uses Def. 10.1, Def. 10.2 · *Used by* Cor. 10.9, Ass. 13.5, Ass. 13.7 · newly derived · accepted

In the fixed-length post-processing model the two parties accept with different keys with probability at most ε_EV, for every attack and every output length. It is the correctness half of the protocol's security.

The argument reads only the verification step of the model: two different raw strings pass it only if their short hashes collide, and the hash is chosen after, and independently of, the strings it compares.

Lemma 10.4 (Correctness from error verification)**.** Let the protocol run the fixed-length post-processing model (Definition 10.1) in the honest-authentication setting, let Assumptions 13.2, 13.3 and 13.5 hold, and let $\varepsilon_{\mathrm{EV}} \in (0, 1]$. Then for every attack and every output length ℓ the protocol is $\varepsilon_{\mathrm{EV}}$-correct on the accepted state, in the sense of [6, Sec. 2.3.1, Def. 2.2, Eq. (4)]:

$$\Pr\left[\Omega_{\mathrm{acc}} \wedge K_A \neq K_B\right] \leq \varepsilon_{\mathrm{EV}}. \tag{10.2}$$

Reading: This is the correctness half of security. Before the key is formed, the key holder announces a $\lceil \log_2(1/\varepsilon_\mathrm{EV}) \rceil$-bit hash of its raw key, computed with a hash function drawn at random independently of the raw keys, and the partner compares it with the hash of its own corrected copy; a mismatch aborts the run. Two different raw keys pass only through a hash collision, so the chance that both parties accept yet hold different keys is at most ε_EV, whatever the attacker does and whatever key length you choose. It needs a two-universal hash family (A-HASH), seeds independent of the raw keys (A-SEED) and an implementation that aborts as specified (A-CONFORM), and takes messages to arrive unaltered; Corollary 10.8 adds real authentication. It does not promise that error correction succeeds, only that a failure is caught, except with probability ε_EV.

Uses Def. 10.1 · *Assumes* A-HASH, A-SEED, A-CONFORM · *Used by* Cor. 10.8, Cor. 10.9, Cor. 10.18 · newly derived · accepted

Proof. Both parties abort when the verification flag is rej and output the keys the model computes (Assumption 13.5 (iii)–(iv)). On $\Omega_{\mathrm{acc}} \wedge \{K_A \neq K_B\}$ the hashed keys differ, so the raw strings differ, yet their $\lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \rceil$-bit two-universal hashes agreed. (If the raw strings agree, $K_A = K_B$ because the same public G is applied: in the honest-authentication setting the partner receives the holder's G unaltered.) The verification function h_{EV} is selected by S_{EV}, which Assumption 13.3 makes independent of the pair $(Z_1^n, \hat{Z}_1^n)$, and its family is two-universal (Assumption 13.2 (ii)). Conditioned on any pair of distinct strings, the probability that their hashes agree is therefore at most $2^{-\lceil \log_2(1/\varepsilon_{\mathrm{EV}}) \rceil} \leq \varepsilon_{\mathrm{EV}}$, and averaging over the pair gives (10.2). □

Every attack on a run, its memory of any dimension and its signals taken in any order, is approximated in trace norm by attacks with finite memory that take every signal at the start. A trace-distance bound that holds for those therefore holds for every attack with the same ε.

The bound of Theorem 10.7 is proved for the attacks the round model of Definition 2.3 represents: their registers are finite-dimensional, as every register of that model is, and in trusted-source mode their first channel receives every transmitted signal, the type the reductions transfer (Lemma "Exactness of the source map in trusted-source mode" (reductions/lem:source-transfer)) and

the timing the published analysis moves to [5, Lemma 8.1]. A physical adversary may keep a memory of any dimension and receives the signals as the source releases them; where a signal lives in an infinite-dimensional space, a finite memory cannot hold it for a later round. The lemma closes that gap by a limit, the route of [5, Def. 8.2, Sec. 8.2, Thm. A.1, Rem. A.2]: compressing the adversary's registers, and each signal released after the first round, onto finite-dimensional subspaces (Lemma "Compressing a register onto a finite-dimensional subspace" (foundations/lem:register-compression)) turns every attack into one of the analysed type whose output state is as close as wanted, and a bound on a trace distance, which no dimension enters, passes to the limit. Corollary 10.8 applies it. The compression touches neither the measured system nor the transmitted system in transit, so no finite dimension of either is needed; that is why the limit is taken at the physical run rather than at the analysed protocol. Unlike the transfers of the reductions module, which hold for every criterion because they are exact (Remark "Exactness of the reductions and generality of the transfer" (reductions/rem:why-exact)), the limit needs a criterion continuous in the output state, and the trace distances of the security statement are.

Lemma 10.5 (Approximating an attack by all-at-once attacks with finite memory)**.** Read a run of the protocol and its attacks as follows.

(a) *(the run)* Its quantum phase is an n-round protocol of Definition "Measurement-factorizing n-round protocol" (reductions/def:protocol), whose measured system B_j of round j is separable and of any dimension, followed by the fixed-length post-processing model of Definition 10.1 in the honest-authentication setting. In trusted-source mode its source releases, for $i = 1, \dots, n$, a separable transmitted system A'_i of any dimension, prepared together with Alice's letter register X_i in a state ρ_i on $X_i A'_i$ that depends on nothing else, and the run reads X_i from round i on; in untrusted-source mode there are no X_i and no A'_i. Every system of the run other than the B_j and the A'_i is finite-dimensional, and its public messages are classical registers it keeps.

(b) *(attacks)* An *attack* consists of a release schedule $0 = k_0 \le k_1 \le \cdots \le k_n = n$ with $k_j \ge j$ for every j, a state ζ on the adversary's initial register E_0, and channels, in the sense of Lemma "Compressing a register onto a finite-dimensional subspace" (foundations/lem:register-compression),

$$\mathcal{A}_j \in \begin{cases} \mathrm{CPTP}\big(A'_{k_{j-1}+1} \cdots A'_{k_j} E_{j-1},\ B_j E'_j\big), & \text{trusted-source mode,} \\ \mathrm{CPTP}\big(E_{j-1},\ A_j B_j E'_j\big), & \text{untrusted-source mode,} \end{cases} \tag{10.3}$$

for $j = 1, \dots, n$, where E'_j is the adversary's memory after round j, E_j is E'_j together with a copy of the public messages of round j, and

$\mathcal{H}(E_0)$ and every $\mathcal{H}(E'_j)$ are Hilbert spaces of any dimension. Under the attack the run starts from ζ and its own registers, prepares ρ_i in round j for $k_{j-1} < i \leq k_j$ before $\mathcal{A}_j$ acts, takes B_j, and in untrusted-source mode Alice's system A_j, from the output of $\mathcal{A}_j$, and places a copy of the public messages of round j in E_j. Its *output state* is the state of the run's output registers, its public messages and E'_n. The attack is *all-at-once* if $k_1 = n$, and has *finite registers* if E_0 and every E'_j are finite-dimensional.

Then:

(i) *(approximation)* for every attack and every $\eta > 0$ there is an all-at-once attack with finite registers whose memory after round n is a subspace of $\mathcal{H}(E'_n)$, such that the output states ρ and ρ° of the two attacks satisfy $\|\iota(\rho^\circ) - \rho\|_1 \leq \eta$, with ι the inclusion of that subspace into $\mathcal{H}(E'_n)$, acting as the identity on every other register;

(ii) *(trace-distance criteria)* let L_1 and L_2 be completely positive maps, not increasing the trace, from the run's output registers and public messages to a finite-dimensional system, put $L := L_1 - L_2$, and for a state σ of those registers and the adversary's memory after round n put $f(\sigma) := \frac{1}{2}\|(L \otimes \mathrm{id})(\sigma)\|_1$; if $f(\rho^\circ) \leq \varepsilon$ at the output state ρ° of every all-at-once attack with finite registers, then $f(\rho) \leq \varepsilon$ at the output state ρ of every attack;

(iii) *(processing by the adversary)* let, in (ii), each L_t keep the public messages and act on the other registers given their value m, $L_t(|m\rangle\langle m| \otimes x) = |m\rangle\langle m| \otimes L_{t,m}(x)$, and let $\Phi = \sum_m |m\rangle\langle m| \otimes \Phi_m$ act on a state that is classical on the public messages by a channel Φ_m on the adversary's register, given their value m, keeping them. Then $f((\mathrm{id} \otimes \Phi)(\sigma)) \leq f(\sigma)$ for every state σ classical on the public messages.

Uses Def. 10.1, foundations/lem:register-compression, reductions/def:protocol · *Used by* Lem. 10.6, Thm. 10.7, Cor. 10.8, Cor. 10.18, Cor. 19.5 · newly derived · accepted

Proof. Idea. Written as a sequence of channels, the run is compressed at finitely many points: the adversary's initial register, her memory after each of her channels, and every signal released after the first round, as soon as it is released. The compressions cost at most η in trace norm (Lemma "Compressing a register onto a finite-dimensional subspace" (foundations/lem:register-compression) (ii), (iii)), and the compressed run is the run under an all-at-once attack with finite registers: the adversary takes every signal at the start, compresses those she would have received later and keeps them, compressed, in her memory until their round. A trace distance of the output state moves by at most the distance of the states, which gives (ii); a channel on the adversary's side commutes past the criterion's maps and does not increase a trace norm, which gives (iii).

Step 1 (the run as a sequence of channels). Fix an attack. The run's own registers are finite-dimensional and start in fixed states, which we take as written by the channels that first write them. Round j is the composite of three channels: the preparation $\mathcal{Q}_j$, which appends $\bigotimes_{k_{j-1}<i\le k_j}\rho_i$ (the identity in untrusted-source mode); $\mathcal{A}_j$, extended by the identity on the run's registers; and the round map G_j of Definition "Measurement-factorizing n-round protocol" (reductions/def:protocol), extended by the identity on E'_j. After round n the run applies K, which includes the post-processing, extended by the identity on E'_n. These extensions are channels acting on products as the named map tensored with the identity. A channel extended by the identity on a finite-dimensional register applies it to every entry of a block matrix, which is a channel by complete positivity. G_j is the measurement channel of the measured system followed by processing of finite-dimensional registers; the measurement channel $\tau\mapsto\sum_v \mathrm{tr}[M_v\tau]\,|v\rangle\langle v|$ extends to $\tau\mapsto\sum_v |v\rangle\langle v|\otimes\mathrm{tr}_{B_j}\left[(\sqrt{M_v}\otimes\mathbb{1})\tau(\sqrt{M_v}\otimes\mathbb{1})\right]$, a sum of conjugations followed by partial traces and the appending of states, and a completely positive map Ψ between finite-dimensional systems is a finite sum $\sum_t K_t(\cdot)K_t^\dagger$, which extends to $\sum_t(K_t\otimes\mathbb{1})(\cdot)(K_t\otimes\mathbb{1})^\dagger$. The output state is the result of these channels applied in order to ζ.

Step 2 (the compressions). The compression points are: E_0 before round one; E'_j right after $\mathcal{A}_j$, for every j; and, for every $i>k_1$, the transmitted system A'_i right after the preparation $\mathcal{Q}_j$ that appends ρ_i, before $\mathcal{A}_j$ receives it, one signal at a time with identity channels between. There are at most $2n+1$ of them. At each point, Lemma "Compressing a register onto a finite-dimensional subspace" (foundations/lem:register-compression) (ii), applied to the run's state there under the given attack, gives a non-zero finite-dimensional subspace of the register at which the bound of that part, $2\sqrt{\delta}+\delta$, is at most $\eta/(2n+1)$; we write E_0°, $E_j'^\circ$ and $A_i'^\circ$ for them. Inserting at each point the compression onto its subspace, and the identity at every other point, part (iii) of that lemma gives $\|\rho^{\mathcal{C}}-\rho\|_1\le\eta$ for the output $\rho^{\mathcal{C}}$ of the run so compressed.

Step 3 (the compressed run under an attack with finite registers). Write each compression as $\iota\circ\mathcal{P}$ (Lemma "Compressing a register onto a finite-dimensional subspace" (foundations/lem:register-compression) (i)). Each inclusion ι of E'_j commutes with the maps that follow it up to $\mathcal{A}_{j+1}$, the round map G_j, the preparations of round $j+1$ and the compressions of the signals they release, all of which act on other registers: for the channels this is part (iv) of that lemma, and for an appended state ρ it is $\iota(y)\otimes\rho=(\iota\otimes\mathrm{id})(y\otimes\rho)$, the product form of part (i). Likewise the inclusion of E_0 moves to $\mathcal{A}_1$, the inclusion of each $A_i'^\circ$ to the channel $\mathcal{A}_j$ that receives A'_i, and the inclusion of $E_n'^\circ$ past G_n and K to the end. Each projection $\mathcal{P}$ stays where it acts: after $\mathcal{A}_j$ it joins it, and after the preparation of ρ_i it gives $\rho_i^\circ:=(\mathrm{id}_{X_i}\otimes\mathcal{P})(\rho_i)$, by the product form $\mathcal{P}_{R_1R_2}(x\otimes y)=x\otimes\mathcal{P}_{R_2}(y)$ of part (i) with $R_2=X_i$

and R_1 the other systems. So $\rho^{\mathcal{C}} = \iota(\rho^{\mathrm{fs}})$, where ρ^{fs} is the output state of the run that prepares ρ_i for $i \le k_1$ and ρ_i° for $i > k_1$, on $X_i A_i'^\circ$, in the round of the schedule, against the initial state $\mathcal{P}(\zeta)$ on E_0° and the channels

$$\mathcal{A}_1^\circ \coloneqq \mathcal{P} \circ \mathcal{A}_1 \circ \iota, \qquad \mathcal{A}_j^\circ \coloneqq \mathcal{P} \circ \mathcal{A}_j \circ \iota \quad (j \ge 2),$$

the projection being that of E_j', and the inclusion that of E_0 for $j = 1$ and of E_{j-1}' and every $A_i'^\circ$ received in round j for $j \ge 2$, each acting as the identity on the other registers. Each $\mathcal{A}_j^\circ$ is a channel between finite-dimensional adversary's registers, with the signals of the schedule and B_j, and A_j in untrusted-source mode, as before.

Step 4 (an all-at-once attack with the same output). Put $E_j^\star \coloneqq E_j'^\circ A_{k_j+1}'^\circ \cdots A_n'^\circ$, finite-dimensional, so that $E_n^\star = E_n'^\circ$. The all-at-once attack has the initial state $\mathcal{P}(\zeta)$, the first channel

$$\mathcal{A}_1^\star \coloneqq \big(\mathcal{A}_1^\circ \otimes \mathrm{id}_{A_{k_1+1}'^\circ \cdots A_n'^\circ}\big) \circ \mathcal{P}_{A_{k_1+1}'} \circ \cdots \circ \mathcal{P}_{A_n'} \in \mathrm{CPTP}\big(A_1' \cdots A_n' E_0^\circ,\ B_1 E_1^\star\big),$$

each $\mathcal{P}_{A_i'}$ the projection of A_i' onto $A_i'^\circ$ with the identity on the other systems, and, for $j \ge 2$, $\mathcal{A}_j^\star \coloneqq \mathcal{A}_j^\circ \otimes \mathrm{id}_{A_{k_j+1}'^\circ \cdots A_n'^\circ}$, from $E_{j-1}^\star$ with the public messages of round $j-1$ to $B_j E_j^\star$; the input of $\mathcal{A}_j^\circ$ is a factor of that register. In untrusted-source mode it is the attack of Step 3 itself. Its run prepares $\rho_1 \otimes \cdots \otimes \rho_n$ at the start, and $\mathcal{A}_1^\star$ turns each factor ρ_i with $i > k_1$ into ρ_i°, by the product form of part (i). From then on no map acts on $X_i A_i'^\circ$ before the round that reads X_i, which is not earlier than the round of the schedule since $k_i \ge i$, or before the channel that receives $A_i'^\circ$; every map before them acts on other registers, and a map Ψ extended by the identity on the finite-dimensional $X_i A_i'^\circ$ takes $y \otimes \rho_i^\circ$ to $\Psi(y) \otimes \rho_i^\circ$. Appending ρ_i° at the start or in its round of the schedule therefore leaves every later state the same, and the output state ρ° of the all-at-once attack is ρ^{fs}. With Steps 2 and 3, $\|\iota(\rho^\circ) - \rho\|_1 = \|\rho^{\mathcal{C}} - \rho\|_1 \le \eta$, which is (i).

Step 5 (part (ii)). The maps $L_1 \otimes \mathrm{id}$ and $L_2 \otimes \mathrm{id}$ are completely positive and do not increase the trace, by the Kraus form of Step 1, so each contracts the trace norm of a Hermitian operator: for $Z = Z^+ - Z^-$, $\|(L_t \otimes \mathrm{id})(Z)\|_1 \le \mathrm{tr}(L_t \otimes \mathrm{id})(Z^+) + \mathrm{tr}(L_t \otimes \mathrm{id})(Z^-) \le \|Z\|_1$. Hence for two states σ, σ',

$$|f(\sigma) - f(\sigma')| \le \tfrac{1}{2}\big\|(L \otimes \mathrm{id})(\sigma - \sigma')\big\|_1 \le \tfrac{1}{2} \sum_{t=1,2} \big\|(L_t \otimes \mathrm{id})(\sigma - \sigma')\big\|_1 \le \|\sigma - \sigma'\|_1.$$

Given an attack and $\eta > 0$, take the all-at-once attack of (i). By part (iv) of Lemma "Compressing a register onto a finite-dimensional subspace" (foundations/lem:register-compression), with $G = L$, $(L \otimes \mathrm{id}_{E_n'}) \circ \iota = \iota \circ (L \otimes \mathrm{id}_{E_n'^\circ})$, and ι preserves the trace norm, so $f(\iota(\rho^\circ)) = f(\rho^\circ) \le \varepsilon$. So $f(\rho) \le \varepsilon + \eta$ for every $\eta > 0$.

Step 6 (part (iii)). Write $\sigma = \sum_m |m\rangle\langle m| \otimes \sigma_m$, with σ_m positive on the other registers and the adversary's. Then

$$(L_t \otimes \mathrm{id})\big((\mathrm{id} \otimes \Phi)(\sigma)\big) = \sum_m |m\rangle\langle m| \otimes (L_{t,m} \otimes \Phi_m)(\sigma_m) = \Phi'\big((L_t \otimes \mathrm{id})(\sigma)\big),$$

with $\Phi' \coloneqq \sum_m |m\rangle\langle m| \otimes \Phi_m$ acting on the output of L_t, which keeps the public messages: both sides apply $L_{t,m}$ to the run's registers and Φ_m to the adversary's in each block, which commute by part (iv) of Lemma "Compressing a register onto a finite-dimensional subspace" (foundations/lem:register-compression). So $(L \otimes \mathrm{id})(\mathrm{id} \otimes \Phi)(\sigma) = \Phi'((L \otimes \mathrm{id})(\sigma))$, and Φ' is a positive trace-preserving map on block-diagonal operators, which contracts the trace norm of the Hermitian $(L \otimes \mathrm{id})(\sigma)$ as in Step 5. Halving gives the claim. □

Whatever an adversary does in the honest-authentication setting, adaptively and with a memory of any dimension, the output state is that of an all-at-once attack of Lemma 10.5 followed by her own processing of her register.

An adversary of the honest-authentication setting is not bound to act once per round: she may act at any time, decide adaptively when to deliver which system, hold the signals she receives, read her copies of the messages when they reach her, and process her memory after the last round. The lemma shows that none of this leaves the all-at-once attacks of Lemma 10.5 (b), up to a final processing of her own register, as the timing modification of [5, Lemma 8.1] does for the published analysis: she is handed every signal at the start and ignores each until it would have reached her. That needs every party to measure in round order, A-SEQ (v), the measurement part of the timing condition of [5, Sec. 8.1.1, Protocol 2], read for each measuring party. Corollary 10.8 uses the lemma to state its bound at every output state of the setting.

Lemma 10.6 (Adversaries of the honest-authentication setting as round attacks)**.** Let the run be as in Lemma 10.5 (a), with A-ALICE-MEAS (Assumption "Alice's declared measurement where no source certificate is demanded" (protocol-compiler/ass:alice-meas)) (i) for Alice's measurement where no source-replacement certificate is demanded, and let A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)), its clause (v) included, Assumption 13.8 (A-LAB) at each of the two parties and Assumption 13.5 (A-CONFORM) hold. Let an adversary act on the run in the honest-authentication setting (Definition 10.1) by any operations, at any times, each chosen given the classical outcomes of her earlier ones, on her own registers, of any dimension, on each transmitted system once the source has released it and on her copies of the public messages once she holds them, delivering the measured system of each round, and in untrusted-source mode Alice's system, for the round's measurements. Then there are an all-at-once attack in the sense of Lemma 10.5 (b), with output state ρ, and a channel $\Phi = \sum_m |m\rangle\langle m| \otimes \Phi_m$ that applies a channel Φ_m to her register E'_n given the value m of the run's public messages and keeps them, such that the state of the run's output registers, its public messages and the adversary's final register is $(\mathrm{id} \otimes \Phi)(\rho)$.

Uses Def. 10.1, Lem. 10.5, foundations/lem:register-compression, reductions/def:protocol · *Assumes* A-SEQ, A-LAB, A-CONFORM, A-ALICE-MEAS · *Used by* Cor. 10.8, Cor. 10.18, Cor. 19.5 · newly derived · accepted

Proof. Idea. Every signal is handed to her at the start and passed on untouched until she would have received it; her operations between the deliveries that complete consecutive rounds, with their classical branches, form one channel; every measurement of a round is made, in the representation, right after the delivery that completes it; and neither the timing of message delivery nor the time a signal is prepared changes anything.

Step 1 (what each side acts on). By Assumption 13.8 (ii) at each party, the adversary acts on a laboratory only through the systems delivered into it, as its declared measurements receive them, and through the public messages it is delivered, and by (i) she learns nothing of its records except through the systems handed out of it and the public messages. So every operation of hers acts on her registers, on transmitted systems the source has released, on her copies of public messages, and on a system before she delivers it; once delivered, a system is acted on only by the measurement that receives it. Every operation of the run acts on its own registers and on the systems delivered to it. In the honest-authentication setting every classical message reaches its recipient exactly as sent, in the order sent and after it was sent, and the adversary receives a copy. By Assumption 13.5 (ii) the messages after the rounds are those of Definition 10.1, and by (i), (iii) and (iv) the run applies that model's processing.

Step 2 (what the run does not depend on). The run's quantum operations, the preparations of the states ρ_i, which depend on nothing else, the one measurement channel of every round (Definition "Measurement-factorizing n-round protocol" (reductions/def:protocol)) and, in untrusted-source mode, Alice's measurement of A_j in the round's processing, the same in every round and on that round's A_j alone by A-ALICE-MEAS (i), read no message. Its classical operations compute functions of its records and of the contents of the messages it has received, and a message arrives unaltered and in order. So the run's registers at its end, and the content of every public message, are the same whenever each message is delivered before the operation that reads it, which the run waits for, and whenever each ρ_i is prepared before its signal is touched and its letter read.

Step 3 (the cuts s_j). The measurements of round j are those of every party that measures in it, Alice's measurement of A_j in untrusted-source mode among them (A-SEQ (i)). For $j = 1, \dots, n$ let s_j be the latest delivery, among the adversary's operations, of a system that a measurement of a round $i \le j$ receives. It is an operation of hers, so it is fixed by the branch of her classical outcomes: each of the finitely many systems of the rounds $i \le j$ is delivered exactly once on every branch, so at each of her operations her branch so far decides whether it is the last such delivery. And s_{j-1} precedes

or is s_j. Every public message of a round $j' \geq j$ is sent after s_j: the system delivered at s_j is received by the measurement of a round $i \leq j$ of some party P, which completes after the delivery; by A-SEQ (v) P's measurement of round j' completes no earlier, P measuring in every round (Definition "Measurement-factorizing n-round protocol" (reductions/def:protocol)); and by A-SEQ (i) every message of round j' follows that measurement. So a message the adversary holds before s_j is one of a round $j' < j$.

Step 4 (moving the measurements to the cuts). Every measurement of round j receives its system at or before s_j, and its outcome is read by nothing before the messages of round j, which follow s_j (Step 3). By Step 1 the adversary does not touch the system once delivered, and by Assumption 13.8 (i) none of her operations depends on the outcome before those messages; every operation of the run between the measurement and s_j acts on other systems and reads nothing it outputs. Moving the measurement to just after s_j, whether it was made before or after, therefore changes no later state, operations on disjoint systems commuting (Lemma "Compressing a register onto a finite-dimensional subspace" (foundations/lem:register-compression) (iv)), with the extensions of part (v) of that lemma. From now on every measurement of round j is made right after s_j, on the systems delivered for round j.

Step 5 (the attack). Hand the adversary every transmitted system at the start: by Step 2 preparing ρ_i there changes nothing, and she passes each on untouched, her operations extended by the identity on it (Lemma "Compressing a register onto a finite-dimensional subspace" (foundations/lem:register-compression) (v)), until the operation at which she received it. Let ζ be the state of her registers before the run starts. For $j = 1, \ldots, n$, let $\mathcal{A}_j$ be the composite of her operations after s_{j-1} (s_0 the start) and up to s_j: on each branch of her classical outcomes the composite of the operations on that branch, the record of the branch kept in her memory, summed over the branches; it is a channel, every operation being one and her branching being on recorded classical outcomes. It acts on her memory, on the signals she holds and on her copies of messages, which by Step 3 are of rounds before j, those she has not yet received being passed on unread. The time at which a copy reaches her is set by the run and may depend on its devices' timing; she records each arrival as a classical outcome of hers, so it is part of her branch. By Assumption 13.8 (ii) at each party, a party's devices, and so the timing of its messages, are influenced by nothing but the systems and messages delivered to it, whose delivery times are in her branch; any further dependence on the quantum systems in its laboratory passes through a measurement, whose outcome is a record. So, given her branch, the timing is a function of the parties' records and local randomness, and by Assumption 13.8 (i) at each party, the timing of the parties' messages carrying no information about their records, the arrival times are independent, given her branch, of every register other than her branch: her quantum memory, the signals she holds and the parties' systems and records. Their law given her branch is

therefore a fixed function of that branch; $\mathcal{A}_j$ samples them from that law with randomness of its own, which leaves the joint law of every register unchanged. It outputs the measured systems of round j, those she delivered before s_{j-1} having been kept untouched in her memory until then, Alice's system A_j among them in untrusted-source mode (Step 3), together with her memory E'_j, which holds everything she still passes on. With $k_1 = n$, $(\zeta, (\mathcal{A}_j)_j)$ is an all-at-once attack in the sense of Lemma 10.5 (b). By A-SEQ (ii) the post-processing follows the n rounds; let Φ be the composite of her operations after s_n, which deliver nothing and act on her register and on her copies of the public messages; the public messages are classical, so Φ applies, given their value m, a channel Φ_m to her register and keeps them, the arrival of each copy it reads being sampled inside Φ_m as inside the $\mathcal{A}_j$.

Step 6 (the same output). The given execution, with its measurements moved as in Step 4, and the run under the attack of Step 5 followed by Φ consist of the same channels, of the run and of the adversary. They differ in the order of channels acting on disjoint systems, which commute (Lemma "Compressing a register onto a finite-dimensional subspace" (foundations/lem:register-compression) (iv)); in the time at which each message reaches the run and each ρ_i is prepared, which changes nothing by Step 2; and in the time at which the adversary holds a signal, a delivered system or a copy, which the passing on of Step 5 leaves without effect, the arrival of each copy being sampled inside the $\mathcal{A}_j$ and Φ as Step 5 says. So the two output states agree. □

For a Rényi order α in (1, 2), a key of length $\ell \le n{\cdot}h_low - \lambda_EC - \lceil log_2(1/\varepsilon_EV)\rceil - \alpha/(\alpha-1){\cdot}log_2(1/\varepsilon_PA) + 1$ is ε_PA-secret on the accepted state at every all-at-once finite-register attack the reduction chain carries onto the round model (A-LINK). Corollary 10.8 extends it to every attack.

The accumulated entropy bound feeds the fixed-length post-processing model: error correction with a transcript of fixed length, error verification whose one-bit outcome is announced, and privacy amplification by an ideal two-universal hash. Announcing the verification outcome costs one bit and makes the secrecy statement hold on the accepted state (Remark 10.14).

The theorem is stated at the attacks the analysis is carried out for: those with finite registers whose first channel receives every transmitted signal, which the reduction chain carries onto the round model, as hypothesis (H-CHAIN) says. Its link part is proved, not attested, on the identity link of Band A (Theorem "The Band-A route" (protocol-compiler/thm:band-a)) and for a chain the reductions module emits with status exact (Theorem "The two-sided reduction in trusted-source mode" (reductions/thm:two-sided), Corollary "Output-state equality against adaptive attack sequences" (reductions/cor:sequential), Theorem "The attack-set licence" (reductions/thm:licence)), where A-LINK is discharged (Remark "Discharge of the chain obligations by the reductions module" (protocol-compiler/rem:obligations-discharged));

for any other chain it rests on A-LINK. With constraints, the part that makes the attacks admissible for the constraint list rests on A-ATK on every route, Band A included, wherever the list carries an entry no theorem licenses. Corollary 10.8 extends the bound to every attack, whatever the adversary's memory.

Theorem 10.7 (Secrecy of the key from the accumulated entropy). Let the protocol run the fixed-length post-processing model (Definition 10.1) in the honest-authentication setting, with the registers (10.1). Let one of the two accumulation branches hold: *(unconstrained)* the hypotheses of Theorem 2.10 with $\mathfrak{L} = \varnothing$; or *(constrained)* the hypotheses of Theorem 3.1 ($\mathfrak{L}$ licensed for the declared source mode, Definition 2.2; A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk))), in either branch save the condition $\Pr[\Omega_{\mathrm{AT}}] > 0$ that the accumulation theorem places on the attack, which the attack below need not meet. Let the run be as in Lemma 10.5 (a), as A-SPEC (Assumption "Specification faithful to the apparatus" (protocol-compiler/ass:spec)) (i) attests for its measurement, with A-ALICE-MEAS (Assumption "Alice's declared measurement where no source certificate is demanded" (protocol-compiler/ass:alice-meas)) (i) for Alice's measurement where no source-replacement certificate is demanded, and A-ENS (Assumption "Emission of the specified ensemble" (transmitter/ass:ens)), or the premise a source model declares in its place, for its source states; let it be attacked by an attack in the sense of Lemma 10.5 (b) that is all-at-once and has finite registers; and let the reduction chain from which the declaration was compiled carry it onto the round model, as A-LINK (Assumption "Soundness of every reduction link" (protocol-compiler/ass:link)) (i)(a), (i)(b), (ii)(a) and (iv) attest, the existence of each link's channel, the pull-back identity, the source identity and complete positivity, with A-ATK in the constrained branch:

(H-CHAIN) the marginal on $Z_1^n \widehat{C}_1^n I_1^n E_n$ of the run's state after the n rounds is the state the round channels of Definition 2.3 produce under an attack sequence of that definition, $\mathfrak{L}$-admissible in the constrained branch, from an input state as in the branch's accumulation theorem; the run's state is an extension of it, holding the parties' further records.

Let Assumptions 13.1, 13.2, 13.3, 13.4, 13.5 and 13.8, the last at each of the two parties, and A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)) hold, let $\alpha \in (1,2)$ and $\varepsilon_{\mathrm{PA}}, \varepsilon_{\mathrm{EV}} \in (0,1]$. If the output length satisfies

$$\ell \;\leq\; n\,\underline{h}_{\alpha} - \lambda_{\mathrm{EC}} - \left\lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \right\rceil - \frac{\alpha}{\alpha-1} \log_2 \frac{1}{\varepsilon_{\mathrm{PA}}} + 1, \tag{10.4}$$

then, under that attack, the protocol is $\varepsilon_{\mathrm{PA}}$-secret on the accepted state:

$$\frac{1}{2} \left\| \rho_{K\,E_{\mathrm{tot}} \wedge \Omega_{\mathrm{acc}}} - \frac{\mathbb{1}_K}{2^{\ell}} \otimes \rho_{E_{\mathrm{tot}} \wedge \Omega_{\mathrm{acc}}} \right\|_1 \leq \varepsilon_{\mathrm{PA}}.$$

Reading: Over n rounds the attacker's uncertainty about the raw key accumulates to n times the per-round bound h_low, less a term for how unlikely the acceptance test is to pass, which cancels over all runs; the bound h_low holds over every attack compatible with your declared devices and with passing statistics. Subtracted are the error-correction transcript of λ_EC bits, the verification hash of $\lceil \log_2(1/\varepsilon_\mathrm{EV}) \rceil$ bits and a finite-size penalty of $\alpha/(\alpha{-}1) \cdot \log_2(1/\varepsilon_\mathrm{PA})$ bits; one bit is added back. The remaining key of ℓ bits is secret,

counted over all runs: it is within the secrecy parameter ε_PA of a uniform key independent of the attacker's memory and every public message. It is proved for a finite-memory attacker receiving every signal at once; Corollary 10.8 covers every attacker. The Rényi order α in (1, 2) is fixed before the run (A-FIXED) and changes the key's length, never its secrecy.

Uses Def. 1.2, Def. 2.2, Def. 2.3, Thm. 2.10, Thm. 3.1, Def. 5.1, Thm. 5.4, Cor. 5.6, Def. 10.1, Lem. 10.5, foundations/def:renyi · *Assumes* A-KEYREG, A-HASH, A-ATK, A-SEED, A-FIXED, A-CONFORM, A-LAB, A-SEQ, A-LINK, A-ENS, A-SPEC, A-ALICE-MEAS · *Used by* Cor. 10.8, Cor. 10.9, Def. 10.13, Rem. 10.14, Rem. 10.15, Prop. 10.17, Cor. 10.18, Cor. 10.19, Rem. 10.20, Prop. 11.1, Req. 16.1, Req. 16.4, Req. 16.5, Req. 16.6, Req. 16.7, Req. 16.8, Req. 16.10 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength:key_length · *Oracles* registry.py

Proof. Idea. Secrecy is first proved on the state conditioned on the acceptance test: the chain rule for a classical register charges the whole public transcript L against the accumulated entropy, the public seeds cost nothing because they are independent of everything else, and the Rényi leftover-hashing lemma turns the result into a trace-distance bound, whose acceptance-probability factor cancels. A contraction then passes from the acceptance test to the accepted state, on which the announced flag F_{acc} is a register of the adversary's side information. An attack under which the test never passes is disposed of first.

Step 0 (an attack that never passes the test). The event Ω_{AT} is a function of $\widehat{C}_1^n$, so by (H-CHAIN) its probability under the attack is the one in the round model. If $\Pr[\Omega_{\mathrm{AT}}] = 0$, then $\Omega_{\mathrm{acc}} = \Omega_{\mathrm{AT}} \wedge \{F_{\mathrm{acc}} = \mathrm{acc}\}$ (Definition 10.1) has probability zero, so $\rho_{K\,E_{\mathrm{tot}} \wedge \Omega_{\mathrm{acc}}} = 0$ and $\rho_{E_{\mathrm{tot}} \wedge \Omega_{\mathrm{acc}}} = 0$, and the left-hand side of the secrecy criterion is $0 \leq \varepsilon_{\mathrm{PA}}$. From now on $\Pr[\Omega_{\mathrm{AT}}] > 0$, and the hypotheses of the branch's accumulation theorem hold in full.

Step 1 (secrecy on the acceptance test). Abbreviate $B \coloneqq \widehat{C}_1^n I_1^n \widehat{E}$. By (H-CHAIN) the rounds of the run under the attack leave, on $Z_1^n \widehat{C}_1^n I_1^n E_n$, the state the round channels of Definition 2.3 produce under an attack sequence of that definition, to which the accumulation theorem of the branch applies, the run's further records being an extension of it, and the register $E_{\mathrm{tot}} = B\,S\,L\,G$ is what the adversary holds at the end of the protocol. By A-SEQ (Assumption "Sequential round structure" (transmitter/ass:seq)) (i), (ii) and (iv), every message of a round is sent after every measurement of that round, no operation on a quantum system of a round depends on that round's messages, and the classical post-processing is applied after the n rounds, so each round is the measurement of its round channel followed by announcements that the model hands to the adversary, and the post-processing acts on the state the n round channels produce. By Assumption 13.8 at each of the two parties, during the protocol nothing about the raw string, the keys or the parties' local randomness leaves either laboratory, other than as the protocol's output, except through the systems the

round model hands to the adversary and the public messages, and the adversary acts on neither laboratory except through the systems and messages the model routes to it. By Assumption 13.5 (ii) the classical messages after the rounds are exactly those of the model and of the authentication post-processing (Definition 10.2 (c)). In the honest-authentication setting no message fails authentication, so the messages of the authentication post-processing are accept or abort decisions that are functions of Ω_{acc}, hence of $\widehat{C}_1^n$ and F_{acc}, which E_{tot} holds; appending them is a channel acting on E_{tot} alone, which does not increase the trace distance of the secrecy criterion and keeps its ideal term of the same form, so they cost nothing. By Assumption 13.4, n, $\mathcal{S}_{\mathrm{acc}}$, ℓ, λ_{EC}, $\varepsilon_{\mathrm{PA}}$ and $\varepsilon_{\mathrm{EV}}$ are constants of the run, so the event Ω_{AT}, the dimension 2^λ of L and the output length do not depend on the run's data, and by Assumption 13.5 (i) and (iv) the run applies that test and outputs the ℓ-bit key the model computes, or $\perp$. The post-processing is executable: the key holder computes Z_1^n from its own readout and the announcements (Definition 1.2 (viii)), and every object the post-processing forms is a function of registers present in the state that Theorem 2.10 bounds, together with the public seeds.

The state after the public phase, conditioned on Ω_{AT}, is classical on L; by the chain rule for conditioning on a classical register [4, Lemmas 5.3–5.4, Eq. (5.94)], applied with the classical system L of dimension 2^λ and the conditioning system $B\,S$ (the α-domain restriction to $(1,2)$ is this module's, inherited from the leftover-hashing lemma applied next; the cited chain rule is stated for the sandwiched family of [4, Def. 5.2, Eqs. (5.18)–(5.19)] on its data-processing range $\alpha \in [\frac{1}{2},\infty]$, the range on which [4, Thm. 4.1, Eq. (4.82)] grounds data processing, and $(1,2) \subset [\frac{1}{2},\infty]$),

$$\begin{aligned} \widetilde{H}_\alpha^{\uparrow}\big(Z_1^n\big|B\,S\,L\big)_{\rho_{|\Omega_{\mathrm{AT}}}} &\geq \widetilde{H}_\alpha^{\uparrow}\big(Z_1^n\big|B\,S\big)_{\rho_{|\Omega_{\mathrm{AT}}}} - \lambda \;\geq\; \widetilde{H}_\alpha^{\uparrow}\big(Z_1^n\big|B\big)_{\rho_{|\Omega_{\mathrm{AT}}}} - \lambda \\ &\geq n\,\underline{h}_\alpha - \frac{\alpha}{\alpha-1}\log_2\frac{1}{\Pr[\Omega_{\mathrm{AT}}]} - \lambda. \end{aligned} \tag{10.5}$$

The second inequality holds because the seeds cost nothing. By Assumption 13.3 they are drawn after the rounds, independently of every other register, so that $\rho_{Z_1^n BS|\Omega_{\mathrm{AT}}} = \rho_{Z_1^n B|\Omega_{\mathrm{AT}}} \otimes \omega_S$ for the law ω_S of S, Ω_{AT} being a function of $\widehat{C}_1^n$. For every $\sigma_B \in S_=(B)$ the state $\sigma_B \otimes \omega_S$ is admissible in the infimum that defines $\widetilde{H}_\alpha^{\uparrow}(Z_1^n|BS)$ (Definition "Sandwiched Rényi divergence and conditional entropies" (foundations/def:renyi)), and with $\gamma := \frac{1-\alpha}{2\alpha}$, powers taken on supports,

$$(\mathbb{1}_Z\otimes\sigma_B\otimes\omega_S)^\gamma(\rho_{Z_1^nB}\otimes\omega_S)(\mathbb{1}_Z\otimes\sigma_B\otimes\omega_S)^\gamma = \big((\mathbb{1}_Z\otimes\sigma_B)^\gamma\rho_{Z_1^nB}(\mathbb{1}_Z\otimes\sigma_B)^\gamma\big)\otimes\omega_S^{1/\alpha};$$

the trace of the α-th power of a tensor product is the product of the traces, $\operatorname{tr}\big[(\omega_S^{1/\alpha})^\alpha\big] = \operatorname{tr}\omega_S = 1$, and the support condition holds for the product

exactly when it holds for the first factor. Hence $\widetilde{D}_\alpha(\rho_{Z_1^n B}\otimes\omega_S\|\mathbb{1}_Z\otimes\sigma_B\otimes\omega_S)=\widetilde{D}_\alpha(\rho_{Z_1^n B}\|\mathbb{1}_Z\otimes\sigma_B)$, and the infimum over σ_B gives $\widetilde{H}^{\uparrow}_\alpha(Z_1^n|BS)\geq\widetilde{H}^{\uparrow}_\alpha(Z_1^n|B)$.

The third inequality of (10.5) uses Theorem 2.10 with Theorem 5.4 ($h^{\uparrow,\mathrm{QKD}}_\alpha\geq\underline{h}_\alpha$) in the unconstrained branch, and Theorem 3.1 with Corollary 5.6 — that is, Corollary 10.19 — in the constrained branch; here and in the rest of the proof $\underline{h}_\alpha$ is (5.4) over the declared $\mathcal{W}$, so in the constrained branch it is the constrained value $\underline{h}_\alpha(\mathfrak{L})$, and the remainder of the proof consumes only the displayed bound, supplied by either branch. The chain rule holds for an arbitrary jointly distributed classical L of dimension 2^λ — no functional dependence of L on Z_1^n is required — so it covers F_{acc}, which depends also on the partner's reconstruction $\hat{Z}_1^n$. The third inequality of (10.5) applies to the post-processed state because $\widetilde{H}^{\uparrow}_\alpha(Z_1^n|B)$ is a functional of the (Z_1^n,B) marginal only, and the post-processing leaves that marginal unchanged: the registers traced out by the round channels (the partner's outcome and announcement copies; Definition 2.3) still exist in the physical protocol as an extension of the modelled state, and every appended object — S drawn independently, L_{EC} and T_{EV} from classical copies of Z_1^n and the public seeds, F_{acc} from T_{EV} and the partner's data (Assumption 13.1) — is generated by channels acting trivially on Z_1^nB. Hence the accumulation bound of the branch in force, Theorem 2.10 or Theorem 3.1, applies verbatim. Now apply the Rényi leftover-hashing lemma [2, Thm. 4, Eq. (13)] (stated there for $\alpha\in(1,2)$; this is what confines the theorem to that domain), under the ideal-family hypothesis of Assumption 13.2 (i) ([5, Lemma 6.5, Rem. 6.1]; a merely two-universal family costs one further bit there) and with G drawn with the law of its family independently of Z_1^nBSL (Assumption 13.3), to the normalized state $\rho_{|\Omega_{\mathrm{AT}}}$ with side information $E\coloneqq B\,S\,L$ and hash register G:

$$\frac{1}{2}\Big\|\rho_{K\,EG|\Omega_{\mathrm{AT}}}-\frac{\mathbb{1}_K}{2^\ell}\otimes\rho_{EG|\Omega_{\mathrm{AT}}}\Big\|_1\;\leq\;2^{\frac{2(1-\alpha)}{\alpha}}\,2^{\frac{1-\alpha}{\alpha}\left(\widetilde{H}^{\uparrow}_\alpha(Z_1^n|BSL)_{\rho|\Omega_{\mathrm{AT}}}-\ell\right)}.$$

Multiply both sides by $\Pr[\Omega_{\mathrm{AT}}]$ to pass to the subnormalized $\wedge\Omega_{\mathrm{AT}}$ state, and insert (10.5): with $\kappa\coloneqq\frac{\alpha-1}{\alpha}>0$, the factor $2^{\kappa\log_2(1/\Pr[\Omega_{\mathrm{AT}}])}=1/\Pr[\Omega_{\mathrm{AT}}]$ cancels the prefactor $\Pr[\Omega_{\mathrm{AT}}]$, leaving

$$\frac{1}{2}\Big\|\rho_{K\,E_{\mathrm{tot}}\wedge\Omega_{\mathrm{AT}}}-\frac{\mathbb{1}_K}{2^\ell}\otimes\rho_{E_{\mathrm{tot}}\wedge\Omega_{\mathrm{AT}}}\Big\|_1\;\leq\;2^{-2\kappa}\,2^{-\kappa(n\underline{h}_\alpha-\lambda-\ell)}.\tag{10.6}$$

Requiring the right side to be at most $\varepsilon_{\mathrm{PA}}$ is equivalent to $n\underline{h}_\alpha-\lambda-\ell\geq\frac{1}{\kappa}\log_2\frac{1}{\varepsilon_{\mathrm{PA}}}-2$, i.e. to (10.4) after inserting $\lambda=\lambda_{\mathrm{EC}}+\lceil\log_2\frac{1}{\varepsilon_{\mathrm{EV}}}\rceil+1$.

Step 2 (from Ω_{AT} to Ω_{acc}). The flag F_{acc} is a register of E_{tot} (it is part of L). Let $\Pi\coloneqq|\mathrm{acc}\rangle\langle\mathrm{acc}|_{F_{\mathrm{acc}}}$, acting on E_{tot} only. The selection map $X\mapsto(\mathbb{1}_K\otimes\Pi)X(\mathbb{1}_K\otimes\Pi)$ is trace-norm contractive ($\|AXB\|_1\leq\|A\|_\infty\|X\|_1\|B\|_\infty$ with $\|\Pi\|_\infty=1$), maps $\rho_{\wedge\Omega_{\mathrm{AT}}}\mapsto\rho_{\wedge\Omega_{\mathrm{acc}}}$, and maps the

ideal term to $\frac{\mathbb{1}_K}{2^\ell} \otimes \Pi\rho_{E_{\mathrm{tot}}\wedge\Omega_{\mathrm{AT}}}\Pi = \frac{\mathbb{1}_K}{2^\ell} \otimes \rho_{E_{\mathrm{tot}}\wedge\Omega_{\mathrm{acc}}}$, preserving the ideal form because Π does not act on K. Applying it to (10.6) gives the claimed accepted-state secrecy. □

Secrecy and correctness compose to (ε_PA + ε_EV)-security with honest authentication; over an authentication scheme within ε_auth that consumes k_auth bits of pre-shared key, the protocol is (ε_PA + ε_EV + 2ε_auth)-secure; when both parties output a key, its net key gain is ℓ − k_auth bits.

Secrecy (Theorem 10.7) and correctness (Lemma 10.4) are the two halves of the security of the fixed-length protocol in the setting where authentication behaves honestly. A real authentication scheme can abort asymmetrically and has its own error; the authentication post-processing reduces the first to the honest setting [5, Sec. 5, Thm. 5.1], and the scheme's error enters the security parameter, here through the triangle inequality, which charges it twice. That it enters is the motivation of [5, Sec. 4, footnote 8] and [6, Sec. 5.6.1]; (iii) is derived without them.

Corollary 10.8 (Security of the protocol and its net key). Let the hypotheses of Theorem 10.7, the run as in Lemma 10.5 (a) and (H-CHAIN) among them, hold at every attack in the sense of Lemma 10.5 (b) that is all-at-once and has finite registers, with $\ell \geq 1$ satisfying (10.4), A-SEQ with its clause (v) among them; let those of Lemma 10.4 hold; and put $\tau_{K_AK_B} := 2^{-\ell}\sum_{k\in\{0,1\}^\ell} |k\rangle\langle k|_{K_A} \otimes |k\rangle\langle k|_{K_B}$.

(i) In the honest-authentication setting, at every output state the setting admits, whatever the dimension of the adversary's registers and the order in which the source releases its signals, the protocol is $\varepsilon_{\mathrm{PA}}$-secret on the accepted state as Theorem 10.7 states it, and $(\varepsilon_{\mathrm{PA}}+\varepsilon_{\mathrm{EV}})$-secure in the sense of [6, Sec. 2.3, Def. 2.1, Eq. (1)]:

$$\frac{1}{2}\left\|\rho_{K_AK_B\,E_{\mathrm{tot}}\wedge\Omega_{\mathrm{acc}}} - \tau_{K_AK_B}\otimes\rho_{E_{\mathrm{tot}}\wedge\Omega_{\mathrm{acc}}}\right\|_1 \leq \varepsilon_{\mathrm{PA}}+\varepsilon_{\mathrm{EV}}, \tag{10.7}$$

and on the complement of Ω_{acc} both parties output $\perp$.

(ii) Run with the authentication post-processing (Definition 10.2 (c)) over the authenticated classical channel (Definition 10.2 (a)), the protocol is $(\varepsilon_{\mathrm{PA}}+\varepsilon_{\mathrm{EV}})$-secure in the sense of [5, Sec. 3.3, Def. 3.8], which admits asymmetric aborts, for every output state that channel admits.

(iii) Let Assumptions 13.6, 13.4 and 13.5 hold, with $\varepsilon_{\mathrm{auth}}$ and k_{auth} the parameters of Assumption 13.6. Run with the authentication post-processing over the classical channel of Assumption 13.6, the protocol is $(\varepsilon_{\mathrm{PA}}+\varepsilon_{\mathrm{EV}}+2\varepsilon_{\mathrm{auth}})$-secure in the sense of [5, Sec. 3.3, Def. 3.8]: for every attack on that run, with output state ρ' on the key registers and the adversary's final information,

$$\frac{1}{2}\left\|\rho' - \mathcal{R}_{\mathrm{ideal}}(\rho')\right\|_1 \leq \varepsilon_{\mathrm{PA}}+\varepsilon_{\mathrm{EV}}+2\varepsilon_{\mathrm{auth}}, \tag{10.8}$$

with $\mathcal{R}_{\text{ideal}}$ the map of [5, Sec. 3.3, Eqs. (13)–(17)]. It consumes at most k_{auth} bits of the pre-shared key, and when both parties output an ℓ-bit key its *net key gain* is $\ell_{\text{net}} := \ell - k_{\text{auth}}$ bits; for $\ell_{\text{net}} \leq 0$ the run yields no net key.

Reading: A run either aborts or outputs a key of ℓ bits that is a perfect key up to the security parameter ε: identical for both parties, uniform and independent of all the attacker holds, public messages included. One party may abort while the other gets a key. No test, whatever the attacker knows, tells the run's output from an ideal protocol's with advantage above $\varepsilon = \varepsilon_\text{PA} + \varepsilon_\text{EV} + 2\varepsilon_\text{auth}$. It holds against every attack, of any memory, if every premise in the run record holds, A-AUTH, A-CONFORM and A-FIXED included. A first-order run also needs its bound to hold, computed h_lb $\leq$ h_low, which no premise covers (Remark 22.4). ε averages over aborted runs too. The criterion is composable. Separate executions provably add ε (Corollary 10.9). Not every later use of the key is covered (Remark 10.10). A run where both get a key nets ℓ − k_auth bits; an abort uses up to k_auth. Abort rates: Remark 10.11.

Uses Def. 10.1, Def. 10.2, Lem. 10.4, Lem. 10.5, Lem. 10.6, Thm. 10.7 · *Assumes* A-AUTH, A-FIXED, A-CONFORM · *Used by* Cor. 10.9, Rem. 10.10, Cor. 10.18, Req. 16.8 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength:composed_security

Proof. (i). We prove the two bounds at an all-at-once attack with finite registers first, and then at every output state of the setting.

At an all-at-once attack with finite registers. Theorem 10.7 gives the secrecy there. The rest is the statement that correctness and secrecy together imply security [6, Sec. 2.3.1, Thm. 2.1], which is [3, Sec. III.B.3, Thm. 2]; we give the argument in this contract's notation. In the honest-authentication setting both parties learn Ω_{acc}, a function of the announced $\widehat{C}_1^n$ and F_{acc}, so on its complement both output $\perp$, where the real and the ideal state coincide; it remains to bound (10.7). Write $\sigma := \rho_{K\hat{K}E_{\text{tot}}\wedge\Omega_{\text{acc}}} = \sum_{k,k'} |k\rangle\langle k|_K \otimes |k'\rangle\langle k'|_{\hat{K}} \otimes \sigma^{k,k'}$, with $\sigma^{k,k'} \geq 0$ on E_{tot}, and let C be the channel on K that measures K in its classical basis and writes the outcome into K and into a copy $\hat{K}$, $C(X) := \sum_k \langle k|\, X\, |k\rangle\ |k\rangle\langle k|_K \otimes |k\rangle\langle k|_{\hat{K}}$. Put $\sigma' := (C \otimes \mathrm{id}_{E_{\text{tot}}})(\rho_{KE_{\text{tot}}\wedge\Omega_{\text{acc}}}) = \sum_k |k\rangle\langle k|_K \otimes |k\rangle\langle k|_{\hat{K}} \otimes \sum_{k'} \sigma^{k,k'}$. The difference $\sigma - \sigma'$ is block diagonal in k, the terms with $k' = k$ cancel, and in block k it is $\sum_{k'\neq k} |k'\rangle\langle k'|_{\hat{K}} \otimes \sigma^{k,k'} - |k\rangle\langle k|_{\hat{K}} \otimes \sum_{k'\neq k} \sigma^{k,k'}$, whose trace norm is $2\sum_{k'\neq k} \operatorname{tr} \sigma^{k,k'}$ because every $\sigma^{k,k'}$ is positive. Hence $\frac{1}{2}\|\sigma - \sigma'\|_1 = \Pr\left[\Omega_{\text{acc}} \wedge K \neq \hat{K}\right] \leq \varepsilon_{\text{EV}}$ by Lemma 10.4. Since C maps $\mathbb{1}_K/2^\ell$ to $\tau_{K\hat{K}}$ and a channel does not increase the trace norm, $\frac{1}{2}\|\sigma' - \tau_{K\hat{K}} \otimes \rho_{E_{\text{tot}}\wedge\Omega_{\text{acc}}}\|_1 \leq \varepsilon_{\text{PA}}$ by Theorem 10.7. The triangle inequality gives (10.7), read for (K_A, K_B), which is $(K, \hat{K})$ in the order the key holder fixes; τ is symmetric under exchanging the two registers.

At every output state of the setting. With Π the projection onto Ω_{acc}, a function of the public messages, the secrecy distance of Theorem 10.7 is the criterion f of Lemma 10.5 (ii) for $L_1(\sigma)$ the selection $\Pi\sigma\Pi$ followed by the partial trace over every register other than K and the public registers of

E_{tot}, and $L_2(\sigma) \coloneqq \frac{\mathbb{1}_K}{2^\ell} \otimes \mathrm{tr}_K\, L_1(\sigma)$; the left side of (10.7) is the same with $K_A K_B$ in place of K and $\tau_{K_A K_B}$ in place of $\mathbb{1}_K/2^\ell$. Each L_t is completely positive, does not increase the trace, keeps the public messages and acts on the other registers given their value, as part (iii) of that lemma asks, being composed of the selection by Π, which is diagonal in the public messages, partial traces over other registers and the appending of a state. By the first part both bounds hold at every all-at-once attack with finite registers, so by Lemma 10.5 (ii) at every attack in the sense of its (b). By Lemma 10.6, whose premises A-SEQ, its clause (v) included, A-LAB at both parties, A-CONFORM and A-ALICE-MEAS (i) where no source-replacement certificate is demanded are among those of Theorem 10.7, every output state of the honest-authentication setting is $(\mathrm{id} \otimes \Phi)(\rho)$ for the output state ρ of such an attack and a channel Φ of the adversary given the public messages, and by Lemma 10.5 (iii) neither criterion increases under Φ. That both parties output $\perp$ off Ω_{acc} holds at every output state, being the run's processing of its public messages.

(ii). Write $\widetilde{\mathcal{P}}$ for the protocol of the fixed-length post-processing model and $\mathcal{P} \coloneqq \mathcal{P}_{\mathrm{APP}} \circ \widetilde{\mathcal{P}}$ for it followed by the authentication post-processing. The model is an instance of the protocol of [5, Sec. 4.1, Protocol 1, steps 2–5] with the key holder as Alice: the length and error-correction decisions are $\ell(\hat{c}_1^n) = \ell \cdot \mathbf{1}[\mathrm{freq}_{\hat{c}_1^n} \in \mathcal{S}_{\mathrm{acc}}]$ and the constant λ_{EC}, so the messages that carry them are public functions of $\widehat{C}_1^n$; error correction is one-way with at most $2^{\lambda_{\mathrm{EC}}}$ transcripts; error verification and privacy amplification are those steps with the hash lengths of Definition 10.1. By (i), $\widetilde{\mathcal{P}}$ is $(\varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}})$-secure, for every output state of the honest-authentication setting, in the sense of [5, Sec. 3.3, Def. 3.9] at fixed length [5, Eq. (23)]: its key lengths always coincide, and E_{tot} holds every public message of $\widetilde{\mathcal{P}}$ (the announcements, the acceptance values, the seeds, L and G) together with the adversary's quantum memory. That is the hypothesis of [5, Sec. 5, Thm. 5.1], whose honest-authentication setting is this contract's and whose practical authentication setting is the authenticated classical channel of Definition 10.2 (a), properties (A1) and (A2) being its Eqs. (37)–(38). The theorem concludes $(\varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}})$-security of $\mathcal{P}$ in the sense of Def. 3.8 for every output state of that channel; by the paragraph following it, it applies to an arbitrary protocol followed by the authentication post-processing of that work, which is Definition 10.2 (c): step 6 of [5, Sec. 4.1, Protocol 1] together with the rule of its preamble that a party receiving an authentication failure aborts, with the key holder in the role of Alice and the partner in that of Bob.

(iii). By Assumption 13.5 the run is $\mathcal{P}$: it applies the acceptance test and the verification abort, sends no message beyond those of the model and of the authentication post-processing, outputs the ℓ-bit keys the model computes or $\perp$, uses no key before the end of an execution (Definition 10.3), and aborts on an authentication failure. By Assumption 13.4, $\varepsilon_{\mathrm{auth}}$ and

k_{auth}, with every other parameter, are fixed before the run, and by Assumption 13.6 (i) the run uses one scheme, so the scheme and the protocol are fixed systems. If the classical channel is itself the authenticated classical channel (Assumption 13.6 (i), second case), the run is one of those of (ii), $\varepsilon_{\mathrm{auth}} = 0$, and (10.8) is (ii). Otherwise fix an attack on the run over the scheme, with output state ρ'. By Assumption 13.6 (iii) and Definition 10.2 (b) there is a simulator σ at the adversary's interface such that no distinguisher tells the scheme, run on two identical copies of a uniformly distributed key independent of everything else and the unauthenticated channel, from the authenticated classical channel with σ with advantage above ε_0 [3, Sec. II.E, Def. 1; Sec. II.F, Eqs. (4)–(6)]. The run, the attack and any distinguisher after them form a channel applied to the joint state, before the run, of the two parties' copies of the pre-shared key with every other register and the adversary's information, together with fresh systems independent of it; by Assumption 13.6 (iv) and because a channel does not increase the trace distance, every final state against the scheme run on the actual copies lies within δ of the one against the scheme run on the ideal copies. So the scheme run on the actual copies is told from the authenticated classical channel with σ with advantage at most $\varepsilon_0 + \delta \leq \varepsilon_{\mathrm{auth}}$ (Assumption 13.6 (v)). Consider the distinguisher that plays both parties of $\mathcal{P}$, every quantum round included, and the given attack, which reaches the classical channel only through the adversary's interface of the system the distinguisher is given, and which then holds the parties' key registers and the adversary's final information. Its final state is ρ' against the scheme and, against the authenticated channel with σ, the output state ρ of $\mathcal{P}$ under the attack that the given one forms with σ; that attack reads messages and controls their delivery only through the authenticated classical channel, so ρ is an output state that channel admits. Any measurement on the final state completes this distinguisher, and the distinguishing advantage between two states is their trace distance [3, Sec. III.B.1], so $\frac{1}{2}\|\rho' - \rho\|_1 \leq \varepsilon_{\mathrm{auth}}$. By (ii), $\frac{1}{2}\|\rho - \mathcal{R}_{\mathrm{ideal}}(\rho)\|_1 \leq \varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}}$. The map $\mathcal{R}_{\mathrm{ideal}}$ is a channel on the key registers [5, Sec. 3.3, Eqs. (13), (17)], so it does not increase the trace norm and $\frac{1}{2}\|\mathcal{R}_{\mathrm{ideal}}(\rho) - \mathcal{R}_{\mathrm{ideal}}(\rho')\|_1 \leq \varepsilon_{\mathrm{auth}}$. The triangle inequality over ρ', ρ, $\mathcal{R}_{\mathrm{ideal}}(\rho)$ and $\mathcal{R}_{\mathrm{ideal}}(\rho')$ gives (10.8). The run consumes at most k_{auth} bits of pre-shared key (Assumption 13.6 (ii)) and, when both parties output a key, produces an ℓ-bit key, which is the accounting of key consumed against key produced in [3, Sec. VII.D]; the net key gain is then $\ell - k_{\mathrm{auth}}$. □

N executions that share no key and no device state and leave their keys unused until all have ended, run against one adversary, are together (ε_1 + ... + ε_N)-secure for the joint ideal map. A key reused from one execution in another is not covered.

Composable frameworks add the errors of protocols composed in sequence or in parallel [3, Sec. II.E]; that is the motivation for the trace-distance statement below, which is derived without it. Each execution's parameter holds against every attack on it, and the other executions, which touch none of its registers, are part of one such attack. The per-execution ideal maps are then applied one at a time, each step costing that execution's parameter.

Corollary 10.9 (Security parameters of separate executions add)**.** Let executions $1, \ldots, N$ of the protocol run against one adversary, in sequence, concurrently or interleaved, the adversary carrying information from each to the others. For each i let the hypotheses of Corollary 10.8 (iii) hold for execution i, with its parameters $\varepsilon^{(i)}_{\mathrm{PA}}$, $\varepsilon^{(i)}_{\mathrm{EV}}$ and $\varepsilon^{(i)}_{\mathrm{auth}}$. Write $K^{(i)} \coloneqq K^{(i)}_A K^{(i)}_B$ for its key registers, $\mathcal{R}^{(i)}$ for the map $\mathcal{R}_{\mathrm{ideal}}$ of [5, Sec. 3.3, Eqs. (13)–(17)] on $K^{(i)}$, and $\varepsilon_i \coloneqq \varepsilon^{(i)}_{\mathrm{PA}} + \varepsilon^{(i)}_{\mathrm{EV}} + 2\varepsilon^{(i)}_{\mathrm{auth}}$. Let Assumption 13.7 hold. Then for every attack on the N executions under which each of them reaches its end of an execution (Definition 10.3) on every branch, with ρ the joint state of $K^{(1)} \cdots K^{(N)}$ and the adversary's information E at the end of the last of them,

$$\frac{1}{2}\left\| \rho - \left(\mathcal{R}^{(1)} \otimes \cdots \otimes \mathcal{R}^{(N)} \otimes \mathrm{id}_E\right)(\rho) \right\|_1 \leq \sum_{i=1}^{N} \varepsilon_i. \tag{10.9}$$

Reading: Running several executions against the same attacker, one after another or at the same time, costs the sum of their security parameters. If each execution i is secure with parameter ε_i and A-SEPARATE holds, whose conditions include that their number and parameters are fixed before the first starts, that they share no key material and no device state, and that their keys stay unused until all have ended, then all their keys together are within ε_1 + ... + ε_N of independent ideal keys. Parameters chosen from an earlier execution's data are not covered, nor is a key from one execution reused in another, for example to authenticate it.

Uses Def. 2.3, Thm. 2.10, Thm. 3.1, Def. 10.3, Lem. 10.4, Thm. 10.7, Cor. 10.8 · *Assumes* A-SEPARATE · *Used by* Rem. 10.10 · newly derived · accepted

Proof. Fix an attack on the N executions under which each of them reaches its end on every branch. For each i let t_i be the end of an execution (Definition 10.3) for execution i, and let $t \coloneqq \max_i t_i$. For a time s and an execution i, write $Y_i(s)$ for the adversary's registers at s together with every register of the honest parties of every execution $j \neq i$ at s. By Assumption 13.7 (iii), at t the register $K^{(j)}$ of every execution holds that execution's output.

The times t_i and t are random and may depend on the attack. The joint process is therefore read branch by branch over the classical record of when each party ends (where the end times are bounded, equivalently with idle steps that give every branch one schedule); the states named below are

averaged over the branches. That record lies in Y_i: the key holder of execution i ends by sending a message the adversary sees, and the partner when the adversary delivers it (Definition 10.3). By Assumption 13.7 (ii) and (iii), the evolution after execution i has ended depends on that execution's history only through Y_i. The restriction to attacks under which each execution reaches its end on every branch is the conjunction of the restrictions that Corollary 10.8 (iii) already makes for each execution, whose bound concerns the output state of a run that ends.

Step 1: one execution against the others. Fix i. Let $\mathcal{E}_i$ be the adversary whose initial register holds the given adversary's initial register together with the initial registers of the honest parties of every execution $j \neq i$ (their pre-shared keys, their local randomness and the initial states of their devices), in their joint initial state, and who carries out, at the times and in the order of the given attack, the given adversary's operations and every operation of the honest parties of the executions $j \neq i$. By Assumption 13.7 (ii), no operation of the honest parties of another execution acts on a register of execution i's honest parties, none of theirs acts on a register of another execution, and execution i meets the other executions only through systems and classical channels the adversary has access to. The given adversary's own operations reach execution i only through those systems and channels, its attack interface, as in any attack on it. So $\mathcal{E}_i$ is an attack on execution i alone, whose registers at time s are $Y_i(s)$, and execution i run against $\mathcal{E}_i$ is the same channel, applied to the same initial state, as the N executions against the given adversary, with the registers grouped differently.

Corollary 10.8 (iii) holds for every attack that its premises admit, whatever the dimension of the adversary's registers, and it restricts the adversary's initial register and its state by nothing else. Through Theorem 10.7 it rests on Theorem 2.10, whose initial state ω^0 is arbitrary on the adversary's initial register E_0 of the round model (Definition 2.3), with Alice's marginal fixed in trusted-source mode; in the constrained branch of Theorem 10.7 it rests on Theorem 3.1, whose hypotheses include those of Theorem 2.10 and so leave ω^0 as arbitrary, the attack sequence being $\mathfrak{L}$-admissible. It also rests on Lemma 10.4, which holds for every attack. The initial register of $\mathcal{E}_i$, however it is correlated with the given adversary's and the other executions' systems, is such an E_0. The premises are the assumptions on which Corollary 10.8 (iii) rests for execution i, and by Assumption 13.7 (iv) each of them holds for execution i against $\mathcal{E}_i$. The registers of $\mathcal{E}_i$, which hold the other executions' registers, may be of any dimension, which the corollary allows. Where a constraint list is declared, A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk)) states its reduction over the licensed attack sets of Definition 2.3, defined for finite-dimensional registers, and is asked, with the corollary's other premises, of the all-at-once attacks with finite registers through which the corollary reaches $\mathcal{E}_i$. Clause (iv) asserts each of these, and the proof uses them only through Corollary 10.8 (iii). By

Assumption 13.7 (i) the parameters of execution i, and with them ε_i, are fixed before any execution starts and do not depend on the attack.

Corollary 10.8 (iii) therefore applies to execution i against $\mathcal{E}_i$, whose information when execution i has ended is $Y_i(t_i)$. With σ_i the state of $K^{(i)}Y_i(t_i)$ at t_i, (10.8) gives

$$\frac{1}{2}\big\|\sigma_i - (\mathcal{R}^{(i)} \otimes \mathrm{id})(\sigma_i)\big\|_1 \leq \varepsilon_i.$$

From t_i to t, nothing acts on $K^{(i)}$ and no information about another register of execution i's honest parties leaves their laboratories (Assumption 13.7 (iii)); those parties, having ended, act on nothing; and no operation of the honest parties of another execution acts on a register of execution i (Assumption 13.7 (ii)). What happens between t_i and t therefore acts on Y_i alone and takes no input from a register of execution i: it is a channel Φ_i from $Y_i(t_i)$ to $Y_i(t)$, and the state of $K^{(i)}Y_i(t)$ at t is $(\mathrm{id}_{K^{(i)}} \otimes \Phi_i)(\sigma_i)$. The registers $K^{(j)}$, $j \neq i$, and E at t belong to $Y_i(t)$, so $\rho = (\mathrm{id}_{K^{(i)}} \otimes \Lambda_i)(\sigma_i)$ with Λ_i the channel Φ_i followed by the partial trace onto them. This channel acts on registers other than $K^{(i)}$, so it commutes with $\mathcal{R}^{(i)} \otimes \mathrm{id}$, and a channel does not increase the trace norm of a Hermitian operator. Hence

$$\frac{1}{2}\big\|\rho - (\mathcal{R}^{(i)} \otimes \mathrm{id})(\rho)\big\|_1 \leq \varepsilon_i \qquad (i = 1, \ldots, N). \tag{10.10}$$

Step 2: the hybrid. Put $\mathcal{T}_0 := \mathrm{id}$ and $\mathcal{T}_i := \mathcal{R}^{(1)} \otimes \cdots \otimes \mathcal{R}^{(i)} \otimes \mathrm{id}$ for $i = 1, \ldots, N$, the identity acting on every other register. The maps $\mathcal{R}^{(i)}$ act on the disjoint registers $K^{(i)}$, so $\mathcal{T}_i = \mathcal{T}_{i-1} \circ (\mathcal{R}^{(i)} \otimes \mathrm{id})$, and $\mathcal{T}_N$ is the map of (10.9). By linearity,

$$\rho - \mathcal{T}_N(\rho) = \sum_{i=1}^{N} \big(\mathcal{T}_{i-1}(\rho) - \mathcal{T}_i(\rho)\big) = \sum_{i=1}^{N} \mathcal{T}_{i-1}\big(\rho - (\mathcal{R}^{(i)} \otimes \mathrm{id})(\rho)\big).$$

Each $\mathcal{R}^{(i)}$ is a channel [5, Sec. 3.3, Eqs. (13), (17)], so each $\mathcal{T}_{i-1}$ is a channel and does not increase the trace norm of the Hermitian operator it is applied to. The triangle inequality and (10.10) give (10.9).

By [5, Sec. 3.3, Eqs. (14), (16), (17)], $\mathcal{R}^{(i)}$ replaces, for each pair of key lengths of execution i, its key registers by the ideal keys of that pair, independent of every other register. Applied for one execution after another, $\mathcal{T}_N(\rho)$ is therefore the state in which, for each tuple of the executions' pairs of key lengths, with its probability under ρ, the key registers of every execution hold the ideal keys of its pair, independent of each other and of E: the ideal of the N executions together. □

One execution's parameter covers one run, and separate executions that share no key add their parameters (Corollary 10.9). A reused authentication key enters through ε_auth, charged twice, and the earlier run's parameter does not bound its distance in general.

Remark 10.10 (One execution's security parameter and a reused key). The parameter $\varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}} + 2\varepsilon_{\mathrm{auth}}$ of Corollary 10.8 (iii) bounds one execution: one run of n rounds and its post-processing. It is not a rate per round or per bit, and it does not shrink when executions are repeated.

A run whose pre-shared authentication key is not fresh, for instance part of the key of an earlier execution, enters through Assumption 13.6 (iii)–(v). If, before the run starts, the key jointly with every other register and the adversary's information is within trace distance δ of the state in which it is replaced by a uniform and independent key, then running the scheme, the protocol and any distinguisher on it is a channel applied to that joint state, together with fresh systems independent of it; a channel does not increase the trace distance, so the outputs differ by at most δ from those on the ideal key, and by the triangle inequality the scheme realises the authenticated classical channel within $\varepsilon_0 + \delta$, where ε_0 is its error on an ideal key. The declared $\varepsilon_{\mathrm{auth}}$ includes δ, and Corollary 10.8 (iii) charges it twice, 2δ in all. For a key taken from an earlier execution's output, that execution's parameter does not bound δ in general. That parameter is a sum over the pairs of key lengths [5, Sec. 3.3, Def. 3.8, Eqs. (11), (17)]: it bounds the key's distance from an ideal one where both parties hold a key only weighted by the probability of that event, which the adversary controls [6, Sec. 2.3, Remark 2.1]. Where one party holds a key and the other none, as after a blocked final accept of the authentication post-processing, the ideal state of that definition also leaves one party without a key, so the parameter does not bound the probability of that event; there no shared key exists to reuse, and δ, the distance of the whole state on which the next run starts, conditioned on its starting, counts that event where the next run starts on it, and otherwise enters δ through the conditioning on its start, which the adversary influences.

For executions that share no key material and no device state, and whose keys stay unused until the last of them has reached its end of an execution (Assumption 13.7), the parameters add: Corollary 10.9 bounds N such executions, run against one adversary in any order, by $\varepsilon_1 + \cdots + \varepsilon_N$ for the joint ideal map, which is what composition gives for constructive statements [3, Sec. II.E]. Not covered are a key used before the last execution has ended, since the ideal map applied at the end would then replace a key whose use the adversary has already seen; parameters chosen from an earlier execution's data, which would make each ε_i depend on the attack; and a chain in which a key of one execution authenticates the next. For such a chain the hybrid would have to replace a key by an ideal one at a moment both parties have ended the earlier execution, which the authentication post-processing does not provide, and the earlier parameter does not bound δ (the paragraph above). That the literature expects such chains to add [6, Sec. 2.3, footnote 2] is cited as motivation only, and a bound on the failure probability accumulated over every use [3, Sec. II.F] lies outside what this contract proves.

Reading: The security parameter ε covers one execution: one run of n rounds and its post-processing. It is not a per-round or per-bit figure, and it does not shrink when you repeat the protocol. Several executions are together secure with the sum of their ε (Corollary 10.9) under A-SEPARATE. The sum does not cover parameters chosen from

an earlier run's data, such as the error-correction budget or the acceptance test. The conditions of A-SEPARATE include: (i) their number and parameters are all fixed before the first starts; (ii) they share no key material and no device state; (iii) their keys stay unused until the last has ended. Reusing key from an earlier run to authenticate the next is not covered either. Such a key enters only through the next run's declared authentication error ε_auth, counted twice, and the earlier run's ε does not by itself bound how far that key is from an ideal one.

Uses Cor. 10.8, Cor. 10.9 · newly derived · accepted

The completeness parameter ε^* of the statistics module bounds an honest run's failure of the acceptance test only. Aborts in error verification and in authentication are bounded by no claim, so ε^* does not bound the honest abort probability of the whole protocol.

Remark 10.11 (Completeness covers the acceptance test only). The completeness statements of the statistics module, Theorem "The completeness bound" (statistics/binomial-box/thm:completeness-bound) with its parameter $\varepsilon^\star$ and Theorem "Completeness of a singleton acceptance set" (statistics/singleton/thm:singleton-completeness), bound the probability that an honest run fails the acceptance test, the complement of the event Ω_{AT} of Definition 10.1 (i). An honest run of the fixed-length protocol can also abort later: in error verification, when the partner's reconstruction $\hat{Z}_1^n$ differs from Z_1^n, and in authentication (Definition 10.2). That the error-correction scheme lets the partner reconstruct Z_1^n from L_{EC} and its own data on honest runs is a premise of no security claim here: it bears only on how often an honest run aborts. No claim of this contract bounds those abort probabilities, so $\varepsilon^\star$ bounds the honest abort probability of the acceptance test and not that of the whole protocol. Security is unaffected: Corollary 10.8 holds for every abort pattern.

Reading: The completeness figure ε^* bounds one thing only: the chance that an honest run, with no attacker and devices behaving as the declared honest model says, fails the acceptance test on its statistics. An honest run can also abort later: in error verification, when error correction leaves the two raw keys different, and in authentication. No result bounds those aborts, so an honest run's overall abort chance can exceed ε^*. Whether your error-correction code reconstructs the key on honest runs is yours to check; it affects only how often you obtain a key. Completeness never affects the security parameter ε, which holds for every abort pattern (Corollary 10.8) and so says nothing about how often runs abort.

Uses Def. 10.1, Def. 10.2 · newly derived · accepted

If every round's acceptance value, given the earlier ones, follows the statistics of a feasible state, the run passes the acceptance test with probability at most 2^(−n D_min). A lower bound b at one order thus caps that probability at 2^(−n (α₀−1)/α₀ · b).

Proposition 9.5 (iii) says that where $D_{\min} > 0$ no single feasible state has its statistics in the acceptance set; it says nothing about a finite run, whose frequencies can still fall in $\mathcal{S}_{\mathrm{acc}}$. This proposition bounds that event. A state ω on AQ put through the single-round protocol map yields an acceptance value distributed as $\mathbf{\Phi}[\omega]$ (Theorem 1.5), so the hypothesis below describes a run without an adversary in which the devices behave as the declaration says: each round prepares and measures a state of the feasible set, which may depend on the earlier rounds. That is a hypothesis about the devices

the declaration describes. It is no consequence of the security model: the round model and the accumulation theorems hold against every attack and say nothing about how often a run passes, and the security claims of this contract do not use the hypothesis. Read the other way, it is a statistical test of the model, for an acceptance set fixed before the run: where the bound is tiny and real devices pass, that is evidence at the level of the bound, not a proof, that the declaration does not describe them, which is the reading of Assumption "An honest law produced by a feasible state" (statistics/ass:reach). An acceptance set built from the same run's frequencies passes by construction and supports no such reading. A round state that depends on a richer record of the past than the earlier acceptance values, with finitely many values, is covered: given the acceptance values the round's law is then a convex combination of statistics of feasible states, which is the statistics of their convex combination, a feasible state, $\mathbf{\Phi}$ being linear and $\mathcal{W}$ convex. The proof is a Chernoff bound whose exponent is read off a minimising pair of the divergence, with the statistics set mixed with the uniform distribution so that every outcome has positive probability.

Proposition 10.12 (The pass probability of devices the declared model admits)**.** Let $\mathfrak{D}$ and $\mathcal{S}_{\mathrm{acc}}$ satisfy the hypotheses of Proposition 9.5, let $D_{\min}$ be as in (9.7) and $\mathcal{D}_{\mathrm{all}}$ the dropped set of every pair there, and let $n \geq 1$ be an integer. Let Pr be a probability distribution on $\widehat{\mathcal{C}}^{\,n}$, write $\widehat{C}_1, \ldots, \widehat{C}_n$ for the coordinates and $\widehat{C}_1^{\,t} := (\widehat{C}_1, \ldots, \widehat{C}_t)$, and suppose:

(H) for every $t \in \{1, \ldots, n\}$ and every $h \in \widehat{\mathcal{C}}^{\,t-1}$ with $\Pr\left[\widehat{C}_1^{\,t-1} = h\right] > 0$ there is a state $\omega_t(h) \in \mathcal{W}$ with

$$\Pr\left[\widehat{C}_t = \hat{c} \;\middle|\; \widehat{C}_1^{\,t-1} = h\right] = \mathbf{\Phi}[\omega_t(h)](\hat{c}) \qquad \text{for every } \hat{c} \in \widehat{\mathcal{C}}, \tag{10.11}$$

where $\mathbf{\Phi}[\omega](\hat{c})$ is the entry of $\mathbf{\Phi}[\omega]$ at $\hat{c}$ and, for $t = 1$, h is the empty sequence and the left-hand side is $\Pr\left[\widehat{C}_1 = \hat{c}\right]$.

The case of independent rounds with one state $\omega \in \mathcal{W}$ is $\omega_t(h) = \omega$ for all t and h. Let $\Omega_{\mathrm{AT}} = \{\mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}\}$ be the event of Definition 10.1 (i), where $\mathrm{freq}_{\hat{c}_1^n}(\hat{c}) := \frac{1}{n}\,|\{t : \hat{c}_t = \hat{c}\}|$.

(i) *(The bound.)* For every real number $\delta \leq D_{\min}$,

$$\Pr[\Omega_{\mathrm{AT}}] \;\leq\; 2^{-n\delta}. \tag{10.12}$$

In particular $\Pr[\Omega_{\mathrm{AT}}] = 0$ if $D_{\min} = +\infty$.

(ii) *(From a bound at one order.)* Let $\alpha_0 \in (1, \infty)$ and let b be a real number with $b \leq \underline{h}_{\alpha_0}^{\mathcal{D}_{\mathrm{all}}}$. Then $\Pr[\Omega_{\mathrm{AT}}] \leq 2^{-x}$ for every real number $x \leq n\,\frac{\alpha_0 - 1}{\alpha_0}\,b$. For every $\alpha' \in (1, \infty)$ the number $f(\alpha') := \frac{\alpha'}{\alpha' - 1} \cdot \frac{\alpha_0 - 1}{\alpha_0}\,b$ satisfies $\frac{\alpha' - 1}{\alpha'}\,n\,f(\alpha') = n\,\frac{\alpha_0 - 1}{\alpha_0}\,b$.

Reading: This result bounds how often devices that behave as your declared model says would pass your acceptance test. Suppose that in every round, given what happened before, the acceptance value is distributed as the statistics of some state the declaration admits as feasible. Then the run passes the test with probability at most 2^(−n D_min), where D_min is the minimal divergence of Proposition 9.5, and the engine's proven lower bound on D_min gives a computable cap. This is a statement about the declared model, not about your real devices and not about an attacker. If the acceptance set was fixed before the run, the cap is tiny and your real devices do pass, that is evidence at the level of the cap, not a proof, that the model does not describe them (Assumption "An honest law produced by a feasible state" (statistics/ass:reach)). A set built from the same run's frequencies passes by construction.

Uses Def. 1.2, Def. 1.4, Def. 2.5, Def. 5.1, Def. 9.2, Prop. 9.5, Def. 10.1, foundations/def:kl, foundations/lem:kernel · *Used by* Req. 16.2 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.consistency:model_pass_exponent

Proof. Idea. If a weight $w \geq 0$ on the acceptance values has mean at most one under every statistics of a feasible state, the product of the weights of the rounds has expectation at most one, and on the acceptance event its logarithm is n times the mean of $\log_2 w$ under the observed frequency. Markov's inequality then bounds the event. The weight is the ratio q^*/p^* of a minimising pair of the divergence: the first-order conditions of the two minimisations give the two properties. Mixing every statistics with a little of the uniform distribution makes every outcome positive, so that the derivatives exist; the mixing is removed in a limit.

Write $C := \widehat{\mathcal{C}}$, a non-empty finite set, Δ for the probability simplex on C, u for the uniform distribution on C, and

$$\mathcal{P} := \{\mathbf{\Phi}[\omega] : \omega \in \mathcal{W}\}.$$

Every $\mathbf{\Phi}[\omega]$ with $\omega \in \mathcal{W}$ lies in Δ, as in the proof of Proposition 9.5, so $\mathcal{P} \subseteq \Delta$ and, by (9.7), $D_{\min} = \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \inf_{\mathbf{p}\in\mathcal{P}} D(\mathbf{q}\|\mathbf{p})$. The map $\mathbf{\Phi}$ of (1.16) is linear (Definition 1.4), hence continuous, and $\mathcal{W}$ is convex, compact and non-empty (Definition 5.1; Definition 1.2), so $\mathcal{P}$ is convex, compact and non-empty. $\mathcal{S}_{\mathrm{acc}}$ is a closed convex subset of Δ (Definition 2.5), hence compact. For $\mathbf{q}, \mathbf{p} \in \Delta$, $D(\mathbf{q}\|\mathbf{p}) = \sum_{c\in C} d(q(c), p(c))$ with d the kernel of Lemma "Extended divergence kernel" (foundations/lem:kernel), whose values at vanishing arguments are the conventions of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl); d is lower semicontinuous with values in $(-\infty, +\infty]$, so D, a finite sum of such functions composed with the continuous coordinate maps, is lower semicontinuous on $\Delta \times \Delta$ with values in $[0, +\infty]$ (non-negative by Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl)).

If $\mathcal{S}_{\mathrm{acc}} = \varnothing$, then $\Omega_{\mathrm{AT}} = \varnothing$, $\Pr[\Omega_{\mathrm{AT}}] = 0$, and (10.12) holds for every δ. Assume from now on $\mathcal{S}_{\mathrm{acc}} \neq \varnothing$. Since $2^{-n\delta} \geq 2^{-nD_{\min}}$ for $\delta \leq D_{\min}$ (with $2^{-\infty} := 0$), (i) follows from

$$\Pr[\Omega_{\mathrm{AT}}] \leq 2^{-nD_{\min}}, \tag{10.13}$$

which steps 1–5 prove.

Step 1 (the mixed sets). For $\varepsilon \in (0,1)$ put $\mathcal{P}_\varepsilon := \{(1-\varepsilon)\mathbf{p} + \varepsilon u : \mathbf{p} \in \mathcal{P}\}$, the image of $\mathcal{P}$ under an affine map of $\mathbb{R}^C$ into Δ; it is convex, compact and non-empty, and every $\mathbf{p}' \in \mathcal{P}_\varepsilon$ has $p'(c) \geq \varepsilon/|C| > 0$ for every c. Put $D_\varepsilon := \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \inf_{\mathbf{p}'\in\mathcal{P}_\varepsilon} D(\mathbf{q}\|\mathbf{p}')$. On $\Delta \times \mathcal{P}_\varepsilon$, $D(\mathbf{q}\|\mathbf{p}') = \sum_c \big[\eta(q(c)) - q(c)\log_2 p'(c)\big]$ with $\eta(x) := x\log_2 x$ for $x > 0$ and $\eta(0) := 0$, continuous on $[0,\infty)$, and $\log_2 p'(c)$ continuous as $p'(c) \geq \varepsilon/|C|$; so D is finite and continuous there, and on the non-empty compact set $\mathcal{S}_{\mathrm{acc}} \times \mathcal{P}_\varepsilon$ it attains its minimum at some pair $(\mathbf{q}^*, \mathbf{p}^*)$, with $D(\mathbf{q}^*\|\mathbf{p}^*) = D_\varepsilon \in [0,\infty)$. Define

$$w(c) := \frac{q^*(c)}{p^*(c)} \in [0,\infty), \qquad S^* := \{c \in C : q^*(c) > 0\} = \{c : w(c) > 0\}.$$

Step 2 (the acceptance side). We show: every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ vanishes off S^*, and

$$\sum_{c\in S^*} q(c)\log_2 w(c) \;\geq\; D_\varepsilon \qquad \text{for every } \mathbf{q} \in \mathcal{S}_{\mathrm{acc}}. \tag{10.14}$$

Fix $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and put $\mathbf{q}_s := \mathbf{q}^* + s(\mathbf{q} - \mathbf{q}^*)$ for $s \in [0,1]$, which lies in $\mathcal{S}_{\mathrm{acc}}$ by convexity, and $g(s) := D(\mathbf{q}_s\|\mathbf{p}^*)$. As p^* has no zero, $g(s) = \sum_c \varphi_c(q_s(c))$ with $\varphi_c(x) := \eta(x) - x\log_2 p^*(c)$ on $[0,\infty)$, which is differentiable on $(0,\infty)$ with $\varphi_c'(x) = \log_2 x + \log_2 \mathrm{e} - \log_2 p^*(c)$. By minimality of the pair, $g(s) \geq D_\varepsilon = g(0)$ for every $s \in [0,1]$.

Suppose $q(c_0) > 0 = q^*(c_0)$ for some c_0. Consider $\frac{g(s)-g(0)}{s} = \sum_c \frac{\varphi_c(q_s(c)) - \varphi_c(q^*(c))}{s}$ as $s \downarrow 0$. For $c \in S^*$, $q_s(c) \to q^*(c) > 0$ and the quotient tends to the finite number $\varphi_c'(q^*(c))\,(q(c) - q^*(c))$. For $c \notin S^*$ with $q(c) = 0$, $q_s(c) = 0$ and the quotient is 0. For $c \notin S^*$ with $q(c) > 0$, $q_s(c) = s\,q(c)$ and the quotient is $q(c)\log_2 \frac{s\,q(c)}{p^*(c)} \to -\infty$; c_0 is such a c. So $\frac{g(s)-g(0)}{s} \to -\infty$, and $g(s) < g(0)$ for some $s \in (0,1]$, a contradiction. Hence $q(c) = 0$ for every $c \notin S^*$.

Then for $c \notin S^*$ the summand of g is $\varphi_c(0) = 0$ for all s, and for $c \in S^*$ and $s \in [0,1)$, $q_s(c) \geq (1-s)q^*(c) > 0$, so g has the right derivative $g'(0^+) = \sum_{c\in S^*} \varphi_c'(q^*(c))\,(q(c) - q^*(c))$ at 0, which is ≥ 0 because $g(s) \geq g(0)$ for $s > 0$. Since $\varphi_c'(q^*(c)) = \log_2 w(c) + \log_2 \mathrm{e}$ and $\sum_{c\in S^*} q(c) = 1 = \sum_{c\in S^*} q^*(c)$ (both vanish off S^*), this reads $\sum_{c\in S^*} q(c)\log_2 w(c) \geq \sum_{c\in S^*} q^*(c)\log_2 w(c) = D(\mathbf{q}^*\|\mathbf{p}^*) = D_\varepsilon$, which is (10.14).

Step 3 (the model side). We show

$$\sum_{c\in C} p(c)\,w(c) \;\leq\; \frac{1}{1-\varepsilon} \qquad \text{for every } \mathbf{p} \in \mathcal{P}. \tag{10.15}$$

Let $\mathbf{p}' \in \mathcal{P}_\varepsilon$, put $\mathbf{p}_s := \mathbf{p}^* + s(\mathbf{p}' - \mathbf{p}^*) \in \mathcal{P}_\varepsilon$ for $s \in [0,1]$ (convexity), so that $p_s(c) \geq \varepsilon/|C| > 0$, and $\rho(s) := D(\mathbf{q}^*\|\mathbf{p}_s) = \sum_{c\in S^*} q^*(c)\big[\log_2 q^*(c) - \log_2 p_s(c)\big]$. Each $p_s(c)$ is affine in s and positive, so ρ is differentiable on $[0,1]$ with $\rho'(0) = -\log_2 \mathrm{e} \sum_{c\in S^*} q^*(c)\frac{p'(c)-p^*(c)}{p^*(c)}$. By minimality of the pair, $\rho(s) \geq \rho(0)$ for $s \in [0,1]$, so $\rho'(0) \geq 0$, that is $\sum_{c\in S^*} q^*(c)\frac{p'(c)}{p^*(c)} \leq$

$\sum_{c\in S^*} q^*(c) = 1$. As w vanishes off S^*, $\sum_c p'(c)w(c) \le 1$ for every $\mathbf{p}' \in \mathcal{P}_\varepsilon$. For $\mathbf{p} \in \mathcal{P}$ take $\mathbf{p}' = (1-\varepsilon)\mathbf{p} + \varepsilon u$; since $\sum_c u(c)w(c) \ge 0$, $(1-\varepsilon)\sum_c p(c)w(c) \le \sum_c p'(c)w(c) \le 1$, which is (10.15).

Step 4 (the process). Put $Z_0 \coloneqq 1$ and $Z_t \coloneqq \prod_{s=1}^{t} w(\widehat{C}_s) \ge 0$ for $t = 1, \dots, n$, and write $\mathbb{E}$ for the expectation under Pr. For $t \ge 1$, grouping $\widehat{C}_1^t = (h, \hat{c})$ by its first $t-1$ entries h, and using $\Pr\left[\widehat{C}_1^t = (h,\hat{c})\right] = 0$ where $\Pr\left[\widehat{C}_1^{t-1} = h\right] = 0$,

$$\mathbb{E}[Z_t] = \sum_{h:\,\Pr\left[\widehat{C}_1^{t-1}=h\right]>0} \Pr\left[\widehat{C}_1^{t-1} = h\right] \prod_{s=1}^{t-1} w(h_s) \sum_{\hat{c}\in C} \mathbf{\Phi}[\omega_t(h)](\hat{c})\, w(\hat{c}) \;\le\; \frac{1}{1-\varepsilon}\,\mathbb{E}[Z_{t-1}],$$

by (10.11) and by (10.15) for $\mathbf{p} = \mathbf{\Phi}[\omega_t(h)] \in \mathcal{P}$. By induction, $\mathbb{E}[Z_n] \le (1-\varepsilon)^{-n}$. On Ω_{AT}, put $\mathbf{q} \coloneqq \mathrm{freq}_{\widehat{C}_1^n} \in \mathcal{S}_{\mathrm{acc}}$ and $N_c \coloneqq nq(c)$. Every value $c = \widehat{C}_t$ has $q(c) \ge \frac{1}{n} > 0$, so $c \in S^*$ by Step 2 and $w(c) > 0$; hence $Z_n > 0$ and

$$\log_2 Z_n = \sum_{c:\,N_c>0} N_c \log_2 w(c) = n \sum_{c\in S^*} q(c) \log_2 w(c) \;\ge\; nD_\varepsilon$$

by (10.14), the summands of S^* with $q(c) = 0$ being 0. So $\Omega_{\mathrm{AT}} \subseteq \{Z_n \ge 2^{nD_\varepsilon}\}$, and Markov's inequality for the non-negative Z_n gives

$$\Pr[\Omega_{\mathrm{AT}}] \le 2^{-nD_\varepsilon}\,\mathbb{E}[Z_n] \le (1-\varepsilon)^{-n}\,2^{-nD_\varepsilon} \qquad \text{for every } \varepsilon \in (0,1). \quad (10.16)$$

Step 5 (the limit). Let $\varepsilon_j \coloneqq \frac{1}{j+1}$. The sequence $(D_{\varepsilon_j})_j$ in $[0,\infty)$ has a subsequence, which we relabel, converging to some $L \in [0,+\infty]$. Then $(1-\varepsilon_j)^{-n} \to 1$ and $2^{-nD_{\varepsilon_j}} \to 2^{-nL}$, so (10.16) gives $\Pr[\Omega_{\mathrm{AT}}] \le 2^{-nL}$. We show $L \ge D_{\min}$, which gives (10.13). If $L = +\infty$ there is nothing to show. Otherwise, for each j pick $\mathbf{q}_j \in \mathcal{S}_{\mathrm{acc}}$ and $\mathbf{p}_j \in \mathcal{P}$ with $D(\mathbf{q}_j\|\mathbf{p}'_j) \le D_{\varepsilon_j} + \frac{1}{j}$, where $\mathbf{p}'_j \coloneqq (1-\varepsilon_j)\mathbf{p}_j + \varepsilon_j u$. By compactness of $\mathcal{S}_{\mathrm{acc}} \times \mathcal{P}$ a further subsequence has $(\mathbf{q}_j, \mathbf{p}_j) \to (\mathbf{q}, \mathbf{p}) \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{P}$, and then $\mathbf{p}'_j \to \mathbf{p}$. Lower semicontinuity of D gives $D_{\min} \le D(\mathbf{q}\|\mathbf{p}) \le \liminf_j D(\mathbf{q}_j\|\mathbf{p}'_j) \le \lim_j \left(D_{\varepsilon_j} + \frac{1}{j}\right) = L$. This proves (i); where $D_{\min} = +\infty$, (10.13) reads $\Pr[\Omega_{\mathrm{AT}}] = 0$.

(ii). By Proposition 9.5 (iv), (9.9), $\frac{\alpha_0-1}{\alpha_0}\,b \le D_{\min}$. For real $x \le n\frac{\alpha_0-1}{\alpha_0}b$ put $\delta \coloneqq x/n$; then $\delta \le D_{\min}$, and (i) gives $\Pr[\Omega_{\mathrm{AT}}] \le 2^{-n\delta} = 2^{-x}$. The identity is $\frac{\alpha'-1}{\alpha'} \cdot \frac{\alpha'}{\alpha'-1} = 1$. □

The maximal key length ℓ_max is the right-hand side of the key-length bound rounded down to an integer and clamped at 0. The floor and the clamp are both in the safe direction.

Definition 10.13 (Maximal key length)**.** The *maximal key length* is

$$\ell_{\max} \coloneqq \max\{0, \lfloor n\underline{h}_\alpha - \lambda_{\mathrm{EC}} - \lceil \log_2 \tfrac{1}{\varepsilon_{\mathrm{EV}}} \rceil - \tfrac{\alpha}{\alpha-1} \log_2 \tfrac{1}{\varepsilon_{\mathrm{PA}}} + 1 \rfloor\},$$

the right-hand side of (10.4) under a floor and a clamp at 0; the floor and the clamp are both in the safe direction: an integer at most the floor of a number is at most that number, and a key of length 0 is trivially secret and correct and claims no key.

Uses Thm. 10.7 · *Used by* Prop. 10.17, Prop. 11.1, Req. 16.4, Req. 28.1 · newly derived · accepted

The key-length bound carries the constant +1 where its source derivation and the published fixed-length statement carry +2: one announced bit buys secrecy on the accepted state by an elementary contraction. The source's secrecy definition, conditioned on the acceptance test alone, does not spend it.

Remark 10.14 (The constant in the key length and the source's convention)*.* The derivation this module's key-length theorem generalises proves (10.6) with $\lambda = \lambda_{\mathrm{EC}} + \lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \rceil$ and stops there, i.e. its secrecy statement is for the $\wedge\Omega_{\mathrm{AT}}$ state and its constant is +2; the published fixed-length statement of [2, Thm. 7, Eq. (44)] reaches the accepted state by a different route (conditioning on the verification event), with the same constant. This module spends one deterministic bit (the announced flag) to reach the accepted state by an elementary contraction argument instead, avoiding any further import. The cost is exactly one bit in (10.4), which the $\wedge\Omega_{\mathrm{AT}}$ secrecy definition of that derivation does not spend; what an instance adopting that definition declares is Requirement 16.9.

Uses Thm. 10.7 · *Used by* Req. 16.9 · newly derived · accepted

With a privacy-amplification family that is only two-universal, collision probability at most 2^−ℓ, the key-length bound holds after tightening it by one further bit. Toeplitz hashing is ideal two-universal and needs no such bit.

Remark 10.15 (One further bit for a merely two-universal family)*.* The Rényi leftover-hashing step of Theorem 10.7 is taken under the reading of [5, Lemma 6.5], which requires an ideal two-universal family, collision probability exactly $2^{-\ell}$ for distinct inputs; that is Assumption 13.2 (i). For a family that is only two-universal, collision probability at most $2^{-\ell}$, the requirement can be relaxed at the cost of one bit of key length [5, Rem. 6.1]. Toeplitz hashing is ideal two-universal [5, Rem. 6.1].

Uses Thm. 10.7 · *Used by* Req. 16.5 · published in [5, Rem. 6.1]; [5, Lemma 6.5] · accepted

Every key-length statement takes the Rényi order in (1, 2), the domain of the Rényi leftover-hashing lemma, while the entropy, program and gradient results hold on (1, ∞). On (1, 2) the key-length statements and the results they consume apply together.

Remark 10.16 (The Rényi order of the key-length statements). Every key-length statement of this contract, among them Theorem 10.7, Corollary 10.18 and the key-length corollary of the first-order results (Corollary 19.5), takes $\alpha \in (1, 2)$, because the Rényi leftover-hashing lemma their secrecy argument applies is stated there [2, Thm. 4, Eq. (13)]. The results on entropy accumulation, the program, its reduction and its gradients are stated on $(1, \infty)$, where Definition "Sandwiched Rényi divergence and conditional entropies" (foundations/def:renyi) defines the sandwiched entropies, so the intersection of the two domains, on which the key-length statements and the results they consume apply together, is $(1, 2)$; a result stated on a narrower range of α holds on that range only. The accumulation import itself extends to $\alpha = \infty$, an endpoint that Definition "Sandwiched Rényi divergence and conditional entropies" (foundations/def:renyi) does not define and no claim uses (Fact 2.1, locator note (vi)).

Uses foundations/def:renyi · *Used by* Req. 16.6 · newly derived · accepted

The maximal key length does not increase with the reconciliation leakage, does not decrease as either error budget grows, and is unchanged by any change of the declaration that keeps h_low. More leakage or a tighter budget therefore never lengthens it.

Proposition 10.17 (Monotonicity of the maximal key length)**.** The maximal key length $\ell_{\max}$ of Definition 10.13 is

(i) non-increasing in λ_{EC};

(ii) non-decreasing in $\varepsilon_{\mathrm{EV}}$;

(iii) non-decreasing in $\varepsilon_{\mathrm{PA}}$;

(iv) unchanged under any modification of the declaration that leaves $\underline{h}_\alpha$ unchanged.

Consequently increasing leakage or tightening either error budget can never increase $\ell_{\max}$.

Uses Thm. 10.7, Def. 10.13 · *Used by* Cor. 10.18 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength:key_length · *Oracles* registry.py

Proof. $\underline{h}_\alpha$ does not depend on $(\lambda_{\mathrm{EC}}, \varepsilon_{\mathrm{EV}}, \varepsilon_{\mathrm{PA}})$. In (10.4): $-\lambda_{\mathrm{EC}}$ is decreasing in λ_{EC}; $-\lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \rceil$ is non-decreasing in $\varepsilon_{\mathrm{EV}}$ (the ceiling of a decreasing function of $\varepsilon_{\mathrm{EV}}$ is non-increasing in $\varepsilon_{\mathrm{EV}}$); $-\frac{\alpha}{\alpha-1} \log_2 \frac{1}{\varepsilon_{\mathrm{PA}}}$ is increasing in $\varepsilon_{\mathrm{PA}}$ ($\frac{\alpha}{\alpha-1} > 0$). The floor and the clamp $\max\{0, \cdot\}$ are monotone. □

Any $h \le h_low$ may replace h_low in the key-length bound through $\max\{0, h\}$, with the same security statement. This is where every solver obtains its key length; a bound $h \le 0$ yields no positive key when $\varepsilon_EV < 1$.

This corollary is where a solver obtains its key length, and the only place: a solver submodule establishes a lower bound $h \le \underline{h}_\alpha$ for the h it computes, under its own trust root and in its own claims, and the Frank–Wolfe linearisation bound, with both linear minima bounded from below, is the instance this repository carries. The core asserts nothing about how h was obtained, in what arithmetic, or how close it lies to $\underline{h}_\alpha$.

Corollary 10.18 (Key length from any lower bound on the value). Let $\alpha \in (1, 2)$, let the hypotheses of Theorem 10.7 hold in its unconstrained or its constrained branch, the run as in Lemma 10.5 (a) and hypothesis (H-CHAIN) included, at every all-at-once attack with finite registers, A-SEQ with its clause (v) among them, and let those of Lemma 10.4 hold. Let $h \in \mathbb{R} \cup \{-\infty\}$ satisfy

$$h \;\le\; \underline{h}_\alpha, \tag{10.17}$$

with $\underline{h}_\alpha$ the exact value (5.4) of the declaration (for the constrained branch: with $\mathcal{W}$ carrying the declared $\mathfrak{L}$). If the output length satisfies

$$\ell \;\le\; n\,\max\{0, h\} - \lambda_{\mathrm{EC}} - \left\lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \right\rceil - \frac{\alpha}{\alpha - 1} \log_2 \frac{1}{\varepsilon_{\mathrm{PA}}} + 1, \tag{10.18}$$

then the protocol is $\varepsilon_{\mathrm{PA}}$-secret and $\varepsilon_{\mathrm{EV}}$-correct on the accepted state, hence, for $\ell \ge 1$, $(\varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}})$-secure in the honest-authentication setting; under the further hypotheses of Corollary 10.8 (iii) it is $(\varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}} + 2\varepsilon_{\mathrm{auth}})$-secure over the classical channel of that part, with net key gain $\ell - k_{\mathrm{auth}}$ when both parties output a key. The secrecy and the security statements hold at every output state, whatever the dimension of the adversary's registers and the order in which the source releases its signals, as Corollary 10.8 states them, and the correctness statement at every attack. Moreover:

(a) if h depends on none of $(\lambda_{\mathrm{EC}}, \varepsilon_{\mathrm{EV}}, \varepsilon_{\mathrm{PA}})$, the four conclusions of Proposition 10.17 hold verbatim for the maximal key length (Definition 10.13) built from (10.18), with $\max\{0, h\}$ in place of $\underline{h}_\alpha$;

(b) if $h \le 0$ and $\varepsilon_{\mathrm{EV}} < 1$ then no positive ℓ satisfies (10.18) and no key is claimed (at $\varepsilon_{\mathrm{EV}} = 1$ the correctness statement is empty and the corner carries no security claim).

Uses Prop. 5.8, Lem. 10.4, Lem. 10.5, Lem. 10.6, Thm. 10.7, Cor. 10.8, Prop. 10.17 · *Used by* Req. 16.4, Req. 16.6, Req. 16.7, Req. 16.8, Cor. 19.5, Cor. 21.3, Cor. 21.4, Rem. 23.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength: key_length_from_bound · *Oracles* registry.py

Proof. (10.18) is (10.4) with $\underline{h}_\alpha$ replaced by $\max\{0, h\}$ and *nothing else*

changed: no additional subtrahend, no floor beyond the one $\max\{0,\cdot\}$ already supplies, no arithmetic requirement of any kind. By (10.17) and $\underline{h}_\alpha \geq 0$ (Proposition 5.8 (i)) one has $\max\{0,h\} \leq \underline{h}_\alpha$; the right-hand side of (10.4) is non-decreasing in $\underline{h}_\alpha$ and $n \geq 1$; so (10.18) implies (10.4) and Theorem 10.7 applies unchanged, as does Corollary 10.8, whose hypotheses are those of that theorem at an ℓ satisfying (10.4) and those of Lemma 10.4, which holds for every ℓ and every attack; the corollary states the secrecy and its three security statements at every output state, whatever the adversary's memory. (a) Under the stated independence, the proof of Proposition 10.17 carries over word for word with $\max\{0,h\}$ in place of $\underline{h}_\alpha$: the only property of $\underline{h}_\alpha$ that proof reads is that it does not depend on $(\lambda_{\mathrm{EC}}, \varepsilon_{\mathrm{EV}}, \varepsilon_{\mathrm{PA}})$. (b) If $h \leq 0$ then $\max\{0,h\} = 0$ and the right-hand side of (10.18) is $1 - \lambda_{\mathrm{EC}} - \lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \rceil - \frac{\alpha}{\alpha-1} \log_2 \frac{1}{\varepsilon_{\mathrm{PA}}}$, and for $\varepsilon_{\mathrm{EV}} < 1$ the ceiling is at least 1 and the other two subtrahends are non-negative, so the right-hand side is at most 0 and no positive ℓ satisfies (10.18) — the conservative outcome. The hypothesis $\varepsilon_{\mathrm{EV}} < 1$ is not removable: at $\varepsilon_{\mathrm{EV}} = 1$, $\varepsilon_{\mathrm{PA}} = 1$ and $\lambda_{\mathrm{EC}} = 0$, all inside the hypotheses of Theorem 10.7, the right-hand side equals 1 and $\ell = 1$ satisfies (10.18) under a $(\varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}})$-security statement that is empty; the corner carries no security claim, which is why the clause excludes it rather than claiming it. □

Under the attack-set licence the accumulated entropy is at least n times the value of the program under the licensed list, less the acceptance-probability penalty. That is the form the key-length theorem reads, so its key length holds with the constrained value.

Corollary 10.19 (Key length from the constrained value)**.** Under the hypotheses of Theorem 3.1, with Corollary 5.6,

$$\widetilde{H}_\alpha^{\uparrow}\big(Z_1^n \big| \widehat{C}_1^n I_1^n \widehat{E}\big)_{\rho_{|\Omega_{\mathrm{AT}}}} \;\geq\; n\,\underline{h}_\alpha(\mathfrak{L}) - \frac{\alpha}{\alpha-1} \log_2 \frac{1}{\Pr[\Omega_{\mathrm{AT}}]},$$

which is the form Theorem 10.7 consumes; under A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk)) and the remaining hypotheses of Theorem 10.7, the key-length statement of Theorem 10.7 therefore holds with the constrained $\underline{h}_\alpha$ (its constrained branch).

Uses Thm. 3.1, Cor. 5.6, Thm. 10.7 · *Assumes* A-ATK · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength: key_length_permitted · *Oracles* registry.py

Proof. Theorem 3.1 gives $\widetilde{H}_\alpha^{\uparrow}(Z_1^n|\widehat{C}_1^n I_1^n \widehat{E})_{\rho_{|\Omega_{\mathrm{AT}}}} \geq n\, h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L}) - \frac{\alpha}{\alpha-1} \log_2 \frac{1}{\Pr[\Omega_{\mathrm{AT}}]}$ under its hypotheses, and Corollary 5.6 gives $h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L}) \geq \underline{h}_\alpha(\mathfrak{L})$ for the same licensed list; since $n \geq 1$, substituting the second inequality into the first yields the display. That display is the accumulation bound the secrecy step of Theorem 10.7 reads, with $\underline{h}_\alpha(\mathfrak{L})$ in the place of $\underline{h}_\alpha$, so the constrained branch of that theorem applies under the hypotheses of Theorem 3.1, which

include A-ATK, together with the remaining hypotheses of Theorem 10.7: its post-processing model and the assumptions it names, $\alpha \in (1, 2)$ and $\varepsilon_{\mathrm{PA}}, \varepsilon_{\mathrm{EV}} \in (0, 1]$. □

In either format the licensed-class half of the constraint gate is a genuine type check in trusted-source mode at d_A ≥ 2, vacuous in untrusted-source mode, and at d_A = 1 rejects only what a producer can trivially avoid. In the last two cases the key length rests on the attack-set licence alone.

The numerical machinery — Definition 5.1, Theorems 5.10, 6.2, 7.5 and 8.4 — is valid for arbitrary $\mathfrak{L}$, with $\underline{h}_\alpha(\mathfrak{L}) \geq \underline{h}_\alpha(\varnothing)$; what the two-part gate on $\mathfrak{L}$ (Requirement 16.10) decides is only whether its output may reach the key length of Theorem 10.7.

Remark 10.20 (Degeneracy of the licensed-class half of the gate). Part (i) of the two-part gate — "the list lies in the licensed class of the declared source mode" — is the half a program can check. How much it checks depends on the source mode and on d_A.

Trusted-source mode with $d_A \geq 2$: the restriction is genuine — the multiples of τ_A are a one-dimensional subspace of the d_A^2-dimensional real space of Hermitian operators on A, so a general entry is in general not a licensed one — and a declaration can fail part (i). *The check is nevertheless a* type *check*: an entry declares either its scalar c_k or an operator Θ_k (Definition 2.2), and part (i) inspects which, rather than comparing the operator against $c_k\tau_A$. What is being rejected is a declaration that *says* it is general, never one whose numbers came out near a multiple; the class is the same and the decision is exact.

$d_A = 1$: it is a type check that rejects nothing a producer cannot trivially avoid. An entry declared general fails it, as at every d_A (Definition 2.2); but every 1×1 Hermitian Θ_k equals $\Theta_k \cdot [1] = c_k\tau_A$ (Proposition 5.2 (iii)), so such an entry could always have been declared homogeneous with the same condition, and no declaration *need* fail part (i).

Untrusted-source mode, any d_A: it is *vacuously true*. The entry *type* is scalar by Definition 1.1, so part (i) degenerates from a mathematical condition to a type check that a declaration passes by construction whenever it is well formed.

In both degenerate cases the constrained branch of Theorem 10.7 therefore rests *entirely* on part (ii), the attack-set licence of A-ATK (Assumption "Attack-set licence" (protocol-compiler/ass:atk)), which no program can verify. Arguably it is the correct degeneration: in the no-marginal model the licensed class *is* the scalar class (Remark 3.4 (a)), so there is no wider class for part (i) to exclude, and the honest statement is that the attestation is doing all of the work. What an implementation reports of a passing part (i) in these cases is Requirement 16.11.

The general format. The scalar class named for the no-marginal model is the licensed class of the basic format. In the general format of Definition 1.1 the licensed class of Definition 2.2 contains, in *either* mode, the entries declared party-local beside the mode's own form; so in the no-marginal model the licensed class is the scalar form *together with* party-local. *And those two are exactly the forms that are expressible there*: a general right-hand side does not exist in untrusted-source mode at all (Definition 2.2; Definition 1.2 (vi) in its general-format reading), so it is refused by well-formedness before part (i) is reached. *The licensed class therefore grew by exactly what the expressible class grew by, part (i) excludes nothing new,*

and the degeneration stands, along the same two readings: *in the no-marginal model* part (i) remains a type check that a declaration passes *by construction* whenever it is well formed, both expressible forms being licensed — two forms there, not three; and *at* $d_A = 1$ it remains the weaker statement that it rejects nothing a producer cannot trivially avoid, a general entry being expressible and unlicensed in trusted-source mode while every 1×1 Hermitian Θ_k equals $c_k \tau_A$, so that such an entry could always have been declared homogeneous instead. In both readings the honest statement is still that the attestation is doing all of the work. In trusted-source mode at $d_A \geq 2$ part (i) is not degenerate, in either format. *What the third form does change* is what part (ii) carries: a tagged entry makes the attestation attest *more* — the certificate on the scoped factor's own channel (Remark "Soundness of a party-local entry" (protocol-compiler/rem:party-local-soundness)) — so the rules of the gate bind it exactly as they bind the others (Requirements 16.10 and 16.11).

Uses Def. 1.1, Def. 1.2, Def. 2.2, Rem. 3.4, Def. 5.1, Prop. 5.2, Thm. 5.10, Thm. 6.2, Thm. 7.5, Thm. 8.4, Thm. 10.7 · *Assumes* A-ATK, A-SRC · *Used by* Req. 16.11 · newly derived · accepted

11 Perturbations of the declared measurement

Declared data are binary64 numbers, and a declaration is exactly admissible when the conditions of Definition 1.1 hold of those numbers themselves, among them operators exactly Hermitian, measurements and marginal exactly positive semidefinite, and completeness sums, formed in the declared order with no addition rounding, exactly the identity. One might hope that two such declarations whose measurements on Q differ by at most ϵ have nearby rates; Proposition 11.1 shows that they need not. On data of fixed small dimensions their rate bounds can lie $\alpha/(\alpha-1)\cdot$N bits per round apart, for every positive integer $\mathrm{N} \leq 53 + \log_2 \epsilon$, which is up to 52 (Remark 11.2). The gap sits in the acceptance-test penalty, can point either way and at the larger N is far above anything a key-length claim could absorb (Remark 11.3). The engine therefore admits declared data only when they are exactly admissible (Requirement 14.1), and the declared measurement model is itself a premise of every reported rate; where a reference of it is held, the first-order bound is taken on the reference and the snap onto the lattice only steers the search, unless a number certified on the snapped declaration carries the key length, and no claim bounds the gap between the model and the device (Remark 11.4).

Imports foundations/def:evaluated-off-snap, foundations/def:kl, foundations/def:snap, foundations/rem:noise-floor, foundations/req:snap-reference

Two exactly admissible measurements on Q within ϵ of each other can have rate bounds $\alpha/(\alpha-1)\cdot N$ bits per round apart, for every positive integer $N \le 53 + \log_2\epsilon$ (up to 52). So closeness of the declared measurement cannot be relied on to carry a key-rate claim to a nearby declaration.

One might admit a declaration whose measurement operators are only close to exact ones, with an error allowance depending on the distance. The following proposition shows that no allowance computed from the declared dimensions, alphabet sizes and α can be useful. It compares values $\underline{h}_\alpha$ of (5.4), the infimum of the objective (5.3) over the acceptance set and the feasible states. That objective adds to its entropy terms the acceptance-test penalty $\frac{\alpha}{\alpha-1}D(\mathbf{q}\|\mathbf{\Phi}[\omega])$, where $\mathbf{q}$ is a frequency distribution from the acceptance set and $\mathbf{\Phi}[\omega]$ the distribution of the test statistic under ω; the penalty grows without bound as $\mathbf{q}$ demands an outcome that ω makes improbable. The hypothesis $2^N \le \epsilon\, 2^{53}$ is what the binary64 number model allows (Remark 11.2). Part (iii) is arithmetic on the formula $\ell_{\max}$ of Definition 10.13, which Theorem 10.7 reads as a key length on $(1,2)$ under its own assumptions.

Proposition 11.1 (Instability of the rate under measurement perturbations)**.** Let $\alpha \in (1,\infty)$ and $B > 0$, write $N := \lceil B \rceil$, and let $\epsilon \in (0, \frac{1}{2}]$ satisfy $2^N \le \epsilon\, 2^{53}$. Write $\mathfrak{D}^{(i)}$ for the declaration obtained from a given one by replacing its measurement $\{F_v\}$ on Q by $\{F_v^{(i)}\}$, and $\underline{h}_\alpha[\,\cdot\,]$ for the value (5.4) of a declaration.

(i) There is a declaration $\mathfrak{D}$ satisfying Definition 1.2 in full, with $\{F_v\}$ itself exactly admissible, and there are two exactly admissible families $\{F_v^{(1)}\}$, $\{F_v^{(2)}\}$ with $\max_v \|F_v^{(i)} - F_v\| \le \epsilon$ for $i = 1, 2$, such that

$$\underline{h}_\alpha\big[\mathfrak{D}^{(1)}\big] - \underline{h}_\alpha\big[\mathfrak{D}^{(2)}\big] \;\ge\; B. \tag{11.1}$$

The dimensions and alphabets of $\mathfrak{D}$ are $d_A = d_Q = 1$, one sector, $|\mathcal{X}| = 1$, $|\mathcal{V}| = |\widehat{\mathcal{C}}| = 2$ and $d_Z = 1$, the same for every B and every ϵ.

(ii) Let err be a function of the declared dimensions, alphabet sizes and α whose value on the data of (i) is less than $\frac{\alpha}{\alpha-1}N$. Then the bound $\big|\underline{h}_\alpha[\mathfrak{D}^{(1)}] - \underline{h}_\alpha[\mathfrak{D}^{(2)}]\big| \le \mathrm{err}$, with err evaluated on the data of the declaration, fails for some declaration and some pair of exactly admissible families within ϵ of each other.

(iii) For $\alpha \in (1,2)$ and every number n of rounds, the maximal admissible key lengths $\ell_{\max}$ of $\mathfrak{D}^{(1)}$ and $\mathfrak{D}^{(2)}$ (Definition 10.13), computed with the same λ_{EC}, $\varepsilon_{\mathrm{EV}}$ and $\varepsilon_{\mathrm{PA}}$, differ by at least $nB - 1$ bits whenever the smaller of the two is positive.

Uses Def. 1.1, Def. 1.2, Def. 5.1, Prop. 5.2, Prop. 5.8, Thm. 10.7, Def. 10.13, foundations/def:kl · *Used by* Rem. 11.4, Req. 14.1, Rem. 22.4 · newly derived · accepted · *Oracles* registry.py

Proof. Idea. The instance has one-dimensional systems, so a single state; the measurement on Q has two outcomes, 0 with probability g_0 and 1 with probability g_1; the acceptance test demands outcome 0 in every round; and no round contributes key, so no acceptance value is key-live. Then $\underline{h}_\alpha$ is the acceptance-test penalty alone, $\frac{\alpha}{\alpha-1}\log_2\frac{1}{g_0}$. Dividing g_0, the largest power of two at most ϵ, by 2^N raises $\underline{h}_\alpha$ by $\frac{\alpha}{\alpha-1}N$ and moves each entry of the family by at most ϵ, so the declared family and its perturbation lie within ϵ of each other.

Step 1 (the instance). Take the instance of Proposition 5.8 (iii): $d_A = d_Q = 1$ (so both source modes coincide, Proposition 5.2, and $\mathcal{W} = \{[1]\}$), one sector, $\mathcal{X} = \{0\}$ with $M_0 = [1]$, $\mathcal{V} = \{0,1\}$, a_A constant and $a_B = \mathrm{id}$, $\phi(\kappa) = c_B$ so that $\widehat{\mathcal{C}} = \mathcal{V}$, all-fail sifting ($\varsigma \equiv$ fail, $r \equiv \perp$, $\mathsf{h} = \mathsf{B}$ with $r_{\mathsf{B}} \equiv \perp$, hence $\widehat{\mathcal{C}}_{\mathrm{key}} = \varnothing$), $\mathfrak{L} = \varnothing$, the default party partition, and the singleton acceptance set $\mathcal{S}_{\mathrm{acc}} = \{(1,0)\}$. For scalars $g_0, g_1 \geq 0$ write $\mathfrak{D}_{(g_0,g_1)}$ for the declaration whose measurement on Q is $F_0 = [g_0]$, $F_1 = [g_1]$. Every clause of Definition 1.2 other than the $\{F_v\}$-clause of (iii) holds for it: all systems and alphabets are finite, sifting is announcement-determined, the key map is party-computable with $r_{\mathsf{B}} \equiv \perp$, $\mathcal{W} = \{[1]\} \neq \varnothing$, $\mathcal{S}_{\mathrm{acc}}$ is closed and convex, and $\dim \mathcal{H}(\widetilde{E}) \geq 1$. That clause holds exactly when $g_0 + g_1 = 1$, and the family is exactly admissible in the sense of Definition 1.1 exactly when in addition both entries are binary64 numbers and the one addition of (1.3) is exact. Since $\widehat{\mathcal{C}}_{\mathrm{key}} = \varnothing$, the objective (5.3) keeps only its first sum, the acceptance-test penalty $\frac{\alpha}{\alpha-1}\sum_{\hat{c}} q(\hat{c}) \log_2 \frac{q(\hat{c})}{P_{\hat{c}}(\omega)}$, and $\underline{h}_\alpha$ of (5.4) is its infimum over $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}} = \{(1,0)\}$ and $\omega \in \mathcal{W}$. $\mathcal{W}$ is the single state $[1]$, at which $P_{\hat{c}_0} = g_0$ and $P_{\hat{c}_1} = g_1$; with the convention $0 \log_2 \frac{0}{v} = 0$ of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl) this gives

$$\underline{h}_\alpha\big[\mathfrak{D}_{(g_0,g_1)}\big] = \frac{\alpha}{\alpha-1}\Big(1{\cdot}\log_2\frac{1}{g_0} + 0{\cdot}\log_2\frac{0}{g_1}\Big) = \frac{\alpha}{\alpha-1}\log_2\frac{1}{g_0} \qquad (g_0 > 0,\ g_1 \geq 0), \tag{11.2}$$

a computation that uses no property of (g_0, g_1) beyond those two inequalities, in particular not $g_0 + g_1 = 1$.

Step 2 (part (i)). Let ϵ' be the largest power of two with $\epsilon' \leq \epsilon$, so $\epsilon' = 2^{-k}$ for an integer $k \geq 1$. The hypothesis says $2^{N-53} \leq \epsilon$, and 2^{N-53} is a power of two, so $2^{N-53} \leq \epsilon'$ as well. Take $\mathfrak{D} := \mathfrak{D}_{(\epsilon',\,1-\epsilon')}$, $\{F_v^{(2)}\} := \{F_v\}$ (distance 0) and $\{F_v^{(1)}\}$ the POVM of $\mathfrak{D}_{(\epsilon' 2^{-N},\,1-\epsilon' 2^{-N})}$, whose distance to $\{F_v\}$ is $\epsilon'(1-2^{-N}) \leq \epsilon' \leq \epsilon$ in each of its two entries. All three families have non-negative entries summing to 1; each has $|\mathcal{V}| = 2$, so (1.3) performs one addition. By $2^{N-53} \leq \epsilon'$, $\epsilon' 2^{-N} = 2^{-k-N}$ is at least 2^{-53}, so $\epsilon' 2^{-N}$ and ϵ' are powers of two in $[2^{-53}, \frac{1}{2}]$, hence binary64 numbers and multiples of 2^{-53}; $\epsilon = \frac{1}{2}$ with $N \leq 52$ is one such choice. Then every entry is exactly representable ($1-\epsilon'$ and $1-\epsilon' 2^{-N}$ are multiples of 2^{-53} in $[\frac{1}{2}, 1)$, where consecutive binary64 numbers are 2^{-53} apart), and the single addi-

tion returns the representable 1 and is therefore exact, which is the clause of Definition 1.1 that no addition of (1.3) rounds. So all three families are exactly admissible and $\mathfrak{D}$ satisfies Definition 1.2 in full. By (11.2),

$$\underline{h}_\alpha\big[\mathfrak{D}^{(1)}\big]-\underline{h}_\alpha\big[\mathfrak{D}^{(2)}\big]=\frac{\alpha}{\alpha-1}\Big(\log_2\frac{1}{\epsilon' 2^{-N}}-\log_2\frac{1}{\epsilon'}\Big)=\frac{\alpha}{\alpha-1}\,N\;\geq\;N\;\geq\;B,$$

using $\frac{\alpha}{\alpha-1}>1$ for $\alpha>1$.

Step 3 (part (ii)). Let err be as in (ii). The declaration $\mathfrak{D}$ of Step 2 has the dimensions and alphabets listed in (i), which are the same for every B and ϵ, so err takes one value on its data, and that value is less than $\frac{\alpha}{\alpha-1}\,N$. The families $\{F_v^{(1)}\}$ and $\{F_v^{(2)}\}=\{F_v\}$ of Step 2 are exactly admissible and lie within ϵ of each other, and their values differ by $\frac{\alpha}{\alpha-1}\,N$, which exceeds the prescribed B for every B the representability condition admits and exceeds that value of err. So the bound fails for this pair. Since the value of err does not depend on B, the argument may be run at the largest N the representability condition admits.

Step 4 (part (iii)). Let $\alpha\in(1,2)$. By Definition 10.13, for n rounds

$$\ell_{\max}=\max\Big\{0,\Big\lfloor n\underline{h}_\alpha-\lambda_{\mathrm{EC}}-\Big\lceil\log_2\frac{1}{\varepsilon_{\mathrm{EV}}}\Big\rceil-\frac{\alpha}{\alpha-1}\log_2\frac{1}{\varepsilon_{\mathrm{PA}}}+1\Big\rfloor\Big\};$$

the terms after $n\underline{h}_\alpha$ do not depend on $\underline{h}_\alpha$ and are the same for both declarations. Insert the gap (11.1): if the smaller of the two lengths is positive, the two lengths differ by at least $nB-1$ bits. □

The condition $N\leq 53+\log_2\epsilon$ of Proposition 11.1 is what exact admissibility forces on a two-outcome binary64 family. It caps N at 52, attained at $\epsilon=1/2$, and belongs to the number model, not to the map from the measurement to the rate.

Remark 11.2 (The representability ceiling). Proposition 11.1 needs a declared family (g_0,g_1) with $g_0+g_1=1$ and both entries binary64: exact admissibility is an identity of the declared numbers (Definition 1.1, the exact-admissibility paragraph), not of the reals they approximate, and it additionally requires the one addition of (1.3) to be exact. The smaller entry g_0 satisfies $g_0\leq\frac{1}{2}$, so $g_1=1-g_0$ lies in the binade $[\frac{1}{2},1)$, the binary64 numbers with exponent -1, where consecutive numbers are 2^{-53} apart (the unit in the last place); hence g_1 is representable only if g_0 is a multiple of 2^{-53}, and $g_0>0$ makes that $g_0\geq 2^{-53}$. With $g_0=\epsilon 2^{-N}$ this reads $\epsilon 2^{-N}\geq 2^{-53}$, that is $2^N\leq\epsilon\,2^{53}$, the inequality the proposition assumes of N and ϵ; since $\epsilon\leq\frac{1}{2}$, it gives $N\leq 52$. The value is attained: $\epsilon=\frac{1}{2}$ and $N=52$ give $(2^{-53},1-2^{-53})$, with the addition exact. The proof builds its families at the largest power of two $\epsilon'\leq\epsilon$, with $g_0=\epsilon' 2^{-N}$, and loses no N by it: 2^{N-53} is itself a power of two, so the inequality holds at ϵ' exactly when it holds at ϵ.

The ceiling is a property of the number model at these fixed dimensions, not of the map $\{F_v\}\mapsto\underline{h}_\alpha$: in exact real arithmetic the same construction runs at every $B>0$, and it is the number model that stops it. What the number model does is cap how much of the gap a family of finite value, $g_0>0$, can express, at $\frac{\alpha}{\alpha-1}N$ with $2^N\leq\epsilon\,2^{53}$; the exactly admissible family $(0,1)$ has the value $\underline{h}_\alpha=+\infty$ (Proposition 5.8 (iv)) and carries no key-length claim (Requirement 16.1), so it is

set aside. At $\epsilon = 2^{-40}$ the instance still delivers $N = 13$, thirteen bits per round or more, and it runs down to $\epsilon = 2^{-52}$, where $N = 1$ still gives $\frac{\alpha}{\alpha-1} > 1$ bit per round. At the instance's $d_Z = 1$ both exceed $\log_2 |\mathcal{Z}_\perp| = 1$ bit per round, the entropy-part ceiling of Proposition 5.8 (ii): more than the entropy terms of $\underline{h}_\alpha$ can contribute at any declaration with $d_Z = 1$, the rest of $\underline{h}_\alpha$ being the acceptance-test penalty. The comparison is the instance's: at a larger d_Z the entropy-part ceiling $\log_2(d_Z + 1)$ can exceed a gap of that size (at $\alpha = 1.9$, $d_Z = 7$ and $N = 1$ it is 3, against $\frac{\alpha}{\alpha-1}N \approx 2.1$). The representability ceiling shapes part (ii) of the proposition, which rules out allowances below the exhibited scale rather than every allowance. It shapes as well Remark 6.10, which cites part (i) to explain why a tolerance may refuse the deletion of a sector but never license one, and which states the gap there as $\frac{\alpha}{\alpha-1}N$ with N up to 52. The conclusion that matters is what the bounded statement still delivers: on the data of part (i), part (ii) with $N = 1$ refutes, at every $\epsilon \in [2^{-52}, \frac{1}{2}]$, every err below $\frac{\alpha}{\alpha-1}$, so every err at or below that entropy-part ceiling of 1 bit per round. The cap moves the refuted range from every allowance to the allowances below $\frac{\alpha}{\alpha-1} \cdot 52$, still far above that ceiling, and no rule of this module turns on the difference.

Uses Def. 1.1, Prop. 5.8 · newly derived · accepted

The gap of Proposition 11.1 involves no defect, lives in the acceptance-test penalty and can point either way. Among families of finite value the instance leaves an allowance of $53{\cdot}\alpha/(\alpha-1)$ bits per round or more unrefuted; at a key rate below one bit no key-length claim can rest on it.

Proposition 11.1 perturbs only the measurement $\{F_v\}$ on Q, not τ_A, $\{M_x\}$ or the constraint data, and it compares values of $\underline{h}_\alpha$, the number Theorem 10.7 turns into a key length and Theorem 5.4 places below $h_\alpha^{\uparrow,\mathrm{QKD}}$, not values of $h_\alpha^{\uparrow,\mathrm{QKD}}$ itself. Its pair stops at $\frac{\alpha}{\alpha-1} \cdot 52$, while the figure 53 in the remark is the largest value of $\log_2 \frac{1}{g_0}$ at an exactly admissible family of the instance with $g_0 > 0$, since such a family has $g_0 \geq 2^{-53}$ (Remark 11.2); between the two scales the proposition decides nothing. Its part (iii) applies when the smaller of the two key lengths is positive, which depends on n and on the other terms of $\ell_{\max}$; the proposition does not say for which n that happens on the instance.

Remark 11.3 (Scope of the perturbation gap). Proposition 11.1 does not say that a small perturbation always matters: it says that no bound in the declared metadata alone can say that it does not. It does not locate the fragility in any defect (a failure of Hermiticity, positivity or completeness in the declared numbers): both families are exact POVMs, so the phenomenon belongs to the map $\{F_v\} \mapsto \underline{h}_\alpha$ and would be there for a producer that emitted perfect numbers for a slightly different device. It does not depend on the entropy term: the gap lives entirely in the acceptance-test penalty $\frac{\alpha}{\alpha-1} D(\mathbf{q} \| \mathbf{\Phi}[\omega])$ of (5.3), with $\mathbf{q}$ ranging over the acceptance set, a term every declaration carries, and it is the same unboundedness that Proposition 5.8 (iii) records for $\underline{h}_\alpha$ itself, there as a range statement and here as a stability statement.

And it does not refute every allowance: an err whose value on the data of part (i) is at least $\frac{\alpha}{\alpha-1} \cdot 53$ is not refuted at any admissible ϵ by a pair of families of the instance of finite value, that is with $g_0 > 0$, and this module neither exhibits nor rules out one. The family $(0, 1)$ is exactly admissible as well and lies within

$2^{-53} \le \epsilon$ of $(2^{-53}, 1-2^{-53})$, but its value is $\underline{h}_\alpha = +\infty$ (Proposition 5.8 (iv)), which carries no key-length claim (Requirement 16.1); it is set aside here, since a gap to an infinite value is not a gap between two rates. On those data such an err would license a discrepancy of more than 53 bits per round ($\frac{\alpha}{\alpha-1} > 1$ at every $\alpha > 1$, and it diverges as $\alpha \downarrow 1$) against a key rate whose entropy part is at most $\log_2 |\mathcal{Z}_\perp| = 1$ bit per round there ($d_Z = 1$), by the entropy-part ceiling of Proposition 5.8 (ii) (the bound $\log_2 |\mathcal{Z}_\perp| = \log_2(d_Z + 1)$ on each entropy term, the rest of $\underline{h}_\alpha$ being the acceptance-test penalty). For the protocols this module rates, a key rate lies below one bit per round in practice, and against such a rate a robustness allowance of tens of bits per round is not a statement any key-length claim can rest on. At a larger d_Z the entropy-part ceiling grows, and the proposition constrains err on the data of part (i) only.

What the proposition does say is, first, that the direction is not controlled: the gap (11.1) has a sign, and exchanging the roles of the two families reverses it, so a perturbation of a declared family may move $\underline{h}_\alpha$ up (for a key-length claim, the unsafe direction) as easily as down. Second, that no threshold escapes the difficulty by being small: ϵ may be taken as small as the number model permits, the gap being bought by the second parameter of the instance, N, rather than by the size of the perturbation. Third, for $\alpha \in (1, 2)$, that the gap reaches the output: by part (iii) the key lengths themselves differ, by at least $nB - 1$ bits whenever the smaller is positive, so the gap is not confined to an unused intermediate quantity. At $\alpha \ge 2$ the same computation compares two values of the expression of Definition 10.13, to which this module gives no key-length reading (Remark 10.16).

Uses Def. 5.1, Prop. 5.8 · *Used by* Rem. 11.4, Rem. 22.4 · newly derived · accepted · *Oracles* registry.py

A tiny change of the declared measurement can move the key rate by many bits per round, so the model is a premise of the rate. Where a handed reference's companion counts and the key length is formed on it, the snap only steers the search; nothing bounds model to device.

A key rate of this contract is computed from a declaration whose measurement on Q is a model of a device, and Proposition 11.1 shows how much the number rests on that model. A domination weight declared for the device, outright or derived from ranges of its efficiencies and dark-count probabilities, would carry part of the gap between model and device into the flags of a domination link; it is future scope (`spec/future-scope.md` §13).

Remark 11.4 (The declared measurement as a premise of the rate). The value $\underline{h}_\alpha$ is not controlled by a tolerance on the declared measurement $\{F_v\}$ on Q (Definition 1.1) at any tolerance down to 2^{-52}. For every $\epsilon \in (0, \frac{1}{2}]$ and every positive integer N with $2^N \le \epsilon 2^{53}$ there is a declaration on which two exactly admissible families within ϵ of each other in every element give values $\frac{\alpha}{\alpha-1}N$ bits per round apart, and no allowance computed from the declared dimensions, alphabet sizes and α below that scale bounds the change (Proposition 11.1 (i)–(ii)); below $\epsilon = 2^{-52}$ the proposition exhibits no such pair. For $\alpha \in (1, 2)$ the gap reaches the key length of Theorem 10.7: over n rounds the two lengths differ by at least $nN - 1$ bits whenever the smaller is positive (Proposition 11.1 (iii)). The gap is bought by a ratio, not by a size: on the instance of that proposition $\underline{h}_\alpha = \frac{\alpha}{\alpha-1}\log_2 \frac{1}{g_0}$, where g_0 is the probability of the one outcome the acceptance test demands, and moving

g_0 from ϵ', the largest power of two at most ϵ, to $\epsilon' 2^{-N}$ changes the element by at most ϵ and the probability by the factor 2^N. Its limit is a change of support: at $g_0 = 0$ the measurement treats the demanded outcome as impossible, no feasible state gives it positive probability, and $\underline{h}_\alpha = +\infty$ (Proposition 5.8 (iv)), a value that carries no key-length claim (Requirement 16.1). A change of which outcomes the measurement treats as impossible, or nearly so, can thus move the rate by far more than the size of the change, in either direction; it need not, since the proposition exhibits one declaration on which it does and states no property of every declaration (Remark 11.3). The declared measurement model is therefore itself a premise of every number computed from it.

Between the device and the declared measurement lie two moves. The first is the snap of a family read as a measurement onto its lattice, whose displacement, however small, bounds the change of the rate by nothing. Where the declarer holds a reference of the family and hands it beside the snap, a claim establishes for the instance that the construction may run on it (Requirement "Emitting the reference companion beside the artefact" (reductions/req:reference-companion)), the companion is compiled and admitted (Requirement 14.3), and the run's bound on the companion counts (Requirement 24.7 (iii)) and its key length is formed from a number on the companion, not from one certified on the snapped declaration in the fallback of Requirement 28.1 (i), the run need not bound that change at all. Its first-order bound is then evaluated on the compiled reference companion (Proposition 12.1), at the points the search on the snapped declaration produced and at those of a short search continued on the companion from them (Corollary 19.7, Requirement 24.7); the snap only steers the choice of those points, and a bound that holds at any admissible point holds at them. Every block on which the companion carries the reference is evaluated off its snap (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap)), nothing the key length rests on reads its snapped data, and it rests on no premise about the snap. In that fallback the number the key length is formed from reads the snapped data of those blocks, which are then among the blocks the run does not evaluate off its snap, below (Requirement 24.7 (v)). Outside that fallback, the key length is one of the model the reference describes, and what lies between it and the device is the second move below. The evaluation of an enclosed reference at its centres is itself a displacement of the measured family, of the size of the enclosure's radius, to which Proposition 11.1 applies as it does to the snap's: the premise about the snap is gone, the exposure to the radius is not, and the recorded largest radius is the only handle on it (Remark 22.4). The rest of this paragraph concerns the blocks a run does not evaluate off its snap. A block kept because its reference is exact and already on the lattice rests on nothing, the snap being the identity there. Every other block is a moved block of the snap (Definition "Snapping declared data onto a lattice" (foundations/def:snap)), and no claim relates it to the reference: domination, which would bound the snapped family by the reference, is not established for a snap, and an idealised model may dominate no lattice family at a small weight at all (Remark "What domination controls, and why it needs a noise floor" (foundations/rem:noise-floor)). On such a block the run proceeds, unless even the last resort yields no snap, resting on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement): that on the blocks its record names the device measures the snapped family, or, where the snapped family is a producer's

emission, that what the run concludes from the emission holds of the snapped family. No claim then relates its key length to the reference, and a run that declares the strict option refuses instead (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv)). The second move is the gap between the model as declared, or its reference, and the physical device. No claim bounds it: the run assumes, without checking, that the model describes the device, for a receiver that the receiver module models through Assumption "Trusted and exactly characterised receiver" (receiver/ass:trust), and by Proposition 11.1 a small distance between model and device bounds the change of the rate no more than a small distance between two declared families does.

Reading: The key rate is computed from the measurement model you declare for your receiver, and that model is a premise of the number: nothing checks that your device measures what it says (for a receiver the engine models, A-TRUST). A change of the model too small to see in its entries, notably one that makes an outcome nearly or exactly impossible, can move the rate by many bits per round in either direction; no result bounds that gap. The engine also moves your model onto its exact number grid (the snap) for its search. Where your model reaches the engine as a reference, the bound is evaluated on your model itself and needs no premise about the snap, unless the rate is certified only on the moved model, as the report then says; any other block rests on A-SNAP, or the run refuses if it declares the strict option. Validate the model, above all which outcomes it treats as impossible.

Uses Def. 1.1, Prop. 5.8, Prop. 11.1, Rem. 11.3, Prop. 12.1, Req. 16.1, foundations/def:evaluated-off-snap, foundations/def:snap, foundations/rem:noise-floor, foundations/req:snap-reference · newly derived · accepted

12 The reference companion

Where the receiver segment of one or more measured factors carries the reference companion, the same reduction run on the receiver's reference in place of its snap, the compiler forms beside the declaration a second tuple from the same specification, with every such factor replaced. It keeps every component of the declaration but the measured families and the entries of those segments, and so its joint sectors and the points of its program; it is admissible save possibly for a non-empty feasible set, and its entries, rounded with their constants relaxed, only enlarge that set (Proposition 12.1). Where the reference carries factor enclosures and the reduction combines blocks with non-negative weights, the companion's blocks are members of the factor enclosures the reduction and the compiler carry to them (Corollary 12.2). The Frank-Wolfe submodule evaluates the first-order bound on it, at points of the snapped search and of a short search continued on it, and certifies it.

Imports foundations/def:factor-enclosure, protocol-compiler/def:compile, protocol-compiler/def:weighted, protocol-compiler/lem:weighted, protocol-compiler/lem:weighted-factor, reductions/cor:companion-factor-enclosure, reductions/cor:reference-companion, reductions/def:reduction-instance, reductions/lem:row-relaxation

Compiled with reference companions in place of the snapped targets of one or more measured factors, the declaration keeps every other component and is admissible save possibly for a non-empty feasible set. Its relaxed entries only enlarge that set.

A run searches on the declaration $\mathfrak{D}$ compiled from the snapped measurement. Where the receiver segment of a measured factor also carries the reference companion, the compiler forms beside $\mathfrak{D}$ a second tuple from the same specification with the companion in place of the snapped target, for every measured factor whose receiver segment carries one, both stations of an entanglement-based declaration among them (Requirement "The reference companion compiled beside the declaration" (protocol-compiler/req:reference-companion-compiled)). The proposition says that this tuple is a declaration of the kind this contract's results are stated for, so that a lower bound on its value is a lower bound of the ordinary kind, and that it shares with $\mathfrak{D}$ everything a point of the program is made of. It is stated for all the replaced factors at once: the tuple with one factor replaced is not shown to have a non-empty feasible set, so the case of one factor cannot be applied again to it. Its entries are the companions', each of which carries its own certificate at the reference; an entry requested of $\mathfrak{D}$ whose certificate is not established at the reference is left out, which only enlarges the feasible set, so that a bound on it stays a bound on the reference model, if a lower one. Which of its conditions the engine decides, and which it imports, is Requirement 14.3.

Proposition 12.1 (The compiled reference companion is an admissible declaration)**.** Let $\mathfrak{D}$ be an admissible declaration (Definition 1.2) that the compilation map (Definition "The compilation map" (protocol-compiler/def:compile)) emits from a specification, and let $\mathbb{F}^{\mathrm{ref}}$ be a non-empty set of its measured factors (the one measured system in the basic format) such that, for each $f \in \mathbb{F}^{\mathrm{ref}}$, the last link of the receiver segment of f is the emission of a reduction instance $\mathcal{J}_f$ satisfying (H-REF) of Corollary "The reference companion of a reduction instance" (reductions/cor:reference-companion). Let $\mathfrak{D}^{\mathrm{ref}}$ be the tuple the compilation map gives from the same specification with, for every $f \in \mathbb{F}^{\mathrm{ref}}$, that link's target family replaced by the target $\{F_v[A_f^{\mathrm{ref}}]\}$ of the reference companion of $\mathcal{J}_f$ of that corollary, A_f^{ref} its datum, and that link's entries replaced by the relaxed entries (Lemma "A rounded constraint entry with its constant relaxed" (reductions/lem:row-relaxation)) of the companion's entries, each rounded operator Hermitian and with the sector blocks of the entry it rounds. The companion's entries of such a link are the instance's own entries at the companion's datum and the companion forms (Corollary "The reference companion of a reduction instance" (reductions/cor:reference-companion) (vi)) of those entries requested of the link whose companion forms carry their certificates at that datum; a requested entry whose companion form is not established to carry one has no entry in $\mathfrak{D}^{\mathrm{ref}}$. Then:

(i) *(what is shared)* $\mathfrak{D}^{\mathrm{ref}}$ has the systems, sectors and dimensions, the

alphabets, Alice's marginal slot and measurement, the party partition, the announcement and sifting maps, the raw-key map and the index map of $\mathfrak{D}$, and in the general format its factor list and Alice's block lists; it differs from $\mathfrak{D}$ at most in the measured families of the factors of $\mathbb{F}^{\mathrm{ref}}$ and in the entries of their last links. In particular the two have the same joint sectors (Definition 4.8) with the same block dimensions and the same coupling condition;

(ii) *(admissible)* $\mathfrak{D}^{\mathrm{ref}}$ satisfies conditions (i)–(vi) and (viii) of Definition 1.2;

(iii) *(the relaxed entries enlarge the feasible set)* let $\mathfrak{D}^{\mathrm{ref}}_{\mathrm{ex}}$ be formed as $\mathfrak{D}^{\mathrm{ref}}$ but with the companions' entries themselves in place of their relaxed entries; then the feasible set of $\mathfrak{D}^{\mathrm{ref}}_{\mathrm{ex}}$ (Definition 5.1) is contained in that of $\mathfrak{D}^{\mathrm{ref}}$, and for every acceptance set $\underline{h}_\alpha[\mathfrak{D}^{\mathrm{ref}}] \le \underline{h}_\alpha[\mathfrak{D}^{\mathrm{ref}}_{\mathrm{ex}}]$.

Uses Def. 1.1, Def. 1.2, Def. 2.2, Def. 4.8, Def. 5.1, protocol-compiler/def:compile, protocol-compiler/def:weighted, protocol-compiler/lem:weighted, reductions/cor:reference-companion, reductions/lem:row-relaxation · *Used by* Rem. 11.4, Cor. 12.2, Req. 14.3, Cor. 19.7, Lem. 21.2, Cor. 21.4, Req. 24.7 · newly derived · accepted

Proof. (i) The compilation map passes through the declared data it reads and constructs only the raw-key map, the index map and the weighted families (Definition "The compilation map" (protocol-compiler/def:compile)), each measured factor's family from the last link of that factor's receiver segment, weighted, where it is, by that factor's own weighting and applied to nothing else. The raw-key map is formed from the public maps and the alphabets; the index map from the alphabets, the sector set and the declared orders that arrive with them; the weighted families from the delivered families and the declared weights P. By Corollary "The reference companion of a reduction instance" (reductions/cor:reference-companion) (i), for each $f \in \mathbb{F}^{\mathrm{ref}}$ the target of the reference companion of $\mathcal{J}_f$ has the alphabet, with its order, and the sector map of the target of f's last link, so every constructed object other than the weighted families of the factors of $\mathbb{F}^{\mathrm{ref}}$ is formed from the same data as for $\mathfrak{D}$ and is the same, and so is the family of every other measured factor. The constraint list is the concatenation of the specification's entries, the source segment's and the receiver segments', each with its datum and tag; only the entries of the replaced links differ. The joint sectors are the pairs of an Alice sector and a sector of the measured system, with the dimensions of both, and the coupling condition reads Alice's slot and these dimensions; all of them are shared.

(ii) Conditions (i), (ii), (iv), (v) and (viii) read only components that (i) shows to be those of $\mathfrak{D}$, or, for (i), the shape of the measured families, finite families of finite matrices over the shared sectors; $\mathfrak{D}$ satisfies them. Condition (iii): $\{M_x\}$ is $\mathfrak{D}$'s. The family of each $f \in \mathbb{F}^{\mathrm{ref}}$ is the target of the reference companion of $\mathcal{J}_f$, a POVM block diagonal for

the sector map by Corollary "The reference companion of a reduction instance" (reductions/cor:reference-companion) (i), or, where that factor's family is weighted after its chain by that factor's weighting P_f (Definition "The compilation map" (protocol-compiler/def:compile)), the family $P_f(\varphi \,|\, t(v))\, F_v$ over $(v, \varphi) \in \mathcal{V}_{P_f}$, a POVM by Lemma "Properties of the choice-weighted families" (protocol-compiler/lem:weighted) (i), each element a non-negative multiple of a block-diagonal operator and so block diagonal. In the basic format that family is Bob's. In the general format every factor's family outside $\mathbb{F}^{\mathrm{ref}}$ is $\mathfrak{D}$'s, and the product family (1.5) is then a POVM on Q, a tensor product of positive semidefinite operators being positive semidefinite and $\sum_v \bigotimes_{f'} F^{f'}_{v_{f'}} = \bigotimes_{f'} \sum_{v_{f'}} F^{f'}_{v_{f'}} = \mathbb{1}_Q$. Every entry operator is Hermitian and block diagonal for its scope: the shared entries as in $\mathfrak{D}$, and every relaxed entry by hypothesis. Condition (vi): a relaxed entry has a relation in $\{\preceq, \succeq\}$, or is the entry itself where its bound is zero, and a real constant, and it carries the scope of the entry it relaxes and, where that entry is party-local, its party-local tag. The compilation map tags it as it tags every entry (Definition "The compilation map" (protocol-compiler/def:compile)): an entry without the party-local tag by the type of its datum, a real constant, which is the homogeneous form of Definition 2.2, and a party-local entry by the tag field it arrives with beside that constant.

(iii) The two tuples differ only in the entries of the replaced links, each exact entry $(\hat{O}_k, \bowtie_k, c_k)$ of $\mathfrak{D}^{\mathrm{ref}}_{\mathrm{ex}}$ standing against its relaxed entries in $\mathfrak{D}^{\mathrm{ref}}$. Let ω be feasible for $\mathfrak{D}^{\mathrm{ref}}_{\mathrm{ex}}$, with E_k and δ_k as in Lemma "A rounded constraint entry with its constant relaxed" (reductions/lem:row-relaxation), whose part (i) gives $-\delta_k \mathbb{1} \preceq E_k \preceq \delta_k \mathbb{1}$ and hence the same for $\mathbb{1} \otimes E_k$ on any larger system. In trusted-source mode the entry reads $\mathrm{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\omega] \bowtie_k c_k \tau_A$ with $\mathrm{tr}_Q\, \omega = \tau_A$, and part (iv) of that lemma with $X = A$ gives the relaxed entries at their constants. In untrusted-source mode it reads $\mathrm{tr}\left[(\mathbb{1}_A \otimes \hat{O}_k)\omega\right] \bowtie_k c_k$ with $\mathrm{tr}\,\omega = 1$, and $\mathrm{tr}[(\mathbb{1}_A \otimes E_k)\omega] \in [-\delta_k, \delta_k]$ gives them. A party-local entry reads (2.4), the case of part (iv) with X the other systems, and an untagged entry of a factor is read for its lifted operator $\mathbb{1} \otimes \hat{O}_k$, whose rounding error is $\mathbb{1} \otimes E_k$. An equality entry gives both relaxed entries. So ω satisfies every relaxed entry, of whichever replaced link, and is feasible for $\mathfrak{D}^{\mathrm{ref}}$, the other conditions of Definition 5.1 being the same for both. The objective and the acceptance set are the same for both tuples, so the infimum (5.4) over the larger set is at most that over the smaller. □

Where a handed reference carries factor enclosures and the reduction forms its datum as a non-negative combination of blocks, each block of the compiled companion's measured families read with a factor enclosure is a member of it, which the certified bracket over the companion assumes.

Lemma 21.2 reads some blocks of a compiled reference companion through factor enclosures and assumes, as its (H-FAC), that the companion's own block is a member of each. The factor enclosures are formed by the module

that constructs the reference, which establishes that its blocks are members of them, as it does for its entrywise enclosures (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (i)); the reduction and the compiler then carry them, each step keeping the block a member. This corollary composes those two steps for the compiled companion of Proposition 12.1, one replaced factor at a time, each replaced factor's family being formed from its own link alone.

Corollary 12.2 (Factor enclosures of the compiled reference companion)**.** Let $\mathfrak{D}$, $\mathbb{F}^{\mathrm{ref}}$, the instances $\mathcal{J}_f$ and $\mathfrak{D}^{\mathrm{ref}}$ be as in Proposition 12.1, and for $f \in \mathbb{F}^{\mathrm{ref}}$ let G_f be the family of Corollary "The reference companion of a reduction instance" (reductions/cor:reference-companion) from which the datum A_f^{ref} of the reference companion of $\mathcal{J}_f$ is formed. Let $f \in \mathbb{F}^{\mathrm{ref}}$, and read the block on a data sector s of $\mathcal{J}_f$'s target of an element of the measured family of factor f of $\mathfrak{D}^{\mathrm{ref}}$ formed from the target element $F_v[A_f^{\mathrm{ref}}]$ with the triple that part (i) of Corollary "Factor enclosures carried to the reference companion" (reductions/cor:companion-factor-enclosure) forms for the element $A_{f,v}^{\mathrm{ref},(s)}$ of A_f^{ref} on s, where the rule of (I1) of $\mathcal{J}_f$ is a non-negative combination (that corollary) and that part forms one from factor enclosures (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)) that blocks of G_f carry, its core multiplied by $P_f(\varphi \,|\, t(v))$ where the family is weighted after the chain by factor f's weighting P_f and the element is $P_f(\varphi \,|\, t(v))\, F_v[A_f^{\mathrm{ref}}]$ with $(v, \varphi) \in \mathcal{V}_{P_f}$ (Definition "Choice-weighted families" (protocol-compiler/def:weighted), with $P := P_f$); and read every other block of the measured families of $\mathfrak{D}^{\mathrm{ref}}$ with none. Then every block so read with a triple is a member of it, the triple being a factor enclosure of that block.

Uses Prop. 12.1, foundations/def:factor-enclosure, protocol-compiler/def:compile, protocol-compiler/def:weighted, protocol-compiler/lem:weighted, protocol-compiler/lem:weighted-factor, reductions/cor:companion-factor-enclosure, reductions/cor:reference-companion, reductions/def:reduction-instance · *Used by* Req. 14.3, Cor. 21.4 · newly derived · accepted

Proof. By the definition of $\mathfrak{D}^{\mathrm{ref}}$ in Proposition 12.1, the compilation map (Definition "The compilation map" (protocol-compiler/def:compile)) forms the measured family of each factor $f \in \mathbb{F}^{\mathrm{ref}}$ from the companion's target $\{F_v[A_f^{\mathrm{ref}}]\}$ as it forms that of $\mathfrak{D}$ from the snapped target of f's last link: the target itself, or, where the family is weighted after the chain, the family $\{P_f(\varphi \,|\, t(v))\, F_v[A_f^{\mathrm{ref}}]\}_{(v,\varphi)\in\mathcal{V}_{P_f}}$ of Definition "Choice-weighted families" (protocol-compiler/def:weighted), over the sector map of the target in either case; the families of other factors take no part in it. On a data sector s the block of $F_v[A_f^{\mathrm{ref}}]$ is $A_{f,v}^{\mathrm{ref},(s)}$, by the target formula of Definition "Reduction instance" (reductions/def:reduction-instance), and parts (i) and (ii) of Corollary "Factor enclosures carried to the reference companion" (reductions/cor:companion-factor-enclosure), applied to $\mathcal{J}_f$, G_f and A_f^{ref}, make the triple formed there a factor enclosure of it. In the

weighted case the block of $P_f(\varphi \mid t(v))\, F_v[A_f^{\mathrm{ref}}]$ on s is $P_f(\varphi \mid t(v))\, A_{f,v}^{\mathrm{ref},(s)}$ (Lemma "Properties of the choice-weighted families" (protocol-compiler/lem:weighted) (ii)), and Lemma "A factor enclosure through a setting-dependent weighting" (protocol-compiler/lem:weighted-factor) makes the same box with the core multiplied by the weight a factor enclosure of it. A factor enclosure of a matrix has that matrix among its members (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)), so each block read with a triple is a member of it. □

13 Assumptions

The assumptions fix what the post-processing must satisfy for the results of this contract to hold, and what a run must attest about its laboratories, its seeds, its fixed parameters and its authentication, and, for several executions whose parameters are to add, their separation. What the declaration and the acceptance set must satisfy is decided by the engine and is stated as definitions (Definition 1.2, Definition 2.5); the product marginal of Alice's systems in trusted-source mode is a lemma under the transmitter's premises and this module's A-SEED (v) and A-LAB at the party holding A (Lemma 2.9). No assumption restricts the adversary's memory: A-EVE-DIM, which would make it finite-dimensional, is assumed by no claim (Corollary 10.8). The attack-set licence and round homogeneity are the compiler's assumptions, cited where they are used. Each assumption states its conditions only, with a plain reading for the user.

Imports foundations/def:round-model

The post-processing acts on the raw-key string the rounds write, the declared holder computes that string, and reconciliation runs one way from the holder through a padded fixed-length transcript. The leakage term counts the register's full length.

The chain-rule penalty counts the dimension of the transcript register, not the average length of a transcript, which is why the register has a fixed length. Whether the raw-key map is one the holder can compute from its own readout is the party-computability condition of Definition 1.1, decided on the declared tables; what is assumed here is that the run's post-processing acts on that string, one way.

Assumption 13.1 [A-KEYREG] (Dedicated key register and one-way reconciliation)**.** (i) Error correction, error verification and privacy amplification are applied to Z_1^n, the string of raw-key values $r(x, v)$ that the rounds write to the registers $Z_1, \ldots, Z_n$ (Definition 10.1); (ii) the declared key holder h (Definition 1.1) computes Z_1^n from its own outcome record and the public announcements; (iii) reconciliation is one-way from h: its transcript L_{EC} is a function of Z_1^n and public seeds only; (iv) L_{EC} occupies a fixed-length register of λ_{EC} bits, shorter transcripts being padded, so that $\dim(L_{\mathrm{EC}}) = 2^{\lambda_{\mathrm{EC}}}$.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Error correction is one-way. The key holder computes its raw key from its own record (its letters, the detection results of its measuring stations, or both, whichever it holds) and the public announcements, and its messages depend only on that raw key and public random seeds. The transcript always has exactly λ_EC bits, shorter messages being padded. Interactive error correction (for example Cascade) or a variable-length transcript is not covered by this key length.

Uses Def. 1.1, Def. 10.1 · *Category* post processing · *Affects* secrecy · *Assumed by* Thm. 10.7 · *Machine key* `key_register` · newly derived · accepted

Privacy amplification uses an ideal two-universal family, collision probability exactly 2^−ℓ for distinct inputs, and error verification a two-universal family. The first is what leftover hashing reads, the second what the verification collision bound reads.

The Rényi leftover-hashing lemma [2, Thm. 4] is applied under the reading of [5, Lemma 6.5 and Rem. 6.1], which asks for an ideal family; the correctness argument uses the collision probability of the verification family only as an upper bound. The two notions are those of [5, Def. 3.5].

Assumption 13.2 [A-HASH] (Ideal two-universal privacy amplification)**.** (i) The privacy-amplification family of Definition 10.1, applied to Z_1^n, is ideal two-universal: for any two distinct inputs, the probability under the family's law that their ℓ-bit values coincide is exactly $2^{-\ell}$; (ii) the error-verification family of Definition 10.1 is two-universal: for any two distinct inputs, the probability under the family's law that their $\lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \rceil$-bit values coincide is at most $2^{-\lceil \log_2(1/\varepsilon_{\mathrm{EV}}) \rceil}$.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Privacy amplification uses a hash family in which any two different inputs collide with probability exactly 2^−ℓ (an ideal two-universal family, such as Toeplitz hashing), and error verification uses a two-universal hash family. If your privacy-amplification family is only two-universal (collision probability at most 2^−ℓ), shorten the key by one further bit.

Uses Def. 10.1 · *Category* post processing · *Affects* secrecy, correctness · *Assumed by* Lem. 10.4, Thm. 10.7 · *Machine key* `pa_hash_family` · newly derived · accepted

The post-processing seeds are drawn with their declared laws, privately, independently and afresh; in trusted-source mode the source's letters and prepared systems are independent of the adversary, who never gets a purification of them. Leftover hashing and source replacement rest on it.

The parties' local randomness is taken to be ideal and never leaked beyond what the protocol announces [5, Sec. 4, opening paragraphs], [6, Sec. 3]; a generator that is only close to ideal in the composable sense adds its own error parameter [6, Sec. 5.6.1]. The published premise covers the randomness used to choose the signal states as well as the hash seeds: the protocol begins with local randomness in classical registers and "the adversary is uncorrelated to this randomness" [5, Sec. 4, opening paragraphs], and in particular the adversary holds no purification of a prepared signal state [5,

Protocol 1, step 1(a)(i)]. Clause (v) states that part for the source's draws, in trusted-source mode, where the analysis describes the source by them.

Assumption 13.3 [A-SEED] (Private, independent and fresh local randomness)**.** (i) The error-correction seed S_{pub}, the error-verification seed S_{EV} and the privacy-amplification hash G of Definition 10.1 are drawn with the laws that definition gives them, S_{EV} and G with the laws of their hash families; (ii) each is drawn privately: its value is available to the adversary only once the protocol announces it; (iii) each is independent of every other register of the protocol and of the adversary's information at the time it is drawn; (iv) each is drawn afresh in every run; (v) in trusted-source mode, the letters Alice's source draws in the n rounds, and the systems it prepares from them, are jointly independent of the adversary's registers at the start of the protocol: the joint state of the letters, the prepared systems and those registers is the product of its marginal on the letters and the prepared systems together and its marginal on those registers; in particular the adversary holds no purification of a prepared system, and no purification of a prepared system is made available to the adversary during or after the protocol.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Your random choices come from a good private random number generator. The error-correction seed, the error-verification hash and the privacy-amplification hash are each drawn with their intended distribution, kept secret until the protocol announces them, independent of each other, of every other record of the protocol (the raw key included) and of everything the attacker knows, and drawn anew in every run. With a trusted source, the randomness with which it picks each round's letter and prepares its state is uncorrelated with the attacker at the start, and no purification of a prepared state is ever available to the attacker. A seed reused across runs, fixed in the device or disclosed early, or a predictable or pre-shared random source, violates this.

Uses Def. 10.1 · *Category* local randomness · *Affects* secrecy, correctness · *Assumed by* Lem. 2.9, Thm. 2.10, Lem. 10.4, Thm. 10.7 · *Machine key* `post_processing_seeds` · newly derived · accepted

The acceptance set, the round count, the key length, the transcript length, the error budgets, the authentication parameters and the Rényi order are fixed before the run and applied unchanged. The key length is that of a fixed-length protocol, not one chosen from the run's data.

A fixed-length protocol outputs a key of a length determined before the protocol begins, or aborts [6, Sec. 2.2]; choosing the length or the test from the observed data makes it a variable-length protocol, which needs a different analysis. The Rényi order α is a parameter of the proof, not of the protocol: the run never reads it, and it enters only the computation of ℓ. Clause (iii) fixes it before the run all the same, so that the order, and with it the length computed at it, is never a function of the run's data; an order chosen by comparing the lengths that the declared model gives at several orders satisfies it. No proof consumes clause (iii): the key-length theorem holds at each $\alpha \in (1, 2)$, and the clause is stated so that the attestation names every

input of ℓ.

Assumption 13.4 [A-FIXED] (Protocol parameters fixed before the run)**.** (i) The acceptance set $\mathcal{S}_{\text{acc}}$, the round count n, the output length ℓ, the transcript length λ_{EC}, the error budgets ε_{PA} and ε_{EV} of Definition 10.1, and the authentication parameters $\varepsilon_{\text{auth}}$ and k_{auth}, the declared authentication error bound and key consumption (Definition 10.2 (b)), are fixed before the first round of the run; (ii) the run applies them unchanged: none of them is chosen, adjusted or recomputed from data observed in the run; (iii) the Rényi order $\alpha \in (1, 2)$ at which ℓ is computed is fixed before the first round of the run, from data available before it, and is not chosen, adjusted or recomputed from data observed in the run.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Every parameter the key length rests on is fixed before the run starts: the acceptance test, the number of rounds, the key length, the error-correction budget λ_EC, the error budgets ε_PA and ε_EV, and the authentication parameters ε_auth and k_auth. The Rényi order α at which the key length is computed is fixed before the run too; choosing it by comparing the key lengths your declared model gives at several orders, as the run tier's outer search over α does, reads no data of the run and is such a choice. None of them may be chosen or adjusted after looking at the run's data, for example by relaxing the acceptance threshold or picking the key length from the observed error rate. That would be a variable-length protocol, which this analysis does not cover.

Uses Def. 10.1, Def. 10.2 · *Category* protocol structure · *Affects* secrecy, correctness · *Assumed by* Thm. 10.7, Cor. 10.8 · *Machine key* `parameters_fixed_before_run` · newly derived · accepted

The implementation applies the acceptance test and the verification abort, sends no message beyond the model's, outputs the model's key or nothing, uses no key before the execution ends and aborts on an authentication failure. It ties the analysed protocol to the one that runs.

That the devices execute the protocol as prescribed is one of the standard premises of QKD [3, Sec. IV.A]; the post-processing prescribed here is the fixed-length instance of [5, Sec. 4.1, Protocol 1].

Assumption 13.5 [A-CONFORM] (The implementation runs the analysed post-processing)**.** (i) The implementation evaluates the acceptance test of Definition 10.1 (i) and aborts unless Ω_{AT} holds; (ii) after the rounds, the only classical messages either party sends are those of Definition 10.1 and of the authentication post-processing (Definition 10.2 (c)); (iii) both parties abort when the verification flag is rej; (iv) each party outputs the ℓ-bit key that Definition 10.1 (iv) assigns to it, or $\perp$; (v) neither party uses any part of its key before the end of an execution (Definition 10.3); (vi) a party that receives an authentication failure acts as the authentication post-processing prescribes.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Your implementation runs exactly the analysed post-processing. It aborts unless the acceptance statistics lie in the declared acceptance set; after the rounds it sends no

public message other than the error-correction transcript of at most λ_EC bits and its seed, the verification seed, hash and result, the privacy-amplification hash choice and the closing accept or abort decisions; it aborts when verification fails or a message fails authentication (an auth-abort); it outputs the analysed ℓ-bit key, or nothing. No part of the key is used before the execution has ended: the key holder, having sent its final decision, waits until the partner has received it or an auth-abort in its place, and the partner until one of these has arrived. Interactive error correction, a resent syndrome or early key use violate this. Tightening $2\varepsilon_auth$ to ε_auth under this timing is an open question.

Uses Def. 10.1, Def. 10.2, Def. 10.3 · *Category* post processing · *Affects* secrecy, correctness · *Assumed by* Lem. 10.4, Lem. 10.6, Thm. 10.7, Cor. 10.8 · *Machine key* `post_processing_as_analysed` · newly derived · accepted

Every classical message travels through an authentication scheme using at most k_auth bits of pre-shared key and realising the authenticated classical channel within ε_auth, a reused key's distance from an ideal one included. It is the premise by which authentication enters the key's security.

A real scheme authenticates with a short pre-shared key and an unauthenticated channel, for instance by Wegman–Carter hashing; its composable security is the distance of [3, Sec. VII.A, Eq. (36)] from the authenticated channel, and an authentic classical channel is one of the standard premises of QKD [3, Sec. IV.A], [6, Sec. 3].

Assumption 13.6 [A-AUTH] (Authentication of the classical channel by a keyed scheme)**.** (i) All classical messages of the protocol, the round announcements included, are sent through one authentication scheme using a pre-shared key and an unauthenticated classical channel, or all over a classical channel that is itself the authenticated classical channel of Definition 10.2 (a), in which case $\varepsilon_{\text{auth}} = k_{\text{auth}} = 0$; (ii) in the first case, the scheme's key consumption is at most k_{auth} bits; (iii) run on two identical copies of a uniformly distributed key independent of every other register of the protocol and of the adversary's information, the scheme ε_0-realises the authenticated classical channel (Definition 10.2 (b)), for some $\varepsilon_0 \geq 0$; (iv) before the run starts, the joint state of the two parties' copies of the pre-shared key the run consumes with every other register of the protocol and with the adversary's information is within trace distance $\delta \geq 0$ of the state in which those copies are replaced by two identical copies of a uniformly distributed key independent of them; (v) $\varepsilon_0 + \delta \leq \varepsilon_{\text{auth}}$.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Every classical message of your protocol, per-round announcements included, is protected by one authentication scheme: the attacker may read, delay or block any message, but one she alters or injects, or delivers before it was sent, is rejected except within an error ε_auth. It uses at most k_auth bits of pre-shared secret key per run. That error is its composable error on two identical copies of a uniform, independent key plus the distance of the parties' actual copies from such a pair (zero for a fresh key; for a key from an earlier run, a distance you must bound, which that run's security parameter alone does not bound); declare that sum as ε_auth and the bits used as k_auth. No composable proof for this timing-aware channel is published yet; Wegman–Carter authentication with separate

keys per direction approximates it. Both declared zero assert a keyless authenticated channel.

Uses Def. 10.2 · *Category* channel and adversary · *Affects* secrecy, correctness · *Assumed by* Cor. 10.8 · *Machine key* `classical_channel_authenticated` · newly derived · accepted

Several executions, fixed in advance and run against one adversary, touch none of each other's registers, leave their keys unused until all have ended, and keep their own premises with the others counted as the adversary's. It is what lets their security parameters add.

Composable frameworks add the errors of protocols composed in sequence or in parallel [3, Sec. II.E]. For the trace-distance parameter of this contract, the corresponding sum needs the executions to be separate in the sense below, so that each can be analysed with every other one on the adversary's side.

Assumption 13.7 [A-SEPARATE] (Executions that share no key and no device state)**.** For executions $1, \ldots, N$ of the protocol, run against one adversary: (i) N and, for every execution, the parameters fixed before its run (its acceptance set, round count, output length, transcript length, error budgets and authentication parameters) are fixed before the first of the N executions starts; (ii) no operation of the honest parties of one execution acts on a register that the honest parties of another execution use, hold or output: no execution uses any part of another execution's key, local randomness, seeds or pre-shared key, no device, memory or random source carries a state from one execution into another, and the executions interact only through systems and classical channels the adversary has access to; (iii) from the end of an execution (Definition 10.3) until the end of the last of the N executions, no part of that execution's key is used, read or otherwise acted on, and no information about any other register of its parties leaves their laboratories; (iv) for every execution, every assumption on which its key and its security parameter rest holds when every register of the honest parties of every other execution is counted as part of the adversary's information, and every operation of those parties as part of the adversary's actions.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: You run several executions, one after another or at the same time, and want their security parameters to add. Their number and parameters are fixed before the first starts. Each execution has its own key material, randomness, seeds and pre-shared authentication key, such as a separate part of one pool. No device, memory or random source carries information between them. An execution has ended once both parties have fixed their output, key or abort, after the closing messages; no key is used or read, and nothing else leaks from an ended one, until all have ended. The assumptions each execution's run record lists must still hold when an attacker holds and runs every other execution, her memory, which then holds their systems, of any dimension. A key that authenticates another execution, a key used early, or parameters chosen from an earlier execution's data violate this.

Uses Def. 10.3 · *Category* protocol structure · *Affects* secrecy, correctness · *Assumed by* Cor. 10.9 · *Machine key* `separate_executions` · newly derived · accepted

Nothing about a party's records leaves its laboratory but its key, the public messages and, in trusted-source mode at the party of A, the emitted systems; the adversary acts on it only through what its declared measurements receive and the messages. It excludes side channels at both parties.

An information carrier of an implementation that the model omits is a side channel that no security proof accounts for, and sources and photon detectors are known places for one [3, Sec. I.B]; that the devices do exactly what they are instructed to do is one of the standard premises of QKD [3, Sec. IV.A]. The premise is stated for each party of the declaration, whatever that party does in the protocol: what crosses the boundary of its laboratory is read off the round model and the party partition, in both directions.

Assumption 13.8 [A-LAB] (Each party's laboratory is sealed)**.** Let $\mathsf{P} \in \{\mathsf{A}, \mathsf{B}\}$ be a party of the protocol declaration (Definition 1.1). Its *records* are its letters, where the party partition assigns Alice's system A to P; the private settings and the outcomes of the measurements made in its laboratory, the public choices it draws before they are announced, and its readout π_{P}; the inputs of its announcements; its raw or final key; and its local randomness. The *systems handed out of its laboratory* are, in trusted-source mode and where the party partition assigns A to P, the systems its source emits into the quantum channel, which the round model (Definition 2.3) gives to the adversary, A being no output of the attack in that mode (2.5) and staying with P; in every other case there are none. In particular, in untrusted-source mode, where the attack supplies A and Q (2.5), no laboratory hands out any system, and a source the declaration treats as untrusted lies outside every laboratory wherever it physically stands. The *systems delivered into its laboratory* are, in untrusted-source mode and where the party partition assigns A to P, the system A; and, in either mode and in each sector s, the tensor factors $Q_{s,i}$ of the declared factorisation of $\mathcal{H}(Q_s)$ that the party partition labels P (in the general format, every measured factor labelled P), physically the input modes those factors model.

(i) During the protocol, no information about P's records leaves its laboratory, other than its key as the protocol's output, except through the systems handed out of its laboratory and through the public messages of the protocol (the announcements of the round channels of Definition 2.3, the messages of Definition 10.1 and of the authentication post-processing (Definition 10.2 (c)), and, where an authentication scheme carries them, the scheme's own transmissions over the unauthenticated classical channel): no light emitted by its devices other than those systems, no electromagnetic, acoustic, power or timing emanation and no side channel of its classical processing carries such information, and neither the timing of its messages nor any auxiliary or synchronisation signal does;

(ii) the adversary acts on P's laboratory only through the systems deliv-

ered into its laboratory, as the declared measurements receive them, and through the public messages the protocol delivers to P and, where a scheme carries them, the transmissions it receives over the unauthenticated channel: no light or signal sent into the laboratory, through the systems handed out of it or otherwise, reads out any of its records, and nothing else influences the state, configuration or calibration of its devices (its source included, where it has one), its measurements or its classical processing.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Whatever a party's role, nothing about its letters, settings, choices, detection results, keys or random numbers leaves its laboratory but the final key, the public messages and, where the party sends with a trusted source, the signals that source emits; timing and auxiliary signals reveal no more. An untrusted source counts as outside every laboratory, even one it is in. The attacker reaches a laboratory only via the systems its declared measurements receive and the public messages. Light sent in to read out a letter, a setting, a choice or a detector state (a Trojan-horse attack, at a receiver as at a modulator), emanations from any device, detector back-flash and anything but the declared input modes influencing a measurement violate this, a clock or trigger line from an untrusted source included; where it sends with a trusted source, so does light seeding or recalibrating its laser.

Uses Def. 1.1, Def. 2.3, Def. 10.1, Def. 10.2 · *Category* laboratory · *Affects* secrecy · *Assumed by* Lem. 2.9, Thm. 2.10, Lem. 10.6, Thm. 10.7 · *Machine key* `laboratory_sealed` · newly derived · accepted

Every register of the adversary, her initial register and her memories, is finite-dimensional, with no bound on its dimension. No claim rests on it: the analysed models have such registers by definition, and Corollary 10.8 reaches an adversary whose memory has any dimension.

The source of the imported accumulation theorem states its convention as "we will assume that all systems are finite-dimensional, but we will not impose any bounds on the system dimensions unless otherwise specified" [1, Sec. 2, the convention stated below its Table 1]. Read for the adversary's registers, that convention would be a premise about her, and this assumption states it. The analysed models of these contracts keep it instead as part of their definitions, at every fixed finite dimension, and the security statement of a run reaches an adversary whose memory has any dimension by a limit over those dimensions; so no claim assumes it.

Assumption 13.9 [A-EVE-DIM] (The adversary's registers are finite-dimensional)**.** (i) Every register of the adversary in the round model (Definition "Round model, attack sequences and attack sets" (foundations/def:round-model)), her initial register E_0, her outgoing memories E'_j and her registers E_j for $j = 1, \ldots, n$, is finite-dimensional, of arbitrary finite dimension.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: No run rests on this assumption: the security statement of a run holds against an attacker whose quantum memory has any size, finite or infinite (Corollary 10.8). The

assumption states that the attacker's quantum memory is finite-dimensional, of any size she likes.

Uses foundations/def:round-model · *Category* channel and adversary · *Affects* secrecy · *Machine key* `adversary_registers_finite` · newly derived · superseded

Part II
Engine requirements

What an implementation of this module must do, record or refuse; each requirement names the result that motivates it.

14 Requirements: admission and the declared structure

The engine admits declared data only when they are exactly admissible, decides every condition of an admissible declaration but the non-empty feasible set itself, admits a reference companion only beside its declaration, deciding exactly what the two share and nothing of its family, and admits an acceptance set only when it decides that the set is admissible, and an implemented branch map drops only components, of registers other than the raw-key register, that are constant across its Kraus terms. It decides the sector classes by exact tests, flags a sector removable only when an exhibited witness meets the whole removal checklist, and reports every class and every deletion. It changes the declared sectors in no other way than the removal theorem and the composition rule allow, and deletes at most one sector where it does not check the composition hypothesis.

Imports foundations/def:exact-reference, protocol-compiler/req:reference-companion-compiled, reductions/cor:reference-companion, reductions/req:reference-companion

The engine admits a declaration only if it is exactly admissible and refuses every other, with no tolerance on declared data, save a reference companion, admitted on its own terms. Proposition 11.1 shows, for the measurement on Q, that closeness to admissible data could license no key-length claim.

Exact admissibility asks the conditions of Definition 1.1 of the declared binary64 numbers themselves, not of reals they approximate. For a declaration with one Alice sector, one measured factor and Alice's measurement in POVM form, this means that, block by block, each declared operator is exactly Hermitian, each measurement operator and the declared τ_A are exactly positive semidefinite, and each completeness sum, formed in the declared order (1.3), is exactly the identity with no addition rounding; the paragraph of that definition on exact admissibility in the general format says how the conditions read otherwise. How the rule meets an instance that moves its own data within its declared tolerance τ_{num} is Requirement "One declared tolerance per instance" (foundations/req:declared-tolerance), read with Requirement "Exact decision by every consumer" (foundations/req:consumer-decides-exactly).

Requirement 14.1 (Exact admission of declared data). The engine takes no tolerance on declared data and decides no condition on it more loosely

than exactly (Requirement "Exact decision by every consumer" (foundations/req:consumer-decides-exactly)): it admits a declaration only if the declaration is exactly admissible in the sense of Definition 1.1, and refuses it otherwise. No perturbation of a declared family is licensed to carry the key length of the family it approximates. Proposition 11.1 is the reason: it shows, for the measurement on Q, that no tolerance could license a key-length claim. A compiled reference companion is not admitted under this requirement but under Requirement 14.3, which decides exactly what it shares with the declaration and decides none of its family; the binary64 numbers it is evaluated at are no declared family whose key length is claimed, since the bound evaluated there is that of the reference, at the unbudgeted arithmetic of Remark 22.4.

Motivated by Def. 1.1, Prop. 11.1, Req. 14.3, Rem. 22.4, foundations/req:consumer-decides-exactly, foundations/req:declared-tolerance · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.general_validation: validate_general_declaration

The engine computes nothing from a declaration that fails a condition of admissibility other than the non-empty feasible set, each decided exactly on the declared data. The feasible set it leaves to the compiler where constraints are declared.

Seven of the eight conditions of admissibility are conditions on the declared numbers and tables, and the engine decides them itself rather than trusting the compiler's decision (Requirement 14.1 for the operator conditions). The eighth, a non-empty feasible set, holds of every declaration with an empty constraint list, which has an explicit feasible state (Theorem "Feasibility by an explicit or a verified candidate" (protocol-compiler/thm:feasibility) (i)); with declared constraints it is a semidefinite feasibility question that the compiler decides on the declaration it emits, and Requirement 16.7 keeps every other declaration away from a key length.

Requirement 14.2 (Admission of the declaration). The engine decides conditions (i)–(vi) and (viii) of Definition 1.2, in the general format as that definition reads them there, exactly on the declared data, and computes nothing from a declaration that fails one of them, which it refuses. It does not decide condition (vii) on a declaration whose constraint list is non-empty. A compiled reference companion is admitted instead under Requirement 14.3, which decides exactly the components it shares with the declaration it accompanies and none of its measured family.

Motivated by Def. 1.2, Req. 14.1, Req. 14.3, Req. 16.7, protocol-compiler/thm:feasibility · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.general_validation:validate_general_declaration

A compiled reference companion is admitted beside its declaration: shared components are decided equal, the acceptance set is read from it, and the maps are generated from the companion's own family, imported to be a POVM, its binary64 defects measured; only a short polish searches on it.

Exact admission exists so that the binary64 numbers of a declaration denote exactly what was declared, which a run needs whose estimate and key length are read off a solve over them. A reference companion has no run of its own: its value is bounded below at points another declaration's search produced and at the points of a short search continued on it from them, which serve only as points, the hypotheses of the bound being decided at each; and the numbers it is evaluated at are an evaluation of a family that is not on the binary64 grid. Its admission therefore decides exactly what can be decided exactly, the components it shares with the declaration, and imports the rest from the claims that establish it; the binary64 defects of its evaluation are measured and shown, and decide nothing. What is evaluated on it, and when its number counts, is Requirement 24.7, and how its number is certified is Requirement 28.1.

Requirement 14.3 (Admission of the reference companion beside its declaration). Where the protocol compiler emits a compiled reference companion $\mathfrak{D}^{\mathrm{ref}}$ beside a declaration $\mathfrak{D}$ (Requirement "The reference companion compiled beside the declaration" (protocol-compiler/req:reference-companion-compiled)), the engine admits it only beside $\mathfrak{D}$, after admitting $\mathfrak{D}$ (Requirements 14.1 and 14.2), and only as follows.

(i) *Decided equal.* It decides exactly, on the declared data, that every component Proposition 12.1 (i) names as shared is equal to $\mathfrak{D}$'s, entry for entry and label for label: the shapes, the sector maps and the joint sectors, the alphabets and their orders, Alice's slots, the party partition, the announcement, sifting and raw-key maps; that the companion's entries are those of $\mathfrak{D}$ save the entries of the links the companion replaced, one per measured factor whose family it replaces, each of which carries the relaxed mark of Requirement "Emitting the reference companion beside the artefact" (reductions/req:reference-companion) (iii) and a Hermitian binary64 operator with the blocks of its scope; and it reads the acceptance set of $\mathfrak{D}$, the acceptance set being one per run. A companion that fails a decision of this item is not admitted, the reason naming the component, and the run proceeds without it. The statistics map, the single-round protocol map, the branch maps, the objective and its gradient of the companion are generated from the companion's own slots, its measured family among them (Definition 1.4), as for any declaration; no map generated from $\mathfrak{D}$'s measured family is used for the companion. The announcement, sifting and raw-key maps decided equal above are declared data, not generated maps.

(ii) *Imported, never decided.* It decides none of the conditions of Defini-

tion 1.2 on the companion's measured family. That the family it denotes is a POVM, block diagonal for its sector map, is Corollary "The reference companion of a reduction instance" (reductions/cor:reference-companion) (i), which rests on the claims that establish the handed reference (Definition "Reference of a measurement family, exact or enclosed" (foundations/def:exact-reference)) on the blocks where one is handed, on the exact decisions of the declarer's snap on the blocks where the companion carries the snapped block, and on (H-REF) for each instance whose link it replaces; and with (i) above it gives the companion conditions (i)–(vi) and (viii) by Proposition 12.1 (ii). Condition (vii) is not decided on it. Nor does it decide that a block of the companion's measured family is a member of a factor enclosure the companion carries with it, which is Corollary 12.2, on the claims that establish the factor enclosures of the handed reference.

(iii) *Evaluated and measured.* It reads the companion's family at the binary64 numbers nearest to the centres of its enclosure, rounded once, entrywise to nearest, when the family is docked, and records, per block of the measured family, the completeness defect and the least eigenvalue of those numbers, measured and not decided, and the largest radius of the enclosure. A defect so measured refuses nothing and marks nothing inadmissible; what these numbers mean for the bound evaluated on them is Remark 22.4.

(iv) *Nothing else.* It computes from the companion only what Requirements 24.7 and 28.1 ask. The one program solved over it is the polish of Requirement 24.7 (i), a search continued from a point of the search on $\mathfrak{D}$, whose points serve only as points at which those requirements form brackets; no estimate is reported from it, and no key length is reported from it save through those requirements.

Requirements 14.1 and 14.2 bind every declaration but a companion admitted under this requirement.

Motivated by Def. 1.1, Def. 1.2, Def. 1.4, Def. 1.6, Def. 4.8, Prop. 12.1, Cor. 12.2, Req. 14.1, Req. 14.2, Rem. 22.4, Req. 24.7, Req. 28.1, foundations/def:exact-reference, protocol-compiler/req:reference-companion-compiled, reductions/cor:reference-companion, reductions/req:reference-companion · *Used by* Req. 24.7 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.companion:admit_reference_companion

The engine admits an acceptance set only when it decides on the declared data that the set is admissible, through a geometry tag whose every well-formed datum denotes a closed convex subset of the simplex and data well formed for that tag.

Whether a declared set is closed and convex is not read off its points: it is carried by the form in which the set is declared. The two forms of the statistics module's taxonomy, a single distribution and a polytope cut out

of the simplex by linear inequalities, denote an admissible acceptance set whenever their data are well formed, so deciding admissibility is deciding the tag and the data.

Requirement 14.4 (Admission of the acceptance set). The engine admits an acceptance set only if it decides, exactly and on the declared data, that the set is an admissible acceptance set (Definition 2.5): the set carries a geometry tag every well-formed datum of which denotes such a set, and its data are well formed for that tag. It refuses every other declared acceptance set (`acceptance_set_ill_formed` for data ill formed for the geometry they claim).

Motivated by Def. 2.5 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.validation:validate_acceptance_set

An implementation of a branch map may drop a component of a register other than the raw-key register that is constant across its Kraus terms, and keeps the raw-key register and every varying component. By Remark 1.8 the first changes no quantity; the others carry what the derivations use.

By Remark 1.8, a component of a register other than the raw-key register R that is constant across the Kraus terms of a branch map can be removed from it without changing any quantity of this module, while a component that varies across them, and R itself, carry what the derivations use.

Requirement 14.5 (Register components of an implemented branch map). In implementing a branch map $\mathcal{G}_{\hat{c}}$ or $\Theta_{\hat{c}}$ of Definition 1.6, or a sector version $\mathcal{G}_{\hat{c},s}$ or $\Theta_{\hat{c},s}$ of either, the engine may drop a component of a register other than R whose value is constant across all Kraus terms of that map, for example the $C_A C_B$ factor when the branch has a single announcement class, and it retains the register R and every component that varies across the Kraus terms: the class label κ across classes and the outcome labels within a class.

Motivated by Def. 1.6, Rem. 1.8 · accepted

The engine reports the four classes of every sector, and justifies every sector it deletes by the checked witness, the order in which the witness was evaluated and the retention of the witness's targets.

A deletion of a set is licensed by the hypothesis of Proposition 6.5, which is read on the recorded witnesses, and the evaluation order of a witness is not part of the declaration (Remark 6.7); so the record has to carry both for a deletion to be checked.

Requirement 14.6 (The sector report). The engine emits, for every sector, the four predicates of Definition 6.1 in `sector_report`, and justifies there every sector it deletes by the six conditions (W1)–(W6) checked on the exhibited witness, by that witness with its evaluation order of (W6), declared

with the witness and recorded with it as `witness_evaluation_order`, and by the retention of the witness's targets.

Motivated by Def. 6.1, Prop. 6.5, Rem. 6.7 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.general_sectors:general_sector_report

The engine decides state-blindness by an exact identity of declared data: every off-diagonal entry of each block is the datum 0 and every diagonal entry the same datum. A block that is only nearly scalar is not state-blind.

Classification runs only on a declaration already admitted, and admission is exact (Requirement 14.1): every declared block is exactly Hermitian (Definition 1.1, the exact-admissibility paragraph), so a diagonal entry is real and the sign condition of Definition 6.1 (c) compares real numbers.

REQUIREMENT 14.7 (Bit-level test of state-blindness). The engine decides whether a sector s is state-blind (Definition 6.1 (c)) by a bit-level identity of declared data. It reads $c_{v,s}$ off the declared block as its first diagonal entry, $c_{v,s} \coloneqq (F_v^{(s)})_{11}$, and requires $F_v^{(s)} - c_{v,s}\mathbb{1}_{Q_s} = 0$ entrywise and exactly: every off-diagonal entry of the block is the datum 0, and every diagonal entry is the same datum. Likewise, whenever $\mathfrak{L} \neq \varnothing$, it requires $\hat{O}_k^{(s)} - o_{k,s}\mathbb{1}_{Q_s} = 0$ with $o_{k,s} \coloneqq (\hat{O}_k^{(s)})_{11}$. No subtraction, no norm and no window enters: two binary64 data are equal or they are not, equality being equality of the values they denote and not of their bit patterns (Definition "Binary64 numbers, declared values and dyadic lattices" (foundations/def:binary64)). The engine decides the sign condition $c_{v,s} \geq 0$ by comparing that same datum with zero as a real number. A block that is only *nearly* scalar fails, and s is then not state-blind.

Motivated by Def. 1.1, Def. 6.1, Req. 14.1, foundations/def:binary64 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.general_sectors:is_state_blind

The engine flags a sector removable only when a witness it exhibits meets all six conditions of the removal checklist, each decided exactly. Every other outcome retains the sector, and no tolerance licenses a removal.

The flag rests on the verification of the checklist and on nothing about where the witness came from, so the witness may come from any search. A deletion on an approximate match would overstate the value (Remark 6.10), so every doubt retains the sector; and with a party-local entry or several measured factors the checklist has nothing to certify or nothing to read (Remark 6.8).

REQUIREMENT 14.8 (Setting the removable flag). The engine sets the flag `removable` for a sector s^* of a declaration with one measured factor when, and only when, a candidate witness exhibited by the implementation is a witness for s^* in the sense of Definition 6.1 (d): the list $\mathfrak{L}$ carries no entry declared party-local and the candidate meets all six of (W1)–(W6); it sets the flag on no other ground. The witness may come from any search, and it is never read as declared input: a declaration supplies no witness. Each

condition is decided exactly, at no window: (W1) by the bit-level identity of Requirement 14.7, (W2) as a comparison of sector indices, (W5) as the finite combinatorial predicate of key-inertness, (W3) and (W4) on the exhibited data as those conditions state, and (W6) on the residual formed in the exhibited evaluation order; a witness whose evaluation rounds anywhere, or whose residual is not the datum 0, sets no flag. Every other outcome — a nonzero residual, a rounding operation, an absent witness, a witness that names s^*, a target that is not key-inert, a block that is not exactly scalar — retains the sector. The flag is set for no sector of a declaration whose list carries an entry declared party-local, nor of a declaration with more than one measured factor. No tolerance licenses a removal: `sector_criterion` is decided at exactly zero and not at the declared tolerance τ_{num}.

Motivated by Def. 6.1, Thm. 6.2, Rem. 6.8, Rem. 6.10, Req. 14.7 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.general_sectors:general_sector_report

The engine never edits the declared sector list: it evaluates the objective over the key-active sectors and deletes from the variable list only a set of flagged sectors that meets the composition hypothesis. It never merges or averages sectors.

Dropping a constraint-active sector (Proposition 6.3) and averaging sectors (Proposition 6.4) can both overstate the key, while the two operations below are identities.

Requirement 14.9 (Operations on the declared sectors). The engine treats the declared sector list as input it never edits; sectors leave the variable list only by the deletion of (ii). The operations it may perform on the declared sectors are exactly: (i) evaluate the objective over the key-active sectors only (Definition 4.7; an identity, by Proposition 4.6); (ii) delete sectors carrying the flag `removable` of Definition 6.1 (d), set as Requirement 14.8 states — and only those, and among those only a set meeting the hypothesis of Proposition 6.5, every deleted sector's witness naming targets outside the deleted set (an identity, by Theorem 6.2 and that proposition); (iii) nothing else. In particular no simplification step merges sectors, and the only structure-changing simplification is the deletion of (ii).

Motivated by Prop. 4.6, Def. 4.7, Def. 6.1, Thm. 6.2, Prop. 6.3, Prop. 6.4, Prop. 6.5, Req. 14.8 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.declaration.sectors:delete_sectors

An implementation that does not check the set hypothesis of Proposition 6.5 deletes at most one sector per run. A single deleted sector meets that hypothesis automatically.

Requirement 14.10 (At most one deleted sector without the composition check). An implementation that will not check the hypothesis of Proposition 6.5 deletes at most one sector per run, which is the safe reading of that

hypothesis: a singleton $\mathbb{S}_{\mathrm{del}}$ satisfies it automatically, by (W2) of Definition 6.1.

Motivated by Def. 6.1, Prop. 6.5 · accepted

15 Requirements: the Choi program and gradients

The engine constructs every support frame it uses rather than accepting one, checks it exactly, uses the diagonal frame only for an exactly diagonal marginal and records the frame; it forms every Choi block in one fixed convention. It drops the conjugation of a constraint row only where the frame's construction exhibits tightness, and records the form used. It evaluates gradients in split form and only at maximal support, never reports the order-one limit as a bound, and records a dropped set with the value computed from it.

The engine constructs the declared isometry of every support frame it uses and never takes it as declared input. It decides the two identities of the frame on that isometry exactly, at zero, and refuses a frame that fails them.

Lemma 7.3 and Theorem 7.5 hold on any isometry satisfying (7.2), and both constructions of Definition 7.2 satisfy it by arithmetic on 0 and 1, the full frame for every admissible τ_A. A frame is therefore constructed to satisfy (7.2) rather than fitted to it.

Requirement 15.1 (Construction and exact check of the support frame). In trusted-source mode the engine constructs the declared isometry ι of every support frame it uses and never takes ι as declared input: by the full frame, which is always available, by the diagonal frame, or by another construction for which it exhibits (7.2) exactly. It decides both identities of (7.2) on the isometry it constructed, entrywise as binary64 data at exactly zero (`choi_isometry`), and refuses a frame that fails either of them with `choi_isometry_inexact`.

Motivated by Def. 7.2, Lem. 7.3, Thm. 7.5 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.joint:marginal_frame

The engine uses the diagonal frame only when every off-diagonal entry of the declared τ_A is the datum 0, compared at no window. For a τ_A that is only nearly diagonal the diagonal frame is unavailable, and the full frame always is.

The diagonal frame is tight by construction only because τ_A is diagonal. Read at a positive window, the test would pass a singular τ_A with small off-diagonal entries and return a frame that is not tight (Remark 7.7).

Requirement 15.2 (Exact diagonality test for the diagonal frame). The engine uses the diagonal frame of Definition 7.2 (b) only when every off-

diagonal entry of the declared τ_A is the datum 0. It decides this by comparing each off-diagonal entry with 0 as a binary64 value, at no window, as Requirement 14.7 compares the entries of a sector block, and the comparison is a clause of `alice_marginal_exact`. A τ_A that is only *nearly* diagonal is not diagonal: the diagonal frame is unavailable for it, and the full frame always is.

Motivated by Def. 1.2, Def. 7.2, Rem. 7.7, Req. 14.7 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.joint:marginal_frame

In trusted-source mode every run records the support frame it used, with the declared isometry, beside any Choi block it reports. In untrusted-source mode there is no frame, and none is recorded.

The reported Choi blocks and the channel reading of Remark 7.8 are relative to the declared isometry, while $\underline{h}_\alpha$ and every quantity the module outputs within one frame are not (Remark 7.9); so the frame is recorded rather than made canonical.

Requirement 15.3 (Record of the support frame). In trusted-source mode every run records, beside any Choi block it reports, the support frame it used: the construction that produced it, the dimension d_{A_0} and the declared isometry ι, per Alice sector in the general format, as `choi_input_basis` in `optimizer_certificate`. In untrusted-source mode no A_0 and no ι exist (Definition 7.10 (a)), and the field is absent.

Motivated by Def. 7.2, Rem. 7.8, Rem. 7.9, Def. 7.10 · accepted · *Code* certiqs_key.solvers.frank_wolfe.joint_driver:joint_run_metadata

The engine forms and reads every Choi block with the input factor A_0 first and takes the transpose of the channel reading in the declared basis of A_0.

The channel a Choi block represents depends on the factor order and on the basis the transpose is taken in (Remark 7.8).

Requirement 15.4 (Factor order and transpose basis of the Choi blocks). The engine forms and reads every Choi block J_s of Theorem 7.5 with the input factor A_0 first, and takes the transpose of the channel reading of Remark 7.8 in the declared basis of A_0, as that remark declares them.

Motivated by Thm. 7.5, Rem. 7.8 · accepted

The engine imposes a constraint row without its conjugation only where the construction of the frame exhibits a trivial kernel of W_τ without computing a rank, and the conjugated row everywhere else. No numerical test on τ_A licenses the frame-free form.

By Proposition 7.6 the frame-free form of a row is exact on a tight frame and only a restriction on any other, while the conjugated form is exact on every frame (Theorem 7.5). Deciding tightness of a given τ_A would be a

rank decision on a computed spectrum; the construction of the frame can exhibit it instead (Remark 7.7).

REQUIREMENT 15.5 (Frame-free rows only where the construction exhibits tightness). In trusted-source mode the engine imposes a homogeneous row of (7.9) in its frame-free form (7.11), and in the general format the row of an Alice sector t of a homogeneous or a party-local entry in its unconjugated form, only where the construction that produced the frame (of sector t) exhibits $\ker W_\tau = 0$ ($\ker W_t = 0$) exactly, with no rank computed and no window declared. Of the two constructions of Definition 7.2 that is the diagonal frame; on a frame produced by the full frame construction the conjugated form is imposed, whatever τ_A is, and so it is on every frame whose construction does not exhibit $\ker W_\tau = 0$. The engine licenses the frame-free form by no numerical test on τ_A.

Motivated by Def. 7.2, Thm. 7.5, Prop. 7.6, Rem. 7.7 · implemented (effective: accepted) · *Code* certiqs_key.solvers.oracles.joint_rows:prebuilt_rows

In trusted-source mode a run records, for every declared constraint row, whether it imposed the conjugated or the frame-free form, per Alice sector in the general format.

The two forms of a row define the same set on a tight frame and in general not on another (Proposition 7.6), so which one a run imposed is part of what its reported iterates mean.

REQUIREMENT 15.6 (Record of the form each constraint row was imposed in). In trusted-source mode a run records, for every declared constraint row, which of the two forms of Proposition 7.6 it imposed, per Alice sector in the general format, as `constraint_row_form` in `optimizer_certificate`, beside the frame it records under Requirement 15.3.

Motivated by Prop. 7.6, Req. 15.3 · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.joint_driver:joint_run_metadata

The engine evaluates the gradient of the pinched trace power in the split form of Theorem 8.4 (i): divided differences over the positive spectrum, pinched operators on the ambient space. It compresses the whole computation only to a pinching-invariant space containing the branch support.

A branch operator is in general singular, and its maximal branch support is in general not invariant under the key pinching (Lemma 8.2 (iv)); compressing the whole computation to its support then computes a different quantity, as the rank-one witness in the proof of Theorem 8.4 shows.

REQUIREMENT 15.7 (Split-form evaluation of the trace-power gradient). The engine evaluates the gradient of the pinched trace power at a branch operator X in the split form of Theorem 8.4 (i): the divided-difference map $DX^p[\cdot]$ compressed to $\operatorname{supp} X$, its kernel summed over the strictly positive

eigenvalues of X only, and $\mathcal{Z}(X^p)$, $Y^{\alpha-1}$ and $\mathcal{Z}(Y^{\alpha-1})$ formed on the ambient branch space. It compresses the whole computation to a subspace S' only where S' contains the maximal branch support $W_{\hat{c},s}$ of Definition 8.1 and $Z_j S' \subseteq S'$ for every j.

Motivated by Def. 8.1, Lem. 8.2, Thm. 8.4 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.gradient:pullback_trace_power_gradient

The engine evaluates a gradient of the pinched trace power only at an iterate whose support is kept maximal, for instance after an explicit perturbation. At lower support a direction that enlarges the support can have an infinite one-sided derivative (Theorem 8.4 (ii)).

Theorem 8.4 (ii) places every infinite one-sided derivative at a direction that strictly enlarges the support, and at maximal support no feasible direction does (Lemma 18.3). A gradient taken after an explicit perturbation is the gradient of another function (Remark 19.4), and on the first-order path it removes that iterate's bracket from the bound (Requirement 24.1).

Requirement 15.8 (Gradient evaluation at maximal support). The engine evaluates a gradient of the pinched trace power only at an iterate whose support is (kept) maximal, maximality being that of (8.1) for the branch operators, for instance after an explicit perturbation.

Motivated by Def. 8.1, Lem. 8.2, Thm. 8.4, Lem. 18.3, Rem. 19.4, Req. 24.1 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.gradient:deficit_gradient

The engine reports the order-one limit of Theorem 8.9, however computed, as a bound of no kind, and never as the first-order bound. The limit validates an implementation as $\alpha \to 1$ and says nothing about h_low at a finite order.

Remark 8.10 says what the limit h_1 of Theorem 8.9 validates: the $\alpha \to 1^+$ behaviour of an implementation, and no finite-α value.

Requirement 15.9 (The order-one limit as a validation reference only). The engine never reports the limit h_1 of (8.20), however computed, as a bound of any kind; in particular it never reports it as the first-order bound $\underline{h}_\alpha^{\mathrm{FO}}$ of Theorem 19.1.

Motivated by Thm. 8.9, Rem. 8.10, Thm. 19.1 · accepted

A run records its dropped set and reports the dropped value under its own name together with that set; for a non-empty set, never under the name of the undropped value.

A dropped set is a choice of the run and not declared data, and the value it gives lies at or below $\underline{h}_\alpha$ (Proposition 9.3); the number means something only beside the set it was computed for.

Requirement 15.10 (Recording the dropped set). A run records its dropped set $\mathcal{D}$ (Definition 9.2) with the run, and reports the number it computes from (9.4) as $\underline{h}_\alpha^{\mathcal{D}}$ together with $\mathcal{D}$, never, when $\mathcal{D} \neq \varnothing$, under the name $\underline{h}_\alpha$.

Motivated by Def. 9.2, Prop. 9.3 · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.general_keyrate:GeneralKeyRateResult

16 Requirements: values and key lengths

The engine carries an infinite value through as the infeasibility of the acceptance set and emits no key length for it, and on a run whose honest statistics are supplied or whose declaration carries a constraint the caller declared it bounds the value with every pair dropped once, at an order fixed before any order is evaluated, records the cap it implies on the probability that devices the declaration admits pass the acceptance test, and refuses the run where that cap is at most a threshold, one half by default. It reports the maximal key length with its floor and its clamp at 0. It evaluates the objective only at a Rényi order in $(1, \infty)$ and reports a key length only at one in $(1, 2)$, and only on a declaration the compiler emitted. It refuses a run whose security budgets, the authentication error included, sum to at least 1, where the security statement is empty. An instance that adopts the weaker secrecy convention declares it. A key length under declared constraints, or with a declared marginal, is emitted only behind the gates that license it, and a degenerate licensed-class check is reported as no evidence.

Imports foundations/def:tiers

The engine carries h_low = $+\infty$ as a value of its extended-real output and clamps it to no finite range. At $+\infty$ it emits no key length and reports the mismatch between the declaration and the acceptance set.

The value has no finite a-priori ceiling, and it is $+\infty$ when the declared acceptance set is infeasible against the declaration (Proposition 5.8 (iii), (iv)); on such a declaration no attack the declared list admits is accepted and no run produces a key (Remark 5.9). It is $+\infty$ as well when the feasible set itself is empty, (5.4) being then an infimum over nothing; on an admissible declaration that cannot happen (Definition 1.2 (vii)), and the bounds of Theorem 19.1 (v) cover it where condition (vii) is not decided.

Requirement 16.1 (Propagation of an infinite value). The engine propagates $\underline{h}_\alpha = +\infty$, the infeasibility of the declared acceptance set, rather than clamping it to any finite range; its output `h_low` has the extended-real domain $[0, +\infty]$. If $\underline{h}_\alpha = +\infty$ reaches Theorem 10.7, the engine emits no key-length claim and no numeric key length, and reports the mismatch between the declaration and the acceptance set (`infeasible_acceptance_set`).

Motivated by Def. 1.2, Def. 5.1, Prop. 5.8, Rem. 5.9, Thm. 10.7, Thm. 19.1 · *Used by* Rem. 11.4 · accepted

On a checked run the engine bounds the penalty-only value once, before any order is evaluated, and records the cap on the probability that devices as declared pass the acceptance test, proven where b is certified. A run whose cap is at most a threshold, one half by default, is refused.

The engine's own honest models are built to match the declared Alice marginal, but no claim states that the law they produce is the statistics of a state of the feasible set, and a constraint entry the caller declares can exclude every state that produces it; a law or envelope the caller supplies carries no such construction at all. Where no accepted distribution is the statistics $\mathbf{\Phi}[\omega]$ of a feasible state, the value has the positive floor $\frac{\alpha}{\alpha-1}D_{\min}$, which grows without limit as α tends to one (Proposition 9.5 (ii)), while the entropy terms stay bounded (Proposition 5.8 (ii)): a key length formed from it is a bound for the declaration, but part of it is a penalty for the mismatch between the declared devices and the accepted statistics rather than entropy. What a positive bound rules out is a per-round law: no feasible state has its statistics in the acceptance set. It does not rule out that the frequencies of a finite run of such a state fall in the set, but Proposition 10.12 bounds the probability of that event: a run whose rounds follow statistics of feasible states passes the test with probability at most $2^{-nD_{\min}}$. That cap is the completeness statement of the model the declaration states, proven where the bound it is formed from is certified and of the first-order tier otherwise. Where it is at most one half, devices behaving as the declaration says would fail the acceptance test at least as often as pass it: a protocol whose own model expects to abort is not the protocol the caller means to run, and a key length formed under it is withheld. The infinite value is the extreme case of the same mismatch (Requirement 16.1). The penalty-only program is the dropped program with every pair dropped, which the first-order bound (Corollary 19.6) and its certified evaluation (Lemma 21.1 (iv), Requirement 28.1) already cover, so no other solver is needed. By Proposition 9.5 (iv) one solve at an order α_0 bounds $D_{\min}$ from below, and the cap reads only that bound; so the solve is made once, at an order fixed before any is evaluated and below none the run evaluates, the lowest the run evaluates where its order search fixes that order in advance and otherwise the lower end of the search's declared domain, with a bounded share of the wall budget of one evaluation of the run: a first-order number bounds the value from below at any effort, so the share bears on the size of the bound only, never on its soundness. Nor does the sign of the solve's estimate bear on the bound: the first-order bound holds at points of the relaxed feasible set, where the objective may fall below zero, and on statistics that a feasible state produces the value is zero, so the estimate's sign is set by rounding and by the relaxation window, at a relaxed point as low as $-\frac{\alpha_0}{\alpha_0-1}\log_2(1+d_{A_0}\varepsilon)$ (Proposition 17.2 (c″), through (17.3), where $d_{A_0}\varepsilon < 1$). Where the acceptance set is built from an honest envelope, a positive bound shows that the envelope's premise Assumption "An honest law produced by a feasible

state" (statistics/ass:reach) fails.

Requirement 16.2 (The check of the acceptance set against the feasible statistics). A run is *checked* when its acceptance set is built from an honest law or an honest envelope that the caller gives as numbers, such as one measured on a device or read from a table, rather than one the engine computes from the declared devices, or when its declaration carries a constraint entry that the caller declares rather than one the compiler or a reduction derives, unless the run sets the option `statistics_check` to `off` (default `on`). On every run, checked or not, the engine records under `statistics_consistency` whether the run is checked and, where it is not, why (out of scope, or opted out), and the report says so beside every key length; a run that is not checked names the premise Assumption "An honest law produced by a feasible state" (statistics/ass:reach) as unchecked, the caller's to vouch for. On every checked run the engine does the following once, before it evaluates the run at any Rényi order and before it reports a key length.

(i) *The order and the solve.* It takes α_0 to be fixed before any order is evaluated: the run's fixed order, or the lowest order its order search evaluates, where the search fixes that order before it evaluates any and evaluates none below it, and otherwise the lower end of the domain the search declares, which every order the search evaluates is at least. The engine evaluates the program only at an order in $(1, \infty)$ (Requirement 16.3), so the solve runs only where $\alpha_0 > 1$, and at any other α_0 it is refused, as (ii) records; a key length is reported only at an order in $(1, 2)$ (Requirement 16.6), so a fixed order at which one is reported lies there, and that the orders a search evaluates and the lower end of its declared domain lie in $(1, 2)$ is the bound the run tier places on the domain it accepts, which no claim of this contract states. On the declaration the run searches and with the run's acceptance set, it runs the first-order search of the dropped program for the dropped set $\mathcal{D}_{\text{all}}$ of every pair (Proposition 9.5) at α_0, with the method, the iteration cap and the gap target the run applies to one evaluation of its own, and with the share `statistics_check_budget_share` of the wall budget of one evaluation of the run, where it has one, as its wall budget. A smaller share can leave a mismatch unproven: b smaller, or not positive, so that (iv) may leave unrefused a run that a longer solve would have refused, and the cap of (iv) looser; no bound is made unsound by it.

(ii) *The bound.* From that solve it forms the first-order number of the dropped program and, unless the run sets `certified_evaluation` to `off`, the certified evaluation of that number, on the declaration the run searches and at that solve's points, by parts (ii)–(v) of Require-

ment 28.1, whatever declaration the run's key length is formed on. The number b is the certified number where a point is certified and the first-order number otherwise, with its tier. The solve is not refused on the sign of its estimate O_0 of (v), which is no estimate of the run and which Requirement 24.4 does not read: b is formed from the solve's points whatever that sign, and a negative O_0 is recorded as a diagnostic and refuses nothing. Where no bound is formed (the first-order number withdrawn, the solve not completed, or the solve refused at an order outside $(1, \infty)$), the run proceeds with the reason recorded.

(iii) *The proven part.* Where $b > 0$, for every order α' the run evaluates, the proven floor $f(\alpha') \coloneqq \frac{\alpha'}{\alpha'-1} \cdot \frac{\alpha_0-1}{\alpha_0}\, b$ satisfies $f(\alpha') \le \frac{\alpha'}{\alpha'-1} D_{\min} \le \underline{h}_{\alpha'}$ (Proposition 9.5 (ii), (iv)), with $D_{\min}$ and $\underline{h}_{\alpha'}$ those of the declaration the run searches: a part of that declaration's value at α' that the mismatch alone accounts for. Where $b \le 0$, $f(\alpha') \coloneqq 0$. Where no bound was formed, $f(\alpha')$ is not defined.

(iv) *The cap and the refusal.* Let $\tau \in (0, 1]$ be the run's option `min_model_pass_probability`, an exact rational number, default $\frac{1}{2}$. Where b was formed and $b > 0$, the engine forms the exponent $x \coloneqq n \frac{\alpha_0-1}{\alpha_0}\, b$ exactly from the run's round count n, α_0 and b, with the tier of b. By Proposition 10.12 (ii), devices the declaration the run searches admits, run so that each round's acceptance value, given the earlier ones, follows the statistics $\mathbf{\Phi}[\omega]$ of some ω in that declaration's $\mathcal{W}$, pass the acceptance test with probability at most 2^{-x}, the *cap*. Where $2^{-x} \le \tau$, that is $x \ge \log_2 \frac{1}{\tau}$, the run is refused as a whole with `acceptance_set_unreachable`: it is evaluated at no order and reports no key length. The engine decides $x \ge \log_2 \frac{1}{\tau}$ exactly where $\tau = 2^{-k}$ for an integer $k \ge 0$, by comparing x with k; for any other τ it forms a rigorous enclosure $[\lambda_-, \lambda_+] \ni \log_2 \frac{1}{\tau}$ and refuses where $x \ge \lambda_+$, refuses nothing where $x < \lambda_-$, and refuses where $\lambda_- \le x < \lambda_+$, the comparison being undecided there. The decision carries the tier of b: where b is a first-order number not certified, the refusal or its absence is of the first-order tier. Where $b \le 0$ or no bound was formed there is no cap, and this item refuses nothing. With $\tau = 1$ every run with $b > 0$ is refused. Where a reference companion carries the key length, the cap is still that of the declaration the run searches, which no claim relates to the companion's feasible set; the refusal only withholds a key.

(v) *The record.* On a checked run the engine records under `statistics_consistency`: the declaration the solve ran on; α_0 and which rule of (i) fixed it; b with its tier and the lower bound $\frac{\alpha_0-1}{\alpha_0}\, b$ on $D_{\min}$, or the reason none was formed; τ; where $b > 0$, the exponent x of (iv) with the tier of b, rounded downward where the record cannot carry it exactly, so that 2^{-x} at the recorded x stays an upper bound on the pass prob-

ability, and whether the comparison of (iv) was decided exactly, by the enclosure, or left undecided and refused; where $b \leq 0$ or no bound was formed, that there is no cap; whether (iv) refused the run, as (iv) decided it on the exact x and never as a comparison of the recorded or displayed values with τ; where the solve ran, the share of the wall budget and the wall budget it was given, the wall time it took and the cause it stopped on, as the solver reports it (such as the gap target, the iteration cap or the budget); where the solve returned a point, the estimate $O_0 \coloneqq O_{\alpha_0}^{\mathcal{D}_{\mathrm{all}}}$ at that point, whether it is negative, and whether its membership in the relaxed set was established; where b was formed, the gap $O_0 - b$; and, where a bound was formed, $n\, f(\alpha')$ in bits at each order α' the run evaluates, a diagnostic of the part of the value the mismatch accounts for, which satisfies $\frac{\alpha'-1}{\alpha'}\, n\, f(\alpha') = n\, \frac{\alpha_0-1}{\alpha_0}\, b$ (Proposition 10.12 (ii)). The report shows, beside every key length it reports and, on a run refused under (iv), in place of one, the cap as a probability: 2^{-x} at the recorded x together with a decimal approximation of it rounded upward, with the tier of b, as the probability with which devices the declaration the run searches admits, run as (iv) says, pass the acceptance test, both labelled as rounded upward; and whether (iv) refused the run, with τ, as the record carries that decision from the exact comparison, so that a displayed cap a little above τ beside a refusal reads as the upward rounding it is. Where there is no cap it says so and why ($b \leq 0$, or no bound formed and the reason). It states the cap as a probability and not as a number of bits, and it does not show $n\, f(\alpha')$. No key length and no other number of the run reads these entries.

Reading: If you supply the honest statistics or declare a constraint, the engine checks, unless switched off, whether a state the declaration admits has statistics the test accepts. Once per run, before any key length is computed and with a bounded share of one evaluation's time, it bounds the mismatch from below; a short solve can leave a mismatch unproven. From a positive bound it forms a cap on the probability that devices behaving as the declaration says pass your acceptance test, proven where the bound is certified, and the record and report show that cap as a probability. Where the cap is at most one half (adjustable), the model's own devices would fail the test at least as often as pass it, and the run is refused with no key length. The record says whether the run was checked.

Motivated by Prop. 5.8, Def. 9.2, Prop. 9.5, Prop. 10.12, Req. 16.1, Req. 16.3, Req. 16.6, Prop. 17.2, Cor. 19.6, Lem. 21.1, Req. 24.4, Req. 28.1, foundations/def:tiers, statistics/ass:reach · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.consistency:decide_pass_cap

The engine evaluates the objective, its gradient and the program only at a Rényi order α in (1, ∞), where the objective is defined, and refuses any other order.

The objective (5.3) is defined for $\alpha \in (1, \infty)$ (Definition 5.1), the range on which Definition "Sandwiched Rényi divergence and conditional entropies" (foundations/def:renyi) defines the sandwiched entropies and on which the entropy, program and gradient results are stated (Remark 10.16).

REQUIREMENT 16.3 (Rényi order of an evaluated objective). The engine evaluates the objective, its gradient and the program only at a Rényi order $\alpha \in (1, \infty)$, and refuses any other order (`alpha_in_the_profile_domain`).

Motivated by Def. 5.1, Rem. 10.16, foundations/def:renyi · *Used by* Req. 16.2 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.objective:check_alpha_structural

The engine reports the maximal key length built from the lower bound h on h_low it uses, the right-hand side of the key-length bound at max{0, h} floored and clamped at 0, as its conservative integer output.

The engine never has the exact $\underline{h}_\alpha$, only a lower bound h on it that a solver establishes, and Corollary 10.18 is where it obtains its key length from one. The key length is a non-negative integer, both steps that make it one, the floor and the clamp at 0, go in the safe direction (Definition 10.13), and, when h depends on none of $(\lambda_{\mathrm{EC}}, \varepsilon_{\mathrm{EV}}, \varepsilon_{\mathrm{PA}})$, Corollary 10.18 (a) carries the monotonicity of Proposition 10.17 over to it.

REQUIREMENT 16.4 (Reported key length). The engine reports as `key_length` the maximal key length built from the lower bound $h \leq \underline{h}_\alpha$ it uses, that is $\ell_{\max}$ of Definition 10.13 with $\max\{0, h\}$ in place of $\underline{h}_\alpha$ (Corollary 10.18 (a)), with its floor and its clamp at 0: that is its conservative integer-output rule.

Motivated by Thm. 10.7, Def. 10.13, Prop. 10.17, Cor. 10.18 · *Used by* Req. 28.1 · accepted · *Code* certiqs_key.meat_core.finitekey.keylength:key_length

For a privacy-amplification family that is only two-universal, collision probability at most 2^−ℓ, the key-length bound is tightened by one further bit, $\ell \to \ell - 1$.

The leftover-hashing step of Theorem 10.7 is taken under a reading that requires an ideal two-universal family; for a family that is only two-universal that requirement can be relaxed at the cost of one bit of key length [5, Rem. 6.1] (Remark 10.15).

REQUIREMENT 16.5 (Key-length penalty for a merely two-universal family). For a privacy-amplification family that is only two-universal, collision probability at most $2^{-\ell}$, (10.4) must be tightened by one further bit, $\ell \to \ell - 1$.

Motivated by Thm. 10.7, Rem. 10.15 · accepted · *Code* certiqs_key.meat_core.finitekey.keylength:key_length

The engine reports a key length only at a Rényi order α in (1, 2), the order at which every key-length statement holds, and reports none at any other order.

Theorem 10.7, Corollary 10.18 and Corollary 19.5 give a key length only for $\alpha \in (1, 2)$, the domain of the Rényi leftover-hashing lemma (Remark 10.16).

REQUIREMENT 16.6 (Rényi order of a reported key length). The engine reports a key length only at a Rényi order $\alpha \in (1, 2)$; at any other order it reports no key length (`alpha_in_the_profile_domain`).

Motivated by Thm. 10.7, Rem. 10.16, Cor. 10.18, Cor. 19.5 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength:check_alpha_key_length

The engine reports a key length only on a declaration that the protocol compiler emitted, and none on a declaration from anywhere else. Every condition of admissibility is then decided on every route that reports a key length.

The key-length results hold of an admissible declaration (Definition 1.2) that describes the protocol being run. On a declaration that the compiler did not emit and whose constraint list is non-empty, no code path of this module decides condition (vii), the non-empty feasible set, exactly, and nothing in this module relates such a declaration to a specification of the protocol. On a declaration the compiler emitted, the compiler has decided condition (vii) (Theorem "Feasibility by an explicit or a verified candidate" (protocol-compiler/thm:feasibility)), and it has filled the declaration from a specification of the protocol (Definition "The compilation map" (protocol-compiler/def:compile)) whose agreement with the apparatus is what the compiler's own assumptions state. Refusing a key length elsewhere therefore leaves no condition of admissibility undecided, and needs no premise that a declaration from elsewhere describes the protocol. Numerical use of any declaration outside the key length stays open.

REQUIREMENT 16.7 (Key lengths only on declarations the compiler emitted). The engine reports a key length, whether as a key-length claim of Theorem 10.7, Corollary 10.18 or Corollary 19.5 or as a numeric key length, only on a declaration that the protocol compiler emitted; on any other declaration it reports none.

Motivated by Def. 1.2, Thm. 10.7, Cor. 10.18, Cor. 19.5, protocol-compiler/def:compile, protocol-compiler/thm:feasibility · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength:declaration_provenance_failure

The engine refuses a run whose budgets satisfy $\varepsilon_PA + \varepsilon_EV + 2\varepsilon_auth \geq 1$, deciding the sum exactly on the three declared numbers before any program is built. The security statement of Corollary 10.8 is then empty, and Corollary 10.18 (b) does not claim the corner.

A security statement with parameter at least 1 excludes nothing: every protocol meets it. The proof of Corollary 10.18 (b) exhibits such a corner inside the hypotheses of Theorem 10.7, $\ell = 1$ at $\varepsilon_{\mathrm{PA}} = \varepsilon_{\mathrm{EV}} = 1$ and $\lambda_{\mathrm{EC}} = 0$.

The parameter a run reports is the composed one of Corollary 10.8 (iii), which at the default $\varepsilon_{\text{auth}} = 0$ of Assumption 13.6 is equal to $\varepsilon_{\text{PA}} + \varepsilon_{\text{EV}}$, the parameter of (ii) (and numerically of (i)).

REQUIREMENT 16.8 (Refusal of budgets with an empty security statement). The engine refuses a run whose budgets satisfy $\varepsilon_{\text{PA}} + \varepsilon_{\text{EV}} + 2\varepsilon_{\text{auth}} \geq 1$, deciding the sum exactly on the three declared numbers before any program is built, and reports `security_statement_vacuous`: the $(\varepsilon_{\text{PA}} + \varepsilon_{\text{EV}} + 2\varepsilon_{\text{auth}})$-security statement of Corollary 10.8 (iii), at $\varepsilon_{\text{auth}} = 0$ equal to $\varepsilon_{\text{PA}} + \varepsilon_{\text{EV}}$, the parameter of (ii) (and numerically of (i)), is then empty, and Corollary 10.18 (b) does not claim the corner.

Motivated by Thm. 10.7, Cor. 10.8, Cor. 10.18, Ass. 13.6 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength:check_security_statement

An instance that adds back the one bit the key-length bound spends adopts the secrecy definition conditioned on the acceptance test alone, and declares explicitly that it does so. Its key length carries no claim of this module.

The constant of (10.4) spends one bit to reach the accepted state; the secrecy convention of the derivation it generalises does not (Remark 10.14).

REQUIREMENT 16.9 (Declared choice of the source's secrecy convention). An instance that adds the one bit of (10.4) back adopts the secrecy definition of the $\wedge\Omega_{\text{AT}}$ state, the convention of the derivation Remark 10.14 compares, and declares explicitly that it does so. A key length so obtained is not one of Theorem 10.7 and carries no claim of this module.

Motivated by Thm. 10.7, Rem. 10.14 · accepted

A key length is emitted with a non-empty constraint list only if the list is licensed for the declared source mode and the attack-set licence is present. Otherwise the request is refused, and the numbers serve only for exploration.

REQUIREMENT 16.10 (Two-part gate on a key length with declared constraints). A key-length output may be combined with $\mathfrak{L} \neq \varnothing$ *only if* (i) the list lies in the licensed class of the declared source mode (Definition 2.2), a condition that is mechanically checkable and degenerate in the cases of Remark 10.20, and (ii) the declaration carries the attack-set licence `attack_set_license` (Assumption A-ATK, Assumption "Attack-set licence" (protocol-compiler/ass:atk)), an attestation no program can verify. The engine refuses any request that couples $\mathfrak{L} \neq \varnothing$ with a key-length output unless *both* parts are present; without them the numerical machinery may be used on $\mathfrak{L}$ for exploration, but its output does not reach (10.4).

Motivated by Def. 2.2, Thm. 10.7, Rem. 10.20, protocol-compiler/ass:atk · *Used by* Req. 16.11 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength:check_key_length_permitted

Where the licensed-class half of the constraint gate is degenerate, at d_A = 1 and in untrusted-source mode, the engine never reports its pass as evidence of anything, nor as evidence about a party-local entry. The key length there rests on the attack-set licence alone.

In the degenerate cases of Remark 10.20 the constrained key length rests entirely on the attestation, so a passing part (i) there carries no information.

REQUIREMENT 16.11 (A degenerate licensed-class check reported as no evidence). Where part (i) of the two-part gate of Requirement 16.10 is degenerate, at $d_A = 1$ and in untrusted-source mode (Remark 10.20), the engine never reports a passing part (i) as evidence of anything, and it never reports a passing part (i) as evidence of anything about a party-local entry either.

Motivated by Rem. 10.20, Req. 16.10 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength:constraint_gate_report

A declared trusted marginal without the source-replacement licence is refused before any key-length output. At a one-dimensional Alice system the licence is vacuous and is not demanded.

REQUIREMENT 16.12 (Source licence for a declared marginal). The refusal discipline of the two-part gate of Requirement 16.10 governs the marginal as well: the engine refuses a declared τ_A without `source_replacement_license` (Assumption A-SRC, Assumption "Retention of the system A in trusted-source mode" (transmitter/ass:src)) before any key-length output, and at $d_A = 1$, where the licence is vacuous, it does not demand it (Proposition 5.2 (iv)).

Motivated by Prop. 5.2, Req. 16.10, transmitter/ass:src · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength:check_source_license

Part III
Frank-Wolfe first-order bound

The Frank-Wolfe submodule supplies the first-order lower bound on the core's value h_low, and its main results are that bound, its forms for the dropped program and for the reference companion, and the key length from its certified form. At a point of the Choi program where the relevant probabilities are positive and every key-active branch operator has maximal support, the objective is differentiable along every feasible direction, and convexity turns its linearisation into a lower bound on h_low once lower bounds on the linear minima over the acceptance set and over the feasible set are supplied (Theorem 19.1); the same holds for the dropped program, with maximal support asked only of the kept pairs (Corollary 19.6). The acceptance-side minimum is computed exactly on the exact island, and the feasible-side minimum is bounded from below in exact arithmetic from any multipliers, the solver's dual among them, with the rounding of the frame the rows are built with charged, so no bound trusts a solver's optimality (Lemma 20.1). Iterates need only lie in a relaxed feasible set read at declared windows, both minima being taken over the exact sets, and reading the set at a window can only lower the value. The theorem holds at any admissible points, wherever they came from, so a run that holds a reference of its snapped measurement forms the bound on the reference companion at the points its search on the snapped declaration produced and at those of a short search continued on the companion from them (Corollary 19.7). The resulting bound feeds the core's key-length corollary; the objective at the returned point remains an estimate, and for the first-order number the objective, the gradient and the data the engine computes from the declaration are evaluated in binary64 with no rounding-error certificate. At the run's final points the bound is formed again from enclosures of the gradient alone, the objective's homogeneity making a bracket the sum of its two lower bounds, every step from the declaration's data exact or rigorously enclosed; the best certified bracket gives the key length at the certified tier, a run whose companion certifies no point can be certified on its searched declaration under the snap premise instead, unless it declares strict_snaps, and a run that certifies no point keeps its first-order number with the reasons (Lemma 21.1, Corollary 21.3).

17 The relaxed feasible set

A run establishes membership of its iterates only up to declared windows on computed residuals, so the feasible set is read at a window: the marginal or coupling condition and the declared rows are relaxed, the normalisation

with them, and positivity and the block structure not at all (Definition 17.1). Every hypothesis of this module that names the feasible set survives on the relaxed set, and reading the set at a window can only lower the value (Proposition 17.2). What does not survive is the non-negativity of the objective, which falls below zero there by at most an explicit floor.

Imports foundations/lem:pinchmono

At a window $\varepsilon \geq 0$ the relaxed feasible set keeps positivity and the block structure exact and relaxes the marginal or coupling condition, the declared rows and, with them, the normalisation. At $\varepsilon = 0$ it is the feasible set of the program, in the state form and in the Choi form.

The conditions defining $\mathcal{W}$ in (5.1)–(5.2) are of two kinds. Some compare the variable against declared data: the marginal condition against τ_A, and each relation $\bowtie_k$ against Θ_k or c_k. The others, positive semidefiniteness, the sector-block structure and the normalisation $\operatorname{tr}\omega = 1$, name no declared datum. A run establishes membership of its iterates only up to windows on computed residuals, so the set is read at a tolerance; positive semidefiniteness and the block structure stay exact. The normalisation is relaxed with the condition that couples the sector blocks because it is an arithmetic consequence of that condition, and in one mode it is that condition: in the state form in trusted-source mode, $\operatorname{tr}\omega = 1$ follows from $\omega_A = \tau_A$; in untrusted-source mode, $\sum_s \operatorname{tr}\omega_s = 1$ is the coupling condition itself (Definition 7.10 (b)); in the Choi form, $\operatorname{tr}[\chi_Q(J)] = 1$ follows from $\sum_s \operatorname{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$ (proof of Theorem 7.5). Holding the normalisation exact beside a relaxed coupling condition would exclude every point whose coupling residual R has $\operatorname{tr}\big[\iota^\dagger \tau_A \iota\, R\big] \neq 0$ (Proposition 17.2 (c′)): the generic point with a nonzero residual, the residual being the drift over the convex combinations that built the iterate, and a nonzero residual is what the coupling check of a run accepts in the normal case. The dimension factors d_A and d_{A_0} of the normalisation clauses of Definition 17.1 are written into them as Requirement "Explicit conversion between norms" (foundations/req:norm-conversion) requires.

Definition 17.1 (The relaxed feasible set)**.** For $\varepsilon \geq 0$ the *relaxed feasible set* $\mathcal{W}(\varepsilon)$ is the set of sector-block-diagonal $\omega \in \operatorname{Pos}(AQ)$ satisfying, in trusted-source mode,

$$\big\|\omega_A - \tau_A\big\|_{\max} \leq \varepsilon, \quad \big|\operatorname{tr}\omega - 1\big| \leq d_A \varepsilon, \quad \begin{cases} \lambda_{\min}\big(\Theta_k - A_k(\omega)\big) \geq -\varepsilon, & \bowtie_k = \preceq, \\ \lambda_{\min}\big(A_k(\omega) - \Theta_k\big) \geq -\varepsilon, & \bowtie_k = \succeq, \\ \big\|A_k(\omega) - \Theta_k\big\|_{\max} \leq \varepsilon, & \bowtie_k = =, \end{cases} \tag{17.1}$$

where $A_k(\omega) := \operatorname{tr}_Q[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega]$ is the left-hand side of the k-th declared relation of (5.2), and, in untrusted-source mode, the normalisation $\big|\sum_s \operatorname{tr}\omega_s - 1\big| \leq \varepsilon$ together with the scalar forms $a_k(\omega) \leq c_k + \varepsilon$, $a_k(\omega) \geq c_k - \varepsilon$, $|a_k(\omega) - c_k| \leq \varepsilon$, with $a_k(\omega) := \operatorname{tr}\Big[(\mathbb{1}_A \otimes \hat{O}_k)\,\omega\Big]$ and no marginal

condition. In the Choi form $\mathcal{W}^J(\varepsilon)$ is the set of $(J_s)_s$ with $J_s \in \mathrm{Pos}(A_0 Q_s)$, $\left|\operatorname{tr}[\chi_Q(J)] - 1\right| \le d_{A_0}\varepsilon$, $\left\| \sum_s \operatorname{tr}_{Q_s} J_s - \mathbb{1}_{A_0} \right\|_{\max} \le \varepsilon$ and the same relaxation of the $\bowtie_k$ rows.

The normalisation clause of each form is implied by the clause written before it, by the marginal in trusted-source mode and by the coupling residual in the Choi form, and is written out only so that the set can be read without re-deriving it; Proposition 17.2 (c′) proves both implications and derives the two dimension factors. In untrusted-source mode there is no clause before it: there the normalisation *is* the relaxed coupling condition (Definition 7.10 (b)), which is why it carries no dimension factor.

Which predicate is which is part of the definition and is never converted: an inequality between operators is relaxed by a *spectral floor* $\lambda_{\min}(\cdot) \ge -\varepsilon$, an operator equality by an *entrywise* ball $\|\cdot\|_{\max} \le \varepsilon$, a scalar equality by an interval, and these are different quantities against the same number (Lemma "The positivity floor against entrywise closeness to the cone" (foundations/lem:floor-stricter): the entrywise reading and the spectral floor). The factors d_A and d_{A_0} are not such a conversion: each is the dimension factor of an explicit trace bound derived in Proposition 17.2 (c′), written into the clause, and neither reads an entrywise defect as a spectral one.

At $\varepsilon = 0$ both sets reduce to (5.1) and (7.9) respectively: each normalisation clause is then the exact $\operatorname{tr}\omega = 1$, respectively $\operatorname{tr}[\chi_Q(J)] = 1$, which the coupling condition already forces, so re-imposing it changes nothing. We write $\mathcal{W}$ for $\mathcal{W}(0)$ and $\mathcal{W}^J$ for $\mathcal{W}^J(0)$ throughout.

In the general format of Definition 1.1 both sets are read on lists over the joint sectors, with every line of (5.5), respectively of the general-format Choi set of Theorem 7.5, relaxed by the predicate its relation calls for, as in (17.1), the party-local rows included. The marginal condition and the coupling condition are then read per Alice sector: $\| \sum_s \operatorname{tr}_{Q_s} \omega^{(\mathsf{t},s)} - \tau_A^{(\mathsf{t})} \|_{\max} \le \varepsilon$, respectively $\|R_{\mathsf{t}}\|_{\max} \le \varepsilon$ with $R_{\mathsf{t}} := \sum_s \operatorname{tr}_{Q_s} J_{(\mathsf{t},s)} - \mathbb{1}_{A_{0,\mathsf{t}}}$, for every $\mathsf{t} \in \mathbb{T}$. The residual of the single marginal clause of (17.1), respectively of the single coupling clause of the Choi form, is then the block-diagonal operator $\omega_A - \tau_A = \bigoplus_{\mathsf{t}} (\sum_s \operatorname{tr}_{Q_s} \omega^{(\mathsf{t},s)} - \tau_A^{(\mathsf{t})})$, respectively $\bigoplus_{\mathsf{t}} R_{\mathsf{t}}$ with $J_s := \bigoplus_{\mathsf{t}} J_{(\mathsf{t},s)}$, whose entrywise maximum is the largest over t of the blocks' ones; so the clauses per Alice sector together are that single clause. The normalisation clauses are those of the first paragraph of this definition. At $\varepsilon = 0$ the two sets are $\mathcal{W}$ of (5.5) and the general-format $\mathcal{W}^J$ of Theorem 7.5.

Uses Def. 5.1, Thm. 7.5 · *Used by* Prop. 17.2, Rem. 17.3, Lem. 18.2, Lem. 18.3, Thm. 19.1, Cor. 19.6, Cor. 19.7, Lem. 21.1, Cor. 21.3, Rem. 23.1, Rem. 23.2, Req. 25.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.joint_driver:_check_point

The relaxed feasible sets grow with the window, are convex and compact, and keep every hypothesis of this module that names the feasible set, positivity carrying them; the objective falls below zero there by at most an explicit floor. Reading the set at a window can only lower the value.

The relaxed feasible sets keep every hypothesis of this module that names the feasible set, and positivity of the variable is what carries them: (a) and (b) are the monotonicity, convexity and compactness of the sets, (c) the facts the objective needs on them, (c′) the implication behind each normalisation clause, (c″) what relaxing the normalisation costs, (d), (d′) that reading the set at a tolerance is conservative, and (e) that all of this holds in the general format. The window at which a run checks the positivity of a point it consumes is a number this module declares and records (Requirement "Scope of the declared tolerance" (foundations/req:tolerance-scope) (ii)); item (c) *Fourth* gives the reason it reads 0 (Requirements 25.1 and 25.4).

Proposition 17.2 (Properties of the relaxed feasible set)**.** Let $\mathfrak{D}$ satisfy Definition 1.2, let $\varepsilon \geq 0$, and let $\mathcal{W}(\varepsilon)$ and $\mathcal{W}^J(\varepsilon)$ be as in Definition 17.1.

(a) *(Monotone.)* For $0 \leq \varepsilon \leq \varepsilon'$, $\mathcal{W}(\varepsilon) \subseteq \mathcal{W}(\varepsilon')$ and $\mathcal{W}^J(\varepsilon) \subseteq \mathcal{W}^J(\varepsilon')$; in particular $\mathcal{W} \subseteq \mathcal{W}(\varepsilon)$ and $\mathcal{W}^J \subseteq \mathcal{W}^J(\varepsilon)$ for every $\varepsilon \geq 0$.

(b) *(Convex and compact.)* $\mathcal{W}(\varepsilon)$ and $\mathcal{W}^J(\varepsilon)$ are convex and compact. Every $\omega \in \mathcal{W}(\varepsilon)$ has $\operatorname{tr}\omega \leq 1 + d_A\varepsilon$ in trusted-source mode and $\operatorname{tr}\omega \leq 1 + \varepsilon$ in untrusted-source mode, and every $J \in \mathcal{W}^J(\varepsilon)$ has $\sum_s \operatorname{tr} J_s \leq d_{A_0}(1+\varepsilon)$, a bound that reads the coupling clause and never the normalisation clause.

(c) *(What positivity carries.)* $\mathcal{W}(\varepsilon) \subseteq \operatorname{Pos}(AQ)$ and $\chi_Q\big(\mathcal{W}^J(\varepsilon)\big) \subseteq \operatorname{Pos}(AQ)$. *First*, the bounds $0 \leq K_{\hat{c}}(\omega) \leq P_{\hat{c}}(\omega)$ hold on the whole cone. *Second*, $K_{\hat{c}}$ is concave on the sector-block-diagonal part of $\operatorname{Pos}(AQ)$. *Third*, Theorem 5.10 holds verbatim with $\mathcal{W}(\varepsilon)$ in place of $\mathcal{W}$, and O^J_α is convex and lower semicontinuous on $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$. *Fourth*, none of this survives a relaxation of positivity: the objective factors through $X \mapsto X^p$, which is undefined off the cone. One fact of the exact set does not survive: for $\varepsilon > 0$ and $\omega \in \mathcal{W}(\varepsilon)$, $\sum_{\hat{c}} P_{\hat{c}}(\omega) = \operatorname{tr}\omega$ for an exactly admissible $\{F_v\}$, which the relaxed normalisation does not pin to 1, so $\boldsymbol{\Phi}[\omega]$ need not be a probability vector and the range fact $\underline{h}_\alpha \geq 0$ of Proposition 5.8 (i) is not available on $\mathcal{W}(\varepsilon)$ or on $\mathcal{W}^J(\varepsilon)$; it holds on the exact sets $\mathcal{W}$ and $\mathcal{W}^J$.

(c′) *(Each normalisation clause is implied by the clause before it.)* Let $J = (J_s)_s$ be per-block positive semidefinite with $\|R\|_{\max} \leq \varepsilon$, where $R \coloneqq \sum_s \operatorname{tr}_{Q_s} J_s - \mathbb{1}_{A_0}$, and put $G \coloneqq \iota^\dagger \tau_A \iota = W_\tau^\dagger W_\tau$. Then G is a state,

$$\operatorname{tr}[\chi_Q(J)] = \operatorname{tr}\Big[W_\tau^\dagger W_\tau \textstyle\sum_s \operatorname{tr}_{Q_s} J_s\Big] = \operatorname{tr}\big[G(\mathbb{1}_{A_0} + R)\big] = 1 + \operatorname{tr}[GR], \tag{17.2}$$

and $\big|\operatorname{tr}[\chi_Q(J)]-1\big| \le d_{A_0}\varepsilon$; in particular $\operatorname{tr}[\chi_Q(J)] = 1$ holds at a point of nonzero residual R exactly where $\operatorname{tr}[GR] = 0$. In the state form in trusted-source mode, $\|\omega_A - \tau_A\|_{\max} \le \varepsilon$ implies $|\operatorname{tr}\omega - 1| \le d_A\varepsilon$. In untrusted-source mode the normalisation clause is the relaxed coupling condition itself.

(c″) *(What relaxing the normalisation costs.)* At a point of $\mathcal{W}^J(\varepsilon)$ put $\varsigma := \operatorname{tr}[\chi_Q(J)] - 1$, so that $|\varsigma| \le d_{A_0}\varepsilon$ and $\sum_{\hat{c}} P^J_{\hat{c}}(J) = 1 + \varsigma$, and, where $1 + \varsigma > 0$, $\bar{\mathbf{P}} := \mathbf{P}/(1+\varsigma)$, a probability vector. Then for every probability vector $\mathbf{q}$

$$\sum_{\hat{c}} q(\hat{c}) \log_2 \frac{q(\hat{c})}{P^J_{\hat{c}}} = \sum_{\hat{c}} q(\hat{c}) \log_2 \frac{q(\hat{c})}{(1+\varsigma)\bar{P}_{\hat{c}}} = D(\mathbf{q}\|\bar{\mathbf{P}}) - \log_2(1+\varsigma), \tag{17.3}$$

and, provided $d_{A_0}\varepsilon < 1$,

$$O^J_\alpha(\mathbf{q}, J) \;\ge\; -\frac{\alpha}{\alpha-1}\,\log_2\!\big(1 + d_{A_0}\varepsilon\big) \qquad \text{on } \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon), \tag{17.4}$$

whose right-hand side is 0 at $\varepsilon = 0$ and negative otherwise. The state form carries the same statement with $d_A\varepsilon$ in trusted-source mode and ε in untrusted-source mode in place of $d_{A_0}\varepsilon$. At $d_{A_0}\varepsilon \ge 1$ the identity (17.3) still holds wherever $1 + \varsigma > 0$ but carries no bound.

(d) *(Reading the set at a tolerance is conservative.)* With $\underline{h}_\alpha[\mathcal{W}(\varepsilon)] := \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \inf_{\omega\in\mathcal{W}(\varepsilon)} O_\alpha(\mathbf{q},\omega)$,

$$\varepsilon \le \varepsilon' \implies \underline{h}_\alpha\big[\mathcal{W}(\varepsilon')\big] \;\le\; \underline{h}_\alpha\big[\mathcal{W}(\varepsilon)\big] \;\le\; \underline{h}_\alpha\big[\mathcal{W}(0)\big] = \underline{h}_\alpha \;\le\; h^{\uparrow,\mathrm{QKD}}_\alpha. \tag{17.5}$$

(d′) *(The Choi-form quantity.)* With

$$\underline{h}_\alpha\big[\mathcal{W}^J(\varepsilon)\big] := \inf_{\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}} \inf_{J\in\mathcal{W}^J(\varepsilon)} O^J_\alpha(\mathbf{q}, J), \tag{17.6}$$

the two relaxed quantities agree at $\varepsilon = 0$, and

$$\varepsilon \le \varepsilon' \implies \underline{h}_\alpha\big[\mathcal{W}^J(\varepsilon')\big] \;\le\; \underline{h}_\alpha\big[\mathcal{W}^J(\varepsilon)\big] \;\le\; \underline{h}_\alpha\big[\mathcal{W}^J(0)\big] \;=\; \underline{h}_\alpha \;\le\; h^{\uparrow,\mathrm{QKD}}_\alpha. \tag{17.7}$$

At $\varepsilon > 0$ no relation between $\underline{h}_\alpha[\mathcal{W}^J(\varepsilon)]$ and $\underline{h}_\alpha[\mathcal{W}(\varepsilon)]$ is asserted.

(e) *(The general format.)* For a declaration in the general format, with both sets read as Definition 17.1 reads them there and χ_Q the assignment $(J_{(\mathsf{t},s)}) \mapsto (\omega^{(\mathsf{t},s)})$ of Theorem 7.5, (a) to (d′) hold as stated. In (c′) the residual is $R = \bigoplus_{\mathsf{t}} R_{\mathsf{t}}$, $G = \bigoplus_{\mathsf{t}} G_{\mathsf{t}}$ with $G_{\mathsf{t}} := W^\dagger_{\mathsf{t}} W_{\mathsf{t}}$, and $\operatorname{tr}[\chi_Q(J)] = 1 + \sum_{\mathsf{t}} \operatorname{tr}[G_{\mathsf{t}} R_{\mathsf{t}}]$.

Uses Def. 1.2, Lem. 1.7, Thm. 4.1, Lem. 4.3, Lem. 4.9, Def. 5.1, Thm. 5.4, Cor. 5.6, Prop. 5.8, Thm. 5.10, Def. 7.2, Lem. 7.4, Thm. 7.5, Def. 17.1, foundations/lem:pinchmono · *Used by* Rem. 17.3, Thm. 19.1, Rem. 19.2, Cor. 19.6, Rem. 23.2, Rem. 23.9, Req. 24.4, Req. 25.1, Req. 25.5 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.joint_driver:_check_point

Proof. Idea. Every relaxed condition weakens as the window grows, which gives (a), and is an affine preimage of a closed convex set, which with the trace bounds gives (b). The facts of (c) read only positivity of the variable, which the relaxation keeps, and transfer to the Choi form along the linear map χ_Q. Item (c′) is a trace estimate on the coupling residual, (c″) a rescaling of the statistics vector, and (d), (d′) compare infima over nested sets.

(a). For $0 \le \varepsilon \le \varepsilon'$ each condition of (17.1) at ε implies the same condition at ε', and so does each condition of the untrusted-source and the Choi forms, so $\mathcal{W}(\varepsilon) \subseteq \mathcal{W}(\varepsilon')$ and $\mathcal{W}^J(\varepsilon) \subseteq \mathcal{W}^J(\varepsilon')$. At $\varepsilon = 0$ the sets are $\mathcal{W}$ and $\mathcal{W}^J$ (Definition 17.1), which gives the inclusions for every $\varepsilon \ge 0$.

(b). Each relaxed condition is convex: $\lambda_{\min}(B - A(\omega)) \ge -\varepsilon$ reads $B - A(\omega) + \varepsilon\mathbb{1} \succeq 0$, an affine preimage of the positive semidefinite cone; $\|\cdot\|_{\max} \le \varepsilon$ is an affine preimage of a norm ball; each normalisation clause is the preimage of a closed interval under the affine map $\omega \mapsto \operatorname{tr}\omega$, respectively $J \mapsto \operatorname{tr}\chi_Q(J)$, χ_Q being linear; and $\omega \succeq 0$ and the block structure are as for $\mathcal{W}$. Each is closed, so $\mathcal{W}(\varepsilon)$ is closed, and it is bounded because $\omega \succeq 0$ with $\operatorname{tr}\omega \le 1 + d_A\varepsilon$ in trusted-source mode and $\operatorname{tr}\omega \le 1 + \varepsilon$ in untrusted-source mode: the relaxed normalisation enters here as a bound on the trace in place of its exact value 1. $\mathcal{W}^J(\varepsilon)$ is closed for the same reasons and bounded because $J_s \succeq 0$ with $\sum_s \operatorname{tr} J_s = \operatorname{tr}\left[\sum_s \operatorname{tr}_{Q_s} J_s\right] \le d_{A_0}\,(1+\varepsilon)$: the Choi input register A_0 has dimension d_{A_0} (Definition 7.2), and $|\operatorname{tr} X| \le d\,\|X\|_{\max}$ for a $d \times d$ matrix, applied to $X = \sum_s \operatorname{tr}_{Q_s} J_s - \mathbb{1}_{A_0}$, gives $|\operatorname{tr}[\sum_s \operatorname{tr}_{Q_s} J_s] - d_{A_0}| \le d_{A_0}\,\varepsilon$. That bound reads the coupling clause and never the normalisation clause.

(c), state form. $\mathcal{W}(\varepsilon) \subseteq \mathrm{Pos}(AQ)$ by definition, and that inclusion is all that *First* to *Third* need. *First*: this is the cone clause of Theorem 4.1, and its derivation is short: by (4.1) and (1.20), $K_{\hat{c}} = T_{\hat{c}} + \operatorname{tr}\Theta_{\hat{c}}$, both summands non-negative for $\omega \succeq 0$; and $K_{\hat{c}} \le P_{\hat{c}}$ is Lemma "Pinching lowers the trace power" (foundations/lem:pinchmono) applied to $\tau = \mathcal{G}_{\hat{c}}(\omega)$, that lemma being stated on all of Pos() and consuming no trace hypothesis. *Second*: Lemma 4.3 (b), (c), whose hypothesis on $\{F_v\}$, sector-block-diagonal positive semidefinite operators, is implied by Definition 1.2. That lemma, and not Corollary 4.2, is the source of the concavity here: the corollary's domain is $S_{\le}()$, which a point of $\mathcal{W}(\varepsilon)$ need not meet once the normalisation is relaxed, and Lemma 4.3 is stated for exactly this situation, its part (c) being that substitution. *Third*: with those two facts the proof of Theorem 5.10 runs verbatim with $\mathcal{W}(\varepsilon)$ in place of $\mathcal{W}$, since it consumes affinity of $P_{\hat{c}}$, non-negativity, concavity and continuity of $K_{\hat{c}}$, and convexity and compactness

of the feasible set, which are (b), and nothing else. *Fourth*: the objective factors through $X \mapsto X^p$, by way of the functional T of (4.7), and $X \mapsto X^p$ is undefined off the cone, so none of *First* to *Third* survives a relaxation of positivity.

(c), Choi form. Every $J \in \mathcal{W}^J(\varepsilon)$ has $J_s \succeq 0$ per block, the one condition of the set that is not relaxed, and $\chi_Q(J) = \bigoplus_s (W_\tau \otimes \mathbb{1}) J_s (W_\tau^\dagger \otimes \mathbb{1})$ is sector-block-diagonal and positive semidefinite; hence $\chi_Q\big(\mathcal{W}^J(\varepsilon)\big) \subseteq \mathrm{Pos}(AQ)$, the one property of $\mathcal{W}(\varepsilon)$ the state-form argument uses. So the bounds $0 \le K_{\hat{c}} \le P_{\hat{c}}$, the cone concavity of Lemma 4.3 and Theorem 5.10 all apply to O_α on $\mathcal{S}_{\mathrm{acc}} \times \chi_Q(\mathcal{W}^J(\varepsilon))$; and $O_\alpha^J = O_\alpha \circ (\mathrm{id} \times \chi_Q)$ with χ_Q *linear*, so convexity and lower semicontinuity pull back to the convex preimage $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, a pullback along a linear map that needs no injectivity, as in Step 2 of the proof of Theorem 19.1. Convexity and compactness of $\mathcal{W}^J(\varepsilon)$ itself are (b). The transfer uses no bijection.

(c), the range fact. On $\mathcal{W}(\varepsilon)$ with $\varepsilon > 0$, $\sum_{\hat{c}} P_{\hat{c}}(\omega) = \mathrm{tr}\,\omega$ for an exactly admissible $\{F_v\}$ (Definition 1.1): both POVMs being complete, the sum over $\hat{c}$ of (1.16) is $\mathrm{tr}[(\sum_x M_x \otimes \sum_v F_v)\,\omega] = \mathrm{tr}\,\omega$, and the relaxed normalisation no longer pins that to 1. The proof of Proposition 5.8 (i) reads a KL divergence between two probability vectors, so it is not available on the relaxed set; read at $\varepsilon = 0$, on $\mathcal{W}$ itself, it gives $\underline{h}_\alpha \ge 0$ there. In the Choi form the range fact does not transfer either: it is read on the exact set $\mathcal{W}^J(0)$, where $\mathbf{\Phi}$ is a probability vector.

(c′). Choi form. R is Hermitian because each J_s is positive semidefinite, and $G = \iota^\dagger \tau_A \iota = W_\tau^\dagger W_\tau$ by (7.3). Then $G \succeq 0$ and $\mathrm{tr}\,G = \mathrm{tr}\big[\tau_A\, \iota\iota^\dagger\big] = \mathrm{tr}[\Pi_\iota \tau_A] = \mathrm{tr}\,\tau_A = 1$ by (7.2), so G is a state. In (17.2) the first equality is the computation in the proof of Theorem 7.5. Hence $\big|\,\mathrm{tr}[\chi_Q(J)] - 1\big| = |\,\mathrm{tr}[GR]| \le \|R\|\ \mathrm{tr}\,G = \|R\| \le d_{A_0} \|R\|_{\max} \le d_{A_0}\varepsilon$, using $|\,\mathrm{tr}[Y\varrho]| \le \|Y\|\,\mathrm{tr}\,\varrho$ for $\varrho \succeq 0$ and the inequality $\|X\| \le \|X\|_F \le d\,\|X\|_{\max}$ for a $d \times d$ matrix X, here on the $d_{A_0} \times d_{A_0}$ matrix R. This is the explicit operator-norm conversion that Requirement "Explicit conversion between norms" (foundations/req:norm-conversion) asks for, with its dimension factor written into the clause, and it is not a conversion between the spectral and the entrywise relaxation shapes of Definition 17.1: R is measured in $\|\cdot\|_{\max}$ on both sides of the derivation and no spectral floor enters it. By (17.2), $\mathrm{tr}[\chi_Q(J)] = 1$ exactly when $\mathrm{tr}[GR] = 0$. *State form, trusted-source mode.* $|\,\mathrm{tr}\,\omega - 1| = |\,\mathrm{tr}(\omega_A - \tau_A)| \le d_A \|\omega_A - \tau_A\|_{\max} \le d_A\varepsilon$, by $\mathrm{tr}\,\tau_A = 1$ and $|\,\mathrm{tr}\,X| \le d\|X\|_{\max}$ on the $d_A \times d_A$ matrix $\omega_A - \tau_A$. *Untrusted-source mode.* There is no clause before it: $|\sum_s \mathrm{tr}\,\omega_s - 1| \le \varepsilon$ *is* the relaxed coupling condition (Definition 7.10 (b)).

(c″). $|\varsigma| \le d_{A_0}\varepsilon$ is (c′), and $\sum_{\hat{c}} P_{\hat{c}}^J(J) = 1 + \varsigma$ is the sum computed in the proof of the range fact above, read at $\chi_Q(J)$. The identity (17.3) reads only that $\sum_{\hat{c}} P_{\hat{c}} = 1 + \varsigma$ with every $P_{\hat{c}} \ge 0$, never where the displacement came from; so the full-alphabet penalty sum is $D(\mathbf{q}\|\bar{\mathbf{P}}) - \log_2(1 + \varsigma)$ with

$\bar{\mathbf{P}}$ a probability vector. Writing the objective as in the proof of Proposition 5.8 (i), $O^J_\alpha = \frac{\alpha}{\alpha-1}\big[D(\mathbf{q}\|\bar{\mathbf{P}}) - \log_2(1+\varsigma)\big] + \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\text{key}}} q(\hat{c})e_{\hat{c}}$ with every $e_{\hat{c}} \geq 0$ by *First* of (c); where some $P^J_{\hat{c}} = 0$ with $q(\hat{c}) > 0$ the objective is $+\infty$ and the bound is trivial. So, provided $d_{A_0}\varepsilon < 1$, (17.4) holds. The state form is the same computation with the factor of (c′) for its mode.

(d). An infimum over a larger set is no larger, so by (a) the first two inequalities of (17.5) hold; $\underline{h}_\alpha[\mathcal{W}(0)] = \underline{h}_\alpha$ because $\mathcal{W}(0) = \mathcal{W}$; and the last step is Theorem 5.4 (Corollary 5.6 for a nonempty $\mathfrak{L}$).

(d′). At $\varepsilon = 0$ the two quantities agree by Theorem 7.5. The inclusion $\mathcal{W}^J = \mathcal{W}^J(0) \subseteq \mathcal{W}^J(\varepsilon)$ of (a) is seen directly as well: a point of $\mathcal{W}^J$ has $\sum_s \operatorname{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$, hence coupling residual $0 \leq \varepsilon$; has $\operatorname{tr}\chi_Q(J) = 1$ by Theorem 7.5; is per-block positive semidefinite; and satisfies every $\bowtie_k$ row exactly, hence in its relaxed form. An infimum over a larger set is no larger, which gives (17.7), the equality being Theorem 7.5 and the last step (17.5).

(e). In the general format the frame is $\iota = \bigoplus_{\mathsf{t}} \iota_{\mathsf{t}}$, which satisfies (7.2), and $W_\tau = \bigoplus_{\mathsf{t}} W_{\mathsf{t}}$ (Lemma 7.4 (ii)); so $G = W_\tau^\dagger W_\tau = \bigoplus_{\mathsf{t}} G_{\mathsf{t}} = \iota^\dagger \tau_A \iota$ is a state, the coupling residual of $J_s := \bigoplus_{\mathsf{t}} J_{(\mathsf{t},s)}$ is $R = \bigoplus_{\mathsf{t}} R_{\mathsf{t}}$ on $A_0 = \bigoplus_{\mathsf{t}} A_{0,\mathsf{t}}$, and $\|R\|_{\max} = \max_{\mathsf{t}} \|R_{\mathsf{t}}\|_{\max}$. With these identifications each proof above applies as written. (a) and (b) read each relaxed line as an affine preimage of a closed convex set, the party-local rows among them, and bound the trace from the coupling clause, $\sum_{\mathsf{t},s} \operatorname{tr} J_{(\mathsf{t},s)} \leq d_{A_0}(1+\varepsilon)$ with $d_{A_0} = \sum_{\mathsf{t}} d_{A_{0,\mathsf{t}}}$. (c) reads per-block positive semidefiniteness, the linearity of χ_Q and Theorem 5.10, whose inputs in the general format are the convexity of the feasible set and the concavity of $K_{\hat{c}}$ on the joint-sector lists (Definition 5.1, Lemma 4.9). (c′) reads G, R and (7.2): $\operatorname{tr}[\chi_Q(J)] = \operatorname{tr}[G(\mathbb{1}_{A_0} + R)] = 1 + \operatorname{tr}[GR] = 1 + \sum_{\mathsf{t}} \operatorname{tr}[G_{\mathsf{t}} R_{\mathsf{t}}]$ and $|\operatorname{tr}[GR]| \leq \|R\| \leq d_{A_0}\|R\|_{\max} \leq d_{A_0}\varepsilon$. (c″) reads (c′) and the statistics, and (d), (d′) read (a), Theorem 5.4 (Corollary 5.6) and Theorem 7.5, all stated for the general format. □

A run reads the relaxed feasible set at the larger of its two residual windows; every result holds at every window, and each reading is conservative. The relaxation always has a subject, and a supplier's window, the declared tolerance and a run's windows relax the set independently.

Remark 17.3 (Scope of reading the feasible set at a tolerance). *The window a run reads.* A run reads the relaxed feasible set at the larger of its two declared windows on computed residuals, the window on the coupling residual and the window on the declared-row residuals (Requirement 25.1). A per-block positive semidefinite J whose coupling residual and whose $\bowtie_k$ residuals each pass their own window lies in $\mathcal{W}^J(\varepsilon)$ at that larger ε, by Proposition 17.2 (a) from the smaller window to the larger and by its (c′) for the normalisation clause, which the coupling clause implies; so the normalisation clause adds no check to any run. In untrusted-source mode the coupling residual a run measures is $|\sum_s \operatorname{tr}\omega_s - 1|$, the relaxed normalisation clause itself (Definition 7.10 (b)). Every statement of Proposition 17.2 holds for every $\varepsilon \geq 0$, so no claim of this module depends on which window a run reads, and each candidate reading is conservative by its (a) and (d).

What the conservative reading buys. By Proposition 17.2 (d), every lower-bound claim of this module holds verbatim with $\mathcal{W}$ read at any $\varepsilon \geq 0$: the number can only come out lower, never higher, and a run that reads the set at any such ε, in particular at the windows it declares, reports a bound on the same $h_\alpha^{\uparrow,\mathrm{QKD}}$. Theorem 10.7 and Corollary 19.5 inherit this, their right-hand sides being non-decreasing in the bound they consume. The Choi-form quantity of (d′) is conservative by its own inclusion $\mathcal{W}^J \subseteq \mathcal{W}^J(\varepsilon)$, not by a bijection with the state form.

The two relaxed quantities are not compared. At $\varepsilon > 0$ the quantities $\underline{h}_\alpha[\mathcal{W}^J(\varepsilon)]$ and $\underline{h}_\alpha[\mathcal{W}(\varepsilon)]$ need not agree, and no relation between them is asserted: χ_Q carries $\mathcal{W}^J(\varepsilon)$ neither into nor onto $\mathcal{W}(\varepsilon)$ in general. Forward, the coupling residual is transported *conjugated*, $\omega_A - \tau_A = W_\tau\big(\sum_s \mathrm{tr}_{Q_s} J_s - \mathbb{1}_{A_0}\big)W_\tau^\dagger$ by (7.4), and $\|W_\tau Y W_\tau^\dagger\|_{\max}$ is not $\|Y\|_{\max}$; backward, for a singular τ_A the set $\mathcal{W}(\varepsilon)$ contains states not supported on $\operatorname{supp}\tau_A \otimes \mathcal{H}(Q)$, which are the image of no J whatever.

The floor of (c″) and the window of the penalty sum. The floor (17.4) is not the two-sided window $\log_2 \frac{1}{1-d_{A_0}\varepsilon}$ that (17.3) yields on the penalty sum alone ($|\log_2(1+\varsigma)| \leq \log_2 \frac{1}{1-d_{A_0}\varepsilon}$ for $|\varsigma| \leq d_{A_0}\varepsilon < 1$): that window bounds the penalty *sum* on both sides, whereas the floor bounds the objective from below and carries the prefactor $\frac{\alpha}{\alpha-1} > 1$ that the objective attaches to that sum, applied to the one-sided shift $\log_2(1+x)$, $x := d_{A_0}\varepsilon$, which is smaller than $\log_2 \frac{1}{1-x}$ for $x \in (0,1)$. The floor is therefore less than $\frac{\alpha}{\alpha-1}$ times the window, their ratio tending to that factor as $x \to 0$, and it can lie on either side of the window itself: at $\alpha = 1.1$ and $x = 0.01$ the two are 0.158 and 0.0145, at $\alpha = 10$ and $x = 0.5$ they are 0.650 and 1. The objective at a point of the relaxed set may be negative, by at most the floor; what a run does with a negative estimate is Requirement 24.4.

(e) What it is worth, and when it is worth nothing. With $\mathfrak{L} = \varnothing$ and in untrusted-source mode the one condition left to relax is the normalisation, and $\mathcal{W}(\varepsilon) = \{\omega \in \mathrm{Pos}(AQ)$ sector-block-diagonal $: |\sum_s \mathrm{tr}\,\omega_s - 1| \leq \varepsilon\}$, which is strictly larger than $\mathcal{W}$ for $\varepsilon > 0$. The relaxation is never without a subject, and that is why: in untrusted-source mode the normalisation is the single unconditional constraint the feasible set carries (Definition 5.1), a run measures its residual like any other, and a relaxation that exempted it would exempt the only condition there is. With $\mathfrak{L} = \varnothing$ in *trusted*-source mode the marginal row is relaxed and the normalisation follows it by Proposition 17.2 (c′). Where the relaxation has a declared-row subject its content is that a declaration whose constraint rows or whose marginal are satisfied only to within a positive window by any reachable state is not thereby made infeasible.

(f) Composition, and one obligation this places on a producer. A supplier that certifies the non-emptiness of the feasible set from a supplied witness certifies it of $\mathcal{W}(\varepsilon)$ at *that supplier's own* declared window, not of $\mathcal{W}$: under an exact reading a binary64 witness certifies nothing, and a set read at some positive number is what such a witness can be a witness for. That window is a declared number of the supplier's contract, and it is neither the declared tolerance τ_{num} nor either of the two windows at which this module reads its own feasible set; a consumer that identifies any two of the three has performed a conversion this module does not license, and Requirement "Explicit conversion between norms" (foundations/req:norm-conversion) says what such a conversion must carry. What *is* claimed of all of them is (a) and (d) of Proposition 17.2: each is a relaxation of $\mathcal{W}$, so each reading is conservative, and no relation between the three sets is asserted or needed. They can disagree, in a known direction: this module decides every predicate on declared

data *exactly* (Requirement 14.1; Requirement "Exact decision by every consumer" (foundations/req:consumer-decides-exactly)), so a declaration a supplier would admit at some tolerance of its own is refused here. That is the safe direction and the licensed one; what it asks of a producer is that it emit exactly admissible data (Requirement "Exact verification before emission" (foundations/req:exact-emission)), which the compiler decides on the numbers it emits (Definition "Exact admissibility of declared operator data" (protocol-compiler/def:exact)). The normalisation is on neither side of that comparison: it is relaxed with the coupling residual, at that residual's own window and with the dimension factor Proposition 17.2 (c′) derives, and it is compared with τ_{num} nowhere.

Uses Def. 1.2, Def. 17.1, Prop. 17.2 · newly derived · accepted

18 Admissible directions

The first-order argument runs on the Choi program, pairs acceptance vectors over the acceptance support, the outcomes that some accepted distribution charges, and names the two linear-minimisation inequalities it reads (Definition 18.1). Positivity of the probabilities is needed only there (Lemma 18.2), and at a branch operator of maximal support no Hermitian direction leaves its support, so the face derivative covers every feasible direction (Lemma 18.3).

Imports foundations/def:kl

The first-order argument runs on the Choi program, the state form being primary in untrusted-source mode. Acceptance vectors are paired over the acceptance support and operators by the trace, and the acceptance-side and feasible-side linear-minimisation inequalities are named.

The lemmas, the theorem and the corollaries of the first-order bound share one ambient program, one pair of inner products and the two linear minimisations whose minima the bound reads at each iterate, through a pair satisfying the inequalities below or through lower bounds on the minima. The pairing on the acceptance side runs over the outcomes that some accepted distribution charges; Remark 22.3 explains why that restriction loses nothing, and Lemma 18.2 and Remark 22.2 show that the acceptance-side gradient lives on those outcomes.

Definition 18.1 (Ambient program, pairings, acceptance support and linear minimisation)**.** From this definition on, the ambient program of this submodule is the Choi program (7.10) of Theorem 7.5: the variable is $x = (\mathbf{q}, J)$ with $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and $J = (J_s)_{s\in\mathbb{S}} \in \mathcal{W}^J$ of (7.9), and the objective is the function O^J_α displayed in (7.10), so that $\underline{h}_\alpha = \inf\{O^J_\alpha(x) : x \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J\}$ by that theorem. In untrusted-source mode the state form is primary: there J_s, $\mathcal{W}^J$, χ_{Q_s} and O^J_α denote ω_s, $\mathcal{W}$, the identity and O_α (Definition 7.10), and every later statement of this submodule is read with these substitutions, as Corollary 8.5 reads the gradients.

The *acceptance support* of the acceptance set $\mathcal{S}_{\mathrm{acc}}$ (Definition 2.5) is

$$\widehat{\mathcal{C}}_{\mathrm{acc}} \coloneqq \big\{\hat{c} \in \widehat{\mathcal{C}} :\ q(\hat{c}) > 0 \text{ for some } \mathbf{q} \in \mathcal{S}_{\mathrm{acc}}\big\}. \tag{18.1}$$

Pairings are the real inner products $\langle \mathbf{a}, \mathbf{b}\rangle = \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{acc}}} a(\hat{c})b(\hat{c})$ on $\mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$, the acceptance support being where the acceptance-side gradient lives (Lemma 18.2), and $\langle A, B\rangle = \mathrm{tr}[AB]$ on Hermitian operators, the latter real and, on a tuple $(A_s)_s$, summed over s.

For linear forms $g_t \in \mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$ and $G_t = (G_{t,s})_{s\in\mathbb{S}}$, a tuple of Hermitian operators of the block shape of $\mathcal{W}^J$, a pair (v_t, S_t) of a vector in $\mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$ and such a tuple satisfies the *acceptance-side inequality* of (18.2) when the first inequality below holds, and the *feasible-side inequality* when the second holds:

$$\langle g_t, v_t\rangle \le \min_{\mathbf{q}'\in\mathcal{S}_{\mathrm{acc}}} \langle g_t, \mathbf{q}'\rangle \qquad \text{and} \qquad \langle G_t, S_t\rangle \le \min_{J'\in\mathcal{W}^J} \langle G_t, J'\rangle, \tag{18.2}$$

the second minimum being taken over the exact $\mathcal{W}^J$ of (7.9).

Uses Thm. 7.5, Def. 7.10 · *Used by* Lem. 18.2, Lem. 18.3, Thm. 19.1, Rem. 19.2, Cor. 19.6, Cor. 19.7, Rem. 19.8, Lem. 20.1, Lem. 21.1, Ass. 22.1, Rem. 23.1, Rem. 23.2, Req. 28.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.support:acceptance_support

Every accepted distribution vanishes outside the acceptance support, so the terms of the objective indexed there vanish at every per-block positive point. The gradient's positivity hypotheses are therefore needed only on the acceptance support, and for the branch quantity only on its key-live part.

Hypothesis (H-POS) of Theorem 19.1 asks for positive probabilities only on the acceptance support $\widehat{\mathcal{C}}_{\mathrm{acc}}$, because that is where the lemma below shows it is consumed; on all of $\widehat{\mathcal{C}}$ it would be unsatisfiable whenever $\widehat{\mathcal{C}}_{\mathrm{acc}} \subsetneq \widehat{\mathcal{C}}$, a case Definition 2.5 admits and Remark 22.2 handles, so the first-order tier would be unreachable for a whole class of declarations for no mathematical reason. Part (i) names the non-emptiness of $\mathcal{S}_{\mathrm{acc}}$ because Definition 2.5 admits $\varnothing$, where the first clause has no witness to name. That case is degenerate throughout and nothing rests on it: $\widehat{\mathcal{C}}_{\mathrm{acc}} = \varnothing$ by (18.1), no $x_t \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ exists, so $\mathfrak{I}$ is empty and no $\underline{h}^{\mathrm{FO}}_\alpha$ is reported (Requirement 24.1), $\underline{h}_\alpha = +\infty$ as an infimum over an empty set, and the hypothesis $\Pr[\Omega_{\mathrm{AT}}] > 0$ of Theorem 2.10 fails, so no key length is claimed (Remark 5.9).

Lemma 18.2 (Positivity needed only on the acceptance support)**.** Let $\widehat{\mathcal{C}}_{\mathrm{acc}}$ be the acceptance support (18.1) of the acceptance set $\mathcal{S}_{\mathrm{acc}}$ of Definition 2.5.

(i) If $\mathcal{S}_{\mathrm{acc}} \neq \varnothing$, $\widehat{\mathcal{C}}_{\mathrm{acc}}$ is the support of a single element of $\mathcal{S}_{\mathrm{acc}}$; and, whether or not $\mathcal{S}_{\mathrm{acc}}$ is empty, every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ vanishes outside $\widehat{\mathcal{C}}_{\mathrm{acc}}$.

(ii) On $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, for every $\varepsilon \ge 0$ and in particular on the exact $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$, every term of O^J_α indexed by a $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{acc}}$ vanishes identically, so O^J_α

restricted to that set is a function of $(\mathbf{q}|_{\widehat{\mathcal{C}}_{\mathrm{acc}}}, J)$ alone; and for $\mathbf{q}, \mathbf{q}' \in \mathcal{S}_{\mathrm{acc}}$ every feasible direction has $q'(\hat{c}) - q(\hat{c}) = 0$ at every $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{acc}}$. The only property of the J factor the first clause reads is per-block positive semidefiniteness, so the first clause holds at every per-block positive semidefinite point whatever, of which $\mathcal{W}^J(\varepsilon)$ is one family.

(iii) Corollary 8.5 holds on $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, for every $\varepsilon \geq 0$, with its hypothesis "$q(\hat{c}) > 0$, $P^J_{\hat{c}} > 0$ and $K^J_{\hat{c}} > 0$ for all $\hat{c}$" weakened to: $q(\hat{c}) > 0$ and $P^J_{\hat{c}} > 0$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{acc}}$, and $K^J_{\hat{c}} > 0$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}} \cap \widehat{\mathcal{C}}_{\mathrm{acc}}$. Then (8.10) is read as a vector in $\mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$ and the sums of (8.11) are restricted to $\widehat{\mathcal{C}}_{\mathrm{acc}}$, respectively to $\widehat{\mathcal{C}}_{\mathrm{key}} \cap \widehat{\mathcal{C}}_{\mathrm{acc}}$; both restrictions are identities on that set, not truncations.

Uses Def. 2.5, Thm. 2.10, Thm. 4.1, Prop. 5.8, Thm. 7.5, Cor. 8.5, Def. 17.1, Def. 18.1, foundations/def:kl · *Used by* Thm. 19.1, Cor. 19.6, Lem. 21.1, Rem. 22.2, Rem. 22.3, Rem. 23.5 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.gradient:check_gradient_hypotheses · *Oracles* registry.py

Proof. (i) Suppose $\mathcal{S}_{\mathrm{acc}} \neq \varnothing$. Choose one $\mathbf{q}_{\hat{c}} \in \mathcal{S}_{\mathrm{acc}}$ witnessing each $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{acc}}$ and average the finitely many of them with equal positive weights: the result lies in $\mathcal{S}_{\mathrm{acc}}$, which is convex by Definition 2.5, and the support of a positively weighted convex combination of non-negative vectors is the union of their supports, which is $\widehat{\mathcal{C}}_{\mathrm{acc}}$. (If $\widehat{\mathcal{C}}_{\mathrm{acc}} = \varnothing$ the average is over the empty index set and any element of the non-empty $\mathcal{S}_{\mathrm{acc}}$ serves, its support being empty by (18.1).) No element of $\mathcal{S}_{\mathrm{acc}}$ has support outside $\widehat{\mathcal{C}}_{\mathrm{acc}}$, by (18.1), and that clause quantifies over a set that may be empty, so it needs no witness.

(ii) Let $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{acc}}$, so $q(\hat{c}) = 0$ for every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ by (i), whence also $q'(\hat{c}) - q(\hat{c}) = 0$. By the cone clause of Theorem 4.1, $0 \leq K^J_{\hat{c}} \leq P^J_{\hat{c}}$ at every per-block positive semidefinite point, hence at every point of $\mathcal{W}^J(\varepsilon)$ for every $\varepsilon \geq 0$, per-block positive semidefiniteness being the one condition that set does not relax (Definition 17.1); and by the convention $0 \log_2 \frac{0}{v} = 0$ of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl), which is stated for every $v \geq 0$, the case $v = 0$ included, the term $\frac{\alpha}{\alpha-1} q(\hat{c}) \log_2 \frac{q(\hat{c})}{P^J_{\hat{c}}}$ and, when $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, the pair $q(\hat{c}) \log_2 \frac{q(\hat{c})}{P^J_{\hat{c}}} + \frac{q(\hat{c})}{\alpha-1} \log_2 \frac{q(\hat{c})}{K^J_{\hat{c}}}$ are 0 at *every* point of $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, whatever $P^J_{\hat{c}}$ and $K^J_{\hat{c}}$ are there and in particular when one of them vanishes. Deleting those terms changes no value of O^J_α on that set. No normalisation, coupling condition or declared row is read in this argument, which is why the statement is on the relaxed sets and not on the exact set alone.

(iii) By (ii), on $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ the corollary differentiates a sum from which the $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{acc}}$ terms have been deleted, so neither its statement nor its proof evaluates $q(\hat{c})$, $P^J_{\hat{c}}$ or $K^J_{\hat{c}}$ at such a $\hat{c}$; and $K^J_{\hat{c}}$ occurs in (7.10) only inside the sum over $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, so it is evaluated at no $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{key}}$ either: in (8.10) the K-bearing branch is the case $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, and in (8.11) the K-bearing

terms are the sum over $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$. A hypothesis with no consumer in either the statement or the proof may be dropped from both. That the surviving gradient is the right object for every use this submodule makes of it is the second half of (ii): a component of (8.10) indexed outside $\widehat{\mathcal{C}}_{\mathrm{acc}}$ is, in every pairing of this submodule, multiplied by a component of a feasible direction that is 0, so it contributes nothing and its value, finite, undefined or $-\infty$, is never read. □

The image of every Hermitian input direction under a branch map lies in the maximal branch support, so at a branch operator with maximal support no direction leaves its support. The face derivative then covers every feasible direction, and no one-sided derivative is infinite along them.

Theorem 19.1 pairs the gradient of the objective against every feasible direction. The lemma below is what makes maximal support of the branch operators enough for the linearisation to exist along every direction the theorem pairs against: a branch map sends no Hermitian direction outside the maximal branch support.

Lemma 18.3 (Every direction is admissible at maximal support)**.** In the setting of Definition 18.1, let $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and $s \in \mathbb{S}_K(\hat{c})$, let $J_s \in \mathrm{Pos}(A_0 Q_s)$, and put $X \coloneqq \mathcal{G}^J_{\hat{c},s}(J_s)$. Then for *every* Hermitian Δ on $\mathcal{H}(A_0 Q_s)$ the operator $H \coloneqq \mathcal{G}^J_{\hat{c},s}(\Delta)$ satisfies

$$\operatorname{supp} H \;\subseteq\; W_{\hat{c},s} \tag{18.3}$$

with $W_{\hat{c},s}$ the maximal branch support (8.1). Consequently, if X attains maximal support, $\operatorname{supp} X = W_{\hat{c},s}$, for which $J_s \succ 0$ suffices, then $\operatorname{supp} H \subseteq \operatorname{supp} X$ for every such Δ, so that:

(a) Theorem 8.4 (i) applies to the pair (X, H), and $t \mapsto T(X + tH)$ is differentiable at $t = 0$, two-sidedly, the hypothesis being symmetric in $H \mapsto -H$;

(b) the $+\infty$ case of Theorem 8.4 (ii) is *unreachable* from X along the image of any Hermitian direction Δ for which $X + tH \succeq 0$ for small $t > 0$. That is the class that theorem is stated on and the class along which T is defined at all, and it contains every direction $y - x$ that Theorem 19.1 pairs against, since $X + t\mathcal{G}^J_{\hat{c},s}(J'_s - J_s) = \mathcal{G}^J_{\hat{c},s}\big((1-t)J_s + tJ'_s\big) \succeq 0$ for $t \in [0,1]$ whenever J_s and J'_s are positive semidefinite. No such direction strictly enlarges the support of X, because by (18.3) none leaves $W_{\hat{c},s} = \operatorname{supp} X$.

Uses Lem. 1.7, Thm. 7.5, Def. 8.1, Lem. 8.2, Thm. 8.4, Cor. 8.5, Def. 17.1, Def. 18.1 · *Used by* Thm. 19.1, Cor. 19.6 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.gradient: z_invariant_face · *Oracles* registry.py

Proof. Write $\Delta = \Delta_+ - \Delta_-$ for the Jordan decomposition, with $\Delta_\pm \succeq 0$ supported on orthogonal subspaces. The map $\mathcal{G}^J_{\hat{c},s} = \mathcal{G}_{\hat{c},s} \circ \chi_{Q_s}$ is a composition of two completely positive maps, χ_{Q_s} being the conjugation $\Delta \mapsto (W_\tau \otimes \mathbb{1})\Delta(W_\tau^\dagger \otimes \mathbb{1})$ of Theorem 7.5 and $\mathcal{G}_{\hat{c},s}$ being completely positive by Lemma 1.7 (i); hence it is completely positive and in particular positive: $\mathcal{G}^J_{\hat{c},s}(\Delta_\pm) \succeq 0$. Each of $\chi_{Q_s}(\Delta_\pm)$ lies in $\mathrm{Pos}(AQ_s)$ and is of the form Lemma 8.2 (i) considers, so (8.3) gives $\operatorname{supp} \mathcal{G}^J_{\hat{c},s}(\Delta_\pm) \subseteq W_{\hat{c},s}$. For positive semidefinite A, B one has $\operatorname{supp}(A-B) \subseteq \operatorname{supp} A + \operatorname{supp} B$, since a vector orthogonal to both supports is annihilated by A and by B and hence by $A-B$, and the support of a Hermitian operator is the orthogonal complement of its kernel. Applying this to $A = \mathcal{G}^J_{\hat{c},s}(\Delta_+)$, $B = \mathcal{G}^J_{\hat{c},s}(\Delta_-)$ gives (18.3), the right-hand side being a subspace and therefore closed under the sum. In untrusted-source mode $\chi_{Q_s} = \mathrm{id}$, Δ is Hermitian on $\mathcal{H}(AQ_s)$, and the same two steps apply with $\mathcal{S}_A = \mathcal{H}(A)$.

For (a): at maximal support (18.3) reads $\operatorname{supp} H \subseteq \operatorname{supp} X$, which is the hypothesis of Theorem 8.4 (i) with $S = \operatorname{supp} X$; that part concludes differentiability of $t \mapsto T(X + tH)$ at $t = 0$, and its hypothesis holds for $-H$ as well, so the derivative is two-sided. For (b): Theorem 8.4 (ii), whose hypothesis $X + tH \succeq 0$ for small $t > 0$ is the scope clause of (b), locates the $+\infty$ case at directions that *strictly enlarge* $\operatorname{supp} X$, and by (18.3) no image of a Hermitian direction does. That $J_s \succ 0$ suffices for maximal support is the equality clause of Lemma 8.2 (i). □

19 The first-order lower bound

The section's main results are Theorem 19.1, which in exact real arithmetic bounds h_low from below by the best linearisation over the iterates when positivity, maximal support and an unperturbed gradient hold and the two linear minima over the exact sets are bounded from below, by the pairing with a valid minimiser or by any lower bound, with iterates in the relaxed set, and Corollary 19.6, the same bound for the dropped program with maximal support asked only of the kept key-active joint pairs. Read for the reference companion at points a search on the snapped declaration found, the theorem bounds the reference model's value (Corollary 19.7). The key length follows from the core's corollary (Corollary 19.5). The remarks show that a failure of maximal support falls on the unsafe side, that a perturbed gradient bounds another function, and that the theorem holds as written in the general format.

Imports foundations/def:kl

In exact real arithmetic the best linearisation over the iterates, even ones feasible only up to windows, bounds h_low from below given positive probabilities, maximal branch support, unperturbed gradients and bounds on both linear minima. An exactly feasible point's objective bounds it from above.

The *first-order bound* $\underline{h}_\alpha^{\mathrm{FO}}$ is the largest, over the iterates the optimiser passed through, of the value of the linearisation of the objective at that iterate, minimised over the feasible set. Part (ii) proves $\underline{h}_\alpha^{\mathrm{FO}} \le \underline{h}_\alpha$ in exact real arithmetic, at maximal support, with no perturbation of the iterate and given the two inequalities of (18.2); part (iii) admits an iterate that lies only in the relaxed feasible set, which is where a run's iterates lie (Remark 19.2). Part (i) restates Lemma 7.11 (ii) of the core, because it is the whole warrant of the number a run reports as its estimate, and the membership $x \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ is exactly what must be checked for that warrant to hold (Requirement 25.1). That $\mathcal{S}_{\mathrm{acc}}$ is an admissible acceptance set (Definition 2.5) is a hypothesis of the theorem itself and not only of the claims it consumes: convexity of $\mathcal{S}_{\mathrm{acc}}$ is used in Lemma 18.2 (i) and closedness in Step 2 of the proof. Part (iv) lets a lower bound on either minimum stand in for the pairing on its side: on the acceptance side a certified bound (Remark 22.2), on the feasible side the bound of Lemma 20.1, formed as Requirement 27.1 says. Where both stand in, the bracket reads no point an oracle returned, and the theorem asks nothing of an oracle. Part (v) makes the bound independent of condition (vii) of an admissible declaration, the non-empty feasible set, which the engine does not decide on every declaration: on an empty feasible set $\underline{h}_\alpha$ is $+\infty$, and every real number bounds it.

Theorem 19.1 (The estimate and the first-order lower bound). Let $\alpha \in (1, \infty)$, let $\mathfrak{D}$ satisfy Definition 1.2, in the basic format or in the general format of Definition 1.1, and let $\mathcal{S}_{\mathrm{acc}}$ satisfy Definition 2.5. In the general format $\mathcal{W}^J$ is the set of lists $(J_{(\mathsf{t},s)})$ over the joint sectors of Theorem 7.5, with $J_s \coloneqq \bigoplus_{\mathsf{t}} J_{(\mathsf{t},s)}$ for every sector s of the measured system, and $\mathcal{W}^J(\varepsilon)$ is read as Definition 17.1 reads it there.

(i) *(The estimate over-estimates; Lemma 7.11 (ii) of the core, restated.)* For every $x \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$,

$$O_\alpha^J(x) \;\ge\; \underline{h}_\alpha. \tag{19.1}$$

(ii) *(The first-order lower bound.)* Let $\mathfrak{I}$ be a finite non-empty index set, and let $x_t = (\mathbf{q}_t, J_t) \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$, $t \in \mathfrak{I}$, satisfy for each t:

(H-POS) $q_t(\hat{c}) > 0$ and $P_{\hat{c}}^J(J_t) > 0$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{acc}}$, and $K_{\hat{c}}^J(J_t) > 0$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}} \cap \widehat{\mathcal{C}}_{\mathrm{acc}}$, with $\widehat{\mathcal{C}}_{\mathrm{acc}}$ the acceptance support (18.1);

(H-SUPP) every key-active branch operator $X_{\hat{c},s} \coloneqq \mathcal{G}_{\hat{c},s}^J(J_{t,s})$, $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, $s \in \mathbb{S}_K(\hat{c})$, attains maximal support: $\operatorname{supp} X_{\hat{c},s} = W_{\hat{c},s}$, the maximal branch support (8.1), equivalently $\operatorname{rank} X_{\hat{c},s} = \dim W_{\hat{c},s} =$

rank $\mathbf{K}_{\hat{c},s}$ by (8.4), the inclusion $\subseteq$ holding always by (8.3); $J_{t,s} \succ 0$ suffices for it (Lemma 8.2 (i));

(H-NOPERT) the pair $(g_t, G_t) := \nabla O^J_\alpha(x_t)$ is the split-form gradient of Theorem 8.4 (i)(a)–(b) assembled by Corollary 8.5 *at* x_t *itself*: no perturbation of x_t, of any $J_{t,s}$, or of any branch operator built from them is applied before the gradient is taken.

Let (v_t, S_t) satisfy both inequalities of (18.2) at (g_t, G_t) (Definition 18.1): the acceptance-side inequality and the feasible-side inequality. Then, *in exact real arithmetic*,

$$\underline{h}^{\mathrm{FO}}_\alpha \;:=\; \max_{t\in\mathfrak{I}} \Big[O^J_\alpha(x_t) + \langle g_t, v_t - \mathbf{q}_t\rangle + \langle G_t, S_t - J_t\rangle \Big] \;\le\; \underline{h}_\alpha, \qquad (19.2)$$

and $\max\{0, \underline{h}^{\mathrm{FO}}_\alpha\} \le \underline{h}_\alpha$ as well.

(iii) *(The iterate read at a tolerance, the minima over the exact set.)* Let $\varepsilon \ge 0$ and let $\mathcal{W}^J(\varepsilon)$ be the relaxed feasible set of Definition 17.1.

(iiia) *The estimate.* For every $x \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, $O^J_\alpha(x) \ge \underline{h}_\alpha[\mathcal{W}^J(\varepsilon)]$, the Choi-form relaxed infimum (17.6); and $\underline{h}_\alpha[\mathcal{W}^J(\varepsilon)] \le \underline{h}_\alpha \le h^{\uparrow,\mathrm{QKD}}_\alpha$ by (17.7).

(iiib) *The bound.* Let $x_t \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, $t \in \mathfrak{I}$, satisfy (H-POS), (H-SUPP) and (H-NOPERT) as stated in (ii), and let (v_t, S_t) satisfy (18.2) *as written*, its acceptance-side inequality and its feasible-side inequality, with the two minima over $\mathcal{S}_{\mathrm{acc}}$ and over the *exact* $\mathcal{W}^J$. Then, in exact real arithmetic, $\underline{h}^{\mathrm{FO}}_\alpha \le \underline{h}_\alpha$ and $\max\{0, \underline{h}^{\mathrm{FO}}_\alpha\} \le \underline{h}_\alpha$, with $\underline{h}^{\mathrm{FO}}_\alpha$ of (19.2).

(iv) *(Lower bounds in place of the pairings.)* The conclusions of (ii) and of (iiib) hold as well when, at some or all $t \in \mathfrak{I}$, the pairing $\langle g_t, v_t\rangle$ in (19.2) is replaced by a real number b^{A}_t, or the pairing $\langle G_t, S_t\rangle$ by a real number b^{F}_t, or both, with

$$b^{\mathrm{A}}_t \;\le\; \min_{\mathbf{q}'\in\mathcal{S}_{\mathrm{acc}}} \langle g_t, \mathbf{q}'\rangle \qquad \text{and} \qquad b^{\mathrm{F}}_t \;\le\; \min_{J'\in\mathcal{W}^J} \langle G_t, J'\rangle, \qquad (19.3)$$

each inequality asked only where its number is used, the second minimum taken over the *exact* $\mathcal{W}^J$, and the pairings $\langle g_t, \mathbf{q}_t\rangle$ and $\langle G_t, J_t\rangle$ being kept. At such a t the inequality of (18.2) on a replaced side is not asked of (v_t, S_t), and where both sides are replaced no pair (v_t, S_t) is asked at all: the bracket of (19.2) at that t is then

$$O^J_\alpha(x_t) + b^{\mathrm{A}}_t - \langle g_t, \mathbf{q}_t\rangle + b^{\mathrm{F}}_t - \langle G_t, J_t\rangle.$$

(v) *(An empty feasible set.)* In place of the hypothesis on $\mathfrak{D}$ above, let $\mathfrak{D}$ be a tuple of the form of Definition 1.1 that satisfies conditions (i)–(vi) and

(viii) of Definition 1.2, and form $\mathcal{W}$, the objective O_α and $\underline{h}_\alpha$ from it by the formulas of Definition 5.1. Let $\mathcal{W} = \varnothing$. Then $\underline{h}_\alpha = +\infty$, so every real number is a lower bound on $\underline{h}_\alpha$, the value of (19.2) that a run forms at its iterates included. Consequently the conclusions $\underline{h}_\alpha^{\mathrm{FO}} \le \underline{h}_\alpha$ and $\max\{0, \underline{h}_\alpha^{\mathrm{FO}}\} \le \underline{h}_\alpha$ of (ii), (iiib) and (iv), for a real $\underline{h}_\alpha^{\mathrm{FO}}$, hold of every tuple of the form of Definition 1.1 that satisfies conditions (i)–(vi) and (viii) of Definition 1.2, whether or not it satisfies condition (vii): under the hypotheses of those parts where it does, and because $\underline{h}_\alpha = +\infty$ where it does not.

Reading: The headline key rate rests on this bound, which needs no exact optimum. The per-round bound h_low is the minimum of a convex optimisation. At each step whose positivity and full-support conditions hold, the tangent plane of the objective lies below it over the feasible set, so its lowest point bounds h_low from below; the engine reports the best such bound, h_lb. That point is bounded without trusting a solver: one part is computed exactly, the other verified in exact arithmetic from the solver's dual against the exact constraint rows, the floating-point frame W_t's error in those rows deducted. The engine evaluates the objective, its gradient and the maps and frame they are built from in floating point, with no error budget, so a computed h_lb can exceed h_low by an amount no claim bounds. Its points are feasible only within a window, so its reported estimate need not lie above h_low.

Uses Def. 1.1, Def. 1.2, Def. 2.5, Thm. 4.1, Lem. 4.9, Def. 5.1, Prop. 5.8, Thm. 5.10, Thm. 7.5, Lem. 7.11, Def. 8.1, Lem. 8.2, Thm. 8.4, Cor. 8.5, Def. 17.1, Prop. 17.2, Def. 18.1, Lem. 18.2, Lem. 18.3 · *Used by* Rem. 19.2, Rem. 19.3, Rem. 19.4, Cor. 19.5, Cor. 19.6, Cor. 19.7, Rem. 19.8, Lem. 21.1, Lem. 21.2, Cor. 21.3, Cor. 21.4, Rem. 22.2, Rem. 22.4, Rem. 22.5, Rem. 23.1, Rem. 23.2, Rem. 23.3, Rem. 23.4, Rem. 23.5, Rem. 23.6, Rem. 23.7, Rem. 23.8, Rem. 23.9, Req. 24.1, Req. 24.2, Req. 24.3, Req. 24.6, Req. 24.7, Req. 25.2, Req. 25.4, Req. 27.1, Req. 27.2, Req. 28.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.joint_driver:run_joint_frank_wolfe · *Oracles* registry.py

Proof. Idea. At each iterate the objective has a gradient, supplied by Corollary 8.5 with its positivity weakened to the acceptance support by Lemma 18.2, and at maximal support that gradient is a genuine directional derivative in every feasible direction (Lemma 18.3). Convexity of the objective turns the derivative into a linear minorant; its minimum over the product $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ separates into the two linear minimisations of (18.2); and a maximum of lower bounds is a lower bound. In (iiib) the point lies in the relaxed set while the comparison point lies in the exact one, and the chord argument runs in the relaxed set, which is convex.

(i). This is Lemma 7.11 (ii) of the core.

(ii). Fix t and write $x := x_t$, $\nabla O := (g_t, G_t)$.

Step 1 (the gradient exists and is a genuine directional derivative in every feasible direction). By (H-POS) and (H-SUPP), and by Lemma 18.2 (iii), which is what lets (H-POS) be assumed on $\widehat{\mathcal{C}}_{\mathrm{acc}}$ and its K-part only on $\widehat{\mathcal{C}}_{\mathrm{key}} \cap \widehat{\mathcal{C}}_{\mathrm{acc}}$, Corollary 8.5 applies at x and supplies ∇O through (8.10)–(8.12), with $g_t \in \mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$; by (H-NOPERT) that is the object the bracket uses. Let $y = (\mathbf{q}', J') \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ and put $h := y - x$. Its J-component

$\Delta_s \coloneqq J'_s - J_{t,s}$ is a difference of positive semidefinite operators and hence Hermitian, so by Lemma 18.3 every branch operator's direction $\mathcal{G}^J_{\hat{c},s}(\Delta_s)$ is supported inside $\operatorname{supp} X_{\hat{c},s}$, and $t' \mapsto T(X_{\hat{c},s} + t'\mathcal{G}^J_{\hat{c},s}(\Delta_s))$ is differentiable at $t' = 0$. The remaining dependence of O^J_α on x is through the affine maps $\mathbf{q} \mapsto q(\hat{c})$ and $J \mapsto P^J_{\hat{c}}(J)$ and through finitely many compositions of $\log_2$ with quantities that (H-POS) keeps strictly positive, all differentiable there, the sums running over $\widehat{\mathcal{C}}_{\mathrm{acc}}$, respectively $\widehat{\mathcal{C}}_{\mathrm{key}} \cap \widehat{\mathcal{C}}_{\mathrm{acc}}$, by Lemma 18.2 (ii), which deletes the remaining terms identically on $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ rather than bounding them. Hence $t' \mapsto O^J_\alpha(x + t'h)$ is differentiable at $t' = 0$ with derivative $\langle \nabla O, h \rangle$, by the chain and sum rules, exactly as in the proof of Corollary 8.5, whose maximal-support step is Lemma 18.3 applied to that same Δ_s. Theorem 8.4 (ii) is what makes this a *complete* statement rather than an assumption: the only way a one-sided derivative of T can fail to be finite is a support-enlarging direction, and Lemma 18.3 (b) says there is none.

Step 2 (convexity turns the derivative into an inequality). By Theorem 5.10, O^J_α is convex with values in $(-\infty, +\infty]$ on the convex set $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$: Theorem 5.10 states this for O_α on $\Delta(\widehat{\mathcal{C}}) \times \mathcal{W}$, and $O^J_\alpha = O_\alpha \circ (\mathrm{id} \times \chi_Q)$ with χ_Q *linear*, so the composition is convex on the convex preimage, a pullback along a linear map that needs no injectivity, which is Theorem 7.5's own convexity clause. For a convex extended-real function the difference quotient $t' \mapsto \big(O^J_\alpha(x + t'h) - O^J_\alpha(x)\big)/t'$ is non-decreasing on $(0, 1]$: this is the three-chord inequality, and it holds with the value $+\infty$ admitted. $O^J_\alpha(x)$ is finite by (H-POS), and $x + t'h = (1 - t')x + t'y$ stays in the convex set $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ for $t' \in [0, 1]$, so the quotient is defined there. Its limit as $t' \downarrow 0$ is $\langle \nabla O, h \rangle$ by Step 1, and its value at $t' = 1$ is $O^J_\alpha(y) - O^J_\alpha(x)$; monotonicity gives $\langle \nabla O, h \rangle \le O^J_\alpha(y) - O^J_\alpha(x)$, that is

$$O^J_\alpha(y) \;\ge\; O^J_\alpha(x) + \langle \nabla O,\, y - x \rangle \qquad \text{for every } y \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J, \tag{19.4}$$

including every y at which $O^J_\alpha = +\infty$, where the inequality is trivial. Taking the infimum over y and using Theorem 7.5,

$$\underline{h}_\alpha \;\ge\; O^J_\alpha(x) + \min_{y \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J} \langle \nabla O,\, y - x \rangle, \tag{19.5}$$

the minimum being attained because $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ is compact and the map is linear. That $\mathcal{W}^J$ is compact is read off (7.9) directly: it is closed, being an intersection of the positive semidefinite cones with affine equalities and inequalities; and it is bounded, because $J_s \succeq 0$ with $\sum_s \operatorname{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$ forces $\sum_s \operatorname{tr} J_s = \operatorname{tr} \mathbb{1}_{A_0} = d_{A_0}$ and hence $\|J_s\| \le \operatorname{tr} J_s \le d_{A_0}$ for every s. $\mathcal{S}_{\mathrm{acc}}$ is closed in the simplex by Definition 2.5. In untrusted-source mode the same reading applies to $\mathcal{W}$ with $\operatorname{tr} \omega = 1$ in place of the coupling condition.

Step 3 (the minimum separates over the product). The map $y \mapsto \langle \nabla O, y \rangle = \langle g_t, \mathbf{q}' \rangle + \langle G_t, J' \rangle$ is a sum of one term in each factor, and $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ is a

product set, so

$$\min_{y\in\mathcal{S}_{\mathrm{acc}}\times\mathcal{W}^J}\langle\nabla O,y\rangle=\min_{\mathbf{q}'\in\mathcal{S}_{\mathrm{acc}}}\langle g_t,\mathbf{q}'\rangle+\min_{J'\in\mathcal{W}^J}\langle G_t,J'\rangle\ \geq\ \langle g_t,v_t\rangle+\langle G_t,S_t\rangle,$$

the last step being (18.2). The pair (v_t,S_t) is *not* required to be feasible, only to satisfy the two inequalities. Substituting into (19.5),

$$\underline{h}_\alpha\ \geq\ O^J_\alpha(x_t)+\langle g_t,v_t-\mathbf{q}_t\rangle+\langle G_t,S_t-J_t\rangle.$$

Step 4 (a maximum of lower bounds is a lower bound). The display holds for each $t\in\mathfrak{I}$, which is finite and non-empty, so it holds for the largest of the right-hand sides, which is (19.2). Finally $\underline{h}_\alpha\geq 0$ by Proposition 5.8 (i), whose proof consumes only Definition 1.2 and Theorem 4.1; so $\max\{0,\underline{h}^{\mathrm{FO}}_\alpha\}$ is the maximum of two lower bounds on $\underline{h}_\alpha$ and is therefore one.

(iiia). $\underline{h}_\alpha[\mathcal{W}^J(\varepsilon)]$ is by (17.6) the infimum of O^J_α over $\mathcal{S}_{\mathrm{acc}}\times\mathcal{W}^J(\varepsilon)$ and x is a point of that set, which is the whole of (i)'s argument with the set substituted; nothing else of (i) is used, and in particular no property of the set is. That $\underline{h}_\alpha[\mathcal{W}^J(\varepsilon)]\leq\underline{h}_\alpha\leq h^{\uparrow,\mathrm{QKD}}_\alpha$ is (17.7), whose one substantive step is the inclusion $\mathcal{W}^J\subseteq\mathcal{W}^J(\varepsilon)$ proved there.

(iiib). The proof is (ii)'s four steps with one change, made in Step 2: the point x_t ranges over the *relaxed* set while the comparison point y ranges over the *exact* one. Fix t and write $x:=x_t$, $\nabla O:=(g_t,G_t)$.

Step 1 is unchanged, and reads only per-block positivity. By (H-POS) and (H-SUPP) and Lemma 18.2 (iii), Corollary 8.5 applies at x and supplies ∇O. That the corollary applies here, at a point of the relaxed set, is a property of its hypothesis and not of this step: it asks $\mathbf{q}\in\mathcal{S}_{\mathrm{acc}}$ and $J_s\in\mathrm{Pos}(A_0Q_s)$ per block, names no normalisation, no coupling condition and no declared row, and every point of $\mathcal{W}^J(\varepsilon)$ meets it, per-block positive semidefiniteness being the one condition the relaxed set does not relax. And Lemma 18.2 (iii), the statement that weakens the corollary's positivity clause to $\widehat{\mathcal{C}}_{\mathrm{acc}}$, is itself stated on $\mathcal{S}_{\mathrm{acc}}\times\mathcal{W}^J(\varepsilon)$ for every $\varepsilon\geq 0$, so the weakening reaches this point too. By (H-NOPERT) the object so supplied is the one the bracket uses. Let $y=(\mathbf{q}',J')\in\mathcal{S}_{\mathrm{acc}}\times\mathcal{W}^J$, the *exact* set, and put $h:=y-x$. Its J-component $\Delta_s:=J'_s-J_{t,s}$ is a difference of two positive semidefinite operators, J'_s from $\mathcal{W}^J$ and $J_{t,s}$ from $\mathcal{W}^J(\varepsilon)$ (per-block positive semidefiniteness being the one condition the relaxed set does not relax, Definition 17.1), hence Hermitian. That is all Lemma 18.3 asks: its hypotheses are $J_s\in\mathrm{Pos}(A_0Q_s)$ and Δ Hermitian, and it names no feasible set, no normalisation, no coupling condition and no declared row. So every branch operator's direction $\mathcal{G}^J_{\hat{c},s}(\Delta_s)$ is supported inside $\operatorname{supp}X_{\hat{c},s}$ and the map $t'\mapsto T(X_{\hat{c},s}+t'\mathcal{G}^J_{\hat{c},s}(\Delta_s))$ is differentiable at $t'=0$. The remaining dependence is through the affine $\mathbf{q}\mapsto q(\hat{c})$ and $J\mapsto P^J_{\hat{c}}(J)$ and through compositions of $\log_2$ with quantities (H-POS) keeps strictly positive. Lemma 18.2 (ii) deletes the terms indexed outside $\widehat{\mathcal{C}}_{\mathrm{acc}}$ identically here as well: its input is $0\leq K^J_{\hat{c}}\leq P^J_{\hat{c}}$ on the set the points

lie in, which Proposition 17.2 (c) establishes on the whole cone (*First*), from positivity alone and consuming no trace hypothesis, and in the Choi form on $\mathcal{W}^J(\varepsilon)$. Hence $t' \mapsto O_\alpha^J(x + t'h)$ is differentiable at $t' = 0$ with derivative $\langle \nabla O, h\rangle$.

Step 2 (the segment lies in the relaxed set, and the chord argument runs there). $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ is convex (Proposition 17.2 (b); $\mathcal{S}_{\mathrm{acc}}$ by Definition 2.5), and O_α^J is convex on it with values in $(-\infty, +\infty]$ by Proposition 17.2 (c), whose Choi form pulls Theorem 5.10 back along the *linear* χ_Q, needing no injectivity. Both endpoints lie in that convex set: $x \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ by hypothesis, and $y \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J \subseteq \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ by Proposition 17.2 (a). So $x + t'h = (1 - t')x + t'y$ stays in it for $t' \in [0, 1]$, $O_\alpha^J(x)$ is finite by (H-POS), and the three-chord inequality gives, exactly as in (ii), $\langle \nabla O, h\rangle \leq O_\alpha^J(y) - O_\alpha^J(x)$, that is, (19.4) *for every* $y \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$. Taking the infimum over that set, and not over the relaxed one, and using Theorem 7.5, which says the infimum of O_α^J over $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ is $\underline{h}_\alpha$,

$$\underline{h}_\alpha \;\geq\; O_\alpha^J(x) + \min_{y \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J} \langle \nabla O,\, y - x\rangle, \tag{19.6}$$

the minimum attained because $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ is compact and the map linear, compactness being read off (7.9) exactly as in (ii), and non-emptiness following from Definition 1.2 (vii) with Theorem 7.5 mapping $\mathcal{W}^J$ onto $\mathcal{W}$.

Steps 3 and 4 are those of (ii). The map $y \mapsto \langle \nabla O, y\rangle$ separates over the product $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$, and the two minima that appear are exactly the two of (18.2) as the hypothesis of (iiib) states them; substituting and maximising over the finite non-empty $\mathfrak{I}$ gives $\underline{h}_\alpha^{\mathrm{FO}} \leq \underline{h}_\alpha$. Finally $\underline{h}_\alpha \geq 0$ by Proposition 5.8 (i), a statement about the *exact* set that is not claimed at $\varepsilon > 0$ (Proposition 17.2 (c), (c″)) and is not needed there, so $\max\{0, \underline{h}_\alpha^{\mathrm{FO}}\}$ is the maximum of two lower bounds on $\underline{h}_\alpha$ and is one. Part (ii) is the case $\varepsilon = 0$.

(iv). Steps 1, 2 and 4, of (ii) and of (iiib), do not read the pair (v_t, S_t), and Step 3 reads of it only the two inequalities of (18.2), the first of its acceptance half and the second of its feasible half, each to bound one of the two minima into which the minimum over the product separates. At a t where a replacement is made, put $a_t \coloneqq b_t^{\mathrm{A}}$ if the acceptance side is replaced and $a_t \coloneqq \langle g_t, v_t\rangle$ otherwise, and $f_t \coloneqq b_t^{\mathrm{F}}$ if the feasible side is replaced and $f_t \coloneqq \langle G_t, S_t\rangle$ otherwise. The separation of Step 3, with (19.3) on each replaced side and (18.2) on each side not replaced, gives

$$\min_{y \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J} \langle \nabla O, y\rangle = \min_{\mathbf{q}' \in \mathcal{S}_{\mathrm{acc}}} \langle g_t, \mathbf{q}'\rangle + \min_{J' \in \mathcal{W}^J} \langle G_t, J'\rangle \;\geq\; a_t + f_t,$$

and substituting into (19.5), respectively (19.6), gives $\underline{h}_\alpha \geq O_\alpha^J(x_t) + a_t - \langle g_t, \mathbf{q}_t\rangle + f_t - \langle G_t, J_t\rangle$, which is the bracket of (19.2) with the replacements made. Where both sides are replaced, neither v_t nor S_t enters any step. Step 4 then runs unchanged.

(v). No claim is used for $\underline{h}_\alpha = +\infty$. By (5.4), $\underline{h}_\alpha$ is the infimum of O_α over $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and $\omega \in \mathcal{W}$; with $\mathcal{W} = \varnothing$ the inner infimum is over the empty set and is $+\infty$ for every $\mathbf{q}$, and so is the outer one, whether $\mathcal{S}_{\mathrm{acc}}$ is empty or not. Every real number is below $+\infty$. For the consequence: where the tuple satisfies condition (vii) as well, it satisfies Definition 1.2 in full, and with it the two conditions Definition 1.1 places on its raw-key map, which are conditions (v) and (viii); it is then a protocol declaration satisfying the hypothesis of (ii), (iiib) and (iv); where it does not, $\mathcal{W} = \varnothing$, so $\underline{h}_\alpha = +\infty$ and the real number $\underline{h}_\alpha^{\mathrm{FO}}$ is below it.

What (iiib) uses of the relaxed set. Only these: that $\mathcal{W}^J \subseteq \mathcal{W}^J(\varepsilon)$ (Proposition 17.2 (a)); that $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ is convex (*ibid.* (b)); that O_α^J is convex on it (*ibid.* (c)); that every member is per-block positive semidefinite (Definition 17.1: the one condition not relaxed, what Lemma 18.3 consumes, and what the hypothesis of Corollary 8.5 asks of x, the corollary being the statement that supplies (g_t, G_t) and therefore the one whose scope decides whether (19.2) names an object at all); and that $0 \le K_{\hat{c}}^J \le P_{\hat{c}}^J$ on it (*ibid.* (c), *First*, which is the cone clause of Theorem 4.1 and what Lemma 18.2 (ii) consumes). The list does not say that compactness of the relaxed set is unused: no step of this proof reads it, the minimum being taken over the exact set, but the convexity of O_α^J that Step 2 imports is Proposition 17.2 (c) *Third*, whose own derivation consumes the convexity *and* the compactness of (b), the latter through the continuity of $K_{\hat{c}}$; so compactness is used one level down and not in the minimisation. Nor does it say that lower semicontinuity is unavailable: the chord argument needs convexity alone, but the fact imported carries both. Of the *exact* set the proof uses compactness, non-emptiness, and that its infimum is $\underline{h}_\alpha$ (Theorem 7.5). No statement is made about $\underline{h}_\alpha[\mathcal{W}(\varepsilon)]$ here, and none is used: the state-form relaxed set enters this part nowhere.

The general format. No step above reads a constraint row, the number of measured factors or the number of Alice sectors, so the proof applies verbatim with $\mathcal{W}^J$ the joint-sector Choi set. Step 1 reads the hypothesis of Corollary 8.5, per-block positive semidefiniteness of the point, which names no feasible set, no normalisation and no declared row; Lemma 18.3, whose hypotheses are $J_s \in \mathrm{Pos}(A_0 Q_s)$ and Δ Hermitian and which reads only (8.3), stated for the admissible Alice subspace in either format; Theorem 8.4 (ii), a statement about the functional on the positive cone of a branch space; and Lemma 18.2 (ii) and (iii), which read only the acceptance support, per-block positive semidefiniteness and the cone clause. Step 2 reads convexity of $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$, convexity of the objective on it, compactness of $\mathcal{W}^J$ and the identification of the infimum over it with $\underline{h}_\alpha$, all four supplied by the general-format paragraph of Theorem 7.5, including for the party-local rows, which are affine semidefinite conditions there like every other row; that paragraph cites the convexity of the objective from Theorem 5.10, whose inputs, convexity of the feasible set and concavity of $K_{\hat{c}}$, Definition 5.1 and

Lemma 4.9 supply in the general format. Steps 3 and 4 read that the variable is a product and that the gradient pairs term by term over its factors, that $\mathfrak{I}$ is finite and non-empty, and, for the clamp at zero, Proposition 5.8 (i), which consumes only the conventions of the kernel, the objective and the cone clause. Part (iiib) reads in addition (a), (b) and (c) of Proposition 17.2 and per-block positive semidefiniteness of the relaxed set, which hold in the general format by that proposition's (e); and (iv) reads nothing more. □

A run establishes its iterate only in the relaxed feasible set, while the linear minima its bracket bounds are over the exact one, and part (iii) of Theorem 19.1 asks each half of the object that supplies it. Its bound compares h_lb with the exact h_low directly, through no relaxed quantity.

Remark 19.2 (Iterate in the relaxed set, minima over the exact set). Part (iii) of Theorem 19.1 substitutes the relaxed feasible set $\mathcal{W}^J(\varepsilon)$ in different places in its two halves, and the asymmetry is deliberate. A run of this module is always in that part: it verifies the J factor of its iterate's membership at positive declared windows (Requirement 25.1) and holds the $\mathcal{S}_{\mathrm{acc}}$ factor exactly, by forming $\mathbf{q}_t$ on the exact island (Definition "The exact island" (foundations/def:exact-island); Remark 22.2), so what it establishes is $x_t \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ and not exact membership in the product; while the minima of (18.2), and the lower bounds that part (iv) of the theorem puts in their place, are taken over the declared conic program (7.9), which is the *exact* $\mathcal{W}^J$. *Iterate in the relaxed set, minima over the exact set*: (iiib) asks each half of the object that supplies it. Its hypotheses (H-POS), (H-SUPP) and (H-NOPERT) are conditions on the iterate alone and name no set. Substituting $\mathcal{W}^J(\varepsilon)$ into (18.2) as well would ask for a bound against a *larger* set, hence below a *smaller* minimum: since $\mathcal{W}^J \subseteq \mathcal{W}^J(\varepsilon)$, one has $\min_{\mathcal{W}^J(\varepsilon)} \leq \min_{\mathcal{W}^J}$, so the inequality over the exact set is the weaker of the two. It is also the one Lemma 20.1 supplies: the trace identity that lemma rests on is the exact coupling condition, which a point of the relaxed set meets only within its window.

Part (iiib) concludes against the exact $\underline{h}_\alpha$ directly, with no intermediate quantity: $\underline{h}_\alpha[\mathcal{W}^J(\varepsilon)]$ appears in (iiia) and nowhere in (iiib), and no non-negativity of it is claimed; it may be negative, by at most $\frac{\alpha}{\alpha-1}\log_2(1+d_{A_0}\varepsilon)$ (Proposition 17.2 (c″)). The clamp at zero reads the range fact $\underline{h}_\alpha \geq 0$ of Proposition 5.8 (i) on the exact set, which is the only reading it needs; on the relaxed set that fact is not available, and nothing uses it there. Corollary 19.5 consumes the same inequality on this reading as on that of (ii), (ii) being the case $\varepsilon = 0$ of (iiib). Which ε a run uses, the larger of its two declared windows on computed residuals and never the declared tolerance, is fixed by Requirement 25.1, and part (iii) is insensitive to it, being proved for every $\varepsilon \geq 0$.

The relaxed quantity in (iiia) is the Choi-form one and not the state-form $\underline{h}_\alpha[\mathcal{W}(\varepsilon)]$ of Proposition 17.2 (d): at $\varepsilon > 0$ the two relaxed sets are not in bijection under χ_Q, for the two reasons Remark 17.3 gives, and no relation between the two numbers is claimed or needed. Part (iiib) uses neither relaxed value: it compares $\underline{h}_\alpha^{\mathrm{FO}}$ against the exact $\underline{h}_\alpha$ directly, so the chain (17.7) is consumed by the estimate half of part (iii) and not by the bound half.

Uses Prop. 17.2, Def. 18.1, Thm. 19.1 · newly derived · accepted

Where maximal support fails, a support-enlarging direction can give the objective a $-\infty$ one-sided derivative, so the linearisation over-estimates the objective and the bracket can exceed h_low. That is why (H-SUPP) is a hypothesis of Theorem 19.1 and a failed check drops the iterate.

Which way a hypothesis of Theorem 19.1 fails decides what the runtime checks of it are worth (Remark 23.6). For (H-SUPP) the failure is on the unsafe side.

Remark 19.3 (Failure of maximal support lies on the unsafe side). The functional T is *concave* on the positive semidefinite cone (Lemma 4.3), hence so is each deficit $f_{\hat{c},s}$ of (4.2) and so is $K_{\hat{c}} = P_{\hat{c}} + \sum_s f_{\hat{c},s}$ of (4.1). The objective carries $K_{\hat{c}}$ only through the term $\frac{q(\hat{c})}{\alpha-1}\log_2 \frac{q(\hat{c})}{K_{\hat{c}}}$, whose partial derivative in $K_{\hat{c}}$ is $-\frac{q(\hat{c})\log_2 \mathrm{e}}{(\alpha-1)K_{\hat{c}}} < 0$ for $\alpha > 1$, $q(\hat{c}) > 0$, $K_{\hat{c}} > 0$: the objective is *decreasing* in $K_{\hat{c}}$, through a negated logarithm. Consequently, at a point where some branch operator does *not* attain maximal support and a feasible direction h enlarges it, the $+\infty$ one-sided derivative that Theorem 8.4 (ii) permits for T, and hence for $K_{\hat{c}}$, becomes a $-\infty$ one-sided derivative for O^J_α along h. Two things follow. First, no subgradient of O^J_α exists at such a point in such a direction, so (19.4) is not merely unproved there, it is false for the split-form gradient: the objective falls away faster than any linear model. Second, the split-form gradient then makes the linearisation *over*-estimate O^J_α away from the face, so the bracket (19.2) can exceed $\underline{h}_\alpha$. The failure is on the *unsafe* side. That is why (H-SUPP) is a hypothesis of the theorem rather than a convenience of the implementation, and why a failed check of it (Requirement 25.2) removes that iterate's bracket from the bound (Requirement 24.1).

Uses Thm. 4.1, Thm. 8.4, Thm. 19.1 · *Used by* Rem. 23.6 · newly derived · accepted

A gradient taken at a perturbed operator is the gradient of a different function, and no statement here relates the two, so the explicit perturbation route has no accounted instance. Every path here differentiates on a face at maximal support; an iterate with a perturbed gradient adds no bracket.

Remark 19.4 (A perturbed gradient bounds another function). The rules drawn from Theorem 8.4 (i) and (ii), the face rule of Requirement 15.7 and the one-sided rule of Requirement 15.8, are the rule of every path this module specifies, without exception, and there is no second path: no numerical lower-bound certificate is specified here, so no route perturbs into the strictly positive cone instead of differentiating on a face, and neither rule is another path's concern. Requirement 15.8 names two routes to a usable gradient: keep the support maximal, "for instance after an explicit perturbation". Only the first is available to Theorem 19.1, and (H-NOPERT) says so; it is the only route this module supplies an accounting for.

A perturbation is not unsound in principle; a gradient taken at a perturbed operator is the gradient of a *different function*, and this module states no relation between the two. Concretely: replacing X by $X^{(\delta)} = (1-\delta)X + \delta\,\sigma$ for some $\sigma \succ 0$ and evaluating ∇T there yields the linearisation of an objective $O^{(\delta)}_\alpha$ built from the perturbed branch operators. To conclude anything about $\underline{h}_\alpha = \inf O^J_\alpha$ from a bound on $\inf O^{(\delta)}_\alpha$ one needs a *domination* statement, $O^{(\delta)}_\alpha \le O^J_\alpha$ pointwise or an explicit two-sided error, and this module proves none. The phrase "after an explicit perturbation" in Requirement 15.8 therefore has no accounted instance anywhere in this module. It stands there as a statement about what a solver may do to keep

a gradient defined, which is true and is all it says.

The consequence for a run is a *withdrawal*, not a weakening: an iterate whose gradient was taken after a perturbation contributes no bracket to $\underline{h}_\alpha^{\mathrm{FO}}$ (Requirement 24.1), so a run that perturbs the gradient at every iterate reports no $\underline{h}_\alpha^{\mathrm{FO}}$ at all. Reporting a perturbed bracket as a first-order bound would be reporting a bound on a function nobody asked about.

Uses Thm. 8.4, Thm. 19.1 · *Used by* Rem. 23.8 · newly derived · accepted

With max{0, h_lb} in place of the rate, the fixed-length key-length bound gives an ε_pa-secret and ε_ev-correct key for a Rényi order in (1, 2), whether the iterates are exact or relaxed. If h_lb ≤ 0 and ε_ev < 1, no key is claimed.

The first-order bound enters the core's key-length corollary as the lower bound that corollary asks for. Parts (ii) and (iiib) of Theorem 19.1 conclude the same inequality $\max\{0, \underline{h}_\alpha^{\mathrm{FO}}\} \le \underline{h}_\alpha$ against the same exact $\underline{h}_\alpha$, and (iiib), which admits an iterate in the relaxed feasible set, is the part a run of this module is in (Requirement 25.1).

Corollary 19.5 (Key length from the first-order bound)**.** Let $\alpha \in (1, 2)$, let the hypotheses of Theorem 10.7 hold in its unconstrained or constrained branch, the run as in Lemma 10.5 (a) and hypothesis (H-CHAIN) included, at every all-at-once attack with finite registers, with $\mathcal{W}^J$ carrying the declared $\mathfrak{L}$ in the constrained branch, let those of Lemma 10.4 hold, A-SEQ with its clause (v) among the former, and let $\underline{h}_\alpha^{\mathrm{FO}}$ be as in Theorem 19.1 (ii) or (iiib), at some or all iterates with the lower bounds of its part (iv) in place of the pairings. If the output length satisfies

$$\ell \;\le\; n\,\max\{0, \underline{h}_\alpha^{\mathrm{FO}}\} - \lambda_{\mathrm{EC}} - \left\lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \right\rceil - \frac{\alpha}{\alpha - 1} \log_2 \frac{1}{\varepsilon_{\mathrm{PA}}} + 1, \qquad (19.7)$$

then the protocol is $\varepsilon_{\mathrm{PA}}$-secret and $\varepsilon_{\mathrm{EV}}$-correct on the accepted state, hence, for $\ell \ge 1$, $(\varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}})$-secure in the honest-authentication setting, and under the further hypotheses of Corollary 10.8 (iii) $(\varepsilon_{\mathrm{PA}} + \varepsilon_{\mathrm{EV}} + 2\varepsilon_{\mathrm{auth}})$-secure over the classical channel of that part, with net key gain $\ell - k_{\mathrm{auth}}$ when both parties output a key, the secrecy and the security statements holding at every output state, whatever the dimension of the adversary's registers and the order in which the source releases its signals, as Corollary 10.8 states them, and the correctness statement at every attack; the four conclusions of Proposition 10.17 hold verbatim for the maximal key length built from (19.7); and if $\underline{h}_\alpha^{\mathrm{FO}} \le 0$ and $\varepsilon_{\mathrm{EV}} < 1$ no key is claimed.

Uses Lem. 10.5, Lem. 10.6, Cor. 10.18, Thm. 19.1 · *Used by* Cor. 19.7, Cor. 21.3, Cor. 21.4, Rem. 22.4, Rem. 23.3, Rem. 23.9, Req. 24.3, Req. 24.7 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.general_keyrate:key_length_from_first_order · *Oracles* registry.py

Proof. This is Corollary 10.18 of the core with $h := \underline{h}_\alpha^{\mathrm{FO}}$. Theorem 19.1 (ii), respectively (iiib), with its part (iv) where lower bounds stand in for the

pairings, gives $\underline{h}_\alpha^{\mathrm{FO}} \le \underline{h}_\alpha$ in exact real arithmetic, which is that corollary's hypothesis (10.17), and (19.7) is its (10.18). Its clause (a) applies because $\underline{h}_\alpha^{\mathrm{FO}}$ depends on the iterates, the pairs or lower bounds its brackets use and the declaration, and on none of $(\lambda_{\mathrm{EC}}, \varepsilon_{\mathrm{EV}}, \varepsilon_{\mathrm{PA}})$; its clause (b) is the last sentence of the statement. Nothing is added to the core's argument: no additional subtrahend, no floor beyond the one $\max\{0, \cdot\}$ already supplies, and no arithmetic requirement of any kind. □

In either source mode the first-order bound holds for the dropped program (Definition 9.2), with maximal support asked only of key-active pairs (acceptance value, joint sector) whose entropy is kept. It bounds h_low from below, also at iterates feasible only up to windows.

A run that gives up the deficits of a dropped set takes its first-order bound from the dropped program. The corollary is to that program what Theorem 19.1 (ii) and (iiib) are to the undropped one, with the support hypothesis asked of each kept key-active joint pair and positivity asked of the dropped branch quantity, which is never smaller than the undropped one. A fully dropped joint sector needs no maximal support, no rank decision and no matrix function. In the relaxed reading the three hypotheses are conditions on the iterate alone and name no set, and no quantity relaxed at $\varepsilon > 0$ appears in the conclusion: the chain (19.8) compares $\underline{h}_\alpha^{\mathrm{FO},\mathcal{D}}$ against the exact $\underline{h}_\alpha^{\mathcal{D}}$ directly, and no non-negativity of a relaxed infimum is claimed. On an empty feasible set both values are $+\infty$, so the bound needs the non-empty feasible set of an admissible declaration only to be informative. The corollary holds in either source mode: in untrusted-source mode the Choi step is the identity, so the frame and the operator coupling of the trusted-source proof become the state blocks and the normalisation.

Corollary 19.6 (First-order bound for the dropped program). Let $\alpha \in (1, \infty)$, let $\mathfrak{D}$ satisfy Definition 1.2 in the general format of Definition 1.1, in either source mode, in trusted-source mode with a frame per Alice sector as in Theorem 7.5, let $\mathcal{S}_{\mathrm{acc}}$ satisfy Definition 2.5, and let $\mathcal{D}$ be a dropped set (Definition 9.2). Write an Alice sector as $\mathtt{t} \in \mathbb{T}$ in this claim, the letter t indexing iterates as in Theorem 19.1. Put $O_\alpha^{J,\mathcal{D}}(\mathbf{q}, J) := O_\alpha^{\mathcal{D}}(\mathbf{q}, \chi(J))$ and $K_{\hat{c}}^{J,\mathcal{D}} := K_{\hat{c}}^{\mathcal{D}} \circ \chi$, with χ the assignment of that theorem in trusted-source mode. In untrusted-source mode the program is read as Definition 7.10 reads it, with no frame: χ is the identity, $J = \omega$ is the list of the blocks $J_{(\mathtt{t},s)} = \omega^{(\mathtt{t},s)}$ over the joint sectors, $A_{0,\mathtt{t}}$ and A_0 stand for $A_{\mathtt{t}}$ and A, and $\mathcal{W}^J$ is $\mathcal{W}$ of Definition 5.1 in the general format, whose condition coupling the blocks is the normalisation $\sum_{(\mathtt{t},s)} \operatorname{tr} \omega^{(\mathtt{t},s)} = 1$; so $O_\alpha^{J,\mathcal{D}}$ and $K_{\hat{c}}^{J,\mathcal{D}}$ are $O_\alpha^{\mathcal{D}}$ and $K_{\hat{c}}^{\mathcal{D}}$. Let $x_t = (\mathbf{q}_t, J_t) \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$, $t \in \mathfrak{I}$, satisfy

(H-POS$^{\mathcal{D}}$) (H-POS) of Theorem 19.1 with $K_{\hat{c}}^{J,\mathcal{D}}$ in place of $K_{\hat{c}}^{J}$;

(H-SUPP$^{\mathcal{D}}$) for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and every joint pair $(\mathtt{t}, s) \in \mathbb{J}_K(\hat{c})$ with $(\hat{c}, (\mathtt{t}, s)) \notin$

$\mathcal{D}$, the branch operator $\mathcal{G}^{J}_{\hat{c},s}(J_{t,(\mathrm{t},s)})$ of the block $J_{t,(\mathrm{t},s)} \in \mathrm{Pos}(A_{0,\mathrm{t}}Q_s)$ attains the maximal support of that joint pair, the support it has at $J_{t,(\mathrm{t},s)} \succ 0$; nothing is asked of a pair that is key-inert or dropped;

(H-NOPERT$^{\mathcal{D}}$) (g_t, G_t) is the gradient of $O^{J,\mathcal{D}}_{\alpha}$ at x_t itself, assembled as in (H-NOPERT) with the kept summands only and with $K^{J,\mathcal{D}}_{\hat{c}}$ in the scalar prefactors, its blocks on fully dropped joint sectors being (9.5) in Choi coordinates;

and let (v_t, S_t) satisfy (18.2) for these linear forms, or let, at some or all t, lower bounds b^{A}_t, b^{F}_t on one or both of its minima, as in (19.3) for these linear forms, stand in for the pairings as Theorem 19.1 (iv) lets them. Then, in exact real arithmetic, with $\underline{h}^{\mathrm{FO},\mathcal{D}}_{\alpha}$ the expression (19.2) for $O^{J,\mathcal{D}}_{\alpha}$, with those replacements where they are made,

$$\max\{0, \underline{h}^{\mathrm{FO},\mathcal{D}}_{\alpha}\} \;\le\; \underline{h}^{\mathcal{D}}_{\alpha} \;\le\; \underline{h}_{\alpha} \;\le\; h^{\uparrow,\mathrm{QKD}}_{\alpha}(\mathfrak{L}). \tag{19.8}$$

(H-POS$^{\mathcal{D}}$) is implied by (H-POS), and (H-SUPP$^{\mathcal{D}}$) by (H-SUPP) read on joint blocks. For $\mathcal{D} = \varnothing$ and $|\mathbb{T}| = 1$ the statement is Theorem 19.1 (ii), with its part (iv) where lower bounds stand in.

The relaxed reading. Let $\varepsilon \ge 0$ and let $\mathcal{W}^{J}(\varepsilon)$ be the relaxed feasible set of Definition 17.1, read in the general format, with one coupling clause per Alice sector in trusted-source mode and its normalisation clause in untrusted-source mode. Let $x_t \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^{J}(\varepsilon)$, $t \in \mathfrak{I}$, satisfy (H-POS$^{\mathcal{D}}$), (H-SUPP$^{\mathcal{D}}$) and (H-NOPERT$^{\mathcal{D}}$) as stated, and let (v_t, S_t) satisfy (18.2) *as written*, or the lower bounds standing in for its pairings satisfy (19.3), with the two minima over $\mathcal{S}_{\mathrm{acc}}$ and over the *exact* $\mathcal{W}^{J}$. Then (19.8) holds in full, and the case $\varepsilon = 0$ is the statement above.

An empty feasible set. In place of the hypothesis on $\mathfrak{D}$ above, let $\mathfrak{D}$ be a tuple of the form of Definition 1.1, in the general format and in either source mode, that satisfies conditions (i)–(vi) and (viii) of Definition 1.2, and form $\mathcal{W}$, O_{α}, $\underline{h}_{\alpha}$ and $\underline{h}^{\mathcal{D}}_{\alpha}$ from it by the formulas of Definitions 5.1 and 9.2. Let $\mathcal{W} = \varnothing$. Then $\underline{h}^{\mathcal{D}}_{\alpha} = \underline{h}_{\alpha} = +\infty$, so every real number is a lower bound on both, the value of (19.2) for $O^{J,\mathcal{D}}_{\alpha}$ that a run forms at its iterates included. Consequently the first two inequalities of (19.8), for a real $\underline{h}^{\mathrm{FO},\mathcal{D}}_{\alpha}$, hold of every such tuple, whether or not it satisfies condition (vii): under the hypotheses above where it does, and by the preceding sentence where it does not.

Uses Def. 1.2, Def. 2.5, Def. 4.8, Lem. 4.9, Def. 5.1, Lem. 7.4, Thm. 7.5, Def. 7.10, Def. 8.1, Lem. 8.2, Thm. 8.4, Cor. 8.5, Def. 9.2, Prop. 9.3, Cor. 9.4, Def. 17.1, Prop. 17.2, Def. 18.1, Lem. 18.2, Lem. 18.3, Thm. 19.1, foundations/def:kl · *Used by* Lem. 21.1, Lem. 21.2, Cor. 21.3, Cor. 21.4, Rem. 23.3 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.joint_driver:run_joint_frank_wolfe

Proof. Idea. The first inequality of (19.8) follows the four steps of the proof of Theorem 19.1 (ii) with the changes below, preceded by the two things

the dropped program needs that that theorem's proof does not state: the support geometry of a joint pair (Step 0) and the acceptance-support bookkeeping for $K^{J,\mathcal{D}}$ (Step 0′). The relaxed reading changes Step 2 only. The proof is written once for both source modes, with the reading of untrusted-source mode fixed in the paragraph on notation.

The last two inequalities of (19.8) are Proposition 9.3 (iii).

Notation. Write $p = (\mathsf{t}, s)$ for a joint pair. In trusted-source mode put $W_{\mathsf{t}} \coloneqq (\tau_A^{(\mathsf{t})})^{1/2} \iota_{\mathsf{t}}$ and $\chi_p(\cdot) \coloneqq (W_{\mathsf{t}} \otimes \mathbb{1}_{Q_s})(\cdot)(W_{\mathsf{t}}^{\dagger} \otimes \mathbb{1}_{Q_s})$ for the assignment of Theorem 7.5 in the general format, so that $\chi = \bigoplus_p \chi_p$ and $W \coloneqq \bigoplus_{\mathsf{t}} W_{\mathsf{t}}$ satisfies $WW^{\dagger} = \tau_A$ (Lemma 7.4 (ii)), and $S_{\mathsf{t}} \coloneqq \operatorname{supp} \tau_A^{(\mathsf{t})} = \operatorname{ran} W_{\mathsf{t}}$ (Lemma 7.4 (i)). In untrusted-source mode (Definition 7.10 (a)) let W_{t} be the inclusion of $\mathcal{H}(A_{\mathsf{t}})$ into $\mathcal{H}(A)$ and χ_p the conjugation by $W_{\mathsf{t}} \otimes \mathbb{1}_{Q_s}$, that is, the inclusion of $\operatorname{Pos}(A_{\mathsf{t}} Q_s)$ into $\operatorname{Pos}(AQ_s)$ as the subcone of Definition 4.8; then $\chi = \bigoplus_p \chi_p$ is the identity on lists over the joint sectors, and $S_{\mathsf{t}} \coloneqq \mathcal{H}(A_{\mathsf{t}}) = \operatorname{ran} W_{\mathsf{t}}$. In both modes χ_p is linear and preserves positive semidefiniteness, and $\operatorname{ran} W_{\mathsf{t}} = S_{\mathsf{t}} \subseteq \mathcal{H}(A_{\mathsf{t}})$. Write $\mathcal{G}^J_{\hat{c},p} \coloneqq \mathcal{G}_{\hat{c},p} \circ \chi_p$, which is the restriction of $\mathcal{G}^J_{\hat{c},s}$ to the blocks supported on $\mathcal{H}(A_{0,\mathsf{t}} Q_s)$, the reading (H-SUPP$^{\mathcal{D}}$) writes, and $X_{\hat{c},p} \coloneqq \mathcal{G}^J_{\hat{c},p}(J_{t,p})$. Let $\widetilde{\Pi}_{\mathsf{t}}$ be Π^A_{t} acting on the A factor of the branch space $\mathcal{H}(R\,XY\,C_A C_B\,AQ_s)$, and put

$$W_{\hat{c},p} \coloneqq \widetilde{\Pi}_{\mathsf{t}}\, W_{\hat{c},s} \qquad \text{with } W_{\hat{c},s} \text{ of (8.1).} \tag{19.9}$$

Step 0 (the support geometry of a joint pair). (a) By (8.1), $W_{\hat{c},s} = \sum_{\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})} K_\kappa^{(s)}(\mathcal{S}_A \otimes \mathcal{H}(Q_s))$, and $\mathcal{S}_A = \bigoplus_{\mathsf{t}} S_{\mathsf{t}}$. In trusted-source mode $\mathcal{S}_A = \operatorname{supp} \tau_A$, and $\tau_A = \bigoplus_{\mathsf{t}} \tau_A^{(\mathsf{t})}$ is block diagonal along $\mathbb{T}$ by Definition 1.2 (iii) in the general format, so $\mathcal{S}_A = \bigoplus_{\mathsf{t}} \operatorname{supp} \tau_A^{(\mathsf{t})}$; in untrusted-source mode $\mathcal{S}_A = \mathcal{H}(A)$ (Definition 8.1), which is $\bigoplus_{\mathsf{t}} \mathcal{H}(A_{\mathsf{t}})$ by the general format of Definition 1.1. By the computation in the proof of Lemma 4.9 (i), $K_\kappa^{(s)}(\Pi^A_{\mathsf{t}} \otimes \mathbb{1}) = \widetilde{\Pi}_{\mathsf{t}} K_\kappa^{(s)}$, so each $K_\kappa^{(s)}(S_{\mathsf{t}} \otimes \mathcal{H}(Q_s))$ lies in the range of $\widetilde{\Pi}_{\mathsf{t}}$; summing over t and κ exhibits $W_{\hat{c},s}$ as a sum of subspaces of *mutually orthogonal* summands, hence

$$W_{\hat{c},s} = \bigoplus_{\mathsf{t} \in \mathbb{T}} W_{\hat{c},(\mathsf{t},s)}, \qquad W_{\hat{c},(\mathsf{t},s)} = \sum_{\kappa \in \mathcal{K}_{\mathrm{pass}}(\hat{c})} K_\kappa^{(s)}\big(S_{\mathsf{t}} \otimes \mathcal{H}(Q_s)\big). \tag{19.10}$$

(b) For every Hermitian Δ on $\mathcal{H}(A_{0,\mathsf{t}} Q_s)$, $\operatorname{supp} \mathcal{G}^J_{\hat{c},p}(\Delta) \subseteq W_{\hat{c},p}$. Indeed $\Delta \oplus 0$ is Hermitian on $\mathcal{H}(A_0 Q_s)$, so Lemma 18.3, (18.3) gives $\operatorname{supp} \mathcal{G}^J_{\hat{c},s}(\Delta \oplus 0) \subseteq W_{\hat{c},s}$; and $\chi_p(\Delta_\pm)$ is supported on $\mathcal{H}(A_{\mathsf{t}} Q_s)$, $\operatorname{ran} W_{\mathsf{t}}$ lying in $\mathcal{H}(A_{\mathsf{t}})$, so by Lemma 4.9 (i) the image of each lies in the range of $\widetilde{\Pi}_{\mathsf{t}}$ and so does their difference. Intersecting the two inclusions gives the claim by (19.9). *(c)* For $\varrho \succeq 0$ the support of $\sum_\kappa K_\kappa \varrho K_\kappa^{\dagger}$ is $\sum_\kappa K_\kappa(\operatorname{supp} \varrho)$ (the support of a sum, in the proof of Lemma 8.2). At $J_{t,p} \succ 0$, $\operatorname{supp} \chi_p(J_{t,p}) = \operatorname{ran}(W_{\mathsf{t}} \otimes \mathbb{1}) = S_{\mathsf{t}} \otimes \mathcal{H}(Q_s)$, so $\operatorname{supp} X_{\hat{c},p} = W_{\hat{c},p}$ by (19.10), while for any $J_{t,p} \succeq 0$ the inclusion

$\subseteq$ holds. So the "maximal support of that joint pair" of (H-SUPP$^{\mathcal{D}}$) is $W_{\hat{c},p}$, and that hypothesis reads $\operatorname{supp} X_{\hat{c},p} = W_{\hat{c},p}$ for every kept key-active pair.

Step 0′ (the bookkeeping, which is Lemma 18.2 for $K^{J,\mathcal{D}}$). Part (i) of that lemma is a statement about $\mathcal{S}_{\mathrm{acc}}$ alone and is unchanged: every $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ vanishes outside the acceptance support $\widehat{\mathcal{C}}_{\mathrm{acc}}$ of (18.1), and for $\mathbf{q}, \mathbf{q}' \in \mathcal{S}_{\mathrm{acc}}$ every feasible direction has $q'(\hat{c}) - q(\hat{c}) = 0$ at every $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{acc}}$. Part (ii) holds for $O_\alpha^{J,\mathcal{D}}$ on $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, for every $\varepsilon \geq 0$, with the cone clause of Theorem 4.1 replaced by Proposition 9.3 (i), which states $0 \leq K_{\hat{c}}^{\mathcal{D}} \leq P_{\hat{c}}$ for every ω block diagonal along the joint sectors with positive semidefinite blocks; every point of $\mathcal{W}^J(\varepsilon)$ is per-block positive semidefinite, the one condition that set does not relax (Definition 17.1), while χ preserves the joint-sector block structure and positivity (Lemma 7.4 (ii); in untrusted-source mode χ is the identity). So for $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{acc}}$, where $q(\hat{c}) = 0$, the convention $0 \log_2 \frac{0}{v} = 0$ of Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl), stated for every $v \geq 0$ and the case $v = 0$ included, makes the term $\frac{\alpha}{\alpha-1} q(\hat{c}) \log_2 \frac{q(\hat{c})}{P_{\hat{c}}^J}$ and, for $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$, the pair $q(\hat{c}) \log_2 \frac{q(\hat{c})}{P_{\hat{c}}^J} + \frac{q(\hat{c})}{\alpha-1} \log_2 \frac{q(\hat{c})}{K_{\hat{c}}^{J,\mathcal{D}}}$ vanish at *every* point of that set, whatever $P_{\hat{c}}^J$ and $K_{\hat{c}}^{J,\mathcal{D}}$ are there and in particular when one of them vanishes. Deleting those terms changes no value, so $O_\alpha^{J,\mathcal{D}}$ is on that set a function of $(\mathbf{q}|_{\widehat{\mathcal{C}}_{\mathrm{acc}}}, J)$ alone, (H-POS$^{\mathcal{D}}$) is stated on $\widehat{\mathcal{C}}_{\mathrm{acc}}$, and on $\widehat{\mathcal{C}}_{\mathrm{key}} \cap \widehat{\mathcal{C}}_{\mathrm{acc}}$ for the K-part, exactly where it is consumed, and a component of the acceptance-side gradient indexed outside $\widehat{\mathcal{C}}_{\mathrm{acc}}$ is paired against a direction component that is 0 in every pairing below, so its value is never read. No normalisation, coupling condition or declared row is read in this step, which is why it holds on the relaxed sets and not on the exact set alone.

Step 1 (the gradient exists and is a directional derivative in every feasible direction). Fix t, write $x := x_t = (\mathbf{q}, J)$ and $\nabla O := (g_t, G_t)$, let $y = (\mathbf{q}', J') \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ and $h := y - x$, and put $\Delta_p := J'_p - J_{t,p}$, Hermitian as a difference of positive semidefinite operators. By Step 0′ the objective along the segment is the finite sum over $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{acc}}$ of $\frac{\alpha}{\alpha-1} q(\hat{c}) \log_2 \frac{q(\hat{c})}{P_{\hat{c}}^J}$ for $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{key}}$ and of the pair of Step 0′ for $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}} \cap \widehat{\mathcal{C}}_{\mathrm{acc}}$. In it $q(\hat{c})$ and $P_{\hat{c}}^J$ are affine and strictly positive at x by (H-POS$^{\mathcal{D}}$), and

$$K_{\hat{c}}^{J,\mathcal{D}}(J) = P_{\hat{c}}^J(J) + \sum_{p:\,(\hat{c},p)\notin\mathcal{D}} f_{\hat{c},p}\big(\chi_p(J_p)\big) \tag{19.11}$$

by (9.3), with $f_{\hat{c},p} = T \circ \mathcal{G}_{\hat{c},p} - \operatorname{tr} \circ \mathcal{G}_{\hat{c},p}$ and the second summand linear in J_p. Each kept summand of (19.11) falls into one of two cases. If p is key-inert for $\hat{c}$ then $f_{\hat{c},p} \equiv 0$ on the whole subcone (Lemma 4.9 (iii)), hence identically along the segment: the summand is constant, contributes 0 to the derivative, and no support hypothesis on it is consumed, which is why (H-SUPP$^{\mathcal{D}}$) asks nothing of it. If $p \in \mathbb{J}_K(\hat{c})$ and $(\hat{c}, p) \notin \mathcal{D}$ then, by (H-SUPP$^{\mathcal{D}}$) with Step 0(c), $\operatorname{supp} X_{\hat{c},p} = W_{\hat{c},p}$, and by Step 0(b) $\operatorname{supp} \mathcal{G}_{\hat{c},p}^J(\Delta_p) \subseteq$

$W_{\hat{c},p} = \operatorname{supp} X_{\hat{c},p}$; so Theorem 8.4 (i) applies to the pair $(X_{\hat{c},p}, \mathcal{G}^J_{\hat{c},p}(\Delta_p))$ and $t' \mapsto T(X_{\hat{c},p} + t'\mathcal{G}^J_{\hat{c},p}(\Delta_p))$ is differentiable at $t' = 0$, two-sidedly, the hypothesis being symmetric under $\Delta_p \mapsto -\Delta_p$. And the $+\infty$ case of Theorem 8.4 (ii) is unreachable: that case sits at directions strictly enlarging the support, Step 0(b) says no image of a Hermitian direction leaves $W_{\hat{c},p}$, and the directions this proof pairs against are admissible for it because $X_{\hat{c},p} + t'\mathcal{G}^J_{\hat{c},p}(\Delta_p) = \mathcal{G}^J_{\hat{c},p}\big((1-t')J_{t,p} + t'J'_p\big) \succeq 0$ for $t' \in [0,1]$. This is Lemma 18.3 (a), (b) read on the joint pair, with Step 0(b) in place of (18.3). A pair with $(\hat{c}, p) \in \mathcal{D}$ is absent from (19.11) altogether, so no matrix power, no eigendecomposition and no support condition of a fully dropped joint sector enters (Corollary 9.4 (iii)). The remaining dependence is through finitely many compositions of $\log_2$ with quantities (H-POS$^{\mathcal{D}}$) keeps strictly positive, all differentiable there. Hence $t' \mapsto O^{J,\mathcal{D}}_\alpha(x + t'h)$ is differentiable at $t' = 0$ with derivative $\langle \nabla O, h\rangle$, where, by the chain and sum rules exactly as in the proof of Corollary 8.5, whose maximal-support step is Step 0(b) here, the acceptance-side component is (8.10) with $K^{J,\mathcal{D}}_{\hat{c}}$ in place of $K^J_{\hat{c}}$, read as a vector in $\mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$, and the block of the feasible-side component at the joint pair p is (8.11) with the sums restricted to $\widehat{\mathcal{C}}_{\mathrm{acc}}$, respectively $\widehat{\mathcal{C}}_{\mathrm{key}} \cap \widehat{\mathcal{C}}_{\mathrm{acc}}$, with $K^{J,\mathcal{D}}_{\hat{c}}$ in the scalar prefactors, with $\Gamma^{(p)}_{\hat{c}}$ of (8.12) read on the joint pair (the operators $W_{\mathsf{t}}^\dagger M_x^{(\mathsf{t})} W_{\mathsf{t}} \otimes F_v^{(s)}$), and with the deficit piece $\Delta^{(p)}_{\hat{c}}$ carrying, in place of $\mathbb{1}[s \in \mathbb{S}_K(\hat{c})]$, the indicator $\mathbb{1}[p \in \mathbb{J}_K(\hat{c})$ and $(\hat{c}, p) \notin \mathcal{D}]$. That is the object (H-NOPERT$^{\mathcal{D}}$) names, and on a fully dropped joint sector every deficit piece is absent, so the block is $\chi_p^\dagger$ applied to (9.5), which is what that hypothesis says of such a sector.

Step 2 (convexity turns the derivative into an inequality). $O^{\mathcal{D}}_\alpha$ is jointly convex on $\Delta(\widehat{\mathcal{C}}) \times \mathcal{W}$ with values in $[0, +\infty]$ (Corollary 9.4 (ii)), and $O^{J,\mathcal{D}}_\alpha = O^{\mathcal{D}}_\alpha \circ (\mathrm{id} \times \chi)$ with χ *linear*, so it is convex on the convex preimage $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$, a pullback along a linear map that needs no injectivity, and $\mathcal{W}^J$ is convex and compact by Theorem 7.5 in its general format, in untrusted-source mode because it is $\mathcal{W}$ (Definition 5.1). x and y lie in that convex set and $O^{J,\mathcal{D}}_\alpha(x)$ is finite by (H-POS$^{\mathcal{D}}$), so the three-chord inequality for a convex extended-real function gives, exactly as in Step 2 of the proof of Theorem 19.1 (ii),

$$O^{J,\mathcal{D}}_\alpha(y) \;\geq\; O^{J,\mathcal{D}}_\alpha(x) + \langle \nabla O,\, y - x\rangle \qquad \text{for every } y \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J, \quad (19.12)$$

the case $O^{J,\mathcal{D}}_\alpha(y) = +\infty$ included. Taking the infimum over y: by Theorem 7.5 in its general format $\chi(\mathcal{W}^J) = \mathcal{W}$, in untrusted-source mode because χ is the identity and $\mathcal{W}^J = \mathcal{W}$, and surjectivity is all a value needs, so $\inf_{\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J} O^{J,\mathcal{D}}_\alpha = \inf_{\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}} O^{\mathcal{D}}_\alpha = \underline{h}^{\mathcal{D}}_\alpha$ by (9.4); hence $\underline{h}^{\mathcal{D}}_\alpha \geq O^{J,\mathcal{D}}_\alpha(x) + \min_{y \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J} \langle \nabla O, y - x\rangle$, the minimum attained because $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ is compact and the map linear.

Steps 3 and 4. The map $y \mapsto \langle \nabla O, y\rangle$ is a sum of one term in each factor and $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ is a product set, so the minimum separates over the product and (18.2), at the linear forms (g_t, G_t) of this corollary, bounds each

part below by the oracle return; substituting gives $\underline{h}_\alpha^{\mathcal{D}} \geq O_\alpha^{J,\mathcal{D}}(x_t) + \langle g_t, v_t - \mathbf{q}_t\rangle + \langle G_t, S_t - J_t\rangle$ for each $t \in \mathfrak{I}$. Where b_t^{A} or b_t^{F} stands in for a pairing, (19.3) bounds that part below by it instead, and the same substitution gives the bracket with the replacement made, exactly as in the proof of Theorem 19.1 (iv); where both stand in, no oracle return enters. $\mathfrak{I}$ is finite and non-empty, so the largest of those right-hand sides, which is (19.2) for $O_\alpha^{J,\mathcal{D}}$, that is $\underline{h}_\alpha^{\mathrm{FO},\mathcal{D}}$, is a lower bound too. Finally $\underline{h}_\alpha^{\mathcal{D}} \geq 0$ by Proposition 9.3 (iii), a statement about the exact set, so $\max\{0, \underline{h}_\alpha^{\mathrm{FO},\mathcal{D}}\}$ is the maximum of two lower bounds on $\underline{h}_\alpha^{\mathcal{D}}$ and is one. This is the first inequality of (19.8).

The implications, and the degenerate case. (H-POS) $\Rightarrow$ (H-POS$^{\mathcal{D}}$): $K_{\hat{c}}^{J,\mathcal{D}} \geq K_{\hat{c}}^J$ by Proposition 9.3 (i), so $K_{\hat{c}}^J > 0$ gives $K_{\hat{c}}^{J,\mathcal{D}} > 0$, and the other clauses are identical. (H-SUPP) on joint blocks $\Rightarrow$ (H-SUPP$^{\mathcal{D}}$): with $J_s = \bigoplus_{\mathsf{t}} J_{t,(\mathsf{t},s)}$, Lemma 4.9 (i) gives $X_{\hat{c},s} = \bigoplus_{\mathsf{t}} X_{\hat{c},(\mathsf{t},s)}$ along mutually orthogonal subspaces, so $\operatorname{supp} X_{\hat{c},s} = \bigoplus_{\mathsf{t}} \operatorname{supp} X_{\hat{c},(\mathsf{t},s)}$, while $W_{\hat{c},s} = \bigoplus_{\mathsf{t}} W_{\hat{c},(\mathsf{t},s)}$ by (19.10) with $\operatorname{supp} X_{\hat{c},(\mathsf{t},s)} \subseteq W_{\hat{c},(\mathsf{t},s)}$ by Step 0(c); two direct sums over the same orthogonal summands that are equal with term-wise inclusions are equal term by term, so maximal support of the sector gives maximal support of each of its joint pairs. At $\mathcal{D} = \varnothing$ and $|\mathbb{T}| = 1$: $K_{\hat{c}}^{\mathcal{D}} = K_{\hat{c}}$ and $O_\alpha^{\mathcal{D}} = O_\alpha$ by Definition 9.2, $\mathbb{J}_K(\hat{c}) = \{t_0\} \times \mathbb{S}_K(\hat{c})$ and $W_{\hat{c},(t_0,s)} = W_{\hat{c},s}$ by Definition 4.8 and (19.10), so the three hypotheses are (H-POS), (H-SUPP), (H-NOPERT) and (19.8) is (19.2) together with $\underline{h}_\alpha \leq h_\alpha^{\uparrow,\mathrm{QKD}}(\mathfrak{L})$: Theorem 19.1 (ii).

The relaxed reading. The proof is the one above with a single change, made in Step 2: x ranges over $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ while the comparison point y ranges over the *exact* $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$. Steps 0, 0′ and 1 are unchanged and read only per-block positive semidefiniteness of x: Step 0(b), (c) ask that of $J_{t,p}$, Step 0′ asks it through Proposition 9.3 (i), and Δ_p is Hermitian as a difference of a positive semidefinite J'_p from $\mathcal{W}^J$ and a positive semidefinite $J_{t,p}$ from $\mathcal{W}^J(\varepsilon)$, per-block positive semidefiniteness being the one condition that set does not relax (Definition 17.1). In Step 2 both endpoints lie in $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, which is convex (Proposition 17.2 (b); $\mathcal{S}_{\mathrm{acc}}$ by Definition 2.5), and $O_\alpha^{J,\mathcal{D}}$ is convex on it. The transfer is the one Proposition 17.2 (c) performs for Theorem 5.10, and it goes through for the dropped objective for the same reason: the convexity argument of Corollary 9.4 (ii) reads of its domain only that it is a convex set on which $P_{\hat{c}}$ is affine and $K_{\hat{c}}^{\mathcal{D}}$ is non-negative, concave and continuous; Corollary 9.4 (i) states that concavity and continuity on the whole *cone* of operators block diagonal along the joint sectors, while Proposition 9.3 (i) states $0 \leq K_{\hat{c}}^{\mathcal{D}} \leq P_{\hat{c}}$ there; and $\chi(\mathcal{W}^J(\varepsilon))$ is a convex subset of that cone, so $O_\alpha^{\mathcal{D}}$ is convex on $\mathcal{S}_{\mathrm{acc}} \times \chi(\mathcal{W}^J(\varepsilon))$ and pulls back along the linear χ. So the three-chord inequality gives (19.12) for every $y \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$, and the infimum is taken over *that* set, not over the relaxed one; Steps 3 and 4 are then as above, and the clamp reads Proposition 9.3 (iii) at $\varepsilon = 0$, where it holds. What this uses of the relaxed set, in full: that $\mathcal{W}^J \subseteq \mathcal{W}^J(\varepsilon)$ (Proposition 17.2 (a)); that $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ is convex

(*ibid.* (b)); that every member is per-block positive semidefinite; and the convexity transfer just made, whose own derivation consumes the compactness of (b) through the continuity of $K_{\hat{c}}^{\mathcal{D}}$, so compactness is used one level down and never in the minimisation. Of the *exact* set it uses compactness, non-emptiness, and that the infimum of $O_{\alpha}^{J,\mathcal{D}}$ over it is $\underline{h}_{\alpha}^{\mathcal{D}}$. No statement is made about the infimum of $O_{\alpha}^{J,\mathcal{D}}$ over the relaxed set, and none is used.

An empty feasible set. No claim is used for the two values being $+\infty$. By (9.4) and (5.4), $\underline{h}_{\alpha}^{\mathcal{D}}$ and $\underline{h}_{\alpha}$ are infima over $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and $\omega \in \mathcal{W}$; with $\mathcal{W} = \varnothing$ each inner infimum is over the empty set and is $+\infty$ for every $\mathbf{q}$, and so is the outer one, whether $\mathcal{S}_{\mathrm{acc}}$ is empty or not. Every real number is below $+\infty$. For the consequence: where the tuple satisfies condition (vii) as well, it satisfies Definition 1.2 in full, and with it the two conditions Definition 1.1 places on its raw-key map, which are conditions (v) and (viii); it is then a protocol declaration satisfying the hypothesis of the statement above; where it does not, $\mathcal{W} = \varnothing$, both values are $+\infty$, and a real number is below them. □

At any points, such as those of a search on the snapped declaration or of one continued on the companion, the bracket formed with the reference companion's objective, gradient, entries and linear minima bounds the companion's value from below; its key length is the reference model's.

Theorem 19.1 asks nothing about where its points came from: they may be any finite set of points of the relaxed feasible set at which its hypotheses hold. A run may therefore search on the declaration $\mathfrak{D}$ compiled from the snapped measurement and form the bracket (19.2) on the compiled reference companion $\mathfrak{D}^{\mathrm{ref}}$ at the points the search produced. The snap then only steers the search, and its displacement plays no part in the bound (Remark 11.4). The corollary is the theorem read for $\mathfrak{D}^{\mathrm{ref}}$, with the hypotheses that carry over from $\mathfrak{D}$ named; what a run evaluates, and how, is Requirement 24.7, and the arithmetic of that evaluation is Remark 22.4.

Corollary 19.7 (The first-order bound on the reference companion). Let $\mathfrak{D}$ and $\mathfrak{D}^{\mathrm{ref}}$ be as in Proposition 12.1, let $\alpha \in (1, \infty)$, and let $\mathcal{S}_{\mathrm{acc}}$ be an admissible acceptance set (Definition 2.5), the one of the run, read for both. Let $\varepsilon \geq 0$ and let $x_t = (\mathbf{q}_t, J_t)$, $t \in \mathfrak{I}$, be a finite non-empty set of points, produced by any computation, a search on $\mathfrak{D}$ among them, such that every x_t lies in $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^{J}(\varepsilon)$ of $\mathfrak{D}^{\mathrm{ref}}$ (Definition 17.1) and satisfies (H-POS), (H-SUPP) and (H-NOPERT) of Theorem 19.1 for the objective, the branch maps and the gradient of $\mathfrak{D}^{\mathrm{ref}}$, and let (v_t, S_t) satisfy (18.2) at the gradient of $\mathfrak{D}^{\mathrm{ref}}$ at x_t, its feasible-side minimum over the exact $\mathcal{W}^{J}$ of $\mathfrak{D}^{\mathrm{ref}}$ (Definition 18.1), or let, at some or all t, lower bounds on one or both of those minima stand in for the pairings as Theorem 19.1 (iv) lets them, (19.3) read at that gradient and over those sets. Write $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}^{\mathrm{ref}}]$ for (19.2) so formed. Then:

(i) in exact real arithmetic, $\max\{0, \underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}^{\mathrm{ref}}]\} \leq \underline{h}_{\alpha}[\mathfrak{D}^{\mathrm{ref}}]$, and $\underline{h}_{\alpha}[\mathfrak{D}^{\mathrm{ref}}] \leq$

$\underline{h}_\alpha[\mathfrak{D}^{\mathrm{ref}}_{\mathrm{ex}}]$ for the companion with its unrelaxed entries;

(ii) the Choi variables of $\mathfrak{D}$ and $\mathfrak{D}^{\mathrm{ref}}$ are lists over the same joint sectors with the same block dimensions, so every point of the one program is a point of the other; where no link the companion replaces carries entries, as on the encoded route, $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ is the same set for both; and a point with $J_{t,s} \succ 0$ for every sector s satisfies (H-SUPP) for $\mathfrak{D}^{\mathrm{ref}}$, whatever its measured family;

(iii) where the hypotheses of Corollary 19.5 hold for $\mathfrak{D}^{\mathrm{ref}}$, its conclusions hold with $\underline{h}^{\mathrm{FO}}_\alpha[\mathfrak{D}^{\mathrm{ref}}]$ in (19.7): the key length so formed is secret and correct for the protocol whose declaration is $\mathfrak{D}^{\mathrm{ref}}$.

Uses Def. 1.2, Def. 2.5, Def. 4.8, Lem. 8.2, Prop. 12.1, Def. 17.1, Def. 18.1, Thm. 19.1, Cor. 19.5 · *Used by* Cor. 21.4, Rem. 22.4, Req. 24.7 · newly derived · accepted

Proof. (i) By Proposition 12.1 (ii), $\mathfrak{D}^{\mathrm{ref}}$ is a tuple of the form of Definition 1.2 that satisfies its conditions (i)–(vi) and (viii). Where it satisfies (vii) as well it is an admissible declaration, and Theorem 19.1 (iiib) applies to it with the given points and pairs, with its part (iv) where lower bounds stand in, whose hypotheses are exactly those listed: membership in the relaxed set of $\mathfrak{D}^{\mathrm{ref}}$, (H-POS), (H-SUPP) and (H-NOPERT) for its objects, and (18.2), or (19.3) where a lower bound stands in, over the exact sets of $\mathfrak{D}^{\mathrm{ref}}$. The theorem reads of the points nothing but these hypotheses, so it holds whatever computation produced them. Where (vii) fails, Theorem 19.1 (v) gives the same conclusion, $\underline{h}_\alpha[\mathfrak{D}^{\mathrm{ref}}] = +\infty$. The second inequality is Proposition 12.1 (iii).

(ii) The Choi variable of a declaration in the general format is a list of positive semidefinite blocks over its joint sectors, of dimension the product of the Alice sector's and the measured sector's (Definition 4.8); these are shared by Proposition 12.1 (i). The relaxed set of Definition 17.1 is cut out by per-block positivity, the coupling condition and the declared entries; the first two are shared, and the entries differ only in those of the replaced links (Proposition 12.1 (i)), so where none of those links has any the two sets coincide. By Lemma 8.2 (i), $J_{t,s} \succ 0$ gives every key-active branch operator its maximal support, for any declaration, which is (H-SUPP).

(iii) Corollary 19.5 holds for any $\underline{h}^{\mathrm{FO}}_\alpha$ of Theorem 19.1 (ii) or (iiib), with the lower bounds of its part (iv) where they stand in, under its hypotheses, and (i) gives one for $\mathfrak{D}^{\mathrm{ref}}$. □

The first-order theorem holds as written in the general format, since no step of its proof reads a constraint row, the number of measured factors or the number of Alice sectors. In trusted-source mode the linear minimisation splits into one conic program per Alice sector.

Remark 19.8 (The first-order bound in the general format). *(1) Theorem 19.1 is stated for a declaration in the general format* of Definition 1.1 as well, with $\mathcal{W}^J$ the set of lists over the joint sectors of Theorem 7.5 and $J_s := \bigoplus_{\mathrm{t}} J_{(\mathrm{t},s)}$ for every sector s of the measured system, and the last paragraph of its proof checks each

step there. That check is made on the general format itself, since the basic format cannot express the general case: a party-local entry, a transcript protocol of more than two messages and an ensemble form measurement slot are each outside it, so no appeal to "the same declaration with its factor tagged A forgotten" would carry the general case, and none is made. *What Proposition 5.5 does and does not contribute here.* It says that restricting the variable to states block diagonal along Alice's sectors costs nothing, which is why the joint-sector program is the right program to run and not a weaker one; it is not what carries the theorem, and the inclusion of the joint-sector Choi set into a Choi set of the A-forgotten declaration is not used, nor is it available in general, that declaration's Choi feasible set carrying only lifted-operator rows and no party-local row. In the pairing $\langle G_t, S_t - J_t\rangle$ only the blocks of the gradient on the joint sectors enter, S_t and J_t being lists over them. (H-SUPP) is asked of $\mathcal{G}^J_{\hat{c},s}(J_s)$ on the whole sector s, and $J_{(\mathsf{t},s)} \succ 0$ for every t suffices for it, since then $J_s \succ 0$. The weaker hypothesis per joint pair, which does not ask anything of key-inert joint pairs inside a key-active sector, is Corollary 19.6 at $\mathcal{D} = \varnothing$.

(2) The linear minimisation splits. In trusted-source mode $\mathcal{W}^J = \prod_{\mathsf{t}} \mathcal{W}^J_{\mathsf{t}}$ (Theorem 7.5, general format), so $\min_{J' \in \mathcal{W}^J} \langle G_t, J'\rangle = \sum_{\mathsf{t}} \min_{J'_{\mathsf{t}} \in \mathcal{W}^J_{\mathsf{t}}} \langle G_{t,\mathsf{t}}, J'_{\mathsf{t}}\rangle$: one independent conic program per Alice sector, each with its own coupling constraint of size $d_{A_{0,\mathsf{t}}}$. A lower bound on each sector's minimum, summed over the sectors, is therefore a lower bound on the minimum over the product, which is how Lemma 20.1 (ii) forms the feasible side of a bracket, one bound per Alice sector; and a pairing at most each sector's minimum, summed, is at most the minimum over the product, which is the second inequality of (18.2). In untrusted-source mode the single normalisation and every scalar entry couple all joint sectors and nothing splits. What does not split in either mode is the iteration: the gradient couples all blocks through the scalars $q(\hat{c})/P_{\hat{c}}$ and $q(\hat{c})/K_{\hat{c}}$.

(3) The relaxed feasible set per Alice sector. In the general format the relaxed feasible set of Definition 17.1 and the recorded fields of Remarks 23.1, 23.2, 23.3, 23.4, 23.5, 23.6, 23.7, 23.8 and 23.9 are read with the coupling constraint per Alice sector, and no statement of Proposition 17.2, of Remark 17.3 or of those remarks changes; what a run decides and records per sector is Requirement 25.5.

Uses Prop. 5.5, Thm. 7.5, Lem. 8.2, Def. 18.1, Thm. 19.1 · *Used by* Req. 25.5 · newly derived · accepted

20 The verified feasible side

The feasible-side minimum is bounded from below by weak duality: the coupling condition fixes the total trace of every feasible point, so any multipliers, feasible for the dual or not, give a valid bound once the least eigenvalue of the residual operator is bounded from below (Lemma 20.1). The bound reads no point a solver returned, and with a one-dimensional coupling and no rows it is the minimum itself. Rows built with a computed frame in place of the exact one cost a deduction linear in the row multipliers and in the frame's error, after which the bound holds over the exact set again.

Any multipliers for the coupling condition and the declared rows give a lower bound on a linear form's minimum over the exact Choi feasible set, dual-feasible or not, since the coupling fixes every feasible point's total trace. Rows built from an inexact frame cost an explicit deduction.

Part (iv) of Theorem 19.1 admits any real number below the feasible-side minimum $\min_{J' \in \mathcal{W}^J} \langle G_t, J' \rangle$ in place of the pairing with a point an oracle returned. The lemma supplies one by weak duality, from multipliers for the coupling condition and for the declared rows. What makes it hold for *any* multipliers, and not only at a feasible point of a dual program, is that the coupling condition fixes the total trace of every point of the feasible set exactly: the least eigenvalue of the residual operator, however negative, then enters with a known coefficient. Part (i) states this for an abstract set cut out by a coupling condition and affine operator rows, part (ii) reads the Choi feasible set of Theorem 7.5 in that form, and part (iii) shows that the bound is attained on a set with a one-dimensional coupling and no rows. A run can form the rows only with a computed frame, which in general differs from the exact W_{t} of Theorem 7.5; part (iv) says what that costs: a deduction linear in the row multipliers and in a bound on the frame's error, after which the bound holds over the exact set again.

Lemma 20.1 (A lower bound on the feasible-side minimum from any multipliers)**.** (i) *(Weak duality with a trace identity.)* Let $\mathcal{H}_{\mathrm{in}}$ be a Hilbert space of finite dimension $d_{\mathrm{in}} \geq 1$, let P be a finite non-empty set, and for every $p \in P$ let $\mathcal{H}_p$ be a finite-dimensional Hilbert space. Let $\mathcal{R}$ be a finite, possibly empty, set of *rows*; a row r consists of a finite-dimensional Hilbert space $\mathcal{K}_r$, a Hermitian operator Θ_r on it, a relation $\bowtie_r \in \{\preceq, \succeq, =\}$ and, for every $p \in P$, a linear map $L_{r,p}$ from operators on $\mathcal{H}_{\mathrm{in}} \otimes \mathcal{H}_p$ to operators on $\mathcal{K}_r$ that maps Hermitian operators to Hermitian ones. Write $L_{r,p}^*$ for its adjoint, the real-linear map from Hermitian operators on $\mathcal{K}_r$ to Hermitian operators on $\mathcal{H}_{\mathrm{in}} \otimes \mathcal{H}_p$ with $\operatorname{tr}[\Lambda\, L_{r,p}(X)] = \operatorname{tr}\big[L_{r,p}^*(\Lambda)\, X\big]$ for all Hermitian Λ and X. Let $\mathcal{J}$ be the set of lists $J = (J_p)_{p \in P}$ of positive semidefinite operators J_p on $\mathcal{H}_{\mathrm{in}} \otimes \mathcal{H}_p$ with

$$\sum_{p \in P} \operatorname{tr}_{\mathcal{H}_p} J_p = \mathbb{1}_{\mathcal{H}_{\mathrm{in}}} \qquad \text{and} \qquad L_r(J) \coloneqq \sum_{p \in P} L_{r,p}(J_p) \;\bowtie_r\; \Theta_r \quad \text{for every } r \in \mathcal{R}. \tag{20.1}$$

Let $G = (G_p)_{p \in P}$ be a list of Hermitian operators G_p on $\mathcal{H}_{\mathrm{in}} \otimes \mathcal{H}_p$, let Y be a Hermitian operator on $\mathcal{H}_{\mathrm{in}}$, and for every row r let Λ_r be a Hermitian operator on $\mathcal{K}_r$, with $\Lambda_r \succeq 0$ when $\bowtie_r$ is $\preceq$ or $\succeq$. Put $\sigma_r \coloneqq -1$ when $\bowtie_r$ is $\preceq$ and $\sigma_r \coloneqq +1$ otherwise,

$$R_p \;\coloneqq\; G_p - Y \otimes \mathbb{1}_{\mathcal{H}_p} - \sum_{r \in \mathcal{R}} \sigma_r\, L_{r,p}^*(\Lambda_r) \qquad (p \in P), \tag{20.2}$$

and let λ be a real number with $\lambda \le \lambda_{\min}(R_p)$ for every $p \in P$. Then for every $J \in \mathcal{J}$

$$\langle G, J\rangle \coloneqq \sum_{p\in P} \operatorname{tr}[G_p J_p] \;\ge\; b(G; Y, \Lambda, \lambda) \coloneqq d_{\text{in}}\,\lambda + \operatorname{tr} Y + \sum_{r\in\mathcal{R}} \sigma_r \operatorname{tr}[\Lambda_r \Theta_r], \tag{20.3}$$

so that $b(G; Y, \Lambda, \lambda) \le \inf_{J\in\mathcal{J}} \langle G, J\rangle$. No other condition is placed on Y, on the Λ_r or on λ: Y and the Λ_r need not be feasible for any dual program, λ may be negative, and $Y = 0$ with every $\Lambda_r = 0$ is admitted.

(ii) *(The Choi feasible set is of that form.)* Let $\mathfrak{D}$ be a declaration in the general format of Definition 1.1, the basic format included, with its Alice sectors $\mathtt{t} \in \mathbb{T}$.

(a) In trusted-source mode, with a frame and the factors $W_\mathtt{t}$ as in Theorem 7.5, the factor $\mathcal{W}^J_\mathtt{t}$ of $\mathcal{W}^J = \prod_\mathtt{t} \mathcal{W}^J_\mathtt{t}$ (Theorem 7.5, the general format) is the set $\mathcal{J}$ of (i) with $\mathcal{H}_{\text{in}} = \mathcal{H}(A_{0,\mathtt{t}})$, $P = \mathbb{S}$, $\mathcal{H}_s = \mathcal{H}(Q_s)$ and $J_s = J_{(\mathtt{t},s)}$, and with the following rows. For every entry k read through its lifted operator, one row with $\mathcal{K} = \mathcal{H}(A_\mathtt{t})$, $\Theta = \Theta_k^{(\mathtt{t})}$, the relation $\bowtie_k$ and

$$L_{k,s}(X) = W_\mathtt{t}\, \operatorname{tr}_{Q_s}\big[(\mathbb{1}_{A_{0,\mathtt{t}}} \otimes \hat{O}_k^{\uparrow(s)})X\big]\, W_\mathtt{t}^\dagger, \qquad L^*_{k,s}(\Lambda) = (W_\mathtt{t}^\dagger \Lambda W_\mathtt{t}) \otimes \hat{O}_k^{\uparrow(s)}.$$

For every party-local entry k, scoped to the measured factor f, and every s_{f^c}, one row with $\mathcal{K} = \mathcal{H}(A_\mathtt{t}) \otimes \mathcal{H}(Q_{f^c,s_{f^c}})$, $\Theta = 0$, the relation $\bowtie_k$ and, at $s = (s_{f^c}, s_f)$,

$$L_{(k,s_{f^c}),s}(X) = (W_\mathtt{t} \otimes \mathbb{1})\, \operatorname{tr}_{Q_{f,s_f}}\big[(\mathbb{1} \otimes D_k^{(s_f)})X\big]\, (W_\mathtt{t}^\dagger \otimes \mathbb{1}),$$
$$L^*_{(k,s_{f^c}),s}(\Lambda) = \big((W_\mathtt{t}^\dagger \otimes \mathbb{1})\Lambda(W_\mathtt{t} \otimes \mathbb{1})\big) \otimes D_k^{(s_f)},$$

where $D_k^{(s_f)} \coloneqq \hat{O}_k^{(s_f)} - c_k\mathbb{1}$ acts on $\mathcal{H}(Q_{f,s_f})$ at the place of that factor in $\mathcal{H}(Q_s)$, while $L_{(k,s_{f^c}),s'} = 0$ and $L^*_{(k,s_{f^c}),s'} = 0$ at every sector s' whose component on the other factors is not s_{f^c}. The two kinds of map, and the formulas for their adjoints, are defined, and the formulas hold, for any linear map W from $\mathcal{H}(A_{0,\mathtt{t}})$ to $\mathcal{H}(A_\mathtt{t})$ in place of $W_\mathtt{t}$; we write $L^W_{r,s}$ for a row's map so built. A homogeneous or party-local row in the frame-free form of Proposition 7.6, on a tight frame, is a row of the same kind with $\mathbb{1}_{A_{0,\mathtt{t}}}$ in place of $W_\mathtt{t}$, hence with $\mathcal{H}(A_{0,\mathtt{t}})$ in place of $\mathcal{H}(A_\mathtt{t})$ in its $\mathcal{K}$, and with $\Theta = c_k\mathbb{1}_{A_{0,\mathtt{t}}}$, respectively $\Theta = 0$; and $\mathcal{W}^J_\mathtt{t}$ is unchanged when it replaces the conjugated row. Hence, for $Y_\mathtt{t}$, $\Lambda_\mathtt{t} = (\Lambda_r)_r$ and $\lambda_\mathtt{t}$ as in (i) for each $\mathtt{t}$, and with $G_\mathtt{t} \coloneqq (G_{(\mathtt{t},s)})_{s\in\mathbb{S}}$,

$$b_{\text{F}} \;\coloneqq\; \sum_{\mathtt{t}\in\mathbb{T}} b(G_\mathtt{t}; Y_\mathtt{t}, \Lambda_\mathtt{t}, \lambda_\mathtt{t}) \;\le\; \langle G, J'\rangle \qquad \text{for every } J' \in \mathcal{W}^J. \tag{20.4}$$

(b) In untrusted-source mode, $\mathcal{W}^J$, which is $\mathcal{W}$ (Definition 18.1), is the set $\mathcal{J}$ of (i) with $\mathcal{H}_{\mathrm{in}}$ one-dimensional, so that $Y = y$ is a real number, $Y \otimes \mathbb{1} = y\,\mathbb{1}$, $\operatorname{tr} Y = y$ and $d_{\mathrm{in}} = 1$; with P the set of joint sectors, $\mathcal{H}_{(\mathsf{t},s)} = \mathcal{H}(A_{\mathsf{t}}Q_s)$ and $J_{(\mathsf{t},s)} = \omega^{(\mathsf{t},s)}$; and with these rows. For every scalar entry k, one row with $\mathcal{K} = \mathbb{C}$, $\Theta = c_k$, the relation $\bowtie_k$, $L_{k,(\mathsf{t},s)}(X) = \operatorname{tr}\Big[(\mathbb{1}_{A_{\mathsf{t}}} \otimes \hat{O}_k^{\uparrow(s)})X\Big]$ and $L^*_{k,(\mathsf{t},s)}(\mu) = \mu\,\mathbb{1}_{A_{\mathsf{t}}} \otimes \hat{O}_k^{\uparrow(s)}$; for every party-local entry k and every pair (t, s_{f^c}), one row as in (a) with $\mathbb{1}_{A_{\mathsf{t}}}$ in place of W_{t}, its maps vanishing outside the joint sectors $(\mathsf{t}, (s_{f^c}, s_f))$. Hence $b(G; y, \Lambda, \lambda) \le \langle G, J' \rangle$ for every $J' \in \mathcal{W}^J$, the whole list being one set of the form of (i).

(iii) *(A scalar coupling and no rows.)* If $d_{\mathrm{in}} = 1$ and $\mathcal{R} = \varnothing$ in (i), then, with $Y = 0$ and $\lambda = \min_{p \in P} \lambda_{\min}(G_p)$, the infimum in (i) is attained and $b(G; 0, \Lambda, \lambda) = \min_{J \in \mathcal{J}} \langle G, J \rangle$.

(iv) *(Rows built from an inexact frame.)* In trusted-source mode fix an Alice sector t, a linear map $\widetilde{W}_{\mathsf{t}}$ from $\mathcal{H}(A_{0,\mathsf{t}})$ to $\mathcal{H}(A_{\mathsf{t}})$ and a real number $\varepsilon_{\mathsf{t}} \ge \left\| \widetilde{W}_{\mathsf{t}} - W_{\mathsf{t}} \right\|$. For every row r of $\mathcal{W}^J_{\mathsf{t}}$ in (ii)(a) let $\widetilde{L}_{r,s} \coloneqq L^{\widetilde{W}_{\mathsf{t}}}_{r,s}$, a row in frame-free form keeping its map, every other datum of the row $(\Theta_r, \hat{O}_k^{\uparrow(s)}, D_k^{(s_f)}, c_k)$ unchanged. Let G_{t}, Y and the Λ_r be as in (i), form $\widetilde{R}_s$ by (20.2) with $\widetilde{L}^*_{r,s}$ in place of $L^*_{r,s}$, and let $\widetilde{\lambda} \le \lambda_{\min}(\widetilde{R}_s)$ for every $s \in \mathbb{S}$. Then for every $J \in \mathcal{W}^J_{\mathsf{t}}$, the factor cut out by the rows built from the exact W_{t},

$$\begin{aligned} \langle G_{\mathsf{t}}, J \rangle \;&\ge\; b(G_{\mathsf{t}}; Y, \Lambda, \widetilde{\lambda}) - \sum_r \kappa_r, \\ \kappa_r \coloneqq d_{A_{0,\mathsf{t}}}\, \varepsilon_{\mathsf{t}} \Big(2 \left\| \widetilde{W}_{\mathsf{t}} \right\| + \varepsilon_{\mathsf{t}}\Big)\; \|\Lambda_r\| \max_{s \in \mathbb{S}} \|O_{r,s}\|\,, \end{aligned} \tag{20.5}$$

where $O_{r,s}$ is $\hat{O}_k^{\uparrow(s)}$ for a lifted-operator row, $D_k^{(s_f)}$ for a party-local row at $s = (s_{f^c}, s_f)$ and 0 at the other sectors, and $\kappa_r \coloneqq 0$ for a row in frame-free form. Summed over t as in (ii)(a), the right-hand sides bound $\langle G, J' \rangle$ from below for every $J' \in \mathcal{W}^J$.

Uses Def. 1.1, Def. 4.8, Def. 5.1, Thm. 7.5, Prop. 7.6, Def. 7.10, Def. 18.1 · *Used by* Lem. 21.1, Rem. 22.4, Req. 27.1, Req. 28.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.verified:verified_feasible_side

Proof. Idea. The linear form is split into the residual paired with J and two terms the constraints fix or bound: the coupling condition turns the Y-term into $\operatorname{tr} Y$, and each row turns its term into at least $\sigma_r \operatorname{tr}[\Lambda_r \Theta_r]$. The residual term is at least λ times the total trace, which the coupling condition fixes at d_{in}, so a λ of either sign enters with a known coefficient.

(i), Step 1 (two identities). The adjoint exists and is unique: the Hermitian operators on a finite-dimensional space form a finite-dimensional real

inner product space under $(A,B)\mapsto \operatorname{tr}[AB]$, which is real for Hermitian A and B since $\overline{\operatorname{tr}[AB]}=\operatorname{tr}\big[(AB)^\dagger\big]=\operatorname{tr}[BA]$, and $L_{r,p}$ restricts to a real-linear map between two such spaces. Let $J\in\mathcal{J}$. For every operator X on $\mathcal{H}_{\mathrm{in}}\otimes\mathcal{H}_p$, $\operatorname{tr}\big[(Y\otimes\mathbb{1}_{\mathcal{H}_p})X\big]=\operatorname{tr}\big[Y\ \operatorname{tr}_{\mathcal{H}_p}X\big]$, the defining property of the partial trace; summing over p and using the coupling condition of (20.1),

$$\sum_{p\in P}\operatorname{tr}\big[(Y\otimes\mathbb{1}_{\mathcal{H}_p})J_p\big]=\operatorname{tr}\Big[Y\sum_{p\in P}\operatorname{tr}_{\mathcal{H}_p}J_p\Big]=\operatorname{tr}Y.$$

Every J_p is Hermitian, being positive semidefinite, so by the defining identity of the adjoint, for every row r, $\sum_p\operatorname{tr}\big[L^*_{r,p}(\Lambda_r)J_p\big]=\sum_p\operatorname{tr}[\Lambda_r\,L_{r,p}(J_p)]=\operatorname{tr}[\Lambda_r\,L_r(J)]$.

Step 2 (the rearrangement). By (20.2), $G_p=R_p+Y\otimes\mathbb{1}_{\mathcal{H}_p}+\sum_r\sigma_r L^*_{r,p}(\Lambda_r)$, so by Step 1

$$\langle G,J\rangle=\sum_{p\in P}\operatorname{tr}[R_pJ_p]+\operatorname{tr}Y+\sum_{r\in\mathcal{R}}\sigma_r\operatorname{tr}[\Lambda_r\,L_r(J)].$$

Step 3 (the residual term). For every p, R_p is Hermitian, G_p, $Y\otimes\mathbb{1}$ and every $L^*_{r,p}(\Lambda_r)$ being so, and $R_p-\lambda\mathbb{1}$ is positive semidefinite, its eigenvalues being those of R_p less λ, which are non-negative by the choice of λ. For positive semidefinite A and B, $\operatorname{tr}[AB]=\operatorname{tr}\big[B^{1/2}AB^{1/2}\big]\geq 0$. Hence $\operatorname{tr}[R_pJ_p]\geq\lambda\operatorname{tr}J_p$. The trace of J_p is that of $\operatorname{tr}_{\mathcal{H}_p}J_p$, so by the coupling condition $\sum_p\operatorname{tr}J_p=\operatorname{tr}\mathbb{1}_{\mathcal{H}_{\mathrm{in}}}=d_{\mathrm{in}}$ exactly, and

$$\sum_{p\in P}\operatorname{tr}[R_pJ_p]\ \geq\ \lambda\sum_{p\in P}\operatorname{tr}J_p\ =\ d_{\mathrm{in}}\,\lambda,$$

for either sign of λ, the total trace being an equality and not a bound.

Step 4 (the rows). If $\bowtie_r$ is $\succeq$, then $L_r(J)-\Theta_r\succeq 0$ and $\Lambda_r\succeq 0$, so $\operatorname{tr}[\Lambda_r(L_r(J)-\Theta_r)]\geq 0$ by the trace fact of Step 3, which is $\sigma_r\operatorname{tr}[\Lambda_rL_r(J)]\geq\sigma_r\operatorname{tr}[\Lambda_r\Theta_r]$ with $\sigma_r=+1$. If $\bowtie_r$ is $\preceq$, then $\Theta_r-L_r(J)\succeq 0$ and $\Lambda_r\succeq 0$ give $\operatorname{tr}[\Lambda_rL_r(J)]\leq\operatorname{tr}[\Lambda_r\Theta_r]$, which is the same inequality with $\sigma_r=-1$. If $\bowtie_r$ is $=$, then $L_r(J)=\Theta_r$ and the two sides are equal for every Hermitian Λ_r.

Substituting Steps 3 and 4 into Step 2 gives (20.3). As $J\in\mathcal{J}$ was arbitrary, $b(G;Y,\Lambda,\lambda)$ is below every value of $\langle G,\cdot\rangle$ on $\mathcal{J}$ and hence below its infimum, which is $+\infty$ when $\mathcal{J}$ is empty. No step used a property of Y, of the Λ_r or of λ beyond those stated in (i).

(ii)(a). By Theorem 7.5 in the general format, $\mathcal{W}^J$ is the set of lists $(J_{(\mathsf{t},s)})$ over the joint sectors with $J_{(\mathsf{t},s)}\in\operatorname{Pos}(A_{0,\mathsf{t}}Q_s)$, the coupling constraints $\sum_s\operatorname{tr}_{Q_s}J_{(\mathsf{t},s)}=\mathbb{1}_{A_{0,\mathsf{t}}}$, for every entry read through its lifted operator the rows $W_{\mathsf{t}}\big(\sum_s\operatorname{tr}_{Q_s}[(\mathbb{1}_{A_{0,\mathsf{t}}}\otimes\hat{O}_k^{\uparrow(s)})J_{(\mathsf{t},s)}]\big)W_{\mathsf{t}}^\dagger\bowtie_k\Theta_k^{(\mathsf{t})}$, and for every party-local entry the last line of (5.5) at J, conjugated by $W_{\mathsf{t}}\otimes\mathbb{1}$, which is one operator condition on $\mathcal{H}(A_{\mathsf{t}})\otimes\mathcal{H}(Q_{f^c,s_{f^c}})$ for every (t,s_{f^c}); every constraint involves the blocks of one Alice sector only, and $\mathcal{W}^J=\prod_{\mathsf{t}}\mathcal{W}^J_{\mathsf{t}}$ with $\mathcal{W}^J_{\mathsf{t}}$ cut out

by the constraints of $\mathtt{t}$. In the basic format the same holds with one Alice sector, the lifted operator of an entry being its own operator, and no party-local entry ((7.9)). These constraints of $\mathtt{t}$ are (20.1) with the data listed: the row of a lifted-operator entry is $\sum_s L_{k,s}(J_s) \bowtie_k \Theta_k^{(\mathtt{t})}$, and the row of a party-local entry at s_{f^c} is the sum over s_f of $L_{(k,s_{f^c}),(s_{f^c},s_f)}$, which is the sum over all s, the other terms being 0. Each listed map takes Hermitian operators to Hermitian ones: for a Hermitian X and a Hermitian operator O acting on one tensor factor Q' alone, $\big(\operatorname{tr}_{Q'}[(\mathbb{1}\otimes O)X]\big)^\dagger = \operatorname{tr}_{Q'}[X(\mathbb{1}\otimes O)] = \operatorname{tr}_{Q'}[(\mathbb{1}\otimes O)X]$, an operator on the traced factor alone moving cyclically under the partial trace over that factor, and a congruence preserves Hermiticity. Here $\hat{O}_k^{\uparrow(s)}$ is Hermitian, every $\hat{O}_k$ being exactly Hermitian per sector block (Definition 1.1, exact admissibility), and so is $D_k^{(s_f)}$, c_k being real. The adjoints: for Hermitian Λ on $\mathcal{H}(A_\mathtt{t})$ and X on $\mathcal{H}(A_{0,\mathtt{t}}Q_s)$, by cyclicity of the trace and the partial-trace identity of Step 1,

$$\operatorname{tr}\big[\Lambda\, W_\mathtt{t}\operatorname{tr}_{Q_s}[(\mathbb{1}\otimes\hat{O}_k^{\uparrow(s)})X]W_\mathtt{t}^\dagger\big] = \operatorname{tr}\big[\big((W_\mathtt{t}^\dagger\Lambda W_\mathtt{t})\otimes\mathbb{1}\big)(\mathbb{1}\otimes\hat{O}_k^{\uparrow(s)})X\big] = \operatorname{tr}\big[\big((W_\mathtt{t}^\dagger\Lambda W_\mathtt{t})\otimes\hat{O}_k^{\uparrow(s)}\big)X\big],$$

and the operator in the last trace is Hermitian; for a party-local row the same computation holds with $(W_\mathtt{t}^\dagger\otimes\mathbb{1})\Lambda(W_\mathtt{t}\otimes\mathbb{1})$ on $\mathcal{H}(A_{0,\mathtt{t}})\otimes\mathcal{H}(Q_{f^c,s_{f^c}})$ in place of $W_\mathtt{t}^\dagger\Lambda W_\mathtt{t}$ and $D_k^{(s_f)}$ on $\mathcal{H}(Q_{f,s_f})$ in place of $\hat{O}_k^{\uparrow(s)}$. Neither computation, nor the Hermiticity of the maps, reads any property of $W_\mathtt{t}$ beyond its being linear, so both hold for any linear W in its place. For the frame-free forms: on a tight frame a homogeneous row, read per Alice sector, and a party-local row hold for a list in their frame-free form exactly when they hold in their conjugated form (Proposition 7.6 (i) and (iii)), so $\mathcal{W}_\mathtt{t}^J$ is unchanged, and the frame-free row is of the stated kind by the same two computations with the identity in place of $W_\mathtt{t}$. Finally, for $J' \in \mathcal{W}^J$ its part $J'_\mathtt{t} \coloneqq (J'_{(\mathtt{t},s)})_s$ lies in $\mathcal{W}_\mathtt{t}^J$, and $\langle G, J'\rangle = \sum_\mathtt{t}\langle G_\mathtt{t}, J'_\mathtt{t}\rangle \geq \sum_\mathtt{t} b(G_\mathtt{t}; Y_\mathtt{t}, \Lambda_\mathtt{t}, \lambda_\mathtt{t})$ by (i) applied to each factor, which is (20.4).

(ii)(b). In untrusted-source mode the Choi step is the identity and $\mathcal{W}^J$ denotes $\mathcal{W}$ (Definitions 7.10 and 18.1). By (7.13) in the basic format and by the lines of (5.5) that hold in untrusted-source mode in the general format (Definition 5.1), $\mathcal{W}$ is the set of lists of positive semidefinite blocks $\omega^{(\mathtt{t},s)}$ over the joint sectors with $\sum_{(\mathtt{t},s)}\operatorname{tr}\omega^{(\mathtt{t},s)} = 1$, the scalar rows $\sum_{(\mathtt{t},s)}\operatorname{tr}\Big[(\mathbb{1}_{A_\mathtt{t}}\otimes\hat{O}_k^{\uparrow(s)})\omega^{(\mathtt{t},s)}\Big] \bowtie_k c_k$, and the party-local rows, unconjugated, one for every $(\mathtt{t}, s_{f^c})$. With $\mathcal{H}_\mathrm{in}$ one-dimensional, $\operatorname{tr}_{\mathcal{H}_p} J_p$ is the number $\operatorname{tr} J_p$ and $\mathbb{1}_{\mathcal{H}_\mathrm{in}}$ is 1, so the coupling condition of (20.1) is the normalisation. The scalar row's map is real on Hermitian X, a trace of a product of two Hermitian operators, and $\mu\operatorname{tr}\Big[(\mathbb{1}\otimes\hat{O}_k^{\uparrow(s)})X\Big] = \operatorname{tr}\Big[(\mu\,\mathbb{1}\otimes\hat{O}_k^{\uparrow(s)})X\Big]$ gives its adjoint; the party-local rows are those of (a) with the identity in place of $W_\mathtt{t}$. Then (i) applies to the whole list.

(iii). With $\mathcal{R} = \varnothing$ and $Y = 0$, (20.2) is $R_p = G_p$, so $\lambda = \min_p \lambda_{\min}(G_p)$ is admissible in (i) and $b(G; 0, \Lambda, \lambda) = d_\mathrm{in}\lambda = \lambda$ is at most the infimum. Let

p^* attain the minimum and let u be a unit eigenvector of G_{p^*} for $\lambda_{\min}(G_{p^*})$. The list with $J_{p^*} = uu^\dagger$ and $J_p = 0$ for $p \neq p^*$ is positive semidefinite, and, as $\mathcal{H}_{\text{in}}$ is one-dimensional, its coupling condition reads $\sum_p \operatorname{tr} J_p = 1$, which holds; there are no rows. So it lies in $\mathcal{J}$, and $\langle G, J\rangle = u^\dagger G_{p^*} u = \lambda$. The infimum is therefore attained and equals $b(G; 0, \Lambda, \lambda)$.

(iv). Write $W \coloneqq W_{\mathtt{t}}$, $\widetilde{W} \coloneqq \widetilde{W}_{\mathtt{t}}$, $\varepsilon \coloneqq \varepsilon_{\mathtt{t}}$ and $d \coloneqq d_{A_{0,\mathtt{t}}}$, and let $J \in \mathcal{W}^J_{\mathtt{t}}$: its blocks are positive semidefinite, it satisfies the exact coupling condition, and it satisfies every row built from W. *Step 1.* Steps 1 to 3 of (i), run with the maps $\widetilde{L}_{r,s}$ in place of $L_{r,s}$, read of the rows only the adjoint identity, which $\widetilde{L}_{r,s}$ satisfies by (ii)(a) for any linear W, and of J only positivity and the coupling condition; they give $\langle G_{\mathtt{t}}, J\rangle \geq d\,\widetilde{\lambda} + \operatorname{tr} Y + \sum_r \sigma_r \operatorname{tr}\Big[\Lambda_r \widetilde{L}_r(J)\Big]$. *Step 2 (the exact rows).* For each row put $\Delta_{r,s} \coloneqq \widetilde{L}^*_{r,s}(\Lambda_r) - L^*_{r,s}(\Lambda_r)$, a Hermitian operator. By the adjoint identities of both maps, $\operatorname{tr}\Big[\Lambda_r \widetilde{L}_r(J)\Big] = \operatorname{tr}[\Lambda_r L_r(J)] + \sum_s \operatorname{tr}[\Delta_{r,s} J_s]$. J satisfies the row built from W, so Step 4 of (i) gives $\sigma_r \operatorname{tr}[\Lambda_r L_r(J)] \geq \sigma_r \operatorname{tr}[\Lambda_r \Theta_r]$. For a Hermitian Δ and a positive semidefinite J_s, $-\|\Delta\|\,\mathbb{1} \preceq \Delta \preceq \|\Delta\|\,\mathbb{1}$ and the trace fact of Step 3 give $|\operatorname{tr}[\Delta J_s]| \leq \|\Delta\| \operatorname{tr} J_s$, so, the total trace being d by the coupling condition, $\big|\sum_s \operatorname{tr}[\Delta_{r,s} J_s]\big| \leq d \max_s \|\Delta_{r,s}\|$. Hence $\langle G_{\mathtt{t}}, J\rangle \geq b(G_{\mathtt{t}}; Y, \Lambda, \widetilde{\lambda}) - d \sum_r \max_s \|\Delta_{r,s}\|$. *Step 3 (the size of Δ).* For a lifted-operator row, $\Delta_{r,s} = (\widetilde{W}^\dagger \Lambda_r \widetilde{W} - W^\dagger \Lambda_r W) \otimes \hat{O}^{\uparrow(s)}_k$ and $\widetilde{W}^\dagger \Lambda_r \widetilde{W} - W^\dagger \Lambda_r W = (\widetilde{W} - W)^\dagger \Lambda_r \widetilde{W} + W^\dagger \Lambda_r (\widetilde{W} - W)$. The operator norm is submultiplicative and invariant under the adjoint, $\Big\|\widetilde{W} - W\Big\| \leq \varepsilon$ and $\|W\| \leq \Big\|\widetilde{W}\Big\| + \varepsilon$, so the bracket has norm at most $\varepsilon \|\Lambda_r\| \,(2\Big\|\widetilde{W}\Big\| + \varepsilon)$; and the norm of a tensor product is the product of the norms, the singular values of $A \otimes B$ being the products of those of A and of B. So $\|\Delta_{r,s}\| \leq \varepsilon(2\Big\|\widetilde{W}\Big\| + \varepsilon)\,\|\Lambda_r\|\,\|O_{r,s}\|$. For a party-local row the same computation holds with $\widetilde{W} \otimes \mathbb{1}$ and $W \otimes \mathbb{1}$, whose difference has the norm of $\widetilde{W} - W$ and whose norms are those of $\widetilde{W}$ and W, and with $D^{(s_f)}_k$ in place of $\hat{O}^{\uparrow(s)}_k$; at a sector whose component on the other factors is not s_{f^c} both maps vanish and $\Delta_{r,s} = 0$. For a row in frame-free form $\widetilde{L}_{r,s} = L_{r,s}$ and $\Delta_{r,s} = 0$. Substituting into Step 2 gives (20.5). The sum over $\mathtt{t}$ follows as in (ii)(a), $J'_{\mathtt{t}}$ lying in $\mathcal{W}^J_{\mathtt{t}}$ for every $J' \in \mathcal{W}^J$. □

21 The certified bracket

The objective is positively homogeneous of degree one, so at a point with positive definite blocks the gradient paired with the point is the objective there, and a bracket is the sum of its two lower bounds alone (Lemma 21.1). Those bounds need only enclosures of the gradient: an entrywise one on the acceptance side, and on the feasible side a ball around an exact residual, the frame of the rows enclosed with it. Over a reference companion known

through enclosures, entrywise or, for an element of rank below its dimension, through an enclosed factor, one enclosure serves the companion itself where every element the gradient reads has its zero pattern and its rank fixed throughout (Lemma 21.2). The best certified bracket gives the key length as any first-order bound does (Corollary 21.3), on a companion for the protocol it declares (Corollary 21.4).

Imports foundations/cor:least-eigenvalue-over-enclosure, foundations/cor:rank-over-factor-enclosure, foundations/def:exact-reference, foundations/def:factor-enclosure, foundations/def:kl

At a point with positive definite blocks and positive probabilities the gradient paired with the point is the objective, so the bracket is the sum of its two lower bounds and needs only enclosures of the gradient.

Part (iv) of Theorem 19.1 lets lower bounds on the two linear minima stand in for the pairings with an oracle's return. The objective is positively homogeneous of degree one (Lemma 9.6), so its derivative along the point itself is its value there, and the bracket loses the objective and the two pairings with the point: it is the sum of the two lower bounds, part (i). What remains to be bounded is the gradient alone, and parts (ii) and (iii) say how an enclosure of it yields the two lower bounds, the feasible one from a residual known up to a ball around an exact matrix, the frame of the rows enclosed with it, so that no deduction for the frame arises. Corollary 21.3 takes the key length from this lemma, and Requirement 28.1 is how a run forms it. Lemma 21.2 says when one enclosure of the gradient serves a reference companion known through enclosures.

Lemma 21.1 (The certified bracket from enclosures of the gradient)**.** Let $\alpha \in (1, \infty)$, let $\mathfrak{D}$ be a tuple of the form of Definition 1.1, in the basic format or the general format, that satisfies conditions (i)–(vi) and (viii) of Definition 1.2, and let $\mathcal{S}_{\mathrm{acc}}$ satisfy Definition 2.5. Let $x = (\mathbf{q}, J)$ with $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$ and every block of J positive definite, on every joint sector in the general format, satisfy (H-POS) of Theorem 19.1, and let (g, G) be the gradient of O_α^J at x assembled by Corollary 8.5, read with Lemma 18.2 (iii) and, in the general format, on the joint sectors as Theorem 19.1 reads it.

(i) *(The Euler form.)*

$$\langle g, \mathbf{q} \rangle + \langle G, J \rangle = O_\alpha^J(x). \tag{21.1}$$

Consequently, for real numbers b^{A} and b^{F} with $b^{\mathrm{A}} \leq \min_{\mathbf{q}' \in \mathcal{S}_{\mathrm{acc}}} \langle g, \mathbf{q}' \rangle$ and $b^{\mathrm{F}} \leq \min_{J' \in \mathcal{W}^J} \langle G, J' \rangle$, the minimum over the exact $\mathcal{W}^J$, the bracket of Theorem 19.1 (iv) at x with both pairings replaced is $b^{\mathrm{A}} + b^{\mathrm{F}}$, and

$$\max\{0,\, b^{\mathrm{A}} + b^{\mathrm{F}}\} \;\leq\; \underline{h}_\alpha, \tag{21.2}$$

in exact real arithmetic.

(ii) *(The acceptance side from an enclosure.)* If $\hat{g} \in \mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$ and $\delta \geq 0$ satisfy $|g(\hat{c}) - \hat{g}(\hat{c})| \leq \delta$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{acc}}$, then $\min_{\mathbf{q}' \in \mathcal{S}_{\mathrm{acc}}} \langle g, \mathbf{q}' \rangle \geq \min_{\mathbf{q}' \in \mathcal{S}_{\mathrm{acc}}} \langle \hat{g}, \mathbf{q}' \rangle - \delta$.

(iii) *(The feasible side from enclosed residuals.)* For each group of Lemma 20.1 (ii), an Alice sector $\mathtt{t}$ in trusted-source mode, its rows built from the exact frame $W_{\mathtt{t}}$ of Theorem 7.5, or the one group in untrusted-source mode, let Y and the Λ_r be as in Lemma 20.1 (i), let R_p be the residual (20.2) formed with G, and for every block p of the group let M_p be a Hermitian matrix, $\rho_p \geq 0$ with $\|R_p - M_p\|_{\mathrm{F}} \leq \rho_p$, the Frobenius norm of Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure), and ℓ_p a lower bound on $\lambda_{\min}(M_p)$. Then the group's $\lambda := \min_p(\ell_p - \rho_p)$ satisfies $\lambda \leq \lambda_{\min}(R_p)$ for every p, and the sum over the groups of $b(G; Y, \Lambda, \lambda)$ of (20.3) is a number $b^{\mathrm{F}} \leq \min_{J' \in \mathcal{W}^J} \langle G, J' \rangle$.

(iv) *(The dropped program.)* In the setting of Corollary 19.6, with a dropped set $\mathcal{D}$, (i)–(iii) hold with $O_\alpha^{J,\mathcal{D}}$ in place of O_α^J, (H-POS$^{\mathcal{D}}$) in place of (H-POS), its gradient of (H-NOPERT$^{\mathcal{D}}$) in place of (g, G), the kept pairs in place of the pairs that are not key-inert, and $\underline{h}_\alpha^{\mathcal{D}}$ in place of $\underline{h}_\alpha$; and $\underline{h}_\alpha^{\mathcal{D}} \leq \underline{h}_\alpha$.

Uses Def. 1.1, Def. 1.2, Def. 2.5, Thm. 4.1, Def. 4.5, Def. 4.7, Def. 4.8, Lem. 4.9, Thm. 7.5, Lem. 8.2, Thm. 8.4, Cor. 8.5, Def. 9.2, Prop. 9.3, Lem. 9.6, Def. 17.1, Def. 18.1, Lem. 18.2, Thm. 19.1, Cor. 19.6, Lem. 20.1, foundations/cor:least-eigenvalue-over-enclosure, foundations/def:kl · *Used by* Lem. 21.2, Cor. 21.3, Cor. 21.4, Rem. 22.5, Req. 28.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.certified:certified_bracket

Proof. Idea. Homogeneity makes the derivative along the point equal to the value, and the gradient's formulas give the same derivative as the pairing with the point; the two lower bounds then carry the whole bracket. An entrywise enclosure of the acceptance-side gradient moves its minimum over a set of distributions by at most the radius, and a ball around the residual moves its least eigenvalue by at most the ball's radius.

(i), Step 1 (the point is admissible to the theorem). Every block of J is positive semidefinite, and each residual of Definition 17.1 at J, of the coupling condition, of the normalisation and of every declared row, is a finite number, so J lies in $\mathcal{W}^J(\varepsilon)$ for every ε at least each of them, the normalisation's divided by the dimension factor its clause carries; with $\mathbf{q} \in \mathcal{S}_{\mathrm{acc}}$, $x \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$. Every block being positive definite, every key-active branch operator attains its maximal branch support (Lemma 8.2 (i), per joint sector in the general format), which is (H-SUPP); and (g, G) is the gradient at x itself, which is (H-NOPERT).

Step 2 (the value along the ray). By (H-POS) and Lemma 18.2 (i), $q(\hat{c}) > 0$ exactly on $\widehat{\mathcal{C}}_{\mathrm{acc}}$, and $P_{\hat{c}}^J(J) > 0$ there, $K_{\hat{c}}^J(J) > 0$ on $\widehat{\mathcal{C}}_{\mathrm{key}} \cap \widehat{\mathcal{C}}_{\mathrm{acc}}$,

so every term of $O^J_\alpha(x)$ indexed in $\widehat{\mathcal{C}}_{\mathrm{acc}}$ is real, and every term indexed outside it has $q(\hat{c}) = 0$ and is 0 by Definition "Classical divergences and the quantum relative entropy" (foundations/def:kl), the second arguments being non-negative by the cone clause of Theorem 4.1. So $O^J_\alpha(x)$ is real. By Lemma 9.6 (iii), $O^J_\alpha((1+\tau)x) = (1+\tau)\,O^J_\alpha(x)$ for every $\tau > -1$, so $\tau \mapsto O^J_\alpha(x+\tau x)$ is differentiable at $\tau = 0$ with derivative $O^J_\alpha(x)$.

Step 3 (the same derivative from the gradient). The terms indexed outside $\widehat{\mathcal{C}}_{\mathrm{acc}}$ vanish at every $(1+\tau)x$, $\tau > -1$, by Step 2 applied to the scaled point, $(1+\tau)q(\hat{c}) = 0$ there; they contribute 0 to the derivative, and the pairing $\langle g, \mathbf{q}\rangle$ of Definition 18.1 and the sums of (8.11) read with Lemma 18.2 (iii) omit them. For $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{acc}}$ the term is a smooth function of $(q(\hat{c}), P^J_{\hat{c}}, K^J_{\hat{c}})$ near the point, these being positive at x and scaling with $1+\tau$ (Lemma 9.6 (i)). Along τ, $q(\hat{c})$ and $P^J_{\hat{c}}$ move linearly, with derivatives $q(\hat{c})$ and $P^J_{\hat{c}}(J) = \sum_{s'} \operatorname{tr}\left[\Gamma^{(s')}_{\hat{c}} J_{s'}\right]$ ((8.12)). Each deficit of a key-active pair is $f^J_{\hat{c},s}(J_s + \tau J_s) = T(X+\tau X) - \operatorname{tr}(X+\tau X)$ with $X \coloneqq \mathcal{G}^J_{\hat{c},s}(J_s)$, the branch map being linear; the direction X is Hermitian with support $\operatorname{supp} X$, so Theorem 8.4 (i) gives the derivative $\operatorname{tr}[\nabla T(X)\,X] - \operatorname{tr} X = \operatorname{tr}\left[(\nabla T(X) - \mathbb{1})\,\mathcal{G}^J_{\hat{c},s}(J_s)\right] = \operatorname{tr}\left[\chi^\dagger_{Q_s}\left(\mathcal{G}^\dagger_{\hat{c},s}[\nabla T(X) - \mathbb{1}]\right) J_s\right]$, which is $\operatorname{tr}\left[\Delta^{(s)}_{\hat{c}} J_s\right]$ of (8.12); the deficits of key-inert pairs are identically 0 (Definition 4.7, Lemma 4.9 (iii)). The chain rule through $u \log_2 \frac{u}{v}$ at $u, v > 0$, exactly as in the proof of Corollary 8.5, then gives the derivative of the term as its coefficients of (8.10) and (8.11) paired with $q(\hat{c})$ and with the blocks of J, and summing over $\widehat{\mathcal{C}}_{\mathrm{acc}}$, the derivative of $\tau \mapsto O^J_\alpha(x+\tau x)$ at 0 is $\langle g, \mathbf{q}\rangle + \langle G, J\rangle$. With Step 2 this is (21.1).

Step 4 (the bracket). By Step 1, Theorem 19.1 (iiib) with its part (iv) applies to the one point x with both pairings replaced, where $\mathfrak{D}$ satisfies condition (vii) as well; its bracket is $O^J_\alpha(x) + b^{\mathrm{A}} - \langle g, \mathbf{q}\rangle + b^{\mathrm{F}} - \langle G, J\rangle = b^{\mathrm{A}} + b^{\mathrm{F}}$ by (21.1), and the theorem gives (21.2). Where condition (vii) fails, Theorem 19.1 (v) gives the same, $\underline{h}_\alpha = +\infty$.

(ii). Let $\mathbf{q}' \in \mathcal{S}_{\mathrm{acc}}$, a non-empty set as it contains $\mathbf{q}$. It is a probability vector vanishing outside $\widehat{\mathcal{C}}_{\mathrm{acc}}$ (Lemma 18.2 (i)), so $q'(\hat{c}) \geq 0$ and $\sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{acc}}} q'(\hat{c}) = 1$. Hence $\langle g, \mathbf{q}'\rangle = \langle \hat{g}, \mathbf{q}'\rangle + \sum_{\hat{c}\in\widehat{\mathcal{C}}_{\mathrm{acc}}} (g(\hat{c}) - \hat{g}(\hat{c}))q'(\hat{c}) \geq \langle \hat{g}, \mathbf{q}'\rangle - \delta \geq \min_{\mathbf{q}''\in\mathcal{S}_{\mathrm{acc}}} \langle \hat{g}, \mathbf{q}''\rangle - \delta$, the minimum attained, $\mathcal{S}_{\mathrm{acc}}$ being compact (Definition 2.5) and the map linear. The minimum over $\mathbf{q}'$ of the left-hand side is therefore at least the right-hand side.

(iii). R_p is Hermitian, G, $Y\otimes\mathbb{1}$ and every $L^*_{r,p}(\Lambda_r)$ being so (Lemma 20.1 (i), (ii)). By Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure) (i), $\lambda_{\min}(R_p) \geq \lambda_{\min}(M_p) - \rho_p \geq \ell_p - \rho_p \geq \lambda$. So λ is admissible in Lemma 20.1 (i) for the group, whose rows, built from the exact frame, cut out the factor of the exact $\mathcal{W}^J$ (part (ii)(a) of that lemma; part (ii)(b) in untrusted-source mode), and the sum over the groups bounds $\langle G, J'\rangle$ from below for every $J' \in \mathcal{W}^J$, which is (20.4).

(iv). $O_\alpha^{J,\mathcal{D}}$ is positively homogeneous of degree one (Lemma 9.6 (iii)), and Steps 2 and 3 run unchanged with $K_{\hat{c}}^{J,\mathcal{D}}$ in place of $K_{\hat{c}}^{J}$ and the deficits of the kept pairs only, the dropped ones being absent from (9.3); (H-SUPP$^{\mathcal{D}}$) holds at a positive definite block as in Step 1. Step 4 runs with Corollary 19.6, whose conclusion (19.8) admits the lower bounds of Theorem 19.1 (iv), in place of the theorem. Parts (ii) and (iii) do not read the objective. Finally $\underline{h}_\alpha^{\mathcal{D}} \le \underline{h}_\alpha$ is Proposition 9.3. □

Over a reference companion whose measured families, of one or more replaced factors, are known through entrywise enclosures and on some blocks enclosed factors, one enclosure of the gradient serves the companion where every element the gradient reads has its zero pattern and rank fixed.

Lemma 21.1 needs, at a point, only enclosures of the gradient. A reference companion known through enclosures leaves a whole set of families consistent with its data, the companion's own among them, and this lemma says when one enclosure of the gradient serves every one of them: when the zero pattern and the rank of every element that the gradient's formula reads are the same throughout. The companion may replace the measured families of several factors, as it replaces both stations of an entanglement-based declaration, and each of them may be known through enclosures, so that an element of the product family may be a tensor product of several blocks that are only enclosed. An element of rank below its dimension whose support no coordinate vectors span has its rank fixed by no entrywise enclosure with positive radii, and is read instead through an enclosed factor of fixed rank. A block may so be known twice, and the statistics may read it through the one enclosure and the deficits through the other; part (c) says that an enclosure of the gradient formed in that way, over every pair of the two readings, serves the companion, whose own family is read twice among those pairs. Part (b) gives the factor structure that keeps the factors of a pass class of known rank where some of them are only enclosed: a tensor product of factors of full row rank has full row rank, the tensor product of their Gram matrices being positive definite, so blocks of fixed rank multiply into elements of fixed rank. Elements whose blocks known only through boxes are the same blocks have their rows grouped by those blocks, named as blocks and never compared by their boxes, so that an exact dependence between their exact factors is reduced exactly. Corollary 21.4 takes the key length from the lemma.

Lemma 21.2 (The certified bracket over a measured family known through enclosures)**.** Let $\mathfrak{D}^{\mathrm{ref}}$ be a compiled reference companion of Proposition 12.1, with $\mathbb{F}^{\mathrm{ref}}$ the set of the measured factors it replaces (the one measured system in the basic format), let α, $\mathcal{S}_{\mathrm{acc}}$ and $x = (\mathbf{q}, J)$ be as in Lemma 21.1 with $\mathfrak{D} := \mathfrak{D}^{\mathrm{ref}}$, save that (H-POS) of Theorem 19.1 is not assumed of x, and, where (H-POS) holds for $\mathfrak{D}^{\mathrm{ref}}$ at x, let (g, G) be the gradient that lemma reads there, the blocks of the measured families of the factors of $\mathbb{F}^{\mathrm{ref}}$ each known through an entrywise enclosure (Definition "Reference of

a measurement family, exact or enclosed" (foundations/def:exact-reference)), some of them read, besides, with a triple (L_c, t, W) of a matrix box and a core as in Definition "Factor enclosure of a block" (foundations/def:factor-enclosure), the block's *factor enclosure*, and every other datum exactly. Call a tuple $F = (F^f)_{f \in \mathbb{F}^{\mathrm{ref}}}$ of families, each F^f indexed as the measured family of factor f is, a *family of the replaced factors*, its elements $F_v^{(s)}$ being the products (1.5) of its blocks and of the blocks of the other factors of $\mathfrak{D}^{\mathrm{ref}}$; call it an *entrywise member* when each of its blocks is a Hermitian matrix that the entrywise enclosure of that block contains, and a *member* when, moreover, each of its blocks that is read with a factor enclosure is a member of it. Suppose

(H-FAC) each block of the companion's own family of the replaced factors that is read with a factor enclosure is a member of it, the triple being a factor enclosure of that block, so that the companion's family of the replaced factors is a member.

Call a block *decided* when every entry of its entrywise enclosure has centre and radius 0; or some entry's centre C_{il} and radius r_{il} satisfy $|C_{il}| > r_{il}$; or its factor enclosure has no column; or its factor enclosure has $k \geq 1$ columns and there are G, ρ and ℓ as in Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure) (ii) with $\ell - \rho > 0$. Call a block *of fixed rank* when it is of one of three kinds: of the first kind when it is zero for every member; of the second kind when there are a matrix E of full row rank and, for every member, a positive definite $\widetilde{X}$ with the member's block equal to $E^\dagger \widetilde{X} E$, E the same for every member (Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure) (iv) gives such an E); and of the third kind when there are an integer $k \geq 1$, a matrix box of complex $k \times n$ matrices, n the dimension of the block, and a positive definite $k \times k$ matrix $\widetilde{X}$ and, for every member, a matrix E_* of rank k that the box contains, with the member's block equal to $E_*^\dagger \widetilde{X} E_*$ (Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure) (ii) gives such a box, that of the adjoints of the factors, with $\widetilde{X}$ the core). Suppose that every block of a factor of $\mathbb{F}^{\mathrm{ref}}$ that is a factor of an element $F_v^{(s)}$ with (x', v) an outcome pair of a pass class of some $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{key}}$ and $M_{x'}^{(\mathsf{t})} \neq 0$, no other block of which is zero for every member, is decided, and that every such block is of fixed rank at the pairs $(\hat{c}, (\mathsf{t}, s))$ that are not key-inert. Then:

(a) every member gives the same effective outcome sets on every joint sector, the same key-inert pairs and the same joint key-active index sets, those read off the enclosures;

(b) fix, for every block that the hypothesis asks to be of fixed rank, one kind it is of, with the data of that kind, and, where that kind is the

third, at every member one matrix E_* as that kind asks, one for the block and not one for each element it is a factor of; below, the kind of a block is the one so fixed for it. At every pair $(\hat{c}, (\mathsf{t}, s))$ that is not key-inert, every element $F_v^{(s)}$ of an outcome pair (x', v) of a pass class of $\hat{c}$ with $M_{x'}^{(\mathsf{t})} \neq 0$ is, for every member, $E_v^{(s)\dagger} \widetilde{F}_v^{(s)} E_v^{(s)}$ with $\widetilde{F}_v^{(s)}$ positive definite and $E_v^{(s)}$ of full row rank, the case $F_v^{(s)} = 0$ included as $E_v^{(s)}$ with no rows, where $E_v^{(s)}$ is the same for every member if no block of $F_v^{(s)}$ is of the third kind or some block of it is zero for every member, and otherwise its rank is the same for every member and it is the tensor product, in the order of $\mathbb{F}$, of one matrix for each block of $F_v^{(s)}$: the matrix E_* fixed for the block at that member where the block is of the third kind, and a matrix the same for every member otherwise. For an element of the latter case, call its *tuple* $\beta(v)$ the list, in the order of $\mathbb{F}$, of its blocks of the third kind, each block $F_{v_f}^{f,(s_f)}$ named by the triple (f, s_f, v_f), so that two entries of tuples agree exactly when they name the same block, whatever their boxes; and for a tuple β let E_β^{box} be the tensor product, in that order, of the matrices E_* fixed for its blocks, which at each member is one matrix for every element whose tuple is β. Moreover, for every pass class of $\hat{c}$ and any factor forms $M_x^{(\mathsf{t})} = E_x^{(\mathsf{t})\dagger} \widetilde{M}_x E_x^{(\mathsf{t})}$ of Alice's elements with $E_x^{(\mathsf{t})}$ of full row rank and $\widetilde{M}_x$ positive definite, the matrix $\widetilde{E}$ that stacks the matrices $E_x^{(\mathsf{t})} \otimes E_v^{(s)}$ over the class's outcome pairs (x, v) with $M_x^{(\mathsf{t})} \neq 0$, in a fixed order, is, for every member, $\widetilde{E} = \widetilde{B}\, C$ with $\widetilde{B}$ of full column rank and the same for every member, and C a stack of a matrix C_0 and, one for each tuple β of an element of the latter case in the class, of a matrix $(C_\beta \otimes E_\beta^{\mathrm{box}}) U_\beta$, with C_0, every C_β and every permutation matrix U_β the same for every member; where the $E_x^{(\mathsf{t})}$, the $E_v^{(s)}$ that are the same for every member and the matrices the same for every member of which the other $E_v^{(s)}$ are tensor products have entries in $\mathbb{Q}(i)$, $\widetilde{B}$, C_0 and every C_β can be taken with entries in $\mathbb{Q}(i)$;

(c) call a pair (F', F'') of families of the replaced factors, the first read by the statistics and the second by the deficits, *admissible* when every block of F'' that is a factor of an element $F_v^{(s)}$ of an outcome pair (x', v) of a pass class of $\hat{c}$ with $M_{x'}^{(\mathsf{t})} \neq 0$, at a pair $(\hat{c}, (\mathsf{t}, s))$ that is not key-inert, no other block of which is zero for every member, is a positive semidefinite matrix. For such a pair form, at x, $P_{\hat{c}}^J$ and every $\Gamma_{\hat{c}}^{(s')}$ with F', every deficit $f_{\hat{c},s}^J$ and every $\Delta_{\hat{c}}^{(s')}$ with F'', an outcome pair whose element of Alice's is zero, or whose element $F_v^{(s)}$ has a block that is zero for every member, contributing nothing to a branch map or to the statistics, over the key-inert pairs and the joint key-active index sets of (a), and $K_{\hat{c}}^J$ as $P_{\hat{c}}^J$ plus those deficits; where $q(\hat{c})$, $P_{\hat{c}}^J$ and $K_{\hat{c}}^J$ so formed

are positive as (H-POS) of Theorem 19.1 asks, let $(g, G)[F', F'']$ be the pair given by (8.10) and (8.11) with these pieces, read as Lemma 21.1 reads them. Let $\mathcal{P}$ be a set of admissible pairs that contains (F, F) for every member F, such as the set of the admissible pairs of an entrywise member F' and a family F'' of the replaced factors each of whose blocks lies in its entrywise enclosure or is a member of its factor enclosure, the one or the other chosen block by block. Then a set that contains $P^{J}_{\hat{c}}$, or $K^{J}_{\hat{c}}$, formed with every pair of $\mathcal{P}$ contains that of $\mathfrak{D}^{\mathrm{ref}}$ at x; and where (H-POS) holds for $\mathfrak{D}^{\mathrm{ref}}$, enclosures $(\hat{g}, \delta)$ and (M_p, ρ_p) that hold, as in parts (ii) and (iii) of Lemma 21.1, at $(g, G)[F', F'']$ for every pair of $\mathcal{P}$ at which it is formed hold at the gradient of $\mathfrak{D}^{\mathrm{ref}}$, and parts (i)–(iii) of that lemma then give $\max\{0, b^{\mathrm{A}} + b^{\mathrm{F}}\} \leq \underline{h}_{\alpha}[\mathfrak{D}^{\mathrm{ref}}]$;

(d) in the setting of Corollary 19.6, with a dropped set $\mathcal{D}$, (a)–(c) hold, the hypotheses read the same way, with $O^{J,\mathcal{D}}_{\alpha}$ in place of O^{J}_{α}, (H-POS$^{\mathcal{D}}$) in place of (H-POS), its gradient of (H-NOPERT$^{\mathcal{D}}$) in place of (g, G) and $K^{J,\mathcal{D}}_{\hat{c}}$ in place of $K^{J}_{\hat{c}}$, the kept pairs in place of the pairs that are not key-inert, part (iv) of Lemma 21.1 in place of its parts (i)–(iii), and $\underline{h}^{\mathcal{D}}_{\alpha}$ in place of $\underline{h}_{\alpha}$.

Uses Def. 1.1, Def. 1.2, Def. 1.6, Def. 4.5, Def. 4.8, Thm. 7.5, Cor. 8.5, Def. 9.2, Prop. 12.1, Thm. 19.1, Cor. 19.6, Lem. 21.1, foundations/cor:least-eigenvalue-over-enclosure, foundations/cor:rank-over-factor-enclosure, foundations/def:exact-reference, foundations/def:factor-enclosure · *Used by* Cor. 21.4, Rem. 22.5, Req. 28.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.certified:_companion_element

Proof. (a). By Definition 4.8, an element $F^{(s)}_{v}$ of a factorised family is non-zero exactly when each of its factor blocks is, and the effective outcome set of a pass class on a joint sector collects the outcome pairs with $M^{(\mathrm{t})}_{x'} \neq 0$ and $F^{(s)}_{v} \neq 0$. Alice's blocks and the blocks of the factors outside $\mathbb{F}^{\mathrm{ref}}$ are the same for every member. A decided block of a factor of $\mathbb{F}^{\mathrm{ref}}$ is zero for every member when every entry has centre and radius 0, since then every entry of a contained matrix is 0, and non-zero for every member when some entry has $|C_{il}| > r_{il}$, since then $|X_{il}| \geq |C_{il}| - r_{il} > 0$. When its factor enclosure has no column, every member of that enclosure is the zero matrix (Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure) (i)), and so is the block of every member, which is one of them; when it has $k \geq 1$ columns with $\ell - \rho > 0$, every member of the enclosure has rank k and is not zero (part (ii) of that corollary), and so the block of every member is not zero. An element one of whose blocks is zero for every member is zero for every member, whatever its other blocks, of which the hypothesis therefore asks nothing at that element. So each effective outcome set is the same for every member, and key-inertness, which reads only these sets and the raw-key map, and the joint key-active index sets with it, are the same.

(b), the factor forms. Let F be a member. A block of a factor of $\mathbb{F}^{\mathrm{ref}}$ of fixed rank is, at F, by the kind fixed for it: zero, of the first kind; $E^\dagger \widetilde{X} E$ with E the same for every member, of the second; or $E_*^\dagger \widetilde{X} E_*$ with E_* the matrix fixed for the block at F, in the box, of rank k, and $\widetilde{X}$ the same for every member, of the third, so that the block enters every element it is a factor of through that one E_*. Every other factor block B, a block of a factor outside $\mathbb{F}^{\mathrm{ref}}$, is the same for every member and positive semidefinite, and with $B = \sum_{k' \le m} b_{k'} u_{k'} u_{k'}^\dagger$, $b_{k'} > 0$ and $u_{k'}$ orthonormal, it is $E_B^\dagger \operatorname{diag}(b_1, \ldots, b_m) E_B$ with E_B the matrix of rows $u_{k'}^\dagger$, of full row rank. A tensor product of such forms is $(\bigotimes E)^\dagger (\bigotimes \widetilde{X})(\bigotimes E)$, where $\bigotimes E$ has full row rank, the tensor product of injective adjoints being injective: its Gram matrix $(\bigotimes E)(\bigotimes E)^\dagger = \bigotimes (E E^\dagger)$ is a tensor product of positive definite matrices, each $E E^\dagger$ being positive definite for a matrix E of full row rank, and so positive definite, and its rank, its number of rows, is the product of the ranks of the factors; and $\bigotimes \widetilde{X}$ is positive definite. A positive multiple $w E^\dagger \widetilde{X} E = E^\dagger (w \widetilde{X}) E$ keeps the form, and a zero weight or a zero factor gives the zero element; an element one of whose blocks is zero for every member is so zero for every member, its $E_v^{(s)}$ with no rows the same for every member, whatever its other blocks. The weight and every factor of $\bigotimes E$ but those of the blocks of the third kind are the same for every member, E being the same for every member at a block of the second kind and E_B at a block outside $\mathbb{F}^{\mathrm{ref}}$. So, where no block of the element is zero for every member, $\bigotimes E$, taken in the order of $\mathbb{F}$, is the same for every member where no block of the element is of the third kind, and otherwise it is the tensor product of the matrices E_* fixed for its blocks of the third kind and of matrices the same for every member, one per block, as the statement says, and its rank is the same for every member: the product of the ranks of its factors, each the same for every member, a block of the third kind contributing its k at every member, whichever other blocks of the third kind stand beside it.

(b), the reordering. Let v be an element of the latter case, with tuple $\beta = \beta(v)$, let $f_1, \ldots, f_m$ be the factors of $\mathbb{F}$ in their order, and let $E_v^{(s)} = E_{v,1} \otimes \cdots \otimes E_{v,m}$, with $E_{v,j}$ the matrix of its block of factor f_j, of p_j rows and n_j columns; let $\operatorname{pos} \beta \subseteq \{1, \ldots, m\}$, non-empty, be the positions of the blocks of β. Each p_j is the same for every member: the k of the block where it is of the third kind, and otherwise the number of rows of a matrix the same for every member. Each n_j is the block dimension $d_{f_j, s_{f_j}}$ of the joint sector s, the same for every element of the class. For positive integers $q_1, \ldots, q_m$ let $\Sigma_\beta(q)$ be the linear map with $\Sigma_\beta(q)(u_1 \otimes \cdots \otimes u_m) = u_{\sigma(1)} \otimes \cdots \otimes u_{\sigma(m)}$ for $u_j \in \mathbb{C}^{q_j}$, where σ lists the positions outside $\operatorname{pos} \beta$ in increasing order and then those in $\operatorname{pos} \beta$ in increasing order. Defined on the tensor products of standard basis vectors, which it maps bijectively onto those of $\mathbb{C}^{q_{\sigma(1)}} \otimes \cdots \otimes \mathbb{C}^{q_{\sigma(m)}}$, and extended linearly, it is a permutation matrix, so that $\Sigma_\beta(q)^{-1} = \Sigma_\beta(q)^\top$, and the identity holds for all vectors u_j, both sides being multilinear in

them. For matrices A_j of size $p_j \times n_j$, both sides applied to $u_1 \otimes \cdots \otimes u_m$ give $\Sigma_\beta(p)(A_1 \otimes \cdots \otimes A_m) = (A_{\sigma(1)} \otimes \cdots \otimes A_{\sigma(m)})\Sigma_\beta(n)$, and the Kronecker product being associative,

$$E_v^{(s)} = R_v\big(E_v^{\mathrm{ex}} \otimes E_\beta^{\mathrm{box}}\big)\Sigma_\beta(n), \qquad R_v \coloneqq \Sigma_\beta(p)^\top,$$

with E_v^{ex} the tensor product, in order, of the $E_{v,j}$ with $j \notin \operatorname{pos}\beta$, the 1×1 matrix 1 where there is none, the same for every member, and E_β^{box} that of the $E_{v,j}$ with $j \in \operatorname{pos}\beta$, the matrices E_* fixed for the blocks of β, as the statement defines it. R_v reads only $\operatorname{pos}\beta$ and the p_j, and $\Sigma_\beta(n)$ only $\operatorname{pos}\beta$ and the n_j; so R_v is the same for every member, and $\Sigma_\beta(n)$ is the same for every member and for every element of the class whose tuple is β. At a member, every element whose tuple is β has, at each block of β, the one E_* fixed for that block, so E_β^{box} is one matrix for all of them. This is where E_* is fixed per block and not per element, and where a tuple names blocks and not boxes: two distinct blocks with equal boxes are distinct entries, and their matrices E_* may differ at a member.

(b), the stack. Fix a pass class, and let V_1 be the elements v of its outcome pairs (x, v) with $M_x^{(\mathsf{t})} \neq 0$ that are of the latter case. Which elements these are, and their tuples, depend only on the kinds fixed for the blocks and on the zero pattern, the same for every member. The rows of $\widetilde{E}$ that belong to the pairs (x, v) with $v \notin V_1$ form a matrix S_0, the same for every member. Let a be the number of columns of the $E_x^{(\mathsf{t})}$, the dimension of Alice's elements on the sector t, and e_x the number of rows of $E_x^{(\mathsf{t})}$. For a tuple β of an element of V_1, the pairs (x, v) with $v \in V_1$ and $\beta(v) = \beta$ have, by the reordering, the associativity of the Kronecker product and the mixed-product rule $(\mathbb{1}_{e_x} \otimes R)(A \otimes M)(\mathbb{1}_a \otimes \Sigma) = A \otimes RM\Sigma$, the rows

$$E_x^{(\mathsf{t})} \otimes E_v^{(s)} = (\mathbb{1}_{e_x} \otimes R_v)\big((E_x^{(\mathsf{t})} \otimes E_v^{\mathrm{ex}}) \otimes E_\beta^{\mathrm{box}}\big)U_\beta, \qquad U_\beta \coloneqq \mathbb{1}_a \otimes \Sigma_\beta(n).$$

The matrices $E_x^{(\mathsf{t})} \otimes E_v^{\mathrm{ex}}$ of these pairs all have the same number of columns, a times the product of the n_j with $j \notin \operatorname{pos}\beta$; let S_β stack them in the fixed order, the same for every member. Stacking on the left commutes with the tensor product by a fixed right factor, $\left[\begin{smallmatrix} A \\ A' \end{smallmatrix}\right] \otimes E = \left[\begin{smallmatrix} A \otimes E \\ A' \otimes E \end{smallmatrix}\right]$, and the right factor E_β^{box} is, at a member, one matrix for all these pairs; so their rows, in the fixed order, form $D_\beta(S_\beta \otimes E_\beta^{\mathrm{box}})U_\beta$, with D_β the block-diagonal matrix of the $\mathbb{1}_{e_x} \otimes R_v$, a permutation matrix the same for every member. Every matrix S has a rank factorisation $S = BC'$ with B of full column rank: over a field that contains its entries, B its pivot columns and C' the non-zero rows of its reduced row echelon form, an empty factorisation where S is zero or has no row. Take $S_0 = B_0C_0$ and $S_\beta = B_\beta C_\beta$ so. By the mixed-product rule, $S_\beta \otimes E_\beta^{\mathrm{box}} = (B_\beta \otimes \mathbb{1}_{k_\beta})(C_\beta \otimes E_\beta^{\mathrm{box}})$, with k_β the number of rows of E_β^{box}, the product of the k of the blocks of β, the same for every member; nothing else of E_β^{box} enters. Let Π be the permutation matrix that returns

the rows from this grouping, that of S_0 and one group per tuple, to the fixed order. Then $\widetilde{E} = \Pi \operatorname{diag}\left(B_0, (D_\beta(B_\beta \otimes \mathbb{1}_{k_\beta}))_\beta\right) C$ with C the stack of C_0 and the $(C_\beta \otimes E_\beta^{\mathrm{box}})U_\beta$; the first factor, $\widetilde{B}$, is the same for every member and has full column rank, being an invertible matrix times a block-diagonal matrix whose blocks have full column rank, $D_\beta(B_\beta \otimes \mathbb{1}_{k_\beta})$ among them as an invertible matrix times a tensor product of injective maps. Where the $E_x^{(\mathrm{t})}$, the $E_v^{(s)}$ for $v \notin V_1$ and, for $v \in V_1$, the $E_{v,j}$ with $j \notin \operatorname{pos}\beta(v)$ have entries in $\mathbb{Q}(i)$, so do S_0 and every S_β, and the factorisations over the field $\mathbb{Q}(i)$ give B_0, C_0, B_β and C_β there; Π, every D_β and every U_β have entries 0 and 1.

(c). Let F^{ref} be the companion's family of the replaced factors. The family of each factor of $\mathfrak{D}^{\mathrm{ref}}$ is a positive operator-valued measure (Proposition 12.1 (ii), with condition (iii) of Definition 1.2, which in the general format asks it of every factor's family), so the blocks of F^{ref} are Hermitian and positive semidefinite; they lie in their entrywise enclosures, which are enclosures of them, and, by (H-FAC), in their factor enclosures. So F^{ref} is a member, $(F^{\mathrm{ref}}, F^{\mathrm{ref}})$ is an admissible pair, and it lies in $\mathcal{P}$. By (a), the key-inert pairs and joint key-active index sets of $\mathfrak{D}^{\mathrm{ref}}$ are those read off the enclosures, so the $P_{\hat{c}}^J$, $\Gamma_{\hat{c}}^{(s')}$, deficits, $\Delta_{\hat{c}}^{(s')}$ and $K_{\hat{c}}^J = P_{\hat{c}}^J + \sum f_{\hat{c},s}^J$ (Theorem 7.5) formed with this pair are those of $\mathfrak{D}^{\mathrm{ref}}$ at x, every piece being formed from its one family: an outcome pair whose element of Alice's is zero, which (c) leaves out of the branch maps, contributes nothing to those of $\mathfrak{D}^{\mathrm{ref}}$ either, its Kraus factor $\sqrt{M_{x'}^{(\mathrm{t})}} \otimes \sqrt{F_v^{(s)}}$ being zero (Definition 1.6); and so does one whose element $F_v^{(s)}$ has a block zero for every member, which (c) leaves out of the branch maps and the statistics, that element being zero at F^{ref}, a member, so that its Kraus factor and its contribution to the statistics are zero there. A set containing such a quantity formed with every pair of $\mathcal{P}$ contains that of $\mathfrak{D}^{\mathrm{ref}}$. Where (H-POS) holds for $\mathfrak{D}^{\mathrm{ref}}$, these $q(\hat{c})$, $P_{\hat{c}}^J$ and $K_{\hat{c}}^J$ are positive as it asks, so $(g, G)[F^{\mathrm{ref}}, F^{\mathrm{ref}}]$ is formed, and by the formulas of Corollary 8.5 it is the gradient (g, G) of $\mathfrak{D}^{\mathrm{ref}}$ at x as Lemma 21.1 reads it; the enclosures hold there. $\mathfrak{D}^{\mathrm{ref}}$ satisfies conditions (i)–(vi) and (viii) of Definition 1.2 (Proposition 12.1 (ii)), so parts (ii), (iii) and then (i) of Lemma 21.1 apply to it at x, where (H-POS) holds. Finally, a member F is an entrywise member each of whose blocks lies in its entrywise enclosure and in its factor enclosure where one is read, so (F, F) belongs to the set of pairs named in (c), whatever enclosure is chosen for each block, and it is admissible: every block that admissibility reads is a factor of an element no other block of which is zero for every member, at a pair that is not key-inert, so the hypothesis asks it to be of fixed rank, and it is zero or $E^\dagger \widetilde{X} E$ at F, and so positive semidefinite. An element one of whose blocks is zero for every member is zero at F and contributes nothing to a piece formed with (F, F), whatever its other blocks.

(d). Parts (a) and (b) read only the zero pattern and the ranks, the hypothesis that the blocks are decided being unchanged and the fixed-rank hypothesis now only at the kept pairs, and (c) runs with $K_{\hat{c}}^{J,\mathcal{D}}$, formed with the deficits of the kept pairs only, and with part (iv) of Lemma 21.1, whose parts (ii) and (iii) do not read the objective. □

The largest bracket formed by Lemma 21.1 over finitely many points is a first-order bound of Theorem 19.1, so it is at most the value of the declaration it is formed on and gives that declaration's fixed-length key length exactly as h_lb does.

By Lemma 21.1 (i) the sum of the two lower bounds is the bracket of Theorem 19.1 (iv) at its point, so the best of finitely many such sums is a first-order bound like any other, and the key-length corollaries take it unchanged. What makes it certified is how the two lower bounds are formed (Requirement 28.1), and what it still rests on is Remark 22.5. On a reference companion, Corollary 21.4 carries the conclusion over to the protocol the companion declares.

Corollary 21.3 (Key length from the certified bracket). Let α, $\mathfrak{D}$ and $\mathcal{S}_{\mathrm{acc}}$ be as in Lemma 21.1, let $\mathfrak{I}$ be a finite non-empty set of points $x_t = (\mathbf{q}_t, J_t)$ each satisfying the hypotheses of that lemma, and let b_t^{A} and b_t^{F} be real numbers with (19.3) at the gradient at x_t, as parts (ii) and (iii) of that lemma form them. Put

$$\underline{h}_{\alpha}^{\mathrm{cert}} := \max_{t \in \mathfrak{I}} \left(b_t^{\mathrm{A}} + b_t^{\mathrm{F}} \right). \tag{21.3}$$

(i) $\underline{h}_{\alpha}^{\mathrm{cert}}$ is the value of (19.2) at the points of $\mathfrak{I}$ with both pairings replaced as Theorem 19.1 (iv) replaces them, and $\max\{0, \underline{h}_{\alpha}^{\mathrm{cert}}\} \le \underline{h}_{\alpha}$ in exact real arithmetic.

(ii) Where $\mathfrak{D}$ satisfies Definition 1.2 and the hypotheses of Corollary 19.5 hold, its conclusions hold with $\underline{h}_{\alpha}^{\mathrm{cert}}$ in place of $\underline{h}_{\alpha}^{\mathrm{FO}}$ in (19.7): the key length so formed is $\varepsilon_{\mathrm{PA}}$-secret and $\varepsilon_{\mathrm{EV}}$-correct, with the composed-security and monotonicity conclusions of that corollary, the secrecy and security statements at every output state, whatever the dimension of the adversary's registers and the order in which the source releases its signals.

(iii) Under a dropped set, with the lemma's part (iv), $\max\{0, \underline{h}_{\alpha}^{\mathrm{cert}}\} \le \underline{h}_{\alpha}^{\mathcal{D}} \le \underline{h}_{\alpha}$; and where $\mathfrak{D}$ satisfies Definition 1.2 and the hypotheses of Corollary 10.18 other than (10.17) hold, the conclusions of that corollary hold at $h = \underline{h}_{\alpha}^{\mathrm{cert}}$.

Reading: The certified key rate is the best of a few brackets, each formed from enclosures of the gradient at a point the search found, with every arithmetic step from the declaration's data exact or rigorously enclosed. It is a lower bound on the minimum h_low of the declaration it is formed on, and the key length built from it is secret and correct under the same premises as the first-order key length: the declaration's modelling premises remain,

but no premise about how a number was computed is added, the key length itself being computed exactly, its one logarithm enclosed until the integer part is decided. It is not a bound on any other declaration, so a run formed on the snapped declaration still rests where its record says on the premise that the device measures the snapped family.

Uses Def. 1.2, Prop. 9.3, Cor. 10.18, Def. 17.1, Thm. 19.1, Cor. 19.5, Cor. 19.6, Lem. 21.1 · *Used by* Cor. 21.4, Rem. 22.5, Req. 28.1 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.meat_core.finitekey.keylength:certified_key_length

Proof. (i) By Step 1 of the proof of Lemma 21.1, each x_t lies in $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon_t)$ for some $\varepsilon_t \geq 0$ and satisfies (H-POS), (H-SUPP) and (H-NOPERT); every point then lies in $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ with $\varepsilon := \max_t \varepsilon_t$, the relaxed sets growing with ε (Definition 17.1, each clause bounding a quantity by a multiple of ε). By Lemma 21.1 (i) the bracket of Theorem 19.1 (iv) at x_t with both pairings replaced is $b_t^{\mathrm{A}} + b_t^{\mathrm{F}}$, so $\underline{h}_\alpha^{\mathrm{cert}}$ is (19.2) so formed, and Theorem 19.1 (iiib) and (iv), or (v) where condition (vii) of Definition 1.2 fails, give the inequality.

(ii) Corollary 19.5 holds for any $\underline{h}_\alpha^{\mathrm{FO}}$ of Theorem 19.1 (iiib) with the lower bounds of its part (iv) in place of the pairings, under its hypotheses, and (i) shows that $\underline{h}_\alpha^{\mathrm{cert}}$ is one; it depends on the points, the lower bounds and the declaration, and on none of $(\lambda_{\mathrm{EC}}, \varepsilon_{\mathrm{EV}}, \varepsilon_{\mathrm{PA}})$.

(iii) Lemma 21.1 (iv) gives $\max\{0, b_t^{\mathrm{A}} + b_t^{\mathrm{F}}\} \leq \underline{h}_\alpha^{\mathcal{D}}$ at each point and $\underline{h}_\alpha^{\mathcal{D}} \leq \underline{h}_\alpha$ (Proposition 9.3), so $\underline{h}_\alpha^{\mathrm{cert}}$ is a lower bound on $\underline{h}_\alpha$, the hypothesis (10.17) of Corollary 10.18, and it depends on none of $(\lambda_{\mathrm{EC}}, \varepsilon_{\mathrm{EV}}, \varepsilon_{\mathrm{PA}})$, the condition of that corollary's clause (a). Its remaining hypotheses are those the second clause of (iii) assumes, so its conclusions hold there at $h = \underline{h}_\alpha^{\mathrm{cert}}$. □

On a reference companion, the best certified bracket, formed over its enclosed measured family as Lemma 21.2 allows, gives the key length of the protocol whose declaration is the companion, as the first-order bound on the companion does.

Corollary 21.3 applies to a reference companion as to any declaration; this corollary carries its conclusion over to the protocol whose declaration the companion is, as Corollary 19.7 does for the first-order bound. Where the companion's measured family is known only through enclosures, the two lower bounds at a point are formed from enclosures that hold over every family the enclosures leave possible, and part (c) of Lemma 21.2, or part (d) under a dropped set, is what makes them bounds at the companion's own gradient.

Corollary 21.4 (Key length on a reference companion from the certified bracket)**.** Let $\mathfrak{D}^{\mathrm{ref}}$ be a compiled reference companion of Proposition 12.1, and let α, $\mathcal{S}_{\mathrm{acc}}$ and $\mathfrak{I}$, with its points x_t, be as in Corollary 21.3 with $\mathfrak{D} := \mathfrak{D}^{\mathrm{ref}}$. At each $t \in \mathfrak{I}$ let b_t^{A} and b_t^{F} be real numbers formed in one of two ways: as in that corollary, with (19.3) at the gradient of $\mathfrak{D}^{\mathrm{ref}}$ at x_t; or, where blocks of the measured family of $\mathfrak{D}^{\mathrm{ref}}$ are known through enclosures, entrywise or by factor enclosures, as parts (ii) and (iii) of Lemma 21.1 form them from enclosures $(\hat{g}, \delta)$ and (M_p, ρ_p) that hold at every pair of a set $\mathcal{P}$ as in Lemma 21.2 (c), at a point x_t at which the hypotheses of that lemma hold

and (H-POS) holds for $\mathfrak{D}^{\mathrm{ref}}$, the blocks it reads with factor enclosures being read with those of Corollary 12.2, under that corollary's hypotheses (for each replaced factor whose blocks are so read, the rule of (I1) of the reduction instance of its last link a non-negative combination, and the blocks of its handed reference carrying those factor enclosures), so that its conclusion is (H-FAC). Put $\underline{h}_{\alpha}^{\mathrm{cert}} := \max_{t\in\mathfrak{I}}(b_t^{\mathrm{A}} + b_t^{\mathrm{F}})$, as in (21.3).

(i) Where $\mathfrak{D}^{\mathrm{ref}}$ satisfies Definition 1.2 and the hypotheses of Corollary 19.5 hold, the conclusions of Corollary 19.7 (iii) hold with $\underline{h}_{\alpha}^{\mathrm{cert}}$ in place of $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}^{\mathrm{ref}}]$, for the protocol whose declaration is $\mathfrak{D}^{\mathrm{ref}}$.

(ii) In the setting of Corollary 19.6, with a dropped set $\mathcal{D}$ and with the numbers formed at the gradient of the dropped objective and, in the second way, with part (d) of Lemma 21.2 in place of its part (c): $\max\{0, \underline{h}_{\alpha}^{\mathrm{cert}}\} \le \underline{h}_{\alpha}^{\mathcal{D}}[\mathfrak{D}^{\mathrm{ref}}] \le \underline{h}_{\alpha}[\mathfrak{D}^{\mathrm{ref}}]$, and where $\mathfrak{D}^{\mathrm{ref}}$ satisfies Definition 1.2 and the hypotheses of Corollary 10.18 other than (10.17) hold, the conclusions of Corollary 10.18 hold at $h = \underline{h}_{\alpha}^{\mathrm{cert}}$, for the protocol whose declaration is $\mathfrak{D}^{\mathrm{ref}}$.

Uses Def. 1.2, Cor. 10.18, Prop. 12.1, Cor. 12.2, Thm. 19.1, Cor. 19.5, Cor. 19.6, Cor. 19.7, Lem. 21.1, Lem. 21.2, Cor. 21.3 · *Used by* Rem. 22.5, Req. 28.1 · newly derived · accepted

Proof. The numbers are bounds at the companion's gradient. At a point where b_t^{A} and b_t^{F} are formed in the second way, hypothesis (H-FAC) of Lemma 21.2 holds by Corollary 12.2, under the hypotheses on which the reduction and the compiler carry the factor enclosures, so part (c) of that lemma shows that the enclosures hold at the gradient of $\mathfrak{D}^{\mathrm{ref}}$ at x_t, (H-POS) holding for $\mathfrak{D}^{\mathrm{ref}}$ there, and parts (ii) and (iii) of Lemma 21.1, which $\mathfrak{D}^{\mathrm{ref}}$ is admitted to by Proposition 12.1 (ii), turn them into $b_t^{\mathrm{A}} \le \min_{\mathbf{q}'\in\mathcal{S}_{\mathrm{acc}}}\langle g, \mathbf{q}'\rangle$ and $b_t^{\mathrm{F}} \le \min_{J'\in\mathcal{W}^J}\langle G, J'\rangle$, the gradient and the exact $\mathcal{W}^J$ being those of $\mathfrak{D}^{\mathrm{ref}}$: that is (19.3) at its gradient, as in the first way.

(i). Corollary 19.7 (iii) holds for any bound formed as its statement admits, lower bounds on the minima over the companion's exact sets standing in for the pairings at points of its relaxed set at which (H-POS), (H-SUPP) and (H-NOPERT) hold for its objects. Each x_t satisfies the hypotheses of Lemma 21.1 for $\mathfrak{D}^{\mathrm{ref}}$, which gives that by Step 1 of its proof, and the numbers satisfy (19.3) at the gradient of $\mathfrak{D}^{\mathrm{ref}}$, so Corollary 21.3 (i) applied to $\mathfrak{D}^{\mathrm{ref}}$ shows that $\underline{h}_{\alpha}^{\mathrm{cert}}$ is such a bound: the value of (19.2) at the points of $\mathfrak{I}$ with both pairings replaced.

(ii). The same argument with part (d) of Lemma 21.2, which runs part (c) for the dropped objective, and part (iv) of Lemma 21.1 gives the numbers as lower bounds on the two minima at the gradient of the dropped objective of $\mathfrak{D}^{\mathrm{ref}}$, and Corollary 21.3 (iii) applied to $\mathfrak{D}^{\mathrm{ref}}$ gives the two inequalities and, under the hypotheses that corollary's second clause and (ii) here assume, the conclusions of Corollary 10.18. □

22 The two minimisations and the arithmetic

The assumption that a solver's feasible-side return reached the minimum (A-LMO) is superseded, and no claim rests on it: the feasible side is bounded by Lemma 20.1. The acceptance-side minimisation is a hypothesis of the bound that is discharged exactly on the exact island, for a singleton or a polytope acceptance set (Remark 22.2). The remarks show that pairing over the acceptance support loses nothing and say what exact real arithmetic means for a printed bound, and which of its terms a run still computes in binary64. A certified number rests on the claims, the declaration's modelling premises and the correctness of the implementation, and on no premise about arithmetic (Remark 22.5).

Imports foundations/cor:least-eigenvalue-over-enclosure, foundations/cor:rank-over-factor-enclosure, foundations/def:tiers

At every iterate a first-order bound uses, the block list consumed from the oracle, any positivity repair included, puts the linear form of the gradient its bracket consumes at or below its minimum over the exact Choi feasible set. No claim rests on it: Lemma 20.1 bounds that minimum instead.

The first-order bound consumes at each iterate two linear minimisations: one over the acceptance set, which is exact rational data and is minimised exactly, and one over the Choi feasible set, a semidefinite program a backend solves in binary64. The second is what this assumption trusts. A positivity repair the adapter applies to the backend's return lies inside the trusted object, and checking that the return is feasible cannot replace the assumption, since feasibility gives the opposite inequality.

Assumption 22.1 [A-LMO] (Exact linear minimisation over the feasible set)**.** (i) At each iterate x_t at which the engine accumulates a bracket of the first-order bound, the block list S_t it consumes, which is the backend's return after any positivity repair the adapter applies to it, satisfies the feasible-side inequality of (18.2) (Definition 18.1), its minimum taken over the *exact* $\mathcal{W}^J$ and not over the relaxed feasible set, at G_t, the block part of the gradient (g_t, G_t) the bracket consumes at x_t, of the objective, or of that of a dropped program.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: No run rests on this assumption: the engine verifies a lower bound on the solver's linear minimisation in exact arithmetic instead of trusting it. The assumption states that at every step whose bound the engine uses, the semidefinite-programming solver returned a block list at which the linear objective it was asked to minimise, the gradient of the objective the bound is taken for, is no larger than its true minimum over the exact feasible set, after the engine's small positivity repair of the return.

Uses Def. 18.1 · *Category* numerical trust · *Affects* secrecy · *Machine key* `exact_linear_oracle` · newly derived · superseded

For a singleton or a polytope acceptance set the acceptance-side half of the linear minimisation is discharged by an exact minimisation, and a certified lower bound may replace the minimum, the bound staying valid. An acceptance iterate formed exactly stays in the set by induction.

Remark 22.2 (Exact minimisation on the acceptance side). The acceptance half of (18.2), the acceptance-side inequality, is discharged exactly, because its minimisation over $\mathcal{S}_{\mathrm{acc}}$ lies on the exact island (Definition "The exact island" (foundations/def:exact-island) (ii)): $\mathcal{S}_{\mathrm{acc}}$ is declared as exact rational data and the minimisation is a linear program over it. Two geometries are admitted, the singleton (a) and the polytope (b), and each supplies what the bracket needs.

The acceptance support $\widehat{\mathcal{C}}_{\mathrm{acc}}$ is (18.1), and by Lemma 18.2 (i) it is the support of a single element of $\mathcal{S}_{\mathrm{acc}}$ while no element has support outside it. So minimising a linear functional over $\mathcal{S}_{\mathrm{acc}}$ may be done in the coordinates $\widehat{\mathcal{C}}_{\mathrm{acc}}$ alone, which is also where the acceptance-side gradient g_t of Theorem 19.1 lives and where the first pairing of Definition 18.1 is taken — the three statements agree, and the case $\widehat{\mathcal{C}}_{\mathrm{acc}} \subsetneq \widehat{\mathcal{C}}$ is admitted throughout rather than excluded by a hypothesis.

Given an exact vector $g \in \mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$, a number β_{acc} with

$$\beta_{\mathrm{acc}} \;\le\; \min_{\mathbf{q}' \in \mathcal{S}_{\mathrm{acc}}} \sum_{\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{acc}}} g(\hat{c})\, q'(\hat{c}) \tag{22.1}$$

is a certified lower bound on the acceptance-side minimum, and a point attaining the minimum discharges the first inequality of (18.2) outright. *The bracket may consume β_{acc} in place of the pairing with such a point, and the substitution is exact rather than approximate*: at $g = g_t$, (22.1) is the hypothesis on b_t^{A} of Theorem 19.1 (iv), so replacing $\langle g_t, v_t\rangle$ by β_{acc} in (19.2) — and only there, the subtrahend $\langle g_t, \mathbf{q}_t\rangle$ being evaluated as before — keeps the conclusion $\underline{h}_\alpha^{\mathrm{FO}} \le \underline{h}_\alpha$ (a run that does so records which of the two it used, Requirement 26.3). Two instantiations are part of this submodule; an $\mathcal{S}_{\mathrm{acc}}$ that fits neither is refused at admission (Requirement 26.1). *(a) Singleton* $\mathcal{S}_{\mathrm{acc}} = \{\mathbf{q}^*\}$: the minimum is the single evaluation $\sum_{\hat{c}} g(\hat{c}) q^*(\hat{c})$, attained at $v_t = \mathbf{q}^*$, and β_{acc} is any downward-rounded evaluation of it. *(b) Polytope* $\mathcal{S}_{\mathrm{acc}} = \{\mathbf{q} :\ \mathbf{q} \ge 0,\ \sum_{\hat{c}} q(\hat{c}) = 1,\ B\mathbf{q} \le \mathbf{b}\}$ for an exact matrix B and vector $\mathbf{b}$, read in the coordinates $\widehat{\mathcal{C}}_{\mathrm{acc}}$: the minimum is attained at a vertex and is computed exactly over the rationals, which discharges (18.2)'s first inequality with equality. A dual witness certifies it independently: for any $\mathbf{u} \ge 0$ and $v \in \mathbb{R}$ with $g(\hat{c}) + (B^{\mathsf{T}}\mathbf{u})(\hat{c}) + v \ge 0$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{acc}}$,

$$\min_{\mathbf{q}' \in \mathcal{S}_{\mathrm{acc}}} \langle g, \mathbf{q}'\rangle \;\ge\; -\langle \mathbf{b}, \mathbf{u}\rangle - v :$$

indeed $\langle g, \mathbf{q}'\rangle = \langle g + B^{\mathsf{T}}\mathbf{u} + v\mathbf{1}, \mathbf{q}'\rangle - \langle \mathbf{u}, B\mathbf{q}'\rangle - v \ge 0 - \langle \mathbf{u}, \mathbf{b}\rangle - v$, using $\mathbf{q}' \ge 0$ componentwise, $\mathbf{u} \ge 0$ with $B\mathbf{q}' \le \mathbf{b}$, and $\sum_{\hat{c}} q'(\hat{c}) = 1$.

The acceptance-side iterate is on the island too, and that is what makes $\mathbf{q}_t \in \mathcal{S}_{\mathrm{acc}}$ hold by construction. The membership hypothesis of Theorem 19.1 (ii) and (iiib) is on a *product*, $x_t \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, and the three checks of Requirement 25.1 are all conditions on the J factor: per-block positive semidefiniteness, the coupling residual and the declared rows. *Nothing checks the other factor, and nothing needs to*, provided the acceptance-side iterate is formed on the exact island under two conditions (Requirement 26.2):

(α) the acceptance-side iterate is initialised at an exact rational point $\mathbf{q}_0 \in \mathcal{S}_{\mathrm{acc}}$ — one exists whenever $\mathcal{S}_{\mathrm{acc}} \neq \varnothing$, the acceptance data being exact rationals by

Definition 2.5 and Definition "The exact island" (foundations/def:exact-island), so that the single point of geometry (a) and the vertices of geometry (b) are rational; and Lemma 18.2 (i)'s equally weighted average of finitely many of them is rational too, and is supported on all of $\widehat{\mathcal{C}}_{\mathrm{acc}}$, which Remark 23.5 also needs;

(β) every acceptance-side update is a convex combination, with *exact rational* weights in $[0,1]$, of points already established to lie in $\mathcal{S}_{\mathrm{acc}}$ — in the intended use $\mathbf{q}_{t+1} = (1-\gamma_t)\mathbf{q}_t + \gamma_t\, u_t$ with $u_t \in \mathcal{S}_{\mathrm{acc}}$ exact and rational, and in geometries (a) and (b) the acceptance-side minimiser v_t is such a point: the single $\mathbf{q}^*$ in (a), a vertex computed exactly over the rationals in (b).

Under (α) and (β), $\mathbf{q}_t \in \mathcal{S}_{\mathrm{acc}}$ for every t by induction, $\mathcal{S}_{\mathrm{acc}}$ being convex (Definition 2.5) and the arithmetic exact — so the property is *true* rather than tested, there is no check for a run to fail and no residual for it to expose, and (19.2) is evaluated at a point whose acceptance factor is where the theorem says it is. *Why it must be exact and not merely checked*: on the polytope geometry (b) a binary64 convex combination of exact points of $\mathcal{S}_{\mathrm{acc}}$ can violate $\sum_{\hat{c}} q(\hat{c}) = 1$ or a row of $B\mathbf{q} \le \mathbf{b}$, and three consumers break if it does — Lemma 18.2 (ii)'s identical deletion of the $\hat{c} \notin \widehat{\mathcal{C}}_{\mathrm{acc}}$ terms is unlicensed and $O^J_\alpha(x_t)$ may be $+\infty$; the pairing on $\mathbb{R}^{\widehat{\mathcal{C}}_{\mathrm{acc}}}$ loses its justification, since $v_t - \mathbf{q}_t$ need no longer vanish outside $\widehat{\mathcal{C}}_{\mathrm{acc}}$ and $\langle g_t, v_t - \mathbf{q}_t\rangle$ would silently drop terms of unknown sign from (19.2); and Step 2's chord argument needs x_t in the convex set it runs in. The singleton geometry (a) is immune — there $\mathcal{S}_{\mathrm{acc}}$ has one point and every convex combination of it is it — but the polytope geometry is a first-class instantiation of this submodule and is not.

What this leaves. With the acceptance half discharged, the other half of (18.2), the feasible-side inequality, concerns the minimisation over $\mathcal{W}^J$ — a semidefinite program in the block variables, solved in binary64 by a backend. It is not asked of the backend's return at all: part (iv) of Theorem 19.1 takes a lower bound on that minimum in place of the pairing, and Lemma 20.1 supplies one from the backend's multipliers, formed in exact arithmetic over the exact $\mathcal{W}^J$, the rounding of the frame the rows are built with charged (Requirement 27.1). Both sides of a bracket are then lower bounds over the exact sets, certified exactly for the linear forms the run computed; what the binary64 computation of those forms leaves is Remark 22.4's.

Uses Def. 2.5, Lem. 18.2, Thm. 19.1 · *Used by* Rem. 22.3, Rem. 23.1, Rem. 23.2, Rem. 23.3, Rem. 23.6, Req. 26.1, Req. 26.2, Req. 26.3, Req. 28.1 · newly derived · accepted

Restricting the acceptance-side pairing to the acceptance support loses nothing: a feasible direction vanishes outside it, and the oracle term is defined by that pairing. The oracle return therefore need not be feasible, though in both admitted acceptance geometries it is.

Remark 22.3 (Pairing over the acceptance support). Restricting the first pairing of Definition 18.1 to the acceptance support $\widehat{\mathcal{C}}_{\mathrm{acc}}$ costs nothing and is not a choice, and the reason splits in two according to what the pairing is applied to. *Where a directional derivative is taken*, the vector on the right is $\mathbf{q}' - \mathbf{q}_t$ with *both* entries in $\mathcal{S}_{\mathrm{acc}}$: $\mathbf{q}'$ by the quantifier of (19.4), and $\mathbf{q}_t$ because the acceptance-side iterate is formed on the exact island and lies in $\mathcal{S}_{\mathrm{acc}}$ by construction (Remark 22.2, (α) and (β)). So it vanishes outside $\widehat{\mathcal{C}}_{\mathrm{acc}}$ by Lemma 18.2 (i), and the sum over $\widehat{\mathcal{C}}_{\mathrm{acc}}$ and the

sum over $\widehat{\mathcal{C}}$ agree, the second being the one that may fail to be defined. *Where the oracle return v_t enters*, the $\widehat{\mathcal{C}}_{\mathrm{acc}}$ sum is not an abbreviation of the $\widehat{\mathcal{C}}$ sum but the definition: (18.2) is stated in it, (19.2) evaluates it, and the two occurrences are the same functional, so the algebra closes whether or not v_t lies in $\mathcal{S}_{\mathrm{acc}}$; that is why (18.2) can decline to require feasibility of the pair. In both geometries Remark 22.2 admits, v_t *does* lie in $\mathcal{S}_{\mathrm{acc}}$ (the singleton's one point; a vertex computed exactly over the rationals), or the acceptance half is discharged through β_{acc} and no v_t enters at all.

Uses Lem. 18.2, Rem. 22.2 · newly derived · accepted

Both linear minima are bounded exactly over the exact sets, but for the gradient as computed: the objective, the gradient and the maps and frame behind them are binary64, and a reference's centres move the measured family. So a computed h_lb may exceed h_low, and the zero clamp repairs neither.

Remark 22.4 (Exact real arithmetic in the first-order bound). Theorem 19.1 (ii), respectively (iiib) — the part a run is in, its iterates being established only in the relaxed set (Requirement 25.1) — with the lower bounds of its part (iv) in place of both pairings, which is how a run forms every bracket (Remark 22.2, Requirement 27.1), concludes $\underline{h}_\alpha^{\mathrm{FO}} \le \underline{h}_\alpha$ in exact real arithmetic, for the exact objective and the exact gradient. No assumption about a solver enters it. What the theorem still asks, and a run cannot verify, is exact real arithmetic in the terms a run computes in binary64. What that means for a number a run prints is narrower than the theorem, and a reader who takes (19.2) for a runtime invariant is wrong in the unsafe direction.

What is exact, and of what. The two lower bounds that stand in for the minima are certified exactly (Definition "The exact island" (foundations/def:exact-island) (ii), (iii)), and both are bounds over the exact sets, but of the linear form at the gradient as the run computed it. The acceptance-side one is the exact minimum over $\mathcal{S}_{\mathrm{acc}}$ of that linear form. The feasible-side one is at most the minimum over the exact $\mathcal{W}^J$ of that linear form: the run builds its constraint rows with a binary64 frame $\widetilde{W}_{\mathtt{t}}$ in place of the frame $W_{\mathtt{t}}$ of Theorem 7.5, which carries a square root of the declared marginal, and Lemma 20.1 (iv) charges the difference, a deduction linear in the row multipliers and in a bound on $\left\|\widetilde{W}_{\mathtt{t}} - W_{\mathtt{t}}\right\|$; where no such bound is formed the row multipliers are zero and the rows are not read. So no status of the backend, no dual it returns and no rounding of the frame can raise the feasible-side bound above that minimum. Without the deduction it could be raised: Lemma 20.1 (i) applied to rows built from $\widetilde{W}_{\mathtt{t}}$ bounds the minimum over the set those rows cut out, which need not contain $\mathcal{W}^J$ and may be empty, and over an empty set every number is a lower bound.

The conclusion is not a floating-point invariant. The objective $O_\alpha^J(x_t)$, the gradient (g_t, G_t) and the subtracted pairings $\langle g_t, \mathbf{q}_t\rangle$ and $\langle G_t, J_t\rangle$ are computed in binary64 (Requirement "Exact arithmetic on the island, unbudgeted rounding elsewhere" (foundations/req:exact-island)), and so are the data the engine derives from the declaration for them: the statistics map, the branch maps and their supports, and the frame $\widetilde{W}_{\mathtt{t}}$ through which they read Alice's side. The gap between the gradient as computed and the exact gradient is part of that unbudgeted evaluation arithmetic, and no claim of this module bounds it. A computed $\underline{h}_\alpha^{\mathrm{FO}}$ may therefore exceed the computed estimate, and may exceed the true $\underline{h}_\alpha$, by an amount no

claim of this module bounds. Two consequences follow, and neither is a defect of the theorem.

(a) $\max\{0, \underline{h}_\alpha^{\mathrm{FO}}\}$ *does not repair it.* The clamp is sound and it is what Corollary 19.5 consumes, but it corrects only the *low* side: clamping at zero cannot bring down a bracket that came out too high. The clamp is no general safety net, and a run is not to be written, or read, as if it were.

(b) *A comparison of* $\underline{h}_\alpha^{\mathrm{FO}}$ *against the reported estimate is a diagnostic, never an assertion.* The ordering of $\underline{h}_\alpha^{\mathrm{FO}}$ below the objective at the returned point is a consequence of exact arithmetic and of the returned point's exact membership in $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ (Theorem 19.1 (i) and (ii)), neither of which this module establishes for a run: its returned point is established at most in the relaxed set, where the objective is bounded below only by $\underline{h}_\alpha[\mathcal{W}^J(\varepsilon)] \le \underline{h}_\alpha$ (Theorem 19.1 (iiia)), and near convergence, where $\underline{h}_\alpha^{\mathrm{FO}}$ is close to $\underline{h}_\alpha$, that alone can put the estimate below $\underline{h}_\alpha^{\mathrm{FO}}$. The difference $\underline{h}_\alpha^{\mathrm{FO}}$ minus the estimate is recorded with its sign (Requirement 24.5), and a positive difference, a bound above the estimate, is evidence about the arithmetic or about the relaxation, and not by itself evidence that the first-order tier is unsound.

Evaluation on a reference companion. Where the bound is formed on a compiled reference companion $\mathfrak{D}^{\mathrm{ref}}$ (Corollary 19.7, Requirement 24.7), the theorem is applied to the companion, a declaration of the reference model, but what is evaluated is not that declaration. Its measured family is known through an entrywise enclosure, recentred so that every entry whose enclosure contains zero has centre zero (Lemma "Zeroing the centres an enclosure cannot tell from zero" (foundations/lem:recentred-enclosure)), and it is evaluated at the binary64 numbers nearest to the centres. The evaluated family differs from the companion's, entry by entry, by at most the radius and one rounding, and it need not be a POVM: its completeness defect and its least eigenvalue are measured and recorded, never decided (Requirement 14.3 (iii)). The objective, the gradient, the branch maps and their supports, the key-inert pairs and the rank comparison behind (H-SUPP) are all formed from those numbers, and so is the linear form whose minimum the feasible-side bound bounds. Recentring is applied to the handed blocks and again to the companion's datum after the construction has combined their entries (Requirement "Emitting the reference companion beside the artefact" (reductions/req:reference-companion) (i)), so that an entry of the datum whose exact value is a sum that cancels to zero is not left with a centre that is not zero. After it, a centre is not zero only where its enclosure excludes zero; the compiler's weighting multiplies a centre by a weight and its radius by the weight's modulus, and a product of factor families multiplies single entries, so the compiled family keeps that property, and rounding a centre to binary64 can turn it to zero but never the reverse. The evaluated family therefore has a non-zero entry only where the companion's family has one; it may lack one where the companion's entry is no larger than the recentred radius there, and a support, or a key-active pair, that the companion has through such entries alone is not seen by the evaluation. What the theorem asks of the point itself, (H-SUPP), holds of the companion whenever every block of J_t is positive definite (Corollary 19.7 (ii)). This exposure is larger in kind than the binary64 evaluation of the objective and the gradient on any declaration. There the inputs are exact and only the formulas are evaluated in binary64. Here the measured family itself is displaced, by up to the radius and one rounding per

entry, with a relative error up to order one on an entry not much larger than its radius. The displacement from the companion to its evaluation is a perturbation of the measured family, and Proposition 11.1 applies to it: closeness of two measured families bounds the change of the value by nothing, and the change may point either way (Remark 11.3); the evaluated family need not even be a POVM. Its size is set by the radius of the enclosure, and not by binary64 rounding alone, save on an exact reference, whose radius is zero. No claim of this module bounds its effect, and the largest radius, which the record carries, is the only handle on it. Evaluating on the companion removes the premise that the device measures the snapped family (Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement)); it does not remove this exposure, which narrows only with the enclosure. The companion's entries are not part of this exposure: they are emitted in binary64 as the oracle reads them, their rounding absorbed into their constants (Lemma "A rounded constraint entry with its constant relaxed" (reductions/lem:row-relaxation)), so they are the companion's own exact rows, and the feasible-side bound is over the companion's exact $\mathcal{W}^J$, the frame's rounding charged as on any declaration. The first-order number is not evaluated over the enclosure; the certified number of Requirement 28.1 is, where every element the gradient reads has a rank fixed throughout the enclosure (Lemma 21.2), and it then holds of the companion itself.

Where the exposure is. Of the ingredients of (19.2), the objective and the gradient are evaluated by formulas this module states in closed form, at points where (H-POS) keeps every denominator bounded away from zero; the subtracted pairings are inner products; and the two lower bounds are certifications of the exact island over the exact sets, for the gradient as computed. No ingredient is a solver's return: the oracle's point and its status are diagnostics (Requirement 27.1), and a solve that stopped short of its optimum, or failed, loosens a bracket and cannot raise it above the value it has at the exact minima of the computed linear forms. What a run leaves unverified is named: the binary64 evaluation of the objective, of the gradient and of the maps and frame they are built from, of the subtracted pairings, and of the checks whose one-sided exposures Remark 23.6 names; and on a reference companion the displacement to its centres. It is *unbudgeted*: no claim of this module bounds it, and no assumption of this module covers it, the engine computing outside the exact island without an error budget, as Requirement "Exact arithmetic on the island, unbudgeted rounding elsewhere" (foundations/req:exact-island) allows. That is what the first-order tier costs: it is weaker than a verified numerical certificate of $\underline{h}_\alpha$, a first-order number being a lower bound established under the hypotheses that the claim stating it names (Definition "The tiers of a reported key rate" (foundations/def:tiers)), evaluated in arithmetic that no claim of this module budgets. The certified tier of the same definition is the bound formed again with that arithmetic exact or enclosed, and what it still rests on is Remark 22.5.

Uses Prop. 11.1, Rem. 11.3, Thm. 19.1, Cor. 19.5, Cor. 19.7, Lem. 20.1 · *Used by* Rem. 22.5, Rem. 23.8, Rem. 23.9, Req. 24.5 · newly derived · accepted

A certified number rests on the claims, the modelling premises of its declaration and the correctness of the code, its ball-arithmetic library included, and on no premise about arithmetic. It does not reach data computed before the declaration; a receiver weight in floating point withholds it.

Remark 22.5 (What a certified number rests on). A certified number, Requirement 28.1, is by Corollary 21.3 a lower bound on the value of the declaration it is formed on, and it rests on three things.

The claims. Lemma 21.1 and Corollary 21.3, on a reference companion with Lemma 21.2 and Corollary 21.4, with what they use, the hypotheses of Theorem 19.1 among them, each of which the evaluation establishes at the point itself: positive definite blocks, which give (H-SUPP) by Lemma 8.2 (i); (H-POS) with certified signs; and the exact gradient, which is (H-NOPERT). The binary64 checks whose exposures Remark 23.6 names, the oracle's return and the gradient the search computed enter no certified number, and the exposures of Remark 22.4 are those of the first-order number, not of this one.

The modelling premises of that declaration. They are the premises of the first-order number on the same declaration, and certification decides none of them and adds none. Where the number is formed on the declaration the run searched, the blocks that rest there on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) rest on it for the certified number as well. That includes the fallback of Requirement 28.1 (i): where a companion counts but certifies no point and the number is certified on the searched declaration, the blocks the companion carried rest on that assumption for the reported key length, though they would not for the companion's first-order number (Requirement 24.7 (v)). On a reference companion the number holds of the companion itself, not of the centres of its enclosure; and the premises of the source and the receiver remain, on every route.

The implementation. Every arithmetic step from the declaration's data is exact or rigorously enclosed, so no premise is about how a number was computed and none of the category numerical-trust enters through the evaluation (Definition "The tiers of a reported key rate" (foundations/def:tiers)). What remains is that the code computes what the claims state, the library of the rigorous arithmetic included. That is the correctness of the engine, held to review and to tests like every other part of it; no assumption is stated for it, and a defect there is a defect of the engine and not the failure of a premise.

The chain begins at the declaration's data. A supplier's floating-point evaluation that produced those data, under an allowance or an assumption of its own, lies before it. Such an evaluation of the receiver's blocks reaches a number formed on the declaration the run searched only through the snap, where the run rests on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) as the paragraph on modelling premises above says, and the tier is not withheld there. The tier is withheld in the two cases Requirement 28.1 (vii) names: where an assumption of the category numerical-trust is invoked on the run, and where the declaration carries a weight routed from a receiver's high-photon weights into which a per-setting weight enters that the receiver evaluated in floating point, the uniform weight counting every setting, a number that holds of the device only as Assumption "Supplier artefact whose claims hold of the device" (reductions/ass:recv) (ii) attests it and that the receiver records so (Requirement "The evaluation label on every artifact" (receiver/req:evaluation-label)).

A receiver weight evaluated rigorously is a bound by the receiver's own theorem (Theorem "The weight on enclosed bound data bounds the multiclick element" (receiver/thm:enclosed-weight)), and rests on that assumption only for the artifact's provenance and for the device premises of the receiver, where the ledger carries its row; for an artifact the run built itself, exact and with every weight evaluated rigorously, the row is omitted (Requirement "Omission of the supplier row for a receiver artefact the run built" (reductions/req:recv-row-provenance)), and the receiver's own ledger carries those device premises.

Over a companion known through enclosures, the number exists only where every element the gradient reads has a fixed rank (Lemma 21.2). An element whose principal block on the rows its entrywise enclosure does not fix at zero is singular has no positive lower bound on that block's least eigenvalue, the element being among the matrices its enclosure contains, so the rule of Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure) (iv) fixes no rank for it, at any radius. Among the elements whose enclosure carries a positive radius, that is the case of every rank-deficient element whose support is spanned by no set of coordinate vectors, by part (iii) of that corollary, such as the projector of an ideal analyser at an angle whose projector has irrational entries, and of every rank-deficient element whose zero rows the enclosure does not fix at zero. Such an element has a fixed rank all the same where it is carried with a factor enclosure that has no column, or whose columns have a Gram matrix with a positive floor over its box, its rank then being the number of columns at every member (Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure)); so the analyser's projector certifies on the companion where the receiver emits its elements in that form. A factor enclosure fixes no rank where the columns of a member are linearly dependent, its Gram matrix being singular there, and where the member's rank is also below its dimension, neither rule fixes one: a multi-photon element of a receiver with loss and no dark counts can be of this kind. Nor is a rank fixed for a rank-deficient element carried with no factor enclosure, as where the receiver forms its blocks without an exact dilation of its instrument, for a generic component with none, or where a squashing map acts on a data sector after the factor is formed. A run whose companion reads such an element at a pair that is not key-inert certifies no point on the companion, and it reaches the certified tier only through the fallback of Requirement 28.1 (i), on the declaration it searched, which does not run under the strict option *strict_snaps* (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv)), resting there on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) for the blocks the companion carried. An element the companion carries exactly, with radius zero, is factorised exactly and is not subject to these rules.

Reading: Certified means that every number between the declaration and the key rate was computed exactly or with a rigorous bound on its rounding. It does not mean that the declaration describes your devices: the source, receiver and laboratory premises in the report still apply, and a number formed on the snapped measurement rests where the record says on the device measuring it. Nor is the software assumed correct by a premise: its correctness, its interval-arithmetic library included, is checked by review and tests. Ideal analysers with irrational projector entries (at pi/8, say) certify on your model where the receiver gives their elements in factor form; otherwise, as for some lossy multi-photon receivers, the rate can be certified on the snapped measurement, resting there on that premise, unless the run sets strict_snaps. A floating-point multiclick weight keeps the

first-order number.

Uses Lem. 8.2, Thm. 19.1, Lem. 21.1, Lem. 21.2, Cor. 21.3, Cor. 21.4, Rem. 22.4, Rem. 23.6, Req. 28.1, foundations/cor:least-eigenvalue-over-enclosure, foundations/cor:rank-over-factor-enclosure, foundations/def:tiers · newly derived · accepted

23 The reported numbers and the runtime checks

These remarks give the mathematics behind the reporting rules. A run reports its estimate and, where the hypotheses held, the first-order bound, and withdrawal never weakens what the bound means (Remark 23.1). The iterate checks decide the relaxed feasible set on the Choi factor only and serve the membership hypothesis and the estimate's premise (Remark 23.2); maximal support is a rank equality read with one ruler (Remark 23.4); and the checks, computed in binary64, leave one-sided exposures that no claim bounds. A repaired oracle return is a step of the search and enters no bracket, a repaired iterate is a construction step, and a negative bound is clamped while a negative estimate is refused.

A run reports its estimate always, and h_lb only where the hypotheses of Theorem 19.1 held, membership of the consumed point in the acceptance set times the relaxed set among them. No hypothesis is read of the solver, and withdrawal never weakens what h_lb means.

Every run reports its estimate, and where the hypotheses of Theorem 19.1 held the first-order bound $\underline{h}_\alpha^{\mathrm{FO}}$ as well, naming the tier that carries its claim (Requirement "Reporting of the tiers" (foundations/req:tiers)). The requirements of this submodule say when $\underline{h}_\alpha^{\mathrm{FO}}$ is withdrawn and what a run checks, repairs and records; this remark and Remarks 23.2, 23.3, 23.4, 23.5, 23.6, 23.7, 23.8 and 23.9 give the mathematics behind them, one topic each.

Remark 23.1 (Two reported numbers and one first-order claim). The objective at the point a run returned is an upper estimate by Theorem 19.1 (i), respectively (iiia), which is the part a run is in: its point is established, where it is established at all, only in the relaxed feasible set (Remark 23.2), and Remark 23.3 says what a run records of it, which is less than the whole premise. Where the hypotheses of that theorem held, the run reports $\underline{h}_\alpha^{\mathrm{FO}}$ as well; Requirement 24.1 says when it does not.

Among those hypotheses is the membership of the point consumed at an iterate in the *product* $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$: it is a hypothesis of Theorem 19.1 (iiib) as much as (H-POS), (H-SUPP) and (H-NOPERT). Its two factors are established by different means: the J factor by the three checks of Remark 23.2, after any repair of Remark 23.8, and the acceptance factor by forming $\mathbf{q}_t$ on the exact island under (α) and (β) of Remark 22.2. A failure of either leaves the membership not established.

No hypothesis is read of the solver. A return that is merely near the minimum satisfies no inequality in the safe direction, which is why no bracket pairs the gradient with it: both sides of a bracket are lower bounds on the two minima over the exact sets, the feasible one verified from the solver's multipliers in exact arithmetic, the rounding of the frame its rows are built with deducted (Requirement 27.1). So a

solve that stopped short of its optimum, or failed, can loosen a bracket but cannot raise either side above the exact minimum of the linear form at the gradient as computed, and neither its return nor its status withdraws one; what the gradient's own binary64 evaluation leaves is Remark 22.4's. Withdrawal is never a weakening of what $\underline{h}_\alpha^{\mathrm{FO}}$ means; there is no reduced form of the claim.

Uses Def. 17.1, Def. 18.1, Thm. 19.1, Rem. 22.2 · *Used by* Req. 24.1 · newly derived · accepted

The three iterate checks decide every condition of the relaxed feasible set, all of them on the J factor; the acceptance factor holds by construction. They serve the membership hypothesis and the estimate's premise; no check of a solver's return would serve the bound.

Remark 23.2 (What the iterate checks establish). Requirement 25.1 decides, on the point consumed at each iterate, every condition defining $\mathcal{W}^J$ in (7.9): per-block positive semidefiniteness, the coupling condition $\sum_s \mathrm{tr}_{Q_s} J_s = \mathbb{1}_{A_0}$, *and the declared constraint rows* $\bowtie_k$. Checking two of the three is not checking the condition. *No such check is made on a point a solver returned, and none would serve the bound*: verifying $S_t \in \mathcal{W}^J$ gives $\langle G_t, S_t\rangle \geq \min_{J' \in \mathcal{W}^J} \langle G_t, J'\rangle$, the *opposite* inequality to (18.2). Feasibility bounds the linear form from below, which is exactly the side the first-order claim cannot use; one can verify *which set* a solver minimised over, never that it minimised. The feasible side of a bracket is therefore a lower bound formed without any point of the solver (Requirement 27.1). What the checks do buy is the membership hypothesis of Theorem 19.1 (iiib) on its J factor, and Theorem 19.1 (i), respectively (iiia): the point the run reports its estimate at is a convex combination of oracle returns; (19.1) holds only for points of $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$, which is what a run does *not* establish, and (iiia) is the part it is in, giving $O_\alpha^J(x) \geq \underline{h}_\alpha[\mathcal{W}^J(\varepsilon)]$ at the relaxed membership the checks do establish — on the point the run returned, the membership the end-of-run measurement of Remark 23.3 establishes, which is of the J factor only. So the checks serve the bound's membership hypothesis and the premise of the estimate tier, which is the tier reported on every run. Their windows are numbers of their own and not the declared tolerance τ_{num}, the object checked being one the optimiser computed (Requirement "Scope of the declared tolerance" (foundations/req:tolerance-scope) (ii)).

Between them the three checks are what membership in $\mathcal{W}^J(\varepsilon)$ at the larger of the two declared windows asks for, and they cover all of it: per-block positive semidefiniteness, which the family does not relax and which is decided on the point consumed at exactly 0 and at no window (Requirement 25.4 says what may be done to a block that fails it, and the record it must carry); the coupling residual and each $\bowtie_k$ residual, each inside its own window, hence inside the larger of the two by monotonicity (Proposition 17.2 (a)); and the family's normalisation clause with them, that clause being implied by the coupling clause and adding no check (Proposition 17.2 (c′)). No fourth check is needed.

All three are conditions on the J factor. The hypothesis they serve is on a *product*, $x_t \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$, and nothing in these checks touches the other factor: *the three checks establish the J half and no more, and the* $\mathcal{S}_{\mathrm{acc}}$ *half is not checked at all.* It is instead true *by construction*, the acceptance-side iterate being formed in exact rational arithmetic on the exact island under (α) and (β) of Remark 22.2: a convex combination, with exact rational weights, of exact points of a convex set, so $\mathbf{q}_t \in \mathcal{S}_{\mathrm{acc}}$ by induction. Remark 22.2 names the three consumers of Theorem 19.1

that break if it does not hold. A run that forms $\mathbf{q}_t$ in binary64 instead has not met the membership hypothesis (Requirement 26.2); there is no check for it to pass on that factor, and none is offered. The coverage claim of the previous paragraph is therefore a claim about one factor of a product.

The window on the positivity check decides whether the coverage claim is true. At a positive window w the positivity check would accept a point with $\lambda_{\min} \in (-w, 0)$, which lies in *no* $\mathcal{W}^J(\varepsilon)$ for any ε — the family relaxes the coupling and the $\bowtie_k$ rows and relaxes positivity not at all (Proposition 17.2 (c), *Fourth*) — so at such a point O^J_α is not even defined and neither (i) nor (ii) nor (iii) of Theorem 19.1 applies. That is why positivity is decided at 0, and the drift a declared window would let pass silently is repaired and recorded under Requirement 25.4. By Theorem 19.1 (iiia) that reading is the conservative one: membership in the wider $\mathcal{W}^J(\varepsilon)$ gives $O^J_\alpha(x) \geq \underline{h}_\alpha[\mathcal{W}^J(\varepsilon)]$, the *Choi-form* relaxed infimum (17.6), and $\underline{h}_\alpha[\mathcal{W}^J(\varepsilon)] \leq \underline{h}_\alpha$ by (17.7), so the estimate over-estimates a quantity that is itself no larger than $\underline{h}_\alpha$. (The state-form $\underline{h}_\alpha[\mathcal{W}(\varepsilon)]$ is a different number and is not what the checks buy; Proposition 17.2 (d′) says why the two are not interchangeable at $\varepsilon > 0$.)

Uses Thm. 7.5, Def. 17.1, Prop. 17.2, Def. 18.1, Thm. 19.1, Rem. 22.2 · newly derived · accepted

The point a run returns is checked once more, on its J factor only, so a pass establishes that factor of the estimate's premise and a failure leaves the premise unestablished. Neither h_lb nor the key length rests on that point, which bears on a run only through the refusal of a negative estimate.

Remark 23.3 (What the measurement of the returned point establishes)*.* The checks of Remark 23.2 are made on the point consumed at each iterate, and the point a run returns need not be one of those: a run that stops at its iteration limit returns a point formed after the last point it consumed, and Requirement 24.2 applies the three checks to it once more. The measurement changes nothing about the point, and it is binary64 arithmetic like every application of those checks (the third exposure of Remark 23.6).

What the record says is about the J factor only. The three checks are the J-side checks of Remark 23.2; the premise is membership in the *product* $\mathcal{S}_{\text{acc}} \times \mathcal{W}^J(\varepsilon)$, and its $\mathcal{S}_{\text{acc}}$ factor is not checked but holds by construction, and only where the acceptance-side point was formed on the exact island under (α) and (β) of Remark 22.2. Whether the acceptance-side point the run returned was so formed is *not* part of this measurement: the run's record states that property of the points consumed at the iterates (Requirement 26.2) and does not state it of the returned point as such; where that point is the last one consumed, the record's entry for that iterate carries it.

When the point passes all three, its membership in $\mathcal{W}^J(\varepsilon)$ is established as far as those checks establish any membership; the premise of Theorem 19.1 (iiia) is then established at that point exactly when its acceptance-side component was formed on the exact island, a property the record does not state of the returned point as such. *When it fails any of the three*, the J half of that premise, and so the premise of Theorem 19.1 (i), respectively (iiia), is *not established at that point*, so the reported objective is not established to be an upper estimate even of $\underline{h}_\alpha[\mathcal{W}^J(\varepsilon)]$, and the record says so. The estimate is still reported, as on every run: it is never a bound, so there is no claim to withdraw from it, and the record names the premise it lacks

rather than hiding the number.

Neither the first-order bound nor the key length rests on any property of that point. $\underline{h}_\alpha^{\mathrm{FO}}$ is the maximum in (19.2) over the iterates of $\mathfrak{I}$, each of which met every condition of Requirement 24.1; under a non-empty dropped set $\mathcal{D}$ the bound reported instead is $\underline{h}_\alpha^{\mathrm{FO},\mathcal{D}}$ of (19.8), the same expression read on the dropped objective over the iterates at which the hypotheses of the dropped program held (Corollary 19.6). The key length reads this submodule only through $\underline{h}_\alpha^{\mathrm{FO}}$, respectively $\underline{h}_\alpha^{\mathrm{FO},\mathcal{D}}$ (Corollary 19.5, respectively Corollary 10.18 at $h = \underline{h}_\alpha^{\mathrm{FO},\mathcal{D}}$ by Corollary 19.6); the measurement of the returned point enters neither the bound, nor the tier, nor the key length. The one way that point bears on a run is Requirement 24.4: a negative objective there refuses the run, and no key length is emitted on a refused run — a refusal, never a changed value.

Uses Cor. 10.18, Thm. 19.1, Cor. 19.5, Cor. 19.6, Rem. 22.2 · newly derived · accepted

(H-SUPP) is equivalent to rank X = dim W, never to positivity of X on its ambient space, whose spectrum always contains 0. Both sides are counted relative to a spectral radius at one cutoff, the reference rank at its square root, since a direction of singular value σ enters X with weight σ^2.

Remark 23.4 (Maximal support as a rank equality read with one ruler). (H-SUPP) is decided by a rank comparison at one declared cutoff, not by a proof. (H-SUPP) says $\operatorname{supp} X_{\hat{c},s} = W_{\hat{c},s}$, and since $\operatorname{supp} X_{\hat{c},s} \subseteq W_{\hat{c},s}$ always ((8.3)), it is equivalent to the *rank equality*

$$\operatorname{rank} X_{\hat{c},s} = \dim W_{\hat{c},s}, \tag{23.1}$$

equivalently to strict positivity of the compression of $X_{\hat{c},s}$ to $W_{\hat{c},s}$. *It is not, and must not be read as, strict positivity of $X_{\hat{c},s}$ on its ambient space*: $X_{\hat{c},s}$ lives on $\mathcal{H}(R\,XY\,C_A C_B\,AQ_s)$ while $\dim W_{\hat{c},s} \le |\mathcal{K}_{\mathrm{pass}}(\hat{c})| \cdot \dim \mathcal{S}_A \cdot d_s$ by (8.4), so the ambient spectrum always contains 0 and an ambient positivity test is never satisfied. Requirement 25.2 computes both sides of (23.1) and reads both with one declared cutoff: the left side as the count of eigenvalues of $X_{\hat{c},s}$ above the cutoff, *relative to that block's own spectral radius*; the right side as $\operatorname{rank} \mathbf{K}_{\hat{c},s}$ of (8.2), read at the square root of the cutoff relative to that matrix's largest singular value, equivalently the count of eigenvalues of $\mathbf{K}_{\hat{c},s}\mathbf{K}_{\hat{c},s}^\dagger$ above the cutoff relative to its spectral radius, which is the left side's own count taken at the identity input ($J_s = \mathbb{1}_{A_0 Q_s}$, respectively $\omega_s = \mathbb{1}_{AQ_s}$ in untrusted-source mode). The relative reading is right rather than convenient, because (8.3)'s condition is a rank condition, invariant under scaling the block, and ∇T is positively homogeneous of degree zero, so a uniformly small block is not a degenerate one.

Why one ruler. The branch operator is $X_{\hat{c},s} = \mathbf{K}_{\hat{c},s}(\mathbb{1} \otimes J_s)\mathbf{K}_{\hat{c},s}^\dagger$, the identity acting on the index of the pass classes (with ω_s for J_s in untrusted-source mode), so $\lambda_{\min}(J_s)\,\mathbf{K}_{\hat{c},s}\mathbf{K}_{\hat{c},s}^\dagger \preceq X_{\hat{c},s} \preceq \lambda_{\max}(J_s)\,\mathbf{K}_{\hat{c},s}\mathbf{K}_{\hat{c},s}^\dagger$: a direction along a singular value σ of $\mathbf{K}_{\hat{c},s}$ carries weight of order σ^2 in $X_{\hat{c},s}$, never of order σ, and $\operatorname{supp} X_{\hat{c},s} \subseteq \operatorname{supp} \mathbf{K}_{\hat{c},s}\mathbf{K}_{\hat{c},s}^\dagger = W_{\hat{c},s}$, which is (8.3). A window at the cutoff's own value on σ would count in $\dim W_{\hat{c},s}$ directions whose relative weight in $X_{\hat{c},s}$ is below that value at every iterate with J_s not far from a multiple of the identity — for a direction at relative singular value σ_{rel}, every iterate at which $\lambda_{\max}(J_s)/\lambda_{\min}(J_s)$ is at most the cutoff divided by σ_{rel}^2 — and the comparison would then fail at every such iterate however exactly (H-SUPP) holds; read at the square root, both sides count

eigenvalues of operators of the one form $\mathbf{K}_{\hat{c},s}(\mathbb{1} \otimes J)\mathbf{K}_{\hat{c},s}^{\dagger}$ at one relative window.

Uses Def. 8.1, Lem. 8.2, Thm. 19.1 · newly derived · accepted

An iterate with positive weight on a point of positive definite Choi blocks and an acceptance vector positive on the acceptance support meets (H-SUPP) and (H-POS)'s first clause; such a point exists if the declared rows admit one. No optimiser is fixed: a run without such a component may lack h_lb.

Remark 23.5 (Reachability of positivity and maximal support). What makes (H-SUPP) and (H-POS) reachable at all is a separate question from what discharges them. $J_{t,s} \succ 0$ suffices for (H-SUPP) by (8.3), and q_t strictly positive on $\widehat{\mathcal{C}}_{\mathrm{acc}}$ gives the first clause of (H-POS); neither is automatic, and an oracle return is typically an extreme point of its feasible set and so neither strictly positive nor of full support on $\widehat{\mathcal{C}}_{\mathrm{acc}}$. What supplies both is the *formation* of the iterate: a point that carries a strictly positive weight on some $x_0 \in \mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$ with $J_{0,s} \succ 0$ for every s and $q_0(\hat{c}) > 0$ for every $\hat{c} \in \widehat{\mathcal{C}}_{\mathrm{acc}}$ inherits both properties, a sum of a positive definite operator and positive semidefinite ones being positive definite. Such an x_0 exists whenever the declared rows admit one — with $\mathfrak{L} = \varnothing$, $J_{0,s} = |\mathbb{S}|^{-1}\,\mathbb{1}_{A_0} \otimes d_s^{-1}\mathbb{1}_{Q_s}$ is feasible and strictly positive, and Lemma 18.2 (i) supplies a $\mathbf{q}_0 \in \mathcal{S}_{\mathrm{acc}}$ supported on all of $\widehat{\mathcal{C}}_{\mathrm{acc}}$. This module prescribes no optimiser and does not require one: the statement is what the first-order tier costs, not an algorithm. A run whose iterates carry no such component may legitimately fail (H-POS) or (H-SUPP) at every iterate, in which case $\mathfrak{I}$ is empty, $\underline{h}_{\alpha}^{\mathrm{FO}}$ is absent, and the run reports its estimate, correctly: the tier is conditional on how the iterates are formed, not unreachable.

Uses Lem. 8.2, Lem. 18.2, Thm. 19.1 · newly derived · accepted

The rank comparison for (H-SUPP) and the three iterate checks are computed in binary64 and leave three one-sided exposures, two on the unsafe side, that no claim of this module bounds. A reference rank computed too large is harmless, and the acceptance factor, exact by construction, adds none.

Remark 23.6 (One-sided exposures of the runtime checks). The rank comparison of Remark 23.4 buys a refusal on the safe side in the common case, a rank-deficient block being refused rather than differentiated. It does not buy the hypothesis, and there are *three* one-sided residual exposures, the first two on the unsafe side of Remark 19.3 and the third on the membership hypothesis. *First*, a block whose true support is a proper subspace of $W_{\hat{c},s}$ but whose smallest relevant eigenvalue is reported above the cutoff by the arithmetic of the spectral decomposition is differentiated on a non-maximal face silently. *Second*, a $\dim W_{\hat{c},s}$ computed *too small* — a direction of $W_{\hat{c},s}$ that is genuinely present but whose relative weight in $\mathbf{K}_{\hat{c},s}\mathbf{K}_{\hat{c},s}^{\dagger}$, σ^2 over the largest σ^2, is at most the cutoff, and which is therefore not counted — lowers the bar in (23.1) and passes a rank-deficient block for the same reason. A $\dim W_{\hat{c},s}$ computed too *large* is harmless for the claim: it can only fail the comparison, which removes the iterate's bracket from $\underline{h}_{\alpha}^{\mathrm{FO}}$ — at every iterate, so that $\underline{h}_{\alpha}^{\mathrm{FO}}$ is withdrawn, if it counts a direction no $\operatorname{rank} X_{\hat{c},s}$ can count, which is why the two sides are read with one ruler. *Third*, and belonging to the checks of Remark 23.2 rather than to the rank comparison: each of the three quantities those checks compare is itself computed in binary64. $\lambda_{\min}$ of an iterate block is computed, so a block whose true minimum eigenvalue is negative within the eigensolver's own backward error passes at exactly 0 and is not repaired, and the reassembly of a block

that *is* repaired is itself rounded; and the coupling residual $\|\sum_s \mathrm{tr}_{Q_s} J_s - \mathbb{1}_{A_0}\|_{\max}$ and each $\bowtie_k$ residual are formed by partial traces and sums that round, so a point whose *true* residual exceeds its window can be measured inside it. So the consumed point lies in $\mathcal{W}^J(\varepsilon)$ by construction only up to that arithmetic.

The acceptance factor adds no fourth exposure to this list: it is not measured at all, being exact by construction on the exact island (Remark 22.2), so the count is three because the three are the J-side checks, and the product's other factor is established in a way that has no residual to expose. Nor does the feasible side of a bracket: its value is a certification computed exactly from the numbers it is given (Requirement 27.1) and decides nothing at a window, the rounding of the frame its rows are built with being charged within it, and how far the gradient it is given lies from the exact one is the binary64 evaluation of Remark 22.4, not a check. No claim of this module bounds any of the three errors; what the declared thresholds and the exact-0 check buy is that each comparison is well posed, that its reference is fixed by this module rather than chosen by an implementation, and that no window *this module declares* admits a point at which the theorem does not apply.

Uses Thm. 19.1, Rem. 19.3, Rem. 22.2 · *Used by* Rem. 22.5 · newly derived · accepted

Repairs are permitted on computed objects and never on declared data. A repaired oracle return feeds only the search and a recorded diagnostic, and its record bounds nothing: a window on the block bounds no error on a pairing whose gradient can grow without limit.

Remark 23.7 (A repaired oracle return as a step of the search). This remark and Remark 23.8 concern two positivity repairs, on two different objects. Both are on objects this module computed, so both are permitted where no repair of declared data ever is (contrast Requirement 14.1: declared data is exactly admissible or refused, and this module repairs none of it). The first, on an *oracle return*, touches only the search; the second, on a *formed iterate*, is a construction step that makes a hypothesis of Theorem 19.1 hold by construction.

On an oracle return: a step of the search. A backend may return a block whose spectrum is slightly negative, and Requirement 25.3 lets the engine reject the return or project the block within a window of its own. The repaired return is a direction for the search and the point of the recorded diagnostic pairing; no bracket of $\underline{h}_\alpha^{\mathrm{FO}}$ reads it, the feasible side of every bracket being a lower bound formed from the backend's multipliers (Requirement 27.1), so the repair needs no account in any claim of this module. Nor could its record supply one: a declared window bounds $\|S_t^+ - S_t\|$, but converting that into an error on $\langle G_t, S_t\rangle$ needs a bound on $\|G_t\|$, and (8.10)–(8.11) carry the ratios $q/P_{\hat{c}}^J$ and $q/K_{\hat{c}}^J$, which Proposition 5.8 (iii) exhibits declarations driving up without limit.

Uses Prop. 5.8, Cor. 8.5, Thm. 19.1 · *Used by* Req. 25.3 · newly derived · accepted

A block repaired while the iterate is formed makes the repaired point the iterate, at which every term of the bracket is evaluated, so no perturbation that (H-NOPERT) forbids occurs. The repair moves the coupling residual by at most the dimension times its size and says nothing about the backend.

Remark 23.8 (A repaired iterate as a construction step). *On a formed iterate: a construction step, and the only repair this module licenses on an iterate.* The point consumed at iterate t is formed by the engine — in the intended use a convex

combination of oracle returns — and Requirement 25.1 requires it to be per-block positive semidefinite at exactly 0; a block failing that check is rejected or projected (Requirement 25.4). *The repaired point is x_t.* The repair belongs to the formation of the iterate, upstream of the point the theorem is applied at, and it is a different act from the oracle repair of Remark 23.7, which touches only the search; everything in (19.2) — the objective $O^J_\alpha(x_t)$, the gradient (g_t, G_t) and the pairings with x_t — is then evaluated at that one point. *This is not the perturbation route Remark 19.4 closes, and (H-NOPERT) is not violated*: that route takes the gradient at a perturbed operator and concludes something about the objective at the unperturbed one, which needs a domination statement this module does not prove, whereas here no second function enters the argument at all — Theorem 19.1 concludes about the infimum over the whole set and asks of x_t only that it lie in $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$. What (H-NOPERT) forbids is a perturbation applied *to* x_t, between the point being fixed and the gradient being taken; a repair applied *before* the point is fixed is a different act, and it is the one Requirement 25.4 licenses.

The repair moves the residuals by a bounded amount. A projection moves the coupling and $\bowtie_k$ residuals of Remark 23.2: if the repair moves each block by at most δ in the operator norm, the coupling residual moves by at most $d_Q\,\delta$ in $\|\cdot\|_{\max}$, since $\|\operatorname{tr}_{Q_s} \Delta_s\| \le d_s \|\Delta_s\|$, $\sum_s d_s = d_Q$ and $\|\cdot\|_{\max} \le \|\cdot\|$, a bound that carries its dimension factor as Requirement "Explicit conversion between norms" (foundations/req:norm-conversion) asks. The requirement nevertheless measures the residuals again rather than bounding them in advance. What the repair establishes is a hypothesis of Theorem 19.1, on an object the engine itself chose, and it says nothing whatever about the backend.

What it does not establish. $\lambda_{\min}$ is itself computed, so a block whose true minimum eigenvalue is negative within the eigensolver's own backward error passes the check unrepaired, and the reassembly of a projected block is itself rounded. That residue is a third one-sided exposure of the kind Remark 23.6 names, and it is named there with the other two. Its *nature* is the unbudgeted arithmetic this module declares everywhere (Remark 22.4), and not a window this module chose while knowing it admitted points at which its own theorem does not apply.

Uses Thm. 19.1, Rem. 19.4, Rem. 22.4 · *Used by* Req. 25.4 · newly derived · accepted

A negative computed h_lb gives way to max{0, h_lb}, itself a valid lower bound on h_low. The objective is non-negative on the exact feasible set but not on the relaxed one, so a negative estimate is evidence that a hypothesis fails, and it is refused rather than replaced by zero.

Remark 23.9 (Opposite treatment of a negative bound and a negative estimate). A negative number can arise on each path, both cases in floating point, and the two are handled oppositely.

On the first-order path. A computed $\underline{h}^{\mathrm{FO}}_\alpha$ may be negative — the linearisation is weak far from the optimum, or the arithmetic crosses zero — and what reaches Corollary 19.5 is $\max\{0, \underline{h}^{\mathrm{FO}}_\alpha\}$ (Requirement 24.3), which is a valid lower bound on $\underline{h}_\alpha$ by Theorem 19.1 (ii), respectively (iiib) — the second being the part a run is in, since (ii) needs exact membership and Remark 23.2 says no run establishes it; the two conclude the same inequality against the same exact $\underline{h}_\alpha$, (ii) being the case $\varepsilon = 0$ of (iiib). This is not a repair of anybody's data; it is the substitution of one valid lower bound for another, and Remark 22.4 (a) says what it does not fix.

On the estimate path. A computed objective at a point of the *exact* feasible

set $\mathcal{W}^J$ cannot be negative: writing O_α^J as in the proof of Proposition 5.8 (i), it is a non-negative multiple of a KL divergence between two probability vectors plus a non-negative combination of the branch terms $e_{\hat{c}} = \frac{1}{1-\alpha}\log_2 \frac{K_{\hat{c}}}{P_{\hat{c}}}$, each non-negative because $K_{\hat{c}} \le P_{\hat{c}}$ (Theorem 4.1, *cone clause*) and $\frac{1}{1-\alpha} < 0$; that argument is pointwise and needs no infimum. *At a point of the relaxed set $\mathcal{W}^J(\varepsilon)$ it can be negative*, and the statement is written on the exact set for that reason: there the *second* argument of the divergence term is not a probability vector, and by Proposition 17.2 (c″) the objective is bounded below by $-\frac{\alpha}{\alpha-1}\log_2(1 + d_{A_0}\varepsilon)$ and by nothing larger. A run reads its feasible set at the two declared windows of Requirement 25.1, so it has a floor below zero, and the refusal of Requirement 24.4 can fire against a point that is otherwise in order. That is intended: the refusal is on the safe side, no key length is emitted on a refused run, and what is refused is an estimate and not a bound. So a negative computed estimate is *evidence that a hypothesis of this module does not hold* of the point it was computed at — most likely that the point is not in the exact set (Remark 23.2), that its distance from it exceeds the scale just named, or that the declaration was not exactly admissible, which its admission excludes (Definition 1.1). Unlike $\underline{h}_\alpha^{\mathrm{FO}}$ the estimate is not a lower bound, and zero cannot stand in for it: replacing it by zero would report a plausible number in place of a detected inconsistency.

Uses Thm. 4.1, Prop. 5.8, Prop. 17.2, Thm. 19.1, Cor. 19.5, Rem. 22.4 · newly derived · accepted

24 Requirements: reported numbers and withdrawal

A run reports its estimate and, where the hypotheses held, the first-order bound; an iterate at which a hypothesis fails or is not established adds no bracket to the bound, with a reason, and with no iterate left the engine withdraws the bound. The backend's return and status withdraw nothing. Where a reference companion is admitted, the run's number is the bracket on the companion at the best iterates of the search and its last and at the points of a short search continued on the companion, the bound on the searched declaration a diagnostic beside it. It clamps the bound at zero with a flag, refuses a negative estimate, and imports no penalty from another formulation. It measures the returned point once more and records the difference between the bound and the estimate, resting nothing on either.

Imports foundations/def:evaluated-off-snap, foundations/req:snap-reference

An iterate at which a first-order hypothesis fails or is not established contributes no bracket and its reason is recorded; with none left h_lb is withdrawn and the run falls to the estimate tier. The oracle's return and its status withdraw nothing.

Every run reports its estimate and names the tier that carries its claim (Requirement "Reporting of the tiers" (foundations/req:tiers)). The first-order bound $\underline{h}_\alpha^{\mathrm{FO}}$ of Theorem 19.1 holds only where the theorem's hypotheses hold at the iterates its maximum runs over, and a gradient taken after a perturbation bounds another function (Remark 19.4). No hypothesis of

that theorem is read of the backend: the feasible side of a bracket is a lower bound formed without its return or its status (Requirement 27.1), so neither withdraws a bracket. Withdrawal is never a weakening of what $\underline{h}_\alpha^{\mathrm{FO}}$ means (Remark 23.1).

REQUIREMENT 24.1 (Withdrawal of the first-order bound). An iterate at which any of the following holds is not a member of $\mathfrak{I}$: it contributes no bracket to the maximum in (19.2), and the engine records the machine-readable reason.

(i) (H-POS) fails (`gradient_hypothesis_positivity`);

(ii) (H-SUPP) fails (`gradient_hypothesis_support`);

(iii) a gradient perturbation was applied, that is, (H-NOPERT) fails (`gradient_perturbation_applied`);

(iv) the point consumed at that iterate fails one of the three checks of Requirement 25.1 and is not repairable under Requirement 25.4, or its acceptance-side component $\mathbf{q}_t$ was not formed on the exact island under (α) and (β) of Remark 22.2, so that its membership in the product $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J(\varepsilon)$ is not established (`iterate_not_feasible` in either case).

Neither the point the oracle returned at that iterate nor its status removes the iterate from $\mathfrak{I}$. If no iterate is left, $\mathfrak{I}$ is empty: the engine withdraws $\underline{h}_\alpha^{\mathrm{FO}}$ and the tier falls to the estimate, with a reason from the same machine-readable vocabulary.

Motivated by Thm. 19.1, Rem. 19.4, Rem. 22.2, Rem. 23.1, Req. 25.1, Req. 25.4, Req. 27.1, foundations/req:tiers · *Used by* Req. 24.7, Req. 28.1 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.reasons:AssuranceReason

Every run applies the three iterate checks once more to the point it returned and records whether that point passes, with the three quantities measured. Neither the bound, nor the tier, nor the key length rests on that measurement.

The checks of Requirement 25.1 are made on the point consumed at each iterate, and the point a run returns, at which its estimate is taken, need not be one of those: a run that stops at its iteration limit returns a point formed after the last point it consumed. What the measurement establishes of that point, the J factor of the estimate's premise and no more, is Remark 23.3.

REQUIREMENT 24.2 (Measurement of the returned point). Every run applies the three checks of Requirement 25.1 once more, at the end of the run, to the point it returned: per-block positivity at exactly 0, the coupling residual at `iterate_coupling` and each $\bowtie_k$ residual at `constraint_rows`. It records whether that point passes all three, together with the three quantities measured, in `optimizer_certificate`. The measurement enters neither the first-order bound, nor the tier, nor the key length.

Motivated by Thm. 19.1, Rem. 23.3, Req. 25.1 · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.joint_driver:run_joint_frank_wolfe

What reaches the key length is max{0, h_lb}, and the engine reports the unclamped h_lb beside a flag recording the clamp. The clamp substitutes one valid lower bound for another and repairs nobody's data.

A computed $\underline{h}_\alpha^{\mathrm{FO}}$ may be negative, the linearisation being weak far from the optimum or the arithmetic crossing zero, and $\max\{0, \underline{h}_\alpha^{\mathrm{FO}}\}$ is still a lower bound on $\underline{h}_\alpha$ by Theorem 19.1 (ii), respectively (iiib) (Remark 23.9); the clamp corrects only the low side (Remark 22.4 (a)). The unclamped value lets a reader see what was computed and not only what was used.

Requirement 24.3 (Clamping of the first-order bound at zero). The engine passes $\max\{0, \underline{h}_\alpha^{\mathrm{FO}}\}$ to the key length of Corollary 19.5, and records the clamp: it emits the boolean `first_order_clamped` beside `h_lb`, and `h_lb` carries the unclamped value.

Motivated by Thm. 19.1, Cor. 19.5, Rem. 22.4, Rem. 23.9 · *Used by* Req. 24.7, Req. 28.1 · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.general_keyrate:key_length_from_first_order

A run whose search returns a negative estimate is refused with a typed reason naming the value and the point, and the estimate is never clamped. Unlike h_lb it is no lower bound, so zero cannot stand in for it.

The objective is non-negative at every point of the exact feasible set, by a pointwise argument (Proposition 5.8 (i)), while at a point of the relaxed feasible set it is bounded below only by $-\frac{\alpha}{\alpha-1}\log_2(1+d_{A_0}\varepsilon)$, where $d_{A_0}\varepsilon < 1$, and in the state form with $d_A\varepsilon$ in trusted-source mode and ε in untrusted-source mode in place of $d_{A_0}\varepsilon$ (Proposition 17.2 (c″)). A negative estimate is therefore evidence that a hypothesis does not hold of the point it was computed at, and unlike $\underline{h}_\alpha^{\mathrm{FO}}$ it is no lower bound (Remark 23.9). No key length is emitted on a refused run.

Requirement 24.4 (Refusal of a negative estimate). The engine refuses a run whose computed estimate, the objective at the point the run's search on the declaration it searches returned at an order the run evaluates, is negative, with the machine-readable reason `negative_objective` naming the value and the point, and never clamps the estimate to zero. A bare exception is not a conforming implementation of this rule; the reason must be readable by the caller. The rule binds that search alone: the polish of Requirement 24.7 (i) and the solve of Requirement 16.2 (i) form no estimate of the run, and a negative objective value of theirs refuses nothing.

Motivated by Prop. 5.8, Req. 16.2, Prop. 17.2, Rem. 23.9, Req. 24.7 · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.general_keyrate:NegativeObjectiveError

The engine records h_lb minus the reported estimate, with its sign, as a diagnostic, and asserts nothing of it. Their ordering holds only in exact arithmetic and at an exactly feasible point, neither of which a run establishes.

The ordering of $\underline{h}_\alpha^{\mathrm{FO}}$ below the objective at the returned point is a consequence of exact arithmetic and of the returned point's exact membership in $\mathcal{S}_{\mathrm{acc}} \times \mathcal{W}^J$, neither of which a run establishes; a positive difference, a bound above the estimate, is evidence about the arithmetic or the relaxation, and not by itself evidence that the first-order tier is unsound (Remark 22.4 (b)).

REQUIREMENT 24.5 (The recorded difference between the bound and the estimate). The engine records the difference $\underline{h}_\alpha^{\mathrm{FO}}$ minus the reported estimate, with its sign, as a diagnostic, and asserts nothing of it.

Motivated by Rem. 22.4 · *Used by* Req. 24.7 · accepted · *Code* certiqs_key.solvers.frank_wolfe.general_keyrate:GeneralKeyRateResult

The engine imports no relative-entropy penalty onto the objective: it subtracts no penalty term taken from a von Neumann entropy key-rate formulation. The objective is no relative entropy, and no claim here relates the two at finite α.

The objective of (7.10) is not a relative entropy: its K-terms are Rényi-order-α trace powers, and no statement of this module relates the two at finite α, so a penalty term taken from a von-Neumann-entropy key-rate formulation, however standard there, would claim a theorem this objective does not have. Face handling here is the compression of Theorem 8.4 (i), unbudgeted, which is why this tier is weaker than a certificate.

REQUIREMENT 24.6 (The program's own objective, with no imported penalty). The engine imports no relative-entropy penalty onto O_α^J from any source: it subtracts from it no penalty term taken from a von-Neumann-entropy key-rate formulation.

Motivated by Thm. 7.5, Thm. 8.4, Thm. 19.1 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.joint:joint_objective_value

Where a reference companion is admitted, the run's first-order number is the best bracket on the companion at the best points of the snapped search and of a short search continued on the companion from them, each assessed by the usual gate; the snapped bound is recorded as a diagnostic only.

The search runs on the snapped declaration; the bound is formed on the companion. Any points at which Corollary 19.7 holds give a valid bound, so the choice of points bears on its size only. The companion and the snapped declaration differ only in the measured families of the factors the companion replaces and in the entries of their last links, so the points the snapped bracket ranks highest are the natural candidates, and a few of them keep the extra oracle calls few; the last point is added because it is the point the search ended at. Those points can still sit badly on the companion. The search fits the acceptance-side component to the snapped statistics, and where the

snap moved an entry that is small beside its outcome's probability, the companion's statistics at the same point differ from that component by a relative amount of the move's size, which the factor $\alpha/(\alpha-1)$ of the objective magnifies: the bracket on the companion can then fall far below the snapped one, though the two values are close. The polish, a short search continued on the companion's own program from the best of those points, moves the point to where the companion's bracket is large. Its points are points like any other, since Corollary 19.7 reads of a point nothing but its hypotheses, and the gate decides them at each point whatever produced it: in exact arithmetic the points it adds can only raise a maximum of valid brackets. That is not the whole of it at the first-order tier. Where the companion carries a positive radius, its first-order bracket is formed from the binary64 numbers nearest the centres of its enclosure, which are displaced from the companion itself by an amount no claim bounds (Remark 22.4, Proposition 11.1); a search that maximises that evaluation makes it more likely that such an unbudgeted excess reaches the first-order number than points found on another declaration do. The polish's points count at the first-order tier all the same, as the search's do; where every radius is zero only the rounding of the evaluation is exposed, as at any point; and the certified tier, formed from the exact centres and radii (Requirement 28.1), is unaffected. Which blocks then rest on no premise about the snap follows from what the companion carries, block by block (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap)). It follows also from which number the key length is formed from: where the companion's bound counts but its certified evaluation certifies no point, a number certified on the snapped declaration may carry the key length instead (Requirement 28.1), and the blocks the companion carried then rest on the snap premise after all.

Requirement 24.7 (The first-order bound evaluated on the reference companion). Where the engine has admitted a compiled reference companion $\mathfrak{D}^{\mathrm{ref}}$ beside the declaration $\mathfrak{D}$ it searched on (Requirement 14.3), it forms the run's first-order number as follows, after the search on $\mathfrak{D}$ has ended.

(i) *The points.* It evaluates the points of two sets. The first holds at most five points of the search on $\mathfrak{D}$: the at most four members of the index set of $\mathfrak{D}$'s bound (Requirement 24.1) with the largest brackets on $\mathfrak{D}$, ties broken by the order of the run, and the last point the search assessed; where $\mathfrak{D}$'s index set is empty, the last point alone. The second holds the points kept by the *polish*, a search on the Choi program of $\mathfrak{D}^{\mathrm{ref}}$ that continues the search on $\mathfrak{D}$ after it has ended. The polish reads the objective, the gradient, the maps and the entries of $\mathfrak{D}^{\mathrm{ref}}$ (Requirement 14.3 (i)) and no map generated from $\mathfrak{D}$'s measured family; it starts at the point of the first set with the largest bracket on $\mathfrak{D}$, at the last point where $\mathfrak{D}$'s index set is empty, and forms its acceptance-side iterate as Requirement 26.2 requires, from that point's

exact acceptance-side component. Where that component was not formed on the exact island (Requirement 26.2), the polish does not run, the second set is empty and the record names that reason. It assesses each point it consumes by the gate of (ii) and keeps the at most four of them that contribute with the largest brackets on $\mathfrak{D}^{\mathrm{ref}}$, ties broken by its own order, and the last point it assessed; a polish that assesses no point keeps none. It runs within the share `reference_polish_share` of the wall budget of one evaluation of the run, where the run has one, set aside for it before the search on $\mathfrak{D}$ begins, and ends where that share is spent, where an iteration cap it records is reached or where its own stopping rule holds, whichever comes first; a polish that fails or is cut short keeps, as above, the points it assessed until then. The share so set aside shortens the search on $\mathfrak{D}$, which bears on the run's estimate, on $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}]$, on a number certified on $\mathfrak{D}$ in the fallback of Requirement 28.1 (i), and on the soundness of none of them. Its objective values are not the run's estimate: none is reported, and none refuses the run (Requirement 24.4 reads the estimate of the search on $\mathfrak{D}$ alone). Each point $x_t = (\mathbf{q}_t, J_t)$ of either set is taken as the search or the polish consumed it, $\mathbf{q}_t$ included, and is a point of the Choi program of $\mathfrak{D}^{\mathrm{ref}}$ (Corollary 19.7 (ii)).

(ii) *The gate.* At each such point it forms the bracket of (19.2) for $\mathfrak{D}^{\mathrm{ref}}$: the objective of $\mathfrak{D}^{\mathrm{ref}}$ at x_t, its gradient at x_t with no perturbation, the acceptance-side minimisation of Requirements 26.1 and 26.2, and the feasible side of Requirement 27.1, from a fresh feasible-side oracle call on the linear form of that gradient over $\mathfrak{D}^{\mathrm{ref}}$'s entries. It decides (H-POS) and (H-SUPP) on $\mathfrak{D}^{\mathrm{ref}}$'s maps (Requirement 25.2) and makes every check of Requirement 25.1 against $\mathfrak{D}^{\mathrm{ref}}$'s entries, on the point; a point at which Requirement 24.1 withdraws it on these terms contributes no bracket, with that requirement's reason.

(iii) *The number.* Where some evaluated point contributes and the bracket is not withdrawn by (iv), the run's first-order number is $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}^{\mathrm{ref}}]$, the largest bracket so formed, its clamp and key length are those of Requirement 24.3 and Corollary 19.5 at it, with the parameters of $\mathfrak{D}$'s run, and its tier is *first-order*. The bound on $\mathfrak{D}$, $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}]$, is recorded beside it with its brackets as a diagnostic, and no key length is formed from it. The difference $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}^{\mathrm{ref}}] - \underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}]$ is recorded with its sign, as a diagnostic, beside the difference of Requirement 24.5, which is taken against the estimate of $\mathfrak{D}$. A point of the second set that contributes counts as any evaluated point does: where no point of the first set contributes, it makes this item apply in place of the fallback of (iv), so that the run's first-order number moves from $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}]$ to the companion.

(iv) *The bracket withdrawn.* The bracket on the companion is withdrawn

for one of two causes: no evaluated point contributes; or the certified evaluation finds the two enclosures of a block the companion carries inconsistent, the cause `factor_enclosure_inconsistent` (Requirement 28.1 (iv)). An inconsistency shows that a module carrying the enclosures is defective, and the first-order number on the companion reads its entrywise enclosure, which may be the defective one; the second cause is decided after that number was formed, and the run then forms its first-order number again as this item says, before any key length is reported. Where the bracket is withdrawn, the run's first-order number is $\underline{h}_{\alpha}^{\mathrm{FO}}[\mathfrak{D}]$, formed as without a companion, with the reason `reference_bracket_withdrawn`, its cause and the reasons of every evaluated point; no block is then evaluated off its snap. The record names in the group *reference bracket withdrawn*, with the cause (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)(b)), every moved block on which a declaring instance whose link the companion replaces handed the reference and every moved class of each snapped emission of such a link whose normalizer the companion bypasses; it names as it does without a companion every other moved block or class that rests on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement), so that a moved block on which a declaring instance whose link the companion does not replace handed a reference rests in the group *reference bracket not evaluated* (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)(c)). A run that declares the strict option `strict_snaps` refuses there instead and reports no key length (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv)).

(v) *The blocks.* Where (iii) applies and the key length the run reports is not formed in the fallback of Requirement 28.1 (i), the record of the run names as evaluated off its snap (Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap)) every moved block of the snap of each declaring instance whose link the companion replaces, on which the companion carries the handed reference, and every data sector of each snapped emission of such a link that holds a moved class and whose normalizer the companion bypasses; a kept block or sector is named as kept and is not counted among them. It names every moved block on which the companion carries the snapped block as not so evaluated, so that it rests on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)), and a run that declares `strict_snaps` refuses

there (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv)). Where (iii) applies but the key length the run reports is formed from a number certified on $\mathfrak{D}$ in the fallback of Requirement 28.1 (i), no number formed on the companion reaches that key length, and the number it is formed from reads the snapped blocks: the moved blocks on which the companion carries the handed reference and the data sectors whose normalizer it bypasses are then not evaluated off their snap, condition (ii) of Definition "A reference handed beside a snap; a block evaluated off its snap" (foundations/def:evaluated-off-snap) failing for them. The record names them as resting on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) in the group *reference bracket withdrawn* with the cause `companion_not_certified` (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)(b)), beside the moved blocks on which the companion carries the snapped block, which rest as above. The run decides this after the certified evaluation and before any key length is reported, and no block is then named as evaluated off its snap. A run that declares `strict_snaps` forms no such number (Requirement 28.1 (i)).

(vi) *The record.* The record carries, per evaluated point, the set of (i) it belongs to, its bracket on $\mathfrak{D}^{\mathrm{ref}}$ and the withdrawal reasons there, and for a point of the first set its bracket on $\mathfrak{D}$; for the polish, whether it ran and, where it did not, why, the point it started from, the share of the wall budget set aside for it and the wall time it took, its iteration cap, the number of points it assessed and the cause it stopped on; the measurements of Requirement 14.3 (iii), the largest radius of the companion's enclosure among them, which bear on the arithmetic of Remark 22.4 and on nothing else; and, per block, whether the companion carries the handed reference or the snapped block.

A run with no admitted companion forms its first-order number on $\mathfrak{D}$ alone, as the other requirements of this submodule state; a moved block on which a reference was handed, and a moved class of an emission beside which a companion that bypasses its normalizer was formed, then rest in the group *reference bracket not evaluated* (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)).

Motivated by Prop. 11.1, Prop. 12.1, Req. 14.3, Thm. 19.1, Cor. 19.5, Cor. 19.7, Rem. 22.4, Req. 24.1, Req. 24.3, Req. 24.4, Req. 24.5, Req. 25.1, Req. 25.2, Req. 26.1, Req. 26.2, Req. 27.1, Req. 28.1, foundations/ass:snapped-measurement, foundations/def:evaluated-off-snap, foundations/req:snap-reference · *Used by* Req. 28.1 · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.reference:bracket_on_reference

25 Requirements: checks and repairs of the iterates

The engine checks every condition of the relaxed feasible set on each consumed point, decides maximal support by one rank comparison, repairs a block only within a declared window and records every repair, an oracle return's repair serving the search alone. In the general format it decides the coupling condition per Alice sector.

At each iterate the engine decides block positivity of the consumed point at exactly 0, and the coupling and constraint-row residuals of that point at their own windows. The larger window is the ε at which a run reads the relaxed set; no declared tolerance is that number.

Every statement about the relaxed feasible set holds for every $\varepsilon \geq 0$ (Proposition 17.2), so no claim depends on which ε a run reads, and each candidate reading is conservative; an implementation nevertheless needs one number. The set relaxes positivity not at all, which is why positivity is decided at exactly 0 (Proposition 17.2 (c), *Fourth*); a point that passes the three checks lies in $\mathcal{W}^J(\varepsilon)$ at the larger window, by monotonicity from the smaller one (*ibid.* (a)) and because the coupling clause implies the normalisation clause (*ibid.* (c′)). The windows are numbers of their own on a computed object (Requirement "Scope of the declared tolerance" (foundations/req:tolerance-scope) (ii)), and this module takes no declared tolerance (Requirement 14.1). What the checks buy, and why no check of a point the oracle returned would serve the bound, is Remark 23.2.

Requirement 25.1 (Feasibility checks at each iterate). On the point it consumes at each iterate, after any repair, the engine decides every condition defining $\mathcal{W}^J$ in (7.9), as the relaxed feasible set relaxes it, all three and none of them optional:

(i) per-block positive semidefiniteness, at exactly 0 and at no window (`iterate_positivity`);

(ii) the coupling condition, its residual $\| \sum_s \operatorname{tr}_{Q_s} J_s - \mathbb{1}_{A_0} \|_{\max}$, or, in untrusted-source mode, where the coupling condition is the normalisation $\sum_s \operatorname{tr} \omega_s = 1$ (Definition 7.10 (b)), its residual $| \sum_s \operatorname{tr} \omega_s - 1 |$, at the declared window `iterate_coupling`;

(iii) the declared constraint rows $\bowtie_k$, each residual at the declared window `constraint_rows`.

No fourth check is required, and none is required of the point the oracle returned at that iterate. The ε at which a run reads the relaxed feasible set is the larger of the two declared windows of (ii) and (iii). No declared tolerance (Definition "The declared tolerance and the defects it bounds"

(foundations/def:tolerance)) is that number: the windows govern conditions checked on an iterate and never on a declaration.

Motivated by Thm. 7.5, Def. 7.10, Req. 14.1, Def. 17.1, Prop. 17.2, Rem. 23.2, foundations/def:tolerance, foundations/req:tolerance-scope · *Used by* Req. 24.2, Req. 24.7 · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.joint_driver:_check_point

The engine decides (H-SUPP) as a rank equality, both sides computed and read with one declared cutoff relative to a spectral radius. The reference rank is computed once per acceptance value and sector from the declaration and the support frame, never from an iterate.

Hypothesis (H-SUPP) of Theorem 19.1 is equivalent to the rank equality (23.1), and no positivity test on the ambient space of the branch operator can decide it (Remark 23.4). A direction along a singular value σ of the stacked Kraus matrix carries weight of order σ^2 in the branch operator, which is why the right side is read at the square root of the cutoff: both sides then count with one ruler. The comparison discharges nothing; Remark 23.6 names the residual exposures it leaves.

Requirement 25.2 (Rank test for maximal support). The engine decides (H-SUPP) as the rank equality (23.1), both sides computed and both read with the one declared threshold `support_cutoff`:

(i) the left side is the count of eigenvalues of $X_{\hat{c},s}$ above `support_cutoff`, read relative to that block's own spectral radius;

(ii) the right side is $\operatorname{rank} \mathbf{K}_{\hat{c},s}$ of (8.2), read at the square root of `support_cutoff` relative to that matrix's largest singular value, a window derived from the same threshold and not declared beside it; equivalently the count of eigenvalues of $\mathbf{K}_{\hat{c},s}\mathbf{K}_{\hat{c},s}^{\dagger}$ above `support_cutoff` relative to its spectral radius. It is computed once per $(\hat{c}, s)$ from the declaration and W_τ, and never from an iterate.

Motivated by Def. 8.1, Lem. 8.2, Thm. 19.1, Rem. 23.4, Rem. 23.6 · *Used by* Req. 24.7 · implemented (effective: accepted) · *Code* certiqs_key.meat_core.objective.gradient:attains_the_reference_rank

An oracle return with a slightly negative spectrum may be rejected, or projected onto the positive cone within a declared window of its own. Every repair is recorded with its block, iterate and depth, apart from iterate repairs, and the repaired return serves the search alone.

No bracket reads a point the backend returned (Requirement 27.1), so a repair of a return changes only the search's next step and a recorded diagnostic. A window bounds the displacement of the block but gives no bound on a pairing with it (Remark 23.7), and the object repaired is a computed

object, whose window is a number of its own (Requirement "Scope of the declared tolerance" (foundations/req:tolerance-scope) (ii)).

Requirement 25.3 (Repair of an oracle return). The engine may reject outright an oracle return a block of which has a negative spectrum, or project that block onto the positive semidefinite cone within the declared window `oracle_repair_window`, a number of its own, declared beside τ_{num} and never as an instance of it. It records every such repair, with the block, the iterate and the measured depth, in `optimizer_certificate`, kept apart there from the iterate repairs of Requirement 25.4, and uses the repaired return only as a step of the search and in the diagnostic pairing of Requirement 27.1 (vi), never in a bracket and never as a bound on anything.

Motivated by Rem. 23.7, Req. 25.4, Req. 27.1, foundations/req:tolerance-scope · implemented (effective: accepted) · *Code* certiqs_key.solvers.oracles.qics.adapter:QicsSectorChoiOracle

A block of a formed iterate that fails the positivity check is rejected, or projected within a declared window of its own. The repaired point is the iterate, its residuals are measured again, and every repair is recorded apart from oracle repairs.

The repair belongs to the formation of the iterate, upstream of the point Theorem 19.1 is applied at, so it is not the perturbation that (H-NOPERT) excludes; it establishes a hypothesis of that theorem on an object the engine chose and says nothing about the backend (Remark 23.8). Re-measuring the residuals is the one place this repair can cost a run.

Requirement 25.4 (Repair of a formed iterate). When a block of the point consumed at iterate t fails the positivity check of Requirement 25.1, the engine either rejects it, in which case t contributes no bracket and is not a member of $\mathfrak{I}$, or projects it onto the positive semidefinite cone, provided its measured depth is within the declared window `iterate_repair_window`, a number of its own, declared beside τ_{num} and never as an instance of it. No other repair is made on an iterate. A projection is governed by three rules:

(i) the repaired point is x_t, the point at which the objective, the gradient and the pairings with x_t of (19.2) are evaluated;

(ii) the coupling and $\bowtie_k$ residuals of Requirement 25.1 are measured again on the repaired point and never carried over from the point before it, and a residual that then fails its window refuses the iterate;

(iii) every repair is recorded, with the block, the iterate and the measured depth, in `optimizer_certificate`, beside the oracle-return repairs of Requirement 25.3 and distinguished from them.

Motivated by Thm. 19.1, Rem. 23.8, Req. 25.1, Req. 25.3 · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.joint_driver:_check_point

In the general format a run decides the coupling condition of the relaxed feasible set per Alice sector, one residual per sector at the one declared coupling window, and records the frame and the row form per sector.

In the general format the Choi program carries one coupling constraint per Alice sector (Theorem 7.5), and the relaxed feasible set is read with the coupling constraint per Alice sector (Remark 19.8 (3)).

Requirement 25.5 (Coupling residual and record per Alice sector). On a declaration in the general format the engine decides the coupling condition of the relaxed feasible set of Definition 17.1 per Alice sector, one coupling residual per Alice sector, each at the declared window `iterate_coupling`; and the record names the frame and the row form per Alice sector (Requirements 15.3 and 15.6).

Motivated by Thm. 7.5, Req. 15.3, Req. 15.6, Def. 17.1, Prop. 17.2, Rem. 19.8 · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.joint_driver:coupling_residuals

26 Requirements: the acceptance side

The engine solves the acceptance-side minimisation exactly for a singleton or a polytope acceptance set and admits no other acceptance set. It forms the acceptance iterate exactly on the exact island and records, per iterate, whether it did; a run that replaces the acceptance-side pairing by a certified bound records which of the two the bracket used.

The engine solves the acceptance-side minimisation exactly for a singleton or a polytope acceptance set and admits no other geometry. A declaration with any other acceptance set is refused before any solve.

Definition 2.5 admits any closed convex subset of the probability simplex, while Remark 22.2 solves the acceptance-side minimisation exactly for two geometries only. For any other set the bracket would need a certified β_{acc} of (22.1) from elsewhere, which no requirement decides and no assumption carries, so the engine admits no other set. The refusal is this requirement's own: the general admission rule, Requirement 14.4, would admit a third tag whose every datum denotes a closed convex set.

Requirement 26.1 (The admitted acceptance geometries). The engine solves the acceptance-side minimisation exactly for the two geometries of Remark 22.2, a singleton (a) and a polytope (b), and admits an acceptance set in no other geometry: a declaration whose $\mathcal{S}_{\mathrm{acc}}$ fits neither is refused at admission, before any solve, and no first-order bound is formed for it.

Motivated by Def. 2.5, Req. 14.4, Rem. 22.2 · *Used by* Req. 24.7, Req. 28.1 · accepted · *Code* certiqs_key.solvers.oracles.acceptance:acceptance_oracle

The engine forms the acceptance-side iterate on the exact island, from an exact rational start in the acceptance set by exact rational convex combinations of points established to lie in it, and records per iterate whether it did. An iterate where it did not adds nothing to the first-order bound.

Under conditions (α) and (β) of Remark 22.2 the acceptance-side iterate lies in $\mathcal{S}_{\mathrm{acc}}$ at every step by induction, so the acceptance factor of the membership hypothesis of Theorem 19.1 (ii) and (iiib) is true by construction and needs no check; a binary64 convex combination of exact points of a polytope can leave the set, and three consumers of that theorem then break (the same remark). Every computation of the exact island is exact rational arithmetic (Requirement "Exact arithmetic on the island, unbudgeted rounding elsewhere" (foundations/req:exact-island)).

Requirement 26.2 (Exact formation of the acceptance-side iterate). The engine forms the acceptance-side iterate on the exact island, under conditions (α) and (β) of Remark 22.2: it initialises it at an exact rational point of $\mathcal{S}_{\mathrm{acc}}$, and updates it only by convex combinations, with exact rational weights in $[0, 1]$, of points already established to lie in $\mathcal{S}_{\mathrm{acc}}$. An iterate whose acceptance-side component was not so formed has not met the membership hypothesis and contributes no bracket to $\underline{h}_{\alpha}^{\mathrm{FO}}$, with the reason of Requirement 24.1 (iv). The run's record states in `optimizer_certificate`, of the point consumed at each iterate, whether its acceptance-side component was so formed.

Motivated by Thm. 19.1, Rem. 22.2, Req. 24.1, foundations/req:exact-island · *Used by* Req. 24.7, Req. 28.1 · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.acceptance_iterate:AcceptanceIterate

A run that replaces the acceptance-side pairing in the bracket of the first-order bound by a certified lower bound on the minimum records which of the two it used. The substitution is exact and leaves the first-order bound valid.

A certified β_{acc} of (22.1) may replace $\langle g_t, v_t \rangle$ in (19.2), and the first-order bound stays valid (Theorem 19.1 (iv); Remark 22.2).

Requirement 26.3 (Record of the acceptance-side quantity in the bracket). A run that replaces $\langle g_t, v_t \rangle$ in the bracket of (19.2) by a certified β_{acc} of (22.1) records which of the two it used.

Motivated by Thm. 19.1, Rem. 22.2 · accepted

27 Requirements: the feasible side

The feasible side of every bracket is the verified lower bound of Lemma 20.1 over the exact feasible set, formed in exact rational arithmetic from the gradient as computed, the declared rows and the backend's dual, less a deduction for the rounding of the frame, with zero multipliers where the backend returned none and zero row multipliers where the frame's error is

not bounded. The oracle's primal pairing, the dual gap and the deduction are recorded beside it. Beside it stands the superseded requirement that paired the gradient with the oracle's return, which no run follows.

Imports foundations/def:exact-island, foundations/lem:least-eigenvalue-bound, foundations/req:exact-island

The feasible side of every bracket is the lower bound of Lemma 20.1 over the exact feasible set, formed exactly from the computed gradient and the backend's dual, less a deduction for the computed frame, or with zero row multipliers. The oracle's return, status and objective enter no number.

Part (iv) of Theorem 19.1 accepts any real number below the feasible-side minimum over the exact $\mathcal{W}^J$ in place of the pairing with the oracle's return. Lemma 20.1 gives one from any multipliers, and its part (iv) charges the rounding of the frame a run builds its rows with. Formed as below, b_t^{F} is at most the minimum, over the exact $\mathcal{W}^J$, of the linear form at the gradient as the run computed it, whatever the backend's status, its dual or the rounding of the frame: it rests on no property of the backend. A dual objective the backend reports is a different number, which an approximately dual-feasible return can put above the minimum, and it is never used. What this requirement does not reach is the gradient itself, computed in binary64 from maps and a frame computed in binary64; that is named in Remark 22.4 and bounded by no claim of this module. The same formation serves the bracket on a reference companion (Corollary 19.7, Requirement 24.7).

Requirement 27.1 (The verified feasible side of every bracket). At every iterate t at which the engine forms a bracket of (19.2), on the declaration it searched and on an admitted reference companion alike, it uses in place of the pairing $\langle G_t, S_t \rangle$ the number b_t^{F} of Theorem 19.1 (iv), formed as follows, and never the pairing with a point the backend returned, a dual objective the backend reported, or the backend's status.

(i) *The groups and the data.* b_t^{F} is a sum over the groups of the program, one per Alice sector in trusted-source mode and one in untrusted-source mode, read as Lemma 20.1 (ii) reads them. Its G is the exact Hermitian part of the gradient blocks as the engine computed them at x_t, the same gradient whose acceptance-side part the bracket uses; its coupling is the exact coupling condition of the group; its rows are the maps of Lemma 20.1 (ii), in the conjugated or frame-free form the engine imposes, built from the declared data taken exactly: $\Theta_k^{(\mathsf{t})}$ as the declared block or the exact product $c_k \tau_A^{(\mathsf{t})}$, the lifted operators, and $D_k^{(sf)} = \hat{O}_k^{(sf)} - c_k \mathbb{1}$ formed exactly, all Hermitian by exact admissibility; and in trusted-source mode the frame in those maps is the binary64 $\widetilde{W}_{\mathsf{t}}$ the engine constructed for W_{t}, read as the dyadic matrix it is. Every binary64 number is read as the dyadic rational it is.

(ii) *The multipliers.* Y and every Λ_r are read off the dual the backend returned for that same solve, as the operators that the multipliers of the coupling condition and of row r represent in the encoding the oracle used, each replaced by its exact Hermitian part. On a group with a one-dimensional coupling and no rows, and on any group whose solve returned no finite dual, whatever its status, $Y = 0$ and every $\Lambda_r = 0$; on an Alice sector for which (iv) forms no ε_{t}, every Λ_r of its rows is 0 and Y is kept. The multiplier of an inequality row is replaced by $\Lambda_r + \mu_r \mathbb{1}$, with μ_r a dyadic rational at least $\max\{0, -\ell_r\}$ and ℓ_r a lower bound on $\lambda_{\min}(\Lambda_r)$ given by Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound), so that $\Lambda_r \succeq 0$ holds exactly.

(iii) *The eigenvalue.* The group's λ is the least, over the blocks of the group, of the lower bounds on $\lambda_{\min}(R_p)$ that Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound) gives at a matrix V of the engine's choosing, with R_p formed from the rows of (i). The bound holds at every V that lemma admits, the identity ($\theta = 0$) always among them, and the choice bears only on how close λ lies to $\min_p \lambda_{\min}(R_p)$.

(iv) *The frame.* For each Alice sector the engine forms a dyadic $\varepsilon_{\mathsf{t}} \geq \left\| \widetilde{W}_{\mathsf{t}} - W_{\mathsf{t}} \right\|$, established in exact rational arithmetic, where it can: 0 where $\widetilde{W}_{\mathsf{t}} = W_{\mathsf{t}}$ is decided exactly; and where $\tau_A^{(\mathsf{t})}$ is diagonal, the frame is the full frame or the diagonal frame (Definition 7.2), and $\widetilde{W}_{\mathsf{t}}$ vanishes exactly wherever W_{t} does, so that both have at most one non-zero entry in each row and each column and $\left\| \widetilde{W}_{\mathsf{t}} - W_{\mathsf{t}} \right\|$ is the largest modulus of their entrywise difference: a dyadic at least $|\widetilde{w} - \sqrt{\tau_{ii}}|$ at every entry where W_{t} carries $\sqrt{\tau_{ii}}$, $\widetilde{w}$ being the entry of $\widetilde{W}_{\mathsf{t}}$ there (for a real $\widetilde{w} > 0$, $|\widetilde{w}^2 - \tau_{ii}|/\widetilde{w}$ is one, since $|\widetilde{w} - \sqrt{\tau_{ii}}| = |\widetilde{w}^2 - \tau_{ii}|/(\widetilde{w} + \sqrt{\tau_{ii}})$). Where every row of the sector is imposed in frame-free form, no deduction is formed: those rows read no frame, every κ_r is 0, and the sector's bound is that of Lemma 20.1 (i). In every other case no ε_{t} is formed. Where it is formed, the sector's bound is that of Lemma 20.1 (iv), $b(G_{\mathsf{t}}; Y, \Lambda, \widetilde{\lambda}) - \sum_r \kappa_r$, with each κ_r formed from dyadic upper bounds on $\left\| \widetilde{W}_{\mathsf{t}} \right\|$, $\|\Lambda_r\|$ and $\|O_{r,s}\|$ (for instance on their Frobenius norms, by an integer square root rounded up). Where it is not, the row multipliers are zero by (ii) and the sector's bound is that of Lemma 20.1 (i), which then reads no row. In untrusted-source mode there is no frame and no deduction.

(v) *The arithmetic.* The residuals, the shifted multipliers, every eigenvalue and norm bound, every deduction and b_t^{F} are computed in exact ra-

tional arithmetic on those dyadic data, as a certification of the exact island (Definition "The exact island" (foundations/def:exact-island) (iii)), the backend's dual entering it only as a candidate (Requirement "Exact arithmetic on the island, unbudgeted rounding elsewhere" (foundations/req:exact-island)). b_t^{F} enters the binary64 bracket rounded toward $-\infty$, the direction that weakens a lower bound, as Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction) rounds an emitted one.

(vi) *The record.* Per bracket the engine records b_t^{F} (`feasible_side_bound`); the sum of the deductions κ_r (`frame_deduction`) and the sum of the bounds on $\|\Lambda_r\|$ it used (`row_multiplier_norm`); where the backend returned a point, the pairing $\langle G_t, S_t\rangle$ with it (`oracle_pairing`) and the difference $\langle G_t, S_t\rangle - b_t^{\mathrm{F}}$ with its sign (`dual_gap`); and, per group, whether its multipliers were the backend's, zero, or the backend's with the row multipliers set to zero (`feasible_side_multipliers`). None of these but b_t^{F} enters any number.

Motivated by Def. 7.2, Thm. 19.1, Cor. 19.7, Lem. 20.1, Rem. 22.4, Req. 24.7, foundations/def:exact-island, foundations/lem:least-eigenvalue-bound, foundations/req:exact-island, foundations/req:rounding-direction · *Used by* Req. 24.7, Req. 28.1 · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.verified:verified_feasible_side

The bracket of the first-order bound pairs the gradient with the oracle's primal return, never with a dual objective the backend reports. No bracket uses either: Requirement 27.1 puts a verified lower bound in place of the pairing.

A backend that solves the linear subproblem by an interior-point method reports a *dual* objective, and it is tempting to read that number as a lower bound on $\min_{J' \in \mathcal{W}^J} \langle G_t, J'\rangle$ and to use it in place of $\langle G_t, S_t\rangle$ in (19.2). Weak duality bounds the primal minimum from below by the dual objective at a *feasible* dual point; an interior-point return is only approximately dual-feasible, and for an infeasibility residual R the corrected bound carries a term of size at most $\|R\| \cdot \sup_{J' \in \mathcal{W}^J} \|J'\|_1$, of *unknown sign* without R itself. A scalar convergence indicator does not supply R, and this module requires no oracle to report it. Consequently a reported dual objective can lie above the primal value at the returned point, and substituting it would put (19.2) on the unsafe side of the very quantity it is supposed to bound.

Requirement 27.2 (The primal pairing in the first-order bracket). In the bracket of (19.2) the engine uses the primal pairing $\langle G_t, S_t\rangle$ at the returned point S_t and nothing else; in particular it never uses a dual objective reported by the backend in its place.

Motivated by Thm. 19.1 · superseded

28 Requirements: the certified evaluation

By default the engine forms the bracket again at the points of the run's first-order number from the declaration's exact data, the maps in factor form and every step exact or rigorously enclosed, and reports the best certified bracket at the certified tier (Requirement 28.1). An element of a companion's measured family whose rank neither its entrywise enclosure nor an enclosed factor fixes, or a block not shown positive definite, leaves a point uncertified. Where a companion certifies no point, the evaluation runs once more on the searched declaration, and a number certified there is reported at the certified tier, the blocks the companion carried resting on the snap premise; a run with no certified point keeps its first-order number with the reasons, or refuses under the strict option.

Imports foundations/cor:least-eigenvalue-over-enclosure, foundations/cor:rank-over-factor-enclosure, foundations/def:factor-enclosure, foundations/def:tiers, foundations/lem:least-eigenvalue-bound, foundations/lem:matrix-function-perturbation, foundations/lem:recentred-enclosure, foundations/req:exact-island, foundations/req:rounding-direction, foundations/req:snap-reference, foundations/req:tiers

The engine forms the bracket again at the final points, every step exact or rigorously enclosed, and reports the best at the certified tier; if a companion certifies none, on the searched declaration under A-SNAP unless strict_snaps is set. Else the first-order number stays, or a strict run refuses.

The first-order number is formed in binary64, from maps, a frame and a gradient that no claim bounds (Remark 22.4). Lemma 21.1 needs, at a point, only enclosures of the gradient, and Corollary 21.3 turns the best certified bracket into a key length. The gradient's maps can be built from factors whose rank is fixed (Lemma 8.6), every matrix function then acts on a positive definite matrix whose spectrum an enclosure bounds (Lemma "Perturbation bounds for matrix functions" (foundations/lem:matrix-function-perturbation)), and an enclosure of an element of the measured family fixes its rank where its non-zero part is positive definite throughout (Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure)) or, for an element carried with an enclosed factor, where the factor's columns are independent throughout (Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure)). A computation in ball arithmetic is rigorously enclosed; it lies outside the exact island and is no error budget (Requirement "Exact arithmetic on the island, unbudgeted rounding elsewhere" (foundations/req:exact-island)). Of the search the evaluation reads the points, and the multipliers of its solves as candidates of which nothing is assumed; any point at which the lemma's hypotheses are established serves, and a point that fails anywhere is dropped, never repaired. Certifying only a few of the points the first-order number reads keeps the evaluation a small fraction of the run's cost; a maximum over

fewer certified brackets is still a lower bound. Where a companion certifies no point, the evaluation is made once more on the declaration the run searched: a number certified there bounds that declaration's value, so the blocks the companion carried rest on the snap premise (Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement)), and the number is never combined with the companion's. Remark 22.5 says what the resulting number still rests on.

Requirement 28.1 (The certified evaluation of the first-order bound). On every run that forms a first-order number, unless it sets the option `certified_evaluation` to `off`, its default being `on`, the engine forms a certified number as follows, after the first-order number is formed and before a key length is reported.

(i) *The declaration and the points.* It works on the declaration the run's first-order number is formed on, at the points that number is formed from: on an admitted reference companion where Requirement 24.7 (iii) applies, at the at most five of the points part (i) of that requirement evaluates, of the search and of the polish together, that contribute there with the largest brackets on the companion, a point that is in both sets counted once and ties broken by the sequence in which the search and then the polish consumed them, each with the solve part (ii) of it makes there; otherwise on the declaration the run searched, at the at most four members of the index set of its bound (Requirement 24.1) with the largest brackets, ties broken by the order of the run, and the last point the search assessed, each with the search's solve there. *The fallback*: where it has worked on a companion by (ii)–(v) and certified no point there, it works once more, by (ii)–(v), on the declaration the run searched, at the points and with the solves of the second case, unless the run declares the strict option `strict_snaps` of Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv); the companion's results are kept for the record (ix). The fallback runs nowhere else: not where a point of the companion is certified, not where no companion's bracket counts, not where the companion's bracket is withdrawn by (iv), and not where the arithmetic the evaluation needs is not available. Where the companion's bracket is withdrawn by (iv), the run's first-order number is formed on the declaration the run searched (Requirement 24.7 (iv)), and the evaluation works once more there, by the second case above, as on any run whose first-order number is formed there. Each point is taken exactly: its acceptance-side component as the exact rational iterate formed on the exact island (Requirement 26.2), a point whose component was not so formed being not certified (`acceptance_iterate_not_exact`), and each Choi block as

the exact Hermitian part of the dyadic matrix its binary64 entries are.

(ii) *The data.* Every quantity is formed from the data of the declaration it works on, taken exactly, every binary64 number as the dyadic rational it is and the acceptance set as its exact rational data, and, for the blocks a reference companion replaces, from the exact centres and radii of the entrywise enclosures its compilation carries, recentred as Lemma "Zeroing the centres an enclosure cannot tell from zero" (foundations/lem:recentred-enclosure) recentres them, and from the exact centres and radii of the matrix box and the exact core of every factor enclosure its compilation carries (Definition "Factor enclosure of a block" (foundations/def:factor-enclosure)). No map, frame, gradient, support, rank, eigenvalue or check value the engine computed in binary64 enters it. In trusted-source mode the frame of each Alice sector is W_{t} of Theorem 7.5 for the support frame the engine constructed (Definition 7.2), held as an enclosure formed by a rigorously enclosed computation; where the engine forms none, every point is not certified (`frame_not_enclosed`).

(iii) *The maps in factor form.* At every joint pair that is not key-inert, every kept one under a dropped set, the branch operators and their pinched blocks are formed through factors S'_{xv} with the Gram matrices of Lemma 8.6 (v), built from factor forms $E^{\dagger}\widetilde{X}E$ of the elements of Alice's measurement or ensemble and of the measured family that enter them, $\widetilde{X}$ positive definite and E either an exact matrix of full row rank (Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure) (iii)) or, on a block whose rank the factor rule of (iv) fixes, the adjoint $L^{\dagger}$ of a factor L of its factor enclosure, held through the box of the adjoints, of full row rank at every factor (Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure) (ii)), with $\widetilde{X}$ the exact core: on exact data from an exact factorisation, on a block a companion carries with a positive radius by the rule of (iv). In the general format an element of the measured family is the tensor product of its blocks, one per measured factor, and its factor form is the tensor product of theirs (Lemma 21.2 (b)), E and $\widetilde{X}$ the tensor products of the blocks' E and $\widetilde{X}$, however many of its blocks a companion replaces and however many of them are held through boxes: E is then held through an enclosure of the Kronecker product of their enclosures, which contains it at every member, and has full row rank at every member, $EE^{\dagger}$ being the tensor product of the blocks' $EE^{\dagger}$, each positive definite. No rank is read off a computed spectrum, and a matrix function applied to an operator A, a branch operator X or a pinched block $(Z_j\otimes\mathbb{1})X^{p}(Z_j\otimes\mathbb{1})$ of one, acts on its compression $Q^{\dagger}AQ$ by an isometry Q, held through

an enclosure, as follows.

(a) *Containment.* The range of Q is established to contain $\operatorname{supp} A$, exactly or by an argument on the factors and the enclosures that holds at every value they enclose. The range of the side-by-side listing of the pair's pass-class factors contains the support of every branch operator they form, and the pinching projector applied to a subspace containing $\operatorname{supp} X$ contains the support of the pinched block (Lemma 8.6 (iv)(a)). Where the listing factors as $T_{\mathrm{out}} \widetilde{B} T_{\mathrm{in}}$, with $\widetilde{B}$ an exact matrix of full column rank, formed exactly, T_{out} the maps applied after it, certified injective by a Gram matrix certified positive definite, and T_{in} the maps applied before it, an orthonormal basis of $\operatorname{ran}(T_{\mathrm{out}} \widetilde{B})$ is established from $\widetilde{B}$. On exact data $\widetilde{B}$ comes from an exact rank factorisation of the exact matrix of the factors of the pass classes, and T_{in} holds the rest of that factorisation and the frame $W_{\mathsf{t}} \otimes \mathbb{1}$ in the Choi form. Where an element is held through the box of the adjoints of the factors of one or more of its blocks, $\widetilde{B}$ is, for each pass class, the exact factor of Lemma 21.2 (b), formed by exact rank factorisations of the rows whose factors are exact and, per tuple of the blocks held through boxes, of the stack of Alice's factors tensored with the exact factors of the elements whose blocks held through boxes are that tuple, the kind that lemma fixes for a block being the third for a block that the factor rule of (iv)(b) decides not zero, fixing its rank, the first for a block decided zero, and the second otherwise, an exact block included. The rows of such elements are grouped by their tuples, each block named by its factor, its sector and its letter, and never by the equality of their boxes: elements whose tuples name the same blocks share a group, whatever their boxes, and two distinct blocks with equal boxes are distinct entries of the tuples. Where the form the engine holds of such an element is not of this shape (its parts and the blocks naming them differing in number, a split into exact and boxed parts held where no block of the element is held through a box or missing where one is, or one tuple of blocks at two positions or of two dimensions), no claim admits it, and every point is not certified (`boxed_form_inconsistent`), a defect of the engine and not of the declaration. T_{in} holds that lemma's matrix C, the stack of C_0 and of each tuple's $(C_\beta \otimes E_\beta^{\mathrm{box}}) U_\beta$, the columns of each such block permuted by its own U_β and by no other permutation, which lies in an enclosure containing it at every member, and the frame. That range contains the range of the listing at every member, and equals it at every member where the product of the maps in T_{in} is established to have full row rank at every member; other-

wise the basis spans a subspace containing the range, which (a) admits and the floor of (b) decides. T_{out} is block diagonal over the outcome pairs, so the pinching projectors commute with it, and $(Z_j \otimes \mathbb{1})\operatorname{ran}(T_{\text{out}}\widetilde{B}) = \operatorname{ran}\left(T_{\text{out}}(Z_j \otimes \mathbb{1})\widetilde{B}\right)$, whose basis comes in the same way from an exact rank factorisation of $(Z_j \otimes \mathbb{1})\widetilde{B}$. A point at which containment is not established is not certified (`support_not_established`).

(b) *The floor.* A positive lower bound on $\lambda_{\min}(Q^\dagger A Q)$ is then established, over the enclosures at every value they contain, which with (a) proves $\operatorname{ran} Q = \operatorname{supp} A$ (Lemma 8.6 (iv)(b)) at every member and makes the compression that of part (iv)(c) of that lemma. A point at which it is not is not certified (`floor_not_certified`).

(iv) *The rank rule over an enclosure.* A block a companion carries with a positive radius is decided, and its rank fixed, by the first of two rules that does so.

(a) *The entrywise rule.* The block is decided when every entry of its entrywise enclosure has centre and radius 0, then zero, or some entry's centre exceeds its radius in modulus, then not zero. Its factor form is that of Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure) (iv): the rows and columns whose every entry has centre and radius 0 are dropped, and the rest must have a positive lower bound on its least eigenvalue over the enclosure.

(b) *The factor rule*, where the compilation carries a factor enclosure of the block and the entrywise rule leaves the block undecided or fixes no rank for it. With no column the block is decided zero (Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure) (i)). With $k \geq 1$ columns and a floor of part (ii) of that corollary established, G and ρ formed as its part (iii) forms them or by a rigorously enclosed computation of $L^\dagger L$ over the box, the block is decided not zero and its factor form is $L^\dagger$, held through the box of the adjoints, with the core as $\widetilde{X}$.

(c) *The cross-check*, wherever the compilation carries a factor enclosure of the block, whichever rule fixes its rank. The engine encloses $LWL^\dagger$ entry by entry over the box of the factor enclosure, in exact rational arithmetic or by a rigorously enclosed computation, so that the enclosure contains $LWL^\dagger$ for every factor L, every member of the factor enclosure; and it decides, as (b) decides them, whether the factor enclosure has no column and, where it has columns, whether a floor of part (ii) of Corollary "Rank over a factor enclosure" (foundations/cor:rank-over-factor-enclosure)

is established.

A block that neither rule decides is undecided, and a block whose rank neither fixes has no certified rank. The two enclosures of a block are *inconsistent* where the enclosure of $LWL^{\dagger}$ of (c) and the block's entrywise enclosure are disjoint at some entry, or where one of the two decides the block zero and the other decides it not zero, as (a) and (c) decide them. The companion's own block lies in its entrywise enclosure and, by (H-FAC) of Lemma 21.2, is a member of its factor enclosure, so it lies in both enclosures and both decisions hold of it: an inconsistency shows that one of the two was not formed or carried as the claims that establish it say. Where the two enclosures of a block are inconsistent, the factor rule is set aside for the block, every point of the companion is not certified (`factor_enclosure_inconsistent`), whatever joint pairs the block is a factor at, key-inert or not, and whichever rule would fix its rank, and the companion's bracket is withdrawn with that cause (Requirement 24.7 (iv)), its first-order number reading the entrywise enclosure, which may be the defective one. On a recentred enclosure, the entrywise rule leaves a block undecided exactly when its centre vanishes and some radius does not, its enclosure then deciding neither that it vanishes nor that it does not. Where an undecided block is a factor of an element of an outcome pair of a pass class of a key-live acceptance value whose element of Alice's is not zero, or a block with no certified rank is such a factor at a joint pair that is not key-inert, every point is not certified (`rank_not_certified`), save where another block of that element is decided zero, an exact block that is zero or a block a rule decides zero: the element is then zero for every member, whatever its other blocks. What the rule establishes is the hypothesis of Lemma 21.2 that such blocks are decided and of fixed rank, in the sense of that lemma; its hypothesis (H-FAC), that the companion's block is a member of the factor enclosure carried with it, is established by the claims of the modules that form and carry that enclosure, and decided by no rule here.

(v) *What is established at a point.* At each point the engine establishes, each step in exact rational arithmetic or in a rigorously enclosed computation at a working precision of at least 128 bits:

(a) every Choi block positive definite, by a positive lower bound of Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound) on its exact Hermitian part (`block_not_positive_definite`), which gives (H-SUPP) by Lemma 8.2 (i);

(b) (H-POS), with $q_t(\hat{c}) > 0$ decided exactly and the lower ends of the enclosures of $P^J_{\hat{c}}(J_t)$ and $K^J_{\hat{c}}(J_t)$ positive (`positivity_not_c`

`ertified`), on the exact acceptance support $\widehat{\mathcal{C}}_{\mathrm{acc}}$ of (18.1), which is also where $\hat{g}$ of (c) lives: for a singleton the support of its one point, and for a polytope the union of the supports of its exact vertices (Remark 22.2);

(c) an enclosure $(\hat{g}, \delta)$ of the acceptance-side gradient and, for every block p of every group, an exact Hermitian M_p and a rational ρ_p with $\|R_p - M_p\|_{\mathrm{F}} \leq \rho_p$ for the residual of Lemma 21.1 (iii), both formed by a rigorously enclosed computation of the gradient of Corollary 8.5 at the point (of the dropped objective under a dropped set) and of the frame: the enclosure of R_p is made Hermitian-centred with symmetric radii by Lemma "Zeroing the centres an enclosure cannot tell from zero" (foundations/lem:recentred-enclosure) (iv), M_p is its centre rounded entrywise to dyadic rationals and kept Hermitian, and ρ_p is a rational at least the radius that Corollary "Least eigenvalue and rank over an enclosure" (foundations/cor:least-eigenvalue-over-enclosure) (ii) gives for that centre plus a rational bound on the Frobenius norm of the rounding, so that the triangle inequality gives $\|R_p - M_p\|_{\mathrm{F}} \leq \rho_p$. On a companion, the computation reads each block of the measured family through its entrywise enclosure where it forms $P^J_{\hat{c}}$, the term $P^J_{\hat{c}}$ of $K^J_{\hat{c}}$ and the $\Gamma^{(s')}_{\hat{c}}$, and, where it forms the deficits and the $\Delta^{(s')}_{\hat{c}}$, through the enclosure that holds the block's factor form of (iii), its entrywise enclosure where the entrywise rule of (iv) fixed the rank and its factor enclosure where the factor rule did, so that the enclosures of (b) and (c) hold at every pair of a set as in Lemma 21.2 (c). The multipliers Y and Λ_r are read off the dual of that point's solve as Requirement 27.1 (ii) reads them in these respects only: the operators the dual's multipliers represent, each replaced by its exact Hermitian part; zero where the solve returned no finite dual; and the exact shift that makes the multiplier of an inequality row positive semidefinite. The rule of that item that zeroes the row multipliers of an Alice sector without a bound on its frame's error is not read, the rows here being built from the enclosed exact frame;

(d) b^{A}_t as the exact minimum of $\langle \hat{g}, \cdot \rangle$ over $\mathcal{S}_{\mathrm{acc}}$ for a geometry of Requirement 26.1 that it solves exactly, less δ, a point of any other geometry being not certified (`acceptance_geometry_not_certified`); and b^{F}_t by Lemma 21.1 (iii), each ℓ_p from Lemma "Least-eigenvalue lower bound from an approximate eigenbasis" (foundations/lem:least-eigenvalue-bound), in exact rational arithmetic;

and the point's certified bracket $b^{\mathrm{A}}_t + b^{\mathrm{F}}_t$. A point at which any of these is not established is not certified, with the reason named, and

contributes nothing.

(vi) *The number, its key length and its tier.* Where some point is certified, the run's certified number $\underline{h}_\alpha^{\mathrm{cert}}$ (`certified_bound`) is the largest certified bracket, (21.3), an exact rational, entering every binary64 use rounded toward $-\infty$ as Requirement "Rounding direction of emitted bounds" (foundations/req:rounding-direction) rounds an emitted bound; the run reports it at the certified tier (Requirement "Reporting of the tiers" (foundations/req:tiers)). The key length it reports is that of Corollary 21.3 (on a reference companion, for the protocol it declares, Corollary 21.4) at $h \coloneqq \max\{0, \underline{h}_\alpha^{\mathrm{cert}}\}$, with the parameters of the run, and it is the maximal key length of Definition 10.13 with h in place of $\underline{h}_\alpha$, computed exactly: $c \coloneqq n\,h - \lambda_{\mathrm{EC}} - \lceil \log_2(1/\varepsilon_{\mathrm{EV}}) \rceil + 1$ is formed exactly, a rational, $\lceil \log_2(1/\varepsilon_{\mathrm{EV}}) \rceil$ being the least integer k with $2^k \geq 1/\varepsilon_{\mathrm{EV}}$, decided by comparing the rational $1/\varepsilon_{\mathrm{EV}}$ with powers of two; the order α is taken exactly, as the dyadic rational its binary64 value is, so that $\frac{\alpha}{\alpha-1}$ is a non-zero rational; the term $\tau \coloneqq \frac{\alpha}{\alpha-1} \log_2(1/\varepsilon_{\mathrm{PA}})$ exactly where $1/\varepsilon_{\mathrm{PA}}$ is a power of two, τ being then rational, and otherwise through an enclosure $[L, U] \ni \tau$ formed in exact arithmetic or by a rigorously enclosed computation and refined until $\lfloor c - U \rfloor = \lfloor c - L \rfloor$, which is then $\lfloor c - \tau \rfloor$; the refinement ends, τ being irrational there, as a non-zero rational multiple of $\log_2(1/\varepsilon_{\mathrm{PA}})$, which is irrational for a dyadic rational $\varepsilon_{\mathrm{PA}}$ whose reciprocal is not a power of two, so that $c - \tau$ is not an integer. The key length is $\max\{0, \lfloor c - \tau \rfloor\}$, and the rule of Requirement 16.4 is met as written. The clamp is recorded as Requirement 24.3 records it. The first-order number and the key length formed from it, by Requirement 24.7 (iii) or without a companion, are recorded beside it (`first_order_key_length`). In the fallback of (i), the certified number is the largest certified bracket on the declaration the run searched, a lower bound on that declaration's value and not on the companion's, and its key length is that of Corollary 21.3 on that declaration, never that of Corollary 21.4; it is reported at the certified tier, and the blocks that Requirement 24.7 (v) names for the fallback rest on Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement). The companion's first-order number stays the run's first-order number, recorded beside it with its key length, and no number is formed from the two together: the key length is that of the fallback's certified number alone, whether the companion's first-order number gives a larger key length or a smaller one. Where the companion's bracket is withdrawn by (iv), the run's first-order number is that of Requirement 24.7 (iv) on the declaration the run searched, a number certified there is a lower bound on that declaration's value with the key length of Corollary 21.3, and the blocks the companion carried rest on As-

sumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) with the cause `factor_enclosure_inconsistent`; a run that declares `strict_snaps` refuses there (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iv)). Where no point is certified, or the arithmetic the evaluation needs is not available to the engine (`certifier_unavailable`), the run's number, key length and tier are those it has without this requirement, recorded with the reason `not_certified` and the reasons of every point.

(vii) *Premises and evaluations upstream of the declaration.* A run is not reported at the certified tier where (a) an assumption of the category numerical-trust is invoked on it, reason `numerical_trust_premise`: an assumption of its assumption closure counts as invoked unless the module that owns it records that it was invoked nowhere on the run, as the receiver records its eigenvalue-slack assumption beside the settings at which it was invoked (Requirement "Invoking settings on the eigenvalue-slack row of the ledger" (receiver/req:eig-ledger-row)); or (b) the declaration it is formed on carries a weight routed from a receiver's high-photon weights (Definition "The weight route" (reductions/wpfss/def:weight-route)) into which a per-setting weight enters that the receiver's artifact records as evaluated in floating point (Requirement "The evaluation label on every artifact" (receiver/req:evaluation-label)), the uniform weight of a slot entry counting the weight of every setting as entering it, reason `receiver_weight_floating_point`. Its certified number is then recorded with the reason, and the run's number, key length and tier are those it has without this requirement.

(viii) *The strict option.* A run that sets `certified_evaluation` to `strict` and is not reported at the certified tier by (vi) and (vii) refuses (`certification_refused`), naming the reasons, and reports no key length. A run reported at the certified tier through the fallback of (i) is so reported under this option too and does not refuse: the premise it rests on there, Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement), is named in its record, and the option that refuses a run resting on it is `strict_snaps`, under which the fallback does not run.

(ix) *The record.* The run records the option; per point, whether it was certified, b_t^{A}, b_t^{F}, its certified bracket, the largest radius of its enclosure of the gradient and its reasons; and per run $\underline{h}_\alpha^{\mathrm{cert}}$, the largest such radius over the certified points (`max_gradient_radius`), the working precision (`working_precision_bits`), the library of the rigorous arithmetic with its version (`arithmetic_library`), and, as a diagnostic, $\underline{h}_\alpha^{\mathrm{cert}}$ less the first-order number on the same declaration, with its

sign (`certified_minus_first_order`), which in the fallback is the bound on the searched declaration that Requirement 24.7 (iii) records and is absent where that declaration has none. It records further the declaration the per-point entries were evaluated on (`declaration`: `reference_companion` or `searched`); the premise at the snap the certified number rests on (`premises`): the short id of Assumption "The device measures the snapped family where the run reads the snap" (foundations/ass:snapped-measurement) wherever a moved block or class of the declaration the number is formed on rests on that assumption (Requirement "Snaps of a measurement family: kept, or moved and disclosed" (foundations/req:snap-reference) (iii)), among them, on the declaration the run searched, every moved block or class it has, in the fallback, without a companion and where the companion's bracket is withdrawn alike, and, on a companion, every moved block on which it carries the snapped block and every moved block of another snap it shares with the declaration the run searched, such as that of another measured factor; and none otherwise; with it, the blocks and classes that rest on it (`resting_blocks`), with their groups and causes as the record of the run names them where the key length is formed from a number on that declaration, those of Requirement 24.7 (v) among them in the fallback, with the cause `companion_not_certified`; and, wherever the fallback ran or the companion's bracket was withdrawn by (iv), the companion's points, each with whether it was certified and its reasons (`companion_points`), which say why. The fields `premises` and `resting_blocks` describe the declaration that `declaration` names, whatever the status: they name what a number certified there rests on, also where none is reported at the certified tier. In a fallback not reported at the certified tier, the status being `withheld` or `not_certified`, they therefore name the blocks of Requirement 24.7 (v) with the cause `companion_not_certified`, under which that item names them where the fallback's number forms the key length, while the run's key length is formed from the companion's first-order number and the record of the run names those blocks as evaluated off their snap. None of these but $\underline{h}_{\alpha}^{\mathrm{cert}}$ enters a number.

Reading: After the search, the engine recomputes the bound at the few points the reported number came from, in an arithmetic that encloses every rounding, from the declaration's data. Where that succeeds at some point, the key rate is certified: no step between the declaration and the number rests on floating-point arithmetic being accurate enough. Where it fails on the model the engine holds of your device, it is tried again on the moved measurement the search ran on, unless the run sets strict_snaps; a rate certified there is reported as certified, and the report names the premise that your device performs the moved measurement on the blocks concerned (A-SNAP). Where both fail, for instance because a detector element has a rank no enclosure fixes or a multiclick weight was computed in floating point, the report keeps the first-order number and says why; a strict run is refused instead.

Motivated by Def. 7.2, Thm. 7.5, Lem. 8.2, Cor. 8.5, Lem. 8.6, Def. 10.13, Req. 16.4, Def. 18.1, Thm. 19.1, Lem. 20.1, Lem. 21.1, Lem. 21.2, Cor. 21.3, Cor. 21.4, Rem. 22.2, Rem. 22.4, Rem. 22.5, Req. 24.1, Req. 24.3, Req. 24.7, Req. 26.1, Req. 26.2, Req. 27.1, foundations/ass:snapped-measurement, foundations/cor:least-eigenvalue-over-enclosure, foundations/cor:rank-over-factor-enclosure, foundations/def:factor-enclosure, foundations/def:tiers, foundations/lem:least-eigenvalue-bound, foundations/lem:matrix-function-perturbation, foundations/lem:recentred-enclosure, foundations/req:exact-island, foundations/req:rounding-direction, foundations/req:snap-reference, foundations/req:tiers, receiver/req:eig-ledger-row, receiver/req:evaluation-label, reductions/wpfss/def:weight-route · *Used by* Rem. 22.5 · implemented (effective: accepted) · *Code* certiqs_key.solvers.frank_wolfe.certified:evaluate

Interface

The declared inputs and outputs of the module and its submodules: each key with its symbol, when it is present and the claims that fix it. The domains, types and refusal rules are in the interface files themselves.

Inputs: MEAT key-rate engine (meat-core)

From `interface.yaml`.

Key and symbol	Presence	Fixed by
`alice_dim` d_A	always	Def. 1.1, Def. 1.2, Rem. 5.3
`alice_marginal` tau_A, or null	always	protocol-compiler/def:declaration, Def. 1.1, Def. 1.2, Rem. 5.3, Def. 7.2, Lem. 7.3, Rem. 7.9
`sector_index_set` S	always	Def. 1.1, Def. 1.2, Def. 5.1
`sector_dims` (d_s)_s	always	Def. 1.1, Def. 1.2
`sector_block_kind`	always	protocol-compiler/def:declaration, Def. 1.1
`alice_povm` {M_x}_{x in X}	always	Def. 1.1, Def. 1.2, Thm. 1.5
`bob_povm` {F_v}_{v in V}	always	Def. 1.1, Req. 14.1, Def. 1.2, protocol-compiler/def:exact
`constraint_list` CList = {(hat O_k, bowtie_k, c_k)}_{k=1..m}	always; m >= 0, and m = 0 is the default route	Def. 1.1, Def. 1.2, Def. 2.2, Rem. 10.20, Rem. 3.4

Key and symbol	Presence	Fixed by
`attack_set_license`	optional; required by part (ii) of the two-part gate whenever m >= 1 and a key-length statement is wanted. Absent there, the engine refuses the key-length output (req:constrained-key-gate): the key-length entry points raise SecurityGateError naming the failing part, and a run emits no key length, its reason naming the failing part unless h_low is infinite, which is reported first. The program is still solved and its numbers are reported for exploration; they do not reach eq:keylength.	protocol-compiler/ass:atk, Thm. 3.1, Cor. 10.19, Rem. 10.20, Rem. 3.6, Req. 16.10
`source_replacement_license` the transmitter's source-replacement licence record, or absent	read in trusted-source mode at d_A >= 2; at d_A = 1 it is vacuous and must not be demanded (prop:trivial-alice-system (iv); req:marginal-licence-gate; transmitter/req:licence-scope)	Req. 16.12, Rem. 5.3, Prop. 5.2, transmitter/ass:src, protocol-compiler/def:eligible
`party_partition` Party	always	Def. 1.1, Def. 1.2, Ass. 13.8, transmitter/ass:src
`public_structure` (a_A, a_B, phi, r, h)	always	Def. 1.1, Def. 1.2
`label_index_map` idxm	always	Def. 1.1, foundations/req:declared-order
`acceptance_set` Qacc	always	Def. 2.5, Def. 5.1, Rem. 5.9, Lem. 18.2
`rounds` n	always	Thm. 10.7, Cor. 10.18, Def. 1.2
`sector_factorisation` the ordered factor list of (DX1)	general format only; absent is one measured factor and no A factor	protocol-compiler/def:declaration, Def. 1.1, Def. 5.1, Thm. 7.5, Def. 4.8
`alice_form` 'povm' or 'ensemble'	general format only; absent is 'povm'	protocol-compiler/def:declaration, Lem. 1.3, Lem. 7.4
`alice_ensemble` {N_x}_{x in X}	general format, ensemble form only; in place of 'alice_povm'	protocol-compiler/def:declaration, Def. 1.1, Lem. 1.3, Lem. 7.4, Thm. 7.5

Key and symbol	Presence	Fixed by
`bob_povm.per_factor` ({Fˆf_v_f})_{f in F}	general format only; one factor is the ‘bob_povm‘ row	protocol-compiler/def:declaration, Def. 1.1, Def. 4.8, reductions/thm:emitted-product
`transcript` ((pi_m, C_m, a_m))_{m = 1..L}	general format only; L = 2 is the pair ‘announce_alice‘, ‘announce_bob‘	protocol-compiler/def:declaration, Def. 1.1, Def. 1.4, Def. 1.2
`constraint_scope_and_party_local_tag` (O_k, rel_k, c_k; f_k)	general format only	protocol-compiler/def:declaration, protocol-compiler/def:homog, Def. 2.2, Def. 5.1, Def. 6.1, Rem. 6.8, Req. 14.8, Thm. 6.2
`reference_companion` Dˆref, beside D	where the protocol compiler emits one beside the declaration (protocol-compiler/req:reference-companion-compiled)	Prop. 12.1, Cor. 12.2, Req. 14.3, Req. 14.1, Req. 14.2, Cor. 19.7

Outputs: MEAT key-rate engine (meat-core)

From `interface.yaml`.

Key and symbol	Presence	Fixed by
`h_low` h_low	always	Def. 5.1, Lem. 7.11, Rem. 5.9, Thm. 5.4, Cor. 5.6

Key and symbol	Presence	Fixed by
`h_lb` h_lb	The submodule's row, restated here only as far as a consumer of this module needs it: present exactly when at least one iterate contributes a bracket, every hypothesis of frank-wolfe/thm:firstorder having held at it; absent otherwise, with a machine-readable reason naming the hypothesis that failed (frank-wolfe/req:first-order-withdrawal). The oracle's return and its status remove no iterate, the feasible side of every bracket being a lower bound verified without them (frank-wolfe/req:verified-feasible-side). The domain, the withdrawal vocabulary, the clamp and the recorded repairs are spec/modules/meat-core/frank-wolfe/interface.yaml's.	Thm. 19.1, Req. 27.1, Req. 24.1, Req. 24.3, Rem. 23.1, Rem. 23.9, Rem. 22.4
`reported_tier`	always	foundations/def:tiers, foundations/req:tiers, Rem. 23.1, Cor. 19.5, Req. 28.1, Cor. 21.3
`key_length` ell	Emitted only on a run whose 'reported_tier' is 'first-order' or 'certified', and formed from the number of that tier. Absent – not zero, not clamped – when 'h_lb' is absent, when h_low = +infinity, and on a declaration the run tier did not pass as the compiler's emission (req:compiled-declaration-only; 'key_length_reason' then opens with 'declaration_not_compiled'); in every case the reason is machine-readable and no key-length claim is made.	Cor. 10.18, Thm. 10.7, Cor. 19.5, Cor. 21.3, Cor. 21.4, Prop. 10.17, Def. 10.13, Req. 16.4, Req. 16.6, Rem. 10.16
`security_parameter` eps_PA + eps_EV + 2 eps_auth	emitted exactly when 'key_length' is	Cor. 10.8, Ass. 13.6, Cor. 10.9, Rem. 10.10
`net_key_length` ell - k_auth	emitted exactly when 'key_length' is	Cor. 10.8, Ass. 13.6

Key and symbol	Presence	Fixed by
`sector_report`	always	Def. 6.1, Def. 4.5, Thm. 6.2, Prop. 6.3, Prop. 6.4, Prop. 6.5, Rem. 6.10, Req. 14.6, Req. 14.8
`sector_weight_report` (w_s)_s	always	Thm. 7.5, Def. 17.1, Prop. 17.2, Def. 6.1
`map_validation_report`	always	Def. 1.4, Thm. 1.5, Lem. 1.7
`optimizer_certificate`	always	Rem. 23.3, Rem. 23.2, Rem. 23.7, Rem. 23.8, Req. 27.1, Req. 15.3, Req. 15.6, Def. 7.2, Rem. 7.8, Rem. 7.9, Def. 7.10, Prop. 7.6
`refusal`	present exactly when this module refuses	Req. 14.1, Def. 1.1, Req. 24.1, Req. 24.4
`assumption_ledger` A-...	always; empty on a run that reports neither a first-order bound nor a key length	—
`assumption_bindings`	always; one entry per ‘assumption_ledger‘ row, in its order	Ass. 13.1, Ass. 13.2, Ass. 13.3, Ass. 13.4, Ass. 13.5, Ass. 13.6, protocol-compiler/ass:atk, transmitter/ass:src, protocol-compiler/ass:rhom, transmitter/ass:seq
`mathematical_spec_version`	always	—
`sector_report.joint_pairs` J_K(c-hat) and the key-inert joint pairs	every run (one Frank-Wolfe driver: an accepted-format declaration is solved as its degenerate general embedding and writes the general report)	Def. 4.8, Lem. 4.9, Def. 6.1
`sector_report.records[].removable` the ‘removable‘ flag	always	Thm. 6.2, Req. 14.8, Rem. 6.8
`dropped_set` D, and h_low^D	only when the run chooses a non-empty dropped set	Def. 9.2, Prop. 9.3, Cor. 9.4, Req. 15.10, Cor. 19.6
`statistics_consistency` b, alpha_0, x, tau, n f(alpha’)	always	Req. 16.2, Prop. 9.5, Prop. 10.12, statistics/ass:reach

Inputs: Frank-Wolfe first-order bound

From `frank-wolfe/interface.yaml`.

Key and symbol	Presence	Fixed by
`choi_program` the feasible set Feas^J and its objective	always	Thm. 7.5, Def. 7.2, Lem. 7.3, Def. 7.10, Def. 5.1
`gradient` the split-form gradient	at each iterate at which a bracket is accumulated	Thm. 8.4, Cor. 8.5, Lem. 8.3, Lem. 18.2
`acceptance_set_exact_data` Qacc	always	Def. 2.5, Def. 18.1, Lem. 18.2, Rem. 22.2, Req. 26.1
`docking_claims`	always	Lem. 7.11, Cor. 10.18
`linear_oracle_return` the primal block list and the dual the backend returns at an iterate	at each iterate at which a bracket is accumulated, where the backend returns them	Req. 27.1, Lem. 20.1, Thm. 19.1, Def. 18.1, Rem. 23.7, Rem. 23.2
`dropped_set` D	optional; empty is the run without a dropped set	Def. 9.2, Prop. 9.3, Cor. 9.4, Cor. 19.6

Outputs: Frank-Wolfe first-order bound

From `frank-wolfe/interface.yaml`.

Key and symbol	Presence	Fixed by
`h_lb` h_lb	Present exactly when the index set of contributing iterates is non-empty – when at least one iterate satisfied (H-POS), (H-SUPP) and (H-NOPERT) and lay in the relaxed feasible set with its acceptance factor formed on the exact island. Absent otherwise, with a machine-readable reason from ‘withdrawal_reasons‘ naming what failed; if no iterate survives, the index set is empty and the number is absent for that reason, with the same vocabulary (req:first-order-withdrawal). The oracle’s return and status decide nothing here.	Thm. 19.1, Req. 27.1, Req. 24.1, Req. 24.3, Rem. 23.1, Rem. 23.9, Rem. 22.4, Cor. 19.7, Req. 24.7
`reference_bracket` h_lb[D^ref], beside h_lb[D]	wherever a compiled reference companion was handed to the run (protocol-compiler/req:reference-companion-compiled), whether or not it was admitted; null otherwise. A companion not admitted (meat-core/req:companion-admission (i)) is recorded here with status not_evaluated and the admission’s refusal.	Req. 24.7, Req. 28.1, Cor. 19.7, Rem. 22.4, Req. 14.3, foundations/def:evaluated-off-snap, foundations/req:snap-reference
`first_order_clamped`	always, on a run that emits the bound	Req. 24.3, Rem. 23.9, Thm. 19.1, Cor. 19.5
`key_length` ell	the core’s row; emitted only on a run whose tier is the first-order or the certified tier, from the number of that tier	Cor. 19.5, Cor. 21.3, Cor. 21.4, Req. 28.1, Cor. 10.18, Thm. 10.7, Prop. 10.17

Key and symbol	Presence	Fixed by
`optimizer_certificate`	always	Rem. 23.3, Rem. 23.2, Rem. 23.7, Rem. 23.8, Req. 25.4, Req. 25.3, Req. 24.2, Req. 26.3, Req. 27.1, Rem. 22.2, Req. 26.2, Req. 15.3, Req. 15.6, Rem. 7.8, Def. 7.10
`h_lb_dropped` h^FO,D_alpha	present exactly when a non-empty dropped set was used and the index set of contributing iterates is non-empty	Cor. 19.6, Def. 9.2, Req. 15.10, Cor. 10.18
`certified_bound` h^cert_alpha	Present exactly when the certified evaluation ran and certified at least one point (req:certified-evaluation (vi)); absent where the option is ‘off‘, where the run formed no first-order number, and where no point was certified, with the reasons under ‘certified_evaluation‘ below.	Req. 28.1, Cor. 21.3, Cor. 21.4, Lem. 21.1, Lem. 21.2, Rem. 22.5, foundations/def:tiers, foundations/ass:snapped-measurement

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